Introduction to Cognitive Psychology

Complete Course Transcripts

IIT Guwahati — Cognitive Psychology Course

*Important terms and concepts are highlighted in **bold** throughout the transcripts.*

Lecture 1: A Brief History of Cognitive Psychology-01

■Kind: captions hello I'm the winter chef I teach psychology at Indian Institute of Technology Gatti one of the subjects in psychology that I teach here is **cognitive psychology** and so I will be teaching you this course this course on **cognitive psychology** will be covered through eight modules we'll start with introduction to the history of psychology **cognitive psychology** then moving on to the methods of studying **cognitive psychology** we will deal with **perception** the basic process of taking in information then moving to **attention** the filter or the barrier which which lets information gets encoded from there we will move into how whatever information has been perceived or taken in into our system how that is stored as memories further on we'll look into what are memories what are the type of memories which are there how many processes operate things like how processes of **encoding** and **retrieval** really work and what is the way in which information is organized in in **memory** once you are done with **memory** we'll move into something called thinking which is a higher order cognitive processes and which draws on from **memory** now thinking is a process which will lead on to something called this isn't making another important proper process in **cognition** which is central to the idea of how people actually make decisions in real life or how people decide or make choices in real life we will also be venturing into a little bit into something called **language** which is how people you signs and symbols to communicate ideas between them because communication of ideas or whatever one thinks whatever **cognition** ones have sharing that **cognition** with other people is also central to the idea of psychology so let's then move quickly into a brief history of **cognitive psychology** psychology developed from philosophy it came out of philosophy and philosophers we're the people who actually first proposed how knowledge is stored how knowledge is dealt and how knowledge is acquired and so they led to the development of the science which is psychology and so before moving on to the history of **cognitive psychology** let's look at a brief history of the philosophically inputs to **cognitive psychology** so one of the brief inputs of philosophy or one of the brief debates of philosophy which led to the science of development of **cognitive psychology** was the debate on something called nativism and empiricism what is this debate empiricism explains that knowledge the idea of knowledge or the idea of how knowledge gets stored it comes from individuals experiences so individual so their own experience to their own interaction with people around them with society around them or through memories are true or other kind of experiencing systems generate knowledge whereas the nativist believed that the rule of heredity the role of biology or the role of constitutional factors are responsible for the generation of knowledge so what was the debate all about on the empresa side we have people Aries total and people like John Locke now John Locke for one who was an English philosopher and from 1632 to 1704 he talked about knowledge or the basic of **cognitive psychology** in terms of contents of the mind so what is the content of mind and he said that mine compose of thoughts it comprises of thoughts and these thoughts are mental images our series of mental images which are associated together and they are what mind is all about or knowledge is all about also John Locke proposed the idea that people at the time of birth ba are born with a blank state of mind so there's nothing written in the mind itself and this debate that this blank state or this idea that is Bank State with experience or true experience get written up and that makes knowledge or that develops knowledge or cognitions similarly Aristotle had a similar viewpoint where he believed that personal experiences of people is the reason how this knowledge develops on the other hand people like Plato or people starting with Plato other people like Rene Descartes they believed that knowledge actually comes from heredity or it comes from the composition or biology and so they basically are the descendants or they basically progress from Darwin's point of view they believe that knowledge is inbuilt or it comes as an inbuilt mechanism and the fact that this inbuilt mechanism is as the reason for knowledge specific to **cognitive psychology** Rene Descartes a who was a French philosopher he gave this idea of my then body Plato believed that **memory** is the first use of Plato where that memories were actually writings which were on a wax tablet and he believed that this mind that people have is actually a

wax tablet and the writings on this tablet is what our memories and so what he goes ahead and says is that these memories the way these memories are written the way these writings are the wax tablet the strength of these writings actually explain or actually tell you how strong or weak a **memory** is also the content of this what is written on this Rex tablet will tell you about the content of memories the idea of John Locke that this mental images the series of mental images are actually what are the contents of mind was contested by someone called George Baker birthday now what is George Burkley say he contested the idea that thought processes or contents of the mind cannot be mental images which are associated together the **reasoning** that he gave to this was that abstract ideas like **judgment** or like truth this cannot be conceptualized as mental images as they cannot lead to this idea of the mental images and so these were some of the some of the basic philosophically passions of philosophical view points of what psychology in general and **cognitive psychology** in special or in in particular are actually comprised of should comprise all with the coming of someone con William weren't in the Year 1879 the first scientific school of psychology developed William bond developed this school at a place in Leipzig in Germany and with his school the first structural school of psychology came into being the basic things or the basic ideas that William actually went ahead and studied where the nature of consciousness what is consciousness in the nature of consciousness now consciousness as such has been one of the basic materials or one of the basic inputs which are the which are the subject matter of **cognitive psychology** so what William wind was trying to do as a scientist as a scientific school of psychology was studying what consciousness is all about for the first time William Volt was able to explain psychology with a scientific viewpoint now what was the scientific viewpoint and what was a limb would actually trying to do we lived won't believe that any psychological process can actually be broken down into sub processes now this idea of breaking a **mental process** into his basic process is actually a hangover which comes from the physical sciences and since William wound came from the basic sciences this breaking of the **mental process** into sub processes well actually hangover which he had and what did he believe no William volt basically approached or described consciousness as to it a two-part system one he said that cons describing a consciousness will require two things characterizing the basic sensations which is what are the sensations which characterize the **mental process** things like the heat the color the texture of the of the particular sensation and also characterizing the feeling what is the feeling associated with now as you can see in this particular slide what William won't would want is to study mental processes or mental ideas using the basic process of hard sciences like chemistry and physics on this slide you can see that there is a structure of a chemical or organic chemical compound and as you can see this organic chemical compound is basically composed of several parts and this chemical component itself can be described by the parts of our elements which comprises it what William want was attempting to do is to study the idea of seeing this Apple the consciousness of staying this Apple and break it into processes really into basic processes of how people see this Apple as you can see on the right hand side there's an apple and there is a process to it now the variability the basic idea of Apple what William Kuhn said is comprised of these five or six ideas what is the idea for example it can be described the Apple can be described in color of in in terms of its color in terms of its taste in terms of its shape in terms of its shine or in terms of how you feel after eating it and that is what I was talking about so what William will did was the idea or the consciousness or the **concept** of Apple the mental image of Apple the mental construct of Apple he described can be described over can be broken down into its physical features which is the shape color texture and things like that and the feeling that we have after eating an apple or how do we feel after it being eating a particular Apple and the second thing is he also wanted they also included so one thing was characterizing the feeling and the basic sensations and the second part of this was to deal with the rules through which these sensations were combined together what are the rules which through which we combine the perceptions the idea of an apple and the feeling of an apple and so he related that this could be done through **perception** where in **perception** we actually not only see the color shape and size we combined them together to see a whole Apple so in all what William will was

trying to do is trying to explain mental processes in terms of the sensations or the the sensation and feelings and also how these sensations and feelings are combined together to give you the actual construct or the actual image of an Apple this notion of taking a mental image or taking a mental feeling and breaking it into its part this led to two major things one he showed that mental activity mental images can be broken down into basic operations so when I see something or when I feel something the feeling of a particular sunrise or sunset can be broken down into its physical part which is the heat that I feel or the coal that I feel or the kind of feeling that I feel and also the basic factors the basic temperature the basic light intensity which the Sun gives off a sunset I'd combine these two two things together and then I get this mental activity or the mental feeling of what a sunset is and this is the second thing that William wind was able to do was to develop an objective method of studying mental activity up till now most philosophers are talking about mental activity as a process as icing on a tablet or it could be in terms of consciousness but none of these things could actually be studied objectively could actually be studied or could actually be measured as changes but what William don't want did was take a mental activity was to take from this mental activity and divide it into its parts which can be measured and which can which are observable in nature so this is the this is the contribution of William bolt the problem with what William Boone suggested was that mental images does not gain guarantee a mental activity and so the study of mental images in no way will guarantee the study of mental activity for example let's say I give you different weights and ask you to compare two weights one is a lighter or whether one is lighter than the other or when one is heavier than the other it's an easy task and most people will be able to do it but the question is do people actually realize that they're doing this comparison or are people able to basically Center on those mental activities which are required for doing these comparisons doing this comparisons of mental weight and the answer to this is no we do not have any idea of what is the mental activity which is going on what is the actual mental activity is going on when comparing this face obviously we will get answers on superficial terms from people of whether two weights are lighter or harder but what is the actual mental activity which is acquire which is required for doing this kind of calculation is not possible let's take another example suppose I asked you to describe the ear of a cat what is the ear of a cat look like most people will be able to do that but when I ask you this question that when I said or when I asked you what does the ear of a cat look like did you actually bring forth an image of the cat in your mind for answering this question not many people will be able to answer it truthfully saying that they do actually bring up the idea or the image of a cat why is that one reason is because we do predictions and in terms of predictions we know how a year looks like and so how the cat's ear would look like it comes from semantic knowledge and we really do not need to imagine a cat and then look at its head and then look at it here in terms of mental activities to actually come up with the idea of a ear of a cat or ascribed in the year texture or shape of a cat's ear so these are some of the problems with his theory or with his **structuralism** which was a solution to that was attempted through something called a new school which was **functionalism** so the school of **functionalism** was started by someone called **William James** and so **William James** was off the idea that the nature of mental activity is not the only solution to studying **cognition** but rather the functions of spatial specific mental activities are what require or are what is required what do I mean by this the functionalist perspective suggests that the components of a mental activity are not important but how does that mental activity actually functions or actually is described in a process is what is of interest for example let's assume that we go to a lecture now in a lecture for some people writing a lecture down making no thought of a lecture down works that is how they learn and for some other people listening to a lecture when a lecture is being delivered is what is the key is what is the key to to to understanding of the contents of the lecture now the people for whom listening a lecture or hearing a lecture is the key to them for them to learn what these people can do is find out what aspects of listening in a lecture not writing legend down actually helps them in understanding the subject and let's suppose that they find out or the reasons why they they subscribe to listening a lecture is because when they go into a lecture a listen to a

lecture they get a chance to ask questions or they get a chance to see visual aids and learn things these might be the reasons why listening to a lecture is beneficial what they could do is in future if they subscribe to a lecture if they want to attend the lecture what they can do is basically look at those lectures which gives you opportunity to ask questions they can look at teachers which give you opportunities to ask more number of questions and then select that particular lecture and learn from it so basically the functionalist approach says that it is not the nature of mental activity that is important what is important is how the function of this is when this nature activity actually takes place or when this when this nature activities are actually taking place in in the real sense how does this really go ahead and an explained **mental process** so **mental process** is not just components of mental images it is it is in terms of how this **mental process** or how this the the acting out of this **mental process** actually lets you gain knowledge **functionalism** derives a lot from someone called Charles Darwin who proposed that evolution is the reason or the idea behind gaining of knowledge so basically there is the difference between **functionalism** and **structuralism** in terms of explaining this recovery psychology is that **structuralism** basically thought of mental processes thoughts of mental images or prescribed the idea that mental images are there and these mental images can be broken down and then can be studied and that is how **cognition** or knowledge is is generated whereas **functionalism** talks about are not these mental processes but rather how this **mental process** operates the operation of these mental processes is actually the way how people generate knowledge a third school which led or a third school which led to the formation of **cognitive psychology** was the get start school now this is a very simple school and this school proposed two rules one that the whole is always more than the sum of its part what does it really mean it subscribes to the idea that if we see something as a whole and if we see a part the same thing as parts they may not give us the combined experience think of it in terms of a movie when we go to a movie we not only see the pictures but we also see the sounds the which is coming from the pictures the whole movie experiences combined together to give you the idea of what a movie is well as you take take the movie apart the picture the music the sound the other effects everything taken apart from it and then show you show them one by one to you will you have the same experience no you will have entirely different experience take another example the idea of a song no song composed of lyrics it composes of music which is that it composes of special effects and so many other things into it now if I take this apart if I first make you hear the lyrics and later on add-on or make you hear the instruments which play with it and later on come out show you or make make you hear the basic effects or are the basic special effects will you have the same experience as when these combined together to form the song no they're different and that is what **cognitive psychology** or **Gestalt psychology** proposes are **cognitive psychology** they say that the idea of studying **cognition** is that the greatest psychology is that the experience that you get from studying a whole process is different from breaking the process apart and so this school was in direct opposition to William Boone school who proposed **structuralism** where they said that the mental activity any mental activity for that matter can be broken down it's true spot studied and how knowledge is gained her knowledge progresses can be understood another school which adds up or which basically goes ahead and explains some of the basic ideas about **cognitive psychology** or someone one of the basic ideas for a psychology is **behaviorism** now **behaviorism** Vicious was a school which strictly proposed the idea that there is a stimulus and there is a response and each stimulus is bound to a particular kind of a response so stimuluses leads to response as the moon are made the optimal response is always adapted as the effect of a stimulus what we've reasoned did was it didn't subscribe to the idea of hidden mental states raising **behaviorism** said since we are not able to understand the mind here since we are not able to understand what is inside somebody's head so let's look at what is observable and so what are the things one of the main propositions of **behaviorism** was that let's look at what is observable so if there is a stable and it creates a response which can be observed and this response could lead to a consequence this is what it should be studied so study a stimulus study a response and study a consequence to the response

and this is what should be the subject matter of **cognitive psychology** cognitive psychologists should not waste its time on what are the mental processes what are the mental images what is inside somebody's mind which and how they direct the notions of knowledge gathering Clark Holly one of the behaviorist he proposed that although mental events cannot be studied through **behaviorism** but their existence can be deduced by their effect on behavior and so what he says is that although things like motivation which is basically internal behavior or is an internal even they it cannot be studied but what can be done is that its effect of motivation can be studied on behavior can be studied can be studied through what people do when they are motivated so if somebody is motivated he will produce most optimal responses and based on that his efforts on doing that particular job or doing that particular act will increase and that is why this increase in the response when he gets rewarded by doing that response basically shows the presence of motivation Skinner who was another behaviorist he went so far ahead and said that look there are thing called internal events there is nothing called mental crosses and so the basic idea or the basic subject matter of **cognitive psychology** should be stimulus response and consequences and studying the relationship with what stimulus leads to what response and optimizing understanding what is the optimal response to a particular stimulus and then studying how these responses are strengths or weaken and that basically happens through studying of consequences the problem with behaviorist are the problem why behaviorists were not successful in understanding **cognition** or in developing **cognitive psychology** was that they couldn't go ahead and explain some of the most interesting behaviors or the existence are some of the most interesting behaviors for example the idea of **language** or how 3d **perception** and things like **language** or depth **perception** initiation people this led to the fall of **behaviorism** as a proponent or as a school which explains you what the subject matter of **cognitive psychology** is or what **cognitive psychology** is all about since most of these schools were trying to understand what should we the subject matter of **cognitive psychology** or what convey psychology is all about none of the schools could actually come up with a definitive answer of what is the true subject-matter of **cognition behaviorism** to some extent did give some very good inputs for developing of a theory of **cognition** but they to fail in some ways because they were not looking at external events what **behaviorism** contributed was that they developed some of the most rigorous idea or experimental techniques of studying **cognition** also they for the first time went ahead and studied animal **cognition** and they said that cutting the way animals behave could lead to understanding how people behave or how people gathered knowledge or how information or how **cognition** really works so they although didn't provide with enough matter to the development of **cognitive psychology** but they did provide enough impetus to the development of a new school of how knowledge is gathered and what is mental activity questions like these a new school came in and they started explaining what is the subject matter of the mind and how is knowledge gathered and this school was called the cognitive school or the **cognitive revolution** now up till this point of time we have been dealing with psychology we have been doing with experiments in psychology we are we dealing with people who have worked in psychology and they have been giving several definitions of what psychology is what it should do what **cognitive psychology** should do what should be subject matter of psychology psychology how knowledge is gained and those kind of things but the new school that came in and explained what is **cognitive psychology** or what is the basic way in which mental events or mental processes a function was the cognitive school this school started with the development of the first computers and this school came up or the school was proposed or came out from the ideas of people like Herbert Simon Alan Newell and norm Chomsky for the first time with the development of computers psychologists got an opportunity to model the working of the human mind in terms of sets in terms of items which taken information and process it and to understand how thought processes are how human thinking or mental processes are actually taken care of by the mind now the computer gave the cognitivist an ideal medium to study the internal mechanism of how information knowledge is grasped is processed and is developed and is used so as I said before people like Herman's Herbert Simon and nuwell who give the

idea of computers gave the idea of computer processing systems provided a chance for the development of the field of **cognitive psychology** not the basic to understanding **cognitive psychology** is studying the mind right up till now we have been looking at schools or we have been looking at viewpoints from various schools of psychology and what do you think about or how they affected the development of **cognitive psychology** is a field from now on we will be looking at what is the subject matter of **cognitive psychology** and how it should be studied what are the processes and things like that so one of the things that led or that is school to understanding **cognitive psychology** or human **cognition** is understanding the working of the human mind now with the development of computers or the **cognitive revolution** there was a chance to compare human mind working with the working of the computers but the problem was the working of the mental activities which led to the working of the mind of how mind generated knowledge or how mine generated any activity and comparing that to a computer program is not a correct solution the reason is because the analog of mental activities as computer programs are wrong is because the mind and/or the brain and the computers are made up of in two different kind of things or two different kind of elements whereas the brain is made up of neurons and biological structures the computers are made up of hardware electronic circuits and several other things which are not comparable on any dimension also the mind the analogy which says that the minds and computers are same it proposes that the mind is the software which runs on the hardware which is the brain no as I said the comparison of the mind and brain to an operating system and the computer is not correct what are the reasons that I gave is because they are made up of different different kind of elements whereas the brain is made up of neurons and neural structures and biologic based materials the computers on the other hand are made up of electrical circuits also if you look at the mind and the brain the brain the mind works on top of the brain right which means that the operating system of the brain and the brain itself are one and the same thing or is a spot off of the same structure whereas computers don't have this specificity the idea here is that computer programs can be run on different kind of computers if I have an operating system no matter what computer I have no matter what kind of hardware that I have I can run this operating system on it and so the hardware the operating system the mind in terms of a computer an analog II can run on different different hardware's whereas the mind that a person has the brain which generates the mind this cannot be transported to other person up till now this is not possible where I can take in somebody's mind and and Transplant it to somebody else's hardware brain and make these people similar now these are some of the this is actually one of the most periodical questions or I would say taught experiments in philosophy where it's described that if I take the mind the brain of someone a part of brain or someone and implant it to someone else who who was having a malfunction brain as brains can can function half brains can function on their own will they be the same person now that's that's a thought experiment in philosophy and so this is the question here so if I take your mind and put this mind let's let's assume that I can take your mind and put this on somebody else's hardware the brain will two of you function similarly because the brain is more or less similar in most people and so it doesn't but on the other hand in terms of computers what really happens is that if I take an OS operating system and I take it to any hardware it keeps on functioning or it keeps on working in the same way also some software programs are so basic that at times we take these software programs and convert them into some kind of hardware some kind of hardware programs for example the basic IC chips which allows certain currents or certain signals certain kind of gates which are there in electronics engineering so at times simple programs computer programs can we actually convert it into a chip and this chip can be later on used for use as in hardware but basic human programs or basic human processes cannot be converted into a part of the brain and can be shifted so that's what this says that sometimes deputy programs can be cut into convert into electrical circuits whereas for humans is not possible so then what is the how do we differ between the brain in the computer or the brain and the mind this can be understood in terms of the various levels of analysis what does this really mean levels of analysis actually means the point from where you are seeing something the point from where you are

analyzing a particular feature for example in a computer program the two levels of analysis can be in terms of the basic electrical current the basic circuits that the brain has and the physics of the computer so one level of analysis will be how current flows through several hardware's in a computer and how these leads to the functioning of the computer on us second level on another level of analysis computers can also be understood in terms of what various programs do how these various programs go ahead what kind of processes the dual that they have what kind of functions that they do and how they go ahead and and actually create an output from the input given so what level is the physical level in in terms of computers understanding levels of analysis could be understanding computers in terms of the physics of it in terms of the in terms of the electrical circuits that it has or in terms of the programs that run on it and that describe the computer know for understand this with a more easy example let's say we are looking at a building so what I'm trying to tell you is what are different levels of analysis so let's think of a building one level of analysis of a building would be the architecture would be the way the building is constructed the design of the building how it is how it looks what kind of rooms and so on and so forth so this is a high-level analysis a lower level of analysis or another level of analysis of this building could be in terms or the bricks that is used to make this building the cement that is used the steel rods which are used and so in terms of the basic components we take of the make up the building although both levels of analysis are talking about the same building but they have different different viewpoints of it and so this building can be explained with the analysis with the level of the architecture architecture input or it could be understood in terms of the level of the basic building materials so then this is how the distinction of the brain and the mind can be that through different levels of analysis what this would mean is that when we are looking at any **mental process** we can look at a **mental process** from both a functional view as well as a structural view let's look at emotions what is the functional view of an emotion the functional view of an emotion is it's an arousal state it's a state of arousal which quickly clients and gets back to normal and through which people experience feelings this is more of functional but what is the structural view of looking at an emotion the structural view of an emotion could be expressed in terms of the brain regions which led to the emotion and probably the areas which are related to amygdala or areas near to it which leads to generation of an emotion so the same emotion that I'm talking about the same emotion that people have can we express both on a functional level in terms of what people feel and also in terms of the structural level of which the brain lead to the development of this particular feeling or with this particular emotion another interesting thing or another input matter subject matter of **cognitive psychology** our **mental representation** what a **mental representation** whenever we think about something maybe it could be a job it could be thinking about a friend a particular vacation whenever you think about this a lot of mental activities takes place in the brain and so these mental activities actually excess or actually use something called mental representations so generally what is a **mental representation** a **mental representation** is basically a physical state that conveys information specifying an object event or category or its characteristics in terms of paper and pen a **mental representation** would be the ink which writes a particular letter the way the particular letter is written in terms of the brain it could be the neural connections which leads to a particular kind of network connection a particular kind of mental state disease generated or in terms of computers it could be a particular kind of magnetic field so **mental representation** basically is a notion is a represent a is the way something is stored each **mental representation** or most mental representations have two things in common one is a form of the **mental representation** and the second is the content of a **mental representation** what is the form of a **mental representation** the form of a **mental representation** states the way a particular message is conveyed the format in which it is conveyed the way in which it is encode and the content of a **mental representation** is basically the meaning the knowledge the idea which is being conveyed to that representation so basically **mental representation** is think of it is a unit of knowledge and this unit of knowledge has a form which is in which format it has been encoded so it could be encoded in a visual format it could be encoded in a auditory format it could be encoded in some other

format a haptic format and the content is what is the data bytes into it what is the knowledge into it so mostly representations are like that let's understand what mental representations are and when I say that a **mental representation** is a unit of knowledge and it can it has a form and a particular kind of a form and a particular kind of content let me explain this a little bit more for example I want to convey to you the following sentence the ball is on the box this sentence means or this idea means that there is something on something a subject an object and a relation between a subject an object or it represents an idea now this idea has two parts one is what is second is where and the third is on what so the bodies of the Box the idea of a ball being in a box can be conveyed through two kinds of representations we can use a propositional representation and we can use a quasi pictorial representation what is the meaning of that prepositional response representations will tell you two ideas and how are these ideas related together whereas a quasi pictorial representation is going to tell you show you a picture of how these ideas are related together so when I look at the proposition of representation looking into it I have the ball which is on the box right so ball and box are two ideas or two representations which are connected by a representation on which basically means that something on something and in this case there is a ball which is a circular in shape and which is on top of a box which may be a rectangular in shape so then what this representation proportionally presentation explains is a relationship first of all so the ball is in the box if I represent it through prepositions it could be in terms it could it could have in terms of proportional disposition it should have a relation how the two ideas are connected together that undulation here is on second and argument what representations or what ideas are connected I have a connector on which is connecting two ideas but what are the ideas which is being connected the ball in the box what is the **syntax** of connecting now if I just write on ball box it is not going to make any meaning there is a **syntax** to it a **syntax** a is that one ball a ball it could be anything on which is the connector or box on one thing and the then the connector is so this ball is basically on the box lift up when I am saying that I am talking about the particular ball the particular box then the abstract what is the abstract the ball in the box the shapes of the balls in the box then the propositional representation it doesn't happen in a special media in a special media in the reason that it doesn't have a pin in space it doesn't explain you in space how these are related so in space you will you will generate the idea in your head that there is a ball and there is a box and the ball is on the box but you cannot see the picture of the ball on the box although you will it will generate the same idea of the ball being on the box and are very related to replace represent objects it can be are very when I use a quasi pictorial representation for explaining the same idea the ball is in the box I can draw a ball which is spherical in shape and a box now you have to be very specific I cannot draw a triangular ball neither can I draw cubical box hecta bundle box obviously it's allowed but then most box will have rectangular or square ish surface and so I have to be very strictly to that also the preposition of on suggest that the spherical shape has to be on top of the shape of the box and so in this case there is no distinct relation there is no distinct arguments no clear **syntax** no concrete ideas but then what happens is it happens this can be expressed in a spatial medium what does these mean what is the meaning of this the meaning of this is mental representations or mental ideas can not only be expressed in terms of words or in terms of prepositions in terms of ideas and connector connectors connecting these ideas but it will also be expressed in terms of just images and so images are one way of conveying an idea let's take another example ROM is running if I say the sentence ROM is running the two ways of showing it I can use running which is a verb or I can and then I can use ROM which is the noun and then show the relation between that running is done by ground this is a propositional so making the preposition or I can draw somebody running doing the act of running or make a cartoon of it and running and then I can show that ROM is running by that / - really both these representations both these formats will actually convey the same meaning so coming back to the idea that mental representations have a particular format and a particular idea or a particular content as you can see both these wings of representations have a format the first one has a propositional format and the second one has the quasi representational format which is a special

format of **encoding** ideas what is the content the content of the first is there is a an object a which is on top of object B similarly with the pictorial representations also the same idea or the same information is being conveyed that a is on top of B so then these mental representations that we have can actually or how do these mental representations work the working of these mental representations takes place through a process a processing system and **algorithm** so once we see something once we form a **mental representation** this **mental representation** is taken by the mind and this then this mind makes meaning out of it to using a process or processing system and an **algorithm** what does the process do the process takes in basic information and produce a specific out of it so the process will look at the **mental representation** whether it is in propositional form or in special form encode it through up through a system of cross **perception attention** that kind of thing a processing system will compose of several processes and these several processes will combine with one another to give a output to give us the idea that a is on B or that this is what is being conveyed so a process is basically one part of a processing system a process is basically a format or is is basically a process through which something is some information is taken and this information is transformed into specific type of output this specific output is later on combined through or it's is froth is passed through several processes and these several processes generally are comprised in something called a processing system so for example going back to the original thing looking at the quasi representational description looking at this makes you understand this knowledge that a ball is on the box now for this for you to understand that the ball is on the box there are several processes responsible and there is a processing system which is processing it one process the first process I would say which is responsible as **attention** you have to pay your **attention** to this particular image the special image the process of **attention** will take out everything else from the vicinity of this particular picture and what it will do is it will only concentrate your mental concentration onto this box and and this ball the process of **perception** will let you understand this or encode this in a visual format and then there will be process of **memory** through which what will happen is there will be a comparison of the size of this the shape of this the relation between this and then there will be a process of thinking of this isn't making will not be a process here but then this is how you will develop the idea that the ball is on the table so the processing system comprises of several processes and these several processes one after another gives you the final order an **algorithm** is basically a step by step procedure of how the processing system should function so any processing system should function in a particular way in a particular in in a particular direction and this algorithms tell you how to where to start where to go from there what to expect out of it and then where to go from there again for example in the idea of perceiving the ball and the box the **algorithm** tells the mind or the brain to first look at the shapes pay your **attention** to the shapes cut out the or separate the background from the foreground perceive this in the through the eyes in the visual format hold it into **working memory** let the **Central Executive** and episodic before of **working memory** talk to **long-term memory** and make a comparison of what you are seeing and based on that generating this output that this particular figure shows a ball on the table so this is how the **algorithm** says so it starts the **algorithm** says that it starts with **attention** goes to **perception** goes to **memory** those to this kind of **memory** really travels and then make making the decision of how this ball is on the table or how the what this particular picture explains so up till now what we have seen is what is the history of **cognitive psychology** and we've looked at the schools of area schools which gave their inputs or which led to the development of the field of **cognitive psychology** and we've also looked at what is the subject matter of **cognitive psychology** we looked at what are mental representations how they can be encoded what is the format in which we can encode and things like that we'll continue our journey by understanding why should we study the brain at all if mind is of what is importance why should be studying the brain and then following on following this we'll also look at the several paradigms of **cognitive psychology** or what are the basic bodies of knowledge of **cognitive psychology** thank you

Lecture 2: A Brief History of Cognitive Psychology-02

■Kind: captions so in the last lecture we looked at some of the basic processes of **cognitive psychology** we looked at what **cognitive psychology** is composed of we looked at the schools which led to the development of **cognitive psychology** is a field we look at the subject matter of **cognitive psychology** in terms of mental representations what are mental representations what do they represent how do they combine together to find knowledge and how mental representations are manipulated the way they can be expressed the way they can be understood in this lecture we look at why brain is to be brought into the idea of **cognitive psychology** up till now we have been looking at mental processes and we'll be looking at mental representations and the functions of mental representations we're looking at how mental representations function but the idea of **cognitive psychology** in terms of what behavior is said or in terms of what cognitive is said proposed that mental representations explain functions the process of how they combine together explained functions of a of how mental activities take place is not enough so at the inception of **cognitive psychology** it was all about the functional aspect of how mental representations make up mental activity and mental knowledge now one of the core challenges one of the main challenges of **cognitive psychology** is to understand what happens in the mind and how does the mind does what it does the conception of the mind have been differently put by different people philosophers have given one way of mark mind is psychologists have understood mine in more in terms of consciousness understanding consciousness and go through the **cognitive revolution** training of the mind was put down to or brought down to the understanding of mental representations understanding mental processes mentally activities and so on and so forth up till now till the **cognitive revolution** the study of mental processes descend to study of mental activity the study of **cognition** which is basically the study of mental activities was in terms of functions but the mind cannot work on its own and so it it's the the idea of mind and brain as a computer hardware and software introduces first the first time brain as that system on which the mind works and so one of the most interesting fact that has been bought or that has been thought of as my cognitive psychologists is introduction of the brain in the whole idea of study of mental processes of how mental events takes place the question is why do we study the brain what help will we get looking at the brain it is composed of different neurons neuronal grooves and these neuronal groups have biological structure and so at the sight of it it doesn't appeal or it doesn't give us much information about mental processes or the functioning of the mind so why should we actually look at the brain looking at the brain will help us identify those processes and the correct combination of processes that are required to complete a task what does that mean what it would mean is that since the brain leads to the mind the working of the mind the process of the mind will have a direct impact or would directly project to the brain and so changes in brain structure changes in brain activity can then be correlated to the of the mind think of it in this way and this is a very blow not example suppose a region a of the brain activates when a certain mental processes takes place and it doesn't activate it doesn't show activity when a certain be **mental process** takes place then now we have an observable a quantifiable fact or a quantifiable system on which we can look at changes because these brain activities the the increase and decrease of brain activities will signal whether process a is working or whether process B is working and so one of the reasons of including the brain or bringing up the brain into the whole idea of study of **cognitive psychology** is to identify because brain helps us in identifying what mental processes are taking place if we are able to form correlations between mental activity and brain activity but then there is a catch to it what is the catch different information processes they lead to the same kind of mental experience what I am trying to tell you here think of emotions now there is a correlation between a brain region which is the amygdala and the kind of emotion that people have and so if I'm looking at unhappy emotion or if I am looking at positive emotion what will happen is that most on most have positive emotions the same or nearly same brain regions will actually activate or show activity so how do I go ahead and pinpoint those

brain networks which lead to the happiness or sadness factor also there is no way to tell us how does the brain reduce or give labels to these emotions whether this is excited this is happy and this activity leads to this kind of emotional feeling so then the different sort of information processes can produce similar results or continue similar kind of at in in outputs is one of the basic problems by including of including the brain so one of the problems is that also there is a problem which is called the structure process trade-off why does it mean this is the same thing that I was trying to explain you a little while ago what is the structure process trade-off I try to explain this to you in a very easy way Sall Starnberg Sternberg sorry was trying to identify how people search mentally what is the process people use for doing mental search and for that he gave people students volunteers experiments so in the first conceptualization what Steinbach was trying to do was trying to find out how information is accessed from **memory** any conceptualized so an experiment which he gave to volunteers the experiment was like this certain numbers were given to people for example numbers 1 3 5 7 and 9 and later on these people once these numbers were shown to them later on these numbers were shown to them one by one and they were asked to tell with confidence whether these numbers were in the list or not so it's like a list recognition so what happens in this experiment is this list of number will given to people to remember to learn and once the learning was over one by one these numbers are presented and they were asked to decide they were asked to recognize whether each of these numbers represent in the list or not what he conceptualized is that this kind of list excess **memory** excess happens through serial search and why did he why did he propose that he found out that as the list increases the **verification** time also increases so list increases the length of the list increases the **verification** time also increases what he found out from his studies well that information excess from **memory** was in a serial way so what happens is that when once you saw this list to people people formed a mental image of this and then later on when you show one of these members of the list and ask them to verify from their mental image or from the **mental representation** whether this is present or not we what people tend to do is match this with this if the match of this with this results in a yes response the search is terminated if it is not you get a negative response here and so the search goes to the next item whether this and these are similar it is not so search goes to the third item not match again not match again not match the list returns to the first item verifies it again and then gives an output saying that it was not present in the list and this is a serial search so what happens is one item at the same time list 2 is matched to the list and that gives this kind of a serial search system and so obviously as the number of items will increase the **verification** time will increase because this serial searches have to be more in number this was what Saul Steinberg proposed and this is how what he said is the reason for accessing or how information is system in **memory** but later on people like Townsend gave a different conceptualization of information accessing from **memory** what did Townsend say Townsend said that the same conceptualization that I have the same problem can add that I have can be thought of as balls in a plate and so now I have one three seven four and nine and instead of a serial search what really happens is information is searched through parallel processes information access from **memory** is done through parallel processes not if that is true if parallel processes occur in the **memory** then adding new items into this list for example two and eight should not increase the time of **verification** as list length increases this is not what was found what Townsend found that as the number of balls the number of item increases the time of **verification** also increases so how do I confer it for a serial search it was accounted in terms of the fact that more number of processes are there and more structure is in so as the number of processes increases or sorry the number of searches increases the time will increase but in this case when I have parallel searches here the same structure in the same structure the number of items are there but still the time of **verification** increases how do I confer this the counting further this is done in terms of the number of processes that take place what happens is in serial search although the process of **verification** is single but the structure that stores this information is a bigger structure and so what happens is although the process is a single process since the structure is more or there is a bigger structure which throws this

information the time of **verification** will increase because as I add up more items into it more structure is required on the other hand parallel search although the structure is same but the number of processes that are there the number of comparisons that have to be done the number of processes takes up for the or icons for the more time what will happen is since there are let's let's assume that there are four process processes which run in parallel what will happen is these four processes will have to do multiple evaluations and these multiple evaluations although this is much faster but these multiple evaluations are the reason for the increase in **verification** time and so the inclusion of the brain the inclusion of the idea of the brain gives us a chance to identify what is the reason for the mental activity or what is the reason how this mental activity is taking place another interesting thing is the adequacy theory is the requirement of adequacy what does it say the inclusion of the brain into the understanding of mental processing or mental processes gives us this chance to understand what necessary or what adequate processes or what requirements are there for certain **mental process** to take place what does this mean it means that if four different processes or if four different answers of a particular way a particular process through which a mental experiences fell is being proposed given the structural constraint of the brain which of them will be the most correct what do I mean by this in the first part I have explained that multiple processes will give the same result but how do I go ahead and look at which of this is the most optimal or which of this is the most optimal theory the best theories or best threat of theories because there will be several theories there will be several reasons of how the **mental process** appeared and so what the adequacy Clause will tell us is that given the constraints of the brain even the constraints of what the brain has or what structures of the brain are which of the theory explains the processing or the answer to a **mental process** or the output to a **mental process** and so education law says that when inclusion of the brain will help us in pinpointing which theory works in which theory doesn't and what should be done for theories or what changes could be done in a theory to adjust it so that it explains a mental mental phenomena now there are certain paradigms to the study of **cognitive psychology** first of all let us understand what a paradigm is it's a body of knowledge structured according to what is proponents consider important and what they do not so it's basically what should be the basic idea behind **cognitive psychology** or what what is the basic matter behind **cognitive psychology** and so this these paradigms are assumptions which are made by investigators appropriate types of research methods because these paradigms then when I use a particular paradigm when I use a particular approach to study **cognitive psychology** it will not only give leaders to having a set of assumptions which vary across paradigms but it also gave us different methods of research of the same different processes or the same **mental process** it will also give new questions in the field so the same processes same **mental process** or same mental activities if it is studied by different paradigms if it is studied by different approaches not only will it give us different types of research methods of having same process but it would also give us new questions and also new analog is to it or new similar problems to it of how the solutions can be generated and so in **cognitive psychology** there are four basic paradigm that I wish to make you aware of here the first paradigm is called the **information processing** approach and so this approach started with the **cognitive revolution** the part of this approach this approach comes from the structure list and what this approach says is that **cognition** is basically information passing through a system so there are several systems out there Seraph several process systems in in the mind and these process systems takes in information from a previous process a processing system and passes processes it in certain way and passes it along to the next system and so **information processing** approach says that quality psychology the subject matter of sovereignty psychology should be how how these systems process information the basic **information processing** approach is built in this way it says that any information which comes out or which people encounter from the environment is first detected so this is the environment and this environment provides certain signals or certain kind of information and this information is detected by the mind or by the brain using a process called **attention** this **attention** then separates the different formats of information which are passed on

from the environment so the information which is passed on from the environment could be in visual medium it could be an auditory medium or it could mean olfactory medium or the other two senses which is haptic or the taste gustatory once these different formats or this different types of information is detected by the human system through process of **attention** it is passed on to something called a **short-term memory** and what is **short-term memory** it's a system or it's a place where information which is taken in through a process of **perception** using **attention** is thrown here or is put here for storage now this storage system is a temporary storage and so if people do not rehearse this information do not go ahead and repeat this information it is lost here through **forgetting** if it is rehearsed it goes back to something called **long-term memory** also information which is coming out from the environment is first recognized through **perception** so **perception** has certain filters and these filters are three these basic modules will recognize different information which is coming from the environment I am talking about here the content so based on the content the information is stored in **short-term memory** from the **short-term memory** it is rehearsed it goes to **long-term memory** once the information comes into **long-term memory** this is basically categorized into already so existing information systems or information pages in the **long-term memory** so mostly **long-term memory** has a very categorical structure since **long-term memory** has a lot of information to into it so what it does is it categorizes information into different categories what are these categories these categories are generally formed through something called **schema** if it's episodic knowledge or it if it is semantic knowledge this **categorization** is done through something called hierarchical tree structure what you need to understand here is that this **long-term memory** takes in information from **short-term memory** and it binds this information which is coming from **short-term memory** into already existing **long-term memory** information for example the next time you see an apple you do not need to learn that this is an apple what happens is when an apple is presented to you this Apple is perceived by your perceptual system the perceptual system identifies a scale color shape and these kind of informations texture passes on to **short-term memory** the **short-term memory** takes this information ask the **long-term memory** in terms of given the fact that this is the format and this is the meaning or this is the kind of information that is coming from this structure or this **mental representation** what could could be the **long-term memory** then compares this information and tells you that this is an apple or if tomorrow a black Apple comes in now when I taste it when I see it it looks like an apple but the possibility of black Apple is not there so what the brain will do is take this black colored Apple compare it with features which are existing in in the **long-term memory** and so since the black Apple has most of the features which is similar to any other Apple only the color is different so what will happen is this black Apple now gets stored up or gets included into the category of Apple which are already stored in **long-term memory** with one pointer to it or with one exception to it saying that this is a black color Apple because most apples are green and red in color so this is a unique feature which says that it is black in color so this is what the **information processing** approach says it says that any information comes in or it passes through certain stages and these stages define how an information or mental mental activity defines how an information path is processed through this system now the **information processing** approach tells us about the system's the processes and the goals what is the systems here the various systems which help in the processing of information are the **sensory memory** here the **short-term memory** here the **long-term memory** here what are the processes which help us in in understanding this mental experience or in sending metal activity our processes recognition **rehearsal** recording reorganization manipulation **categorization** and so on and so forth and what are the goals of it the goal is the response the goal is the understanding so in the Apple example the processes which help us and understanding the Apple is the **short-term memory** the **long-term memory** the perceptual systems those are the processes what are the sorry those are the structures those are the systems the processes are the process of **encoding** the process of recognition the process of **rehearsal** the process of response manipulation the process of identification recognition these are the process through which you are into Phi

this Apple and the response the output is storage of a new kind of Apple the response output is the information the knowledge that a black apple also exists this is the goal though the goal is to understand that that can only be L can also be black is achieved through a system of processes the second paradigm which has been used or which is popular in **cognitive psychology** is connectivism what is connectivism now collectivism is also known as parallel distributed processing model and what it says is that there are several small units which are connected together and these connections lead to the formation of mental experience in the **information processing** approach we saw certain systems interacting with certain processes to give a particular kind of a goal in **connectionism** what happens is there are several small units and these small units connect to each other through circle connect connections and these connections later on lead to the formation of mental experience how does this connect to vision works the basic fundamental premise of this **connectionism** theory is that individual neurons in the brain they do not transmit large amounts of symbolic information but instead they compute or they come up with the goal by being appropriately connected to large number of situational units so what is the meaning of this it means that there are specialized systems and these specialized systems are connected to each other and these connections lead to the idea of **mental process** for example let's understand how this particular sentence Robin is a bird the mental experience to this is explained by **connectionism** look at Robin this is our name it's a kind of bird and so if you look at this side of it it falls from the category of birds so there are Robin a canary sunfish Salman all these are birds and these fall under the living things living things have are further divided into plants and animal then into trees and flowers bird fish and so on and so forth so basically on this side you have a system of connections through which in which you have **categorization** in terms of living nonliving birds animals and so on and so forth and so Robin forms Robin goes here the name Robin goes here is or ASA is basically a connector which defines what is the relation of Robin to the word bird and once Robin which is a bird is connected to three to this connection of SR we can then say that if the mental experience of Robin is a bird can then conclude - or can give then can give us the idea that since Robin is a bird it should have wings it should be able to fly why because the idea that bird falls under the category of living things and living things certain features of living things are there and these live certain features should be present on any member of living things so bird is a member of living things and so everything that is there in the living thing should be present in bird so Robin Hood Robin which is basically a bird should be leaving should be locomoting should be doing daily activities and certain features of the bird should also be present in them the basic idea and connection is is that it is not big processes it is not big systems and process a connection between big processes and systems which leads to the formation of mental activity or mental experience but there are several connections between several networks which are there there as there is a network of what a bird looks like there is a network of what if there is a bird what is the basic feature of the word and so on and so forth and this is how connections are so small units small amount of units or small units combined together to give us the mental experience now these units which combine with each other to give us the kind of experience that trees wear that we get when we talk about robbing is a bird it they have certain weights and so weight wise combination will tell us what features should Robin have now we'll discuss **connectionism** the idea of **connectionism** in detail in the chapter of **memory** where we look into what is **semantic memory** and what is the parallel distributed model there and how symbolic the hierarchy structure of symbolic **memory** is and those things there but the basic difference between **connectionism** and **information processing** approaches in **information processing** approach there is one system there is a set of there is a set of system and set of processes which leads to the goal but whereas in **connectionism** several units several independent units they combined together to give us the mental experience the third approach to studying of **cognitive psychology** is called the **evolutionary approach** now what this approach does it says that the study of mental processes through the ways in which the IP model the **information processing** model and connectivism connectivist models look at they basically don't count for emotional features or aversion Airi

approaches evolutionary inputs to our understanding of community psychology so what they say is cognitive psych or subjective matter of **cognitive psychology** is not how mental processes are studying off **mental process** how **mental process** a process but there are certain specialized competence which comes through evolutionary input for example basic cognitive abilities like **language** and three-dimensional **perception** of **perception** of depth are present in very small children and they don't need some kind of mental experience for them to develop that and so what **evolutionary approach** says that human beings they have very specialized ideas of completeness produced by our evolutionary heritage evolution has a role to play in **cognitive psychology** so the subject matter of **cognitive psychology** the view the way of looking at what **cognitive psychology** should study should not only be the study of mental processes and thoughts it should also be how evolution has made us think in certain ways or behave in certain ways what they say the unitary approaches say is that the human mind is basically a biological system that has evolved over generations and the development of the human mind has come through a process of national selection so certain features or certain aspects certain working of the human mind or the human brain is inbuilt is evolutionary in nature and nature has given us this kind of system to process information so basically then the **evolutionary approach** will use evolutionary as one of the reason of mental experience and one of the basic views or one of the basic ways of looking at how mental experiences are processed and similar to that is the last approach which is called the **ecological approach** the **ecological approach** goes ahead and says that mental experiences or **cognition** does not occur in isolation all cognitive activities take place in certain context and so contexts are very important in studying career **cognition** so just studying the mental processes or what is going in the brain what is going on in the mind how the brain or what the brain does or what mental experiences are what kind of processes and systems make up certain kind of certain kind of mental mental experience should not be the only way of looking at **cognition** of how **cognition** happens but one way of looking at **cognition** or one viewpoint of looking at **cognition** is also in terms of the ecology or the culture in which people live because culture has only a large role to play of how information is processed certain cultures do not allow information probe to be processed in certain ways certain cultures allow information to be certain process in certain ways and that leads to the different kind of **information processing** or different kind of thinking think of the individualistic and the cult and the collectivist culture the kind of culture that the Westerners have individualistic in nature and the kind of culture that we have here the Asian countries have is more collectivist nature and so the way the mental processes the **cognition** of people who live in the West are entirely different from the way we live in our country because we have a collectivist culture the reason is that from right on right very beginning they the Westerners are taught to be individualistic taught to be alone taught to be doing things on their own and so they process information in an entirely different way whereas the as Asians since we live in a collectivist culture what really happens is the way we perceive information is always in terms of how the whole society will look into it so basically then the **ecological approach** says is that all correct cognitive activities are shaped by the culture and context in which they occur and so this brings an end to the base the first section of where we looked at the history of **cognitive psychology** the subject matter of **cognitive psychology** the various paradigms of **cognitive psychology** of what it should be the what is the subject matter what is the content what kind of things are studied into it what kind of processes are there and what is the input-output kind of a system this is the basic introduction to the history of **cognitive psychology** in upcoming lecture what we look into is how do we study **cognition** now as I said **cognition** is the **mental process** processing of mental activities or mental representations so how do we study this processing how do we study this **cognition** in action is what we'll do in next lecture and from there on will slowly move on to other cognitive processes or basic cognitive processes such as **perception** things like **attention** things like **memory** this isn't making and so on so forth thank you

Lecture 3: Studying Cognition

■Kind: captions hello friends to be back to the continue from the last class where we studied what is **cognitive psychology** and what is the subject matter of **cognitive psychology** so in that class earlier class we saw the basic subject matter of **cognitive psychology** what are the various inputs from various schools of psychology to **cognitive psychology** what is it that we study in **cognitive psychology** and the various paradigms which are there for the study of **cognitive psychology** how they explain mental events and metal facts in the last class we discuss **cognition** is about understanding mental events and metal facts or how does mental processing take place on mental events and how does the mind come into play and what role the blame has in the mind in today's lecture we'll look at the various ways of studying **cognition** what are the various methods which are available for studying **cognition** so let's begin there from the start of **cognitive psychology** and the or the heydays of **cognitive psychology** there were several methods which was out there to studying **cognitive psychology** the basic method that primitive **cognitive psychology** or artificial intelligence used didn't employ any rule of brain into it so the earlier the first advent of **cognitive psychology** or the first wave of **cognitive psychology** basically banked on **information processing** theory and so **information processing** theory was all about mental processing how mental events get process to save several mental mechanisms and what is the input what is the output that kind of process or that kind of structure was what you're being studied by **cognitive psychology** so information for century as we saw in the last class actually looks at the events and the mechanisms rusting and isms through which mental events or mental activity is generated in contrast artificial intelligence was also looking at how at the process level mental events are studied and both of these methods did not play or did not emphasize the brain in any ways as we know today that the mind is actually a software kind of a software or mind is actually a fact which is made by the brain and so studying of the brain is very important in studying according to psychology now some of the main reasons why the brain should be included in the study of **cognitive psychology** is to understand the **dissociation** and **Association** the process of **dissociation** and associations because these dissociations and associations will tell us what processes are similar if two mental acts takes place and what processes are different if in two mental events or two **mental process** acts to the heart of this debate is the idea of localizing those brain areas defining are those brain areas which actor which you activity when a particular cognitive at a particular cognitive process is initiated and so what the new diet **cognitive psychology** or the new field of **cognitive psychology** is looking at not only the **information processing** or the artificial intelligence approach where mental crosses events and various other mental events are studied but in relation to it the new wave of quality psychology is also looking at brain areas of how brain areas are what brain areas are really involved in producing **cognition** there are multiple methods of studying **cognitive psychology** as I said and so each of these methods use a different measure a different way to study **cognition** now these methods all have their own problems and all have their own limitations and war has to have what to study to you to find out the complete method or perfect method to study **cognition** would take into account the improves from all these methods why should we do that we should do that as multiple methods when they provide information what would happen is that the limitation of one method could become the benefit of other method and the benefit of our method could serve as or could add up to more important information and so when looking at multiple methods when looking at multiple ways multiple ideas or multiple approaches to studying **cognition** this will give us a more complete picture or a more desirable outcome of what **cognition** is all about now the general goal of **cognitive psychology** is to study dissociations and associations at the level of the brain now when I say **dissociation Association** what is it that I actually mean **dissociation** is a fact or a term which explains that one if two processes or if two mantilax give rise to activation in one area of the brain how do we dissociate it simply stating what it would really mean that an event if an event or a variable leads to brain activation during the processing of one task but we do the

brain activation on the other tasks this is what **dissociation** is this will tell us which task requires that particular mental activity or that particular **mental process** and which of the tasks doesn't require that particular mental task let's understand this with an example Allen Bradley developed the idea of **working memory** and so what Bradley said is that there is a whole conceptualization of the **central executive** component of **memory** which executes most of the working process in in **working memory** and within the **central executive** or with the **central executive** or other parts of the **working memory** which is the spatial store and the **phonological loop** the idea of the **working memory** is conceptualized in this way that input or the **encoding** from the environment is thrown into the **central executive** which then assigns the audio input the auditory input into the **phonological loop** and the visual input into the spatial and spatial loop or the spatial store what does this mean it means that at the level of processing itself at the level of **encoding** itself or just after **encoding** the mind takes away two inputs from two different senses and processes them separately now the same region of the brain is involved in storing the **phonological loop** or the phonological matter and the special matter is that true so experiments are done to find out whether this and there as the assigning of the visual input and the auditory input in two different stores or the **working memory** whether they use the same mental **mental process** or the same brain areas and if different brain areas were activated then we could say then would the tasks the tasks of assigning the the auditory input into the **phonological loop** and the visual input into the spatial sketchpad they require different main areas and through this we can then specifically study one of these area of the deficits in one of these areas so Alan Bradley what he did was he did some studies to find out what really happens at the level of the brain so we specifically inhabited or the visual input and he saw that there was no deficit into the auditory input what he did was he gave a visual task to his subject and with this visual task he actually loaded the **working memory** or the **attention** the **attention** span with more data more items and when that happened the subject was not able the subject could not process the visual input but the auditory input was intact let's understand this in a more easy example so what Alan Bradley wanted to study is whether if the visual input is loaded whether the auditory input can function on its own meaning which both the processes are both the processes of visual input and the auditory input in the **working memory** I have dissociative processes or dissociative brain regions involved into it so in his task he gave visual tasks and loaded the **attention** or loaded the **central executive** with more visual information so that selects are not able to rehearse are not able to study back the visual inputs but the auditory input the auditory information which was taught in **phonological loop** got a proper **rehearsal** and so when a test was done it was found out that although the subjects not able to correctly identify visual items or visual matter auditory inputs or auditory tests or auditory information **retrieval** was not at all this stuff this particular experiment shows that **dissociation** is at the core of study **cognition** why because **dissociation** can tell about tell us which processes have similar although they have similar representations but they use different mechanisms similar to it is the study of associations in associations what we actually study is whether two different processes are linked or two different processes actually excite the same brain region for example in this case experiments were done to find out whether the frontal cortex the frontal regions and another area of brain whether both of them have anything to do with spatial representations or of **encoding** or special in entered Earth spatial encodings what they found out is that special in information when encoded through two different processes or two different brain regions gave rise to similar presentations meaning which both the brain processes or Road the brain areas although different but they combined lis give right give rise to same interpretations now why is the study of **dissociation Association** important the study of **dissociation** and **Association** is important to find out which processes all those are emanate from the same structure but they lead to different tasks and which if two different processes which emanate from different brain areas converge on to particular task to find out the specificity to find out the process to this know the methods of studying **cognition** so there are four different methods through which **cognition** can be studied and so these methods are the behavioral methods the correlation neural method the causal

method and the modeling within the modeling we have the box model approach and the neural network model approach so let's start with the behavioral methods what the wavering methods generally do is they look at directly observable behavior like **reaction time** the time one person takes to react to something or **accuracy** of how accurate **retrieval** is from **encoding** so these processes or these events responses are mapped on to behavioral methods the **accuracy** of response or the time taken to response gives us an idea on to what behavioral processes are going through in the processing of a particular mental event now researchers they attempt to draw inferences about what kind of interpretations and what kind of processing leads to the differences in **accuracy** and differences in **reaction time** for popular approaches there have been or for popular methods within the behavioral method has been used so what we look at is what is the banner what are the benefits of teach methods what are the drawbacks and that kind of thing into less so let's go into that comparison as you can see the major behavioral methods and methods used in **cognitive psychology** the major methods are looking at the **accuracy** of how accurate the **retrieval** is from the **encoding** percentages that is percentage of correct responses response times judgments and protocol now let's take this methods one by one the method of a in this method of **accuracy** a very good example to look at is item recall or **memory** recall for item such as remembering a particular job requirements of our particular task the advantage is this it gives an objective idea of the processing effectiveness how effective is the mental mechanism or the **mental process** which is processing that particular even the limit to **accuracy** method or behavioral methods of **accuracy** is something called the ceiling effect and floor effect ceiling effect is when a task is made too easy so that each person gets the highest score and so it will be very difficult to find out the exact works across a mechanism similarly this is the floor effect in which a task is made so difficult that none of the people are able to solve it and due to this will not be able to ascertain what processing mechanisms are at play also they suffer from something called speed **accuracy** trade-off which means that as we want to be more accurate on something the more time it will it is going to take as **accuracy** increases more accurate we want the speed decreases or the time to perform a task decreases it's an inverse relationship the second method here is response time which is the time one takes to come up with a particular response the advantage is this that it is an objective method and suitable for measuring or processing mechanisms that is at play and also those unconscious processing mechanisms now this the limitation to this method is something called experimental expectancy effect and task demands what this really means is that each experimenter when he gets into an experiment he has certain expectancy he has a certain want from the subjects and so this is the transfer of the subject subject identifies this expectancy and actually behaves in a similar way so if x-parameter ones are subject to say yes to most of the things or take more time in responding what would really happen in the subject understands that or judges that from the from the experimenter and even if the processing is fast they take more time the third method is called the **judgment** method here what really happens is to find out what really what really is happening in our how does about any process really work is people are given a **judgment** scale and so there are several these are seven point or eight point nine point scales which are given to people who actually judge or go ahead and rate how they are processing a particular item it is very inexpensive to collect and it access the subjects reaction to a particular cognitive process for example the thing like how are you feeling right now and so our tests can be given to people and what they could do is actually go ahead and on a seven-point scale go ahead and tell on several questions of provide responses to several questions as to how they are feeling the problem with this method is that subjects are not aware of how to use this scale and so may get into difficulty the last two the behavioral methods which are used to study **cognition** is something called the protocol collection the method here allows subject to verbally announce what is going on or what they are doing for example if I ask someone to go ahead and show me how counting of two and three is done this person has to actually start the first the experiment the subject has to actually go ahead and relate how he is how is doing the addition process for example thinking of the number two thinking of the rule for addition thinking of the number three going adding

carrying off and so on the protocol method really uses or really requires the subject to announce what is going on mentally this method can tell you or the advantage of this method is basically it what are the steps basic strips through which a mental processing happens or any task what is the mental requirement for this task what is the **mental process** for this task for complex cognitive processes this cannot obviously be used because people would not be aware of what is going on and they could not or they could not accurately go ahead and describe what is being going on the next method that we use in studying **cognition** or cognitive processes is a correlation or the neural methods no what does correlation really mean or what is this what is the heart of this matter the heart of this method is localization as we discussed in the start of a lecture that the new wave of cognitive science is looking up at the brain towards the brain to explain how these cognitive processes or mental events are being processed or how is the mental event actually being processed what is going on inside the mind what is going on inside the subjects **cognition** how is the processing input and output coming into effect No then I explained to you that the brain is a recent addition why because the brain can objectively tell you the processing the methods or by through which particular mental event is being processed and so this method looks at finding out or localizing that area of the brain when a particular mental event is being carried out what this method does is actually tries to form a correlation between the fact that when a mental event happens which areas of the brain actually show activity and so through correlation it tries to show off or tries to find out which area of the brain risk is responsible for which kind of mental activity this will benefit us as we will be able to specifically now tell that each **mental process** or each event or each mental mechanism is related to a particular brain area the problem here could be that sometimes numerous other brain areas beside the area of interest also get excited and so it is it is a little bit difficult or I would say a little challenging to pinpoint the area that is responsible for the core **mental process** which is being carried out another problem with this method is correlation doesn't lead to causation correlation simply means the two events by chance are related to each other they don't mean that process a leads to process B and so we cannot with confidence say that this brain area excitation of a particular brain area leads to a particular mental even a particular **mental process** a particular cognitive illness now there are a couple of techniques which I hear so let's look at those techniques let's look at the advantages of these techniques and the disadvantages of this technique and discuss how this method really works so the causal neural methods have four different techniques to look at I'm sorry three different techniques to look at and these techniques are the neuro physiological studies the transcranial magnetic stimulation and the drugs so I was saying as I was saying the correlational neural method the problems with the correlation of neural method and so there are two problems with this first that many areas side along or they get along or are activated in addition to the meaning of interest and so what really happens the problem with this method is to localize the area of interest because there are other areas also which inside along gets activated and the second problem with the correlational method is that correlation not causing causation now what really happens in correlation is two events a and B they also show in sync activity but correlation doesn't guarantee that a is the cause of me or B is caused by a and so there is no way to find out what process leads to or what method leads to what structural activation in terms of the brain mind relations it would be very difficult to find out whether the activation in area lead to the mental processing or whether mental processing leads to the activation in an area so there are four different methods here techniques here to study the correlation or the correlational localization of rain areas the first of this technique is the **EEG** or the electroencephalogram the electroencephalograph can also be used for recording something called the ER piece now what the **EEG** or the electroencephalograph records are potentials from the scalp when a particular mental activity takes place it's believed that a mental when a there are changes in neuronal polarity and due to this a weak electrical current actually passes to the scalp surface the **EEG** sensors which are placed in the scalp at record these activities and based on that are able to predict whether there is an activity change from the baseline activity a baseline activity is an activity when no mental processing

is being taking place and and then whenever an activity change place there will be a change and so the change the activity subtracted from the baseline activity will give you the kind of mental activity which is being taken place Iggy's are also used for recording event related potentials which are basically the potentials which are recorded when a particular stimulus is set and so after a particular stimulus is initiated from there on to when the response or the system or when the response of a person takes place the **EEG** the the electrons are left in encephalographic patterns of the brain are recorded and the time course of this **EEG** the time the changes in **EEG** from the stimulus onset to the response is calculated or recorded or studied and this gives us an idea of what processes and what areas or what kind of pattern of activity is being shown by the brain generally there are some very fixed patterns that the brain show there is something called the base activity that mostly happens in everyday life is the Alpha activity which falls in the 8 to 13 Hertz or 13 cycle per second region then if as a person gets more relaxed either at the amplitude of these waves increases and the frequency lowers down and it comes back to something called Delta and so on and so forth so as you get more worked up as you are doing more general activity the amplitude is less the frequency is more and the band pattern or the activity pattern in the brain is generally in the 8 to 14 Hertz cycle / or cycle per second region an example of using the **EEG** is to find out brain patterns during sleep or something like in terms of ERP something like p300 now p300 is actually a component which happens p300 is when I say a component what does it really mean a component is a deviation in an electrical activity which takes place after the start of a event so when a stimulus is given when a stimulus is initiated and the ages are recorded 300 milliseconds after the simulation stimulus initiated on the positive direction if a peak is recorded this is called the p300 now the p300 generally the component p300 generally is associated with novelty so when the brain then the process the mind looks at something which is normal which is new it generates the p300 activity and so these this this can be recorded the spatial resolution of eg is very poor because it cannot locate the area from where or the neuronal group from where this activity is being recorded as current passes or current moves from the base of the brain towards the surface of the brain it forms a chronicle structure and so it loses most of its strength and it is very difficult to to go back in or to reverse-engineer from where the source of this activity is although newer techniques do allow for something called source localization which actually from an **EEG** will point point a gross area where it is coming from but generally speaking the **EEG** is a very poor resolution in terms of temporal resolution and **EEG** is excellent because it can show you within within milliseconds of the activity when within milliseconds of the mental event the mental activity and can show a correlation to that so in terms of time of processing in terms of time course of processing of mental events the **EEG** is very good can tell you what events in in that on the time axis what events process process or what even processes take place that that kind of information can be there gathered from an **EEG** the invasiveness of this particular method of the **EEG** is very low as you don't have to put anything inside the body the surface sensors actually captures the information and so not much of an invasive process and the cost for this particular method is very low if smaller units are available in in very few dollars so you can buy them and start using them similar to the **EEG** is the M eg the magnet to Instagram where electrical activity instead of capturing the electrical activity the magnet of activity is captured an example is detecting activity from the auditory cortex or different picture stones so here there are several sensors which are put around the brain and the magnetic activity that is generated when a particular event happens a particular stimulus on after the particular state was stimulus the particular event which happens are recorded or the magnetic activity or which is generated by the neuronal group which is present the event are captured through it the spatial resolution of the magnet or in circular graph is very good it is under one centimetre in terms of temporal resolution also it is very good is excellent the invasiveness since you are not putting anything inside the body it is very low and so invasiveness wise Remy G's are very good but the problem is buying an M eg or establishing a and M eg requires a very high cost so cost wise it requires a system which produces this magnetic moments of this predator produces magnetic fields which actually are able

to process or which actually are able to record changes in brain activity or neuronal activity and a particular mental event is going on third method that we use as a correlational method or for localizing brain areas that are responsible for a particular mental event is the positron emotion tomography the PE T in **cognitive psychology** which we generally use the 15 oh I saw tok the 15 oxygen isotope as a method for doing PE T so what really happens is that the oxygen the 15 the Oh 15 isotope which is radioactive is injected with normal within the normal body and so this particular isotope is moving around the body so when the brain does it is believed that the when when the brain does a particular function it pulls up more and more blood and with the blood more and more water also transposed to the brain and so when a particular region of the brain shows more 15 the accumulation of more 15 I sort of that is the region which is correlated or which is assumed to be active in a particular process and this is how the **pet** of 15 or oxygen isotope is done generally it is used to study or mostly it is used to study **language** in the brain of how the **language** areas process or what is the processing that happens in **language** area when we actually speak the spatial resolution is very good up to one centimeter but the temporal resolution is very poor because it takes almost 40 seconds to generate an image out of the PT of 15 oxygen isotope or the Oh 15 isotope invasiveness is high as we in as we inject an isotope of oxygen which is radioactive into the blood and so this has radioactive properties which could lead to problems although the kind of radioactivity that happens with an oxygen with Oh 15 I so talk is as less as what a airline pilot would actually get through in half a year of his work so it is it is very very less but still it is radioactive and so it's best to avoid it on terms of the purchased cost it's a high purchase cost because PE t means cyclotron and PD cameras and things for noticing those sensors which can go ahead and record the radioactivity the low radioactivity of these oxygen isotope so it's very costly in that method and improvement on the PT is the MRI and the **fMRI** which was which is a new addition to the study of the brain or brain processes or cognitive processes the MRI on on one hand is actually stool's the structural integrity of how the structure of the brain is the **fMRI** on the other hand shows the blocks blood oxygen level dependent or how much oxygenated blood is taken up by the brain when a particular activity is being being processed by the brain the MRI works on the principle of body magnets or at atoms of the of the body and how they align to an external magnetic field so what the American really does it a uniform magnetic field to which most atoms of the body align and then a radio ad pulse is created or radio pulse is it is generated which disturbs the alignment of most atoms or the of the area which is in the MRI scanner the speed of returning back is calculated when when the magnets dis align when the particular polls or the particular elements in the in the body the particular atoms of elements in the body when a radioisotope dis aligns them from the alien from the alignment that the MRI magnet does it returns back and the speed with which the return back is calculated and from there an image is generated and these images are the speed at which the different atoms get back or they come back is different for different different media like bone skin or that other tissues in the body they have different speeds of returning back or this is calculated and an image is generated in an **fMRI** what we really do is the amount of oxygenated blood in comparison to the deoxygenated blood at a particular brain region which is processing a particular event is taking that is calculated and that gives us an idea of which region of the brain is showing an activity through a sensor the spatial resolution F and both an MRI and **fMRI** is very superb it's very good in the terms of millimeter range the temporal resolution is not that good but it depends upon the level of resolution that you want or generally it is in in terms of several seconds so not like the **EEG** which can give you the millisecond kind of a temporal resolution but not as low as the PE T which gives which takes almost 40 seconds to produce an image so MRIs have this indecent time requirement or this takes decent time produces images the invasiveness is low because MRI doesn't MRI and **fMRI** doesn't inject anything into the body so the invasive is low but here the purchase cost is very high because the kind of magnets that that the MRI and **fMRI** uses are huge electromagnets and they have huge amount of cost involved into them the last method that we use here is called the optical imaging method and here this is a recent addition so it has been found out that the brain is transparent to certain

kind of light waves and when these lights waves are shown into the brain through a laser the speed with which they return back can give you an idea of how much blood is flowing to that particular area and so this can be used this method is now being used to find out the amount of blood traveling through a particular area depending on how much light laser light is being reflected back through various tissues and and various substances in the brain it's the special resolution at present is very poor and the temporal resolution it depends on the level of resolution and several minutes takes up for from forming an image the invasiveness is generally medium to low as lasers are shown on - although the lasers are focused on to your brain so it has some level of and invasiveness but not too high the cost of patching is very low but the temporal and spatial resolution is not appropriate one of the examples is they can awright which is functional optical imaging suit which is available these days another method of studying **cognition** is through using the car general method of training **cognition** what does this method do in this method it is assumed that if certain regions of the brain are required for certain activity what would happen if that region of the brain is somehow not allowed to do any activity or stop from doing activity and so one of the one of the particular Methos within this approach is using patients who have certain brain damage so suppose the Broca's area is responsible for speech and so if somebody has a deficit in the blocker area and if certain type of speech related experiments are given to to them and compared to a particular person it would be easy to understand what kind of what kind of activity into the brain does what is the requirement of this area and so on and so forth so here in this in in the near physiological studies patients with localized brain damage or brain damage to certain areas are use and they compared with people who have no brain damage and this is how the cognitive processes or the mental processes that go on when a particular mental activity is being done is actually studied example could be deficit in understanding then how nouns and verbs are processed and so people who have deficit in this area if you give them a noun in a verb the area since this deficit will note it will show no activity whereas people who don't have deficits will show different activity and so that will be enough to say that this area is responsible for the particular kind of act that is being required the advantages is of this method is testing theories of causal role of specific brain areas testing theories of shared indistinct crossing and use in different tasks relative ly easy and inexpensive to collect the limitation is we do not find too many people who have damaged areas and so the idea is it's it's very difficult to find people with damaged areas also when damage to a particular area is there there will be related areas which might be affected by the damage so it is very difficult to localize whether this damage is to this area or with this damage the damage area also has any relation with other areas the second method in this particular group is called the transcranial magnetic stimulation now in **TMS** there are two types of **TMS** s one is called the single pulse and the other is called the repeated **TMS** or the repeated transfer pulse may transmit transcranial magnetic stimulation what clearly happens here is that a huge coil is put into or near or on the surface of an area that you want to temporarily inhibit and a large current is passed through this coil the magnetic field that is generated by this coil temporarily inhabits a particular area and through through that we can find out whether the disturb how what will happen if a particular area which is known to be associated with a particular activity cognitive activity if it is disrupted what really happens so what we could do or a studies of how **visual imagery** is different from **perception** can be done by using PMS a PMS are using a our **TMS** and blocking the occipital lobe now there is a difference between single person - MS and our **TMS** in single person **TMS** a single pulse is given after the stimulus is is generated and the time course of this pulse is recorded and that gives us an idea of how the or what activity is happening in that area whereas in our **TMS** repeated pulses are given over a particular area so that these weak pulses they add up together to inhabit a particular area the inhibition happens because the neurons under that area becomes too fast to to extended or the the fired so many times that they they go through a phase of non firing and that inhabits the area or that depolarization in that activity in that area so advantages it has the same advantage as the neurophysiological studies but here and the lesions are restricted and also the idea here is that in these regions in in in the **TMS** the the destruction is

for a very blue brief period of time and so patients can come out out of it or subjects can come out of it also **TMS** is can be done from their surface areas it cannot be done for deep areas of the brain and so it is one of the disadvantage or limitation of this particular method another method which can relate or which can show where how a particular brain area destructing about the particular brain area has effects on a particular **mental process** is using specific drugs so you have what happens is first for a particular drug is specifically used and these drugs inhabit a particular area of interest and so once this area of interest is inhabited the same task is given to a person who has taken this drug which inhabits area and a person who has not had drugs which inhabits this area and comparisons are made these comparisons give us an idea of what really happens or how mental processes what brain areas are involved what mental processes has been processed and what is the time course of process of these mental events the problem example whose the disruption of destructive action of not adrenaline which is crucial for the operation of hippocampus over not literally an injection is given to the hippocampus and the hippocampus actually becomes temporarily disturbed and so when you give a **memory** task to this person he will not be able to process this **memory** task and not come up with better of how or this particularly travel paradigm in comparison to somebody who has no disruptions and this will show how much or to what extent does which part of hippocampus relates to a particular **memory** type drugs been used to inhabit a particular area affects not only the area of interest but also affects other area which is related to this area and so this if can also be a can can can be generalized to other areas similar similar to this effect the **TMS** also has this particular disadvantage where activation of a particular area can lead to activation or subsequent areas and so what would happen is the disruption will not be in that particular area but also move out or generalize to other areas of brain and so to stay with confidence that this is this destruction of this area is responsible particular motive lag is becomes a little bit dicey also the limitation is that many drugs can create many other problem and so this is this has to be used under supervision the last method that we discuss here are studying **cognition** recovery psychology is using modeling so in psychology models are implemented as computer programs which are meant to mimic the underlying **mental representation** and processes that produces different specific types of human performances so generally models are made and this models try and explain the different processes events that actually happen when a particular mental event or a particular **cognition** happens there are two types of models which are studied in **cognitive psychology** one of these is the process model and the second is the neural network model so what is the process model the process model specifies a sequence of processes that converts an input into output and can these kind of processes can be used to generate flowcharts so basically it's it's a box diagram kind of a thing where you get from one for one phase to another fun event to another event and the processes which relate these events together coming back to understanding the process model let's recall the structure process trade-off that we saw in the last class and so in the last class we saw Sternberg's example in which what he did was he he wanted to study the **mental process** of how comparisons are made in in the comparisons and made in **cognition** and so what he did was he gave he subjects a particular number of items to remember and then these items were tested the **retrieval** of these the **memory** for these items were tested to a pro and what Strandberg proposed is that this search for the target in the list of item that is being studied other that is being remembered happens to a serial process generally speaking this idea that a civil process looked through a list of items which have been strained in **memory** can be explained through a process model and this is the process model which explains how this serial processing takes place so generally how this experiment would work is when a probe is given the probe is first encoded prior to this people are given a list to remember and the list has several numbers or several letters into it and what the job is the subjects look at a probe and then find out whether this probe is present in this list or not and to explain this process a process model can be developed how will the process model really works look at this in finding out or scanning whether the probe is present in the list which is remembered the first step is **encoding** of the probe this is followed by scanning and comparing

their processes internal processes take the probe and match it with each item which is represented which is stored into the **memory** for each item after a comparison a decision has to be made so let's say this if we have one two three four and five stored list of items stored or five numbers stored in our **memory** and if you want to compare whether three is present or not how this model we really work on how this process will really work in terms of mental effect is that three is first encoded later on three is matched scanned and compared with the first item this which is one and so a decision has to be made here whether one is three or not since both the representations for three and one are different responses to be executed here the responses know and the process moves to the second item now three is taken in compared with two since both the representations are different response here has to be generated which says the three is not equal to two and so we don't find a match and we move to the third item which is three now here three and three I have the same representations and so we get a yes response make a decision the yes response comes in and so in terms of serial processing the process doesn't stop here it continues continues to the end of the list and then gives the final answer so that is how the process model explains the process model X actually goes ahead and explains how this simple job or simple task of comparing a probe with list which has been already remembered actually goes ahead the problem with this process model is it tells you about the processes that goes through or the processes that happen when this list comparison or probe list testing is done but it doesn't tell you how does the brain or what neural factors play a role in to this particular whole structure so this has this is basically this kind of process models are basically influenced by the behaviors which believe that the input output sequence is what is of important to us but what is inside is not of importance to us and so it doesn't give you the brain structures the brain processes which actually go ahead and lead to the final final answer of comparisons or final comparisons that are done and so to counter that neural network models were done or thought of now what is the neural retina Network model it relies on interconnected units which are not just input/output units but it has a hidden hidden unit which is generally called the brain unit into it and so most inputs are connected to a hidden unit and the hidden unit are further connected to an output unit so if an **encoding** if a input unit is given this input unit then goes ahead and makes connections with this hidden units which are generally neurons now this type of the neural network structure that I'm showing you is a simple feed-forward Network what happens here is there are three units I1 I2 and three three inputs which have been taken and these inputs are then compared or their process through h1 h2 h3 three different brain networks or neuronal networks now each of this network can have only two activity one is either it can be excitatory which is the eye response or inhibitory or response so when a particular input a particular representation from the environment is encoded this particular network of this particular responder on network will either be excitatory will either pass along with either move along this information to the next record and so for connection or it could show inhibitory whereas it will not process this information further also these connections that the input layer or the input item has with the hidden layer has weights into it and depending on what weight it is what is the strength of this connection will decide whether this in half wins whether this response of excitation or inhibition will these new will this hidden units actually show and based on the excitatory or inhibitory input or mode of the hidden layer and the weights of the input layer the output layer is finally or the outputs are finally determined and so that is how we see that I1 or I 1 represents gives to give rise to o2 and the hidden layer the neural network or the group of neurons which actually process it how does it really function so this particular model shows us gives us several benefit for example it it shows us what are the neural codes which are there which are which are which are responsible for this kind of an output and it distinguishes these neural codes from mental representations so it doesn't just tell you **mental representation** of how event is represented in **memory** but also what are the brain states or what are the neuronal states which are processing this particular neuronal unit so that brings us to the end of this session and to do a quick recap in this session what we did was we looked at the different methods of studying **cognition** and we looked at the heart of this study the use of different methods that the the basic

point being that as we use different methods we get seven whatever the the Leupold from one method turns out to be the benefit from another method and so we got a more comprehensive picture if one method explains process a but cannot explain process me maybe there is another method which can explain process B and comes up with another limitations and so looking at the benefits and limitations on all these methods gave us a more complete picture of how cognitive Authority processes develop and so within that we looked at four different types of methods or four different techniques of studying **cognition** right from behavioral techniques where we look at **accuracy** response time judgments and protocol to coming up with techniques which actually go ahead and localized or pinpoint a very brain area which is active when a particular **mental process** is being carried out and methods like eg PT MI MRI or **fMRI** on optical imaging - looking at Kaushal methods which actually destruct a brain area which is believed to be active when a particular event has been taking place a vertical mental event is being taking place - the methods of modeling where a process is designed a computer stimulation is designed which mimics what the brain does so instead of studying the brain its I'm a mimic or model of that is done and studied through computer simulations to find out what will happen if a particular an input or a particular group of brains brain areas which which are known to respond in particular ways if they take in an information how what is the possible output that can come in thank you

Lecture 4: Perception:Basic Principles

■Kind: captions hello friends welcome to this lecture on **perception** which is making meaning out of sensations one of the basic problems with psychology is to take in the physical environment to understand the physical environment interpret it and make meaningful observable interpretations out of it so one of the basic questions the psychology faces or human beings face is to interact with the human environment to interact with the external environment now psychology is the science which makes us understand which makes us observe or realize how physical inputs are actually taken in are taken in by our senses and interpreted into some meaningful interpretations so we can understand this by a simple diagram so as I was saying the basic problem or the basic idea about psychology is to integrate hello friends this lecture is focused on **perception perception** is about making meaning out of sensations what are sensations sensations are the raw input which the brain receives from various sensory organs sensations are the stimulus which has been engaged on different sensory organs and **perception** is the process which makes interpretations of these sensations in this section we'll be looking at how do humans interpret sensations which is the basic information which is gathered by different senses and how do we go about making meaning from them it's a very important part in **cognition** or in **cognitive psychology** because this is the first part which are the first step in the cognitive process think of this in terms of the computer analog a **perception** is equivalent to inputting something to the computer system **perception** can be thought of as the processes which take it input from various attachments to the computer various input device to the computer and makes an interpretation sort of it sensations are equivalent to things like key presses on the keyboard push of the mouse button or writing of the right pen onto a computer screen which leads to an electrical current and this electrical current are then interpreted by the computer system in terms of meaningful stimuli **perception** is an important process as it should be able to quickly verify what is being inputted and also should be able to distinguish between different inputs for example if you press the W key and the V key the computer should be able to know when the W key is will pressed and when the V key is been pressed **perception** is a process akin to similar to this but it takes place in the human beings the **perception** is actually a very complicated process that's what most artificial intelligence scientists would tell you for us it's easy because it's a process which is very natural to us now before understanding require up the process of **perception** let's understand a little bit about what is sensation and even before understanding sensation let's talk a little bit about what psychology does the basic aim or the primary aim of psychology is to take in to understand the physical environment and convert the physical environment into a psychological environment psychology is the process through which physical input is converted into psychological input what do I mean by that in the environment in the physical environment around us we have physical stimuli like light waves which are packets of photons are sound waves which are pressure differences created by sound moving to the air not these stimuli this physical stimuli is converted into or is transferred into the psychological realm and the process which converts these physical stimulus into psychological stimulus is **perception** but even before **perception** the fifth the physical stimulus has to be read in the correct format and inputted into the human system the human computer and that is what sensation does so sensation is basically a system which takes in a physical stimulus like a light wave or packets of photons which are impinging into the eye and then convert them into electrical signals which are then passed on to certain brain regions which pick up these signals and depending on the strength of the signal depending on the type of signal it interprets what the signal means now let's look at one basic sense organ and try to understand how does this really happen a basic sense organ or a primary sensor again is the human eye now if you look at the human eye what really happens is there is a lens with a lot of structures around it and this is how the structure of the eye looks like there is a physical stimulus here and this physical stimulus reflects light which is falling onto it onto the lens the lens then redirects this light into or onto the retina which has photoreceptors onto it these photoreceptors then

look at or understand the strength of the light and pass these along through optical fiber nerves onto the occipital lobe or the occipital region of the brain the occipital region of the brain then parses the stimulus and tries to make meaning of what the stimulus is the process through which this light wave is picked up by this retinal part of the eye is sensation and after that anything after that is **perception** and so this is what **perception** or difference between **perception** and sensation is all about so in this particular section we look at object **perception** how does **perception** take place so the basic theories are what **perception** is about now since you're not in this course we are not focusing on to the brain too much or looking at **cognitive psychology** from the brain point of view we'll look at cognitive processes from more of a mind point of view so we'll focus more on to those processes those systems are those viewpoints which tell you about the cognitive processes about the mental processes which goes on in the mind when some mental event takes place so what is **perception perception** is a process through which sensory inputs are gathered and interpreted perceptions can happen in several formats for example **perception** could be visual **perception** it could be auditory **perception** it could be haptic possession construct early **perception** so four or five different senses and four or five different types of perceptions **perception** which basically require us to understand or to localize where an object is what is the object look like and what is the meaning of our object now in this particular section we'll focus more on to visual **perception** since this is an area which has been looked or which has been diversity studied by most cognitive psychologists so in terms of object **perception** what does **perception** really mean in terms of object **perception** **perception** means understanding where an object is in the environment what is the shape of the object what is the color of the object what is the texture of the object what is the form of the object and things like that this is the first step of **perception** in a later step these basic information which was gathered by sensation or true sensation are then mixed together and interpreted into a meaningful whole into a meaning and this meaning comes from something called a process called **pattern recognition** not an interesting question when a **perception** happens when we see something does it imply that we also know about it functions let's look at **perception** from more basic view think of any summer day you're sitting in the classroom looking out through the window of the classroom and what do you see you see a green tree a bunch of bicycles which are parked outside as it would happen mostly in outside my classroom some students talking a little bit of grass a little bit of building and so on and so forth the idea here is that when you actually look out or when your eye actually looks out of the window it is not the tree that you see it is not the cycles bicycles that you see the vehicles that you see it is not the not the building that you see or the shape of the building that you see what you actually see when you see when you look outside the window is some kind of information about various light waves which are affected from these basic objects into the environment so **perception** is a step or is a process through which these basic information about the color about the texture about the form of what you're looking outside and making interpretation out of it in terms of what is a building and how is the building and what is a person and how is it different from each other so that's what **perception** will involve and let's come back to the original question which I just asked you is when we perceive something when we look at something and when we identify something do we also go ahead and understand it function now we'll leave this for a time being and answer this a little bit later this particular question has its answer on something called the Gibbs Ronen approach proposed by Jay Jacobson who believes that as we see an object we also see its function so the next time when you see something new a new model of a car do you also actually perceive its function or something which you which you have not seen before a new model or a new type of object out there in in the visual field in in the market do you also perceive its function and there's a lot of debate which is the Newtonian view or Gibson believes that most organisms when they look at something they not only see the object classify the object or understand the object and make meaning out of it they also have gathered some idea about the function of this and that is this debate between something called the classical approach and the Gibson approach motapod Gibson an approach a little bit later so let's again start studying about the classical approach to

perception classical approach to **perception** the classical approach to **perception** starts with identifying something called the **distal stimulus** and something called the **proximal stimulus** look at the figure which we have here as you look at it what we see there are two objects on the field or on the environment these objects have a particular shape a particular texture a particular form and light is reflected from these objects onto the retina the retina looks at this light and converts them into electrical impulses even before that the light which is reflected from these objects fall on to the lens of the eye the image of the object these objects into the environment which forms on the retina is inverted in shape so as you see these objects an inverted image of this object falls onto the retina this image which is which is on to the lens is then straightened up by through certain processes and these processes send this image back to the retina the retinal cells are very specialized cells they are not only able to tell you or correct the orientation of this object they can also at this very point at the first step of **perception** look at or make the the optical areas of the brain or the optical centers of the brain provide information about the various shadings the various lights the various textures the kind of intensity of light which falls onto it the various edges and so on and so forth about the object these informations take it together when passed on to the brain an interpretation of it is made and this interpretation is then compared with some kind of already stored template and that is how **perception** happens so basically the objects in the environment as envisioned by the classical approach is generally called the **distal stimulus** and the one the image the inverted image which falls on to the retina is the **distal stimulus** the interpretation of the **distal stimulus** now understand the **distal stimulus** is just packets of photon or intensities of light which are actually falling on the lens and the layer later on to the retina the interpretation of these lights which are falling into a tree or a box is what is called a percent so the percept is basically the psychology itself a psychological **concept** it's a it's an interpretation which the brain does of the **proximal stimulus** so basically what I am trying to say here is the **distal stimulus** is the object in the environment the **proximal stimulus** is the image of the object in the environment or the inverted image or the object in the environment but recognizing an object or what an object is or the meaningful interpretation of our object or distribution object as a tree as an animal as a child as anything else as humans as buildings is basically what is the person now the percept actually gets compared with fixed patterns which are stored into the brain and this is called **pattern recognition** once the occipital areas gather information from the eye the optical nerves about basic informations about the **proximal stimulus** such as the shading of the stimulus the contours of the stimulus the kind of intensity of light which is falling on the stimulus from there a meaning is interpreted and this meaning is interpreted in terms of comparing what the occipital lobe has as an input from the eye and comparing them to pre-stored concepts to pre-stored input and representations which are already in in the brain this process of comparing the input from the eye to priest or deeper representations in the brain and making meaning out of it is called **pattern recognition** which is the rig so **pattern recognition** is the recognition of a particular object or event as belonging to a particular class or even **pattern recognition** is the process which makes you differentiate between two kinds of trees or two kind of people in computer science **pattern recognition** is very important as **pattern recognition** lets you identify clusters or visible patterns in huge amounts of data what are the most prominent problem in computer science is having a lot of data but not finding patterns in it not finding clear identify patterns into a lot of data and what are the solutions of interpreting a chaotic data or huge data is to find similarities finding similarities between items is what is called making classifications or making patterns and then identifying these patterns with restored pattern is what is the process of **pattern recognition** so a similar process happens here as you can see in Figure B and C the actual retinal image is always inverted but what really happens is the eye reinterprets this or we correct this particular thing into the actual image now one of the basic debates here is whether **distal stimulus** which is the image that is in the environment and the **proximal stimulus** which is what forms into the eye the debate that most psychologists our **perception** psychologists have is where the distance simulus is equivalent to what a percept is so whether the **distal stimulus** the stimulus in the environment

in in our case if this is the eye and this is the lens and this is an object in the environment and this is the light falling from the Sun which is being reflected here this is called the **distal stimulus** this is called the **proximal stimulus** and this is the brain interpreting this image as a percept now the debate has been whether this percept is equivalent to the **distal stimulus** many a psychologists have debated this and the idea is that they are not clear do a small experiment to see what do I mean by this the percept the meaning that the brain does or the brain makes of anything in the environment is entirely different from what actually is in the back and so brain recreates or adds on to certain things adds on to certain values - or certain elaborate interpretations to what the **proximal stimulus** is a small experiment to basically tell you the difference between what a percept and water **distal stimulus** is extend your hand in front of you and slowly bring the hand towards you what do you think you see your hand moving towards you but do you see or when you see your hand moving towards you actual physics says that as the hand moves away from you it's size on the retina is small and you should see a smaller hand whereas when it moves towards you the size of this : hand increases as the distance between your eye in the hands increases so you should see a bigger image but is that what you see is that what you feel no when the hand is extended and when it is near the high at both the times the hand your hand has the same shape or the percept of the hand has the same shape which basically proves the fact that the **proximal stimulus** the changes in **proximal stimulus** do not make changes into the percept and the percept is entirely different from what the **proximal stimulus** is this particular feature is called size constancy a feature of a perceptual system which maintains sizes of things more about size constancy in later units now one of the basic things in **perception** is to identify what is in the environment and one of the basic approaches for that is the **gestalt approach** let's start with the **gestalt approach** look at the figure displayed on your screen what do you see there are two interpretations of this figure it could be that white birds are flying towards your right or it could be the black birds are flying towards your left these interpretations depend upon which part of the figure which part of the image you consider a figure and which part of the image you consider as the background and that is what **Gestalt** is say the basic interpretation basic idea of get guest artists in terms of **perception** is what is figure and what is background what part of an image is figure and what part of an image is background and this is the first distinguishing that happens in terms of making the person these kind of images is called are called reversible images look this image what do you see most people when they see this image they see an old woman but if I tell you that this is a reversible image and you also can see a young woman not of it you'll be surprised but then I can show you how to change this figure or how to change this picture into a young and old woman this part over here that I'm highlighting is what creates the difference the red color part becomes the mouth of the old woman whereas it becomes the necklace of the young woman and as you see this part over here becomes the nose of the the young woman and the eyes of the old woman so basically then forceps or meaning that I made from physical stimuluses depends a lot about subjects motivation subjects expectations and a basic process about understanding what is the figure and what is the background so let's look at jump into one of the first approaches in understanding **perception** there are several theories which is out there which help you understand **perception** one of the basic theories that we are going to talk today about is **Gestalt approach** to **perception** now another first primary interpretation of **Gestalt approach** is they start approach it interprets visual **perception** as how as interpretation of stimulus arrays into objects and backgrounds waters gets that approach start with what is the starting point of a **gestalt approach** the astark approach believes that processing of a percept start starts by parsing or interpreting the visuals the stimulus the visual stimulus or electrical patterns which are transferred from the eye into foreground and background so interpretation of an image interpretation of a visual field we'll start by first distinguishing that into foreground and background so the first step into the understanding of figure calling to **Gestalt approach** is basically parsing or interpreting the figure in the background as we saw in these figures as you change the object as you reclassified a figure as a background and background as a figure the meaning of the image changes look at this figure this is again

a reversible figure now depending upon what you're looking at or where you are focusing your background and foreground is that will define define what the figure actually means if you consider bad white as the background and black as the foreground you actually tend to see start seeing two people facing each other but reverse this if you make black as the background and white as the foreground you start seeing immediately start seeing the image off of us this particular feature of distinguishing an object of perceiving an object as a figure in background is called foreign **perception** according to the guest artist what does this help us with understanding or breaking something into figure in background breaking an image into **figure and background** helps us with several things first the figures tend to have show better shapes and tend to be remembered better than the backgrounds as soon as you interpret a visual environment two figures in backgrounds what really happens is the contours of the figure the idea about the figure or the part of the back of of the image which you consider as figure is remembered much better with much better **perception** then the background now from **perception** happens when we actually are able to classify part of it this image into figure and into background now distinguishing the figure enter back into distinguishing a visual field into **figure and background** requires us to understand or markup subjective contours now these contours will actually define what a figure is and water background is for example in this particular figure as you can see these lines are the contours which define what a figure or an background is the **gestalt** psychologists they believe that perceivers follow certain laws or principles of organization in interpreting any percept or any kind of **perception** and one of the first laws that the guest artists were hooked up to or they proposed is something called the law of whole and part what is the law of Golden Path guest artists believe that means when we perceive something when we perceive an image when a **perception** of an object or event is done the **perception** is in terms of the whole object or even it is not in terms of parts of the event or object and that is why the guest artists are in j-just direct opposition to what the structure is believe now the structurally school of psychology believe that any mental event any **mental process** can be broken down into its part and it could be studied that way for example what guest artist would say if you're looking at a cold lemonade the cold lemonade can be broken down into the idea of a cold lemonade the **perception** of a cold lemonade can be broken down into its physical form which is in terms of its temperature sweetness bitterness which are all physical quantities and also in terms of the idea of what this lemonade would taste like in terms of sarna's in terms of sweetness so the psychological part of a lemonade or the psychological content of a lemonade is the taste and the feeling that you get after drinking it whereas the physical part of a lemonade we'll move its coldness its temperature its sweetness its saltiness and so on and so forth so a basic opposition to this particular aspect to this particular school was provided by the guest artists who believe that it is not possible or it is next to impossible to interpret perceptions to interpret meanings about objects in terms of holes in holes and parts in terms of parts what they believe that any **perception** has to be in terms of holes and also that holes the hole of an object is always different from a part remember in the first lecture on introduction we looked at briefly on the **gestalt approach** and there I said to you that the holes are not different from its part is one of the prepositions of the **gestalt approach** there I brought for what for a foreword an explanation and I'll borrow that explanation again if you are hearing a song the song has lyrics has music and several other things included into it so the feeling that you get after hearing a song will not be equivalent when you hear the song in parts meaning which if I first make you hear the lyrics later on I make you hear the songs instrument playing and several other things one by one for the come the meaning of the song or the feeling that you get after hearing the song in parts will not be equivalent to what it will be in terms of holes and that is what guest art is safe guest artists say that **perception** is generally in terms of holes and these holes are divided into figures and backgrounds part **perception** is not possible and one of the reason for dividing a perceptual environment in the figures and background is because when you divide a particular environment into **figure and background** the figure is the one which is perceived and the back the one which is not perceive is is the one which is supporting you and as soon as you change meaning changes a

good example to bring here is the image on to my right bottom look at this image what do you see it's a dot image which has been created by psychologists to understand how **perception** works and as you will see that this image has several dots into it it's this was created by a fine art painter and if you look closer if you go closer to the image all you see are dots and patterns several dots black dots and on on white backgrounds but as you move yourself away from this image and as you put start put putting something called top-down processing start thinking about it or start deliberately making meaning out of it sooner or later you will realize that you see a dog a shape of a Dalmatian here this Dalmatian the shape of this Dalmatian here is what is the meaning that we get from this particular visual field and this happens only when we are able to separate this shape which now becomes a figure and everything else which is the background now guest artists they came up with or they proposed several principles are several organization principles which make people help in making successful perceptions we will discuss some of this although there is a list of perceptions of perceptual principles organization principles the guest artist what we'll discuss some of them here there are five major perceptual organization principle and guest artist the first is called **proximity** or nearness principle it says that when objects are close together they are perceived as together then when objects are far away from each other look at the figure on the bottom of your screen what do you see you see and when I ask you to interpret this what most people are going to interpret is that there are dots blue and red dots columns of blue and red dots although nowhere it is clear whether these are called amps of blue and black dots but people Club these blue and black dots since they are close together since they are proximal together into one and now they say there are most people try to explain this figure as four columns and of repetative blue and black dots and this is what the **perception** of Prague the principle of **proximity** really means principle of **proximity** says that when two objects are brought together or when multiple objects brought together they are combined into to get into togetherness or they are interpreted as together and this is one of the principles of organization of perceptual organization which helps us in making **perception** in understanding or making interpretation of the visual field a second principle that the construct is proposed is the principle of **similarity** what does this principle say it says that in if in a visual input or a visual field we have objects which are similar together we tend to Club those items which are similar as grouped together and so when I ask people to interpret what this is and when I asked my students to interpret what this is most people come up with answer saying that there are two rows alternating rows of red and white dots now when they say two alternating rows of red and white dots they Club the red dots together and white dots together a third principle is a principle of **good continuation** this principle suggests that when we group objects together whose contours form a straight line or a curved angle they are grouped together so when I asked most people to look at this particular image since these objects share a curved contour it's most people actually go ahead and say that this is a wave pattern whereas most of us can disagree today saying that it is not wave pattern as this and this part this in this part this and this part can be separately different but most people see these as wave pattern because the contours of these lines are similar are forming an arc a fourth principle is a principle of **closure** looking at this most people tend to interpret this as a circle whereas this is not a circle now the brain what it does is using a principle of **closure** it tries to make or close this figure to make it to make the easiest interpretation out of it and the easiest interpretation of this figure is the circle the brain feels of the gap which is there into the image the fifth principle is a principle of common faith this principle is difficult to explain on 2d surface but then a basic idea is that objects which move together in a particular direction are grouped together as you can see since these objects tend to move on the upward direction and these objects tend to move on the downward direction they group together as moving together and that's why we say there are groups of balls which are moving up and which are moving down what is the need of the **gestalt** principles one of the basic needs of the **gestalt** principle is that it helps us into organizing the but the perceptual details the details from the visual environment and make meaning out of it the style principles help us into organizing the information which is coming on from the senses together

to make some meaningful interpretation and **Gestalt** principles the organization principles of **Gestalt** also help us in a quick **pattern recognition** another important **gestalt** principle is something called the law of branch it's against general principle which all other principle which holds all of the principle and the interpretation of this principle is that the simplest meaning is always the most correct meaning what this principle says that looking at anything the most simplest explanation that is available for a particular image for a particular event or idea is the most correct idea or most correct interpretation of that event or or image looking at this particular figure here most people can define it in number of ways but the most simplest explanation of this is squares which form chain as you can see these squares actually tend to form a chain in the most simplest interpretation of this is that squares which are arranged in chain now the **gestalt approach** is not without limitations there are a number of limitations to this particular approach one of the most basic limitation to the **gestalt approach** is that these are this particular principle or gastral test view cannot be translated into cognitive or physiological process there is no one to an agreement of how these principles of organization of the **gestalt** gets translated into cognitive processes or what cognitive processes actually come into plate what physiological processes or brain processes come into play when these principles are in operation another interesting or I would say another limitation of this approach is the circularity of the pranks if the simplest explanation of this figure is the triangle this happens because it is the simplest figure or the most eligible stable figure that I can see I can go circular about it saying that since this is the most stable image I can see about this the most actually actually the effect that I see about this is the triangle and so this the simplest interpretation into it so it's it's circular in nature the prangs law is circular in nature and without additional information the law of pranks doesn't actually go ahead and interpret anything or give any meaningful input thank you

Lecture 5: Models of Perception-01

■Kind: captions hello in the last lecture we looked at what is **perception** and we also looked at the classical approach to **perception** in addition we went ahead and saw the **gestalt approach** to **perception** the organization principle of the gas style and the law of plants and the limitations of the **gestalt approach** now **perception** which is making meaning of visual stimuli is believed to be a two-part process in terms of the human brain **perception** happens through something called a **bottom-up process** and something called a **top-down process** the **bottom-up process** is a data-driven process and it moves from top to bottom so if I consider a triangle representing how the process of **perception** really work moving from general to specific is basically what turbulent processes are most opt-in processes bank upon may 6th immunise properties add up these ways to make these stimulus properties and come up with a meaningful percept a meaningful interpretation in contrast the top-down processes are generally motivation driven processes or Theory driven processes what it means is coming from specific to general most **perception** require the **top-down process** and the **bottom-up process** and so the next particular thing that we are going to understand in this lecture is the **bottom-up process** and the **top-down process** think what really happens when you see a tree like this the process of **perception** of visualizing this tree of making meaning of this tree starts by the creation of a **proximal stimulus** from the **distal stimulus** by the eye the lens of the eye projects this **proximal stimulus** which is the inverted image of the tree the inverted image of the lights falling from the tree on to the retina which then depending on the intensity and the type of **illumination** the type of electrical current that is generated by the first the the **proximal stimulus** sends this information on to the occipital lobe the occipital lobe then integrates various kinds of information which is sent on by the retina it's probably right time to introduce that the retina does not only passively transmits the intensity of light waves which are falling onto it it transmits several other information about the **distal stimulus** onto the occipital lobe for example some information regarding the motion of objects information regarding the contours which divide the background and the foreground and several other basic information in terms of the color in terms of the texture are transmitted by the retina itself these informations are then taken up by the occipital areas are the visual processing areas now this process which starts with light falling off the tree being reflected on to the lens and the retina and then there is not passing up specific bits of information onto the occipital lobe or the occipital visual processing areas is called the **bottom-up process** this percept which is formed this interpretation which is formed from the primary inputs from the retina is then compared with pre-stored notions or pre-stored representations into the brain and then a meaningful interpretation a meaningful idea is made about the particular object in the environment in this case the idea that this is the tree composed of two parts the **bottom-up process** will include those processes are those events in through which the greens color the brown color of the trunk the green color the leaf the shape size of the tree the texture of where the tree is standing how far it is from some other object in in the visual field these kind of informations which leaders are the kind of structure of the leave the idea that it has a leaf and the idea that it is it has a certain length and it is green in color the bark is brown in color it has roots it has fruits these kind of information which leads us to identify a particular visual stimulus as a tree is the the **bottom-up process** whereas taking all these information to get this percept together and comprising them to approach the type of a tree to the idea of a tree which is stored into the **long-term memory** is the **top-down process** basically most perceptual processes or most perceptual events require the integration of both the top-down and the **bottom-up process** so let's start with understanding what are the bottom of processes in **perception** and looking at some of the bottom-up processes in **perception** which are used for understanding the perceptual process the bot the term **bottom-up process** or data-driven process essentially means perceiver starts the punt first person who's perceiving the brain which is perceiving the image starts with bits and pieces of information of the object in the environment integrating them together to form in a percept as I was explaining if you're looking at a

tree the idea that green leaves are their ground trunk is their roots are there flowers are there it is of certain height it is of turtle texture all this information which is passed on from the retina onto the occipital visual processing area is what is the **bottom-up process** so basically in this case in this particular image I have brought an image I have got an image which will explain to you or which will try and explain to you what is the how do we interpret the meaning of this particular image through a bottom-up and **top-down process** so looking at this particular thing this particular image what do you see most people think of this as a corridor of a kettle of old building or of a European building or some kind of a corridor out there the idea that this is a corridor can be generated by both the top-down and the **bottom-up process** from for the **bottom-up process** what would happen is you would see into this picture and based on the kind of shading based on the kind of light that is have that it is having based on the kind of architecture that it has the kind of curves that it has the kind of our architecture the kind of bricks that are making this the diverging lines inside this this image the diverging messages inside there all these information basic information the kind of steps that you see the kind of arches that you see these and and the Lighting's the pattern the shading the color all these integrate together to give you the idea that this isn't hall and this process of using these information from the visual field to tell you that this is in Hall is what is the **bottom-up process** in contrast it the **top-down process** we'll look at this particular picture and compare it with something that you have seen before and we'll try to match up so if you have ever been into a hall like this before or if you have been to a cat riddle before which has similar interpretations or which were similarly interpreted or which was seen similar to it or how close that image which is in your mind that representation of a holiday is in your mind how close that fits to this that gives rise to or that if if you're using that to interpret that this is a corridor of a cat riddle if you're using that particular image or if you're using that approach to to make giving meaning to this image that is called the **top-down process** so this picture through a **bottom-up process** is basically coming up from facts like the depth the figure the ground in texture semicircular arch a semicircular blocks doors and so on and so forth now how do we understand the **bottom-up process** an easier way to understand the **bottom-up process** is that bottom-up processes for both most bottom-up processes as you move along the chain as processes move along from lower dimension to the higher dimension there is no possibility of correction so in most bottom-up processes as you move higher along the dimension as you move up higher along the processes there is chances of Corrections are very low now how do we visualize a **bottom-up process** think of a class in which the person who's sitting on the last bench is given the task of writing a word anybody wishes and then the next person to him writes adds up another word our job here is to interpret or basically make a story of what students write onto it and so the person next to the person who wrote the word first writes another word and this way this keep from what he will soon see that the person at the front of the class has not much option what he has to do is he has to look at whatever has been written before and based on that make an interpretation whatever processes are like that basic stimuli or first processes in are the first inputs are taken in and from there how inputs are made or higher configurations are made but it is not possible to go back to the basic inputs from which higher inputs are made no that's how we visualize the **bottom-up process** also most **bottom-up process** are automatic in nature as they don't require some kind of control by the mind they don't require some kind of cognitive control so they are very automatic process as you see something the brain interprets the eye retina of the eye basically breaks the information which is coming into its similar path in terms shading color and so on and so forth interprets this takes in this and sends it to the occipital visual area for processing also these are very reflexive processes what am i processors are generally a reflexive process now there are three or four bottom-up processes that we will discuss in this particular lecture the first is called the **template matching** process it's a very simple process and what it says there are pre-configured template into the brain and what really the **bottom-up process** really does is it takes in information from the visual world integrates them together and compares them across this template with this template and when a match is recognized an

interpretation a correct interpretation is done when a match is not there another template is looked to a match is made and so on and so forth now think of templates as tencel's these stencils in general use most tencel's are used to make figures or to make words or to make events which are equivalent to the census in **template matching** model or according to a **template matching** matching model a reverse of what the census does happen so we have a fixed stencil into the brain a fixed idea into the brain and whatever the eye or whatever the visual system is sending whatever information it is sending into the brain it is taken in and compared to preconceived stencils out there as the information matches a stencil it is recognized a percept test form and it is recognized as as it is and this is called **pattern recognition** but if a match is not found a next stencil looked into and so a large number of comparison has to be done also think of **template matching** or understanding **template matching** in terms of checks now all of you would have seen what a check looks like so most checks have a certain area where person can write his name and amounts and then he has a signature area and then below you see certain numbers what is the use of these numbers these numbers are generally used by check sorting machines these checks slotting machines which are installed in different banks and what these banks actually do is that based on various numbers and letters it then forwards a check to a particular branch it's a very simple interpretation of what the **template matching** model does so banks otters or bank cheques otters are equivalent to a **template matching** scheme in which several templates are there with the blank sorting machine and what this machine does based on which numbers and letters you have on the cheque it will pass your cheque along to a particular bank it's a very comprehensive model and the main proposal of this model there are prefix templates into the brain and incoming information from the eye is made into a percept and a template it's matched to preconceived templates into into the brain and if there is a match and interpret is a minute if it's not a second abilities looked about and so on and so forth now there are several obvious limitations of the **template matching** model what are the first **template matching** model limitation is the requirement of a huge number of templates now by its the design itself the **template matching** model requires that we have to have a huge number of templates into the brain now since the brain capacity the idea that the brain has a fixed capacity and it cannot have a large number of templates so this model lacks a particular feature lacks a particular gain over other models also then a new template or when a new image or new object comes into **perception** in the perceptual field how does the **template matching** model account for that think of Addison in this way the phone industry started with button phones and then came in the first resistive touchscreens and the capacity touch screens no matter we move from the button phones to the resistive touch screen or the capacitive touch screen phones people didn't have any difficulty at all the bar kinda phone with the buttons had a particular mental template this is what is supposed to do but then when people migrated to capacitive touchscreen or touchscreen phones they didn't have too much difficulty into identifying touch screen phones or the functions of the touch screens corn which means that the template of the button phone got transferred or somehow accommodated the new template of the capacitive phone or it extended yourself to the capacitive phone now how does that happen that's not that's not explained by the **template matching** model so it doesn't say that if a new model comes in or if a new form of object comes in which is I'll be different from the template how do you account for that also another interesting thing or limitation of the **template matching** modulus handwriting's now a lot of people recognize a lot of symbols as meaning the same thing so how does the template mapping model actually account for that on your left on the slide you see a text return and so this text has been very nicely written in all capitals and it is very easy to read and most people are able to read it on the left you see a very bad handwriting tell me this is it difficult to read the text on your right it's not most images or most letters are preferably out there and it makes meaning to you but **template matching** model says that fixed templates are available to the brain so if a is written in this way any deviation of a into like this will not be matched and will be an outlier or will be not perceived by the brain as we can see handwriting's whereas as you see here this day and this a here this a here and this oh here and this F here they don't match to the general way of how f AE NC

should be written but still we are able to read this which basically means that the existence of fixed templates is something which is a far off cry which is not true and so to basically fill in the gap which was made by the **template matching** model to counter the limitations of the **template matching** model as we saw in the study of **cognition** that several methods are used for studying **cognition** because one method we have if one method of the limitation a second method goes ahead and plugs this limitation or basically proposes something which accounts for this limitation and so a better idea of how of how a cognitive process really works or how mentally means a process that gives us a great idea and so to plug in through this limitations to account for these limitations or the **template matching** model a new model which was called the **feature analysis** model was proposed now what is the **feature analysis** model **feature analysis** model says that when exception happens when we perceive things from the visual environment we look for units basic units which are called features we look for features certain features on to **proximal stimulus** which which is coming which is being inputted true from the retina onto the visual visual processing area so then when looking for when looking at a stimuli looking at a complex environment or stimuli we generally do not perceive the stimuli at as a whole but we break the stimuli into its several part into stimulus into its different features and we look at these features perceive these features integrated back and that is how we go ahead and recognize a particular particular object when I enter my classroom how do I make this idea or how do I perceive this classroom the idea of a classroom or the **perception** of this is a classroom is made by several features of the classroom for example there is a door there is a certain kind of edges and angles which suggest which is a door or there is blackboards and certain ideas about what features a blackboard should have what is the background what is the foreground and that kind of a thing is there so when perceiving something or when looking at and any environment finding out the basic features or breaking a hole or breaking a hole in visual environment it's too intense different features helps us in identifying an object much successfully and much faster than gobbling up on or then interpreting the whole of the object as a whole there are several support which is out there for the **feature analysis** model for example the idea that while perceiving something the first step in **perception** is identifying pictures starts by works on frogs on retina frogs and so what scientists what researchers did was they was they implanted mic electrodes into the retina of these frogs and they found out that these micro electrodes they responded according to how the retinal cells responded and so on where they found out that there are certain specialized cells onto the retina which goes ahead and responds in certain ways in in different ways for example to cells or to groups of retinal detectors have been found one is called the edge detector the other is called the bug detector now the edge detector is that group of cells in the retina which become active or which fire more when a frog retina or a when a frog actually sees or forms a border between dark and light area and this this group of cells are called the edge detectors so within the retina itself they're a group of cells which actually respond to the visual environment to the input from the visual environment and separate them into its different features and so one of the features of any percept is the contours or the edges of where the figure in the background is banking up to what what the **gestalt** is said the first step or the first process in any **perception** is figure background effect and so this particular experiment gave ideas that in the retina itself there are group of cells which actually make you distinguish between the edges the light and dark find the four edges similarly retinal cells in France have also been known to identify movements so if there is a movement these retinal cells actually go ahead and respond more vibrantly more actively than other cells and these are called the bug detectors as you can see when when we actually see this woman this picture of a woman here who this is what the retinal image is if you look into it this is how the retina responds to and this is been captured from a human eye through micro electrodes and what what you actually see is we don't see colors we just see this kind of an interpretation this kind of black and white contours in to it similarly bug detectors have been found so when you see somebody running the bug detector which is there in also in the human eye it responds something like this as you can see it actually goes ahead and responds in this way it is responding to the motion of this

person whereas in this case it is responding to the edge detectors are responding to the edges the differences between what the figure background would be or what are the contours which basically makes this this woman there so this is basically the direct output from edge detector and a bug detector into the human eye and into the human eye and what does this really mean it means that at the level of the eye itself the features are separated out now what other basic idea or what are the basic support which was which was provided to the feature and Isis model was by it someone called living Bateman look at the image that I am drawing what does this image look like and so most of you will be able to tell me it is some kind of an cart with wheels into it and and so these are the wheels this is the body of the cart and this is a handle for pushing it and so on and so forth so basically this image that I have done is basically an inconclusive image it doesn't make any conclusion doesn't fit into any template pre-existing template but even when this image makes no meaning most people or people like the viewers that I have are able to make at least some features extract some features out of it for example this is a cart it has real so bases which basically means that it can move it has a handle which can make it move and then it has some kind of nose I don't know why it is that is the kind of things basic support by Biderman was provided that even if an image an inconclusive image like this was presented to people people were able to identify this and this provides support the fact that people respond to features and these features are what gives rise to two interpretations to the **bottom-up process** of how **perception** really takes place now it Biderman propose the conceptualization or something called jion's and what Biderman says that these jion's are something called geometric ions so bitumen gives the idea or suggests the idea that unlike the ions in chemistry there are something called geometrical ions or some basic shapes which are available and according to bitumen there are 36 basic shapes which are available in in the visual field and most perceptions that we have are actually interpretations of these 36 basic shapes so that's what's a what's our vitamin says and he says that any kind of interpretation or any kind of change in **perception** can occur by changing the arrangement of these jion's so these are the beta means jion's if you look into it this is the shape of a Q this is a wedge pyramid cylinder barrel expanded handle handle cylinder cone arch these are some of I'm showing you some of it but there are different jion's available and what Biderman says these are the basic Gian these are the primary giannis which are required geometrical lines which are required and if you look into this most visual field or most interpretations most **perception** that we have is actually a combination of these jion's and to say that these jion's if you change a Gian you can change the meaning of an object what do I mean by that let's look at this so this is the rectangular Gian this is the semicircular Gian so when I look into it right now the rectangular Gian and the semicircular jiron it looks like income but as soon as I change the arrangement of this jion's I put this here and then this cups now become a basket or a bag and this is what the diamond says all I need to do is change this Gian from here to here to move this position argument of Gian and then you start seeing these two things together and that's what bitumen says so as as you interpret it's basically features that make you make a **perception** and bottom-up processes generally use the **feature analysis** model for making any kind of interpretations now beethoven's can say that these 36 jion's are necessary for forming all basic shapes into the world he analogues this into the idea or he contrasts this into an idea of sound **perception** and he says that there are a compare his theory of 36 jion's with the 44 basic fundamental so as you would know that there are 46 44 different phonemes and these phonemes the basic sound structure will study phonemes into the in the chapter on sounds on and so there will will study what fun amazed and but right now what the idea here is that this 444 basic sounds they combined together they get together to give all the English songs which are out there and so what what Bitterman says is that this idea of looking at these forty-four phonemes and making all the sound is similar to his idea of having 36 jion's making all the shapes which are out there and he says that if you interpret a **phoneme** differently to phonemes differently a different sound would be interviewed be produced for example the da and tough on you are most confused whereas the da in Saffron you are less confused and this is one of the basic one of the basic tenants of the or support of of this model of **feature**

analysis model of **perception** or top-down processing so Biderman shows us of a demand shows us this particular this particular thing and he says that when we see this particular image most of us are at least able to tell us that this looks like clop the reason why this looks like a wrist watch or a clock is because it has mechanical parts which tends to move has a strap and so on and so forth although we all know that this is not a watch and this is no no idea of a watch now support from Elna gives us 1969 theory of **perception** of **perception** of similar letters also support the idea of the **feature analysis** model now what I found out they found out that items are worse letters or words which have similar shapes which has similar features are confused more than words which are different shapes and so what Gibson said is that it is easy to confuse between visual inputs of G and C then to confuse between G and F the reason being that G and C both have a circular G on which which which is attached to another small G on which is there but if we look into G and C they the circular G on is not there here we have a circular G on where we have some other kind of a G on and as the reason how how these confusions occur so confusions occur because G and C smalls they share the same features or with each other another support for this **feature analysis** model comes from the work of nicer who says that in a visual search task so what nicer did was he gave a visual search task to his subjects and what was the visual start starts the visual search task was easy he had letters like this written and within these letters of having a semi-circle he hid a letter with straight angles so as you can see there are gcq and something like this where and on the other hand he had letters like like this now when he showed these two displays two subjects he found out that finding a Z in this case was easy whereas in this case what it was difficult why was that because in this case Z was very distinct that has features which are very is distinct from all other letters in this percept whereas in this case what really happens is that Z says some of the basic features with all other letter Z a TA tkl and so it is very difficult for for somebody to quickly identify Z into this this particular visual input the reason being that features are important to **perception** he says that the more closely two items share a feature the more difficult it will be for us to separate but the more distinct features are the easier it is for us to to understand the percept and what this all combines to to make interpretation is that features are important for bottom-up processing a major major idea in whatever processing is that features make a person another interesting model or another interesting study that we look into in the **feature analysis** model is called the Selfridge spend ammonium model what is the Sailfish pendemonium model is a very interesting model to look at self it's pendemonium model will again discuss it in another chap another chapter in another coming chapter in this course now what self is smarter model actually tells is how does **perception** of letters happen for example there is a letter a and there is a letter B and there is a letter error how does the visual system perceive this now prior to this I explained to you how the I have specialized regions which can the retina has specialized regions which can process information regarding lines angles curved edges and so on and so forth and this information is passed along to the brain the visual processing areas of the brain which takes this information and creates the percept or creates the **perception** now how are these letters perceive now what pendemonium model of self it says is that the **perception** of these letters for example a starts by there are several demons he calls them demons and I call them sub processes which are involved in the **perception** of a so if I have to perceive the letter a it starts at a very **basic level** now since whatever process starts at very basic levels or the first inputs that I get from the eye is that a when an a is displayed in front of you you have something like this arched line you have another arch line and you have a line like this now the basic demons actually read this kind of input and they forward this input into the letter daemon which will then combine this into the different interpretations for example to arch line and a single line can mean three things in the English **language** I can mean an a I can mean a we although the we will not have this between thing and I will have a hatch now the thing is whether this is a V or edge the interpretation of that takes place through this kind of a **bottom-up process** at the level in the first level I have the line in edge demand which will then which will provide an input to the letter demand not the problem with selfish model is the letter demand cannot ask back or cannot call back the edge

daemon for asking any questions also these demons director demand or the curve demon or the the demons which interpret land lines and angles these demons keep on shouting so there will be several dementia will be demand of a straight line where you have a different demon of a slightly curved line on this side a line and so on and so forth so those demons which actually process this input when when I show an a now the demon this Damon is the one which is excited and so it will shout the most and all other Damon will shout the least will not shout at all and so input from this is basically taken in and fed into a letter processor now the letter processor will take the idea we'll get the idea that there are two lines one which is slanted to the left the other to the light and run a straight line in between that is what is that is what the information is coming from the I if we look into this combine this to basically produce three inputs three things can happen with these lines three different english-**language** letters can happen either we have a a we have a V or we have an H and there is a sec and another demand the highest demand which is out there which will then look at compare this with pattern recognitions with patterns which have been stored in the brain previously and say that a is the only percept and F and H are not now this kind of model selfish model is very close to the connectivist model which we saw in the first classes or in the first lecture throughout this series where will we looked at how this connectivist model really works we'll come back to the connectivist model a little later into the chapter of **language** and into the chapter of **memory** so this is how the model really works when image a is presented there is an image diamond which will first start shouting because this is an image that is there and then from later on then we have feature diamond the feature diamond is there are two lines and those these feature diamonds will start sorting then there are cognitive demands will interpret what this is all about then there are word demands and then there is a decision lemon which is the highest demand which says that this is an a and not not and so on and so forth good example is to look at it this is the B image demand if you look here that then these are several other diamonds which represents various lines which are there and so those lines those diamonds which represent the features of an B will shout the most and which do not represent features of B will not shout and then there are angled amends so we have the image demand dang think of the demon as a process of how the processing takes place so how does the processing takes place first we divide B into a line like this into a line like this into a line like this and then combination of this so that is what it is and this is how first the image diamond excise then the line Domenic's i'ts then the angle diamonds excites and then the pattern diamonds because we could also mean it is closer to R it is also closer to B it is also closer to P so these are the pattern diamonds which are there where because these are all the possibilities with this this and this into being and then there is a disease and MN which then compares this this this and this and says that the image that I have the input that I have is the closest match with me so this is how the selfridge model really works there are several diamonds and these diamonds are the inputs which you get from the retina so this kind of a model has been designed by Selfridge and it's it's provides another greater support to the **feature analysis** model that when we perceive something it is the feature that is important the **feature analysis** model suffers from some very basic shortcomings one of the first thing is that there is no good definition of what a feature is and what is to be expected so the idea of what a feature should have what is a feature should not have is not very explicit and so once a definition of what a feature should and should not have is not it is not out there we do not know what a feature what should be a feature and how these features are identified another problem is there are different sets of features for different objects and how does the perceiver know which ones to use to perceive an object for example for for different objects we have different different set of keys for a face we can look into the eye and define a face we can look at the nose and define a face we can look at other features of the face for the part of the face and define a face so which are the ones which are the prominent and which are the one features which is the most for example some people define a face define a person based on his eyes and some people look at the cut off her face the kind of hairs are somebody has and so all of them are features of the face so which is the one which is the most important is really difficult and it's it's one of the main limitations of the feature

nemesis model another interesting model which adds up or which has been discussed as a bottom up model is the **prototype** model now the product model is very similar to the template model but then it is an improvement of the template model what does it do the protec model says that whenever an input is given to us whenever an input is given by the visual system to the visual processing system this is compared to a stored representation of information into the brain and so what template model loses on to the **prototype** model gains on to that so so in instead of so what the template model what the **prototype** model says is that instead of comparing the total visual image which has been provided by the eye instead of taking this and comparing 800% with templates which have been stored in a brain we do some kind of close **proximity** matching what we do is how closely what we try to find out is how closely this fits on to a **mental representation** which is out there so we do not look for hundred percent matches we look at how closely they fit the more closely a particular image a particular visual scene comes to a stored representation the more chances of it being categorized as that particular representation now the basic feature of the brain the basic desire of the brain is to categorize the brain doesn't has a lot of free energy so what does the brain generally do is takes up any stimulus which has been given to it any kind of input which has been given to it and then categorizes the first thing that it does is categorizes makes meaning out of it the quickly can categorize something and makes meaning out of it the lesser it work it is required to do so that is what the job of the brain is and one of the things that the **prototype** matching model does is eases the the function of the brain or the task of the brain by making an assumption that the input which is coming from the visual system is matched in in in terms of in terms of problem similarities rather than in terms of complete similarities so instead of being as a whole pattern that must match how closely and that is a **prototype** now what is a **prototype** the idea is what water **prototype** is **prototype** is basically an abstraction which is generated from several instances of event or an object for example for developing a **prototype** we need to understand that a **prototype** is develop from several interpretations or several instances of a particular even let's take a look at this all four of these images are actually cars right this is a car which is a highly developed car this is a very basic car as small currents on a school but all of them are car now when I say car what comes to your mind the idea that what a car should have is the **prototype** idea of a **prototype** of a car so when I say a car the basic features that a car should have is what is a **prototype** and if I look at this forecast what is the **similarity** that I find into it most cars most cars tend to have four wheels so the first feature is four wheels it should be driven by an engine and so these are all driven by an engine and so no matter how difficult or how new a car model I come up with my **prototype** of a car will actually try to match it closely with it try to match the new model of a car which is out there it's closely with it and the kind of **similarity** the kind of closeness the kind of close approximations the close **similarity** is that the new model has with my car will tell how close the new model of a car is to my **prototype** of a car so when making a **prototype** I look at all the similarities in all instances of this image and from there I create a **prototype** for the example in this case the **prototype** of a car my **prototype** of a car or anybody's protector of a car should be something a vehicle first of all they should be able to transport you from player plays a to B then it should also be able it should also have four wheels at least so most of them have four wheels it should have a body where you should be able to sit and a cover or something as you see most of these images have have that thing that **prototype** now in addition to a **prototype** is something called an **exemplar** and in extension of the **prototype** in example now **exemplar** is basically the best **prototype** which is available so the dog is dog which is available is most people believe that it is the dog yes dog is the the common home dog which is called the Alsatian or the small dog which keeps in Barking which most people tend to have that is the dog yes dog it should have a tail it should have a far it should bark and that is what the idea is and so this is how the dogs are compared into so a non session is the dog is dog which is available and so that is an **exemplar** when I think about a **exemplar** an **exemplar** is the best example of a category of a **prototype** which is out there and so the most classic car for me would be the BMW because that car has all the features are in in terms of the

Indian standard the most **prototype** car of the **prototype** of a car would be a Maruti Alto is the most **prototype** of a car because it has all the features of a car it gives you all the features although it may not have some of the highest versions but it is that is the **prototype** of a car so generally prototypes are abstractions which are made these are mental representations which are made from several instances of a particular of a particular category of a particular class of events according to protocol mathematical model matching model when a sensory device registers a new stimulus the device compares it with pre previously stored prototypes an exact match is not required for as closely a match of what you are seeing is to the **prototype** the easily the more closely the way the visual input matches to the pro tonight the more easily it is understood or most easily I it is identified as belonging to that particular class now the question is where do these prototypes actually come from where do we develop this **prototype** and an elegant answer to this comes from a work done by Posner and key what poster and qu'il says that the development of a **prototype** they found out how does **prototype** how do people develop **prototype** and they found found out that **prototype** development is an automatic process and that's how **prototype** develops it is by mere exposure so what was the experiment in their experiment what they and what they did was initially they created a nine dot pattern like this which they called up there which they called as the **prototype** and then with that in the first version or in the first instance they created our nine dot pattern in the second instance they took this pattern and placed it in a grid with patterns which are similar to this but distortions of this pattern so they gave this kind of input to subjects so first subjects were called in initially before the subjects were called in nine dot pattern like this was created which was the **prototype** later on subjects were called in and they were shown the various distortions of the **prototype** with something which is closer to the **prototype** and they were asked to classify these distortions which were there for example this is as you can see it will form a triangle here it will form a triangle this is kind of a triangle I cannot draw it right here but then this is how it should look like forms like this and so dot patterns are like this which closely resembles a triangle but not a triangle later on so various arrangement distort chance of this **prototype** was done in a third phase of the study several other distortions of the nine dot pattern was given to subjects which were new distortions and people were asked to classify or to basically group these new distortions what really happened people were able so within this third experiment in this third experiment we included the original **prototype** the distortions from the old distortions into it and some new distortions what was found out that the chances of identifying the **prototype** which the subjects have not seen was 85% and the identifying of identifying of old arrangements distortions of the original **prototype** we're almost 87 percent whereas classifying or identifying these new distortions which were nowhere near the distortions of the this kind of new distortions which were there which were never close to these distortions new older distortions which are there was only 65 percent now what is of interest to us is this 85 percent so all those subjects was not shown the **prototype** figure at all they were able to identify the **prototype** figure on their own how does this happen this happens because subjects when classifying something they have create a **mental representation** they create a mental abstraction and these abstractions are used for classifying things or taking new things and classifying according to these representations which are this and this is how things really work so this is how the prototypes really come from so in the next class then we go ahead we will start discussing about the **top-down process** this is the another process model so as in today's lecture we went ahead and saw what are the **bottom-up process** model those models which take in data and this the witch actually taking data and this data build up accumulate together to actually form the percept in the next leg possible in the next lecture we'll go ahead and discuss about the top-down model those models which are theory driven which a motivation driven which help us in **perception** thank you

Lecture 6: Models of Perception-02

■Kind: captions hello we're back again with this session on **perception** now in the last session we looked at some models of bottom-up processing where we define **perception** through a data-driven process so bottom-up processes actually help in **perception** or predict **perception** by looking at how **perception** gets built from stimulus single stabilized and addition of the single stabilized gives rise to the **perception** another modular **perception** which is prevalent in **cognitive psychology** is the top-down model of **perception** now at the beginning of this lecture I gave you the difference between what is the top-down and the **bottom-up process** of **perception** the turbulent process is a theory deviant process where expectations of people generate **perception** whereas the **bottom-up process** is a data-driven process where characteristics of the stimuli certain features of the stimuli lead to whatever **perception** we are looking for or whatever percept or the meaning that we are generating from the stimulus which is coming from sensation so let's begin a study of the top-down processes of **perception** the **top-down process** of **perception** also are also called the theory driven or conceptually driven process of perceptions here expectations of the perceiver is the main responsible feature for the development of meaning or the development of percept from the sensations chapathis turn through the visual system so the whole process of **perception** right from the meaning of the percept to **pattern recognition** has on the back side or on behind these processes the expectation of the viewer look at this le way the same thing that we saw in model R in the first model of bottom-up processing now they're the explanation of this visual scene was given in terms of the structure of the le way in terms of the shading the light which is falling on this le way the kind of windows it has the kind of background it has and all those features combined together gave a meaning of what this picture depicted in the **top-down process** we again look at the same elevate and we try to explain the meaning that is generated through the **top-down process** to things that are evident that are really the main frame or the backbone of the **top-down process** is the expectancy effect and the context effect so basically the context effect the effect of context in which a **perception** is is made and the expectancy of the viewer are responsible for the **top-down process** so coming back to the explanation of this visual scene using the **top-down process** this visual scene is explained in terms of **memory** or in in terms of the idea that people would have viewed a structure similar to this or maybe this particular structures on there so basically in terms of in in terms of the **top-down process** the experience that are this is an archway with margulies comes from the fact that you expect certain kind of certain kind of visual features a certain kind of objects in the visual field based on your **memory** based on your previous experience with visual scene like this which you have seen in the real world as I said the two factors which the **bottom-up process** cannot explain in terms of **perception** are expectancy effect in context effect let's look at context effect let's look at the figure on the top as you can see if I ask you the question whether the size of the middle ball the ball or the circle at the middle of what the figures are the same and most people would actually get up with this answer that they are not same but amazingly the size of both the balls are both the circles which are in the middle of the black circle is same the reason being that the one on the Left has bigger balls and in the context of that it appears the circle in the middle is smaller in size whereas it is not and on the right since he has smaller balls surrounding the circle it seems to be different and so context where an object is in the environment has a lot to do with **perception** similarly the figure on your right if I show this figure most people would go ahead and read this figure and the actual meaning of this figure would be wonders of **perception** now amazingly word wonders of **perception** is not clear but what really happens is the brain sees that or from your past experiences you see that this is the most plausible this is the most plausible answer which is out there and so one does the **perception** is being is being interpreted because you expect the brain expects and and the brain sees this this particular idea that it should be wonders of **perception** also since this is a class of **perception** so we do see the bottom half of this figure to be meaning **perception** because the context in which we are studying is **perception** and so

should be wonders the **perception** and so these are the two figures are these are two examples of how **perception** progresses or meaning of **perception** is done through the top-down model the context in which a pattern or object appears apparently sets up certain expectations in the perceiver as to what objects will occur know both **accuracy** and the length of time needed to recognize an object vary with the context so the more complex the context the more time it will require for somebody to go ahead and make the meaning of the **perception** now elegant and computational model of **top-down process** of **perception** was given by David Marr now what this Mars theory or mass idea is that the model of **perception** proceeds Bekins **perception** proceeds by constructing mental representations or mental schemas no **perception** Mars beliefs is a special-purpose computational mechanism which progresses through certain modules and these modules are the module for analyzing color the module for analyzing motion the module for analyzing shading and this part of the theory or this part of his idea is the **bottom-up process** he further says that with these modules the modules for basic information about the stimulus combines with the module for expectancy and context and that generates **perception** so in in brief Davis Marr model is basically a model which combines both bottom-up and top-down processes to explain **perception** what David Marr believes that visual **perception** is are actually **mental representation** or mental sketches based on three kind of representations which exists within the observer within the person who's perceiving now there are three steps to it there are three different steps to it when you perceive something when somebody sub peers perceive something the first step takes place and this is called a perceptual or the **primal sketch** in the **primal sketch** depicts our from eight from the final sketch one can get information related to the relative brightness and darkness to the images localized geometric structures and boundary conditions so this kind of a basic information is related to the two-dimensional world related to the basic brightness and darkness and geometrical structures as what what is gathered up from the **primal sketch** so according to Davis math model **perception** progresses through a complete through a list of steps even the first step the **primal sketch** is made and this **primal sketch** is actually a two dimensional model or a two dimensional percept which has information on to it relate relating to brightnesses and darkness and geometric structures now once an which which basically helps people in detecting boundaries between objects and figures within figures and backgrounds now once the primary sketch or the **primal sketch** is ready the next step is constructing a to an Hardy sketch now in the two and a half D sketch using cues from the environment again so both step one the **primal sketch** and two and a half D sketch are actually the bottom-up processes of **perception** or the bottom a part of **perception** which this model goes ahead and explains so the first two steps actually uses the data from the stimuli so into an hearty sketch what really happens is that shading in texture and other information for example surfaces and relative depth are gathered from the environment or from the environmental cues onto the percept and that gives an idea about where the object is in the environment where the localization of the object and and the kind of object it seems to be so the first to step in Mars mass's theory of **perception** are **bottom-up process** once a two and a half D sketch is made at the point when this sketch is complete viewers have an idea of where the object is what does it look like what are the basic features of this object what are the basic characteristics like the color the shape the shading the boundaries the brightness the darkness those kind of informations get available to the visual processing area once these two steps are complete a third and final step called the final 3d sketch is produced where recognition of an object is made in terms of the in in terms of the context and expectation here the meaning of the visual scheme and the recognition of object is made and this happens through a **top-down process** so basically Mars model combines both the **top-down process** and the **bottom-up process** for making any kind of percept and in very simple words what his model suggests is during the **primal sketch** and a two and half D Sketch information relating to stimuli from basic 2d information to some sort of two and a half the information which is related to shading localization of object in the world depth **perception** relative depths are gathered in the final step of his process which is a **top-down process** which is a

top-down method what really happens is that viewers expectation and **memory** come into play and leads people to perceive to make a final percept to make a final recognition of out of the object in front of him of other object is and this information is derived from **memory** now some examples of how this **top-down process** really works I'm going to relate now one of the best examples of or one of the best example of the **top-down process** is something called perceptual learn if you look at the figure on the right side when you look it for the first time you won't be able to see anything but as you move along as you go ahead and see it time after time for multiple times what starts to emerge is a figure of a person and happens because as you keep on seeing it a number of times some meaning is generated out of it so what **perceptual learning** really means is that **perception** changes with practice as as we go on practicing as we go ahead and practice a lot what really happens is that we are able to generate **perception** now very recent experiment was done by Gibson to prove **perceptual learning** what Gibson really did was he did an experiment a very simple experiment in which a coil like this was given to subjects this kind of coil was shown to subjects what subjects were required to do is to remember this later on subjects are shown for different coils of this type but these have different orientations so this kind of coils so this is my basic coil and these coils which differ from the basic coil in some or the other way in terms of orientations in the term of number of coils in the term of where the starting point is were shown to people later and and they were asked to report whether these four coils how many of these four coils actually matched the basic white later on a number of different coils with different different orientations were shown two subjects and they were asked and within this this whole picture was hidden for different four versions of the initial basic point and subjects were asked to reproduce how many times times did they actually go how many copies of the original coil was available now it was seen as that with number of practices that the increase in the number of practices subject starts making less and less error and when feedback was taken from subject what happened is most subjects related to the fact that as they kept on practicing as they kept on seeing the basic coil and practicing it they started learning that all started observing those features of the stimulus which they didn't see it at the time at the first time so as they came familiar became familiar with the subjects more and more they started noticing parts of the stimulus which they had not noticed on the first advent and then the first interaction with the stimulus so this is an interesting feature or this is an interesting I'd say an example of **top-down process** where what happens is that the more number of times you interact with something the more features you'd see and the your expectations are developed accordingly and more easier the learning becomes another interesting thing or another interesting example of **top-down process** of learning of **perceptual learning** is something called **change blindness** now it is the inability to detect changes in objects or seen especially when given different views of that object so basically if something is shown to you and many versions are many different versions of this particular object is shown to you from different different angles people are not able to they take this gene and this is what is called **change blindness** now look at the figure which is on your left and you would see that one of the person right here changes but if I don't tell you this that this is going to change you will not be able to similarly if you look at the figure here what happens is the person at the back of the back of the screen will change but if I don't advise you or if I don't give you this clue that is going to change if you don't expect it it's not going to change now **change blindness** is a direct example is one of the best example to show that the brain sees what it wants to see an expectation place a lot of role into into **perception** we have seen a lot of movies and so what really happens is that often you would have found that in movies there are scenes which are inconsistent with the movie and no matter how many times the editor cuts it so there are parts of the movies we still remain but most people do not see these deviations in the scene these deviations in the movie example in one movie in one scene there was a white was and in in in a next scene the wasps disappeared and this was quite quite funny but the thing is that this was this particular thing was not edited I mean there was got away or the what waswas not sewn in the second scene whereas the third scene the wash reappeared and so somehow the editor who was reading the movie

forgot this and people since when they were watching the movie they were actually worth watching the gist of the movie so they they had in a certain expectation and since this was which was there in front of them it was not in their expectation or they were not expecting it or they didn't see that much so they couldn't notice this change and so **change blindness** is another interesting feature of the **top-down process** another interesting example of the **top-down process** is something called the word security effect look at the word which is given to you on the right what is this word it seems like the word is wo RR but it is not as most people if I give - right those people who know the English and most people who are literate if I give them they'll say it is wo RK now since this part that you are seeing is covered by a black spot and so expectation believes that since wo RR is no word and people will not go ahead and write and so this is since I'm talking about word security this has to be a word and it has to be work and so that that is another interesting phenomena what security effect is another interesting phenomena which suggests or supports the idea that top-down processes of **perception** exist now interesting experiment was done to test word security effect so here what happened is people were shown interface like this in which the letter K or D was presented to them for very brief periods of time and later on they were tested the letter K and D was presented to them and they were tested in the sense that whether or in which context do they are they able to unto identify this KND so the experiment ran in this way initially they were letter K or D was present to people for a very brief period of time later on they were shown this kind of three different segments in which in the first segment the blank side was represented and in the slide there were a single letter K or D written on it and people were asked to recall back whether K or D was presented in the second case a word was written a part of the word for example the blue or was eaten and this D or K was presented to people and they were asked to report that whether D or K the whether D or K was presented in the initial design and later on unknown word was presented to them and it was asked whether the D or K occurred in my presentation slide and interestingly it was found out that when this D or K was presented in the context of a word it was produced or more less number of errors are generated by participants which means that in the context of a word **perception** is easy another interesting feature or example of what security if effect or top-down processes of **perception** is something called the connectivist model of word **perception** now what this model says is that first the **perception** of word actually goes through or progresses through several stages whether it is word or whether it is a sound which is perceived the **perception** happens to several stages and these stages can be either excitatory or inhibitory for example what **perception** starts at the level of features where at the first step the features of the word for example whether it is a curved line whether it is a slanted line whether it is a dashed line a circle is presented at or is part of the word is first perceived and from here from the level it goes to the little or the level of letters here these features are combined together to form individual letters for example ACK and so on and so forth and at a higher level there is a word level where the actual word is basically being presented so this kind of **perception** really takes place now what is the what is the reason or what is the meaning of all this it basically says that from the feature level to the letter level a lesser number of connections are there whereas from the letter level to the word level more number of Corrections are there and so what really happens is if the **perception** happens of the word level it's easier for people to perceive letters it are then when it is at the level of features what really it says is that **perception** of a word the activation of relevant node for a word activates the node correspond to the letter within the word thereby facilitating their **perception** so for example if it is able that I'm looking at if a word is able that I have to see what happens is this kind of connections start this kind of this is the feature level so the first identification of first **perception** is the bottom of process process this starts at this level and from there this is called the letter level here this is integrated together so for example if I am looking for a the word a which is not present here so ble is presented to you a is not presented and if I am looking at whether a a is presented or a it should be the word here or not and if a is presented to you what really happens is that these all these features are evaluated now since a doesn't has this feature so this connection from here to here is an average reconnection whereas since this is

present in a this is an excitatory connections and so since these with these many features a number of letters can actually be generated for example a and T GC and so what happens is for each of it we will have an excitatory or inhibitory connections so those connections which suggests that this is the part of the word that you're looking at or part of the feature of the word that you're looking at is is is present will be excitatory connections in part of the word which is not for example this is something which is not present in a and so these connections from here will be inhibitory similarly from these letters the on on this word so if a is not present a is the most apt answer here example n ble is not a word and so a is the most possible answer T ble is another no word and so T is not a fit here G ble is no word and so G is not fit here and SB le is another no word so basically the only word which is available in this case is a and so a is the only **perception** which happens and so this happens to expectation so at this level the first if the **perception** happens from this level it is generally a **top-down process** and if the **perception** happens from this level of it is generally the **bottom-up process** and so what mainly **top-down process** suggests is that it is not only the stimulus features of the words that helps us in **perception** the word **perception** progress is not only through basic features of the word it also progresses through something called the level of expectation that is why when we actually lead in terms of reading when we are eating something even if some word or meaning of word is not clear to us based on the context on where the word is we generate the meaning and we are able to understand the meaning of the sentence and so this is this is an expectancy effect where what happens is the context in which the word is presented where it appears gives us a meaning of that word or some Greek meaning of this word so this is the **top-down process** of **perception** where what we actually looked at those features of the word are those parts of the word where where **perception** happens through people's expectation and context effect in what context award is presented what is the expectation of people that helps us in the overall **perception** another interesting theory which was given or which was proposed by someone called JJ Gibson is called **direct perception** now it is an interesting theory of **perception** which says that the user doesn't have to do anything for **perception** remember most of her idea of **perception** starts with the fact that the user or the perceiver gets information from the **distal stimulus** does something on the **proximal stimulus** generates a percept and dependent on the motivations or dependent on the expectations and context this percept is assigned a meaning which means that in this case up till now the theory suggests that the user does something on to the incoming stimulus and based on that a meaning is generated contrary to this view a theory of **perception** was presented by JJ Gibson which is called the theory of **direct perception** what does the theory say this theory basically goes ahead and says that the world has a lot of information and we do not need to do anything to that information for **perception** to progress so in in a way what Gigi Gibson goes ahead this says is that **perception** is a direct process and the environment that we are in the environment that we are perceiving has enough information onto it enough information onto the onto the **proximal stimulus** which is falling on the retina so the **proximal stimulus** which is part of the **distal stimulus** which is falling on the retina has so much information and the user has to to actually go ahead and perceive he doesn't have to make a per second doesn't has to use context effect it does it has to use some kind of a building a process using or using Mars method Mars theory of **perception** but the environment on its own the user can sit back and relax the environment provides enough information for **perception** to progress so the **top-down process** and **bottom-up process** of **perception** believes that the perceiver does something to the **proximal stimulus** for **perception** to proceed and so it is called the constructivist approach it describes people as adding to and distorting the information in the **proximal stimulus** to obtain a person so what the user does is adds or subtracts information from the **proximal stimulus** to generate a percept which later on through the process of what I'm talkin combine gives up the meaning Gibson adopted an opposite view to this to the connectivist approach and gave up his idea of something called **direct perception** his Gibson believes that the perceiver as I said has nothing to do or very little to doing **perception** according to this view the light hitting on the retina contains highly organized information

and the information that is falling onto the eye onto the retina is is is so great or is has so much information or has so much bits of information that the observer is not required to do anything not to prove whether Gibson was right whether observer doesn't require to do anything or another observer doesn't require any kind of interpretation or any kind of work on to the **proximal stimulus** to generate the percept an experiment was done to to prove this and this experiment was done by someone called Johnson in 1973 so Johnson 1973 he did an experiment he did an experiment with light bulbs what was the experiment so what he did what Johnson really did was that he attached lightbulbs to actors and so these actors actually wore the light bulb so these are two actors and they have light was attached to them and so they are in two different configuration can assume that these are light balls and so when an experiment like this was done these actors were actually taken into a dark room and in this dark room these actors were asked to do several pose poses and take several process do several acts the observers who are observing this particular people or these actors who were wearing light bulb shirts and from just from the way they acted just from the movement of the light bulbs they were able to perceive or they were able to generate meaning or tell exactly what was going on what was the act in doing this was not the only amazing fact about it what actors could also tell or what observers could also tell from the actors who were acting was whether this actors were males and females and kind of other specific stimulus it basically means that with very least information available from the environment with not building any information actors by just looking at observers by just looking at the actors acting or displaying certain forms or displaying certain acts they were able not only detect who the actors were or what are the act they were doing but also the gender of the actor the kind of act they were doing the kind of movement that they were doing and several other features of it and so this particular experiment really proved that we really do not need a lot of **information processing** for **perception** to takes place but there isn't Cavett to the Gipson entity it's not always correct the explanation which has been provided is that the bottom-up and top-down processes actually allow cognitive structures to see what the **perception** is whereas the Gibson in view actually tells you to act in certain ways to adapt that information and act in certain ways so this is the difference between the Gibson in view and the view of the bottom-up and top-down processes so basically what Gibson design or Gibson Theory comes up and says that he developed the Gibson developed his theory of **direct perception** when he was training World War two pilots and what he realized or he gave the **concept** of **optic flow** if you look at the stimulus here what he did was and I mean a number of experiments with pilots who were taking training or who were going ahead and landing planes what Gibson explained that the pilot when he sees information when he sees out of his windshield he gets enough information from the environment which is approaching him the texture the structure the kind of movement which is there the ground movement the ground color all these information was enough for to tell the pilot of how to land the plane and so what Gibson proposed is that the idea of something called optic flows or visual arrays and as you can see here this is a pilot who's landing and when he's landing just before landing he sees all this information about texture structure kind of relative speed for example things which are closer to the pilot will move at higher speeds than the then things which are far away and these kind of informations which the pilot gets is enough for him to land the plane he should not be or he need not be concentrating on information which is really providing by his display system and so that is where he gave this idea of **optic flow** there so the arrow in this particular thing represent perceived movement the parent motion of clouds ground a sec traffic respect to the pilot now there is a texture of the motion namely near as things will appear faster and and directions in which object seems to move upward opposite depends on the angle of plane and movement and so this particular example was one of the best example which gives some use to propose the idea that there is something called **direct perception** the observer doesn't need too much information the observer doesn't need to work on the information which is being which is being presented to him the environment presents a lot of information to him and that information the object the observer only has to understand that information not even make meaning out of it and just follow that

information and that is what Gibson's idea of **perception direct perception** was was there so what Gibson says is that the information that is available from the environment for the observer exists in an animal environment ecosystem so basically the environment process provides some information to the animal and the animal provides certain kind of cognitive structures or selects which information from the environment to take to take and based on these two kinds of interactions the **perception** progresses so the animal decides the animal only has to decide what information to take from the environment and this decision process is the core of **perception** whereas the environment on its own can provide a lot of information and this decision process is the main key to act actually any kind of **perception** according to Gibson Yin views now for Gibson the central question of **perception** is not how we make an interpreter stimulus array but how we see and navigate among real things so it's what Gibson believes is it's all about functions is not important to study in terms of **perception** of how and what we interpret from stimulus what a particular stimulus really means that's not important what really important is is known that an object is there how do we navigate through it so navigation in the environment is more important than having any percept then realizing or then understanding what the percept really means so the basic idea of Gibson in view is to understand what the environment how to navigate the environment rather than knowing what the environment has now important idea of Gibson in theory is the information available in an organism exists not merely in the environment but in the nmi environment in an animal ecosystem as I was just explaining to you what really happens is that the job of the animal here the job of the person who is perceiving here is to decide what to take in from the percept which is available what he has to do is that the environment gives enough of information to the person the person only has to decide what information to take and what information not to take and **concept** which is related to Gibson in theory or the gift running idea of **perception** is the **concept** of affordance now what really it means is that organisms directly perceived shape or whole object and then objects gives have something called an affordance now what are affordance really means is that an object which you are perceiving it directs you to do certain act on it and that is called an affordance for example think of a chair now when you look at a chair or when you think of a chair the idea that the chair gives to you the affordance that the chair gives to you is sitting similarly if you look at a button a light button now a light button gives you the affordance to be player to be pressed down or looking at if you have printers at home and if you think about the printers power button the red light the green light which is on it is off when it is off when you press it it gives you the affordance to be press down similarly doors handle or the handle of glass doors affordances or tell you what to do for example if I have a handle like this it gives you the affordance to put your palm on to it and push whereas if I have a handle like this it gives the affordance to to basically hold it and pull it towards yourself because this requires you to wrapping your hand here whereas this requires you to push it with this hand now I think this is a little bit confuse or people people confuse with this kind of affordances or affordances is what an object requires or desires to be done so basically here the definition is given as affordances are the acts or behaviors permitted by an object place or even for example donuts to tell you two there is a round or nap so it it the doorknob tells you to grab it and to turn it down similarly the affordance of a chair for sitting or any other affordances of a light ball to push through pushing or light switches to be pushing it down or affordances of door handles to be either pulled or pushed and these are what affordances are and so what Gibson says is that objects in itself affordances objects have affordances and these affordances guide people's behavior or what should be done and what should not be done now the obvious question is on one hand we have gifts only in view and on the other hand we have the bottom-up and top-down the classic culpers approaches to **perception** so is there a merger between them how can this theory coexist what is the way in which these theories can co-exist and so a model was proposed by nicer in the year 1976 which takes care of both the theories which in which which adds up both the theories to develop the final percept and this history is called the perceptual cycle now what this theory or what this approach goes ahead and tells is that this perceptual cycle says there are the integration of the get run in view and the

view of the classical approach to **perception** combines to this way for example the information of the world is here and this is modified by cognitive structures through **perception** that forwards or directs exploration in certain way and this will further suggest what should be taken from the environment or sample from the environment so what does this theory really say it says that the information of the world whatever information perceptual information is available into this world that is modified or that is being perceived or that gets perceived perceived through a classical approach of **perception** using different cognitive structures so the visual cognitive structures of **perception** rigid structures of **attention** these structures or higher-order structures leads to **perception** either using the bottom of the bottom-up or top-down approach it leads to **perception** it leads to the generation of the percept when the percept is generated this percept leads us to explore the environment to explore to do certain acts when we do certain acts when we do a particular act based on the **perception** we present a behavior we do a behavior they commit a behavior and when I come behavior is there we explore this exploration tells us what is possible and what is not possible once a behavior is done it directs which type of behavior or which type of actions can be done through the percept so if I perceive something what that particular thing when when behaving with it what particular thing information what behavior is allowed and what behavior is now and once we know what behavior is a lot in what we were is not we feel this again to the environment and from there it again gets modified in this way so basically information from the world is actually gets modified and generates a percept which is later on this percept is tested in the environment in terms of whether what behaviors are possible on this percept what behaviors can disperse percept generates and which of this behavior allowable or not those behavior which is allowable gets added up into the information in the world of a particular thing and whatever behavior is not allowed is deleted here or is not added and that information is also added for example think of a tea cup know the kind of a tea cup if you have there are certain behaviors which is allowable when we see it we see a tea cup and so when we are looking at a tea cup then it there is a certain orientation with it and there are certain feature that it allows example a lousy 50 to drink tea but if we try to do something else with it by moving the Gion in certain way that is not allowed in certain behaviors for example drinking is a behavior which is allowed and other behaviors are using this as something else or using this for looking at things or doing something else with it the behaviors are not allowed and that is fed in environment and based on that our **perception** is improve or enrich so the next time we deal with this cup or this kind of a percept we know what is to be done and what is to be not done and the **perception** then becomes automatic the whole of this perceptual processes is done so as to make this whole idea of **perception** automatic so that something something what what is allowable is accepted and what is not allowable is not accepted and it goes into **memory** so that a next time a new instance of this behavior a new instance of this percept is generated it doesn't take that much time for **perception** so basically **perception** in this section of **perception** we looked at we started with the classical approach where we looked at some classical theories the **gestalt** ideas of how what perceptual organization is and how this perceptual organization really works what is the figure background **concept** and how the **figure and background** decides how what should what is to be Percy what information is to be generated and what is information is to be left behind no this idea was added on to some models or added on to this there's some models of **perception** we started up in the in the next lecture on some models of top-down processing we're sorry' bottom-up processing where we looked at models of **template matching prototype** matching and so these models actually showed how **perception** progress through a data different data-driven approach in which we look at basic stimulus features is take this feature combine them together and once we have combined all these features together how does the percept they will have out of it in this particular lecture we looked at another approach of **perception** in which the data-driven approach is supported by something called the theory divin approach so **perception** is not only taking in data from the environment combining together and generating a percept this **perception** process is also held or is also influenced by something called the expectation in and something called the the context in

which a particular object is present and so context and expectation adds on to the data-driven approach to give you the final **perception** for example if you look at something which is new which you have not looked before a part of information which is coming to you you will look at its shading you will look at the kind of brightness the figure and all those basic informations plus you will also look at your **memory** to look at to understand what is this particular thing and based on that you combine these two information together to generate this idea of **perception** now in addition to that there is an idea or there is a whole theory of **direct perception** which says that the object by itself by by being where it is in the environment offers you some information as to what it is and where is the place at which is present the kind of information that is related when you are looking at it it should also in addition provides you what kind of things you should do with it and you should not do with it take an example let's say we have UFO which has landed somewhere and crash-landed somewhere so when you see further obviously we have some idea of what an UFO is so next time when you see something which is circular and flying which has landed somewhere you both you're the turbulent process and **bottom-up process** will work together so when you go to the crash site you see something which is circular something which has which has stands on it from where there is some kind of air movement coming in where the fans would be on an on an UFO where it would take off and things like that so you combine these informations together to to get the **concept** or what our UFO is plus in terms of the fact that in numerous movies you have seen that a disc life shape which has fans and which has doors and which is of a certain way should look like a UFO should be UFO that information combined with you actually presently seeing that particular UFO in a particular environment combines together to give the **concept** that this is a UFO what **direct perception** would come in and say that being where it is being where the UFO is in in itself allows you to do certain acts for example from your experiences you you understand that there is a door and so in the UFO where from where people from other environments will come so subtle behavior you it will actually need you doing that behavior and certain behavior it will it will not allow you to do for example if there is no door in in in and you know or there is a crack in in in the UFO it will give you an affordance to basically go ahead and push it and realize the door because most of the movies don't tell you where is the door in in the so this affordance will tell you if the crack is there from where the light is coming on from the UFO you will have this affordance that this is the door and that itself should be enough for you pushing in and your expectation of what should be inside will guide you of what you are going to see when you go inside that place when you go inside that UFO maybe Green man's coming in and then and maybe taking you for a probe aerial probe or whatever you wish to come up with but this is how **perception** progresses so the whole idea of **perception** is what I have presented in this lecture in the next lecture we will go ahead and look at what is **attention** and look at the theories of **attention** thank you

Lecture 7: Basic Attention Processes

■Kind: captions hello in this class we'll be looking at another cooperative variable another gullible factor which is **attention** now **attention** is a very interesting cognitive factor because it signifies the mental concentration the mental effort somebody puts it to and so **attention** is important why because **attention** decides whether **perception** takes place or not so even before something is perceived even before something is registered by the brain and analyzed **attention** decides what message is to be analyzed what stimuli is to be analyzed and what stimuli is not to be analyzed now why is **attention** really necessary think of a scenario where you could hear a lot of things you could hear almost all the stimuli which are impinging on to you at any government movement it would be very difficult to process all the stimuli at any if every stimuli is being imprinting on you you are able to hear everything and so focusing or concentration will be difficult and that is why the cognitive system or our human brain has the system of **attention** so **attention** is just like a sieve or a filter which separates those information which needs to be processed in those information which need not be process and so basically then the question comes in that how is **attention** directed so in this particular lecture we will look into **attention** we look into what are the factors which control **attention** how is intention manipulated and whether this **attention** that we are talking about whether this could be improved so does **attention** span improve we'll also look at some of the theories of **attention** and talk about something called automatic T of **attention** in addition to it we'll also look at something called the psychological reflector period which is basically a carrot which is basically out process of **attention** so let us then begin our section or detention think of a scenario a car driving scenario look at the picture on the screen now when an expert driver car drives a car he doesn't need to pay **attention** to a lot of variable to a lot of things he is an expert and so driving is kind of a by-product to him and so he doesn't not need to focus on all the systems which are there in the car like the gear the steering and so many other things so what he could do an expert could do is not only drive the car but also start a conversation follow a conversation fiddle with some gadgets take up a phone call and do several other things but reverse the scenario or look at look at the scenario in the picture in the picture you can see a no wise driving a car remember the first time you started driving a car or a vehicle now at that point of time you didn't know what the controls were and you didn't knew so many other things about the car or about the vehicle that you were drawing or any new thing that you learn and so you have to put your **attention** or your mental consideration into it because this meant will consider concentration decides how you should be doing a particular job into incase of the car the no vice user pays **attention** to whatever is happening around him learns about the control learns about how to move the vehicle and later on when he practices for some time when he has enough practice for some time he can then use something called **divided attention** where he doesn't need to focus all his mental effort onto the controls in car and with the driving can do other jobs as well so basically **attention** is a mental effort kind of a cognitive resource which makes you focus on two things remember in earlier classes in smaller classes our teachers is to say pay **attention** and the Kabat here was that no matter how hard we paid **attention** it used to jump one object to another even if I ask you right now to pay **attention** to something or some object what would happen is that for a few seconds you would be able to focus your **attention** onto that thing and it jumps on which basically means that **attention** is very fiddly it keeps on jumping from one thing to another and that is why we have something called an **attention** span which basically defines the amount of time for which a particular stimulus can be attended to so then what is basically **attention attention** basically requires the study of cognitive resources and the limitations of the cognitive system so basically those cognitive psychologists who are studying **attention** actually look at what are the cognitive resources how are the cognitive resources channelize into making a cognitive **decision making** a cognitive act like **perception memory** and what are the limitations of this this **attention** this filter that we are talking about so **attention** is some time also synonymously are related or synonymously considered equivalent to mental

concentration now **attention** is something which is very prominent without **attention** it would be really difficult and so as I said as you do more and more practice what you could do is you could take the **attention** that you have the amount of **attention** that is your cognitive resources that is available to you and put it into other jobs which means that people can simultaneously then do what any other job so one basic fact to know here is that with practice **attention** span or **attention** can be diverted you could use your **attention** for doing multiple jobs that is the question which I have here and that says that thus people's concentration change with practice the question is does people's **attention** actually go ahead and change with practice with doing are we doing the job over and over again all learning the job over and over again now one of the first person to talk about **attention** was **William James** when he said that **attention** cannot be focused on one on more than one stimuli he he predicted that **attention** can be only paid on our **attention** can be studied for only one stimuli but if we are die working or if we are playing around with many stimuli if we are focusing ourselves or we if we are able to do a number of jobs at the same time it is not **attention** that we are studying it is happiness that we are studying and so this first definitions are these first Clues basically give us something about or tell us something about Artemis City which is another outgrowth of **attention** so as you become more practiced as you have more practice you do not need a lot of **attention** on to jobs and so most of the jobs then become automatic think about any morning any unusual morning which is there and so when you get up you go to doing your brushing and there are several other things but if I ask you this question do you remember doing any of these or do you pay **attention** to any of these jobs very seriously and your answer would be no the reason being that these are so routine jobs that there is a process of doing it and so what you tend to do when after you get up is pay less and less **attention** to it and the process itself becomes automatic till the point of time when you either miss a toothpaste or there is a one element from the whole routine that is not existing and if that happens your **attention** is focused back on to the situation and solving a problem with the situation so the study of **attention** is also important because at any given point of time we have limited cognitive resources the brain has limited cognitive resources and these cognitive resources have to be diverted is to be used in a number of jobs that we have to do and so we have to use it wisely and that is my **attention** came comes into play where it allocates very tells in allocating these mental concentration into getting more eminent or more important jobs done and making those jobs which are routine as more automatic and more sequential in nature now the question is how do we study **attention** a problem is how do we study **attention** and a more relevant question here is that can **attention** be paid to multiple stimuli let's deal with the first question how do we study **attention** the moment you say somebody the moment you ask somebody not to pay **attention** on to something his **attention** goes back into that particular job and so it's really difficult to study **attention** because if you ask someone the only way I see of studying **attention** is taking two groups one group which is paying **attention** and the other group which is not paying **attention** but as soon as you release this instruction as an experimental psychologist I would say that as soon as you give this instruction to people not to pay **attention** there goes it he starts paying **attention** back to whatever job you are not asked him to pay **attention** and that is because humans means a curious they will do what they are not asked to do so how do we study **attention** the only way to study is to have two groups in which one group doesn't pay **attention** and so a noble design was used to study **attention** we'll talk about this design in a moment but before that let's look at this question of can we pay **attention** to two things at the same time there are several answers to it and several theories will discuss into it think of a class a class which is filled with a number of students who was actually listening to a professor who's professing at any point of time there are so many stimuli which is being imprisoned on your sense organs you have a number of stimuli interactions a number of stimuli which is trying to capture its **attention** but what you tend to do is focus yourself on to the lecture and this particular thing or this particular job of focusing your **attention** on to one particular thing is what is called **selective attention** so **selective attention** basically is **selective attention** is basically using the filter to focus on one job and to eliminate

other job to eliminate putting your focus on two other jobs so as soon as the teacher stops teaching you will soon start hearing your class wrestling against yourself your skin you can hear the noise that the person who's sitting right next to you is making you can hear the fan the noise of the fan which was there well amazingly when you were actually looking at the professor when he is teaching you actually didn't pay any **attention** to all the other songs which were there and this particular thing this interesting thing of focusing our **attention** on to one particular job or one particular event is called **selective attention** and so what we'll be looking at now is something called **selective attention** but even before looking at **selective attention** let the question is how do we study **attention** and so a very ingenious design was used to do that and that design uses something called the the **dichotic listening** task so as I said **selective attention** is the term which refers to the fact that we usually focus out **attention** on too few tasks and events rather than many of this particular task and help assure one of the famous figures in **attention** research states that **attention** is basically equivalent to any given movements people's awareness and it encompasses it comprises of only a tiny proportion of the stimuli which he is receiving from the environment now let's do a very quick thing a very quick study to find out how **selective attention** really works and so on the display in front of you you see two rows of animals now what he earlier has happened is most of the figures that you see are actually a combination of a dog and a cat and there are some pure figures when I say pure figures you what I really mean is a pure dog and a pure cat take some time and do tell me or you do do this exercise of finding out which is the cat pure cat and which is the pure dog now once you are given this job you will soon realize the only pure dog is this and the only pure cat is this now how did you come up with this answer for finding out this answer what you really needed to do is to scan each of these image one by one and for that what you needed to do is to focus all your **attention** or mental concentration focus yourself into this image and basically compare the **prototype** as we saw in the perceptions section **prototype** is basically the most abstract abstraction that you have the most clear abstraction that you have of any category use that **prototype** to compare each of this image with a pure dog image in a pure cat image and once you do that you'll realize that the fifth image from the left on the top row is the dog of your dog and similarly the third image from the left bottom on on the bottom row is a pure cat and what you did right now when you did this job of finding the pure cat in pure dog is something called **selective attention** so the idea here was how do we go ahead and study **attention** and as I mentioned a few minutes before the oil the best way to do that is using something called **dichotic listening** task now what is this task art all about so this dichotic task listening task involves a listener a person who listens through a headphone another on the cavity or the hidden part here is that both the heads or both the channels or both the ears that this person has will be receiving two different inputs so one year will be seeing a different input and the other year will be receiving a different input when I mean input what it really means is that they will be receiving two different messages so it's a situation like this where you can see the one-year is receiving stimulus is like God googy kind of thing and the other here is using stimulus like Barbie something like that so basically for the people who take part in the **dichotic listening** task are given a earphone and the two years then receive different messages now there are some obviously there are some assumptions here or there are some clues there are some points to be noted here instructions that is followed the first thing is that the people have to shadow one here now what does **shadowing** actually really mean the **shadowing** means to repeat aloud whatever you're hearing in a particular year that the experimenter is asking you to do so in the **dichotic listening** task what you really have to do is both the years that that you have received two different inputs from two different microphones turn to two different headphones and all you have to do is to shadow one ear now **shadowing** is repeating aloud whatever is here on one other year so if it is right here that you're **shadowing** whatever is coming to your right ear you're going to repeat that aloud now the rate of presentation of stimuli in in a **dichotic listening** task is very fast it is nearly 150 words per minute at the end of the task people are asked to reveal what information that they have gathered from the message

from both the years or from a single ear and so the deal here is that this experiment or this **dichotic listening** task was designed by someone called cherry in the year 1953 in a classic study cherry define this **dichotic listening** task now the question was there are two different years on one year you are doing **shadowing** the other year you are not doing during the question is whether you are able to hear words from the year that you are not **shadowing** which means that whether you are able to hear from the ear that you are not paying **attention** to the answer to this cherry god was that yes most people were able to hear some information some messages or some kind of stimulus property from the other year and this information was basically two words are two words stimulus properties like what is the what is the message type whether it is noise or whether it is words which has been repeated or who is speaking whether it is a female or a male who's speaking whether it is the message is in a correct order which is a forward order or whether the messages are in reverse order this kind of information basic stimulus information you can gather from the non in intended or non attended year but from the right hear you you tend to hear everything so basically this question here or this property here tells us that people are able to hear some kind of information from the other year so basically attending to one stimulus doesn't mean that the information from the other year will not come in or will not be attended to so basic information or basic facts from the other year is also attended to like stimulus properties related to basic message properties that is what is attended to and so when a chairing later this this particular thing he he said that the unintended year or he found out that his subjects from the unintended year although could not analyze the message the meaning of the message from the unintended year but they could tell the voice of the person who was repeating it the kind of message that it was whether it was noise white noise which has been presented or it was words and letters that was being repeated and for another backward direction now what they couldn't tell of the message which was being played and the nun intended year was whether what was the **language** in interest it was being played and what was the content of the message and so this was the first demonstration or this is the kind of a first demonstration a very ingenious design to study **attention** now as I related before studying **attention** is difficult because as you say don't pay **attention** to this people start paying **attention** and so this design has been used in a lot of **cognitive psychology** tasks or a lot of experimental tasks which measure **attention** so the question here is what kind of information do we get from the intent on intended or the unattended here and and whether some information and the meaning level can also bypass the attentional filter the question to be looked at is whether **attention** is it is an all-or-none kind of a filter which means that when you're not when you are not focusing yourself or when you are not using attentional filters whether the one the messages which are not being attended to whether they also pass and at what level what kind of level do they pass no cherries experiment very clearly says that some information or some idea about the non intended message you can hear but then to what level is what several other theories will argue and so there are theories of **attention** which has been put forward which go ahead and actually talk about what is being attended to and what are the matters being attended to and what part of the non attended message or what is it from the non-degenerate message that people can actually hear remember in this case in this whole scenario we are talking about focusing our **attention** on to one year and we are attending and **shadowing** it so when you're **shadowing** it it becomes really difficult job because **shadowing** requires you to you to repeat aloud whatever is being said and so puts a lot of mental effort on to the **working memory** and so it will be very difficult to actually go ahead and process anything from their nan in a tendon here so the question whether things from the nan attended hear stimuli from the nan engineer also moves back or are also heard another paradigm that we'll go into further is whether **attention** is fixed or does it fluctuate and so one of the reason that has been given in one of the theories is that **attention** is never 100% and so it keeps on fluctuated fluctuating from one ear to the other year and that may be the reason why we are able to hear from the other year but we'll look at these debates one by one when we move into it so let's start with the first theory of **attention** which is called the **filter theory** of **attention** so let's start with the **filter theory** so

what does this theory actually say it's a very simple theory which was proposed by someone called Broadbent in the year 1958 and he proposed that there are limits the cognitive limits of how much information a person can attend to now if the information available at at any given time exceeds the capacity of the mental the mental limit what will happen is a filter is designed which will block most information and let let things pass or let those stimuli pass on to which we are paying our **attention** and so it's a in a simple demonstration I had tried to look at the or tell you about this theory or the **filter theory** now another question is on what properties does the filter work what is the variable switch control the filter so let's first look at what happens here so there are a number of messages 1 2 3 4 5 6 7 these are seven messages which are here and which are impinging on to our cognitive system and this is any cognitive barrier which is there this is the barrier the psychological and physical barrier which I was talking about and as I said in in previous lectures that the most important thing in psychology is to define how physical stimulus has entered the psychological domain so this is the barrier which is there it could be here it could be I it could be anything and so the number of messages which I am painting on to it but the amount of mental concentration the limitation the limit is very less the mental concentration available is very less and so what happens a filter a quilter like this is available which basically takes in one message and lets it pass and filters all this message away now the filter well and so the answer to this has been given in terms of physical properties of the stimulus is what so physical properties of the stimuli guide the filter so the filter the use of the filter or the variables which control filter is the physical property for example the basic acoustics in the DLT that decides whether you attend to it or not and these basic caustics could be location pitch loudness and so on and so forth so then the **filter theory** goes ahead and says then that any message which you're not paying **attention** to will be blocked only those message which have not been put to a limitation - will only get passed on and this is the message which you require and then the filter which which filters out these messages is based on something called the physical property of the stimuli which is impinging on to it now the **filter theory** explains while there is so little of the meaning of the unintended message in Shelley's experiment happened the meaning from the unintended message was simply not processed and that is why we only get the basic physical features of the unintended you remember says experiment that we talked a while ago and as I said from the unintended year things like the basic property of the message whether it is in forward or backward direction or whether it is a male or a female which is which is speaking this message these way process but nothing beyond that and so that is what the **filter theory** goes ahead and says it accommodates that and it says that these are the only things which can be processed or which is processed through this particular theory because other pictures are not being attained now the question comes that can be not here or can we not pay **attention** to to message at the same time can we pay **attention** to to message at the same time is it that if we are paying paying focusing on one message is it that the other message gets passed on our only physical basic physical features of the stimulus a register broadband propose that two messages that contain very little information or that information or if the information is presented very slowly in this case we can process both the messages so what broadband goes ahead and proposes that there are situations in which we can focus our **attention** or even if you are focusing **attention** on one message we can process the other message and that happens in only those cases where either both the years contain very basic minimal information the messages contain very basic in minimal information or the presentation rate is very slow now in these two cases and in these two cases alone both the years can be tended to or what the messages can be actually read or perceived but an interesting thing was pointed out by someone called Marie in 1959 and he stated something called the **cocktail party** phenomenon now what is the **cocktail party** phenomenon no in the **cocktail party** phenomena look at the figure on your right as you see there's so many people sitting here and it's a large party which is happening everybody is enjoying this large sounds loud music everywhere and so even in this cases and you are focusing yourself on to what your friend next to you is talking so this person or these people are actually listening to listening to somebody

who is talking among their growth now in the **cocktail party** phenomena what Mireille suggests is that if your name is said or if somebody's name is uttered this message even if we are not attending to will pass this filter will pass the filter which filter he talks about and that raises a problem now what he says what more he goes ahead and says in the **cocktail party** phenomena is that **shadowing** performance is disrupted when one's own name is embedded in either the unattended or the attended year now this happens as important materials can penetrate the filter set up by setter to block unintended messages now bachelors experiments say that only all although only thirty percent of people could actually when could actually hear their name when they were not informed before but then this really happened which basically means that there are certain stimuli out there where which can be processed when your name is being processed from the unintended ear which means that it is being processed not at the level of the stimuli but the level of meaning because you know your name has a meaning and when your name is being processed when somebody is speaking your name it is a meaning so at the neck even at the level of meaning your name or words like fire or words which are more important to your adaptation are process and so what the cocktail for a party phenomena goes ahead and says that whatever broadband suggested in his theory **filter theory** which says that only those messages we are tending to gets passed and processed from meaning whereas other messages if ever process a process for basic properties is actually wrong and so he demonstrated that through this phenomena where he where he was very shown that people's own name or some important words which are adaptive in nature can be processed even from the unintended here so even if in a big party where you are enjoying yourself listening very hard to your friends and somebody calls your name it is found that most people turn around and look at who is calling their name which means that they are able to attend within that huge voice when that huge noise into their name and the **reasoning** that was provided for this particular feature of the **cocktail party** phenomena was that **attention** fluctuates since **attention** fluctuates since since the attended year never takes up 100% of **attention** that tension keeps on fluctuation in between there an Indian and an intended year and that could be the reason why the **cocktail party** phenomena happens so **shadowing** in Westham **shadowing** doesn't take 100% of **attention** since **attention** fluctuates between the attend agent on a 10-day dr and that could be the reason why you are able to hear at the message level the information from the unattended year now a nitrous pen actually goes ahead and hits back at or demonstrates that whatever morris says in terms of **attention** fluctuation or not or the fact that hundred percent **attention** is not being paid on to that in nature ear is wrong and so anna transmen designed and very ingenious experiment to test whether the attended channel the channel the year that is being used for **shadowing** that has been used for reading aloud the words at which is being put into that year whether 100 percent **attention** is being is devoted to this year and so in her experiment what she did was she used the same **dichotic listening** task in which two years were there or two the two different years got two different kind of messages and what really happened is that the message one year was to be shadowed and the other year was not to be shadowed not in the only interesting thing or the only new thing in this experiment was that in the middle of in the middle of the the whole listening task the messages very diverse so the message on the right here went to the left here and the message from the left here right i went to the right here what really happened the results of the experiment suggest that people who were paying **attention** or **shadowing** the left ear and as soon as the message from the left ear shifted to the right here they continued hearing the message from the left ear and repeated a couple of words also of the message from when even even after the message got reversed into the right here which means that they were clearly focusing hundred-person attendant so at the point when the switch happened people didn't know the switch people didn't follow the switch and they moved or they were able to repeat a couple of words from the unintended year or the messages from the unintended year after the switch it basically means that people were paying 100% **attention** into the attended year and later on as time progressed or after the couple of seconds people started repeating after the switch messages from the unintended year which

basically proves that people are able to to pay 100% **attention** and that the the the **reasoning** that **attention** fluctuates is is not so correct a similar experiment was done by someone called wooden Conan and so they wanted to test similar experimentation by wooden Conan and Conway Conan and bunting in 2001 resulted in showing that messages from the untreated year was also processed that's challenging the **filter theory** and so what wouldn't Conan's experiment was that wooden Cornyn use the same **dichotic listening** task for their experimentation now they have they gave on both the years two different film clips to be heard on one year you have the grapes of what dialogue from The Grapes of what which was being processed and on the other year from 2000 Space Odyssey dialogues were from that particular movie was being repeated and so people they were to group of people who were able or who were given this kind of a task to be done and a control group was also there no people were asked to shadow the grapes of our dialogue for the grapes of what movie and the words represented 175 words per minute so basically the setup is like this the the two years here two different movie dialogues the left here's something from the The Grapes of what movie and the right here here's of dialogues from 2000 Space Odyssey to group of people do that there is a control group also the witch which is there and the word presentation is 175 words Munro words per minute and people are asked to shadow or repeat aloud the message which comes on the left ear or the year which is which is here in the grapes of what now five minutes into the experiment five minutes after the experiment started the unintended year the year which was here in 2001 Space Odyssey dialogue there there was a reversal so the message goes reverse or black test a back messaging was done so that words were repeated from front to back a bachelor doing or a backward message was played into it and that happened only for 30 seconds and later on after that one group heard the or followed on to the original experiment for two and a half minute one group followed the experiment for one and a half minute and the control group actually never saw or never heard the back of messaging into the unintended year now it was the result what would be the result of the experiment and so then experiment like this the result is that people who notice the backward problem shadow the people who actually go ahead and notice that there is a backward **shadowing** on one of the year they they they create or they do more number of errors and this error exit actually gets generated after the backward messaging stops whereas people who actually do not people who actually do not hear this backward messaging they never notice that the backward messaging was done also the third group who didn't hear any kind of backward message they never heard any any kind of thing and so there was that was what was comparable and so the idea from this is that this at this idea of **attention** shifting or **attention** fluctuating between between two years is not so correct and that was what was relate so that was the main idea of wouldn't condense theory so they went ahead and this particular experiment suggests supported the **filter theory** that when only those messages you are tending to actually go ahead and a process but those messages you are not attending to are not process that is the bottom line of the wooden Condon experiment now these are very interesting experiments or these are a very interesting theory of **attention** but later on another attentional theory or theory of **attention** of how the **attention** works or how **attention** goes **attention** filter goes ahead and looks at cognitive resources or the location of cognitive resources two different factors are different cognitive tasks and one of the limitation of this was proposed by someone called Anna Treisman so ni Treisman what she did was the night rise when 1960 proposed a modified theory of **attention** which is called the **attenuation theory** it's a very simple theory to look at now the theory proposes that messages from the unintended year is not blocked so what happens then messages from the unintended year according to Anna tradesmen is actually toned down is brought down or volume down is attenuated and so what she says is the volume of the unintended message that the tone of the unintended message is attenuated or a brought down and that is the reason why this kind of an attentional filter the early works so incoming messages what an address meant but then extended arteries to that any incoming message any incoming information on to the sense organ before the the attentional filter works on to gets cross through attach our three stage attentional filter and what are these attentional

filters so incoming messages what she says are subjected to three kinds of analysis there is a physical analysis there is an analysis at the word level and there is an analysis at the meaning level we quickly look into that in a minute now some messaging units are processed quite easily because they have more adaptation or more adaptive value right and sometimes messages are also processed from the unintended year so some messages from the unintended year a process because they have adaptive value like your name your your words like fire danger that kind of words because they have adaptive value there or they help you negate a threat some of the messages are also processed from the unintended year because they are in context for example if an end ended here you are being played this message - is chased by the dog and from the attended year if you have words like cat into it you will quickly go ahead and hear the word cat it is something called the context effect so sometimes context in which the message is being played the context of the message also helps us in attending or in processing messages from the unintended year so what tradesmen and a tradesmen says that any message which has been processed which is new process to the attentional filter goes through three stage of filtering three extensive stage of filtering so I have a drawing here to basically go ahead and show you that how does it happen so this is the incoming message any incoming message and the first level of analysis are the physical properties of the sound of in terms of sound BIC since we are looking at the **dichotic listening** task for I but sound here what happens is the first level of analysis F is at the level of physical property and so here physical properties of the sound or the message which is being relayed ordinarily to you in terms of its pitch and loudness is what is being focused on so this is where the focus is if messages pass through this stage it goes through a second level of analysis where a linguistic or parsing as syllables and words is done so if the messages are not filtered at level one of **attention** which is the **attention attention** filter dependent or based on the physical property a second level is our second level analysis the second level filtering is initiated and in this level what really happens is that the filter is dependent on so the linguistic parsing in terms of words and syllables and so if in terms of words and syllables the filter operates here in terms of the linguistic property of the world so what kind of linguistic property a word has or whether it is in which context how many words are there the differentiation in terms of da ba kind of one thing it is it happens at this level even if at this level the filtering cannot be done if there is a problem in filtering or if the filter doesn't work in this level a third level of filtering is or level three analysis is is is run on to a level three filter is run on to the incoming messages which is the semantic level and here if the messages are not blocked or cannot be blocked at level one or level two a semantic level filter is applied where the messages are blocked in terms of the meaning and so here the filtering happens in terms of what the message has or what is the meaning of the message on the basis of that basically go ahead and make the the filtration or the blocking of messages are done in terms of the filtering and what an a tradesman goes ahead and says is that the **cocktail party** phenomena happens because your name or certain other words which are adaptive value have lower thresholds when they have lower thresholds they are get processed from level one level two and level three and add the level three even at the levels here the semantic level it is the meaning of your name and so it since it has very less threshold it gets processed now what Treisman na Treisman goes in and says is that people will use these levels of analysis based on at what level they can separate the attended and the unattended year so let's say the messages of similar physical property is coming at both the years so filtering will not be done here and so a second level will be applied where people will be looked at in both the messages will be looked at in terms of the linguistic properties in terms of the words in terms of the phases that is that suppose both the messages have the same tone the same physical property as the say as well as the same linguistic property only in those terms only in those cases a semantic level analysis will be done now to prove this analyze men did an experiment a **dichotic listening** experiment where both the years were hearing a message and in both the in in the attended year the message was was about a particular seashore and the word Bank came in some here and on the unintended year the financial structure Bank was being produced or something related to the finance

structure was being put into and so for distinguishing when Bank the word Bank came in the years they were having the same physical properties the same speed the same word properties for words same kind of a thing and so a semantic level analysis a semantic level filter was applied and because the context in which the tended message was working on the bank what they were disclosed being described in the attended message was in terms of Ashish or a bank of ash or a bank of a see kind of a thing bank of a river kind of a thing the water body and so that was the message it was grabbed and the financial institution which was being grabbed which was employed on the unintended year which also had the word bank was not taken forward and so at this level the kind of analysis or the kind of interpretation was done so basically then these are the theories which have been proposed for using or for **attention** studying **attention** or how does the location of **attention** really works another theory which is out there which looks into which adds on to the the theories that we are discussing is something called **late selection theory** now this theory is much similar to what the **filter theory** has in it was proposed by someone called duration 2h in 1963 and what does this theory and the theory was of course modified by Norman in 1968 and so what does the theory say well the theory takes up some information from the **filter theory** and some information from an attachment theory and comes up with a new theory and what this theory says is that the filter that Broadman talks about doesn't happen at the initial stage of processing of a message what happens at a later stage at a meaning stage and that is what the theory is all about so what this theory goes ahead and says that all messages are routinely processed for at least some aspects of meaning and the bottleneck the filter since in the in the **filter theory** the bottle name or the filter the filter is equivalent to the bottleneck think of a bottle and the neck of the bottle basically prevents spillage of information so the bottle neck or the neck of the bottle is thought of as a filter and so in the **filter theory** this bottle neck is towards the top of the bottle and so since it is there what happens is much information doesn't rush in and so the spillage of information from the inside to the outside is reduced now what - the Russia's Russian versus 1963 theory of late selection says is that this bottleneck appears at a later phase of time at a later stage in message processing and the bottleneck comes in a later point of time so a message important depends on many factors what is what makes a message important according to this theory so basically what they say is that most messages are processed through all the stages to process through most of the features and to our end to some level of meaning and only at the level of the meaning we have this filter or bottleneck coming in so if this is how I have the **filter theory** where this is the bottleneck and later on this is the processing stages this is if I can describe the **filter theory** the **late selection theory** says that this is how it looks like so all messages are processed early on and later there is a filter a bottleneck which ad through the meaning using the meaning as the reason for the filter or meaning being the variable which controls the filter through which a message is stopped or block and so what they have again go ahead and say that there are several other variables which determine how the filter will work and they say that the content and personal significance of messages are also responsible for this filter addition what is the content the message and what is the the the personal significance of a message for example names of people for example certain words which are personally significant to people they have very less threshold and they are not blocked by the filter even at the late stage and for process for meaning it is led in that kind of information is letting whether you are focusing it or not and another thing that they bring into the picture another interesting variable that bring into the picture of **attention** is something called the observers level of alertness how alert somebody is that defines how you will process an information or how much information you will you process the more alert you are the better processing of messages you can do the more **attention** you can pay the lesser alert you are the lower alertness you are showing for example alertness in sleep or if you are in a sleep stage if you are sleeping you have very low lower level of alertness and so the message will not be processed at all an additional factor is the complexity of the message if complex messages are there then these messages will take more resources and so processing will be difficult but if easy messages are there then even the non

intended messages actually get processed so in this lecture we looked at what is **attention** what are the various factors which determine **attention** what are the theories which go ahead and explain **attention** we looked at three theories of **attention** and try to compare these theories of course we will do a more comprehensive comparison of these methods or these theories at a later Slater stage or later point of time but this is what we did in the structure we looked at what is **attention** what factors control **attention** what is the definition of it and best of all how do we study **attention** it's an interesting thing because once as I discussed before if I say don't pay **attention** you start paying **attention** to it so how is it solved an answer to this is something called the **dichotic listening** task so by by we'll meet again in the next lecture and continue with this thank you

Lecture 8: Models of Attention

■Kind: captions hello so we are back today with a section on **attention** and to start this session we'll do a quick recap of what we did in the last lecture and from there on we'll continue with the today's lecture so in the last lecture what we focused on or what we did was we looked at what is **attention** what does it really mean and we also looked at the various factors which control **attention** and the various theories of what **attention** is so a quick recap onto the concepts which we discussed in the last class of **attention** basically is filter it's a process or it's a mechanism of the brain which lets the brain control the kind of stimuli which is impinging on to the brain from the sensory organs and **attention** is very beneficial because **attention** limits the capacity of the mental resources which should be diverted to various cognitive processes which are taking place in the brain so basically **attention** is an important factor now the question that we asked in the last class was whether **attention** can be increased or whether there is something called diverted **attention** so with practice whether we can use the **attention** of capacity that we have on different tasks and as an answer to that we saw that as you practice more and more what we get is that we can divert our **attention** and this is called **divided attention** we can use our resources mental resources on several activities so basically while doing one job we can also put our **attention** onto other job and this basically when we are able to divide **attention** to several other jobs to a number of jobs then some of the jobs which are less important which are routine becomes automatic and the ones on which we are focusing **attention** presently is the one which is at the attentional system which is at the center of **attention** another interesting thing which we saw was that **attention** is basically a mechanism with fluctuate so **attention** basically keeps on fluctuating from movement to movement there is something called **attention** span which basically defines for how long **attention** would be put on to something or for how long **attention** is being captured now in addition to that we saw something called **selective attention** base is basically focusing our **attention** on to something specific how do we focus our attentions on two specific things and look into specific things which is basically equivalent took something called mental conscious concentration in addition to this we also looked at a couple of theories so brief review of the theories we looked at something called **filter theory** which defines the fact that on **filter theory** says that only those processes only those stimuli which on which we focus our **attention** to our process and other stimulus are not processed so our focus of **attention** besides what stimuli gets processed and what stimuli doesn't in addition to that it also points out that some features of the unprocessed stimuli are processed but these are physical features and so nothing of much value and so this there was an experimentation then by Chery using something called **dichotic listening** task to prove how **attention** should be studied later on and a tradesmen went ahead and challenged the **filter theory** why because Mori came up with the idea of something called cocktail what a phenomena and he was able to show that some materials from the unintended here or from and from those processes should not attended to could also pass for example somebody's name and so Ana tradesmen came up and gave her gave her theory of **attenuation theory** which says that **attention** basically what **attention** does it doesn't filter out things but it attenuates or volume downs the material which is attended which is unattended which we are not putting our **attention** to and if needed we can actually go ahead and process that in addition to that we also looked at something called the **late selection theory** where we looked at the bottleneck which was defined by the **filter theory** which exists right at the beginning of an incoming message nor shapes toward the back of the message which basically means that most messages goes ahead and are processed for meaning for some level of meaning not all level of meaning but seven level of meaning and depending on the kind of processing that that you require kind of processing being happening or the kind of differentiation that you want the the place at which the differentiation happens the messages are processed what I mean by this is that at what level two messages differ we'll decide to what extent messages a process so let's continue with our discussion and focus on to our look into a new theory of **attention** which is called the **multimodal theory**

of **attention** so it's a very brief theory it's a very short theory which was proposed by someone called Johnston in Heinz in 1978 and it is called the **multimodal theory** of **attention** and what they say is **attention** is actually a very flexible system which allows for the selection of message over others and this selection of message or this filtration of message from the incoming stimuli happens at three stages so most of the messages pass through stage 1 stage 2 stage 3 of processing if the message is filtered at stage 1 it is called early selection then a message is filtered at stage 3 it is a late selection now so how is this Theory little bit different the only thing which is different in this theory is the variables which controls stage 1 or which direct stage 1 stage 2 and stage 3 which basically describe stage 1 stage 2 and stage 3 and as you can see stage 1 is sensory representations where the sensory representation is constructed so if a message is processed early on it is processed at the level of the representation with the messages creating onto the neural structure so if the representative itself and so basically it will be physical properties of the message which makes us apply that racial filter or which directs the attentional filter for processing here if both the messages are a number of messages are approaching us or the incoming stimuli has a number of messages different messages which are very similar in terms of the physical property then they can the more or less have the same kind of representations so how are they processed in this case the message are differentiated the filter is applied based on the semantic representations of the message so at the level of meaning so what this theory goes ahead and says is that most messages which differ so the first factor which is used for processing messages for separating messages by the attentional filter happens at the level of the physical stimulus or at the level of the representation itself and as you know that sensory representations are generally the kind composed of the physical features these are the first features or these are the first representations which are made onto the brain or the mind now in the second step if messages which are approached are of more or less same kind a meaning wise differentiation happens separation of messages is done in terms of what the meaning of the messages a fact where most messages or two messages have the same meaning or nearly same meaning and they have the same kind of physical features what happens then in those cases are deeper level analysis a deeper level of filtration a deeper level of attentional processes really work and here the separation or the filter is dependent on both the semantic and sensory representation into takes takes care of or it takes includes both the semantic representation and other senses representation so if most messages are similar in terms of outwardly similar physical features and **semantics semantics** meaning then a third level of analysis where happens it's a deeper level of analysis is a deeper level of filtration we're very very minutely the sensory features are looked into and the semantic features are looked into and in combination the sensory features and the semantic features are in combination with each other then become the basis or differentiation of two messages so this is what the theory is and this is a very simple theory of Hines and Johnston the next theory that we talked about is Daniel Kahneman theory even before we go ahead and look at Daniel's kind of a theory of how **attention** really works let's talk about **attention** then the capacity of attentional systems and the mental effort which is required for moving or using **attention** so more psychologists are actually looking at in terms of **attention** those messages the factors which make a message being processed or what makes what factors of a message makes something being processed and they are not looking at those factors which are not making a mess being processed to make it more simple since in attentional as in attentional theory or in terms of an attentional experiment most psychologists are now focusing on the fact that what makes a message being selected for processing so they are not looking at those factors which which makes a message not get processed but they're looking at those factors which make a message being processed and they believe most psychologists nowadays believe that looking at those factors which basically give this idea of how something is in process or what is the way in with something using process that should be and that should be the the the center of **attention** of attentional studies our center of focus of **attention** certain status now one particular experiment or one particular view which defines how attentional focus or what is the capacity of **attention** is the viewpoint of

those theories which look as look at **attention** as a spotlight so a group of theorists believe that **attention** is like a spotlight on a stage and so what is this center of the stage is what focuses **attention** and what they believe is that most spotlights that that is there on a stage or any particular area the spotlights and fuzzy boundaries it basically means that part of messages which are towards the boundary of the spotlight also gets processed now Daniel can men he gave a very interesting theory of **attention** where he used something called things like a location policy things like user disposition things like physical states and so many other things or so many other factors or defining how **attention** the alliteration filter really works or how the process of **attention** really works and so what Daniel Kahneman says is that **attention** is a set of cognitive processes which is used for categorizing and recognizing stimulus so Kalman proposes a model which views **attention** as cognitive process which helps in making categories in making categories of the incoming stimulus or taking incoming stimulus and assigning them into pre recognised category and while doing this because this helps in the recognition or in the identification of the stimuli what he believes is the more complex stimuli and incoming stimuli is the more mental resources is needed the lesser complex I enter in an internal stream an external stimuli is the lesser that national resources are needed so essentially this model then depicts the location of mental resources to various cognitive tasks now if on the right hand side you actually see the model itself of how this canvas model really work now as if let's decipher this model a little bit and see how this model really works and as you see what can man believes is that there are a number of mysterious determinants from in the environment which decides whether a message is processed or whether our message should be processed or not and this miscellaneous determinants are something which is physical in nature and this comprises the physical property of any incoming messages but once a message enters into or impinges on our sense organs it is the level of arousal which decides whether a particular message will be will be attended to or not so basically the first and important factor in his theory Daniel Kahneman theory of **attention** is the level of arousal and we have seen this level of arousal also in previous theories which says that level of arousal is an important factor which determines whether a message gets processed or not and so what Daniel Kahneman says is that with high levels of alertness high levels of arousal a chances of processing the chances of attentional filter working is very high also he says that not only the ll the arousal levels but also the available capacity how much capacity or how much mental country the concentration is left that defines how the filter will work if a very less mental capacity is available if most of the mental capacity is taken up then the attentional filter will not work as required and so as wanted but then if the available capacity is high and there also is less the filter works perfectly or these are the terms on the filter actually works and another factor an interesting factor which defines the functioning of the attentional system or the attentional filter is the evaluation of demands of the capacity how much demand is being or made on the attentional capacity if some other job or most jobs which are happening with the new incoming stimuli which needs to be filtered are automatic in nature is the the eval if the demand on the attentional capacity or the attentional resource is less then an effective filter will work or the filter working will be effective but if a second job which is given to you or the secondary job that you are doing in addition to focusing your **attention** on to something new what would really happen is that both jobs will compete with each other and lesser **attention** is required and the chances are a filter will block the new message or block the important important contents of the message also in addition to the arousal in the available capacity the allocation policy is important term for Daniel Kahneman's theory now what is allocation policy according to a Denman Daniel Kahneman allocation policy in the model is affected by individuals in enduring dispositions and momentary intentions and the evaluation of demanded mental capacity so what he says is the the way the kind of system that we use for allocating or for distributing mental concentration or mental effort on two different tasks depends upon three things one is the enduring disposition it is like the interest so if a person is interested in something he is bound to put much **attention** to that particular thing and want to put lesser **attention** to other things so if somebody is doing something interesting chances are that a lot of his **attention** policy will be allocated to

that job and very less of **attention** is left over but if somebody is doing a boarding job a job which which which is not of interest to you and which is not something which you endure it's not an enduring disposition something not of your liking what would happen is that you would basically most of your **attention** attentional system is available resources are available and so what would happen is that most of the since it is available it can be it can be made to use also of interest our momentary intentions our momentary intentions is things which happens at the moment so at the time of processing at the time of when attentional filter really works when something is a new stimuli comes in what happens then so if if the task demands are something or if at the time of processing and even to occurs which requires you to focus yourself on to something else when to some some other event or some other job or some other some other feature of a stimuli and those reasons or those things could be also responsible for our allocation of how much allocation you can do our mental resources to a particular you even incoming stimuli and so this how much demand is being made how much of the processing is going on presently what are your likes and dislikes and water is a momentary intention these these these factors will define how a person will define or will divide his **attention** so basically looking at it the first factor which basically decides whether **attention** would be paid to something or not is basically somebody's or a person's arousal level and available capacity well the person is aroused enough and whether he has a huge or not if not huge but a substantial amount of available cognitive resources if the resources are pre using something is being used in something then the person will not attend to any new messages or will not at potential will not come into play given the fact that the enough attentional capacity is available and illa Feroz 'el is there or more than maximum level of arouse seneschal capacities available even then your **attention** onto a particular job will depend on the allocation policy of how things are being located and/or how somebody wants to allocate that or how somebody wants to put their **attention** onto something and this will depend upon your likeness and liking unlikeliness or some things your motivations towards a particular job or towards a particular event which is happening at the present also on what happens now so some sudden event might occur which might take up a lot of your **attention** so think of yourself counting notes in in a bank or people counting bank tellers counting notes and a sudden loud bang comes in now when a sudden loud bang comes in or a sudden loud bang happens the it's very difficult for the person who's doing this job of counting the notes to focus on to it and his **attention** will save so momentary intentions are those things which happen in the moment and so those are those demands of those environmental demands are those movement demands which shift **attention** other thing of interest are the evaluation of demands on the available capacity so how much demand is right now how much tasks are you doing in addition to the present task which is in hand and these will decide all possible activities that somebody can actually go ahead and do we all look a location boss policies will decide how much **attention** you will put to something how much **attention** you're going to put on to something else and this finally will decide the responses so basically this theory goes ahead and says that **attention** depends not only on arousal and available capacity it also depends upon people's motivations liking and likings momentary features and the current demand on a particular current demand on a particular system or on a particular processing which at the next theory that we look into is something called the **schema theory** of **attention** and so this is a very interesting theory of **attention** because this is our totally a different view of **attention** and so what this theory actually goes I didn't says so this theory was first of all proposed by someone called logic myself in the year 1976 and this is called the **schema theory** so what is the theory actually goes ahead and says it's a very interesting theory a very interesting approach which says that people's **attention** doesn't go ahead and filter or unfilter something so people's **attention** is not a process which actually separates things which gets filtered through it and something which doesn't get filtered to it what he says is that things which are not filtered things which we won't put put our **attention** on to we actually don't go ahead and perceive it in any way so basically what this theory says is that the **schema theory** says is that only those things which we focus our **attention** to uh perceive but those things which we won't focus our **attention** to are not

perceived at all it's not taken in at all and so attentional system doesn't work like if in in or if then kind of a system but it works as all-or-none kind of a system so if every so if attentional systems basically decide if everything is processing whatever is not processed is left over and nothing no kind of even or no kind of mental resources are allocated to it now what he says is that **attention** the process of **attention** is more or less like apple picking from trees it is like picking messages from trees so when he says that when we go apple picking think about we go picking mangoes or apples from trees the ones that we pick are the ones which are available to us and the ones that we don't see remain on the tree and similar to that **schema theory** basically says that the one message the messages that we put our **attention** on to are the only ones that we should be interested in or that we care about the ones that we do not look into that we do not pay our **attention** to that will not focus on to actually remain outside the system and there are or there are there is no way to basically go ahead and process them to prove that this kind of approach is true that what we don't process or what we are not being **attention** to is not process at all to look at a system like this nicer and blacken conducted an experiment in 1975 to provide evidence to this theory so it's a very simple experiment which was done now there were two different videos which were shown to a group of subjects on the first video you see people are hands slapping so this is the first video in which you be the subjects in the experiment saw people hand slapping and they had to count the number of hand slapping so this is an easy job to do and people were able to count the number of hand slaps which were being made on the second video people went ahead and saw the number of ball passes which were happening between people so there are three people who are playing a ball and a number of ball passes are happening and so the job of the subject or the person hosting taking part in the experiment is to count the number of passes which the ball is making from one person to another or the number of hand passes that is being happening so both easy jobs to and so people were able to do them a third video was made where people were made to look into this video and the sport from what this video had is a superimposed image so put the videos were superimposed on to one another and so people were actually asked to shadow one of these videos the video that they had to shadow was the one which is ball passing so now what happened is a third video is presented so in video one you count the number of hand passes sorry the number of hand slapping which is there in number two you look at the number of ball passes and in number three what you tend to do is the soup in the superimposed video you actually have to share we actually have to read aloud or tell aloud how many ball passes are there so there are two images there are two superimpose videos there on one you have the hand stepping going on on the other you have the ball passing going on and so what really happens then is that people are so as we forgot focus to put their put their mind onto the hand passes people don't really pay **attention** to what is happening in the hand slapping experiment and soon enough if you're five minutes into the video the people who are hand slapping actually stop doing that and starting passed and started passing balls to each other but it people most people most experimenters actually 80% of the people who view this video we're not able to recognize how this or when does this switch happens they don't realize that people who were doing this hand slapping this changed to ball throwing this kind of thing was not noticed and so this video and even when the experiment ended and a brief recap before view of the video was overview of what this on the video was taken 80 to 85 percent people are not able to tell whether this hand slapping video in the hand slapping video the change happened in terms of people stopped hand slapping and started throwing balls which basically provides evidence to the fact that that **attention** could be a system which is and if and and all or non kind of a system where what you perceive is what you perceive and what you don't perceive actually doesn't get into the system at all now similar to the study done by a nicer and back pain an interesting fact which is of this or which is present in our environment is something called **inattentional blindness** no **inattentional blindness** is basically a phenomena where things in front of us we don't see it now often the most of you would have had a scenario in which I think you are looking for is just in front of you and you don't it and this particular phenomena is called **inattentional blindness** and this is basically

even if something is in front of you you don't see it and so this provides **inattentional blindness** provides support to the **schema theory** so to study in attentional blindness to kind of experiments were done in one experiments people were passing ball among each others and so there were 40 hand passes of ball which were there at the rate of 40 hand passes per minute and so most subjects we're advice to actually go ahead and count the number of hand passes and so people actually endorsed onto that and so they looked at the number of hand passes which happening to look at we do this and we do this and so in this week people look at a number of hand passes which are actually going along and then very equally or very nicely they're able to count the number of passes that the ball made from one object to another no hidden from the subjects **attention** where two events which most subjects actually didn't notice only twenty percent twenty five twenty five percent people actually went ahead and knew that this happened this even happened so within this scenario of ball passing in one case a person carrying an umbrella came in between these ball passes or came in so look at this person this person I'm talking to he came in between and he passed these people whose hand passing similarly in one case a woman a short woman wearing the dress of a bear came in and passed this particular thing so if you want you can actually go ahead and see this video of **inattentional blindness** it's available on YouTube and you can see that when engorged when somebody is engorged onto a job and when it a difficult job like this example how many hand passes are being taken placed count the number of males who are passing the ball and Contin umber of females from whom the ball is passing and if this kind of task load is put onto your attentional system is put on to your processing system what happens is people are not able to focus on on stimuli which is just in front of them as you see that these two stimuli these two people are the bear clad lady and here the umbrella taking man passed in front of people's eye but they were not able to recognize this they were not able to see this the reason is that as more and more load our pole is put into the attentional system you do not process things in front of you we basically suppose a partly a **schema theory** that what you process that if you do not process something it is like after picking so what what you don't process remains order system what you process is what remain in the system so an interesting experiment to support **schema theory** now the question that we had at the beginning of this section was the safe **attention** is such a hard thing to look at whether there is something called **automaticity** whether there is something called automatic processes is there a fact that if **attention** can become automatic is there a fact that can **attention** become automatic is and if it becomes automatic with practice can systems become automatic and pro-human processing or mental processing become automatic so that more **attention** is available for other jobs now we have seen that this is possible because the more learner drivers with a lot of practice can divert their **attention** to multiple other things for example talking listening to music and so on and so forth so we look at to that feature of **attention** and that feature is that with a number of practice with a lot of practice most jobs become automatic in nature and so **attention** is freedom the attentional capacity is freed up so I will become well practice as we can well practice doing something that takes less upper tension to perform a good example is typing as I said so well you have been typing on your keyboards for a lot of time for a lot of years and so you realize that as you practice as you learn this type skilled more and more what really happens is that things become automatic it becomes the processing becomes automatic and so while typing earlier when you when when you were learning typing you have to find out where the cue weit word is where the F and D word is the switch and all those kind of things but when as soon as you develop this habit or style you practiced more no more tension is available now you can type talk to people rain coffee and do a lot of other jobs at hand so basically typing has become a lot of is a good example of how **automaticity** with practice **attention** becomes attentional spans is freed up and jobs become cognitive George Jacobs automatic in nature so if one is killed at typing he can carry out fairly accurate amount of information and jobs quickly and carry out a conversation as I just said so what affects the capacity of any task at capacity any given task require what is the reason or what are the factors which affect the capacity any task is going to require the answer to the above question are two factors one is task difficulty now the

more difficult a toward a task is the more attentional capacity it will require and the more time it will require for you to finish this task so for example if a simple counting task is given to you a number counting task is given to you or if a simple count counting of money is given to you it's a simple job and doesn't require too much **attention** but if I load your **memory** with other demands for example not only the fact that so I give you a bunch of notes to down let's say all of them 100 rupees notes and so you go ahead and count and so if the task demand is just counting the number of 100 rupee notes it's an easy task but then suppose I say that this job of counting the money you have to do till two things you have to not only it tell me how many nodes are there but also tell me which of the nodes are forward facing and which other nodes are the worst facing which I have notes in the bundle which are on which are forward-facing which has the front side up or the back side up now the task becomes a little bit complex and requires more of **attention** and similarly I can ask you several the other load your **memory** or working capacity with several other facts and so as task capacity increases the task difficulty increases the capacity required is more or more more more and more in demand so the demand of **attention** capacity becomes high an individual's familiarity with the tasks was the second factor which decides the capacity any task is going to require is the individual **similarity** with the task now individuals who are very familiar with a particular task who know what a task looks like or what is the requirement of a task they are able to do a task very nicely and very easily in comparison to those individuals who do not know a task for example look at bank tellers now bank tellers are very familiar with this counting task they know how to count and they know a lot of other things about the counting how the note looks like what is the crispness of the note the motor **memory** which means the the hand **memory** of counting is better and so they can do this job easily but if you take these people the same people and into some other job it is very difficult to them to pay **attention** look at coders people like in our Institute people students who do coding now when they're doing coding they are very familiar with the coding job and it's in a particular **language** let's say somebody is learning C is very familiar with that and coding is easy for him but he moves from one **language** to the other **language** or tries to do something else in terms of data mining or something else in those cases in a in an unfamiliar territory it becomes really difficult so individual **similarity** how much familiar you are with a particular task decides how much effort or how much capacity of **attention** you will require and so two important factors that I discussed now a star's difficulty and the individual's familiarity with the task so practice generally is believed to decrease the amount of mental effort a task requires and makes this automatic so as I discussed briefly with you that the more amount of practices that you do the more familiar you get with a particular task the more easy it becomes for you to do the task and the more automatic in becomes the more automatic task becomes the more our automatic many tasks become the more easier your life is think of your daily routine now if your daily routine if you look into it from getting up in the morning to come into the to the mess for having food coming to the place of study which is the lecture theatres in our Institute and similarly and riding a bicycle or based processes which are there all of these are automatic right you don't have to put too much mental concentration into it you do not have to each day think about how to ride a bicycle or what should I do in riding a bicycle or how do I eat my food you know even a people who walk here don't even have to think how how and why my foot is pushing the how do I walk about pushing the earth down and that makes me walking they also don't have to think about simple task of going brushing or this kind of job these are fairly automatic and if you ask these people these students or what happened today what kind of food you have most people don't even realize what kind of food you have it is an automatic process yes but then there's with practice now why has this become automatic is that they do this routinely they do these tasks routinely over and over again 365 days a year and so it becomes more and more automatic water my city actually states it goes ahead and saves your mental concentration so you can put this concentration so that you can put this **attention** to other jobs which is of interest and so you can or other jobs which is more demanding and you can put your **attention** so if students are get engrossed into basic things like what food they eat or how the cycle bicycling works or

how do they brush and those of daily activities they will not be left with mental consideration to read to understand lectures to understand what is happening here and so this practice making automation is an important fact remember that as we practice more and more more of your attentional system becomes free and becomes available to put into tasks which require so a very interesting task or a very interesting experiment which goes ahead and proves that autumn acity or being automatic and doing things with practice the jobs become automatic or as you practice more and more a task more and more more attentional system or more attentional resources are available and this availability of more resources makes makes you focus on to other jobs so which basically means that number practice you do the more **automaticity** becomes and more attentional resources are available and so a task which was defined which is called the Stroop task not just a very very interesting task so let's do this task and see how does it really work and in the later half and in the end of this task I'll also go ahead and show you the meaning of how this test really works so Jonas readily Stroop in 1935 made a famous demonstration to prove that autumn SAT is basically the reason or it's it's a by-product of practice and **automaticity** frees up a lot of **attention** so let's do the task I'll give you some couple of minutes to go ahead and do the task so basically focus on to your screen and start doing what I am asking you to do so this is a demonstration of others to task so look at this demonstration what you have to do is state the color as fast as possible so I'll give you one minute to basically do this task self task of this and look at this three rows and quickly name this color or quickly go ahead and name this color and write it if possible somewhere so look at this demonstration and name the color now I'm pretty sure most of you would have accurately named the color bars which are presented here and so we move to the next part of this demonstration but I will give you around about 30 seconds to 45 seconds to finish this task so in the demonstration to what you really have to do is state the colors as fast as you can just read what is the colors which is we're reading written on this demonstration just go ahead and read the colors so do that and I'll give you 30 seconds for doing it okay so I given enough time to read the colors and now we'll shift to third demonstration but over the advice in the third demonstration you have to do the thing that I'm asking you to do as fast as possible so don't cheat basically what you need to do is to take enough as fast as possible the job which is being given to you and so what you have to do in the third demonstration is to name listen very carefully to name the color of the ink color of the ink in which the particular word is written so all you have to do is to go ahead and name the color of the ink they linked the color of the ink in which it is written so I'll present to you and I'll give you a 30 second period to do as this job so let's turn up do it what you have to do is to name the color of the ink in which state the colors of the ink in which this was a pigment so let's start so host of you would have completed and did you see what happened what really happened is most people actually are not able to correctly identify the color of the ink in which this particular word is written the reason is that red is it and in green as most people tend to read red instead of green which is what the correct servers are for example here you see that blue is written in yellow so when given a task of reading as fast as possible what you see you will read blue here but not the question was the thing that I asked you to do was to name the color of the ink so your response here should have been yellow and similarly here the response should have been red not green and in this case the response should be blue not yellow now why does this happen why is it difficult for you to to name the color of the ink and instead of which you can very fast tell me what the what the display what are the things are written or the what are the words written in the display what do you think happened what is the reason why this happened and another question is given enough practice can people lower the time of response here so if enough time is given can people do this task fast so let's find the answer now stoops task presents with a series of color bar as i said red blue green or color word went in on conflicting colors and people have to name the color of the ink but then as I said most people are not able to perform the Stroop task according to its true why this happened is that adults are literate participants has so much practice reading the task requires the attend that the task requires little **attention** is performed rapidly now since as illiterates we are so familiar so familiar with reading we have so much

practice with reading that it is difficult not to read and so when the question asks is to read to basically name the color of the ink it becomes a difficult job why does it become a difficult job it becomes a difficult job because reading is natural to us it's something which we know and so the first response our system so we have so much practice with it when the first response our cognitive system does is read the letters and that is why you cannot name the color of the ink but then what really happens is that with practice this could be reduced so if I give this task to you let's say for five days and if you practice 20 times on five days amount of time it takes for you to read the color of the ink it comes easy and so does this become this conference with the item consisting of word pairs and now this type of response when they take little **attention** and effort is called automatic and so on a funny cartoon on the right hand side you can see that our way obviously dogs don't see it but then this puppy is saying to the other puppy let's see for some reasons human find this stroke starts really tricky they do not believe their eyes they don't believe that humans are so stupid because you know job dogs cannot talk and so when dogs cannot talk it's very easy for them also people who are not literate people who have very literacy or less less literacy are not familiar with the learning paradigm it is very easy for them to do this job only for people who are literate people who have enough literacy or people who have enough practice with reading it becomes difficult for them now an interesting thing that I want to bring in here is whether the neural structure responds to **attention** so whether there are something called neuro physiological evidences of **attention** so basically what I mean here is that in the beginning of this lecture I told you I said to you that we were looking at not only how the cognitive processes work or what is the reason of how the mind works I will also look at something called the brain responses to it so in the earlier experiments in the earlier part of this chapter we have looked at that attended years produce things from attended years are processed more than from the unattended year so an ancient materials or stimuli from the unattended here is either tuned down or not processed at all or not picked up at all and the several theories which go ahead and say that attended years differ from an idea in terms of how much is process so what we want to see is whether there is an ERP marker very there is a brain response to this whether this is true at the level of the brain and so someone called banished banished I'm sorry in 1997 he studied or he designed an experiment to study how does the ERP markers of the attended and unattended years in Cherry's experiment I would say modified Chinese experiment really evolved so his task was like this there is this two-year headphone that we talk about and so both the headphones actually go ahead the right the the left headphone and the right headphone they hear soft and heart tones short and long tones long and short tones on both the years so I have long tones and short tones which shall be presented in both the year and what people have to do is to shadow one of the year in this case the left here and to tell me aloud how many short tones or long tones rugby was being presented and the design of the experiment generally most designs of fear if experiment in works in this way so basically this is the time axis and this is the voltage axis the current axis of how much potential is generated what is the amplitude of the ERP ways or what is the amplitude of the event related potential wave so basically we discussed in the first section the TRP is a method of doing a **cognitive psychology** and yet people measures through sensors on the on the surface of the head what kind of electric potential is generated and so after stimulus presentation so let's say that this is T equals to zero where the stimulus is presented from here on 1,000 millisecond event or 1000 milliseconds section was cut so eg G's recorded from different regions of the brain this eg that I'm talking about is from the auditory cortex since it's the auditory job and so this our number of trials so er P is what you have to do is you have to take a large number of trials and basically go ahead and calculate something called the mean of all these trials so a large number of trials are taken and I mixed together because human beings respond in different way there is a baseline risk there is a baseline waveform which is there and so these baseline bombs may have may differ so to find out that unique signature that we are looking for in erp we have to take a number of trials so number of trials were taken and from these number of trials ERP was cut from the moment the stimulus was initiated to one hundred to a thousand milliseconds or a response to

thousands a millisecond was the response time which was given so within thousand millisecond you have to go ahead and response whether it was a short Tolan long tone and what really happened so when it was done it was found that an ERP like this was generated no this ERP here is something called the N 1 and this ERP here is something called the P 1 now what is the N 1 and P 1 N 1 is 100 milliseconds so that you see that the the potential this is the voltage axis and this is the time axis so what really happens is that you see a peak a peak in voltage on the negative side amazingly most ERP that are in this way so the upside is generally negative and the bottom side is not generally positive off of my graph and so what happens here is that attended years so this is the ERP affair attended here this is the ERP of it unattended here so the one which is thick is the ERP of an unattended year and so what is found is that in the attended year 80 milliseconds so this is basically hundred millisecond but 80 milliseconds following the the onset of the stimulus I see a peak and this peak is called the n1p this happens this n1 peak is basically a peak which shows novelty it's basically a peak which shows new new things into the stimulus or it's it's basically our attentional peak most books we'll talk about this that attentional peak something which captures the **attention** something which is not present in in the system so basically this 80 millisecond is a time which is required for a stimulus to basically come in and imprint on your or a sense organ so when you're attended year is waiting or when they attended here is actually preparing itself within the first 80 millisecond of peacfully will register a peak will be registered and this peak will demonstrate that it is attend that ended year the year which is being attended to shows a brain response or shows our brain dynamics and this says that it is attended whereas on the other end here you do not see this peak similarly you will find B 1 PP 3 peak and so on and so forth but we'll discuss those Peaks somewhere else at some other point of time when we were looking at what are our P's and our ERP is done so in the next coming lecture upcoming lecture we will continue with the autumn a CT the **concept** of autumn a CT of how autumn SAT really works what are the factors which makes attentional systems available and jobs automatic and how is **attention** of capacity allocated you to different resources and we'll also talk about something called the **psychological refractory period** which is a very interesting thing to look at thank you

Lecture 9: Automization and Attention

■Kind: captions hello so we are back again today with a section on **attention** a brief review of what we have done up till now in **attention** so in the previous classes we looked at what is **attention** and the various factors which go ahead and define how attentional really works or - so specifically we looked at this brain process or this **mental process** called **attention** how it goes ahead and distributes the cognitive energy sources which is available and how **attention** is equivalent to mental concentration we also looked at a method of going ahead and performing attentional studies and in that we saw a brilliant experiment by Jerry which is called the **dichotic listening** task the reason why this experiment was designed was because attentional studies are difficult to carry out and so after defining what sherry looked into or what share his experiment brought out it the common result from the study was that those stimuli which we focused our **attention** to metal concentration - those what processed but those stabilized to which we were not putting any mint metal concentration from the unattended year they were not processed but amazingly the study did bring out this fact that some physical features of messages from the unintended year was also processed now Marie came up and gave counterexample or basically a problem with the sherry study which was the **cocktail party** effect and threw his **cocktail party** effect phenomena he showed that **attention** even if we are not paying **attention** to some stimuli they passed the threshold so they are registered on to our attentional system so working on this assumption that even if we are not paying **attention** to some civilized they are processed several theories were developed and so the first theory was the **filter theory** which explained **attention** as filter and the filter works on physical properties of the incoming stimuli from **perception** and so they find that only those events to which we perceive our **attention** gets passed out and all other stimuli to which we would focus our **attention** gets excluded at the attentional filter and at Richmond added a new theory into this whole idea of **attention** she div defined or showed that those stimuli which which were not processed or which were at the unattended year were actually it's it's not that they were not processed the volume we're toned down they were attenuated and through a clever experiment she showed that people's **attention** actually work as all-or-none kind of a phenomena and her task was the yes switching tasks so basically on a tradesman showed that it's not that the unattended here the information from the unattended ear doesn't get processed they are toned down and so if need be or the threshold of the incoming message is such that it is it is low they to get processed and so an additional theory a new theory was also proposed which is called a **late selection theory** which argued against the **filter theory** saying that the bottleneck or the filter that we talked about in terms of processing of incoming messages doesn't exist at the very beginning of a message but at the late part of a message which basically means that most messages are processed till the meaning level till a basic meaning level in addition to this we had the **multimodal theory** and the theory which talks about the stages of processing of any stimuli through the attentional filter and so the model Theory defines that there are three stage of processing or there are three attentional filters which exist and depending on at which stage two messages appear different to to the cognitive system that is the point at which the filter is applied so at the first level what happens is two messages are differentiated on in terms of the **phonology** I'm sorry not the **phonology** but in in terms of the basic physical properties like what is the tone of fate how intense it is and so on and so forth and the second level it is the **semantics** or the **phonology** and **semantics** of the message which basically goes ahead and designs the filter or is responsible for filtration and then there is a third stage if the messages are same in terms of a basic meaning and and the basic physical properties a third level or higher dimensional level is is chosen for distinguishing two messages and this dimension uses both the physical properties phonological properties and the semantic properties of the message so all three are considered and differentiations of messages are done so this is about the attentional theory now in addition to this we also looked at something called automation the question was whether if we do enough of practice can we go ahead and make a task automatic which basically means

that it requires lesser and lesser **attention** and so more **attention** is available or more resources are available for putting ourselves to or doing some other work and we define certain limits to it we define certain features of how preferences of **attention** really work and what is the basic factors which govern this autumn a city that we are talking about and in one of the interesting examples that we took in in the last class physically in the last class was the Stroop desk so we saw that if enough practice is done I finished a fitness practice is applied **attention** can be made automatic and so just the slides that you're seeing right now it is the last sight from the last class which shows that with enough practice autumn ization happens so in today's class what we are going to see is the optimization how does this automation really work this automation of **attention** really works and how is it helpful I will also look at what is the difference between an automatic system and **automatic processing** of message and a control transfer processing of message so attentional systems how our attentional system two systems which use an **automatic processing** in a control processing how are they different and what are the factors governing them in addition to it we'll also look at some of the examples or some of the good outgrowth of this attentional study why not growth that we look into is something called the psychological refractive period so let's continue with a discussion of automatic of **attention** so a basic question which is in front of us right now is what does it mean to perform a task automatically authorization of task performance what is the meaning of that particular thing and an answer was given by Snyder and Posner in 1975 and what they said was attentional processing is automatic or there is an automation of attentional process only when it fulfils any process fulfills three different things or three different features the first is that the task that is at hand the task which is getting process must occur with and intention which means that an task is said to be getting processed automatically with automatic attentional systems with the help of automatic as natural systems it requires no intention at all so as soon as we develop an intention as soon as we think of a task and we develop an intention to do that particular task then this particular task doesn't become automatic so Intermatic task processing requires no intention and I'll give an example to see how this optimization really works so many a times in your everyday routine you would have seen that the part from your University or the part from the place of your work to your home is somewhat automatically remembered to you and so at time what happens is when some other job for some other task is getting into your head for example some other work-related or university related problem is bugging you in some way if that that is what is happening then what would happen is in those times you you will automatically drive towards or automatically move towards your home and this movements instance you the part between your home and your work is so well learned that if another task is taking up your **attention** you don't even realize that you had to go to the market but then what has happened is since the your attentional capacity of mental resources were exerted we're engaged somewhere else you automatically turn towards your home so which basically means that turning towards the act of turning towards your home is without intention and that is why it is an automatic process because you didn't decide to go towards your home you were deciding to go somewhere else but since the mental effort the attentional processes were not available and so since this task is too practiced this way from your home to the University or unis to your home is so well practice so that that unintentionally there without an intention you move towards your path because this is what you every day now the second process or the second feature that should be involved in an **automatic processing** or **automatic processing** of tasks is something called conscious without the task should happen without conscious awareness so if you make in mind or if you keep in mind the fact that I have to do a particular task and you do it with that formal intervention consciously which means that you remind yourself somehow they'll do this task then doing that task they will require **attention** and will not be automatic example in our like in the one plot that I just gave to you where you move from the since your **attention** is so occupied you move from your University to the home and you in in that path you automatically go towards your home in that particular example the act of moving or taking the turn which which separates the way to the market to the way from the way to

the home the act of conscious awareness so you have not decided it you have not form an intention and it is without conscious awareness you never planned it before and so it is automatic in nature and the third thing is that these tasks should not be or should not have any kind of interference from other mental activity the meaning of this is that if another mental activity which is similar to the automatic task visit to the task which has requires **automatic processing** or which goes to **automatic processing** if they are same what would happen is the task will no more be automatic so in our example since our problem our university problem has engaged you in certain way and it has nothing to do with motor movement it has nothing to do with thinking about going home or anything related to you going to home this other task which you are thinking takes up a lot of your mental activity a lot of your mental resources and this mental activity is very different from the walking task or writing or whatever way you communicate from universe with the home and so since the very different in nature the second task of in evidently of unintentionally moving towards your home or taking the turn towards your home becomes automatic in nature so basically summing it up any automatic tasks should have three features into it it should be done it shouldn't be done with with an intention it should not have a conscious awareness behind it which basically means that you should not be consciously doing it and the third is that this task should not be interfered with any mental activity should a mental activity interfere with it it the task will not be becoming automatic or the start a task would not become automatic so next let's take a look at what what does it mean to have a controlled process tasks an automatic process task what I mean by here is that those tasks which require at a control processing attentional processing and those tasks which require onion unintentional processing or intentional processing which are automatic in nature what is the difference between tasks like this now so one of the interesting things is before discussing this is something called **attentional capture** we will come to that in a moment so look at the display in front of you now in this display if I asked you to find out the one item which doesn't belong to the display your answer is very quick that tree is the word or tree is the letter which pops out or which is not equivalent to what the display is and so this is called the pop-up phenomena and this becomes this is an automatic phenomena so those but if I ask you to find out the specific letter for example say the letter I or let us see it will take more time so three throws out itself to you and so it becomes automatic and whereas searching for I or C in this display is a more controlled task because it will require you to go ahead and basically do control processing attentional processing of in this letter so distinguishing this how this control and **automatic processing** works and what are the factors which control it Snyder's Schneider and Shiffrin in 1977 designed a very ingenious experiment to test whether or what are the factors which control or which go ahead and define an attentional and uh non-intentional processing so basically it's a search letter it's a it's a search task and the search is either of a letter or of a digit in a frame so it's a very simple task a certain target is given and you have to search this target in a particular frame now there are two different versions of this particular task one is called one version is called the varied mapping version and the other is called the consistent mapping version of this task so what is the task like the task has a certain target so these in the varied mapping version of it you would have targets as both letters or a digit or a letter as a target so you can have I1 three-peat as a target whereas in the consistent version of the task you either have letters or you have digits so you can either you have one two four six as a target because this is what you need to search in the frame so this is called the **memory** set this is what subjects have to commit to **memory** and then there is a frame that you have to search these targets so one difference between Maryland and consistent mapping is that in varied mapping we could have both numbers and letters as in the **memory** set the set for to which you need to compare whereas in the consistent mapping either have numbers or you have letters then you have a display like this which is called the frame and what you need to do is you need to search this in the frame so in the very condition you would have either letters or numbers on to the frame of such though so what you need to do is to find out whether this is present in this frame or not so it could the **memory** set could be one letter it could be two letter could be chill it depending on different versions of the task it could be

anything and the task is to search whether this **memory** set is present in this frame or not the difference is I also go along telling you the differences between two versions the task in the varied version you have either letters or digits that need to be searched where is in the consistent for a version you all only have either letters or digits similarly in the frame set also there is a difference the difference here is that in the case of varied mapping version the search in the frame set what you are searching in the frame the frame could also have both letters and digits whereas in the consistent were a version if I am searching for numbers or if I am searching for digits you only have digits which are reported here another interesting thing or another change which distinguishes these two versions of the task is that in this version in the varied version one of these letters one of these item and the **memory** set could become a distractor in to the next frame for example I have P here in the **memory** set this P can very well appear in our different **memory** set and so what would happen is since P is available here now it is a target in this case these this P because the next search item is q3 to 8 and so in this case Q is the letter here and P which has been used previously in this **memory** set which was an item of the **memory** Novica now is a present here with distractor whereas this is not true for the varied and the consistent mapping of it so three differences first of all what happens is that **memory** set there is a difference in terms of **memory** set **memory** set in the varied version could be both letters and numbers whereas in the consistent version it is either letters and numbers second the frame set could have both letters and numbers in the varied condition whereas in the consistent condition it will either be letters or it will either be numbers both is not true and third that one item or any item from the **memory** set can become a distractor in the next frame in the next trial of the task whereas each trial is new or each new each new presentation of frame is actually new it has nothing from the older **memory** set so an experiment was done like this and people were supposed to find out this or basically go ahead and tell whether this particular **memory** said is present in this frame or not wearing these three other factors were varied in addition to whatever we have discussed here so one factor which were varied was called the frame size size of the frame so basically how many letters do appear here so frame size is equivalent to how many letters or how many digits are appearing in the frame the second thing that was there is for how long so frame size and then frame time how long these frames were presented so you see a target and the frame is shown to you and so the length of time for which the frame was presented to you so it was a fast frame presentation or a slow frame presentation and the third thing was the **memory** set itself so how many letters were present in the **memory** said that you need to search it could be one it could be two it could be four so depending on and four was the highest number which was there so I have displayed the highest number it could also be just one letter one that you need to search so these three factors were actually used and this kind of a search task was design the results were very interesting so there is a good amount of work or there is a good amount of results which are came out of the study and it proposed some of the very good outputs in **cognitive psychology** and I refer you I'll basically suggest you to read this Snyder and schriften experiments show SCH n I de R and s hifi1609 so let's focus on to some of the most consistent resources the study in through this study we are trying to see how does consistent mapping or thus controlled sorry how does control processing and at an **automatic processing** work so let's look at the the results which are akin to these two problems so what are the results then and so the results that we got or the results that these psychologists got to us in the varied condition all three factors right the frame time the frame size and the **memory** set had a role to play in to how fast you are actually passing now if you look into it it's actually the consistent mapping is representing something called **automatic processing** and varied mapping requires something called control processing because what could happen is that the **memory** suit could maybe it could have numbers and letters growth and similarly the frame put had numbers and letters so you need more attentional system to process and so the results suggested that the varied mapping condition all three factors actually went ahead and showed their effects on two results whereas in the consistent mapping condition only you have guessed correct it is the frame nine how much time the frame was present

because one of the things is that is **encoding** so if a stimulus is not encoded at all control processing cannot take place so since it is letters verifying letters and numbers verifying numbers the only factor the only variable which went ahead and and and affected or this consistent mapping paradigm was the frame time whereas inconsistent in the result for very varied mapping condition you have the frame size the frame time as well as the **memory** set of how many letters are being searched all three of them made their marks into it or made their effects or showed their effects on to this search process so the thing is attentional is a cognitive capacity and just before **attention** to start or part of **attention** also works in terms of **perception** so we seen a sectional **perception** and it is a good time to ask the question what role does **attention** play in **perception** and this is the exact question which was asked by enna tradesmen she asks the role of **attention** in autumn a city of **attention** of control an automatic **attention** that plays in **perception** and so what very briefly what the theory says is that **attention** has a lot of role to play in **perception** and what is it Arnett has been investigated through the future integration model and what we find out is that any **perception** to take place has two stages so **perception** is basically a two-stage process it starts so this is a **bottom-up process** it starts this way in stage one what happens is it's called the pre-attentive stage and in stage one what really happens is that the physical features of a stimulus that is incoming into **perception** into perceptual system into the perceptual module for processing those basic features are integrated together and so we register features of object color and shape into the **creativity** stage so **perception** is a two-part process because **attention** plays a role into it in the first part of **perception** the basic features of any stimulus are recorded or map or integrated and then there is a attentive stage a second stage so first stage basic features are integrated together or a whole is combined through various features of them all the incoming stimuli here **attention** allows to glue the features together so basically this is as you can see there is a curvature depth motion orientation color these are the physical features and so these features this is the pre-attentive stage in these features are actually integrated together under the attentional spotlight to discuss the attentional spotlight paradigm and only this attentional spotlight only after this is integrated together here does the attentive stage works through which a meaning is generated to object **perception** and object recognition is takes place at the attentive stage so think of it in this way as you as we see something as soon as we see something initially in the first stage things like color shape texture gradient this kind of things are available to us but if we if that particular image or if whatever we are seeing is available more time they are glued together to make a meaning which basically suggests that if something is presented to very fast people will be able to tell the physical features but may not be able to tell what does the thing mean or what is the object in terms of our recognition and this particular feature where if a stimulus is presented very fast so that the attentional stage so that the attenti stage is somehow hindered people report something called illusory correlation so if if a stimulus if two features of a stimulus is present it very fast what really happens is the at the pre-attentive stage automatic stage recognizes the features of the stimulus and so late but the gluing is not correct you don't know what is what think of it in this way if I quickly show you four different cars in four different colors in four different model when I show you for one thousandth of a second kind of for one hundredth of a second kind of display and very quickly I so I show you this and very quickly I take it away and then ask you what do you actually see in terms of tell me the colors of the car tell me the models of the car and also tell me which car was rich model amazing you will find out that you could tell me the color of the car you could also tell me how the model looked like because the model is basically the model of car is displayed by the logo of the car so you can perceive the features of the logo or the the image of the logo and so you can tell me what car was present so you could go ahead and tell me that the blue car was present and in a car which is Honda I was present because Honda has a logo specific kind of a logo a Maruti was present but if it is very fast what would happen is you will mix up everything and so you tend to tell me that a blue molecules present whereas the truth is that of course a blue car was present of course a marque was present but what happened is they were not the same and this is what is called a diversity

correlation so tradesmen in scheme 8th 1982 interesting study showed that when **attention** is diverted or overloaded participant makes something called illusory correlation so if I give you more than one job to do is in the previous example as I defined if I give you two job for example if I ask you very simple questions of what was the car color or what was the car model it is an easy answer but when I mix them together which model which car and which color then the problem becomes more and since it was presented for very very brief period of time this irizarry correlations or do happen so in terms of in terms of an attachment this kind of display was used knowing the first display it's very easy because people were asked to report an S right just a letter S or or a particular color of or double X or s or a tea so tell me those letters which appear in the pink color or tell me those letters which are different than the display so most of the most of the display as X's and T's and s is the only one which is different and so this kind of questions can be very easily answered and this requires automatic nearly **automatic processing** or no **attention** at all because this pops up and this is what is call **attention** of capture what happens is that this particular thing pops up it moves now I've seen these websites several websites which are there when you visit a website what really happens is that certain things keep moving and the bottom are certain things quickly change color and they catch your **attention** but other things don't basically it is a tsunami nough like this what happens is that this with the very idea that it is different in terms of shape and shape is a very basic feature at the pre attentive stage itself these get captures but look at this day number two now in display number two if I ask you to find an X which is green in color so I'll ask you to find a letter which is of different color so a letter in a color so not two features to be looked at a different color has to be looked at on a different letter has to be looked at any of these displays per represented very fast you can of course see two colors and a different color is there and a different name is there so which is the digit which is in a different color two features required when two features required this becomes more of a **controlled processing** so when more than one because here what would happen is if an illusionary correlation can happen I'll tell you how how it happens what happens is if this display is presented very fast then you tend to say yes a tea with a green color was there right or if it is very very fast you would see a V because it looks more or less like this but then what really happens is that if two features are are asked a letter with a different color and a different image is what you need to find out if that is what you need to do then a problem exists because there are two features that you are looking into and so the two piece features has to be somehow combined single feature presentation or signal single feature identifications happens in the **creativity** stage whereas this multiple **feature comparison** and object recognition require that interface and so at the attentive face so this this particular task requires the control processing or attentive face to really work and this requires the automatic phase so the answer to the question of how **perception** or tension affects **perception** is that if paths are automatic in nature or tasks require a very physical features basic physical features for identification they are or nearly automatic but if tasks have were stimulus tasks have more features and they require some more processing then **attention** comes into play and so attentional systems are needed so very simple tasks don't require **attention** or they are more or less automatic in nature a related feature is something called **attentional capture** that thing they were discussing in in in in the studies before so visual search tasks they if they use this or they have this pop-up phenomena remember the example that I gave you in websites there are certain features there are certain animations out there which quickly change color and form and so they capture your **attention** and this is called the pop-up phenomena or papad phenomena and so experimental psychologists call this phenomena as **attentional capture** which means that they mean to imply a stimuli that cause an involuntary shift in **attention** so these features are such that they somehow threw themselves out of the stimulus set which is in a work and they desire your **attention** or they move out of of the display catching your **attention** psychologists have defined **attentional capture** to be a **bottom-up process** driven entirely or almost by properties of the stimulus rather than perceivable so it is basically it it is a stimulus feature which makes something pop out but it has nothing to do with whether the observer has anything to do with it where the

person perceiving has anything to do with it so if we look into here if I give you a second to look into here and tell me what is the problem so most of you would point out that this is not possible from 1840 to 2007 is something which is wrong and this captures our **attention** most people's **attention** it should be 1942 and this is an error so basically this system which pops out which says that which somehow tells you that this long somebody cannot leave that is what what on **attentional capture** so an interesting experiment was done with **attentional capture** so what was this experiment basically this experiment had this kind of circles six or seven circles and at the center of these circles you had the number eight returned on to them and all these circles were gray in color so this kind of circle was then people were asked to look into this circle and so at certain predefined time all but one circle remember all of these are gray colored circles with the letter 8 into it so at a predefined time let's say after 1 second or after 5 seconds all but one circle they change their color so most of the circus change the color from gray to red and so the inside became C right so one circle let's say this circle remained gray and this circle all these circles became red and so the ins and also the inside of the circus changed from A to C now the question was asked that first of all tell me which of the circles didn't change the color that thing and what is written inside that circle not unknown to the observer an extra red circle appeared either here or somewhere in the display so an extra circle came up into this display and popped up from some from somewhere and the task is taxing because people have to find out how many circles change color which ones which was the one which didn't change color and which what is the shift in terms of letters and so when a display like this was used and a sudden letter a certain new circle with red circle appears somewhere in this display with with C written on it or something written on it you would what is the result you would assume most people almost 85% people actually saw the red circle appearing what happened is that irrespective of the fact that people had their attentional blocked up by the task itself but this circle suddenly appearing somewhere other than this this particular slot it captured their **attention** so a red circle in the middle or somewhere here there it captured the **attention** and this is what is called **attentional capture** think of it in terms of those websites those things which capture your **attention** things which come out or it come out because they have that the stimulus the those things or those events has certain features they jump out of the of the screen or jump out of any environment on to you and with the fact that you have nothing to do with it you are not expecting it so kind of an unexpected event and so this is what this particular theory of **attentional capture** basically goes ahead and tells us now so based on what we have learned up till now Gordon Logan and Joseph E Turton they combined this idea of what **attention** is when it becomes automatic when it becomes control what are the factors combined all these two things together to give and auto attentional hypothesis of **attention** now what is this hypothesis this hypothesis is very simple nature what it says is that **attention** is only required at the practice phase of any task and so as soon as you have started practicing as soon as you have enough practice that task becomes automatic but if something changes in in the stimuli in which it is being presented again a practice is required or the fact that so basically attentional hypothesis of atomization stage that **attention** is needed just during the practice here is the fat ask and determines what gets laundering practice and **attention** also determines what will be remembered from practice so simply say it says that basically learning is a by-product of **attention** but if something else changes into the stimuli then what happens is you don't learn it and so you need a attentional stage to further follow so just at the time of learning a task you need **attention** and once you have learned it enough the task somehow becomes automatic it is basically what experience would suggest so if you are a tennis player at the initial hours you need all kind of **attention** to learn what is the backend what is the foreign hand what is your serve and so on and so forth and how to do the footwork in these kind of things but once you know this task very well playing tennis is almost automatic in nature and then unless something changes and they did a very good experiment to prove that it is practice that require this kind of **attention** so they use something called they had associate learning to prove that this kind of autumn ization hypothesis appears which means that practice requires autumn ization and after practice practice requires a tangent sorry and after that the task

becomes automatic so they use this kind of a pair associate tasks in their task what they had they had this kind of pairs so you had the word steel which is attached to a word let's see Canada right and this kind of pairs were presented to people and so people had to learn this kind of a task so this kind of pair associate has to be learned and in one version of the task this steel was never paired to any other word and so they were very specific in terms of this in another version of the task the word steel was sometimes paired to Canada whereas sometimes it was Pat - let's say broke only another word which is there now in this case you see that performance has increased and tasks become automatic or the process thing becomes automatic since steel on Canada are related only one so if I do a **retrieval** of this task it is more or less automatic but in this case we're still sometimes appear with Canada the task is paired associated to most para to associate tasks required to learn both the pair's and actually travel you are given the Q this is the Q so in this case steel is called the Q and you when I say the Q steel the person who's responding the person who's retrieving back has to tell me the target so Canada here is called the target and the person has to the person who's responsible Canada so this is the kind of job so when steel is specifically related to Canada the performance becomes rather automatic and so what happens is that only during the practice of it you need to put more focus and so task learning is very fast but in terms when it is in the second case when steel is related to Canada somehow another terms in another version of its steel is also related to broccoli when I stay steel the performance doesn't become automatic and so you have to actually think use your brain and use more **attention** to come up with the particular answer to the question also it was shown that even if one part of this pad associate learning if the Q was colored in some way if the Q was green in color and this was right in color even in those cases this learning was very slow and it was not automatic it required a lot of **attention** which means that failures were huge it took more time corresponding so this is more faster this is more slower and even changing one part of this the Q part of it required people more time so basically this task somehow this experiment goes ahead and proves that only during practice you need **attention** whereas once the practice is over it is automation or the **automatic processing** starts happening you don't need **attention** that much but only in those cases where things changed from the original display if something is changing the original display is some form of changes made then **attention** will again be plotted back or required back an interesting thing or an interesting feature of this **attention** or this task of **attention** is something called a psychological reflective period so you would have seen that the premise of this whole thing is that with **attention** with more practice this **attention** becomes more or less automatic things become automatic so the question is are there tasks where no matter how much practice you do it doesn't become automatic are the tasks which are there and so to prove that principle something called **psychological refractory period** was defined so **psychological refractory period** the existence of this period basically says that there are certain kind of tasks or there are certain kind of scenarios I no matter how much practice you do with a particular task what really happens is that you tend to have no **automaticity** or very less automatically into it so how does the thing really work so psychological reflective period refers to the delay observed in execution of a or the second of the two tasks which must be in close temporal succession with this task so basically I will be finding the experiment and then let you know what this particular thing is so two tasks were actually given let's first look at the pattern of how it happens so there is a first task if two tasks are there and so what we are trying to see is what is the reason why the two tasks are given and enough practice is given in to both the task where both the tasks are well done still we do not see them being automatic why this happens so we're seeing that so let's say this is the first task now there are three stars three stages to a task one is the perceptual analysis one is the response selection then the response processing so if a job is given to a task is given to the first thing is you have to look at what the task is the intuiting phase then select a response to particular that task and then go ahead and respond to it now if a second task is given to you in Succession right in this SOA means stimulus onset and crony how close this is a time dimension so del t how much close these stimuluses are I'll explain to you in the in the next figure and so in a second while

the first task is being preceded a second task starts now this particular slack which is there which says that the delayed response selection if a second task comes in when the first task is being processed this is called the psychology called refractive period what does it really mean it means that if two tasks are presented to you and even if they are very well learned tasks but as the first task is being processed a second task is also introduced what will really happen that the second task will get delayed in terms of processing and this delay that you see is what is called the psychological reflective period now in the original experiment of this PRP what happened is our stimulus was presented as first stimulus was presented to people and this stimulus was a stable simple stimulus people had to hear a tone so we had two tones we had a high tone and we had a low tone and people had to hear this tone and basically so this is the hearing part so you hear this tone and then categorize it whether it is a height or not note on this is stimulus number one a second stimulus was given which was delayed a little bit so once this stimulus was given once the tone task started a little bit ad LT time after that a second task was given to you a second task was in terms of a letter identification so an image was presented to in front of you and what you had to do is look at this image so if an image of a was presented to you what you had to do is look at this a so basically this is the display you see an a and when you see a letter vowel or a consonant you have to press a different button so a button pad is given to you you have to press the left button if it is a wobble and the right button if it is a consonant so two tasks are there one task is the tone matching task so you have to tell whether it is high or low and when you and the other is a letter identification task when you see a letter you have to press a particular button now what is the experimental paradigm say so the experimental paradigm says that if enough time if delta t right the time between s 1 and s 2 is high then performance of the second task doesn't require or happens flawlessly second task proceeds as usual but as you start narrowing this delta T if you start bringing this delta T this T or rather R instead of delta T at say T here so T is the time difference between s 1 and s 2 but as soon as you start narrowing this T as soon as you start presenting this s 1 and s 2 in quick succession if the time difference between presentation of this task s 1 and s 2 is high what would happen is that response delay would happen which means that response for the second task will get delayed more and more and which basically means that the response to the first task hardly after the response to the first task a certain delay will come in before the response of the second does coming so quickly understand the paradigm of this 2 tasks are given to you task 1 tone matching task 2 is identifying images and so what happens is as the difference between these two tasks are more if the task 1 completes and the second task starts after that no delay is there or even even between if it starts no delays there and so now **psychological refractory period** but if the first task and second task happen near to simultaneous if first task is started and just within a delta T time or very less time after the start of the first task a second task is reduce the response to the second task delays and delay is called the psychological reflective period so this delta T or this response this slack the time that it takes for response are to to happen is what is called the psychological reflective period so a general interpretation of the PRP effect assumed the presence of a bottleneck when initiating a response to stimuli now in very simple words you detected a stimulus and and are processing that information while the second stimulus comes along we aren't able to attend to the second stimulus and that is what is the **psychological refractory period** now the question is why does PRP happen or where is this bottleneck this delay the water neck is the delay of which which is there in this context the second task so where is this delay coming from where is this delay happening and so three answers have been given by pal co19 pashler 1963 and was bachelor 1960 says three stays that there are three interpretations of it it could be at the stage of presentation the second stimulus what could happen is the bottleneck have happens because the second stage the presentation of the second stimulus is very peak and so this delay occurs because the presentation of the second stimulus is in quick succession or it could be the stage of which a response is selected for the first emulous so basically or it could be at the stage of making a response so three scenarios can exist where I have as I said this is my stimulus presentation this is my response selection

and this is my response for stimulus one whereas for the second case this is my s2 this is my response selection to s2 and this is my response to s2 so response to s1 and response to s2 and there are three scenarios the bottleneck can appear because of because of the fact that this second stimulus appears because the presentation second stimulus or it could be because the response selection of this happens and so since the response selection has not happened a slack happens here or it could be because of the response giving out of the response or omitting the response creates this kind of a bottleneck which is there and the most prominent example or most prominent answer which has been given the bachelor and on well foots theory is that at the stage of response is selected so what happens here is that at the stage of selection of a response if a second stimulus comes in if a second stimulus comes in when the first response are the first stimulus is still being processed a response of it is being selected what will happen is you will observe psychological reflective period but if the response has already been selected but the response has not been given in those cases the **psychological refractory period** will not occur think of it in terms of example you go to a bank teller who is working on a job so cause you so customer one comes in and asks a question now as soon as the question is put into this person this bank teller is searches for an answer now a second customer comes in and asks the question psychological reflective period will only occur if the second person the second customer puts a question and this question interferes or comes in or is appears at the time when the bank teller or when the bank person is still searching for an answer is doing a response selection still searching for an answer for the cost customer in case the bank teller has already found the answer and it is only the response that he has to give no kind of slack will happen and both responses will be automatic you could answer to both the persons at the same point of time but then in cases where the person the bank teller is still working at an answer and you put your question quickly what really happens is that second stimulus get processed delayed get processed at a later point of time and the delay that it takes from response selection for a small selection of stimulus one - there is - the response selection to the static of response election of item two is what is called the psychological defective period so in very simple terms the time slack which happens or the time lag which happens for the processing of a second stimulus when s1 and represent s2 are presented very in equal time that and that happening because the response election for stage 1 or simulus 1 is still happening and s 2 arrives at a time when s ones response election is still being taking place is what is called psychological reflective period or psychological reflective period in very very simple terms is a delay on response to stimulus 2 because stimulus 1 is still getting processed but if the processing has happened if the answer has is present then occur so in this section in this particular class we looked at what is autumn ization what does the factors fertilization how to test out normalization what is **controlled processing** and automate and **automatic processing** in **attention** and we also looked at what is called how **attention** is a part of **perception** how **attention** plays apart into **perception** we looked at **attentional capture** and also something called **psychological refractory period** which means that even if two jobs even if two tasks are very automatic if it is well done task still the idea that one task comes in quick succession or it comes quickly after the other task and the response selection or or the kind of answer to the first task has not been selected from from the **memory** then it creates so basically **psychological refractory period** is a **memory** related phenomena and so that creates a time delay in response of the second stimulus and that is what is called the psychological reflective period so this this is an end to the section on **attention** will meet again with a new section on **memory** thank you

Lecture 10: Memory Introduction

■Kind: captions hello friends in today's lecture we will introduce a new topic which is called **memory** so continuing from what we learned in the last classes **perception attention** the next step into any cognitive process is **memory** so what is **memory** I have given definition of it not kind of a definition but I I have given what **memory** comprise off in the title of this slide so it says **memory** is something which is equivalent to knowing something or you may bring them something and this is very cleverly put here because one of the debates in **memory** is something called know or remember debate which is debate about how **long-term memory** stores things or what is the way in which **long-term memory** processes information so what is **memory** before we go into explaining what **memory** is I have some funny cartoons here for you to look at which will give you some idea or which will slightly entertain you and understanding what memories and so you can see on the right hand of your slide a person who was into a chamber of a therapist and therapist asks something about this therapist is doing psychotherapy so he asked something about whether he remember the person who's the patient remembers a therapist or not and the patient denies it and the therapy is only worry is whether he remembers the therapist because if he doesn't remember the therapist and no amount of money is what comes out of the pocket of the person and the therapy which basically means that the therapist doesn't get paid and so in a very brief way this is what **memory** and compresses off on the right hand side you see another interesting figure interesting cartoon which tells you a little bit about what **memory** is and so this is a demonstration of the famous remember no paradigm by and will tell me and what this is is that the the person in question is asking the cat whether the cat remembers what she saw was a mouse or whether she knows what she was there it was a mouse or not and so there's distinction between remembering and knowing something and remembering something is a famous debate in **memory** studies and will take up this debate when we come to the section on **long-term memory** so let's start with understanding what **memory** is so let me put a question to you do you remember the last ball which was hit for a six in the 2011 Cricket World Cup and as soon as I put this question to you most people do remember because it was a proud movement for India India won the World Cup in cricket in 2011 and the last ball was Mahendra Singh Dhoni hitting a six to win the World Cup and so when when I ask you this question many information comes in front of you like the process that I'm doing right now when I am trying to tell you what happened on that day is actually **memory** so mostly the way we look at **memory** the we most laypersons look at **memory** the **concept** of **memory** is about **retrieval** so **memory** is a process of the travel and when we say do you remember something or do you think you know something these are all facts about **memory** so then the main idea about **memory** or the main the main feature of **memory** is **retrieval** so how do we define **memory** now think about the first day you came to your present college wherever you're studying you came to the college you made a lot of new friends you did a lot of things and when I asked you to remember the first day that you came to your college several experiences will be competing with each other to come to your consciousness to come to your mind and these ideas are these facts which are competing with each other as what is called **memory** and so **memory** is that particular thing or **memory** is that particular feature of **cognition** which brings to mind or which focuses on or puts into consciousness information which has been learned before and the information that has been learned before has passed has already passed to two important cognitive stages which is the first stage of **perception** where information is coded on or information is processed from raw stimuli to the to the process of sensation and **perception** and the next stage where **attention** another cognitive process is applied on to the drawing information which comes up to **perception** and then whatever is to be remembered at whatever has to be stored is put onto the focus so then **memory** is the process of basically storing something and do it driving it so we've because before you retrieve something it has to be stored somewhere and so another dimension or another fact that can be added to **memory** is the storage fat so basically speaking then **memory** in general layman's term is a process of storing

something retrieving something and we add one more feature to it which is basically called to process something so basically **encoding** something storing something and retrieving it back is what is called **memory** in very generic terms so the definition in a slide would say that it is **memory** is a most basic process which is used in our daily life right from the time that you wake up to the time that you go to bed you do a lot of things you undergo a lot of acts then they take a lot of work and most of this work requires you to access information from your mental storage and this meant to storage the process of storing and forces of retrieving information from your mental storage is what is called **memory** now the question is why is **memory** needed so as we saw **memory** is a three-part process it starts with **encoding** at least to storage and it further leads to something called **retrieval** of information from the store value where people store it so why is **memory** needed at all what is the reason why **memory** is needed and so the appreciation of **memory** the real regard for **memory** will not come to anyone until then unless he sees someone or a person see someone who has no **memory** and so people who don't have **memory** are called M Enix so you have something called amnesia or disorder of **memory** where people cannot or are not able to remember anything and there are two kinds of Emin easier to look at one is called retrograde and another is called antegrade in which most of these ammonia's 4mn eazy-e's focus on the fact that in one of it you people are not able to form present memories which basically means they are not able to form memories of whatever is happening right now and in retrograde happening there are people not able to retrieve **memory** from the mental store so in one case there is an **encoding** failure so people are not able to form **memory** in other case they are not able to retrieve **memory** and so until and unless you come or you see a person who has no **memory** the appreciation of **memory** is very difficult I'm saying this because **memory** is something which most people take for granted it is there and so we don't appreciate it any more till the point of time that it goes away and so a beautiful case study is defined by alan bradley of a person called life-bearing who had no STM no **short-term memory** we basically mean that he had a type of amnesia in which this person was not able to form **memory** off to encode something new and so this person some means yeah well Bradley goes on to say it was so dense that even if you talk to this person even if you're talking to this person he will talk to you for two minutes and within two minutes he will forget you so think of a life of a person where this / where anybody cannot remember you or cannot remember any information for more than two or three minutes it is really hell of a life because it will be very difficult to live a life like that because what would happen is that you wouldn't be able to generate you won't be able to encode or you won't be able to store something new and this person had a kind of amnesia where he where the initial **memory** or the present **memory** could not be encoded so if questions were asked to him we put to him about facts which were prior to the phase when he suffered this ebenezer so his amnesia developed because of accident so those facts or those information which was stored which he was aware of before his accidents they were perfectly intact but he was not able to go ahead and and actually remember something which was happening right now and so this particular case could be used as a reference for what is the highlight of **memory** or what is the meaning of **memory** the question is what are the various ways in which **memory** has been explained by philosophers now since psychology moved out of philosophy and a lot of philosophers a lot of views of philosophers are carried on to a so a lot of philosophers have also said a lot of things about what **memory** is so there are several models there are several definitions which have been there which has been explained which have been used to explain what **memory** is and so let's quickly discuss some of these metaphors or some of these models which has been proposed by various philosophers or various scientists working into the science of **memory** and so Plato any 1988 points out that Plato wrote about **memory** by comparing it to something called an every or a wax tablet on which an impression are made so what Plato said is **memory** is like a wax tablet where you make certain impression and when you don't need it these impressions are taken away or it's molded back and so most **memory** is like a wax and so you take a pen or a chisel and heat it and make some kind of mark onto these facts and that is what **memory** is similarly in the Middle Ages several other

conceptualizations of **memory** were done and some of these conceptualizations led to the idea that **memory** could also be compared to a cave and empty cabinet a storage house and so on and so forth so **memory** mostly these studies are these viewpoints of **memory** they are talking about **memory** as a storage system because the tribal is a process so **retrieval** accesses a storage so when I say think about something the process that you use the way that you get information is what is called a triangle but from where you get is what is called storage part of **memory** and so the **memory** in itself is defined in terms of the silikal studies or in this in terms of these old studies as a system which stores information so basically then the question here is whether **memory** is a single store system which means that our information stored in one typical way or whether there are different stores and the **memory** is stored in different ways so from very early on this philosophers knew that there are five sense organs and since the number of sense organs are different is basically five all of them would produce a different input which is not similar to our input from another sense organs which basically means that a number of **memory** stores have to be more than one and so that is one of the debate that how many stores are there and so we'll quickly come to that idea of when how many have been defined in terms of the storage idea also how is **memory** arranged so how do you access **memory** so is **memory** like a box which is full of a number fight em so think of your closet back in your hostel now most of you would have a disorganized closet and so what would happen is on one particular garment the other garment would go on and so a number of dresses will go over it and in the middle of it there will be books and so many things would be there in your closet which means that a disorganized closet now if a disorganized closet is there searching anything from it is difficult so then how is **memory** organized what is the way in which **memory** is organized so are there ordered structures of organizing **memory** or are they under structures and so if there are order structures how are they access how what is stored how much information is stored all these questions we look into in the upcoming several weeks in several different definitions of what **memory** is and so we'll discuss it there but for now the idea here is that people who followed from the trends of philosophy or people in the Middle Ages they had this idea of cave or cabinet or empty house kind of a thing of **memory** which suggests that they thought of **memory** as a storehouse now in the 1960s and in the nineteen sixties in the nineteen fifties particularly the telephone came in Graham Bell has already invented the telephone system and so **memory** was then compared as to the telephone system which was supposed of interconnected wires from a central node and so on in those days in the 1950s the telephone system was compared to the **memory** to house one connection leads to another connection and there is a central node with goes ahead and makes this connection is what **memory** was thought of so different bits of information comes from **encoding** and different bits of information are a tribe and the one area which is connecting it is called the central node and that is where or or it's it's it's it's called the control center where people or telephone operators sit and manage telephone calls the similar way **memory** is sort of similar there is there is a central operator who actually sits there takes in information which which comes in from through **perception attention** and this is called the incoming call kind of a issue and this is **encoding** then the telephone operator takes this information and whenever needed passes this information out which is equivalent to the **concept** of retrial and so this operator then manages this information and so those days this is how that **memory** was thought of and this is how the analog II was developed now in the 1960s the first computers were generated and in those days from that idea on was **memory** started being thought of as a computer system which has an input which has a throughput and which has an output so right from the point where it was thought of for the cave to the development of the telephone system to the development of the computers the idea of **memory** kept on changing from one generation to the another generation no in the 1670s when the first computerized model or computer model of **memory** came in a lot of information was gathered by people working in this area and some of the primary information that they gathered was that there were multiple kind of memories so for the first time people came to know that there was different kind of memories so even within a particular sensory organ so I'm not talking about

whether difference as the sensory organ brings in different kind of **memory** what I'm talking here is that within a specific sense organ also we had different kinds of memories and also the fact that not only there are different kind of memories in terms of a particular sensory organ it is also that these memories have a particular duration a duration for which they last so there are certain stores or there are certain terms for a certain kind of **memory** which lasts for a very short period of time and then there are certain kind of embrace which longer which are there which are available for longer period of time and so these kind of options are these kinds of studies started developing and that was the time when **memory** studies the full form of my studies came into being so with the advent of the computers and the video and with the additional knowledge of the fact that memories are of different kind and they have different time periods to be stored they require different time series to be stored they are accessible for different time periods an important breakthrough happen in the research of **memory** and a model was developed which is called the **modal model** of **memory** as you can see now this model of **memory** suggests that information which is passed on from the perceptual processes and through the attentional processes these arrive at something called the sensory register kind of a buffer where which has a property of storing a lot of information and from there on depending on certain features of the sensory register depending on certain other mental cognitive processes these information are passed on to another store which is called the short-term store and from there depending on certain features or the short-term store and certain active processes that has to we take in that has to be taken up their **memory** is passed on to another kind of store which is called the long-term story that is what it talks about so basically a three store system was developed which is the short term store long term store and the sensory register so briefly speaking then this is how **memory** would look like so the model model of **memory** the **modal model memory** talks about a very short store which is called the **sensory memory** or some books will talk about this tour as the sensory register now this is a buffer store buffer store in the sense that it has several properties so we will look into the idea of sensory registers **sensory memory** long term a very short term **memory** in a while but for now let me introduce you to the **modal model** and so what it says is that at the beginning the information which has been coming so information which is coming from **perception** and **attention** land up at this sensory register now it's a buffer as I said and it's a buffer of a unique status why it is unique it is unique because it can take a lot of information it can store a lot of information but the information duration for which it can store information is very very brief and so depending on certain features here certain processes here mostly attentional processes mostly **attention** or some person deleted factors also this information is passed on to another **memory** we look at as I said in to most of these models or into most of these stores one by one and in the upcoming lectures and so the short the short term store does is it stores information for a very very brief period of time so mostly cents a day just passes on information depending on certain features are certain requirements and these requirements are determined by people's **attention** or intention and so on and so forth and so this information then passes on to something called the **short-term memory short-term memory** is a store it is like a cave it is like a box which can store a lot of information but this information is very very limited because this **attention** what has this attentional filter over here has limited the number of information which passes to the short-term store so limited number of information can be assessed to this store but then the idea here is that this information can stay here for longer period of time it is equivalent to remembering a phone number while talking to a friend so that kind of a thing and so from here there is another store where information can move out too so depending on whether you repeat the stimulus sub vocally which basically means whether you do a **rehearsal** of the stimulus so whatever information comes from sensory register to the short-term store whether you rehearse this information or not whether you go ahead and repeat this information or not that will decide whether it goes to long-term store or not the long-term store has a property that can store a number of information into it and so a number of information for extensive period of time so that kind of structure is what is called a long-term store and this is the model model of **memory** it is akin to or similar to

act insurance reference model but then at keychain sniffing has several more addition to it we'll talk about that kitchen chef and model and another model which is a fake and Oakheart model in a while so let me just introduce you these things here so there are two models one is called a create a low cut model and the air conditioner shipping model now the question was whether **memory** is a single store system or a multiple store system and that was what the debate is and that is the debate which will be solved by that constraint reference to model and they can look at model so we'll come to that can we from there so the question now is how do we experiment ly prove that this is right that this kind of a store appears that this kind of a store structure is there in **memory** and for their that particular thing a number of experiment a number of free recall experiments none and what was these experiments like so in this field called experiment a list of words were given to people so these are number of words which are given to people and they were asked to remember these to learn these by heart and later on they were asked to retrieve this back these words back and to basically go ahead and retrieve retrieve or to tell me these words back or repeat this words back to me and they don't have to follow any sequence that is why it is called a fini call there there is no sequence that did they follow so anyway even if the fact they can remember what number 10 first and what about at two second and so on and so forth so they are free to recall the way they want to the only thing here is that they have to first learn a list and then retrieve a list back now an interesting thing was found out here when this **retrieval** the percentage **accuracy** of **retrieval** and this is the time dimension when this kind of thing was plotted something like this was found and this basically so these are the I'm sorry this is not the time dimension this is the list items items of the list and so what was found out is that this kind of curve which is called the **serial position curve** was what was evident from the results so what is the serial poison curve so on this axis you have the probability of recall and this on this axis you have the list of items which has to be recalled and this is what you tend to get so what does this list actually tell you so this list basically tells you or this kind of a graph basically goes in and tells you that these are items which are at the beginning of the list and these are items which are at the end of the list so let's say in a 30 word list these are item number 30 29 28 27 and these are items number 1 2 3 4 and 5 and here is item number let's say 18 17 15 16 kind of a thing what's the meaning of all this the meaning of all this is that items present at the top of the list those items which are presented towards the beginning of the list are remembered more and similarly items which have been presented towards the end of the list I remembered more which basically means that these two kind of items have a preferential or have three more better **memory** why they have better **memory** this kind of effect when items which have been learned at the beginning of the list are remembered more as called the **primacy effect** and this kind of thing where items which have been remembered from the end of the list is called the **recency effect** so what psychologists when they did experiments like this they found out that this kind of effect was there and this kind of effect basically means that there are two **memory** stores these **memory** stores the one which is storing this and the one which is storing this I entirely different the reason being that this words are being remembered because people rehearse so when a list is given to you the top of the list items these are being rehearsed these are being repeated so item number one you first learn it you repeat it then you learn item two then you repeat item one and then item two again you learn item number three so you repeat item number one item number two and you repeat item number three then fourth item comes in and then you repeat item number one two three four so number of repetitions that item number one gets is multiplied by this factor and so they what happened is in this primary if **primacy effect** have is happening because of **retrieval** whereas the the effect is happening because people tend to say when people were questioned why do you remember list or what happened actually when they remembered items from the end of the list they thought or they basically replied back saying that they could still hear the list they could still hear the words ringing in the ears which basically means that some kind of reasons you are happening and so this proposal then says that there are two different kinds of memories one **memory** which stores information which has been repeated and another **memory** which stores information which has not been repeated and

since these items have not been repeated at all there's not been processed at all they form another class or another domain and these form another domain or another class and so this is called the **short-term memory** in this part is called the **long-term memory** and this is one of the basic demonstrations or one of the basic experiments to show that a structure of short-term and **long-term memory** exists and which basically means that **memory** is a multi-modal store so another thing was found out that to basically verify recency was that another interesting thing was done and what was done is that a list of words so whether recency happens because people hear words or people hear the feeding of your words whether that is the reason for recency or not to test this a list of words were read very very fast so a list of words were read very fast so that there was no time for herself and so when there was no time for yourself in those cases there was no **primacy effect** but still **recency effect** which means that people did remember words toward the end of the list you do here at time that something some information it broadcasted very fast so the end of the list information still stays with you and that is what is called recency but then if time enough is given then you have both the primacy in the **recency effect** coming in to this and that basically is the idea of something called the **modal model** of **memory** or the idea that there are different stores of **memory** and so you can see as this particular graph shows you this is about in **memory** system so you see there is something called a sense a register so stimuli from the world it impinges on to this sensory register it's a buffer and these are the properties of it there is quick scan for importance and precoding these are the features of this kind of a thing the information also in addition information stays there for a very very brief period rhymes or something around one second and then the information here which is stored is huge information so a huge amount of information is stored here also there is something called **short-term memory** or presently in these days it is called **working memory** so we'll introduce you I'll introduce you to the **concept** of what is **working memory** and what is the difference between working in **sensory memory** in the later half of this chapter so the idea here is that there is another store which is called **working memory** and the features of this store is coding **rehearsal** recording and so on and so forth and similarly there is a third store which is called the long-term store which has the features of processing storing it recalling and so we will discuss this whole idea of this is basically a quick view of attrition - reference view of idea of **memory** we'll come back to this in in a while when we discuss the occasion is different views about **memory** so very briefly then if you look into here we have something called sensory register **working memory** and **long-term memory** so then let's for the sake of consistency start with the first type of **memory** if let me explain you a little bit about what it is how does it work and what are the features of this and then we'll quickly move into **working memory** so in the presence section will look into sensory register in **working memory** in the next section we'll look into **long-term memory** so let's quickly go into what is **sensory memory** so what is **sensory memory** most people generally look at **sensory memory** as a record of your percept let me give you an example of what sensory most of you have looked at displays or your professors who are teaching you and so suppose a professor is going ahead and testing whether the slide projects or not and so we quickly project something and then takes it back you do see some form of information or you do see an image of the slide fading away very fast this percept or this particular feeling where you see images fading very fast is what is called the **sensory memory** and this is called this is kind of an iconic **sensory memory** it's visual in nature it is called an iconic **sensory memory** so basically that is what and so **sensory memory** is basically what is equivalent to the percept or what we will call it in terms of **perception** now it's a record of a percept because it refers to initial brief storage of information now you might retain after a quick glance to an object's field so think of looking or quickly glancing so quickly glance towards the room and then move back this just like something I did and so when you start thinking about what you saw you will see or you basically what comes to mind is a fading image of something which lies behind you and that is what is called the **sensory memory** so it's it's a **memory** which is there with you for very very brief period of time holds a lot of information but the problem is that most information that it holds is unprocessed form and so cannot be processed and so **attention**

and **perception** decides whether information passes from here to the next stages of **memory** so **sensory memory** exists and another interesting thing is that we have different kind of **sensory memory** so five different senses five different **sensory memory** are there although the **sensory memory** for taste is a little bit gustatory **memory** is a little bit I would say that fast but then most memories are follow some kind of a rule or some kind of inherent to each other and so for visual we have the icon as a **sensory memory** for a tree we have the Eco and for touch we have the haptic and for the sake of consistency for the sake of easiness since this is not a very extensive course so we will just look at the icon and eco as a demonstration of **sensory memory** so the first kind of **sensory memory** that we have is called the icon now icon is a **sensory memory** for image or it is in an visual **sensory memory** so then what is the icon so nicer 1967 called icon a very brief visual **memory** it's a **sensory memory** storage system for visual materials holding information up to one second horizon as I said sensory memories have very very less time availability so of information is available in sensory memories for very brief period of time but it can hold huge information a huge number of information which comes up to the **perception** process and so what happens here is that most information is available but since the time period for which it is available is very less you cannot look at it so one of the features of an **sensory memory** is that it can hold a lot of information for what for a very very brief period of time and so the demonstration that I just told you in terms of looking at something a quick glance on to something or quickly glancing up or looking at something which is presented for a very very brief period of time the fading image is what is an icon so icon is equivalent to that fading image which which moves out which leaves you and it fades away quickly that is what icon is and that represents the the **iconic memory** so the best demonstration of **iconic memory** was done using something called Sperling's partial report technique in nineteen 60 and avirulent Carell ever ever back in Carell showed that icons can be raised by the stimuli presented immediately after icon will will I'll show you how this happens through something called masking basically means that an icon or a **sensory memory** or the visual nature it is there it is in front of you for a very brief period of time but if something in the time period that the icon is presented is something else is presented a masking stimuli a stimuli which goes ahead and hides this icon for for a brief period or time and an interference will happen and the icon disappears but we'll come back to that in a minute so then how do I test for the presence of an icon how do I test that **iconic memory** is present or not and for this a very interesting experiment was done by Sperling and what this experiment was in very brief terms I will try and explain you is called the whole versus part report whole whole versus partial report technique so the demonstration looks something like this in this demonstration what happens is most subjects will come for this demonstration first sees a central fixation point like this so you this is the central fixation point people are asked subjects for doing the experiment are asked to put their eye on to it and then later on they see a matrix like this we have which have a number of let us return on to it and then a blank slide is presented so what you really have to do after that after the blank side is presented a tone is presented which cues you which tells you to report or to write this particular thing back so very simple experiment you start with focusing on this particular fixation point for 500 millisecond then a display like this is presented to you for 50 millisecond then a blank slide is presented for 0 to 5 seconds this is a masking stimuli this is done so that we are sure that more stimulus that the image of the stimulus is is moved out of your buffer is moved out of the sensory buffer and then a tone is there to tell you that you have to report it back so there are two versions of it one is called the whole report technique knowing the whole report technique what happened is that subjects we're asked to report the whole of this and so what was seen I'll quickly jump to the result so what was seen is that when that was done then people could add the best remember the first row of letters or if they are remembering it for this so maximum four letters is what they remembered so in this kind of demonstration people remembered the four letters and by the time they were up to the fifth letter the image disappeared and that was what was happening so by the time they were to the fifth and six the whole image of the four stimuli disappeared as I said is a feeling experience the whole thing faded and they could

not report it back so the question was can this be helped can this be improved and so something called a partial report technique was designed now it is just an addition into it and so what was the partial report technique the same kind of structure was used for doing the experiment and the only thing here was that three different tools were used so this was a high tone this was a middle tone and this was a low tone so three different kind of tones were used and now the subjects interested in were instructed in this way if there is a high tone you have to just remember the first row if there is a middle tone you have to remember the second row and if there is a third is a low tone you have to remember the third row so retrieve back the items from the third row what do you think happened in a situation like this subjects very accurately remembered the letters they were very accurate in remembering so when a high tone was presented they very accurately remembered all the four letters it basically means that if the first thing that it's it's is that the **sensory memory** stays for a very very brief period of time and all the information of **sensory memory** is available too but what happens is the storage you have so two basic proposals of **sensory memory** that it can huge a lot of infirmity it can store a lot of information but for a brief period of time and so we tested both of them here the thing is that if I present a low tone and the same image you are able to go remember back the third row it basically means that the whole of the slide is available to you but what happens is when you start to reading it back it disappears so the thing is when a technique was used where you have to just remember one part of it you were able to do it which basically means that the whole of slide information the informations on the slide the whole of slide was available to people and what why they were not able to complete the whole report technique is because they were trying to read everything so one problem the other problem solved is that time duration now the thing is if cue was given a particular mid high medium low kind of a cue was given that time was very less and people were able to accurately finish it so then but then when a whole report technique was used since that time period since the time that you require to move from one word to write these words we're constantly increasing so you were actually **forgetting** and so this is a very clever demonstration or a very good demonstration of something called the **iconic memory** now the thing is what kind of cues can be used so here I have used or here in the classic experiment Sperling has used something called the tone cue so the question was to find out whether this **iconic memory** the icons what is the nature of the icons so is icon does icon contain categorized information which means that what is the level of information which is stored in the icon and so to test that cueing was done first by something called physical properties so brightness color these kind of cues were use this kind of factors were used for cueing this kind of a whole report and partial report so with physical properties it was very easy but then if people were asked to report back from this kind of a structure from this kind of a slide whether they saw vowel or whether they saw consonants into this structure people were not able to attain this task or finish this task which basically proves that on the information which is available here is at the very very physical level it is highly unprocessed and so this kind of a meaning related thing can be generated cannot be generated out of it so basically what you can do is that this kind of **categorization** cannot be used or the category based distinction cannot be used for cueing another thing another interesting thing was that can people also generate meaning from it and so as I explained that cannot be done so then what kind of cues can be used so very basic cues can can be used for example you have brightness you have tone you have intensity of the stimulus and these can cue people for the whole report in part report technique but then things like **categorization** if you see a vowel reported so report all the work for example report all the vowels that you see here in this that is the question which is given to you it's a dual task first of all you have to remember what know whatever well is and then categorizes and later on report it so if that kind of a task is given to you where it has certain rules certain kind of meaning information attached to it that information was very difficult to access that kind of task was very difficult to access and and accomplished by get accomplished ly done by this particular thing also one interesting thing if you are interested is is to know that need and super event in 2003 they found out that newer studies say that people are able to prop provide category information this kind of a display which

means that category information can also be extracted from these displays so very recent study and then the time course of an icon is not once again it is handed and not say 100 milliseconds it is 150 to 200 milliseconds that is how icon is basically present in order it is available so let's quickly then look into a counter part of the icon the auditory icon as I would say it is called the Eco so Eco is a contra is is a is a kind of **sensory memory** of the auditory dimension and so nicer 1967 called it the Eco so very interesting experiment intelligent experiment was name done by someone called Maury Bates and Barnett in 1965 to show what an eye equal is so what they did was they presented subject with something called for eared listening task which is similar to **dichotic listening** task so subject was centered or subjects were made to hear to the two years four different channels of audio information and so these information and this is similar to Spelling's tasks so these information was coming from the right rough forward the left forward the right backward and left backward so similar to this think of it in this way so here what happens is that a person sits here and so he is wearing an earphone and so this is my person who's sitting here and this is the headphone that he is wearing so he hears two tones one is forward right forward right forward left for channel so estrella a stereo mixing equipment was soon used and this is backward left and this is backward right so he is you is hearing these four tones now the thing is in the whole report technique the subject was asked to report whatever he heard from all these different different so one string was presented here one string of information represented here one string here one string here so in the whole technique the person has to report everything that he hears from four different so a four channel system was there where is in the partial we put technique what the person had to do was to report from one of these channels whether it is for forward right forward left and so on and so forth so how do you do it the experiment design in such a way that this person is actually hearing this for different different sound and then in front of him are four bulbs which represent front right front left back right and braclet and so one of when one of these bulbs this is the cueing method one of these bulb eliminates this person has to tell what he heard from the forward right or say backward left so the elimination of this bulb will tell you to attend to this particular channel and report back what you hear and so Darwin theory and Crowder 1972 they replay replicated Moyes experiment with better controls and found a much smaller partial to put advantage so in this case also the experimenters found a large large benefit for the partial report case then for the whole report case and so some facts which were revealed was the equate **memory** has larger capacity than **iconic memory** and so this is one interesting thing that the echo stores more information than the icon the equals stores more information one reason could be plausible reason is that echo is a single channel system so the words have to be stayed for more period of time and if I write something let's say I write on a piece of paper round goes to village this if I show you for 100 different millisecond you will be able to understand but until I unless I spell the whole sentence round goes to the village you will not have an idea of what is happening who is Ram where is he going what kind of a thing is there so the whole sentence has to be played and so visuals auditory systems store more information because it has to it's a single channel system to go from one word to the other word to the other word to the other word for making meaning and so that could be the reason why it stores more information also eco eco scan last 20 seconds longer than icons that is what it is and so the so if icon stays for 100 millisecond eco can stay for 120 milli second or maybe hand if eCos can if icons can stay 450 millisecond they equals nearly about 200 milliseconds so then information in the eco is huge it is unprocessed like the icon but the idea is that it is better or it stores more information than the icon and so this is the next type of our second type of **sensory memory** which is there and this is a demonstration of the Darvin turvy encoder experiment where you have left right and so you you do see that these are the words which have been presented these are the items which are presented to different different through different different channels on the headphone and pair and this personal sitting here has to report so **sensory memory** can currently then what is the definition of current **sensory memory** can para chronically it can be described through these things first sensory memories are modality-specific which means that there is a **memory** for visual that there is a

memory for haptic senses there is a **memory** for auditory senses and so on and so forth also the capacities are huge but the length of time the information can be stored is quite slow so huge capacity but smaller time so information stays for very very brief period of time also the information can be stored up appears unprocessed so as we saw that category tasks or several other tasks which require which I if given two subjects to perform on to the icon of the eco they are not able to do it which means that information is available but it is not processed in any way so it's like simple perceptual information unprocessed the perceptual information so then let's look at some examples of what this is so for example in visual **sensory memory** it could be that quickly take your hand and move it in front of you very quick once you do that you can still see your hand moving back and that is visual **memory** in auditory with your hands beat a quick rhythm on the desk you will hear after eco and that is wall is a equate **memory** in terms of tectile **memory** rub the palms of your hand along a horizontal surface and even when this is done when you are finished with it you will still have some kind of haptic sensation a sharp edge in in this case on the edge of the desk and that is what is the sensory so we'll then let's break the class here for today and we'll continue with other types of **memory** so we just see in today's class what are other that why is **memory** important what are the different types of **memory** and we have evaluated just one type of **memory** also the existence of something called a single and multiple store system of **memory** so in this part of present section we just saw two different kind of sensory memories we'll continue the discussion in the next class thank you

Lecture 11: Short Term Memory

■Kind: captions hello so to the second section of **memory** as they discussed in the last class we saw what is **memory** what is the need for it and what type of 20 process this is we also discussed about model model of **memory** what the model model talks about or how **memory** is multi-store in a single store kind of a **concept** so in the single store **memory** was discussed as processes which work or which use a single store for storing but different processes define different kinds of memories whereas in the multiple store model there are different stores and different processes related to each store and that is how the model is defined we also looked at the first unit or the first part of the model model that we discussed in the last class and that was paused the **sensory memory** so we looked at what is **sensory memory** what is the different aspects of **sensory memory** different requirements of it and how does it work we looked at two basic types of special sensory memories these are the visual **memory** and the auditory **sensory memory** now in today's class what we are trying to finish is **short-term memory** so what is **short-term memory** now looking back at the definition of the model model **memory** is a three part system and it's also proposed by at internship and **attention** shipments model of **memory** the first part of which is **sensory memory** and when an information is processed from **sensory memory** using attentional filters or intentional filters information from **sensory memory** is passed on to something called the **short-term memory** now the **short-term memory** store is basically thought of as a store which has limited capacity but it can store information for longer nine longer than the duration which is available the durability of time which is available for the short-term store or the STS also the main difference between the **short-term memory** and the sensory register the sensory store is the amount of information which can be stored so basically a large amount of information is what the sensory register holds whereas the short-term store can hold only limited information so let's continue our this lecture on short-term store and look at some of the properties of the **short-term memory** so starting with the definition of what it really means **short-term memory** when people mostly talk about is a type of **memory** which people use for temporary holding information think of a scenario like this you're on a call talking to someone and within the call itself number is given to you by the caller the person who's speaking to you gives you a number to dial to now imagine a situation where you have nothing at your hand to write this number down so what do you do how do you remember this number or how do you make sure that the number telephone number which has been passed on to you from the collar stays with you till the call ends the most likely scenario or the most likely thing that most people do is locally rehearse this number and so this number till the point of time that you are vocally rehearsing it stays with you if you stop **rehearsal** then this number goes away and this particular temporary store where this telephone number or the number given to you by the collar stays is what is called the short-term store so here mostly if not rehearsed most information which is passed on from the Census register stays for no longer than a second one or two seconds so that's the timeframe which is there and as we saw a time frame for which the information stores on to or stays on to the sensor history's about 100 250 milliseconds so basically the in short-term store is a temporary store which holds on to information little information but for a longer duration of time now as far as the definition goes short-term our primary inactive stores it is the capacity of holding small informations for some limited period of time now the question again comes to us is what is the capacity of this store as we saw the capacity of the other store which was the short-term store was huge it has a huge capacity in taking a lot of information but then what is the capacity of the short-term store also we talked about processes which are active during or which are inbuilt to these stores so one of the processes which helps in passing information from the short-term store to the **short-term memory** is called the attentional processes so basically **attention** is that process which takes in information from the short-term store and depending on people's motivation and intentions and the requirement of the task move information moves on to a short-term store to a **short-term memory** so as concerning the capacity of the short-term store a paper

was written by Sperling I'm sorry George Miller in 1956 and where he defines the store to be having a capacity of seven plus or minus two chunks of item now when I say seven plus or minus two chunks of item what I am specifically trying to tell you here is that chunk is a word and it's it's it's a conceptual word so seven plus or minus two chunks doesn't mean that you have to have nine or five items into it but a chunks of an item so basically let's go ahead and define water chunk is so chunk is a process or **chunking** is a process where meaningful units or meaningful materials which are similar to each other are clubbed together are categorized together we'll look into this in the chapter of **categorization** how does this process happen but for now **chunking** is a process of basically putting into a similar mean items which are conceptually or meaningfully similar in nature for example letters conformer chunk or digits can form a chunk and certain other for example visual stimuli scan from a chunk and that kind of thing is junking so the capacity which has been defined by georges milies paper is called 7 plus or minus 2 that is the maximum capacity of information which can be stored into the **short-term memory** now the thing is this is also called the magical number and very recent studies my nicer demonstrates that it is not seven plus or minus two items but it is rather four plus or minus two items that is the capacity of the short-term store so one thing is the capacity the next thing to like example of how **chunking** really works so if I show you this string which is at the top of this particular slide and if I ask you to remember this if I show this string to you for let's say a hundred millisecond or for one second and then I ask you back to relate back what are the number of things that you are seeing or what are the number of letters that you are seeing here it becomes a difficult job so I'll demonstrate what **chunking** is what **chunking** really does is to make meaningful sentences or Club meaningful sentences together so one way to remember this whole string is to understand the fact that this string has acronyms from the famous detective agencies or basically security agencies of the world so what I do is if I can break into three chunks FBI is one agency NSA is another agency as an agency CBI is an agency CIA is an agency mi5 is an agency and B and D is an agency so these basically can be chunked into smaller chunks the meaning of the chunk or the meaning of the bin which holds these chunks together is security agencies and that way you will be able to remember this whole number of digits or this whole string of so this is basically what is the process of **chunking** then another thing that we want to know is how are things coded into the STM nitrous area explicit to us that coding in the short-term store was dependent on modality so different modalities had different codes in the short-term store so is it also true for our SDM what is the way in which information is coded into the SDM that is an interesting thing to be looked at now the coding in STM generally happens to be in terms of the caustic feature or in terms of the caustic way so information mostly are held on to your STM in an a caustic trace which basically means that this sub vocal **rehearsal** that we are talking about in our earlier example where we were looking at how this study this number was given to you by your friend over a telephone and so you needed needed to dial this number what you tend to do is mentally rehearse it and so when you are mentally rehearsing it you are doing a a caustic **rehearsal** now just to prove in the fact that it is a caustic trace that is the way of how items are coded or how items are stored on to the short-term store or **short-term memory** koenraad 1964 he conducted a famous experiment and so what his experiment was he gave people some letters to remember and a string of letters to remember later on ask them to remember these letters Pat for example Kay's GC or TBC kind of thing and what he found out from the remembrance that when people try to remember these letters back what he found out that people confused or committed errors by retrieving or in in **retrieval** by those letters which sounded similar together so basically letters which sounded similar or which appeared similar they created more confusions or more errors then for those letter strings in which letters were not similar together for example in terms of when a visual presentation of letters weighed on so to presentation of these letters were done in one case KGC kind of a thing was given to people and in in and we in an auditory manner and or through an auditory stimulus in the other version of the experiment this KGC was shown to people and so when this kind of a stimulus was shown to people it was that people made more confusions when auditory

testing was done and what happened here is that K the K or the G confused more with C people made more confusions with C whereas in terms of in terms of the visual people didn't can confuse with the letters G and C so it is basically the caustic code or it is basically the acoustics are the four names the basic speech sound which made the confusion and that led to the idea that it is coding which happens in the STM is mostly in the caustic and form Nate in 1988 he found that people use a caustic code dominantly for STM storage and recall so basically people do some kind of a sub vocal **rehearsal** and it is not mental images it is not imagery that people make off items in the STM people generally make an acrostic trees or an acrostic image I would say so let's not use the word images or in a caustic trace of any information which has to be put into STM and and basically then remember it that way now the question then which is in front of us right now is what is the retention duration and how does **forgetting** take place in STM not to to the way in which the **attention** is done in STM or how items are retained in STM and what is the way in which **forgetting** takes place in STM several experiments fact discussed and so what I will try and do at present is show you two different versions of the experiment in first version which is called the Brown and Peterson task will see that decay has been named as the reason for **forgetting** whereas in the fog and norman task i will try and show you the same explanations the same results from the walk from the peters peterson and brown experiment or how it is explained in terms of interference so basically there are several theories of how items get moved out or items are forgotten and so there are two popular theories of it one is the **decay theory** which basically goes ahead and says that items which are not used for a longer period of time they decay so they what happens is that they trace the **memory** traces erased and they got they get decayed or without use in the theory of not you says that they slowly be there away with time so that raised itself disappears another theory of **forgetting** is the **interference theory** which says that if items of similar nature they come in contact with each other or if one item is presented in the second item which is similar in nature to the first item is presented alongside then one eye both the items will compete with each other and they will interference so basically in the next section in **long-term memory** when we will discuss we will also discuss on to the interference phenomena but now these are the two modes of **forgetting** so what people were wondering is what is the nature of **forgetting** or how things are forgotten in to esteem and how they are stored so let's then look at these experiments one by one the first experiment is called the Brown and Peterson task so what is the task all about and so this this experiment was done by Brown and Peterson in 1959 so what is the experiment all about now in the drawing Peterson experiments a very experiment to look at a digit string was presented to people so digit string like this was presented to people and what they had to do this digit strings were generally a try again a three letter word or three letter trigram a triangle is basically a three letter word which doesn't make sense because if you have sensible word here then the chances of you recalling it will be better because it makes meaning and so a trigram is used and basically the start starting point of tri gang of or who started the use of Trigon was basically event because what he did was he created three letter words of the format CVC which is the consonant vowel consonant format and so in CVC he combined three words together or three letters of the English **language** together so that they don't make any meaning for example K and C or Q K T now these words don't make meaning and so they're combined together and this is what a diagram is so in this particular experiment an arrangement was done in which people were shown this kind of try GAMP and so after they were shown this try again they were asked to come back this trying m to **memory** so basically someone is shown this dragon and they're asked to commit this to **memory** later on they were said to count back from 1,000 increase so the first really nine nine let's say seven and then the next will be nine nine seven minus three so we'll be nine nine four and so on and so forth so keep on counting like this so our task was introduced so the the working of the task is that I distract this kind of heart like I was introduced people were made to commit this to **memory** to sub vocally remember it so we were given this task and the sub Oakleigh remember it and later on a delay destructor tasks like this was given it is called the end back counting task here what you need to do is that you have to count back in digits of three this

kind of a thing and keep on counting now people were stopped at several points of time starting from twenty seconds delay so within the first twenty second they were worse so people remember this task and they started counting back and after the first twenty second they were stopped and asked to relay it back this this try again so this is called the recoil diagram phrase in which after the first twenty second people were asked to stop so within this twenty second people are counting back now one thing is that that this counting back tasks then background and back counting tasks as it is called this requires a huge amount of **short-term memory** it requires a huge amount of **attention** and so most attentional resources will be diverted to it and so chances of the any chance of people to basically subvocal rehearse this this particular thing is not possible it's not at all possible in this task and so this was to demonstrate the fact that this khr or this recall was basically do to indicate now people were stopped at two seconds time after the counting start started so they were made to recall the tagging after two seconds four seconds six seconds till twenty seconds so after every two seconds they were made to stop this counting task and recall back the diagram what really happened what is the result of this experiment the result of this experiment showed that after 18 seconds so this is what the result looks like if you look into it initially what happened is this is the relative **accuracy** which is there in the recall interval of time so what the first thing that shows or that is evident from this particular result is that at 18 seconds the curve becomes a sort of a simple tat and so which basically means that to the recall **accuracy** doesn't fall anymore so the curve runs parallel to this but if you look into it for the first three seconds there is a maximum fall of **accuracy** so this is the portion if I draw this then this is the kind of portion which is then immediately there is an **accuracy** so just let's say half a second the **accuracy** is almost 1 which means that people are almost highly accurate 80% accurate and recalling the or more than 85% accurate and recalling the trigram back but then as you proceed in time by the fur by the first third second by the third second of still the start of that accounting task people almost lose the **accuracy** by or the **accuracy** drops down by 42 to 43 percent by the next 6 second it drops to almost let's say around 35% and then similarly it keeps on dropping till it becomes stable at 18 second and this idea gave or this task gave two prominent result 1 that decay is the reason for the **forgetting** since the task was so heavy the **working memory** task which was used was for heavy what happened is people didn't get time to repeat and so the trace of **memory** trace of this task went on decreasing till it reached a very low level which is nearly 0.1 here the point one chances of **accuracy** here after 18 second also another important thing that was demonstrated here was that the time period for which any item get stored on to the short term store and the time period which was determined through their experiments by Brown Peterson was eighteen to twenty second this is the time period through which an item will be in your **short-term memory** if not rehearsed and after twenty second a **rehearsal** doesn't happen then most items from the **short-term memory** will go away and so the duration of the **short-term memory** from this experiment is twenty second now Brown and Peterson task which explains decay as the reason for the delay can also be explained by the walk and norman task but before that let's look at the results from broaden Peterson tasks the study interpreted that freely talk recall occurs during the decay of due to the key of **memory** traces and though within the first twenty second no another group is said that vagan norman tried to explain this result in terms of interference and so what they said is that they challenged the decayed here we are **forgetting** in **memory** and proposed a different mechanism which is called interference so interference as I explained to you is a phenomena in which what happens is that two items which are similar in nature they compete with each other and so they block the recall of one item or the important thing two remember is that they have to be two tasks have to be similar in nature and they have to be competing if they're not competing if people are not focusing on both the task then the interference phenomena will not happen so basically then the fearest theory says that some information displaces another information making the farmer hard to retrieve as the second item competes with the first item and since they are similar in nature people confuse and the second item is pushed out or interfered out from the **memory** so an explanation of the same Brandon Peterson task or Varga Norman's

interpretation of what happens in terms of **forgetting** and time delay in SDM another task was design which is called the probe digit task for explaining the interference phenomena or interference for getting from **short-term memory** now what was the task it's a very easy task to look at so it's a 16-digit number task so a 16 digit is presented to you at one go so you have things like 2 4 6 8 4 6 3 4 3 6 8 9 4 3 4 2 3 4 5 6 so this kind of a setup is presented to you so one example is given here the last digit now this digit then serves as a cue so or digit like this a string like this is presented to people and what they have to do is to first recognize the last digit and this serves the cue now what is the task the task is to basically go ahead and name the letter which preceded the first occurrence of the last word here what do I mean by this what I mean by this is that this six which is the last digit in this string this serves as a cue and so that is the first occurrence of this the first occurrence of this is here and so the letter B precedes the first occurrence of the six is four and so what people had to say is four was the correct answer so you had two people have two last digit in the number is a cue for the participant a report a number that the number that first came after the first occurrence the queue in the number and so that is what it is and so far is the answer here so this is the task a 16 digit number is presented you have to remember the last digit remember the first occurrence of the last digit and remember the number which preceded the first office of the last digit now there are two versions of this task in the first version letters represented very fast so either very quickly at the rate of four digit per second so this whole string was presented at the rate of four digit for a second and so people had to then go ahead and so first the string was presented and then the string was presented in a manner in that four digits very quickly were presented to people and then from the last one so for for the sixteenths for presentation are there four digits presented together into four different times makes it 16 so you have to last remember the last one remember the last digit and then remember where the first occurrence or the last digit on the fourth panel was there and then go ahead and tell me the first number which occurred in the the first occurrence of the last cue and or there was another presentation which was one digit for a second so 16 seconds 16 repetitions rather than here four repetitions were done with four digits or a sixteen repetition one number at a time was presented people who remember asked to do the task no walk in Norman the reason that if decay cost **forgetting** in STM then participants receiving a slow rate of presentation should not be able should not be as good at recalling digits from early in the number which basically means that if the first occurrence of the last digit in the 16 digit string comes in the first half or towards the middle of the list then people should not be able to remember this but this is not what happened keeping an Underwood they found that **forgetting** in the brown Peterson tasks doesn't happen until after few trials so what happens is this is not the answer they found they found that they have a similar kind of **forgetting** which was then but then if words they're similar if if DK is the reason then this kind of answer should come but this is not what they found they found that quite similar kind of **forgetting** into both the versions of the task and so now this Kapil and Underwood they found that **forgetting** in the brown and Peterson toss doesn't happen until a few trials had happened they suggested is overtime **proactive interference** brings up so what happens is that well in in in the four into four kind of a set what happens is due to **proactive interference** if they disappear in the later half if the first presentation of the cue happen in the later half **forgetting** was more but if the presentation was in the earlier half **forgetting** was less and so this kind of an result was what was form from the worgen Norman which basically suggests that interference is the reason for **forgetting** so this is how the task looks like this is 16-digit task this is the first occurrence of it and this is what you tend to remember and this is what affects our presentation rate versus number of interfering item is looked at so if four items are there then you tend to have so if you look here the number of interfering items are five if this is there at four second pour the 1 second end the four second presentations are almost similar in terms of relative **accuracy** but then if the number of digits interfering various number of interfering items are more which means that the number of letters which are presented or number of letters which precede the first occurrence of the Q so if my Q is like this let's say sixteen digit and so four digits so a block of four into four what happens is if my Q the first

occurrence of Q falls here somewhere then the chances of **forgetting** the **accuracy** of forget will go down in the for presentation group but not in the one presentation group but then what happens is if I am presenting it in one one word at a time there won't be any interference and so there will be no **forgetting** at all here and so this basically is a concrete example to basically go ahead and say that with interferences the reason for **forgetting** and so this graph in itself explains that a four words per second is there and how it relates to number of interfering items is the number of interfering item is five this is where both the list matches but as it goes to seven the one word has lower **accuracy** then the forward items and this happens because in the one word the interference will be more since one words are presented one after another and so there will be a lot of interference but in this there are only four presentations of the interference will be less then the question is is **forgetting** from STM rdk or interference related phenomena now it's a badly first question why because it rules are the possibility for lost by both the phenomena and so broad Baddeley said in 1990 I used that phrase decay does occur from STM alpha and we also propose that decay does occur in fact an essential to avoid catastrophic throw at proactive interferences basically both these people go ahead and say that it is not one over the other kind of a thing both the factors decay an interference they go ahead and then they basically relate to each other to actually go ahead and show that this kind of interference has happened or both the items are responsible for the interference now these are authors believe that information must be updated frequently in **memory** its current value decays to prevent interference with the previous values so another interesting thing that can be that was looked at is that if interference is there is a reason why **forgetting** happens in **short-term memory** phenomena which is called the release from **proactive interference** now as as we saw in the worgen normal study it was basically **proactive interference** which was the reason why **forgetting** happened and **forgetting** happen in the one word item very clear explanation what happens is since is one word presentation so there will be 16 occurrences of it and so more number of interferences but in the for white a word items or for red presentation there are only four different presentations which are there and so interferences are unless the **proactive interference** is the lesson number and so what we have here an interesting thing that we have here is that a phenomena called proactive release now what is proactive release so we concern born an experiment was done by weekends and born in 1963 and what these people said is that the phenomena of proactive release from **proactive interference** should be evident if interference is the reason for **forgetting** in the brown Peters and tasks now what this phenomena really says is that if interference happens because similar items are presented so if items are different items are presented to people then what would happen is they would show interference or the performance would increase and that is what exactly happen so what these people showed that if similar type of items was used in Brown and Peterson task so if letters only may use then people had lower higher **forgetting** rate and that because it was due to interference but then if on one particular expects presentation you had letters everywhere as well as digits coming in so you have a number and a 16 digit number where you have letters and digits fill in to it and so you have to tell me the last occurrence of either a letter in a digit so the display would look like something like 1 to 4 is C's 6 4 3 D K kind of a thing and let's say e is the last year and so or a is the last year and so you have to tell me then the digit 4 which comes out to the first occurrence of the letter a in this case the **forgetting** was known to unless the reason given was released from **proactive interference** which means that when items were switched so here what happened is items were switched and so some items were letters and some items were numbers in this case the **proactive interference** is well less and so in those cases where you have to remember a letter a digit after letter in those cases the interferences were less but if let's say if the last word here is not K let's say if it is 4 and people have to remember the number which comes or that the item which comes after the first occurrence of 4 which is 2 now since 2 and 4 are both digits and so they will create more **proactive interference** but if let's say the last word is here a or let's say that this is 4 and a comes in before 4 in this case is a is remembered more so if the word which appears after the first occurrence of the Q is a different

word it's not the same as the one which is the Q when productive interferences will be very less because for the phenomena which is interference so how is information retrieved from **short-term memory** that's another thing to be looked at and interesting so we looked at how it is stored what is the manner in which **forgetting** happens what is the time and what is the duration for which an item it stays in STM and what is the kind of coding which happens in SDM so in this particular next step we look into how is information retrieved from SDM what is the way in it so Saul Steinberg he designed in explained to looked at how items are actually retrieved back from SDM and what did he find so what was his task like his task was search for information in serial or parallel form so he had people look at a display like this where seven or few letters where so seven or few letters were given to people to be kept in the **memory** bank to be remembered into **memory** and then later on a cue was presented to people so let's say we have seven or few letters which people have to remember so these letters are a B K T kind of a thing and then they are presented with C people are presented we see and what they have to look at is to find out whether this C is presented here or not and so they have to hold this in **memory** and then later on verify this cue or this probe cue whether it is present here or not and then so from his experiments he came up with a very interesting example or a very interesting proposal and what is this proposal he conducted this kind of an experiment and found some interesting results what was it first something called parallel search happened so when comparison of probe is done with all items stored in the STM at the same time then it is something called parallel search so if C is searched with every of this item every one of this item it is a parallel search but when us for comparison or the probe is done with all items stored in the STM one at a time it is called a serial search so the question here was this kind of two searches could be the result result of an experiment now if you remember the first lecture we talked about something called structure process trade-off and we saw how stalls are strong words the same experiment is described here how solid Starnberg went ahead and described this whole process of serial versus parallel search and he gave reasons for what really happens how the structure and the process compensate for each other and is the same thing now within the serial search Steinberg basically proposed that there are two versions of the serial search the one is called the self terminating search in which what happens is as soon as so this C will be searched with a or compared with a if it is not a match this C will again go into searching with B which is the next letter so this is my letter items which is in the **memory** set and since my probe goes ahead and matches I have seen here and when as soon as the sea matches with this a self termination happens so the list is not further search doc although the list has seven items as soon as the sea is there here I ever know so the match or the matching proceeds to be if it is not match here it will proceed here but as soon as the match is there a seraph termination will happen a sort of termination will take place but in another form of the search which is called the exhaustive search what Strandberg says is that no matter the fact that this sea is matched here this C is again matched to the next item which could be a B or which could be another C so the whole number of items which are there in the **memory** set will be searched one after another with a probe and that is what is called a serial exhaustive search and what Sternberg argued is that the way in which **short-term memory** starts is basically the serial exhaustive search so no matter if even if you get a clear search first of all there are no parallel searches in STM there are always serial searches and even if the serial search is done it is not a self terminating search even if a match is found to the probe even if some item matches the item that you're looking for in your SDM the search will continue till the end of the list this is because the exhaustive search takes place and the one probable reason could be the fact that multiple copies of the item would be there and so the requirement of that makes you to do something called the exhaustive serial exhaustive search and so to test so a review done by hunt 1978 found that people of all sorts showed results consistent with the idea that **retrieval** from STM uses serial exhaustive search although the search rate changes with the group now derozen take a 1976 demonstrated that with certain stimuli people apparently do such STM in a parallel way so basically D Rosa came up with an experiment to show that the both the serial as the parallel search does happen in

STM and so the designed sequence are kind of a stimuli to test that and the results of their study was so let's look at the stimuli that they use for their experiment so in their experiment they gave a situation like this stimulus like this now if you look at the stimulus there are one two three four five different stimuli and what we look at is basically a stimuli in action so there are several natural actions of the stimuli for example look at this person who's playing a god so you see that it moves from frame to frame as you look at his divert so he gets up and prepares themselves make this move pens down launches himself into the air then dies forward makes this move dies forward and then lands below or look at how this word goes and reply so these are the ordinal positions these are the positions in which basically the sequence of action should go and these are the number of actions if these stimuli is done so what is the result of a study like this what happens in a study like this so in a study like this DeRosa found that the stimuli consisted both of as I said it consisted a number of ordered pictures in one after another and so the result of the study says that if the **memory** set consisted of some randomly selected subset of nine pictures from any of the cells the results were similar to Starbucks results so if let's say that I give you a number of positions to verify so what I do is let's say from the first stimuli here I take in ordinal positions 1 3 2 6 9 4 9 and then I ask you to verify whether 5 is there or let's say 4 is there in this set or not so that is what you need to do so this is the fourth position which is there and I have selected the first position the third position the fourth position obvious the the second position then then so this is the way in which it it is the sixth position followed by the fourth position and the ninth position and so they are in an unordered list and so you'll have to go and search whether this is there or not so in this kind of search when the queue has to be searched in a display like this in a situation like this in a feature like this then a serial exhaustive search is what happens here but then interestingly if an ordered set is used so instead of using instead of using this kind of a random sequence if I give an ordered set so if I take positions number 1 3 5 7 8 & 9 and if I give you to find whether 3 is present here or not parallel search exists so items when they are presented randomly if the **memory** said if the test item that you have to find out in the **memory** set and the **memory** set consists an unordered kind of a setup and another kind of a list where all ordinary positions are randomly mixed ordinal positions are present then in those cases what really happens is the search is more or less of the type which is popular to what Starnberg says it is serial exhaustive search in nature but then when if I have an ordered position or if I have there an ordered search in those cases so if I take 1 3 5 7 8 9 and so it's rather 3 is present or not there are parallel searches which results here so that is what this says so results were similar to Strandberg when random selected subject was taken but when in order substrates were taken took participants no longer to search for file item then it then 4 2 and so generally there was paralyzed so this raises suggest that SEM does treat organized material in different form the organized material also may be process apparently work differently as a function of material that has to be remembered so basically the search is both can be both cereal or different different types of searches are there but when it is ordered material it could be cereal exhaustive when it is not ordered maybe it is parallel or it could be a simple sell terminating sells such so depending on the material which is being presented in the CAI type of task which is there the it is decided what kind of search will happen now the problem was that Sdn as we saw here is basically then quickly define what it is esta is basically a short term **memory** esteem is a short term store which has first of all a limited capacity so one of the things which you should be aware of is that it has a limited capacity and the capacity here is 7 plus or minus 2 although ian's need came up with a new idea that it has 4 plus or minus two items that is what it is but will not discuss this right now will focus on to the original Miller's idea then it has seven plus or minus two items to be looked at then the second thing is the way items are coded and so mostly a caustic coding happens in **short-term memory** the type of coding that is happens in **short-term memory** or the way items are encoded onto **short-term memory** is elastic in nature it is basically in terms of verbal and **rehearsal** is the main reason for this kind of storage and then the time period for which any item can remain the short-term store without being reputation is 18 to 20 second and this is an output from the brown and Peterson task also the fact

that **forgetting** mostly in **short-term memory** can happen both from decay and from interference mostly it is interference but then we have decay also as a reason and so mostly it is interplay between decay interference which lets you lose memories from short-term store and the last point that has to be that can we remember here or that has any value here is that **retrieval** from this kind of a store is basically mostly in a serial exhaustive manner so serial exhaustive search basically a serial exhaustive way of replacement of this particular materials but depending on the type of material use this serial exhaustive search could also turn out to be parallel exhaustive search or some other form of such which is needed now we have looked at what STM is and what it does and what is the various ways in which STM works but there was debate which was going on within the community of cognitive psychologists and that was that at times STM was somehow not approved of what do I mean by this if you look at the the first instance of my definition of the short-term store I explained to you there is something called the serial position effect and so what we found out in the serial position effect is it's an effect like this so what happens is a list of numbers are given to you a list of items are given to you and asked to remember then what happens is items at the beginning of the list and items at the end of the list are remembered more than items at the middle of the and those two two things are called the primacy and **recency effect** now it was found out in several experiments that if items which had some kind of a personal meaning to people and even if they were presented in the middle of the list they had a higher chance of remembering and so this played several people or several psychologists to go ahead and question this idea of a single store which doesn't do anything but stores information for some period period of time and no act onto it the question this particular store so the question was whether short-term store was a static store or a dynamic store and a number of questions of time also this interference does interference work on to all kinds of items which is stored so if there is a visual item and if there is an order to the item where whether this kind of interference that we talk about whether it happens on on two different different modalities or different different encodings of items which is there so if different versions of items are presented or different types of items uh presented whether this interference will there be there or not and several studies reported that it was not there and so was conceived the idea of **working memory** so in the coming class we look into what is **working memory** and how does is **working memory concept** go ahead and take up or basically replace the **concept** of **short-term memory** thank you

Lecture 12: Working Memory

■Kind: captions so welcome back to this section on short term **memory** and we basically stopped at the point of what **short-term memory** is in the last section we discussed the whole idea of what **short-term memory** is so in this section we'll look into as promised into a new conceptualization of **short-term memory** which is called the **working memory** so let's first look at why this **concept** of **working memory** was produced or that was presented at all the reason why **working memory** was presented was the fact that the short-term store as we discussed in the last class it was basically problematic problematic in many reasons now the short-term store the idea of the **short-term memory** which is a limited capacity store with seven plus or minus two items as the maximum Intel entitlement of item which is based here and also for a time duration of 20 second this posed a number of problems what was it the first problem that it posed is that this store limits the number of information that I can store here so there were limits on to the number of items stored here also in a standard **serial position curve** people should not be able to retrieve items in the middle of the list so this is how a steal position comes look I'd look like so these this is where the **recency effect** happens this is where the prime is here happens and the items in the middle of the list should not show any kind of better recall and so this is the this on this axis we have the percentage of recall on this axis this is the serial position of items so these are items which is the beginning of the list item number one two three four here item so it's a four if it's a 20 item list item number 20 18 and 19 fault here and so item number 11 12 are what are present here so it was found out that these items when they were familiar to people the items when these items men which were familiar to people or people knew it somehow they were so basically then items which are familiar to people they should pose a problem to people in terms of the memories but what was happening is the item suggests which are familiar to people they were issuing metric memories also in terms of interference that we discussed this interference that we were talking about items of different codes or different types of items for example auditory item and an auditory cue or a tree test item and a visual test item when they were presented to people in a sh s TMR where people were asked to remember these two items in STM people were not showing any kind of interference and so the fact of the matter was this stn or the short term **memory** or the short term store that we are talking about was a little more than the way it was received into that kitchen chef and what we know we have not discussed that in two different model very quickly we look into it that concern a shipment model is similar to the model which is called the model **memory** model and so what **attention** and Shiffrin says is that **memory** is a three part system which starts with something called the sense a register has something called the **short-term memory** and then is followed by something called the **long-term memory** so these are what at each end Chafin calls as the stores the different stores and that is why he says since a we store the short-term store and long-term so with these additionally **attention** a friend also defines certain control on active processes which move information either from one store to another or move information out of these stores and so what **attention** shipment model discusses is there is something called **attention** here which makes information move from the short-term register or the short term store to the STM so **short-term memory** and from the **short-term memory** information is moves to the **long-term memory** depends on a process which is called **rehearsal** so this is how the items move from one store to another also decay and interferences are two phenomena through which items move out of these two stores so they came from Rufina interference happens in terms of **long-term memory** as well as **short-term memory** and so this is what a brief overview of Atkinson and Shiffrin model is we'll discuss this model in the **long-term memory** so basically two positions to look at interference and in terms of items in the middle of the list or those items which people are familiar with they showed higher amount of detriment or certain type of rules or certain type of remembrance which required the activity of the LTM those somehow went ahead and helped people in terms of remembering so with this conceptualization another kind of store was perceived or another kind of store was thought of and that was called the

working memory store so the **information processing** model of **attention** Shipman 68 described **information processing** as a two-part process the information representations being stored either as SDM or LTM and the structure defining an estimate LTM store and also the processes which is moving it now these authors they conceived of STM not as a store for seven a few pieces of information for few seconds but founded the information STM somehow activates relevant information incremental tiem and gathers some of this information into STS so this is what the conceptualization of the STS or the short term store or the **attention** shipment model was but then as I described to the problem these two problems led to the emergence of a new store a new kind of **memory** system which was developed by Bradley and H now Bradley in his 1974 they performed the C's of experiment to test the model which was described over which was a tension shipping model of how the STM works and how it is the way in which STM owes information from itself on to the **long-term memory** or how it receives information from the short-term store now the design was to have participants temporarily store a number of digits while simultaneously performing another task which is a **reasoning language** tasks so two type of tasks were given to people and as the STM store has sent personas to item it was believed that if number of items are increased on to the short-term store some kind of interference or some kind of **forgetting** would happen or the task would have a lower performance for performance or the hypothesis was that s team capacity is taken up by stored digits fewer resources available for another task and so performance on the task would suffer as I just described to you what happens if two kind of tasks are given and one task is being stored onto the STS or the short term store then what would happen is that the next task which is given to you if it's a multi-dimensional task paradigm and if a second task is given to you the second task will not be processed or if you process the performance will be very bad on to it and so this is the conceptualization on which Bradley NH came up with a new dimension or a new kind of **memory** system which was called the **working memory** system now what was this task about so let's quickly look at the cast that Bradley and hitch came up with the idea of that Bradley and H came up in and so what Bradley and hitch task was called the **reasoning** task with a recall letter record so what this task generally had is people had first of all they had to keep some letters store a number of digits so a six digit or eight digit frame was what was what people had to store onto the **memory** so they had to remember this six degrees so let's say the digits are eight two eight six four three seven this kind of a setup was given and they were people were asked to commit to **memory** this particular kind of system which was there and then later on they were given another kind of a logical DC's and making tasks so for example here a B the letters a B are presented to people now what they were required to do while holding onto the six or eight digit task into **memory** is basically tell as quickly as possible verify the statements whether a precedes B people had to say true or false or whether P B is preceded by a so this kind of thing was given to people and they had to keep these eight or six days until their **memory** and later on verify these statements this a B is given to people and then shown to people and then they have to verify this a precedes B or B is preceded by a what is the result of an experiment like this no generally or statement like B does not proceed by a this kind of a statements are true or false statement was given to people now the interesting thing that happened here the result of the experiment says that if the number of digits which were two two three if this is the number of digits which I held in **memory** then the performance of people on the letter **verification** task on the a B **verification** task it was similar to what was there if no digit was given to them so two kinds of people whether or two groups of people whether a control group had no digit to keep in mind or keep in the **memory** and then do this very sentence **verification** task or to let a **verification** task and in the next case people had two or three digits in their **memory** and when needed to do this kind of a **verification** task whereas it was found out that if the number of digits that has to be kept in the **memory** was more than six then the performance went down there was poor performance so if sentences also some of the interesting thing was found out that if the sentence was negative if a negative sentence was used for this kind of a thing so B does not proceed by a look at this this is a negative sentence so if a negative sentence or a

passive sentence like this was used now in these cases also the **verification** task was lower or the **verification** task was somehow less but then a poor performance happened so basically then what happened is one two three items are pre-loaded **reasoning** task was done and the letter recall task was done but in experiment 2 what happened is 0 to 6 items are given to people a **reasoning** task when an annual letter recalled task was done so this is the result of the experiment what really happened so Bradley and H they found out that two codes which are different one is the logical code the other is the digit code now if two codes are different then with lesser number of items our digits to be stored the performance was not and becoming fewer all not hampered in any way what does this suggest is such as that the esteem is conceived in the short-term store that if the number of digits are more they will be cooler performances that is not what's happening which basically means that the short-term store as conceived by **attention** is a friend that it is a store which hire which holds on to similar kinds of seven acrostic code in the caustic code form similar kind of seven plus or minus two items was not true so badly English H in 1984 or Radley alone in 1991 they found out that there was no common system for cognitive processes there was no common short term store which was prevalent or which was available for all kind of cognitive processes used in this also the fact that **memory** load does not disrupt the performances so if lower **memory** load are given if two paths are there and if a number if one task requires some kind of a **memory** load or a given little **memory** load then the performance on the second task was was not hindered in any way and so this is the result that we are looking at so **reasoning** times in terms of **reasoning** time later recalled **reasoning** time the later recall in terms of the load was 0 and 6 and as you can see what happens is if the load was more the **reasoning** time and later recall was higher than in terms of so it is equal stress in this case it is **memory** stress and this is the kind of thing that happened so if the number of load was 1 0 1 & 2 and what do you see that the **reasoning** time was very less it is nearly 3.3 1 second in which people were able to perform the **reasoning** tasks but as it increased the number of the time increase to almost 5 second for **reasoning** which demonstrates two things that parallel task processing can happen in the STM store as it was conceived by what **attention** is taken said also the fact that STM's roar was not the way that it was conceived and in in that in right concern and shipment so STM was now a limited capacity store the way it was looked at into **attention** was no more limited capacities or it was it had more capacity is it had capacities which is beyond which which was sufficiently numerous and then there are certain control processes which go ahead and move it also the fact that the way this Sdn is conceived on to that it is a limited capacity is no more true that the capacity is more than seven plus or minus two the way that is looked into also if multiple tasks are given people are able to complete this task and interference that we talked about does is not the reason or could could be although it is a reason but then if two different types of items are taken then both a can be process which gave the idea also one more thing to be looked at here is that both the tasks require different kind of jobs the a in the a B **verification** task require people to borrow from long term **memory** the rule for **verification** for example let's say this is a and B and if there is a sentence the sentence say that a precedes by B now if this is the sentence or the sentence say that B is preceded by a or things that we is not a is not preceded by B so if a is not preceded a is not preceded by B if this is what you need to verify or B is not succeeded by a this is the sentence that you have to verify first of all this is a negative sentence and then the words have a meaning and the way that the **verification** happens is that these rules are actually stored into something called a **long-term memory** so a **long-term memory** has these rules to stored up so what the conceptualization was that these experiments were badly and H they gave up three the brought to light three new important factors about SDF one that STM can talk to **long-term memory** and can at the time of processing itself can borrow rules from **long-term memory** and that is how this **verification** really happens the conceptualization of Atkinson and Shiffrin was that STM is a store which basically keeps items and works and links them through together through a control process but on itself it does nothing to the item whereas a Baddeley and hitch proof for the first time that there can be this kind of a sentence **verification** task

basically gave the plausibility or laid the foundations for the fact that more integrate or more complex verifications or more complex processes can be done on to this kind of tasks which were presented to people also the fact that if items different items where first of all it is not the caustic code in which most items are stored into **short-term memory** it is different codes which are there and so there are different stores within the **short-term memory** is what the conceptualization was and for the first time it was found out that this idea of limits capacity was not true so proof **short-term memory** was not sort of a limited capacity store other thing that interferences do happen but then if items are a slightly different nature then these interferences are not so prominent if the items to task has little items to get look to interfere than to then the differences will not result also the fact that at the time of the **working memory** execution at the time of an item is in to STM with the control processes there are other processes where LTM rules can be borrowed and some kind of processes some kind of operations can be done on to jobs put into this particular store and so a new conceptualization a new idea of a store was thought of and this idea or this store was what was the **working memory** store and so Bradley and H gave the idea of this **working memory** store now this is a very interesting **concept** realization because they still date this is the best conceptualization of to **memory** which wasn't out there and the best results have been approved of this so what is **working memory** then now **working memory** as conceptualized by Bradley and H is a three-part system what happens is this itself this whole part is now called as the short-term store although a new additional item L is included here which is called the **episodic buffer** which I have not demonstrated here but then this is the third system so it's a basically a three-part store controlled by a central part so the conceptualization or the **working memory** is done in this way through attentional filters attentional processes reaches something called the **central executive** so by **working memory** store is a three-part system it has something called the **central executive** it has something called the **phonological loop** it has something called the visual spatial sketchpad and has something called the **episodic buffer** so information which comes from the short-term store through the attentional process in different different modalities reach generally something called the **central executive** of the **working memory** now what are the activities of this **central executive** the activities are to initiate control and this is in process the central YouTube can be thought of as a manager it doesn't do anything it takes in and then manages this information or basically redistributes this information on to two sub stores which are there and so the two sub stores are the **phonological loop** or the articulatory **rehearsal** loop and the **visuospatial sketchpad** now information which is verbal in nature which is auditory in nature or pushed on to the articulatory loop articulatory **rehearsal** loop or the **phonological loop** where as information which is both special or spatial I would say in nature or which requires visual medium which is in terms of a visual medium are pushed on to the the usual spittle sketchpad so what basically is an executive control system it does this takes in information from any medium and throws it into these two stores which other let's look into one by one these stores so basically the negative control system in our executive store is basically related to or its activities are initiating controlling the same process so basically it and measures control it distribute this information different information coming from STS into these two stores also **reasoning language** and comprehension so basically if a task requires people to do **reasoning** or to basically go ahead and and use **language** then in those cases people if **reasoning** or **language** is the requirement of a job then in those cases what needs to be done is that the **central executive** comes into focus and then it does this job how does this does does this job of **reasoning** what it tends to do is talks to the LTM through an **episodic buffer** now as you know then in education achievement model their team is the store where most information is stored now in the upcoming lectures we'll look into how the LEM is arranged what is the way in which LTM is arranged and what is the way in which LTM really works they look into the parts of the LTM but then most rules facts are the things that we call **memory** are stored on to what is called the LTM so what this particular model does is that executive control whenever it needs to do a task it sends or it makes these **reasoning language** decisions based on rules which are borrowed from LTM for example

let's take a very simple task the idea of being more so if i thing is more than something else let's say if a has more number of if there are two jars and this jar a has four items into it or four balls into it and jar b has eight balls into it and the **concept** of more so who the question is whether jar a or jar B is has more number of balls if that is the question which needs to be verified and then the central YouTube does this **verification** by first taking this information the information that there is a jar there number of balls and pushes this information into the visual system then through the **episodic buffer** it talks on to the **long-term memory** and borrows the rule for what is more the definition of what do you mean by more so what it does is he talks to **long-term memory** and the **long-term memory** then three wheels or relates back the idea of what is more on to the **episodic buffer** the **episodic buffer** transfer this information to the **central executive** which then understands what is more and then does the comparison also it transfers information to **long-term memory** by the Ursuline recording so whatever answers that you get from this kind of control processes this kind of rules which are borrowed from **long-term memory** and the information which is inputted through both the articulatory loop and the **visuospatial sketchpad** these informations when it makes any kind of meaning or if it needs to be stored through earlier process of **rehearsal** it is transferred to a **long-term memory** example let's say if there is an event you see your picture from the fourth or fifth class and so you see a friend of yours and now you know this friend is now married and does a job there so this information update which is called recording of information so the friend which is there with you in the fifth class obviously at that point of time he has he doesn't have these features of getting married or having a particular kind of a job or doing a particular kind of a thing or looking in a different way so these informations are added of up to this event and we transfer it back to LT m **memory** so that LT m is updated onto these information so this is what also the **Central Executive** does and effects like recency a prime is here also which is verified by the central executed now the central curative takes in auditory information and pushes all these information into the fore look now the **phonological loop** has a ID or something called subvocal **rehearsal** so it does audited **rehearsal** as I said the conceptualization of **working memory** is better than short-term store because the short-term store only talks about auditory codes but then **working memory** doesn't talk about audit records it talks about both image imaginal codes which is image code and auditory code the **phonological loop** that we are talking about has oddly code as the way of remembering things and so information which is passed on of the STM is if it is in the auditory nature this pushed from the **working memory** onto the **phonological loop** and then this item is rehearsed back is remembered in terms of sub vocal **rehearsal** also this **phonological loop** has two different systems into it one is called the phonological buffer the other is called the sub vocal **rehearsal** now the phonological buffer keeps information ready for being accessed let's say I tell you the word ROM is going to the village now ROM is going to the village this sentence in an auditory code reaches your year it is passed on from the stn on to the **working memory** the executive **central executive** listens to this one channel information Ram is going to the village and pushes it to the or tree loop or **phonological loop** and the **phonological loop** the buffer then keeps brakes away this LOM is going to the village into its constituent parts for example what is the subject what is the object and what is the connector what is the verb and loops it the sub vocal **rehearsal** or sub vocal yourself is the is the process where this whole sentence is loop because it is needed if it is not needed then it won't do loop and so if it is needed this kind of information is go and the buffer is the place where the information stays for some period of time also recycling of items for immediate recall is done by this phonological of a buffer and articulatory basically articulatory processor Daniel but if an item or if a piece of information which is approaching is not in the vocal dimension if it is not vocal in in any format then what happens is that that kind of information is then passed on to the **visuospatial sketchpad** so in the visual schedule scratchpad the information is then broken down into his visual constituents and so visual imaginary special image visual search tasks are done by this visual such task and so here what happens is that the visual images are stored nothing about seeing a movie how is it that we see a movie when we go to a movie there is a

wise and there is an act which is happening there is a sequence of acts which is happening here is the best example of understanding out of **working memory** happens so generally the audition or the auditory channel is a single-channel process for example when I say Romney it is going to the village you have to listen to each word until unless you come to know the meaning of the word which basically means that Rahm is going to the village has to be completely said or completely relayed back to you if that is not done then you will not understand the meaning of the word so basically then Rahm is going to village if I order to present you each one has to be presented to you and so each word has a meaning and so as words appear to the **central executive** they shall pushed onto the **phonological loop** and a fun little loop then sub bocalee rehearses one art after another so rom is real starter is the ROM is is nor here stupid going and ROM is going is rehearsed with two RAM is going to is rehearsed with village and so this is the kind of yourself and then later on you can take an information but and so this that is why auditory channels or auditory systems are basically a single channel system but if I show you this as a display ROM is going to the village now in one particular brief movement you can look at the whole image you can make meaning out of it so that basically means that video systems are more accurate knowing when you are going to when you are in a movie what happens is the dialogue comes as a one channel system whereas the images that you see the actions that you see is basically in terms of several sequences which are there and so there is where the best example of **working memory** is well the **Central Executive** does it takes in dialogues because riders proceed and so these dialogues are taken in and then pushed on to the **phonological loop** and then it is matched with the sequence of the action so the whole dialogue is taken in and is repeated around through the buffer and the sub vocal in yourself and then later on a meaning is interpreted of the scene based on the the auditory feedback based on the dialogue and the action and they are mashed together with the **central executive** so basically then both the information the auditory and visual information in a movie a portion to the store the visual information goes on through sketchpad visual informations are slower it is basically stored down and articulatory information are speeded up and they are mashed together in real time for giving the meaning of what you hear and that is where a good example is now a third stage of this particular **working memory** is called **episodic buffer** and the job of this **episodic buffer** is basically to talk to the LT M so this F is Erdrich buffer basically does a feed-forward and feedback connection with LT and so it takes in the information to the LT m and takes that information also back from it that is what the the idea of this phonological buffer is all about so basically then an experiment was done to be to test the **phonological loop** the existence of the **phonological loop** and so what happened in this experiment was what is the span or what is the capacity of this **phonological loop** so three sentences were given to people they read very fast and so this integer so three kind of centers around the village Sita is doing something and Ram is doing something this is he sent the different sentences were given to people and people were asked to relate back the last word so Ram is going to the village so you have to read back the word way you have to remember the word village back or tell me the word village that so three sentences are given to people and they have to remember back the sentences so what was found out in this task it was found out that this span the **phonological loop** the efficiency of the **phonological loop** or the span of the philosopher loop depends both on the comprehension and the task complexity the kind of task if the sentences are very simple it is easier for people to remember where the words in the sentences are long and complex in nature or if people are not able to understand the sentences the sentence contains words which have difficult meaning in those cases the **central executive** additional input resultant delegative is required then those tasks are not processed and so basically the recall or the span of this auditory loop depends on both the task complexity and the comprehension the way I think is understood or read back also for visual information or we just catch bad similar experiment was done it was found out the complexity or the visual system depends describes the span so if it's a simple system which is there which people have to look into then it was easier for people to relate that the information but then if the information was complex it was difficult for people to relate back the information

so then what does the **working memory** consists of so as I said it consists of a limited capacity workspace let me divide it between the storage and **controlled processing** and in this term which is the **central executive** that we are talking about this component directs the flow of information choosing which information is to be operated on when and now as I said it's like a manager so it takes a lot of information and then decides what information goes where then there is something called the **phonological loop** which is used to carry out subvocal **rehearsal** and maintain verbal material so this is what we were talking about in terms of vocal and the visuals spatial sketchpad which is that part of the **working memory** which takes in information from the visual area now a very interesting study was done to to comprehend the idea of STM and that was with something called stimulus independent thought the SI t now the study was done by a steel in 1995 and so what was the study the study was done to find out what factors are what stores of the **working memory** produces interference with stimulus independent thought now what is stimulus independent thought let's first look into it and so stimulus into independent thought as is defined is basically a flow of thoughts or images that the content of which is unrelated to the immediate sensory input so basically it's kind of a daydream so basically stimulus independent thought is a thought or is is a kind of thought process which is not related to the job at hand to the task at hand so the thing was the task was to find out if STM disturbs by performing another task if people do another task enough of short-term nature if people are doing some tasks which is basically of some nature whether it is visual and for logical in nature whether stimulus independent thought are affected by it in some way or not and that is what the task was so two type of tasks were loaded on to the **central executive** one task was to verify silly sentences so sentences like you can buy priests in shops or a dog a dog was raised on a flagpole that kind of sentences were there and so you have to verify the sentences whether this is possible or not and another kind of task that I used was more awful looking tasks so in this some kind of simple and complex a field was given and within this field people had to find out known geometrical figures for example triangles and circles so this kind of tasks were given as you can see this task is a **phonological loop** task and these are visuals scheduled sketchpad tasks so visual tasks and phonological task this type of task we're given to people and then people were asked to get engaged in something called stimulus independent thought or basically go ahead and dream so in sessions participations so people then did this task and with that they also daydream and so what happened is that at several points of time so at several points of time people were basically stopped so this is how they are looking at this is the stimulus independent thought and they're also doing some other job so these dusts are loaded here and so at several points of time this is the time dimension so they are stopped here here here and they are asked to relay as what is going on to the through their mind so they have to describe their daydream so the task is very simple in nature somebody is asked to daydream and with that daydreaming they are also given two more tasks either to verify a silly silly sentences which could which had no meaning at all whether it can exist or not and then also to do a looking task which is basically equivalent to looking at something looking at a simple field or a complex field and find out hidden geometrical figures on to it so basically then and in in this particular framework people were stopped at various pioneers points of time since they are also D dreaming they asked to relate back what is it that they're dreaming or what is it that they are thinking about so the sparse **categorization** categorizations of society of task it was found out that basically neither the visual task or this kind of a thing both the task together went ahead and interfered this stimulus and if so if kind of extra tasks doing was done if people were doing some either either sentence **verification** test which is phonological in nature or a simple looking task which is which is basically visually in nature this interfered with the stimulus independent thought it basically means that si T's are something which are not independent of this kind of a loop so if a stimulus independent thought if people are daydreaming and they're also given another kind of task which has to be done then it did interfere with the stimulus independent thought so it's basically then what was the result of the study that basically **Central Executive** is the reason why this kind of a destruction happened also it was a version of this task was

done in which people were able to do tasks with the stimulus independent so people were asked to daydream and with this daydream they were asked to do some kind of a routine task which is they always did now in those cases when people did routine tasks and they daydream then the performance or daydream the performance or relating back daydream was not that poor then in cases where people were loaded up with tasks for people they asked to do tasks which required some part of I have the phonological due or the we just catch a sketchpad so basically then it's not these centers are these these areas of the **working memory** identification or **working memory** system which is responsible for destructing the stimulus independent thought it is the central executed so when it asked has been well practice routine in nature then the **central executive** can devote enough **attention** and civilize independent thought can process but when a task requires some kind of **information processing** some kind of jobs to be done or some kind of matter to be looked at some kind of **verification** which requires some kind of an effort in those cases this stimulus independent thoughts got receded so basically in this particular section what we looked into is two things so breaking up front elastics section a section or thinking from the last section where we look at what is short term **memory** and what are the codes our **short-term memory** how to store what is the conceptualization what is the test for **short-term memory** and what is the kind of **forgetting** happens in this particular section we devoted ourselves on to a new conceptualization of **short-term memory** which is the idea **working memory** and we also looked at how this **working memory** is tested and then what is the different parts of **working memory** and what is the benefit of using value the **working memory** model over the **short-term memory** model so one obvious benefit is that the use of **working memory** doesn't limit ourselves to limited capacities first of all then it doesn't basically bound us on to the **concept** of limited time processing plus it says that the kind of coding that is done **working memory** model of short-term store doesn't limit us to a particular kind of code which should be used for remembering so that's another benefit which is there plus the fact that working model or **working memory** model allows us to think about a conceptualization of a store in which the store can talk to ATM and can get informations and so the explanation of the **serial position curve** that items which are familiar appeal to you more and so there's until a descriptive what it does is if an item appeals to you more if it relates to you in some way if the relative 'no sizz very high it is immediately transferred on to the **long-term memory** or it is immediately verified because this item is already present in the **long-term memory** and so the conceptualization or the question that came in from the results of the **serial position curve** that items in the middle of the list which you're familiar are which are somewhere personal to people how they remembered more that could be explained to this model **working memory** so then **working memory** model is a better model of short-term store or the short-term conceptualization and so in this whole three-part lecture on **memory** introduction to **memory** we saw how this **memory** system really works we looked at the short-term store the basic performance of the short-term store of how what happens and how things happen we look at something called a short-term store what is the capacity of the short-term store what it does and what it doesn't and what kind of information is stored and also on to the **working memory** which is a better conceptualization of the short-term store what are its part what is the feedback it takes in how does it connect to the long-term store and what are the benefits of the **working memory** store over to the short-term store so in the next coming lecture we'll look into something called the long-term store or the **long-term memory** we look into the various parts of the **long-term memory** and then in upcoming lectures we'll also look into parts of or different conceptualizations of **working memory** thank you

Lecture 13: Long Term Memory Encoding

■Kind: captions come back so in today's lecture we are going to study about **long-term memory** this is a part of **memory** which has been discussed in that internship and model and this is the kind of **memory** that people when they collect early or when a layman talks about **memory** this is the kind of **memory** that they talk about so what we're going to do in today's lecture is will look into this kind of **memory** so basically mostly tribals that you have any kind of information that you think of storing any kind of life event knowledge any kind of facts any kind of mathematical rule or any other kind of information which you think is stored in your mind or brain is basically what is stored in the **long-term memory** so let's then look at what is **long-term memory** and in this particular section we look into basic processes of **long-term memory** as in how it is formed and what are the basic processes and kind of **retrieval** and all and the upcoming couple of lectures we look into different kinds of **long-term memory** now continuing from what we saw in the last class I discussed about something called at kitchen and reference model so at kitchen shiven model or the **information processing** model talks about a system which has three parts into it a three part system so what we have is a setup like this and this is called the sensory register in the last C's of last lectures we saw what is sense a register and how does it work and what are the various properties of the sensory register so basically any information which comes out or which comes in or gets processed through the perceptual and rational processes basically arrive at the sensitivity and it is a huge capacity very short duration of time and then we have something called the short-term store this is another thing that we talked about in our last lecture this is the kind of system which shows information from brief periods of time not very brief but we could be periods of time but it can hold only limited information and we saw an extension of this **concept** called the **working memory** in our last class so **working memory** is basically an improvement of the short-term store because it talks about the system which has more capacity then as was discussed in that engine and Chafin model and obviously the last thing that we didn't discuss the last part we didn't discuss was the LTM or the **long-term memory** this is the store which holds most of the information even the **working memory** as we saw the capacity **working memory** executed systems which basically make decisions for the **working memory** system they threw the **episodic buffer** connect to this long-term store so this long-term store then holds our most of the information now that kitchen and schriften model just just doesn't talk about this kind of box and arrow structure but they also talk about processes and so these processes are the ones which actually go ahead and transfer information so here the process here is the attentional processes water so I'll write **attention** allocation here so that is the process which takes an information from the sense a register and moves it to the short-term store and once the information is in the store can store it is basically the **rehearsal** process which moves information from the short-term store to the long-term store so the number of rehearsals that we do a number of repetitions that you do of an information number of times you use an information it basically goes to something called the long-term store it gets stored here now the long-term store is not an active store as we discussed in terms of the short term store we saw the **working memory concept** describe the short-term store is an active store said it can do a lot of stuff it can move around a lot of stuff it can do a lot of activity but the long-term store doesn't do any activity it is just like a kind of storehouse which is there so information are there a number of information is stored here it is stored in a highly organized and categorized format but there is no action as such which basically means that things cannot be acted upon if a particular rule or if a particular information needs some action on it some kind of an action has to be done on this information it has to come back to the short-term store the **working memory** store and the process which moves information from the long-term store to back to the short-term store is basically called the read tribal process so basically then we will look into this part this whole part in this lecture now the question is what is the existence or evidence for existence of this long-term store but again let's look quickly look into the idea of serial position car now what was the experiment for looking at the **serial**

position curve the experiment was that unlist is given to people to actually go ahead and read and once you read this list you are asked to retrieve it back and as you retrieve it back what tends to happen is a curve like this is shown in terms of **retrieval** so if I look at percentage **accuracy** of recall which is on this side and this is the item numbers on the list what I tend to see is that items which are recent tend to be retry more and items which are at the the list tends to be they try more whereas item in the middle of the list are not rape tribe and so this effect was called this was called the **primacy effect** and this is called the **recency effect** and very popular theory we have discussed this before now this base basically experiment shows that and the reason that there are there has to be two systems of **memory** there has to be two stores of **memory** the reason why is that in the items which have been learned at the beginning of the list they receive a number of repetitions and so they had retrived back better is because this goes to a LTM store a store which stores information a kind of structure in the brain which stores information for longer durations of time and the experiments are done where it was found out that the **primacy effect** items which show **primacy effect** they remembered for longer duration of time why because it was processed think of you doing the list what happens is the first item comes in so second item pops in so we repeat the first item and the second item third item pops in you repeat the first item the second item the third item and so the number of repetitions that the first item actually goes through or it receives is n factorial times so kind of the list items is there and make a vector low that and that is the kind of repetition cetera see so with more number of repetitions you are prone to more number of more processing and so shifting to a store so this basically here this diagram here shows you that there are two different kind of stores and this is the short term store and this is the long term storage now another interesting thing is that as we saw that the short term store which is basically then improved into the in **memory** store that is divided into its **central executive** the **phonological loop** the VCO spatial sketchpad and the **episodic memory** similarly the long-term store has or the **long-term memory** has several subdivisions so we look into the subdivision in the and as we move progress through this and each of the subdivision we are going to look at or we are going to study one by one so the upcoming lectures will look into all of these stores so basically there are two subdivisions of it one is called the declarative part and the other is called the procedural part so basically the long-term store and itself is divided into two stores there is a lot of debate we'll also look at the debate of what are the declarative and procedural parties but basically the difference is declarative is that part or the declarative is that type of **long-term memory** which you when you retrieve that you're conscious about it so a task like thinking about your first birthday I'm pretty sure that you cannot remember your first but that but let's say you're 12 class farewell or things like that now another interesting thing why you cannot remember your first birthday is up because there is something called childhood amnesia we'll talk about it means at the end of this section but right now that's the reason that I'm just given to the reason so basically conscious things which you are aware of when you are thinking about it or when you are retrieving it is what is the declarative knowledge and procedural knowledge is those knowledge which are unconscious so you are basically unconscious or you do not know that you are craving it example mostly I give this example in my class is riding a bicycle so how do you teach someone to ride a bicycle so even if I ask you to give me how do you ride a bicycle it's very difficult for you to show how do you ride a bicycle and so there are two different parts now within this declarative and procedural part you have within the declarative part you have something called the semantic type and episode it type of **memory** we look into these two types of **memory** semantic stores world knowledge so basically facts rules truths and those kind of thing and episodic is that kind of knowledge which is basically related to episode which is basically related to some kind of event in your life procedural basically there are four subdivisions but will not look in detail into the procedural part of **memory** so I just give you a brief overview of what can come here so you have basically things which are learned from classical conditioning things like habits right and so this kind of structure is there and also priming priming is another thing which comes under **procedural memory** so things priming is basically a instance or it's basically a phenomena in which if you want to if you want to

identify or if you if you are given the task of perceiving an object just before **perception** of an object some information some very brief information is given to you and that way if information leads you to basically remember or basically recognize that our object that is called priming so basically I think of it in this way so when you're going to meet someone before you meet someone somebody gives you some kind of information about that person and that makes you quickly identify that person now this information which has been given to you is basically called priming and similarly there are two types of priming one is called semantic priming the other is called perceptual priming but we'll come to that when we come to that particular exercise but right now this is what it is so let's then start the usual things let's start looking into the usual thing of long-term so what we have done up till now is we have established that there is something called long term **memory** and we have us to also establish that this long term **memory** has some kind of set up which is that which we look into the upcoming lectures so basically then in this particular lecture if we look into the retrial and **encoding** our long term memories basically long term **memory** it's basically described as a place for storage or large amount of information and as you know from philosophical viewpoints this has been quoted as a **script** book of memories or a treasure chest so we split it isn't thought of a place like a box or a chest which can store a large amount of information so that is what the conceptualization of long term **memory** is these are all philosophically conceptualization now the question arises what is the capacity of LTM so there have been several experiments which have been done and so generally speaking the **long-term memory** has a huge its it has sometimes it's called that it has in finite capacity why because whatever you know up till now whatever you've learned up till now in your life and till the point you die it will all be stored in **long-term memory** so basically meaning that most information that you know in your life so again that the evidence here is that **long-term memory** hasn't in by night storage so basically but there have been some attempts to find out what is the capacity of this long-term storage to look at let's look at this evidence so commerce lender he tried to provide an answer by making two basic estimate one estimate is the cerebral cortex has 10 to the power 13 synapses and so what synapse is at if you look at a neuron and in Iran which is there and these neurons they connect to each other through an empty space so basically this is the neuron this is the dendrite of a neuron dendrites are those portions of a neuron which help in communicating so these are two dendrites which communicate with each other and they communicate through both an electric and electrical messenger and a chemical messengers or chemical messengers are stored here but this is something which comes into the biology of **memory** and so let's not get into that particular framework but then this synapse is the space between these two dendrites which are there and so basically this space allows chemical messengers are basically substances which are called neurotransmitters to basically move from one neuron to another so if this is the neuron which is transmitting and this is the neuron which is receiving information is passed through the synapse so synapse is there that particular space avoid which is there and it is very important in looking at how **memory** is so there's a whole idea of how **memory** is told what is the way it is stored and biological basis of **memory** says that there is something called phenomenas called long-term potentiation and long-term depression which is how memories are stored but we will not discuss on to that really quick please look into this so this is this empty voids or spaces are what are called synapses and so Thomas Lanier he found out that there were 10 to the power 13 synapses in the in cerebral cortex and which is the number of information which is stored in the brain so he basically created it that one particular synapse is trained for only one kind of information and so if it's 10 to power 13 then it is 10 to power 13 bits of information which is stored huge No another estimate was made as that there are 10 to power 20 bits of information which is the number of neural transmission made by an average human lifetime and so this particular deduction is made on the basis of the fact that the liar is the life of human beings in the life of a neuron and based on the life of the neuron and life of the human being it is estimated through computational calculations which show that 10 to the power 20 bits of information pass on from one neuron to the another or that's the number of connections and neuron actually a number of neurons

which are there they make and so this is the kind of information or this is the kind of capacity of LTM but then both of them are genetically an estimate of written as I said in genetic speaking it has infinite capacity now the question is how does it account for this infinite capacity so basically think of the archiving program that you have in your computer so you have either a wind you have something like some other I don't know the names of a lot of those archiving systems but then you have these systems or you have these programs which help you to compress file now similar system helps is there in long term every what it does is it compresses file but then it uses certain features which already advanced feature so what what it does is first it selects what's information has to be stored and based on that it also compresses information and then it provides a key to this compression so let's look at how information is stored and how information is retrieved the duration and all those things in the next few slides so basically another question is we saw that the pudding that happens the way information is stored in **short-term memory** was a caustic in nature and so there was a very good experiment which was there which was done by Bradley and so Bradley showed that if a caustic words were used if two different kind of information was fed into **short-term memory** and both for a caustic in nature in the sense that either they had the same **phonology** which basically means the same speech sound and another ones which didn't have the same speech sound the confusion was more for those words which had the same speech sound so P and B we're confuse more PBE controls were then P and K kind of a thing and so basically that experiment showed us or basically told us that the kind of storage that happens in **long-term memory** is basically a costing now what is the format in which it is stored in **long-term memory** so basically this is in terms of again Bradley didn't experiment and found out that this is in terms of semantic nature so what was the experiment so what Alan Bradley did was he gave two groups first he selected two groups of people for one group he gave similar sounding word for example map made and Men two people for remembering and then he gave on other words for the same place he gave people to group for people to different leads layer lists to remember now there are two versions of the list in one version of the list what happened is people were given similar sounding words to remember for example and had her hair kind of a thing or map madman as he are written here and in the other list you had my words from the same list the presentation list but then these words were not similar sounding so you have things like pandering because they don't sound similar and this was to create a caustic confusion and in the group B people were given same meaning words so you see here what happened is you have the huge word the big word and the great word all of them have the same meaning so it's like a tsunami tsunami generally more synonymous are given and then another list was created where people were given words which didn't have similar meaning so it had different meaning for example foul old and deep and recall was done after 20 minutes participants engage and related tasks for the task while basically and back counting so an N back counting task was used now what is an impact counting tasks we showed you in the last lecture in the Brown Peters and task n back counting is basically a task in which you count back n units so 1,000 and if you have to count three units back it will be 997 994 kind of a thing and so this is n back you have to keep on counting back and those has to what the n-back task is all about and so people did this kind of a task this was given so that they should this kind of attack should take up all their available attentional resources so that the **rehearsal** is not allowed or **rehearsal** is not given so what 20-minute wait was done and after that when experimenters are the subjects were asked to retrieve back the caustic **similarity** produces very little effect on performance as well as semantic **similarity** so when words which were similar were they represented or they were shown to people twenty minutes is enough time for any item to get process so when this kind of **retrieval** was done it was found out that in these people in Group A there was no difference so there was no difference between people who were given map map and man to remember but then more confusions were seen on people who were given similar meaning word and so this experiment basically demonstrates the fact that these words which had similar meaning that was what was creating confusion and this experiment also shows that the kind of manner or the manner in which information is stored in **long-term memory** is basically in terms of

semantics or in terms of meaning another reason or another thing for the existence of semantic evidence I would say for the existing **semantic memory** is basically this is called AVENGERS **forgetting** curve so what is the event cause **forgetting** curve and how did it arrive so I explained to you in earlier lecture that Evan was was a German psychologist and he was testing his limits to **memory** of how much thing he can remember how much items he can remember and for how long the time but if he tried to remember words from a list then there would be some a **long-term memory** or there will be some bias into it so what he did was he created a CVC a consonant vowel consonant kind of a try gap we call this as a try again so this triad was so let's say X V and Q now xvq doesn't mean anything but PA t would mean something or pad or ca t would be cat so these have meaning words and these as a tiger so this kind of a dragon was then he took a lot of try and he remembered them and he then retrieved the these back over periods of time this is a very simple demonstration a very simple thing very simple result from his experiment what he saw that the percentages of mean saving and versus the retention interval what happened is the as inputted arrived or one day's after 31 days people still remember this 20% of items from the list so even when he did experiments he found out that from the list that he remembered 20% of items were still with him he was able to retrieve this 20% of item even after 30 days or 31 days and which kept on going for a number of more days so basically meaning that there is a store which allows you to store this information for longer duration of time and that also this idea that you do not get **forgetting** or do not forget after this long duration is basically another evidence for the existence of a store which stores information for longer duration time with more **rehearsal** so what happens is as time passes the you don't tend to forget more and this is an evidence for **long-term memory** so let's do a quick review of a **long-term memory** so basically the question here is they travel transfer the information from long-term to **short-term memory** as I showed to you and **forgetting** is the inability to basically retrieve information so we people often talk about **forgetting** now **forgetting** from **short-term memory** is very obvious because items to get stored for 20 second and so we cannot think of **forgetting** as a **forgetting** here is that again we looked at how does this **forgetting** happen so in the last class we looked at the brown Peter sent out tasks and the worgen dorwan tasks and both of them said that there is a decay kind of a thing and interference kind of a phenomena which basically leads to **forgetting** from **short-term memory** so let's look at why do we people forget from **long-term memory** how do we forget things from **long-term memory** so is there something called **forgetting** from **long-term memory** or if something is encoded into **long-term memory** does it stay there forever so let's look at that quickly so basically this is how it works so this is the **sensory memory** this is the working more **short-term memory** information is rehearsed here and then it's passed to **long-term memory** from the **long-term memory** it is retrieve back as and when it is wanted back to **long-term memory** so why do people forget there are several **forgetting** theories which have been proposed for **long-term memory** and let's look at some of these theories so some of the theories which have been proposed for **forgetting** from **long-term memory** is something called the poor **encoding** Theory the **decay theory** the **interference theory** and a retry will cue theory so some of these theories are what you find so basically the reclaim and **interference Theory** is what also applies to the short-term store but then for the long-term store you have two additional theory so let's look at these theories one by one so let's look at the first theory first theory say is that **forgetting** happens from **long-term memory** because people never encoded that so it could have happened that a piece of information was given to you and you never encoded that information and that could be one of the reason why **forgetting** happened and that could be one of the fact by **forgetting** happened it could be that you will never know example if as is the question where is the tilde key in your computer keyboard so people who have not used the tilde key they have no never encoded whether tilde key is or where is the function key so people who type some people who never use the function key they will never know because they never encoded it that is called **encoding** failure theory so information which is never encoded into LT M is called the reason for it there is a short term **memory** information never pass from the short term **memory** to the long-term

maybe so no **encoding** and that is the reason for **encoding** failure here so some examples are what letters accompany the number five on your telephone if that is what is given to most of you I mean these days you have these newer smartphones we don't have these numbers with it but the push-button phones that were that were there it had to be had a number of a number written three digits or three letters which were written after every digit that is the question most people would not have ever thought about that right and so you don't tend to remember or vary the number zero on your calculator if this question is asked most people do use it and so it's become so procedural or so automatic as we saw in attentional theories at the become so automatic the pressing zero becomes so automatic that it's very difficult for you to know where it is or so according to this theory objects seen frequently but never but information was never encoded into LT m and that is one reason for **forgetting** from LT m another reason why people tend to forget another theory which proposes **forgetting** from long-term story is called the reach rival the failure theory so what does this theory say it says that not all **forgetting** is due to end coding failure some of the **forgetting** is due to retrial failure also so on one poor **encoding** theory the first theory says that it was not encoded but in the next three theories talk about treat rival failure something happens at the time of the **retrieval** getting information back from long term store and that must be the reason why this kind of **forgetting** really happens so let's look at those theories so sometimes information is encoded in Altium but it doesn't come back and so you see this is **short-term memory encoding** is done but it doesn't come back visible the retry will failure leading to **forgetting** so what are the few is there or what are the some common examples that one common example is reporting phenomena the purity phenomenon is called people now there are times in your life when you know that you know information for example movie song is coming up and so you are there looking at it you know the actresses name you know that which letter it starts now you know which heroes she has worked with when which movie she has work with but suddenly there is a freeze and the name doesn't come your mouth and this is called let they put some na minha what happens is that you're not able to retrieve that and that is called a travel failure also can't retrieve information that you absolutely know is stored in your **memory** because you know the first thing of this actors you know vo machine work with a lot of information will basically mean that there is somewhere also the name stored but you are not able to retrieve it an example I have given so evidence of **forgetting** has an inability to retrieve information why we can't retrieve information that is what we are going to see so three theories from very tribal fear failure section one is a **decay theory** the **interference theory** and then they travel cue theory so let's start with the **decay theory** first and so what does the **decay Theory** actually says the **decay theory** this is basically the average percentage of information retrieved and this is the interval between original learning of nonsense syllables and **memory** tests which is basically some something like a brain goes CVC kind of a paradigm so he looked at try games and this is the curve which is that so basically the key theory says that memories fade away or decay gradually if unused so if some **memory** is not used for a period of time it became and there is a biological explanation to that also will come to the biological explanation in a while but that is what it is so if you do not use information which is stored in your **memory** then there would be a problem and this is something which we're seeing nowadays what has happened is the google has taken up so much of her information bucharest taken up so much of her life that we don't try to retrieve information even if we need some bits of information what we tend to do is we tend to look into google we type that and we get any information and voila it's there so basically that's a problem so we don't use it and so I always keep on telling this to my students that please use information if you don't use information it will decay out and one of the fact is the time how much time is spent between learning and retrieving the information that is one of the reasons of how this particular theory works so time plays a critical role now I ability to retrieve info are the clients with time after original **encoding** so as more and more time happens and you're not receiving this information the K will decay will lead to **forgetting** so let's say at some point of time when you learn something so there is a bit of knowledge that you learn our Sanskrit for example if you learn Sanskrit in your in your earlier classes in class seven eight I have not

used it because you didn't follow it you tend to forget some ass or things like some other **grammar** or Sanskrit you don't tend to remember how the Sun D is made that's basically conjugation of sentences how it is done in sanskrit so basically that's what happened as time passes on information is never used and so you tend to forget it so basically this is what the **decay theory** is so basically **decay theory** is a biological based theory and so it gives a reason why decay really happens and what is the reason the reason which has been provided is that when new memories are formed it creates a **memory** trace a change in the structure of the brain chemistry as a new **memory** is formed from the neurobiology of **memory** you'll come to know that as a new memories form what happens is new connections are developed and so when new connections are developed what happens is there are number the new dendrites get attached to each other and information passes or one neuron gets connected to multiple needles but what happens is if you do not use this connections it will be there away the brain chemistry will take it away so you finally unused normal brain metabolic activities will erode these connections these connections which have been formed between two junctions or two dendrites these connections will erode away because they are not talking to each other so to your on should be talking to each other only then a connection is there but if this talking stops on how the neurotransmitter if it stops passing between two junctions the receptor two receptor neurons what will happen is or some change in metabolic activity within the synapse happens this leads to example one of the interesting thing is known is which is evident is dopamine deficiencies or Department deficiencies in d1 d3 d5 kind of a structure so these are three receptor types on a particular on particular neurons and so when lack of dopamine happens here then people show tendencies which are very similar to neurotic people and so this kind of thing happens so if you don't use any connection for a longer period of time it will be that way think of it in terms of a railway store a railway warehouse so in the olden days people these railway warehouse used to be somewhere and so you would see railway lines basically passing through roads but then as time has progressed nobody uses those railway lines or old stations or whole junctions which are there and so it withers out nobody remembers it and there is no real cares about it example then your New York Central Station below that there is another station which is there it's a popular thing which is there and so there is a whole new network out there but nobody uses it anymore and so it decayed and similarly the brain also decays the information from the brain also decays by not getting you so theory not widely favored today in today's world we know heart yeah follow this theory because there are certain brain chemistry's which say that this kind of degeneration doesn't happen in the brain then you have the print appearance theories and what does the interference theories say so **memory** is interfering with another memories is the reason for **forgetting** so why do we forget we forget because one kind of **memory** interfere with our other kind of **memory** now we looked at what is interference and we looked at how does it happen so what happens is that if two items which are very similar fairly similar together are brought to their brought in close **proximity** to each other one will interfere with the other or one will compete with the other so there is something that there is something called a fan effect which has been explained by understand for this kind of thing farrance so Edison in nearly 1996 they did this work on how interference really works so they what they looked at is that any **retrieval** cue points to a particular target this is how they looked at so if this is a **retrieval** cue or this is how you retry something it project it points to a target but then assuming that another target is added to it which is similar in nature this we travel cue then is shared between these two targets and so if more number of competitions are generated what will happen is an interference will happen because this is a travel qc1 is now cueing these targets T 1 T 2 and T 3 and so what will happen is when C 1 is activated one neuron is activated or one Junction is activated now they are not uniquely connected what will happen is all of these targets will get activated and they will interfere with the with the processing of one now **forgetting** not caused by mere passage of time so this theory basically says that **forgetting** is not basically due to passage of time is caused by one or more competing with or replacing other **memory** so the definition the thing that I showed you what will happen is multiple targets with compete with each other so as soon as

this cue is activated this targets will compete with each other the more similar they are the more chances that they will not be retrieved because **similarity** is one way of how information is retrieved so if let's say there are how is the conceptualization of **working memory** there are different bean bags or think of it in the there are different kind of structures out there and so structures which have similar kind of structures which have similar kind of features are put into this bin back we look into that and when we talk about **semantic memory** and so what will happen is if informations which are of similar type they compete with each other what will happen is that they will not be processed so it's caused by one more competing with or replacing another **memory** so if similar items are there they will compete with each other and so they will not be able trivet there are two types of interference to look at one is called the **proactive interference** of the other is called the retroactive independence and a very simple paradigm explains it so let's assume that an experimental group a group is given to learn a list a B and an and while they are doing that a control group is given an unrelated activity to do after a while the same group both the groups are now given a list a C to learn so a B is a list basically a B is a pair associate list so you have how what is a bad associate list in a bad SC at least you have something called a Q target kind of a set up in Q target what you have is so let's say if we have goat and we have leaf here the goat is what is presented to you the leaf is what you have to remember and this is called the bad associate so a pairing between goat and leaf is done and this is called the Q and this is called a target so the while learning you'll learn goat leaf but while you're driving I will show you good and you have to learn leaf now we have done this earlier on international theory and in **perception** all those are part of **perception** also where we showed basically in **perception** of diversity of **perception** where we showed how Ottawa City really works so basically in this paradigm then what happens is at a later period of time you are given to learn list a see now when that happens god roll group shows no interference but then since list a B and list AC has a in common so this will interfere with each other and so the learning of a the running of a C after a B creates this kind of interference that this is called **proactive interference** so here since no list was done the learning of a B creates interference with learning of AC and that is what the **proactive interference** is all about in **retroactive interference** paradigm what you have is both groups learn list a B and a B but group to learn a C and group the control group doesn't learn anything and so what happens here is this AC list the learning of this particular list it interferes with the learning of the earlier list and that is what attractive and proactive interferences let's quickly look into what these really mean so when a new **memory** interferes with remembering of old information this is called **retroactive interference** example is when a new phone number interferes with the ability to remember your old number so let's say you think you have changed your SIM card you have changed your number and when some deletion the old number the new number interferes with remembering of the old number this is basically what is called your **retroactive interference** so quick paradigm example learning a new **language** interferes with ability remember old **language** so this person here she knows French and these are the French words that she knows and so what happens is after that she learns Spanish no peppier and peple both are for paper liver and/or pluma and bloom both of her flower and so this kind of things are there which are very similar in the two languages and so what happens is if she learns French first studies Spanish second and Spanish interferes with the learning of French that is why she gets an F in her exam and this is what is called **retroactive interference** in fact if interference what happens is opposite of **retroactive interference** of an old **memory** interferes with new **memory** so in in the attractive new **memory** interferes with old **memory** as I saw as we saw that Spanish interferes with French I mean **proactive interference** is the old opposite the old **memory** keeps on interfering with new **memory** that is called the **proactive interference** and so in **proactive interference** what happens is memories off where you parked your car on campus the past with interfaces the ability to park car today so what happens is since you have parked this car your cause how many times near your past parked a bicycle in wherever you are eating your department so many times at so many places that that interferes with the fact that where you park today this is what is called **proactive**

interference this is if so old knowledge hampers because you tend to park at some place and because that was not available you part someone else and so the ability to go back or the **automaticity** which makes you go back into the case where you used to park and not finding a bicycle is what is called **proactive interference** so all knowledge basically interfere as a completes with new knowledge now basically as you saw what happens here is that most both the information or similar type because it is a parking situation in both the cases and so **similarity** is there another theory which talks about the failure or which talks about how they travel failure happens is called the **retrieval** cue theory of **forgetting** and so what is the **retrieval** cue theory the travel cue theory says a clue prompt or hint that can help **memory** to retry so basically when any any **long-term memory** is formed as I said it is a cue target kind of any kind of a **memory** is a cue target kind of a system so **memory** is generally has something colon healed or kind of a system so basically when a **memory** stored somewhere in there in the long-term store a cue or a key is provided to it and this key fits with this lock of the **memory** and it opens it so this cue cue is the clue or the law or the key for the **memory** and the **memory** is called the lock or that is the store which is there and so the travel cue is very important because if you use a wrongly travel cue you may not be able to they try also another feature that you have to know what the travel cue is that sometimes one **memory** can have multiple **retrieval** cues what do I mean by that if you want to remember your let's say farewell the 12th pass farewell now that is what you want to remember there are several ways of remembering it one is the kind of importance you were given there another is the kind of feeling that you have one that they may be in their MB so that embarrassment which could make you remember that day or maybe it was other kind of words people said to you or the kind of feeling that you have that day the kind of shock that you got the kind of emotionality which was generated and so several events or several events can link to that **memory** and can bring forth that **memory** and so multiple cues can link to that **memory** and each of these cue has a certain weight to it the most weighted I mean to say the most potent cue will have the highest weight so basically if my **memory** that I'm talking about is let's say remembering an event and the event is my farewell and there are four cues which can make it remember for example the day in terms of what day it was how I felt that day that is another so my feeling my emotion is one way of remembering it or it could be and let's say party an after party which is making it remember or some other event at home which makes me remember it now let's say this emotionality is whenever I think about whenever I become emotional I remember this farewell so it has the highest weight here right and so these have another and so each one of them can think can make you think about this farewell situation but what would happen is that this is the one which will make you the thing more and this is the highest rate so basically what is there and similar situation can also occur where one queue can retrieve so many memories so there are always all types of things of that let's look at this theory so **retrieval** cue is in this this is water cue looks like so **forgetting** the result of using improper **retrieval** cue if you think about a party and if that is what is retrieving the farewell and what would happen is there you would have so many parties in your life so it could be the farewell party it could be your home party which is there it could be some other party with your friends which is unrelated to the farewell and and so forth so if party is the fact or party is the cue that you're using to try to get **retrieval** this has the lowest weight right because you have so many parties but how you felt that day has the highest weight and so that is the most potent thing and so if using improper cue if thinking of **retrieval** in some other way in which it was not encoded if that is what you are trying to retry then that is not possible and so that might not be a good idea of freight traffic so it rival **forgetting** can be basically happening because people don't have the right they travel queue now with **retrieval** cue there are two things to look at one is called a recall and the other is called recognition there are two **retrieval** types from long term **memory** we will discuss this **retrieval** end and recall in detail in upcoming lectures but for now let us look into it so importance of **retrieval** cue is evident from recall versus recognition and what is the difference so recall test must retrieve information which is learned earlier for example fill in the blank type of test so those tests which ask you to remember something from **memory** right and as as demonstrated

you are filling the blank type of questions are mostly **retrieval** or recall kind of or a tribal system so here you have to retrieve the actual information which is stored and in the **cognition** test what happens is it's like in answers in front of you but then you have to basically go ahead and match them and so as you will see that recognition has recalled in building to it because only if you recall something if you are able to recall something and then you will be able to match it so basically remember vs no kind of a thing remember worse is no kind of system so I remember something versus I know something so for knowing something you have to have this remember for remembering something you have to have the knowing kind of a thing and so multiple choice tests are something called recognition yes so for example look at this question what is the capital of Finland no one's is there and so whenever you try to give an answer she oh she ever give an answer to that it happens so recall because when you search a **memory** for it and look at these four answers which are given I'm pretty sure that you know the answer the answer is as inky and so the fan what is the capital of Finland there are four answers now when you are recalling you are doing a free recall so you are searching your **memory** for any information but in the **cognition** you have to have recall also because you have to recall well where does these things fit and so recall basically is a part of recognition and so this is what a recognition test looks like and so you will put in to fill Helsinki because all of them now have a probability of 0.25 one-fourth probability of being remembering so which was easier recall and recognition psychological exam would you rather have filling the blanks of multiple-choice and it's there it's best that you have the **cognition** item is basically multiple choice questions are what's on which what I've been used nowadays everywhere and so this is the most important kind of examination which has been used and so this is the most prominent kind of recall that has been or kind of a recall system which is used everywhere nowadays so in this lecture we looked into some of the what is how **long-term memory** is designed what is the way in which it is encoded and what is so **forgetting** how it is stored and what is the **forgetting** from **long-term memory** or how does this **long-term memory** store information thank you

Lecture 14: Retrieval from Long Term Memory

■Kind: captions so welcome back to this lecture on **long-term memory** and in the last lecture we looked at what is **long-term memory** how it is stored what is its format and what is the **forgetting** that it happens or how does the **forgetting** happen for **long-term memory** and we looked at several **forgetting** theories like **decay theory** the theory of in improper **encoding** the **interference theory** and so on and so forth so continuing from there the last thing that we did in the last lecture was how does rate rival really work and so this rate rifle failures how does it really process and so one thing which you were saying is in terms of the tribal queue failure so if our improperly travel queue is used how does **forgetting** suffer so how do we make this happens really improve this **retrieval** cue and so what **retrieval** cue works the best so what is the way which we can improve this **retrieval** cue kind of a system so one of the things in which can help people to learn to remember that from long term **memory** is something called the **encoding specificity** principle basically queues that whatever way in whatever format you remembered in whatever way you remembered if I have the same context back in whatever context you people remember if I bring the same context back people tend to remember more nothing of when you are giving an exam what you tend to remember is in which page the answer was there what was the page like what was drawn on top of the page everything comes to mind but the answer doesn't come to mind and so if people are tested where they learn back the chances are that the context because this context service additional cue let's see if this is the cue which is connected to my target there are several other cues from the environment which are present when I am learning it for example the atmosphere which is there the kind of noise which was happening the state that you were there in the amount of the type of people which represent when you are learning and so when you are remembering it back all these additional cues all these additional features will also help you in recalling back things or following that the answer and so this is what is called the **encoding** principle so if the context this is what the context look like if the context is the same here then people tend to remember but and so let's look at what **encoding specificity** principle basically tells us so in **encoding specificity** principle we have two different types one is the context dependent main **memory** so improve ability to remember if tested in the same environment as the initial learning so in the if people are learned in a particular environment in let's say they will learn in a library if they test you back if I test you back in a library it is much better if I if I make you read something in a classroom and test you back on to the classroom the other features of the classroom where you sack the kind of seed that you sat in the kind of people that were there with you all these will help you in more **retrieval** because the number of **retrieval** cues will increase and so number will travel queues will increase which means that the chances of recognition increases as each queue has a weight and so the added weight will increase the efficiency of queuing so better recall if tested in a classroom where you initially learn then you went to move to a new classroom or if running room smells of chocolate or what ball people are better to remember if that the chocolate smell or the chocolate mud ball smell is same so compared to different smells for no reason now another context effect that is popular in most researchers and psychology and particularly the kind of research that I do is basically called priming effect so basically working something for sleep and **memory** and so one of the things that we have looked into is the chronological cycle the basically the ultrathin and the there isn't the chronical reason that people have so people have the set rhythms there are ups and downs which are there and so in within days the system the human metabolism goes up and down so that for example at two o'clock in the morning 2:00 to 3:00 there's a deep in this rhythm technological the chrono biological rhythm which is there so that is what we have to take care of and we are testing people are when we are doing psychological experiments because if you learn in a high in a in a high metabolic state and if you not retrieve it back in a low metabolic state you will have differences so we have to think about that and so this is another factor which you have to think about what time of the day is also important let's say that if we remember something run at 3 p.m. and then it is better to perform that at 3 p.m. there in

9 p.m. they call 3 p.m. is the first nadir is the first dip that is there in in your biological cycle and at 3 p.m. the metabolic rate or the metabolic system is at a very low rate and so if none at this metabolic rate remembering back at that metabolic rate is much better than remembering at any other metabolic rate when it is at 9 p.m.

so an experiment was done with diverse to find out whether this works or not and so the divers women do not a particular kind of a list or list of word were given to people who dived under water and so they learnt the list on land and on water and they again so diverse they use there were two kind of divers one who was sitting on the water over the water and one under the water and they learned lists over the water under the water and later on already tribal was done the trial was done both on land and underwater and so what happened is those divers who learned list underwater the **retrieval** was better underwater and for those divers who learned list over water on land their performance was better on land and so that is what you see water land land water water water land land in both these cases the performances are better than so they sir always better than the performance here so words under hard under water are better recalled underwater in words and cut under land are better equal under land now with context effect you also have in the **encoding specificity** you also have something called state dependent features so basically improved recall related to internal physiology or emotional state contact dependent effect external environmental factors and state dependent effect depends on the internal physiology so what is the body physiology when you're remembering something when you are recalling something that also has a lot to say about how recall really progresses so what you are trying to achieve or what is your body physiology will actually also support you are the tribal and so for example moods or emotions have a lot to think so if you are negatively emotion or if you are feeling negative when you are learning a list and then we are trying to retrieve back this list if at this point of time you are still feeling that negativity this will lead you to learn better so people who learn under certain emotionality when they retrieve it back that happens or for people who it is better or people who learn and a certain mood determining it back under the same mode is better so bipolar depresses information learn and so in experiments were done on bipolar depressive it was found out that informations which people learned these people learned of the manic state they when they try to test it under manic state the important recall was better than when test another depressive so these many peopre see people go under phases so this is the kind of phase this is called the mania phase where they are hyperactive and this is the depression phase where they are not where they are very depressed so no kind of activity again mania and negative depression so this is called bipolar disorder they keep on switching between mania and depression so once once they are very excited if they learn something they are very excited and they try to remember this when they're depressed doesn't work three qualities for if you are drunk when you are learning something and you recall better if drunk then and the sober so basically wouldn't suggest that but then if you if your metabolic state is having alcohol if you have heightened alcoholic content or your metabolic state is such that it's supposed alcohol intake and if you learn something there then recall is better when you are drunk then when you are sober because this is where it was encoded and this is how it should be another feature or another effect which promotes recall is called the **spacing effect** what does it say so better to study for short periods of time than for one long period of time so if there is a list of item that you have to remember or if there is a number of items that you have to remember it is better to space it off with basically means that break it off so what you tend to do is if you are learning a lesson what you tend to do is break that lesson into several parts remember the lessons in two parts learn this lesson in two parts this is called this **spacing effect** and so event was also tested this is called the massed practice versus spaced practice kind of a arrangement in math practice event was tried to learn the whole list in one go so hundred lists with 100 words each in one go and so it was not easy to do but what he did was when he tried to learn only one list take a break again take another less than none it was much better this is called the **spacing effect** a **spacing effect** what you tend to do is take the whole section or whole chapter which you are learning break it into smaller parts and then start

learning with it and then I think that is a good idea to learn that is to learn how to better learn something so one hour per day for eight days leads to better recall than eight hours of cramming in one day and so this is what I see more students doing just before the exam what the to do is that they start climbing or they start losing the sleepover and so they go into this for five hours of learning everything and then coming back and then there is a failure why it's a failure because spacing something gives you enough time for the neurons to basically recuperate back to basically regenerate back and so they can store more information but then if you keep on exciting and you're on over a period of time these neurons also have a kind of an no boat kind of a system where they don't get excited after a period of time there is a show this the dull state act at that point of time and so they cannot excite after a number of excitations and so that is what happens and so they are not able to remember anything so why because **encoding** variability best to encode in a variety of ways to attach a wider variety of **memory** cues to the material spacing practice allows you to have more variability what will happen is that if you break the lesson into several parts and start learning it there will be so many other cues because as you take breaks and you come back learning the context the surrounding will change and the number of surrounding cues will be there the chances are that even if you do not remember one part of information there are several other parts you will remember but if let's say in one context you are learning and if something goes wrong if you forget something from the context the whole information will go away so it's like learning things in in small chunks and when that happens where as they will start learning in different different chunks the environment keeps on changing the number of changes of the environment will give you more number of cues more the number of cues more the chances of remembering and that so that's what really happens and that is what **spacing effect** is all about so we were also discussing about something called the different kinds of **memory** and so that is what I showed to you so there are several subdivisions and **memory** as I said there's something called **semantic memory** which is the **memory** for general knowledge so it stores most information into it stores information like the facts arithmetic rule its stores information like how what is the meaning of bigger what is the meaning of smaller it stores information like what is the capital of Finland what is the capital of united says who are the first u.s. president who was the first Indian president who is the only lady president Prime Minister of India or things like where is America from here or what is a wheel cart what is a cow that kind of information is general knowledge information because these are something which you need to know and we which are facts which today so also informations like where will the Sun Rise when will the Sun set and that kind of information now **episodic memory** those which have events or which remember events so this is things like my 12th class farewell my first birthday party my first fling which was there my first day at school my first day at college those kind of things my first party the most amazing party in my life and all those highlights of your high life events of your **memory** or of your life is what this **episodic memory** is it's like an episode so it's like a film where as **semantic memory** is like a text so it has knowledge into it but no even related to it episode acknowledge or **episodic memory** is like a event so it is like a film so you have all the sounds and lights and everything into it and that is what an **episodic memory** will look like but **semantic memory** are bare facts it's like a newspaper so a very crude comparison is **episodic memory** is like television but then the **semantic memory** is like a newspaper but then whether I say episodic and semantic you do have to remember where **episodic memory** has to have **semantic memory** into it but **semantic memory** need not necessarily have episodic events example when I say what is an Apple define an Apple and when you define me an Apple you do not really need to think about an apple or when you saw your first act first episode with an apple you can say that it's a fruit is a red green blue in color it's not blue in color but then red of green in color or it tastes like this that is the kind of information that you relate but nowhere along the line do you have to think your first encounter with the Apple you do not have to think when you met your first Apple or when you ate your first article or where you learn the word Apple but in **episodic memory** you have to have segmented **memory** embedded into it for example I say your first birthday when I say your first birthday there is a **schema** there is a basic event that you think about so

what happened in your first birthday so in your first birthday you might have got a toy or you might have got a certain kind of cake for that you have to know what is a cake what does it taste like what is the meaning of the word cake and what is the meaning of the word toy and what kind of toy so with movie you have newspaper items also or facts and knowledge also but in a newspaper you do not have these events into it so newspapers are static and TVs dynamic and that is the difference between what are semantic and episodic memories so basically most and then the semantic and **episodic memory** are what are called **explicit memory** because when you remember it you are conscious about it you consciously think about it so when you think about Apple you consciously are searching your **long-term memory** for DiNapoli's and if I ask you to differentiate between what are Apple is and what in our range is or what a mango is you consciously do this task you ask your **long-term memory** what is an Apple what is an orange and what is mango and differentiate between it so you bei are conscious you are aware of what you are doing or if I ask you to tell me your farewell how it went or let's say your first day in college then you will relate events of what happened you came here whom do you meet for the first time what are the events which took place how it unfolded and so on and so forth you can briefly tell me like a movie so you are aware you are interacting with that particular thing and so both of these are consciously nature whereas there is a kind of **memory** which is called **procedural memory** which is implicit in nature how it is impressive in nature for things like how do you ride a bicycle if I ask you how to ride a bicycle or 21 days to bicycle or 7 ask to teaching bicycle most of you will not be able to tell me so most people have tried this in my class and I ask them how do you ride a bicycle most people say you get on to it and start paddling but then how do you get on to it and start pedaling that's very that seems very funny but some people say you sit on it and you start riding if you sit on it on a static bicycle if you fall so basically then there are steps to it but you cannot vocalize it you cannot go ahead and they late it and that is the problem with it so **procedural memory** or non conscious they do not interact you do that so bicycling is something which you do but you're not aware of it and a popular joke is that somebody said someone when he was riding a bicycle of look the tire on your back of your bicycle is not moving he saw and he fell why because he became conscious as soon as he became conscious the bicycle from the bicycle he fell and that's what it's all about in **implicit memory** because you are not consciously aware as soon as he become consciously aware then problem arises so these are the subdivisions of long term **memory** and so also how we will discuss about these **memory** types in the upcoming lecture there are also so as we discuss these explicit and implicit memories explicit memories are called declarative in nature they hold facts information in ideas and then there are procedural memories they they look at things like how things are done how a particular thing happens and so on and so forth so basically these declarative procedural differences is in terms of explicitly or explicit e of it so whether you are aware of it or not and so this is another distinction which is there in **long-term memory** the procedural versus declarative type whether a member is conscious or not and within the declarative as I said you will have three types sorry two types you will have the semantic type which is facts and knowledge and we'll have the episodic type which is basically the events of your life and within the procedural you will have classical conditioning so those things that you learn from classical conditioning you will not know so fascicle conditioning is a way of learning in which something is given to you so it is called reinforcement induced learning so you do a particular behavior because something is given to you for example think of those chips packet which say that buy it because something comes free with it so free is the reward that you get and because of free you are buying something and that is classical conditioning so you don't think too much and that is procedural in nature or you have habits habits are also procedural in nature because if your habit of let's say I have a habit of scratching my head if that is my habit I don't think about doing it or I have some other habit which is there so habits are mostly non conscious forms of **memory** you think about it and similarly the third form is called priming in priming also you don't consciously think about that the information is provided to you so basically conscious the information is not given to you some kind of information brief facts about something is provided to and that

facts help you doneita **perception** is what is called priming and so these are the three types of memories which are out there now there is a level of processing framework which is which has been used or which has been named in addition to the Atkinson and Shiffrin model and so what is this level of processing framework this level of processing framework says that there is no one store two stores of **memory** so that can change - model say is about three stores of **memory** so I have a short-term store I have a sensory register I have a short-term store and I have a long-term store into **memory** so three stores to be looked at and so what happens here in this case is that there are three stores and three process which makes you basically go ahead and process information and so there are processing systems or there are processing process which moves information but then there is a conceptualization of **memory** which sees that it is not three stores which slows **memory** there is one store and that is called the level of processing store so what the level of processing store says is that information there is one single **information processing** or store but then what differentiates different kinds of **memory** is what is the kind of processing that is applied and so they talk about two kinds of processing one is called the **elaborate rehearsal** so what is the kind of **rehearsal** that you do with information that will decide how you have processed an information and the other is called the **maintenance rehearsal** so in maintenance yourself what happens is an information is basically processed or it is basically a it's basically rehearsed for just the maintenance purpose so remembering a phone number from a telephone conversation for dialing it further at some point of time is basically maintenance yourself because you do not want anything in future with that number and so will not process it at a deeper level but elaborately your cell is basically processing something at a deeper level so let's say somebody gives you a phone number and this person you do not want to remember for the rest of your life or any part of your life and so what you will do is you'll dial these numbers on extend your dialing call center or somebody for help which you won't want a further in life that is basically maintaining a secure cell so you will repeat the numbers to the point of time that you dial it once we else connected to it the number is of no meaning to you but let's say that in your first day in college a beautiful girl approaches you and your number and now you are immediately head-over-heels for this person you remember that number each number is now related to a meaning is provided of meaning for example flowers this that all kinds of things are associated with each number which is out there and so that number you might never forget and that is what is called a laboratory **rehearsal** so a laboratory answer is a process where you attach meaning kind of a meaning which makes information **retrieval** easy so more processing you do the more **elaborate rehearsal** it acts out to be and so basically to to prove that this kind of a system exists participants were told to answer questions as quickly as possible on words which are shown on a screen so for example the word which is shown here is screen so three type questions were asked one is what is the capital letter is it in capital letter what does it rhyme with and does it fit with the sentence for example the - job over the mailman so three types of questions were asked quickly they had to relate this question so word was shown to people and they were asked to retrieve back these kind of questions what do you think happened so later on a surprise recall test participant showed better **memory** for words that have fit in a sentence why they remembered better because this the word then was processed from meaning so that - jumped over the fence or jumped over my friend or jumped over to meet me is actually the word dog and so when it is processed for the word Bo G which now has a meaning so when we are asking how many capital letters were there or whether it is capital or not you are not making meaning of the word do G or if I even if I'm asking questions like what does it rhyme with frog or not I'm not interpreting it in a meaning level I am interpreting it is non-caustic level but when I put this new sentence it has to mean something that sentence need a completion and for that the word has to have a meaning and that is why it is required better so better recall because deeper processing for these types of questions which force participants to think about the meaning of that's what I've been saying so this is kind of and so the dog in the last sentence had went through something called **elaborate rehearsal** deliberate **rehearsal** is a system in level of processing which says that information if they are processed for a deeper meaning

for some kind of meaning they go through algebra herself but information like what whether the word was in capitals or not or whether the first letter was capital or not required maintenance yourself because you don't need to process the word for some kind of deeper meaning now a feature of **memory** that we have to talk about is called reconstructing **memory** as we saw a **long-term memory** is generally rika reconstructive in nature which means that it it suffers from several inaccuracies whenever you store something an exact replica of that doesn't come back there is always male ability in it there is always in accuracies in it and that is what is called reconstruction of **memory** so basically retelling of stories leads to distortions of what is remembered and so this kind of **reconstructive memory** was testing of reconstruction of **memory** or that may be tiny accurate were tested long back by Fredrick Bartlett and what Fredrick Bartlett did was he wanted to test how these reconstructions work in terms of everyday life process so what he did was he gave a story called the war of the ghost to a couple of people to read and then later gave them some time to recall back the story now when people recall back the story they changed the story according to their own biases according to what they wanted to so most of the items in the story was there but then the story was somehow change in a manner and this is called reconstruction and this is the first evidence of reconstruction so the kind of **memory** that you have the **memory** system that we talk about is highly inaccurate in nature it is highly reconstructive which means that only gist's of information is remember when you see a scene when you hear a particular audio of something or a movie or this happens with us also if you go to see a movie and when you come back from the movie and ask you what do you see in a movie you will be able to tell in five or six sentence what you find a movie now the movie was three years which means that you have to remember each part of it but that doesn't happen what you tend to remember the gist of it and that is what memories are highly deconstructed also the way you will describe the **memory** base is based on your expectations so if you expected a **memory** to be action the you will highlight the action part and you will not tolerate some other part and so that is basically called reconstruction so **eyewitness memory** is subjected to distortion when leading questions are also there is there is a phenomena in reconstruction of **memory** which is related to something called **eyewitness memory** so whenever an excellent happens is people who are witness to the taxon accident are called eyewitnesses and so it has been found that people who relate accidents or relate events they have highly inaccurate memories so what happened or what could be done is that if certain questions are asked in frame in such a way that they they give them a leading answer or they give them some kind of information they tend to change their **memory** or reconstruct **memory** for example in in an experiment which Loftus did where they showed people two cars collide with each other with certain speed and later on they were asked certain questions questions like this another cost paths are at the stop sign or what sign were actually yell sign a participant later also recognized stop signs 59% of the time because what happened is that in the initial picture Loftus didn't show the there was no sign no stop sign and this is basically something which Loftus added on to the participants and since the question gave them all the questions which are asked the very leading question in match their **schema** it's match the way they saw the incident that leads them to make false recognitions so basically an eyewitness testimony the facts to be remembered is that recall is not an exact replica of the original event what really happens is that original events are not recall and so there is always recall that also recall a construction recall is a construction which is rebuilt from various sources so the different different sources something we can remember from experience with matter into it so whatever you are recalling something back it is basically a reconstruction is basically reconfiguration of several facts and that can come from several sources it could be something which you actually saw something which you actually think you saw something that you read which you actually know and so something at which something somebody else said all this information will add up to form the recall orphan fit memories into existing believes or schemas so there are certain schemas which is there and so memories tend to fit into the schemas a **schema** is basically a container so when I say classroom the classroom **schema** is basically the idea that there will be one teacher when stool some

students they're talking to each other of the class board and this is an expectation so **schema** is basically this kind of a framework which is out there and so when something fits in through the **schema** it is basically easily processed so if I say that student came to the class with red shirt on it is acceptable but when I say a student beside the student in the class how are sitting not possible because I was not in a **schema** for class and so this kind of a thing is that so when things fit into the **schema** so it is perfectly okay to say somebody came with a pink shirt or a pink trouser to the class that's perfectly okay and so people will believe it and that fits a **schema** and so people that's how I witness testimony is basically manipulated so **schema** is a **mental representation** of an object seen or even example **schema** of a countryside may include green grass Hills farm and so that's what I was trying to explain to you similarly there is something called scripts which are mental organization of events in time so what is the **script** a **script** is how mental the how things occur for example let's say dinner not dinner is a **script** because then you expect something to happen in sequence for example you go to a dinner and you start with the first course which is the soup generally or some other hot beverage which is out there then there is a beverage to follow then there is a scores a second course a third course followed by sweet and then there is an ending with an ice cream so we expect a time-related fact to happen or traveling in in in bus or a car and so you expect a certain sequence of event to happen in this sequence of event is what is called a **script** so example is a classroom **script** coming to the class sit down talk to a friend bell rings instructor begins to speak and so on and so forth so is there is a **script** of this particular thing **memory** distortions so **memory** can be distorted as people try to fit new information into existing **schema** so basically if as I said if I say that a person came in a blue trouser a red trouser you expect that in a classroom that can happen so it will fit but if I say a person came who was not wearing trousers or not reading anything in a classroom that's not possible and so that cannot be accepted so giving misleading information after an even causes subjects to unknowingly distort their memories and incorporate new memories into people and this is called the distortion of **memory** or misleading information and this really happens so as I explained to you the Loftus experiment that they were discussing let's look at what the Loftus experiment was all about so subjects are shown a video of an accident between two cars so this is to card there's an accident to it and some subjects were asked questions like how fast were the cars cars going or when they smashed into each other and for the other question people were asked how far the fast way now the key word is smashed here and the smashed means that they not actually touched each other and when they actually went ahead and had on collision and so that is the key word here and so this kind of an experiment when it was done with people when it was asked how fast the cars going when smashed was used most people had a average rating of 41 miles per hour but when it was a contacted word was used the speed came down and so this is how false **memory** is generated in the Loftus paradigm two more types of **memory** which are there in **long-term memory** is the autobiographical **memory** in the false **memory** sorry the flashbulb **memory** into this one by one so basically what is autobiographical **memory** it's a real-world **memory** which are more durable than laboratory memories so basically automatical **memory** is a **memory** of yourself so what do you do or what are your personal memories of what autobiographical memories are so some items are forgotten because they are hard to distinguish from other similar memories what a vital memories generally are memories that people have about themselves so your own **memory** or what happened in your life how it happened what events took place and those kind of memories are quite autobiographical **memory** now see when to single even memories are often combined into extended summarized events so as you progress in life what would happen is that one **memory** could be connected with so many other memories and so the **memory** gets inflating and so your **memory** of who you are as a child as you grew up this adds on to it with several other factors and so what happens is every other members combined with it and then you see your childhood as a particular thing but when your child and when you what **memory** you have and when you become an adult and what **memory** of your childhood that happens or what you remember is basically what is called autobiography so their actions

are more likely to be recalled that frequent action so those features which are rare which never happened too often are what are remembered in more in autobiographical **memory** another kind of **memory** which is out there is called flashbulb **memory** and what is this kind of **memory** so this kind of **memory** was mostly related to emotional events and so flashbulb **memory** was popularized why because it was believed that certain events in your life create certain flashes or create certain emotional events and when a **memory** is that to that that is what is called flashbulb **memory** for example memories of what happened if you are a cricket fan where were you if I ask you this question where were you when India won the 2011 world cup so most people who familiar with this idea of Horace who who like cricket or cricket fans will have very good idea of where they were and so this India winning was a flashbulb item which basically meant that it it created us as situation it created a thing which was an event which was emotional in nature and so most people then go ahead and encode this particular **memory** or encode their own living their own context with this kind of a **memory** and so these are what is called false **memory** now two things to remember that this flashbulb **memory** are generally highly inaccurate in nature and they are related to or they are more prone to **forgetting** it distortions so floral **memory** for example when you went to a shopping trip with your mom and your cousin somehow you wandered away in a store and get lost a security guard found you and you were reunited with your mom about an hour later this is flashbulb **memory** where this event was so emotional that you actually thought that whatever you thought during that period of time whatever you felt become a flash **memory** now this event could have never happened but after repeated question per nine percent of people recalled having having this fault episode so this kind of a structure was given people were shown this kind of a sentence and they were asked to remember the sentence and then every call was done and when people were again shown this there was this particular information again and again people were shown this bit of information again again they falsely recognized that this could have happened and they remember this flashbulb **memory** in this case this is the 9/11 attack on United States so most people do remember where they were during the eleven attack and it is what is the Falls well false memories all about so generally two things about this **memory** it is highly inaccurate and then it is related general related to inaccuracies and **forgetting** than distortions no there is a methodological way to create false memories now we we saw how I witness testimony is one way to do false memories but there is another way to create false memories and that is called the DS Rho dragger and make them a paradigm to show that memories are inaccurate and to show the fact that false memories or memories that never occur can actually be created and embedded into people's mind now for the DRM paradigm a list like this is given to people now these lists are semantically the words in this list are semantically related to each other for example dark dream pillow nap night white and I am very sure that when you look at this list you do know or it it seems to you that all these words are highly semantically related or highly connected to the word sleep it seems like these words are very connected to the word sleep and so that is what it is all about so basically what happens is our list like this is presented to people but the theme word here the theme what under which most of these words fall which is sleep is not presented so it when this happens and people are just given this list and where they're asked to recall this this back most people along among these 80% people also remember the word sleep also recall the word sleep that what does it really mean the theme word was never presented but people falsely recognized this word and first recognized the category theme work so 80 percent of participants falsely recognized sleep as having been on the list as it even if it wasn't there and so this is one way of creating false **memory** or distortions in **memory** now you know that **long-term memory** is highly inaccurate in nature it always on the edges and so you can never trust a **long-term memory** the reason is that this jesting process of extracting only relevant information from whatever information is happening from whatever event is happening if taking only the relevant information is a classic thing for **long-term memory** because that is how it saves space what it does is if whatever information is passed on on the **short-term memory** whatever kind of or whatever context it has or whatever pre information it has it takes in any new whatever is coming from the SS

short-term store and Maps this to the long-term store or to the already known store otherwise if the information that is coming to you for example if you take breakfast every day the things like you make breakfast every day or you sleep every day that kind of information is redundant it is not needed but let's say some day you sleep and a burglary happens in me in your home now that information will be because burglary is not what was not expected in some noble information and something which you never expected to be so this information will be embedded so with sleep information this information will be embedded otherwise your normal sleep information will not be embedded here now there as if memories are important to us so is a process called amnesia which is basically not remembering things or not remembering **long-term memory** and so there are two types of **memory** amnesia one near the dough great in the retrograde amnesia so integrity media is the ability to learn new information after the initial point of **memory** loss or an example is hm so in integrated media they is basically what it leads to is the inability to learn new information so anything with the **short-term memory** is basically not coded at all so you'll learn very well or you remember very well information of the past from the present you do not remember and and that is called integrate or is just old memories in LTM remain intact so you remember everything from the past but new informations don't get encoded there and in this case general knowledge and skills remain intact so you know every fact you know every detail but personal memories from new some new event is never encoded only old memories are encoded in a classic study is a other patient hm similarly the retrograde amnesia of a loss of **memory** for information stored before the point of a **memory** loss so what happens is if you have an accident and you forget everything from before this is called retrograde amnesia in retrograde amnesia key things from the past are forgotten new things can be retrieved new things can be added up but all things cannot be stored or all things cannot be accessing this is retro in India and spares over learned information and skills so those skills which you practice so much are that the procedure and **memory** basically is spared in this case ok so basically then in this section what we did was we looked at several kind of memories long-term memories what are they what kind of **long-term memory** is what is the need of **long-term memory** what is its capacity and we also looked at from spanning from capacity we also looked at the kind of **forgetting** that happens here we looked at different types of **long-term memory** different kind of interferences which is there and whether what is the format which is there and then whether it long-term memories are inaccurate or not and if that is in the **accuracy** in memories if there is some kind of distortions in **long-term memory** what is the reason from the distortions also what these distortions lead to or what these distortions are and how does the distortions really function is what we looked at we also looked at two different kinds of **memory** when the one is a portable **memory** which is the knowledge of oneself and the flashbulb **memory** which is a kind of **memory** which is attached to highly emotional incidence in one's like not American **memory** is basically is very difficult to test the reason being that nobody knows your auto-rifle **memory** and so testing of it is really really difficult but with flashbulb **memory** testings can be done and so a lot of work in threshold **memory** was done where it was found out that the flash is not needed at all basically there is no need for this emotionality even and so this kind of memories are the flashbulb and the auto worker **memory** although they very popular but a lot of testing has not been done on to it and so these are parts of **long-term memory** but lot of questions are not being done into it so they're two classifications or two kinds of **memory** which exist in the long-term store is basically the procedural and declarative in the upcoming lectures will not look at the procedural ones because it's a little bit difficult to look at procedural memories and since this is a very **basic level** course so we look into the declarative parts of **memory** into it so in the upcoming lectures we look into both the semantic type of **memory** what a store how does it store what are the models for it what is the way in which it functions and so on and so forth we'll also look at the **episodic memory** of how imaginary **visual imagery** visual imagination happens authors visual images and so how events are stored how are they are retrieved and that kind of information that kind of feedback is what we will look into in the upcoming lectures and so we look into detail these kinds of **long-term memory** so

overall then in this particular lecture we just looked at **long-term memory** subdivisions how it functions always forgets types of it and the well how if any queries exist how does it exist and how these inaccuracies are basically mapped for or accounted for so let's meet next class and discuss some other types of long term memories and how they function thank you

Lecture 15: Semantic Memory Basics

■Kind: captions dear friends welcome back to this lecture on **memory** continuing from the last class we were discussing about long term **memory** in the upcoming lectures we'll discuss on the types of long term **memory** which are available for humans and we'll see what are the properties of this **memory** how do they function in several the qualities of these memories so let's begin this lecture on **semantic memory** now if you remember from the last class I discussed about the way **long-term memory** is organized and so what we saw in the last class was that **long-term memory** had two separate distinctions so two separate parts of it and I also explained a little bit about the distinctions of these two parts so these parts are the declarative and the procedural type of memories nor declarative if you remember from the last class declarative type memories are those memories which have conscious which have conscious awareness people are consciously aware so when performing this **memory** people know what they're doing and so they basically take part in these memories consciously remember are well aware of what is happening and so memories of events memories of facts memories of other kind of memories which you know that you are acting on or you're thinking of declarative in nature and so the declarative memories are then divided into two parts also declarative **procedural memory** can also be subdivided in terms of explicit so some books were talk of declarative memories as an **explicit memory** and the **procedural memory** as an **implicit memory** and this explicit implicit is basically dependent on the amount of consciousness that you are putting onto it so whether you are aware so if you are aware of a particular **memory** type it is declarative if you are unaware of a particular **memory** when it is functioning this is the implicit type and so there is a debate on these two types of memories of declarative and procedural or two memories area now looking at the subdivisions the **declarative memory** is divided into two subdivisions we have the **semantic memory** and we have the **episodic memory** and so the upcoming few lectures will be basically focusing on the semantic and episodic now procedural type **memory** where people are not aware of that they are accessing the **memory** is basically implicit and basically visit in nature is something we shall not discuss here in this particular introduction our two cycle **cognitive psychology** class but I will give you the distinction of what type it is further divided into so we have habits we have classical conditioning and we also have priming now if you look at the earlier lectures I have briefly defining these habits classical conditioning in priming and what is the meaning of that and also about semantic and episodic so quickly create look into what it is so procedural type memories are those memories which you are not aware of that they exist but they take place for example riding a bicycle playing a piano typing on the keyboard and so on and so forth so you do it with such less intention that you are not consciously aware Petunia doing it and declarative memories are those where your functions arise or thinking about a particular event about the particular fact in life verifying and only verifying a rule or several other kind of memories where you are aware of it is what is declarative within the semantic episodic distinction semantic memories are those memories which are based on facts whose knowledge arithmetic properties things about about objects facts about objects which you can verify color shape that kind of a thing is **semantic memory** so when I say I know what an appellees are questions like who was Christopher Columbus or questions like who is a Donald Trump now where is US where is Europe or facts like two plus two equals to four what is the meaning of addition sign what is the meaning of a division sign what is the integral what is an indefinite integral and that kind of facts which are either arithmetic rules facts in life or facts and knowledge are go all **semantics** in knowledge or **semantics** in type declarative memories are those memories which we basically go ahead and remember about events so your first day of school your first graduation the graduation 90 the prom night that you had after graduation your first farewell of school the second farewell obviously you would have been to ten interest class both so two favors to think about first day in college first day in school any party in your life all those kinds of even say you're personally aware of or where you're personally part of is basically what you're at Masonic **memory** would consist of now in the

procedural type you have habits habits are those facts which basically are something which happens automatically so habits like habits of scratching yourself a bit of touching your hair at age point of time habits of twitching your nose there are several habits that people have and these are automatic kind of things automatic kind of act that people do but they are not aware of similarly in classical conditioning what happens is that doing something not knowing that you are doing that particular act or behavior for a particular reward is something called classical conditioning so buying something because something else is free with it or thinking positively about someone because something has been offered to you for thinking positively about that person is classical conditioning in priming of brief kind of information or a brief information is presented to you and this information leads you to basically go ahead and access **memory** or verify a particular fact so before knowing someone is some kind of information is provided to you very distinctly or very implicitly about this person your first impression of the person would be change according to it and so that is what is priming so in this particular lecture will be mainly focusing on the **semantic memory** so what we look into is what is **semantic memory** got out of the basic what are the basic steps of formation of **semantic memory** what are the various tools which I use for accessing it other models which explain it the various underlying limitations of the various model and what does it explain what does it do for us and so on and so forth so that's what we'll be looking in this first lecture let's continue with our idea of **semantic memory** in the next coming lecture we'll also look on a part of **episodic memory** or racial **memory** so then what is obvious to us is that people have a lot of knowledge they hold on to a lot of knowledge this world or about this world with them at each point of time so from the time when you are born to the time to the present time you have a lot of information with you about the world around you about the environment around you what people around you and so they are stored in your in your personal **memory** and most of this information which is stored may be personal and may not be personally mature for example something which doesn't concern you facts like Mary's America doesn't concern you and so it is stored and there will be at the facts thing something about when is a birthdate or what is the color that you like but also stored on to it so a lot of **memory** is stored a lot of information is told in your **memory** now these contains as I said knowledge relating to definitions of words arithmetic facts procedures historical and scientific journal of acknowledge and so on and so forth and so most of these are cemented in nature so any **memory** which is not concerning you or concerning yourself into it or doesn't form around you a semantic knowledge so facts that can be tested facts that can be verified fact that you learn from people around you and facts that are helpful for you in terms of making you leave this world are basically semantic in nature now basically then how is **memory** arranged what is the way **memory** arranged and one of the earliest metaphor or what is the earliest analog of how **memory** is arranged is basically the bookshelf metal metaphor so it's and this is the **semantic memory** that I am talking about so knowledge facts ideas arithmetic properties and all those how are they arranged one of the basic metaphor ever and the primary metaphor is the bookshelf so basically it's believed that just like when you go to a library how do you search a book so the best way if you have been to a library nowadays there are computer programs like Lipsey's which actually help you into finding a book but if you go to an old library where they don't this software's for helping you to how find out how to locate a book you would find out that there are still the old methods of finding a book and how does that work so you basically go to particular desk which is called the naming catalog so it could be either a naming catalog and/or a place catalog so naming catalog would have names of authors and a place catalog or title catalog will have titles about us so you choose one of this catalog and so let's say I choose the title catalog or the subject catalog so I choose that let's say I choose the naming catalog now suppose I am looking for a particular book I'm looking for the book Alice in Wonderland and so if I am looking for Alice in Wonderland I'll have to search from something called the a so I look for a of all the books which are available in a and within a there will be several variations from a a to a a set and so within a you will have several books a be evil has several books a see EULA several books and so when you go to that Alice in Wonderland is a hey L so I'll go to Al and then I'll search for the

particular book so within Al you will have lots of books so I come to the basic book within that particular book also Alice in Wonderland there will be several versions of it the abridged version the main was and the total version and so I'll look at that and then there will be something called a calling number or basically an accession number and I'll look at the accession number then I will go to the row the main library where books are stacked and then I also search for I look for the particular accession number because books are arranged according to oxidation numbers and look for that particular three-digit four-digit accession number and I'll find that version of the book so this is basically how the bookshelf metaphor really works it is believed that **semantic memory** also follows a metaphor like this this is how most items or most facts are in a **semantic memory** so information in **memory** they consist of knowledge of specific events and **memory** for general knowledge as I said this being most information in **memory** will be either personal or will be either general knowledge think it is of how generalities is facts about general knowledge are stored and so mostly the follower metaphor which is equal to bookshelf no the fact theory is that the question that do we have this kind of a distinction in **declarative memory** now in the last class we saw that the debate over the existence of procedural and seam and **declarative memory** in this particular lecture we'll see our debate or we will discuss a debate on the existence of cement again **episodic memory** so whether there are two kinds of **memory** and if they are then what is the way in which they're arranged and what are their properties so and filming in 1972 1983 he argued that Altium conveyance of two stores basically the episodic in the **semantics** store which are which are very highly distinct they are distinct stores because one store stores facts knowledge and that kind of a things you don't need you to think about it think about anything or think about any personal relevance information there and the other store is the declarative store where I am sorry the episodic store where you have information which is personal in content and so you argue that these are the two kind of store an interesting fact here the fact is that that piece Adak store can not exist without the semantic source or some kind of information is always borrowed from the semantic store and when we end this section will also talk about something called scripts and schemas and six key mother scripts of the way of how the semantic store interferes are basically aids the episodic store now that Pixar extort I think of the episodic store when I am thinking about a particular party or a particular your graduation night thinking about you night if that is what I am thinking about it has a lot of informational to it what information follows a certain rule or follows a certain structure and this structure is so when you think about a party you think about a rupee no what happens in a party and that is what basically scripts are scripts are **schema** it's a basically a **schema** for proteins we'll come to that so basically there are certain steps through a party and that is what you tend to remember in that particular way you tend to remember so although the facts may vary from person to person but the graduation night the what happens in a graduation night would be similar to most people and that is what a **script** is all about and the idea that this is what happens in a graduation night you go there you do something you do a little bit of dancing lot of chatting mr. and mrs. miss graduation is is named then you drink the punch or whatever you do have a little fun and maybe dance with your lady and then you come back and this is the kind of **schema** which is there or this is the kind of structure of a graduation party which is there and so this is what is going to happen and this is what happens in a particular **schema** in a in a particular even and in particularly in the episodic store this is how the **schema** really works so as I was telling you that within the **episodic memory** of your graduation what to do and what not to there will be several schemas and these schemas are part of the **semantic memory** so whatever you do in your graduation party is basically an extension of the **semantic memory** so episodic memories have to have semantic knowledge or semantic facts embedded into it but not the reverse case semantic memories may not have things from the episodic store so basically when you when I'm talking to you when I'm asking you what an Apple is you will need not remember we will learn the word Apple or the first time you were shown an apple and told or an apple is but then with **semantic memory** this is the independence that you have but with **episodic memory** this is not the independence that you have you when you describe an apple you have to know

what an apple is or what defines an apple by the apple a certain shape certain color that kind of a thing has to be there but you can also define or tell me the first time you heard the word Apple that is basically Episode II knowledge so let's look at the distinction or the episodic and **semantic memory** and so quick distinctions **episodic memory** has it said it enables people to travel back in time and become consciously aware of witnessing or participating in events in earlier life so basically what episode of **memory** does is that it gives you this confidence it gives you that freedom to travel back in time so here I can travel in time and so trying travel has not been invented but with **memory** studies you do travel back in time to that particular event it becomes live on to you so **episodic memory** gives you this particular freedom to travel back in time and you can also witness that particular event so although you cannot do anything to the event you can only remember it as a third-person perspective so you can see an event unfold in front of you you see yourself and folding in front of you but you cannot participate into it and that's the main problem with **episodic memory** so you can view you can consciously feel the event but not change it in any way and that is why this time travel has a limitation so you can travel back but cannot do anything do not disturb anything into it and so that is what is **episodic memory** whether the **semantic memory** you have it contains general and world knowledge so basically all knowledge is about where Africa is America is what is Germany where is Germany who was it who was Napoleon who's the Pope who's the Prime Minister particular country who's Barack Obama and all kind of informations story who is basically general world knowledge arithmetic rules like 2+2 4 + 4 2 into 2 and all those kinds of higher arithmetic rules past tense of work like what is the past tense of do what is the present tense of any verb so run makes ran running that kind of a thing so this kind of information which is there which basically is world knowledge basically his knowledge is what is **semantic memory** so quick distinction of these so **episodic memory** has to deal with personal experiences which is everything which is personal to you which you are involved in to which your self is the concern is what is per se my **episodic memory** where a **semantic memory** deals with facts and concepts so it deals with those information which is world knowledge so you may be part of it you may not be part of it but then this is a fact this is something which everybody shares and that is what type somatic memories now when I say remember when it is basically **episodic memory** so I'm talking about the time and that leads to the third definition or the third quality of **episodic memory** but we'll come to that in a minute so when I'm saying remember when so remember when an incident happened I'm actually asking to Texas and **episodic memory** but I remem ask you questions like remember what what is a table what is a chair things like what is an apple what is water and all those kind of information so remember what it is generally a **concept** of **semantic memory** now episodic memories are temporally organized and so it is time analysis or time businesses so you can see an event unfold in real time when you're accessing an **episodic memory** who is just like the Pangolin **concept** in your facebook so there are certain events put onto the timeline **concept** and as I visit it you can see the event fold in real-time that is what is the **episodic memory** but then when you have **semantic memory** it is related or it is arranged in terms of meaning so meaning the related arrangement is **semantic memory** and so this is classic definition or destination between the episodic and **semantic memory** and so the several evidences which were there to basically prove these kind of **memory** systems exist and one of the famous cases or one of the famous evidences was provided by sagittal in 1996 and so he studied a particular brain-damaged person a vein damage patient called gene and we need to studied this brain-damaged person what he found out that a certain region of the brain specifically inhibited **episodic memory** but it didn't inhabit the **semantic memory** so this person was able to access emetic **memory** tell about facts about the world knowledge but personally means this person could not remember back another evidence which was provided for this distinction of episodic **semantic memory** was by tulving in 1989 where he looked at cerebral blood flow and he looked at cerebral blood flow when a person was accessing **semantic memory** and when he was accessing in **episodic memory** and he found out this cerebral blood flow that was measured through a PT scan they were different so basically different kind of blood flows exist in different

regions of the brain and so that is what the third thing is different brain regions are involved with the semantic and episodic memories so the blood flow was different all not only the flow the regions of the brain were all so different from cement can a piece OD **memory** so now we come to know that the **semantic memory** is a huge store it has a lot of information arranged on to it right a huge capacity now from every system is a huge capacity because the world knowledge that you have is a huge you learn everything every someday every day you learn something or that or a huge amount of knowledge today so how are they organized what is the way in which this is organized and so several debates were coming up there were several propositions of how they were organized so in the basis of that the whole idea of semantic models of semantic knowledge were proposed now one of the primary things that we have thing is that researchers working in the field of artificial intelligence and computer science who were modeling **memory** back in the 87 80s and 90s those people they wanted to build a common sense knowledge system a system which has common sense which can explain common-sense things like how to do something how to open a laid-off jar so how does the system like this operate because common sense exists from world knowledge only and so they were thinking of designing a model now what they said is that with a lot of knowledge that we have the kind of knowledge that we have we have a lot of implicit knowledge hidden within the kind of knowledge that we have which we are not aware of now take the example of a shampoo bottle now if you look at a shampoo bottle and if you look at the directions that it says it says that hair take a little bit apply rinse wash and repeat so if a robot is given this to do if a computer is given to do this what it will do is it will keep on doing this cycle of wetting the hair taking a little bird shampoo putting it on raising it washing it and repeating it till the bottle is finished but then with most of us that is not what we do we tend to stop after one or two days and we on continuing so where is the knowledge the implicit knowledge of stopping after one or two rains is is where does it come from what how it is an inn and so that particular store or that particular idea of arrangement was what was people looking at how is this idea or this implicit knowledge where is it stored with the world knowledge because one all it says that you repeat so how many times should repeat that is the other question so implicit knowledge in everyday routine informed you to know a lot of things and a lot of other things which you don't know that you know but you take it for granted so knowledge of **language** has an associate great deal of implicit knowledge also things like common things like this certain other implicit knowledge for example when I say the the dragon had a liver now when I say a dragon had liver you do not have to go to check the whole idea of what a dragon is or the whole structure of a dragon go to a museum and find out whether it has or not you know the dragon is the kind of members in most mammal tends to have liver and so it can be generalized so this is implicit knowledge that if men here have or birds like dragon had liver so the dragon also had liver and so this is implicit knowledge and so people wanted to test how is this impressive knowledge actually arranged so one of the other reasons for making the inferences or making thinking about or modeling presenting models of increase acknowledge of **semantic memory** was how was the mental representations of various knowledge arranged so we could also wanted to know when a fact or a knowledge is stored in a **memory** system what is the way in which these represented representations are stored another interesting thing that was happening is that how are they accessed what is the speed of accessing so basically questions like this were asked that named which has first letter L and name quickly words which held the fourth letter L for example the first word could be livid and the LA and so words with first letter L and the third letter is L Cal q+ and so or the seventh letter word l now if i ask you to name the number of words which are with first letter L and there were number of words which are third letter a fourth letter L you will end up telling me more words which have first letter L and then the number of words which are the seventh or third or fourth or fifth letter L why because this shows the speed with which of how many words you can retrieve can actually tell you how is the mental dictionary organized so basically mental definition is then organized with the first letter remember the metaphor that we used in the bookshelf so basically the first letter of Alice in Wonderland that al is how it is arranged so a is LS and

so you will find a lot of it and so Alice in Wonderland is arranged with Al the first two letters and this is how this things really works so basically then several models are proposed of the **semantic memory** and we look into a few models of **semantic memory** which are here we'll look into couple of models we'll also look at how these modules compare with each other and how one appreciates one model goes ahead and cancels or basically adds on to the limitation of the other model and how a **semantic memory** arrange so basically it was the heretical semantic model was the first model and so this model does a number of things it tries to tell us how limited database is arranged for the huge knowledge so how is the database for a huge knowledge that we have is arranged and or basically the number of knowledge that we have the number of facts and knowledge and things semantic things that we have how they are array also any model of **semantic memory** should be made in such a way that it should not store redundant things right things like let's say a dog has four leg right and a cat has four leg and let's say any other animal has four leg so when I say that so each time when I present a dog has four leg cat has four like animal has four leg so basically it should not be repeated this for like the idea that this particular kind of **concept** this particular kind of object has four legs should not be repeated with each instance which means that the idea that most mammals because anything which falls under the members should have a four leg issue bannin is such a way that the higher category or the the idea that any particular item within a category should have four leg and the support leg should be at the top of the feature at the top level so if I am thinking of a hierarchical semantic model which means that knowledge is arranged in terms of hierarchy any knowledge in Turing in terms of hierarchy the top level hierarchy should have these facts because anything under that hierarchy should have four like so lion tiger and all kind of members I'm talking about all follow under the animal category or the mammal category and so the the model should remain in such a way that the mammal category should have in inbuilt the idea that it has four legs so basically that kind of thing is dance and what it does is it preserves something called **cognitive economy** which where the properties and facts are stored in the highest level possible so basically if a particular category if a particular structure is designed in such a way that properties and facts of the category anything which falls in the category and the main facts of those category should be at the highest level and then within that you will have lower levels on to it and so the first model of **semantic memory** was proposed by someone called Collins and Quillin and they tested the idea the **semantic memory** is analogous to a network of connected ideas so what koalas and Quillin they said is that basically **semantic memory** is something so **semantic memory** is basically an arrangement of network of connected ideas so they define that any word our **concept** is arranged in terms of **nodes and pointers** towards now so these are two nodes this is one node node one and this is node 2 so then two concepts two words are related to each other through a node and the junction or the connection between two nodes is prevented by a pointer so people who are familiar with programming or C programming C++ programming they are familiar with this idea now this in terms of programming this is a **memory** location space which the computer provides you and the pointer points how to **memory** location spaces are basically connected in terms of the hierarchical model by Collins and Colin what it says is that this is the **memory** area our **memory** allocation or this is the node for a particular word and this is the node for a particular word and so if two words are there then they have different different **memory** location or different different structures or different different places are located in the **semantic memory** and they are connected by a pointer which points from the highest node to the lowest node so basically then what they said is the model contains of consists of nodes in this case what the concepts and each node is connected to a related note by means of a pointers and that is what I was also telling you so this is the node let's say this is the node for any kind of dog so maybe German Shepherd this is the node for a dog and so what happens is these two nodes are connected because within dog you will have German Shepherd and within the German Shepherd also you can have another node which is different kind of Shepard's which would be there and so what happens is within dog the German Shepherd will fall and within the German Shepherd there will be others nodes also within the dog category there will be other

nodes which are there so I will have a per million node also and so on and so forth so what is the fact here this is how information is stored and the dog will be again following under the mammals node the node for mammal and so on and so forth so we'll come to that in a in a minute so basically that is what their idea is that this is how our arrangement is nodes and then connection of nodes now there's a note that corresponds to given what our **concept** together with the pointers to other nodes to which the first node is connected constitutes of **semantic memory** so basically **semantic memory** is this kind of a structure and this kind of a structure that I'm was talking about if I know to a lower note kind of a thing is called the **semantic memory** Network so semantic network is the network of mammal then following to dog and then following to a German Shepherd this is a network but say dog and German Shepherd within each other is what is node node to node connection and the pointers are the pointing or what leads to what kind of a thing so that is what this says and so the collection of node associated with all the words and concepts are called the semantic Network so let's look into this particular thing so quickly look into this as you can see here I will first wrap this thing here quickly the whatever is written here to show you what is there and so this is a classic network that I was talking about and so as you can see here what you see here is a very small network and so as you see this is the network of vehicle under a car falls under which truck falls and as you can see the vehicle has several properties which most vehicles which follow under the vehicle note would have and these properties are moves around it means fuel it's man-made so car is also vehicle so it has by default it has moves around needs fuel and man-made these are the properties of the node remember the **cognitive economy** that we talked about and so by putting these three features at the top node which is this is called the **superordinate** node and this is called the **subordinate** node so proudly named node is a node higher than a particular node of concern and super and and sub node is a node which follows under the super ordinate node so these properties then by default should be a property of any member or the vehicle and so when I talk about and this is the link that is there and this is the node so this is the pointer that I was talking about and so we the vehicle will have these three features at the feature of any vehicle and so these these features should also be in cars where I talk about car cause car also moves around Carlos who needs fuel Carlos is man-made and when I talk about car it has four wheels has an engine in a window and so within the car node but then all vehicles may not have four wheels because our two wheeler is also a vehicle so it may not have four wheels but it moves around so these features are common so these features are common features to any item which is here but features which are here may not be here and the four wheels engine in window and within the car then they have two more nodes one is the truck node car follows a truck and then I have a variety of car which is the sports car it runs very fast it is adjustable so I'll have a sports car a sedan and so on and so forth which fall on the corner also you will find out this is this is a sub node and this is the **superordinate** node so any feature of the car will always be here so it always supports car will have four wheels will also have windows and engines but then every car will not run fast and so this kind of a reverse pointer or this kind of a reverse matching is not true but forward matching is always true also truck is kind of a car which has four wheels but it special tall and German highway and transport loads or something which doesn't this note doesn't have so they're separate with each other but then the UH truck will always have four wheels it will have an engine it will have windows onto it but two more features of it which may not be consistent with the car and all of them will have these four features so basically this is how this is the arrangement of the quillion and Collins model of **semantic memory** he says there is a network of nodes and these network of nodes are connected by pointers and so one network leads to the other also facts we just told it the highest network which is the **superordinate** node so the highest node is called the **superordinate** node super ordinate node and the one which is beside it is called the **subordinate** or the **subordinate** node so features which are at the **superordinate** node will always be shared by features of the **subordinate** or feature so super ordinate is shared by features of the **subordinate** but in features a **subordinate** may or may not be shared by the **superordinate** node so this is how : in Chileans define the idea of **semantic**

memory so they tested the principle of **cognitive economy** in their model and as I said one of the things that needs to be tested with most models of **semantic memory** is the idea of **cognitive economy** that properties which are the highest properties which are the highest placing in model it should have the it should have the higher the basic concepts or all properties of the **concept** so words and concepts should be always at the highest note all **superordinate** know so they state the principle of **cognitive economy** with this model and they reasoned that closer in fact the store to a particular node the less time it should take to verify a particular property so if a particular item a particular fact or a **concept** is stored very close to another node very close to another **concept** it should take less time to verify then if it is stored far away and they tested this particular thing so they reported that people took less time to respond to sentences those representations should spend two levels a Cannady is a but then to representation which spencey levels so then let's look at this particular how this arrangement is so basically I have the arrangement of animal and within animal we have mammals and birds and within birds I have the Kennedy the parrot and so on and so forth so when I'm verifying and so they said that since if **cognitive economy** is to be preserved if words which are concepts were higher-order concepts should be stored above a lower order **concept** if that is what the idea of formative economy is because all properties of word has to be shared by Kennedy and our property of animal has to be shared by what if that is how the arrangement is then people should take less time in verifying Kennedy is a bird then verifying Kennedy is an animal and that is what it actually happened with them so they took that people they saw that people took less time to respond to sentences representation of span two levels Henry's a bird then for presentation which was planning three levels so this is one and two level and this is three level so basically **cognitive economy** is preserved here also the model was called the **hierarchical semantic Network** model of semantic memories of this particular thing because what I have is the higher-order concepts or the concepts which have basic features which every items on that particular **concept** or every item within that **concept** should have should be the highest level and within that will have other items and so cons the properties of one chef will be shared by any item which is under that particular **concept** now the nodes in the model are organized in hierarchy as I said they're in the top node then there is a another node and then there is another node and known and all appointed by all are connected by a pointers now a very good example of thinking of it is think of your windows when you think about your windows you have this tree structure and so there are folders within folders within folders within folders if you have ever worked with the Linux system you will find that there is a part that we talk about PA th and this part is basically how nodes are connected so because if you had so let's say there is a in in Linux you have the first part a slash root and then within that you have slash et Cie and then you have something something and then within this you have generally have STC and within STC you have us ad which is the user and then you have the home and then you have your desktop so if you want to reach your desktop you have to go through this part and this is how the semantic network is also connected so root is the one which is the highest level **concept** with the **superordinate concept** within that the etc' is there which is the **subordinate concept** within that the STC is there which is another store which is there within that user is there within that home is there and with that you are a sa and which is basically your desktop similarly they have a system like this so then most nodes have **superordinate** nodes and **subordinate** nodes above a particular node you **superordinate** just as we saw in this particular thing in this particular example so what we had is vehicle is a **superordinate** node and truck is a **subordinate** node or sports car is a **subordinate** also there is what is what it has been saying here now with a model like this where we have this kind of **superordinate subordinate** node and things which are saved in this way a good experiment was done by Marin skin wealth in 71 and they reasoned beautifully they gave this reason that if related words are stored close on one network and they are connected to one another in a semantic Network then when one node is energized energy shift we should actually spread to really tear related notes one reason is that **spreading activation** the idea that exciting is particularly network along a connection of network in the semantic

Network so let us try to understand what this is so we have the animal node and within the animal node will have something called the mammals and within them birds and mammals will have let's say a parrot and another one is let's say a Robin and within the mammal we have a tiger and we have a lion so what he says is that when and so word since birds and members are not connected to each other but then what he says is that when some energy I want some energy is when somebody is accessing their animal node this activation that follows to the bird should should basically go ahead and and excite both parrot and Robin at the same time or with so Robin may not be connected to Tiger in any way but then if they are connected in some way if these nodes because some kind of connection would be there between Robin and Tiger of course very difficult to explain what it is but they they might would be kind coming from animals and so the um kind of connection would be spread between them so when I energize this animal or when somebody starts traveling from this animal down to Birds both this node will be activated right and so within the birds you will have colorful words non colorful words and within the colorful and non colorful even as a talkative words non talkative words and small and so forth so what they said is that when an excitation moves down a node like this what would happen is this excitation will not only spread to one node particularly also spread to excitation spreads along connection of Nagina semantic network so this excitation would also go ahead and connect to tiger so tiger will be weakly excited but then it won't be excited now how Robin is connected Tiger is very difficult to look at but then there will be certain members which may be related to Robin in some way or parrot in some way and so that excitation will always be there or connected there and that is why they said this is because we have always found out that when I say Queen people say King when I say bread people say butter now when you look into it bread and butter they are two different things king and queen are two different concepts which altogether Queen is from the female category working is from the male category but Queen and King are related to each other and this relation is basically can only be explained through the idea of this **spreading activation** what happens is and an activation when a node is energized this energy also spreads no it doesn't only go to the node which is subset with it so we'll have reptiles also for example so when animal is excited reptile nodes also get excited onto it so kin Queen Queen although being this is female and male and so on top of it if we look into the node of animals so within animals we have million female within king and queen so although they are different but some kind of activity relate them and so this was an basic example to show that these river nodes are connected to each other now there are several limitations to this particular the hierarchical model of **semantic memory** one is about **cognitive economy** as a Conrad in 72 1972 they found that people are no faster to verify sentences like a shark can move then a fish can move or and anyone can go so what they found out is that even if we span so if we span let's say two levels or if we spend four levels on a network so let's say it's a four level network four nodes so when we verify a sentence which is four not far from the **subordinate superordinate** node people take almost the same time and so this idea of kinetic on a-- me was not preserved by this **hierarchical semantic network** another problem was the structure of the hierarchy itself so ribbon rips and sha'ban Smith they showed that participants were faster to verify a pig is a mammal nt verify a pig is an animal so basically they quickly identified pig is an animal then Pig is a mammal now if you look into it animal is the high note and then within that you have mammal and within that you have several other things let's say two more nodes are there and then you have the pig which is actually a member so when people verify Pig is an animal which is let's say that it is directly connected here so we have one node and two nodes so when people are verifying one two and three and where people are verifying two levels and people are verifying three levels of people verify three levels faster than two levels and so and this is what the thing would look like so they found out that this idea of hierarchical structure was violated by the hierarchical network model also **typicality effect** was another reason or another limitation of the identical symmetric model what it says is that Birds which were typical for example Robin parrot was of a sparrow these were faster in **verification** so when people verified a Robin is a word or a parrot is a bird they're faster to say yes verifying the sentence then to verify sentences

like ostrich is a bird or a turkey is a bird or a hen is a bird if you look into it both hand turkey and ostrich are all birds but then people took more time to verify these sentences now although if we look into it ostriches and turkeys both have the same kind of structure the same kind of features as Robin or a parrot would have but when we are verifying this when people verify this kind of a structure and better people verify this kind of a typical versus non typical instance of a particular **concept** people are more prone to verifying the more typical car or instance than not driven typical so the idea that all elements of a **concept** are not arranged or not verified in the same manner and so this is typical defect and so they've found that problem with hierarchical semantic model so since these were the problem that were there the violations which are there another model called the **feature comparison** model was actually proposed which would go ahead and solve some of the problems which the a model of hierarchical symmetrical Network was actually doing so what was the model like so split so Marin rapes they proposed an alternative model to the hierarchical semantic model which was called the **feature comparison** model or **semantic memory** assumption was the meaning of any word or **concept** it consists of set of features are elements called features and features come in two types so he said what they said is that **semantic memory** is not arranged in terms of some kind of a structure or some kind of hierarchical structure but they're arranged in in in a in a way in which there are certain features so there are two basic features that we talked about so when I look at a **concept** item for example I'm looking at dog so the dog has two kinds of features one is called the defining features of a dog and then there is a characteristic features so facts like it has four it barks it has four legs it has a tail which bags are defining features of a dog because most dogs will have them and then there are characteristic features which some dogs they have and some dogs may not have for example there is a dog that that's out there which doesn't bark and then there is a dog which has very less steel for example you have a dog which has no tail at all and then you have a dog which is a huge tail or there is a dog which have or think of a dog which has three legs on it so how do we accommodate for that so that kind of a thing is there and so most dogs whatever they have is the defining feature and some dogs may have features which are extra which are called characteristic features and so this is what it is so the meaning of a word or **concept** consists of set of elements called features and we shall generally come up two types defining features features must be present in every example so every dog should have them four legs barking our tail or greeby your man's best friend is what most dogs should have and so these are the defining features and characteristic features are those features which are usually not necessarily which are usually but not necessarily present for example barking as I said or other features example big tail or facts like drooping years-long teens these kind of features maybe there may not be there or a glossy skin may not be there or the height like that person Bernards are dogs which are very high and Chihuahua is a dog which is looks like a cat and so they may not have the same feature so for them this is the kind of network they thought about this is the kind of feature related fact that they thought about and so if you look at attribute or feature of a model example if I'm looking at a Robin and I'm looking at a bird so there are two level concepts bird will follow or birds are the super ordinate node and the Robin is the **subordinate** so if you look into it both are physical object both are living both animate both are feathered so these are the features which are the features which are defining so any bird will have any borders physical organism well as living is animate and his feathered but then this particular thing is called the characteristic feature of a Robin because most bird may not be red feathered and already bested and so this is the characteristic feature and all these are the defining features because defining features is something which most of the elements of this **concept** should have whereas if you look into the comparison with hierarchical model helical model says animal is a bird which has feathers is a Robin which has red breasted and so this kind of thing this network has a problem because they should share so if the word has feathers and Robin should have feathers but red district that is the kind of problem which is there and so this model has this problem which is solved by my feature model or the new feature model of **semantic memory** so what it says is that when why does the verifying sentence problem really occurs or how do sentence **verification** happens according to the

feature network model or the feature model according to the feature model the first step is the presentation of an item so when I show you a new item there's our test presentation is there and we retry features as to north and so basically c1 is the defining feature here and C 0 is the characteristic feature of a particular thing so I present us st. Bernard to you and I ask you to verify this whether this is a mammal or not whether st. Bernard is a ml or not and so when you look into it it's a stage one you look at the characteristic feature in the defining feature of strain but not so what happens is initially the first step is to look at all the features or gather all the features then compare X to the criteria C naught and C 1 if X is less than C naught which is the characteristic feature which is that I'm sorry the defining feature if it doesn't has a defining feature for example it works and so st. but not doesn't work let's say then I can move this way so C naught is the defining feature C 1 is the characteristic feature and so if C naught X is less than C naught or C naught is greater than X then I move here executes a negative response and say it is false so if I have a cow a calf and I am trying to say to you that this is a st.

but not and the calf goes maomao and doesn't say bow-bow which is what a dog shouldn't say this is a defining feature of barking and so I can execute a negative response saying this is false otherwise there could be other things that if I can also compare the characteristic feature or if the match is not of even of a characteristic feature I can or even if some characteristic feature matches so medium in the may not be defining but a characteristic feature for example I said there's a dog which doesn't bark but it loads like a dog it has a glossy screen it has a tail which wags and it looks more or less like a dog droopy has all those characteristic features are there and so if it matches even though if the characteristic features matches and executes a prostitute responds and say it is true but in cases where my response the response that I get by comparing the characteristic and defining features is in between the characteristic and defining features so the amount of if the amount of defining features is very high I say yes I am I new item is a part of this particular list but then if it is very low I say it is not part of a list so if it doesn't have a defining feature like say false if it has very high amount of defining features or some very high characteristic feature I say it is yes but if it is in a mediocre range if it is hiding high having neither a very high characteristic feature nor very less than a defining feature I go ahead and do that I compare the defining features then so defining features doesn't come into play in in comparison there in the feature models until then unless it falls the comparison of features falls in the medical range when the characteristic feature is is is looked at so first step is I gather all the features then I look at the item which has been presented in the category characterized feature is very less than matching the characteristic feature that the new item has matching to the **concept** is very low I'll say no it is not part of this particular category if it is very high I say it is a part of it in circumstances and the characteristic feature is somewhat madacorp it lies between the defining and the characteristic features or not too much of characteristic features present but it has some characteristic features I go ahead and compare the defining features only the defining features only if there is a match with the defining feature execute of to response or I execute a false response and this is the two-stage characteristic **feature comparison** for semantic categories so again very quickly what I do is if a new instance comes in first I compare the characteristic feature hi comparison of erect feature or high number of characteristic features present I say this instant belong or this incident or this item belongs to my category low amount of characteristic feature present I say it doesn't belong if the amount of characteristic feature present is somewhat in between low and high I go ahead and look at the defining features or if the defining features is high I say a match is their defining features in low I say execute a negative response and it is false so **feature comparison** model white is good because it can explain some other or shortcomings of the earlier model first **typicality effect** sentences like a Robin is a bird I verified more quickly than turkeys about because Robin becomes more typical example of birds then thought to share more characteristic features with the bird so Robin stares although it has a number a number of defining features but when you actually look at Robin it has a lot of characteristic features add-on features with the bird and so they are verified much quicker but when I look at a turkey when I look at an ostrich they don't

have a number of characteristic features so a stage two **verification** has to be looked at but what happens is the comparison with Robin is only a stage one it has a lot of characteristic features which is of the word and so it is cat categorize and it's easy to verify Robin is a bird also category size effect what it says is the category goes higher so if there is a **subordinate superordinate** category and within that there is **subordinate** category if the category is small and the characteristic feature list of characteristic features will be small and so **verification** will take less time but if a category is huge if I am talking about mammals and there will be all kind of members which is there and so if I am talking about a system like that what will happen is the **verification** time will be more and so as categories grow as the number of categories grow as the kind of **verification** category is huge it will take more and more time this is called the category size effect now what is the criticism to that the criticism to this theory is that there is no existence of defining features what is a defining feature nobody knows and so there is no there is no existence for what a defining feature should be suppose a bird has clipped wings will it be called the word so what could be called a defining feature is not very forthcoming here or not very present here so in this lecture we looked at what is **semantic memory** we looked at how they are arranged we looked at the episodic semantic distinction and we also looked at the two basic modules the hierarchical semantic model and **feature comparison** model and how these model compare to each other and they go ahead and sort of complement each other in the next lecture we will look at some other models of **semantic memory** and also something of how **semantics** may be really works thank you

Lecture 16: Models of Semantic Memory

■Kind: captions hello friends welcome back to the second lecture on **semantic memory** now in the last lecture we discussed what is **semantic memory** and how it is arranged what are the properties of **semantic memory** and how **long-term memory** is divided into both the semantic type and the episodic type we also looked at what is stored in **semantic memory** the kind of information facts knowledge is arithmetic rules certain kind of information related to everyday world activities every very world knowledge is all are stored in **semantic memory** and so there is it's a kind of **memory** which stores most kind of information that people use in everyday life we then discussed a distinction between the episodic **semantic memory** and we looked at what is the main differences and so some of the main differences I'll repeat here is that **episodic memory** is basically time dependent or temporarily arranged where a **semantic memory** is arranged according to meaning it basically means that as you are as a **episodic memory** unfolds it unfolds in time you can see yourself in time whereas if and **semantic memory** unfolds it unfolds as a meaning now what do I mean by this the meaning of this is that when you unfold in therefore sadiq **memory** like let's say you're a prom night or maybe your graduation night so every event is are in terms of time from morning to afternoon to evening what happened during the whole day whereas if you ask questions like if you think about acts like an ethnic rule of how addition subtraction progresses you really no need to no the time the day when you learned out how they progressed in terms of time the other differentiation between the episodic and **semantic memory** is in **episodic memory** it is when so remembering when it the incident happened but in terms of **semantic memory** it is more about what happened so what is the content of it and so this is one distinction between episodic and **semantic memory** and so this is the focus of the **memory** so **episodic memory** the focus is the self so everything arranged in in **episodic memory** is around self you as a person can see everything going around and you can watch this thing as a third-person observer but in somatic **memory** that's not possible what really happens is that you get gather information and you are not the self the self is not the center stage of this **memory** but rather the world knowledge is the center stage so knowledge is basically what comes out of **semantic memory** so this is something that we saw in the last class now we also saw two models of my brief **semantic memory** that in the last class one model was called the **hierarchical semantic Network** model where we looked at how **semantic memory** our ingest concepts and a world information and we saw that it arranges the world information in terms of nodes so there are several nodes and so nodes are basically concepts and these nodes to nodes which are similar together or where even if they are dissimilar they are connected by a pointer or a linked between them and so this pointer node kind of a thing which is very similar to programming in C is what happens in the hierarchical semantic model it also looks at looked at some of the basic interesting facts about cement the hierarchical semantic model and we also looked at a criticism of how it doesn't follow the **cognitive economy** on how the **concept** of hierarchical structure is basically violated so that kind of thing is what we saw in the last class we also saw the feature integration model or the feature model of **semantic memory** we saw that there are two kinds of features so most information is innate in terms of features and so there are two basic kinds of information that is available with any word **concept** or knowledge and so one of these is called the defining feature that is called the characteristic feature we also saw a model of how these features are tested so that's another interesting thing that that has to be there so what really happens is that in the beginning to start with the feature model takes off in this way the first at first you start comparing the characteristic features and if the characteristic feature of two objects or a characteristic feature of a new event is similar to a proposed **concept** the more **similarity** it is the quicker a person decides the newer instance to be a part of the **concept** if the feature is very low we can go ahead and say that this particular new **concept** which is in front of us this particular new item which is in front of us is not a member of the **concept** and so we can delete this thing now if the characteristic feature is neither very high nor very low

so we proceed to combining or testing the defining features and so when we look at the defining features this is the next step in this case only the defining so if characteristic features and neither high nor very low other defining features are tested and when the defining features we get a match or the number of defining features are yes answer comes in then the new items or the new entrant is basically you know matched to the **concept** and a yes signal is given or if the defining features number of defining features are very low in the new item as compared to the **concept** it our new are and no signal is outputted and basically it is rejected that the item belongs to this **concept** and so this is how the idea of integration model really works we also looked at some of the interesting things about the feature integration model for example how it progresses and what are the how it solves the hierarchy problem or it goes ahead and solves a category problem so what it basically says is as category increases more and more abstraction comes in and so with the increase in category the or increase in the size of the category this process of **similarity** matching becomes more and more difficult now in today's class we look at some of the newer models which are there so the more models which are there and then further on we look into something called how beside the model the **concept** of model how is **semantic memory** supposed to store information so we'll start with model proposed by Collins and Loftus in 1975 and so Collins and Loftus they basically went went ahead and they proposed a model which is very similar to the hierarchical semantic model and so what the basic of this model is it's an elaboration of Pauline villains 1969 model as I said it isn't in extension of the original model proposed by Collins and krillen and so basically this model also states that semantic information or world knowledge is basically integrated together or the world knowledge is stored into the semantic Network in terms of nodes where nodes is the **concept** and pointers the pointers other links between these concepts hierarchical model the only change that this model brings about is that in this model the **concept** of **spreading activation** is a is the center point or is the central feature of this module so then what does this model basically say the model conceives our **semantic memory** as a network with nodes in the network corresponding Network corresponding to concepts so what this model says is that the **semantic memory** or semantic information well information is arranged in terms of a node on a network of nodes and so in in this network the nodes which we are talking about the central point from which communication occurs these are called the nodes the nodes are the concepts and basically the nodes which stores the basic features of the **concept** or the the nodes are always the **superordinate** ones **superordinate** nodes so **concept** storing nodes are always the **superordinate** nodes also they saw religious concepts related by connected in paths of the network so if two concepts are related together they are connected by some kind of a pointer or some kind of a link and this link is basically called the pointer now the more similar two nodes are the more similar two concepts are the higher the weight between these links so basically if two concepts are similar they the path running between these concepts or the pointer running between these concepts will have a higher weight then if two two concepts are connected which have very dissimilar properties are very dissimilar features now how do we think about this so for example let's think about fire and sunset now both fire and Sun said they are not connected together they should not be connected together because fire relates to some other **concept** is some other network and sunset is related to another network but then there is one feature of fire and sunset which are similar and that is that so one deletion between fire and the sunset is red but this since they are not related together the pointer running between fire and sunset should have a lesser weight but then if I'm talking about bread and butter although our king and queen when I'm talking about king and queen these two nodes or these two conner items in a particular Network are originating from two different networks but since king and queen are connected by also through our intermediate network or intermediary pointer and so each time when I say a king a queen really comes in or other ants the in a pair associate testing the Queen is the answer to king most of the time for if you ask people King and then ask them to say the first word that comes to mind that his queen is what the transient and what really happens is these two concepts are connected by at pointer which has very high weight similarly pointers connecting bread and

butter or pointers connecting within shoe and shoe laces are very high weights then pointers which are dissimilar - connecting our desk and an aeroplane so that although they would be related or they would be freely related but the pointer which connects these two nodes or these two items will be very weakly retained so that is what this model says the first further asserted that so in this particular model is further said that when one one mode is not activated the excitation of the node spreads down a path of links of X related notes so what happens it is similar to the **concept** of **spreading activation** and so what taking back from the idea of **spreading activation** what this **concept** says is that as soon as a node is activated if sooner than some energy is provided to a node this energy spreads down through not only through this node but all connected nodes all connected links so if I'm activating the animal node what would happen is the energy will spread down to all kind of animals which are there it could be Birds it could be mammals it could be reptiles it could be all kind of animal and so from there it will further process down or it will spread down to other within a million members could spread down to all kind of members which are there so it may maybe lion tiger and so on and so forth in terms of reptiles it will spread down to snakes to alligators and to so on and so forth incomes of birds it will spread down to different kind of birds whether it is a feathered word non feathered bird and soreness so basically this is how the the idea is what this module says is that when an energy is provided to a particular node this energy spreads down through all connected and Hulk linked nodes now they this particular theories or this theorists they believe that as activities activation spreads down the path or links to related node so the exhibition spreads outwardly and decreases in strength it basically what it means is that as the node as the energy it moves down a node what happens is this energy since its channelized it gets channelizing to so many other nodes the energy becomes decrease in strength so if two nodes are far apart if to connecting items or two connecting nodes are far apart from each other in terms of a number of pointers or fourth or fifth or sixth level relation and on the sixth level relation the energy which starts from the top the node will be very very decreased so when activation spreads outward it decreases in strength and as I said that both aeroplane and desk are related but they are weakly related because the energy which is activated when I say or when I activate the **concept** of an aeroplane or any flying object should be the highest **concept** of a Honda which are open bid fall so when that is activated although desk will be activated in some way but the energy spread will be very low because somehow somewhere they will be related but since they are so further out related the weights are very low and the energy that reaches to desk from an aeroplane will be very low and so what they I also say is that activating related concepts when an activation sparked from the **spreading activation** when any activation or when in OD energized it activates very related concepts with with high amount of energies and distant concepts are our distant nodes are activated with less energy that is what is supposed to be right so nodes which are related together or semantically or perceptually or any kind of relations are there between nodes they would be very closer together and so when energy when a particular node when the top node is energized then similar nodes since they are very close to each other they will receive maximum energy but those nodes which are further apart or which are very dissimilar to the particular node in question then the energy which is the witches which is coming to these nodes will be very less or it will be very low so think of this in this way so let's say that we have a node of animal and so within animal or let's say I have a node of living things so I have a top order node of living things and so within living things I'll have two or nodes which are animals and plants and within animals let's say I'll just cannot put two categories I'll have mammals and burped and within plans I'll have trees and shrubs and so within mammals I'll further have let's say lion tiger is one category and the other mammal is monkey orang utan within words I will have the feathered words so I will have something like a parrot and I will have a non feathered bird so I'll have something like an ostrich and with it this also so within trees I'll have an apple tree or I will have a pine tree and strap similarly a cellular process suppose I have a and BS now what does this model explain and how does this model quite unexplained so let's say if living thing is the node and these are the pointers to the nodes remember these are the nodes and this is the pointer which

shows the spread of energy so what it says is when any node is activated energy spreads down equally to these nodes right and so when a node let's say if our node line is activated now somewhere across this line will be related to this apple tree or this tree right some kind of connections would be there so lion leaves in forests and forest has trees and so they'll be connected somehow or other Saada but what this model particular says is that when this is energized and if lion is the node which is energized and this is how the energy is moving then concepts which are closely related together for example lion man animal and living things they will have a stronger connection stronger weight and so more energy will you spread here or more of them energy is invested into this node then further plants although plants are living things but since the land node energized and so energy will be spread to these nodes so birds will receive lesser energy similarly the monkey will receive less energy similarly feathered and non mother birds will receive less energy also no question of energy distribution here a very less energy distribution here but the most energy distribution will be to the node lion member animal and living things and this is what this module says so if first of all energy spreads it moves down and it moves down it spreads out so energy will be basically moving to all directions right and so here the less less energy is invested in these parts then in this part and this is what the basic proposition of this module is as I was talking about so this is how the model looks like so this is the animal node this is the bird node this is the chicken node and so within the bird node you have the chicken node and so you look into it so let's say if feet or if Robin is activated then this particular node will be energized and so high energy will be or more energy spent into this node then the nodes with chicken or the nodes with ie feathers and so these are also nodes these are also certain nodes which are they which are connected pointer so these nodes will not perceive and enough sufficient energy but these these these nodes will they see sufficient energy and so will these nodes because they are connected to Robin also these nodes which are their legs breathe eyes ears and so on and so forth will not be seeing energy and so that is what the model talks about and this is a way of it so basically this model is just an extension of the hierarchical symmetric model the only difference is that it talks about this shredding activation of how energy spreads when it starts when is activated at the top and how it filters down or it how it moves down from there so in this model the idea is that various our concepts have many connecting links and not placed close to each other so what this model says is that similar concepts concepts which has certain features certain distinctive features in each other they are placed together or they are blocked in some kind of a block which are where which have very close **proximity** with each other right now on each link and connection between two concepts is thought to have a certain weight or set of weights and now as I was explaining now since two concepts are related to each other are they are placed in close **proximity** to each other these links will have very higher weight so as as two links the more similar two links are there are more similar two nodes are the stronger the weight between them and the more energy or more energized they become then two nodes which are dissimilar with each other now the criticism to this model the criticism is that the model is the breadth of the model metal model is so huge that it is difficult to make clear and strong predictions from the model regarding empirical findings so the model is so huge it talks about that everything is connected to everything and there is there's this wait the idea of weight and the idea of how it is connected to each other and another problem with this model is what makes two concepts similar so this reality **concept** or this idea of **similarity** has not been explained by this model and so it's a difficulty it's a genuine difficulty with this model of how do we income pass or how do we study this **similarity** that we are talking about what do we call **similarity** so are we talking about the defining features are we talking about the characteristic features what makes **similarity** similar and so this is one thing or one problem with this model and so in opposition to this model not exactly direct opposition to this model another interesting model was proposed for **semantic memory** and this is called the end essence a CT or a CT our model so what is this model it was proposed by John Anderson in semi 1976 and it is called the **adaptive control of thought** model of **memory** it is also known as a CT a CT star a CTR and so on and so forth so those of you who do modeling or who do computational modeling will be

very familiar with the actr structure the adaptive control of thoughts no that so basically what is it's a process model is a process orientation model and so this process orientation model basically it's it's kind of a box pointer model remember the first class that we did we talked about two kinds of model the box and pointer model and we also talked about the connectivist model where we have unit input link an output in Kenna and hidden link so the next model that we talk about the connectivist model is basically at that time now this is a box and pointer model and so what does this model say so based on analog is two computers a CT gives rise to several computer simulations of cognitive processing in different tasks so basically this model has certain conceptualizations of how information in is stored in our **memory** basically world information or common sense information with implicit knowledge is stored in our **memory** so what are these conceptualization that this modulus is now the first conceptualization that this model talks about is that it distinguishes between the three type of **memory** systems it says that the human **memory** system is basically arranged in three different types or three different ways of looking into it it is arranged in terms of declarative memories non declarative memories is that **memory** which will have more all information facts knowledge arithmetic rule world knowledge everything into it and so there is no distinction of the episodic and the **semantics** here and so this model looks at **declarative memory** is the storehouse of all kinds of information whether it is personal information whether it is self related information or even related information or factor information so the first conceptualization of this model is that there is something called **declarative memory** and this **declarative memory** has all kinds of information that should be present or that is present in this world also there is a conceptualization of **working memory** which is information that is currently at the highest level of activation so it all talks about **working memory** or **working memory** is that kind of **memory** according to a CTR or **adaptive control of thought** model which we are using right now so although **declarative memory** stores a lot of information it has a lot of information and so I'll explain to you the how this **declarative memory** really works so basically **declarative memory** is conceived as a model with nodes it is similar to the hierarchical semantic mal model and so there are nodes and these nodes are connected by weights and these weights are the determined by the strength to the pointer or the **similarity** between two nodes so basically that's how **declarative memory** store but **declarative memory** cannot on its own do anything and so there is a conceptualization of **working memory working memory** is that part of **memory** or that kind of **memory** which is active at any point of time so when I'm accessing **declarative memory** the kind of **memory** which is accessing and doing all Carter alka operations on to **declarative memory** is called the **working memory** it is **memory** at the highest level so information access and **information processing** are all all those operations which are doing the **information processing** and accessing **memory** from declarative store is basically what is called the **working memory** so think of **working memory** is a part of **declarative memory** but it is that part which is active which is doing the processing or which is running the operations or the rules of the game which which it is running is what is the **working memory** and next to it is the idea of **procedural memory** so **procedural memory** is a kind of **memory** which is procedure based but it works on two or it works something like in terms of production rules so there are certain production rules will come to explain that also so there are certain production rules there are certain ways of doing things and the **procedural memory** is basically a kind of **memory** which follow these rules and does a particular kind of an act right so basically then this **memory** talks about **declarative memory** in this way Sanderson eighty-three believed the **declarative memory** stores information in network that contained notes as I said before what **declarative memory** generally tends to do it has a large vast network and these works as concepts which are stored as nodes and so all concepts which have similar features a **similarity** based kind of matching is there so those concepts which are similar a clubbed together and these have the highest weight and they the weight is basically the strength of how close two concepts are and so these have higher weights and they are connected by some kind of a pointer so this is how **declarative memory** is all about now there are different type of nodes including those corresponding to special images and

abstract propositions so basically instead of the conceptualization of **working memory** by allen-bradley very thinks about there is on something called us **visuospatial sketchpad** and **phonological loop** Anderson's model talks about **declarative memory** as network of nodes and these nodes are of two specialized very variation one is spatial image so space related information and because this model is a conceptualization of a computer program so it has to define spaces and so that one way of looking into **declarative memory** is space information or space images basically what in in a **problem space** how does somebody move so you should have to you have to know the boundaries of the space and the kind of problems in a space and that is why the idea of special image or space images there so nodes in **declarative memory** are of two types either it is space related nodes which tell you the boundaries or the kind of problems or the kind of hindrances that you get while self solving a problem and so that is path space relation information where you have abstract propositions so abstract propositions are basically how things are related to another what is the proposition or what is the kind of rule which relates to information together so basically after abstract propositions are how to rules are related to each other because these rules will define how a particular information is excess so a rule a and will be how they are related to together for example let's look at simple row so if if a if a inverse B and we infer C which not not means that seen fuzzy so this backward propagation is not that and so similarly these kind of rules of what relates to what or kind of it's part of a set theory so how these rules are there how these prepositions really work is what is stored in a particular node also a city model allows for both activation of any node and first reading activation of **concept** so it basically encompasses or it basically includes activation of node and **spreading activation** to be true so if any node is active the node will pass this activation on to correlated nodes and not related nodes together but related nodes will have high energy and non related no tree left lower energy and that that's why it also agrees with the idea of excitation passing over to nodes and the idea of what **spreading activation** is all about further this **concept** the actr the deductive control of thought it talks about **procedural memory** so what is **procedural memory** it says representation of a series of production rules so basically the **procedural memory** works in this way then there are certain production rules of how an act should be done now these production rules generally have a goal it has a starting a goal and certain kind of conditions a certain kind of rules to be followed so basically it goes in this way that if you start somewhere now this start and then there is a **goal state** also reaching to the **goal State** al are basically within the **goal state** and the start state are certain hindrances and certain rules that you have to follow and if you follow these rules and overcome these hindrances you will reach the goal States let's try and explain to you what the ceiling is so let's say that it is festival which is going on and so just near where you live in your hostel rooms or wherever you leave the the place near that everybody is singing in a very good mood but then you know that there is an exam right so the idea is that tomorrow you have an exam you are living where you are living and just beside the window of where you are start in your home lies or in your hostel lyza program where people are shouting and marrying together or they are making merry now the thing is that the goal is to pass the exam with good marks the starting point is in a position where you where you are sitting right now and then there are rules and there are hindrances to it the hindrances is if you are there then basically yeah there will be so much sound that you cannot read and the certain and the rules are that you should not disturb them so you have to follow a rule the rule is that certain rules have to be followed so basically you if there's a lot of people so you cannot stop them by rule they are free to do whatever they do but the rule also says that there are certain possibilities which are here the production rule says though there are certain possibilities here for example one part one solution to this problem is going to the library but it also has a rule so the library starts at 9:00 p.m. so wait for 9:00 p.m. then follow the rule of the library where by rule or the library no man can shout that so even if people wander when they cannot shout there and so one way to to reach your goal is to end to get out or get away from these hindrances is back you're back after 9:00 p.m. go to the library study there till the rule permits come back and pass the exam nicely so this way you have enjoyed till 9:00 p.m. whatever is happening

after 9:00 p.m. what are their follow certain rules of get rid of certain obstacles go to the library which is a solution to it and then pass the exam and there so this is basically achieving the **goal state** following certain rules and certain hindrances so basically these production rules or this **procedural memory** generally is the if-then statement which tells how to perform a particular action so if this happens then this is the kind of thing which is there that kind of a thing so if if you want to study for an exam together to four years for tomorrow then you have to find a place which is silent if you have to find a find a place which is sound then you have to do this and so there are certain rules hindrances goals and four-part system which is they we will talk about this into problem-solving into when we come to the section of **problem solving** also production rules specify your goals to achieve one or more conditions that has to be true for the rule to apply and one or more action that results by applying the rule so as they explain to you there is a goal to be achieved there are certain conditions so that should be fulfilled for example if you go to the library they could be silent and the rule says that it has to be silent and so you have to go there and then these are the conditions preconditions and what are more action the action is getting your books going to the library starting there and then coming back and so this is how the conceptualization of **working memory** or the idea of actr really talks about in terms of **procedural memory** so basically in terms of a CTR it's about three **memory** systems which interact with each other and they lead to the formation or storing of **memory** data into the human brain now basically then this is how a simple ACPR model will work and so you as you see this is the **declarative memory** so you have both the motor module and the visual module so two modules are there and these collect information from the environment the motor model can contain spatial information the visual module contains those proposition abstract propositions which are there and combine to something called the actr buffers the actr buffers then also gathers information from **procedural memory** because the **procedural memory** will tell you how to exactly go about the procedure of doing certain things so basically how to go about the storing things so **declarative memory** is these two parts are the **declarative memory** and this information comes to a CGI buffer the ICTR buffer then talks to **procedural memory** which tells you the procedures which tell you what has to be done and so the two procedures are pattern matching and product segregation and based on that you will come to again to the actr so this is the input and this is the output and the output is again fed to the input and this kind of a chain goes on and on so basically any module or any kind of information which comes here is processed through this kind of a thing else this here it is a simple visual task matching program that have this and so as you can see this is how this model really works the actr model really works now in addition to the actr model interesting another model which has been proposed is called a connectivist model now what is this model it's a very interesting model and this model doesn't conceptualize the way we thought about how things are stored either as networks or memories or in terms of concepts or in terms of knowledge network of knowledge or in terms of three kinds of memories they don't conceptualize this in this way so what connectivist model think about is that what is stored they say that what is stored in the **memory** or what is stored as word knowledge is basically a set of changes in the instruction of neurons which sent to each other affecting the pattern of activity that can be constructed from a grave engine what does it really mean it says that what is stored first of all it's it's kind of a copy from the biological model so as as we saw in one of these classes I explained to you how **memory** stories every stored as interconnections between neurons and so what really happens is **memory** is basically a connection or it's basically a path from one neuron to the other neuron and so what this model says is that what is stored between two neurons is not knowledge or concepts now what is stored in the **memory** is not knowledge or concepts it is set of instructions which are stored and which is passed on from one neuron to the another what are these instructions these instructions tell a particular set of neuron which are connected to each other to process how to process data and so there are certain modes of processing and so these instructions basically ask neurons are basically instruct neurons in processing so let's look at I will briefly define this model it's a very huge and complex model so I'll try and very briefly define this model for you so basically this model learns

to develop patterns of activation through many trials which examples for our example back propagation so basically how this model links is how this model really works is there are certain input layer there are certain input features or input units now our unit have has is a pattern is a unit is thought about as item which receives input from the environment and so this unit then passes this information on to a hidden layer or hidden unit or hidden related set of units which are there for example verifying Robin is a bird it was a very simple statement organ is a word the Robin as soon as a Robin unit is activated it will unit it will activate all related units with the bird with it right so things like bird has legs bird has feathers but has weeks but has a different color and so on and so forth so all these units will get activated now how does this model really work so it has an input unit and this input unit when they get an input some of these hidden and that there is a lot of hidden layer there is a lot of in hidden input which is there and so what happens is the strength of activation or there's a number of inputs which are there let's look into this model first of all so basically this is the connectivist model of a robin and so this is a lot of inputs which is coming from the environment so when you see Robin for example this is the bird that you are seeing and so only when Robin is activated all these things which are coming from the environment is deactivated at that point of time only the robin mode is activated and this robin mode northern connects to this hidden layer so robin is so basically if I am activating the word Robin Robin is basically a bird and so basically is is an has this is basically the kind of connector to robin which can get activated basically robin is a bird robin can be a bird robin has a word all kind of connectors are there and so robin Easy's activated or can is activated and so these activations further link to this the hidden model or this hidden layer and the certain layer then basically gets strained and then from there the thing is now Robin is a particular name and kin is also a particular name so what can Robbie do Robin basically can be both a bird and a person right and so when I see a Robin in the environment the node which is activated is Robin if I'm saying if this is activated and Robin can so what can Robin do if it is a man if it is a human thing it can do these many things but if it is Robin can also be an abstract thing and so when it is an abstract thing it can have several other features with it but when it is a bird then it can do all these things it these nodes are activated because the bird can grow the word can move the word can not swim in so this is not activated the bird cannot sing so it is not activated maybe it should activate the word cannot bark and so this is not activated the bird doesn't have branches and so it is not activated so when a set of input node is activated these input nodes has a number of connecting nodes with it right so when when I look let's assume an example where I look outside of my window and when I see several things I put my **attention** on to Robin now this Robin has a bird is a bird and so there are certain features of Robin which has to be activated and so basically this model then say is that if Robin is activated similarly we will be activated the word can Robin can and Robin can do what so you see a Robin or you see a bird which is on the on on where your **attention** is followed but there are several other things which are there so there will be several other nodes which can get activated but your **attention** is focused on two only Robin and so what Robin can do is it can grow it can move it can fly and these three things are the only one which should be activated but could also be activated but then other things are related to all other things that you have seen in the outside environmental world and so basically it will not get activated now how does this model really work so this model says is that the input layer basically it learns so when an input layer is activated it activates related layers related hidden lanes with it and so it at the beginning of any input cycle the strength between the input node and the hidden node will be random weights let's say if zero and one of the two allowable weights which the input layer and the hidden layer would have at the beginning of any propagation or the beginning of any input signal the weights between the hidden layer and the input layer would be 0.25 now as as I said there are a number of factors or the number of items which compete with each other to start the input and since Robin has started the input it will change its weight right though so the weight of Robin will become 0.8 from zero point five will become zero point eight but all other weights for example tree house all other things which have been captured at the same moment but **attention** has not been put to

the weights of these will change slowly change to 0.2 0.3 0.1 why 0.15 or so on and so forth only the robin node is activated so the weight of this will increase which will increase to one zero point eight and so the in hidden layer will learn this now then there will also get some information both from the perceptual system as well as from the **memory** system because robin is activated so the **memory** system will also feed in some information and so for example robin can Robin has Robin kind of a thing so it is present tense and so can would be activated and so can will have a higher weight it will the weights will change to 0.8 and similarly there are certain features of what Robin can do now Robin can fly it oven can has a weak it it can do this it will do sing it can have color and all those things will be activated in all other ways will not be activated now basically this model then what it does is from an input it generates an output and with that input/output system it starts learning right and so well how it what is the meaning of learning each time a particular input is selected and output is generated the connection becomes stronger and stronger more stronger the connection the more faster the learning is this kind of model basically is a learning model so let's look at what this is and so this how do we train this model so first you show the model that Robin is whatever it is to anybody you show the model any but it can do these things and they can basically result in this kind of a feature and so it is a training example from that training example show it the model a bird now you have started with the idea of bird so from Robin change it to parrot and that's how the model behaves now parrot is very similar to la Robin so as soon as it identifies it is closer to the **concept** Robin so it will then adjust the bait between it and the when the the input system and the hidden system as 0.8 it will learn this and similarly output all the features that will output for the **concept** a bird and it goes on doing this and so our training after training or time after time it will then develop that this is the feature of what should be of a bird so next time you present it a bird that it doesn't know which is not a training example it quickly classifies this as a bird and then outputs an list of features that the bird is going to happen so initially connections between nodes are set as I said you said a set at random weights experience leaves these connections to be activated more and less strongly so the more closely each time when it produces an output this output is compared to a set of truth features this output is connected is basically matched to what should be what should be expected as an output and the closer the match the better their training so if it's not closer then the model model adjusts the weights accordingly and we either delete some of the features or increases some of the features and this goes on making the weights or the goes on making the number of connections so training occurs by presenting a specific example to the network with there generates a particular output so you start with let's say parrot and you teach the model then if parrot occurs if parrot comes in these are the features you need to output next time you start the training with say a sparrow this is what it is so slowly the more model or Scalia the input layer will understand that ok once something like this is presented something like a bird is presented with gathers those features then these are the nodes which has to be activated and so the nodes which have to be activated then where there's a higher weight and then the output which it which it should present so also tell it the output so when I show a parrot show me the output that it can fly and wing it has wings it can do this it can do that that kind of a thing and also show me the weights for things it cannot do and so match it with this idea that parrots can fly so this is a verified sentence and this is a true sentence so verify with that now the next after so many trials after say 50 or 100 trials when you show it may be a different kind of bird for example let's say crow and what does the model do when you show it a crow although you do not tell the model this is a crow the model adjust the weights accordingly and says that it has a beak it can fly and so on and so forth because these are the pruning programs so you train it with some examples and then the model learns on its own and starts adjusting the weight so it will also give you with the crow it will also give you inputs outputs like it is black in color it has two eyes beaks these are the generic things but certain other features also it will tell you also training placing epochs so basically these trainings they go about in epochs and each epoch produces an output activation which is compared to a correct target activation this is how this model really works and as I said there's a number of things are there's a number

of input systems which are activated only robin is activated and so what will happen is only these connections are energized and all connections have starts having lower weights and so since this feature is related to this feature so these features add on to this model and so only ABC as you can see is activated here and others are not and so when it gets training then in the next training round on the next epoch it will also learn that it is a bird it will also learn that it is red or green in color it is also along that it it has wings and feathers and so on and so forth and some more number of outputs will be activated so it is not only that through these models we can explain **semantic memory** there are two more concepts through which we can understand how semantic knowledge or world knowledge is Eirene and so toward these concepts one of the famous ways of doing that is what was proposed by Sir Frederick Bartlett and so what Bartlett did was he's read a very interesting study and he showed that there is a lot of reconstruction in **memory** and this reconstruction is how so what he says is that what semantic memories stores is not an exact representation of what happens but a gist a kind of a broken down version of the information that is what is stored in our **memory** and so he in his experiments he presented something called a war of ghosts a story to one a couple of people and the story was a complex story it had many relations and many kind of information into it they were so interfaced and they were so closer to each other that was very difficult to break the story apart and so this story was presented to a number of people and when these people who were presented they were asked to recall the story after a certain period of time but what really happened is that most of the people remember the gist of the story the main idea of the story but they forgot the basic features are the basic some basic things about the story for example names places these very mixed up but the story the outline the story was what was there in Frederick that partly says it there is a way of storing these kind of information agenda involve knowledge information and that is what a **schema** is so then look at what is a **schema** so basically **schema** represents knowledge in the **semantic memory** that's what what les proposition is there is something called a **schema** so what is a **schema** then a **schema** contains general knowledge about the world and information about particular events so it's not only consisting of general knowledge about world but it also has information about particular even so it has both declarative as well as schematic semantic type of information onto it now schemas are large units of organized information which are used for representing concepts situations events actions in memories so basically a **schema** is kind of a huge it's a huge box kind of a thing where a lot of information could be presented into or could be put into and these information when they are put into they sort of act together or they sort work together for example rumelhart and oourtney in 77 they propose that the fundamental building block of **cognition** is this **schema** so this **schema** has information stored into it into all kinds of and world knowledge it's kind of a guideline of what should happen a **schema** is kind of a guideline or what's to be expected for example the **schema** of let's say a village now the **schema** for village basically says that these things for example bollocks feels Greenfield use water bodies people farmers trees animals this is what should you expect in a village you should not or you can not expect a car or modern build building or an aeroplane in a village and so these are the kind of things which is expected with the kind of which is not expecting the **schema** for village will have this kind of an information so basically then schemas are big packets of information that contains both a variable in a fixed part for example if we look at the **schema** of a dog the fixed part of this information is that it has mammals in four legs because the fixed part is sad across all the instance of the **schema** so any dog we look into it it will be a mammal and it will have four legs but then most Chema's also have a variable part and the variable part is where we wear this feature they change across instances or change across variations of the **schema** and so things like breed things like size color temperament these mice change the curd dog so you might have an Alsatian you might have a chameleon you might have a scent Bernard or you can have any kind of dog which is there you could have a very light tempered dog a heavy tempered dog a highly colored or non colored dog a Chihuahua kind of a thing which has a very strange kind of a thing sizes so as I said you can have great dane which is very high in size you can have a chihuahua which looks like more like a cat and so on and

so forth so within dog variations could be there but then the fixed part is that all dogs have are all dogs within that particular category or that sistema should be same also **schema** indicates relationship among various parts of information now what what do I mean by this if I'm talking about a dog now the idea is that the dog should have a head and this head should have two eyes only the tail should not be coming out of the head and or the nose so basically there are certain relations which are there among certain parts of the dog and that is what it is so the rear of the log should have the tail and the forward of the dog should have eyes and the color of the dog should be in this variation the hind leg should be here the fore leg should be here this kind of interpretation should be there so Sigma indicates the relationship among the various pieces of information should be there what is the fixed piece and what is the variable piece and what is the relation to each other so a dog which has our legs coming out of the eyes is not actually your dog an instance of a dog but a dog which has eyes coming out of the head is basically a kind of a dog and so this is the relationship and so the leg should be below the eyes and so on and so forth all school schemata connects to other schemata in a varieties of ways so not just one **schema** a number of schemas can be connected to each other one **schema** of particular thing can also be connected to on the **schema** of some other things now **schema** is also used for filling or in default values of certain aspects of a situation which help us in making an assumption so basically the **schema** tells us as I said the **schema** tells us what is to be done in a particular situation so let's say if you are in in in an exam hall the **schema** says that or if I am telling you a story of how I met my roommate school class roommate for the first time I do not need to tell you that what is my age because most people turn to school around 18 19 years in a graduation school around 18 19 years so this can be auto-filled that this is what your age should be and so when I am talking when referring to a friend as her you know I need to tell the gender of this fail so this can be a **schema** will provide this kind of a thing so when I was talking about a gender I am talking about that my roommate is a her then the idea that this is a female is automatically comes in also **schema** exists in all levels of abstraction thus they can exist for small parts of knowledge also so basically what it means is that schemas can be for very small knowledge abstractions for example certain letters written in certain kind of ink letters or it could be the **schema** could also exist for greater details for example like the theory of relativity so there are certain schemas in the theory of relativity for example the fabric of time the idea that time dilation happens the idea that there is something called the space-time where space means time and then this variation so those schemas and also very small schemas of how certain letters can when combined with each other give us a **perception** of a different letter so that basically our that level of abstraction also **schema** should work so write in this way and it becomes a B or mean this a you have seen those letters right Kelly graph is of certain kind of letters and so they also are a certain level of scheme identification now related to the idea for **schema** is the idea of a **script** now **script** a ski mask for routine events for example going to a restaurant what is a **script** think of it in this way when you go to a restaurant there is basically four part meal system right you start with the soup or some kind of a beverage to start with then there is the first course then there is the main course then there is a suite and then you wash your hands you pay the bill and you move away and so basically this is what the **script** is the **script** says how the **schema** moves in time how the **schema** moves across so what is the production rule for a **schema** or what is the procedure of a **schema** or what is the procedure for routine event is what our **script** is all about now scripts are used across a variety of situations for filling out unknowns for example let's say that I visit a new city right so I visit some some city I'm the Luminess am sitting in am living in India and I go to the US now when I go there I did not think what a restaurant should be in how to sue the restaurant function because I know from India that this is how a restaurant functions so most rational restaurants would function in the similar way it's not that in those days friends something else will happen the same idea of what I have in India should be a capture of an incident there and so scripts certainly help us in organising information or filling up information or basically how we can use it in different scenarios schemas also scripts also help us in making a number of inferences now what do I mean by that when I

say that you have you can get a number of inferences so a particular **script** when define it shows you that what is the result of this particular thing so if a main course is there what is the result of this main course in a dinner what is the result of the or what follows for example so if I say that I went to the restaurant and I ate ate my food and I came back basically you feel in your head that you might have paid the bill and that then you came back and so this feeling of information that you arrive the restaurant with someone went into the restaurant sat at a place when I say that I ate food in a restaurant today basically the all other ideas of you sitting in a restaurant having ordering the food eating itself is filled up and so they this information this inferences can be made from it that you are paid the bill and come and you not run away from things the restaurant did come out and so these inferences can be and also lastly schemas help us in making an order and so to this to test this particular thing how **schema** makes us the things perceive things in order this baauer black and Turner they need a particular study and so in this study what the leaders do is presented a story and in this story the story was made in such a way that there were several concepts in the story and they were in unordered form so some parts of the story the climate's of the story came before and the meaning of the story was in the somewhere in the middle and the middle of the story was somebody in the back and that kind of information was presented to people nobody when I recall was done most people remember the story in the right correct procedure in the right scripted order so although the scripts was deviated people remembered it in the right way from how it began - what was the climax - what was the middle part - what was the climax and how it ended right so no matter if intermix the kind of sequences of a story or sequences of a particular event a routine event this schemas or this scripts will basically help us reorder these things together so basically what we did in today's class is we looked at some other models of **semantic memory** or world knowledge **memory** into it how do they work and what are the pitfalls of this model criticism of this model we also source the connectivist model which is a very interesting model so if you are looking up for taking up computational modeling you can either use a CTR or connectivist modulus and these models are interesting models of how they product because these models are also used for generating artificial intelligence systems or learning systems so basically that's what we discussed there and then from there we also looked at another way of storing information in terms of scripts and schemas and so this brings us to the end of this lecture thank you

Lecture 17: Introducing Concepts and Categories-01

■Kind: captions so welcome friends welcome back to this lecture on concepts and categorizations and basically I've given it the title let me organize so basically that's what our **concept** when **categorization** really means it helps you to organize knowledge now what do we study in this particular lecture so what we do is this lecture is exactly an extension of the **semantic memory** lecture where we saw how world knowledge facts and arithmetic rules and procedures and that kind of knowledge each a which has some element of consciousness into it my my the element of a conscious **retrieval** into it how are they arranged into the **cognition** how it how is it our into the mental **concept** or the mental structure so what concepts in **categorization** are basically these are the process of organizing this knowledge now in the **semantic memory** lecture we saw that there are several models and these models what they do is they go ahead and explain the organization of knowledge into **semantic memory** and one of the famous models which was explained in the **semantic memory** lecture was the hierarchical structure model so but the hierarchical structure model explained is that there are nodes and these nodes are connected by monitors and so there are top level loads and and sub level nodes and so on and so forth so that's kind of arrangement with the **semantic memory** lecture talks about so basically what in this lecture we'll see is this how what is the meaning of this node or what is the feature of this node and so the nodes that we talked about in **semantics** when the lecture is basically concepts so concepts are kind of **mental representation** there of loans which hold a lot of knowledge into it and concepts also help us in categorizing information so why is **categorization** important so as you remember that the human brain has a limited capacity and so it can do only a limited number of jobs now number of neurons is limited we saw in the lecture on **long-term memory** that there is a limited number of information that you can hold on to **memory** even the calculations are given there which basically means that the human brain has to do something in terms of organizing or in terms of archiving data in in the mental lexicon and so one way to do that is basically **categorization** so **categorization** is a process which basically goes ahead and it form some kind of a mental bucket where you put an information based on either **similarity** or some other aspect so basically that is what categorizations really mean and the principle in which **categorization** works is basically the principle of how concepts are formed or what are the guidelines of a **concept** so using the guideline of a **concept** a **categorization** is done so let this gen begin this lecture on **concept** and **categorization** now let's look at this psychology what is it psychology is basically in most Indian universities they are taught under the arts stream they are taught under the arts faculty or sometimes that is the social science faculty and so most people when you talk about psychology they will say there is in subject to the arts or a subject of the social sciences and you imagine social sciences but most all universities will clamp it into arts but if you look into most Western University psychology is not an arts it's not taught under arts it's talking to brain in behavioral sciences or sometimes it is taught in the Medical Sciences or Natural Sciences now this basic classification of a psychology should be is basically demonstration to show how **categorization** happens and these categorizations of placing psychology into social science or arts or into the Natural Sciences or into the medical sciences basically they are done based on some kind of a conceptualization basic basically based on some kind of a rule or some kind of an aspect or some kind of a feature that psychology share with these Sciences and so this is basically formation of a **concept** so there is a **concept** and the **concept** is basically a **mental representation** a mental bucket so the reason why these Western universities go ahead and classify psychology in two Sciences as opposed to what in our country it is done in terms of the arts or social sciences is basically dependent dependent on the **mental representation** of how they look into psychology and so this is basically a demonstration of how concepts and **categorization** happen so they have some **concept** the Westerners have some **concept** of psychology because it uses scientific methods a **verification** of some kind of data falsification some kind of repetitions and experimentations and so they are classified into

science but in terms of our own country they don't look at these aspects and they have some other aspects of psychology for example is social aspect or organizational aspect and because of that they classify them into then the art sciences or into the social sciences group dealing with society and so on and so forth but that's the kind of division visible and so this example then explains what is the difference between the **concept** and **categorization** is but before going into the more details of it let's look at what is basically the proper definition of a **concept** or categories so starting before starting I have some diagrams or as some figures for you which which are quite funny to look at and they explain something about water concepts and water **categorization** and so first figure says that there are a property so a **concept** has certain property and so within the **concept** then you have so here you see the **concept** of a cup and a coffee which is a sign of a **concept** so basically it's an emblem of a **concept** or it's the logo of a **concept** and so when you see something like this you basically believe that it's a coffee shop or it's something which is hot beverage to be drawn and so it has certain properties so when it say it is a coffee this drawing represents our coffee in a certain properties and these are the properties of a **concept** and so further on this is the **concept** so if you look into it this is the emblem or this is the referral or the logo for a **concept** and these are the properties of this **concept** and so basically this is what a **concept** looks like so it is the **mental representation** of what a coffee would look like these are the properties of a **concept** so every type of coffee that we classify here that we put here into this coffee's the diagram should have these properties and as we look into it this is called **categorization** so in **categorization** what you do is you take in items or you take in elements events or instances of objects which are similar to the **concept** and place it under one particular bin and that is basically called categorizations as you look into it so this is coffee and so there are properties of coffee is a shape type color aroma ingredients and so on and so forth and in terms of **categorization** the reference our latte cappuccino espresso a black coffee or the iced coffee or it could be some other form of coffee which oh it could be the Irish coffee or some other form of half coffee which it could be so basically this is the **concept** and this is the property of **categorization** and so **categorization** has done based on these property so as you see that most of these type of coffees have these properties and so this is the process of **categorization** now a funny cartoon to look into it and so here if you look on the right hand side there is this particular cartoon which explains that this lady says I do understand the **concept** of what cooking is I don't know how does it apply to me she doesn't want to cook and so that's the reason why she is putting into these facts here and so that might be the reason why is she saying what she is saying so again coming back to the definition of what concepts are before we understand that let's look at what do humans do by doing **cognition** or what does cognitive processes and actually they do so in general the normative psychologists believe that humans they form mental representations and that these are processed by the cognitive processes so whenever we whenever we store knowledge or whenever we take all knowledge and store into our brain or mind what we do is we form a **mental representation** of it and this **mental representation** is what is processed by the brain or by different cognitive processes so what is basically **mental representation mental representation** is the psychological equivalent of the physical representation right and so remember in the first lecture when we looked at different different kind of mental representations so we saw the spatial arrangement and and the abstract arrangement so in terms of further amplified there are two mental representations that I will show you here and so in one metal representation what really happens is that I can show you a visual or a propositional **mental representation** in the visual representation I draw this kind of a structure and I put a ball on it and so this explains the ball is on the table or in the prepositional this representation I can say ball on n table are two propositions of how the ball is connected to it so remember for the first class both of them represent the same thing the idea is that the ball is on the table and so these two formats of what are called mental representations and so what a **mental representation** these are the psychological equivalent of physical representation so what **cognition** does is that they take in information from the world and convent mental representations and you say well cognitive processes to process this **mental**

representation and so thus all knowledge in humans are stored as mental representations and that they in turn guide our behavior so basically then most knowledge that we have they are stored in terms of these mental representations different kind of metal representations be it is visual beat in order to deform weight in in propositional form and different other different forms of propositional form is a famous form which Anderson goes ahead and talks about he says that most mental representations are stored in propositional form so there's a proposition to it and so that's how basically it is they're stored into so we did some part of propositional thought and propositional representations in the earlier classes so remember from there that's what it is so basically then the idea here is that it is mental representations they told an and his mental representational data processed by human beings and most behavior that humans do is by the processing of these mental representations for mental representations how are they stored they ask Torres concepts and categories so remember the coffee example what we did was the idea the the logo of a cup kind of a Gyan or a cup kind of a Jian and a plate kind of a genre presence of coffee and so this is the **mental representation** of visual **mental representation** and when I show you this there are several knowledge which comes with it for example the property is the type of things so the remembering different types of it you physically call calls in the process of **categorization** and remembering that this is coffee is called **concept** so basically concepts and categorizations are basically play around mental representations so what is a **concept** then what is the meaning of a **concept** the **concept** is a **mental representation** of some object event or pattern that stores in it much of the knowledge typically thought relevant to object pattern or even so basically mentally the **concept** stores in some **mental representation** and knowledge related to **mental representation** so in our earlier example the idea or the coffee logo it stores with it that this is the logo so the material presentation of a coffee in terms of the logo and some knowledge which is relevant to it in terms of object event in pattern so example the different type of coffees which are available right what is the aroma of a coffee what is the taste of the coffee where can I find it what is the price of a coffee different countries from where coffee comes in and so these are all related knowledge right related to the idea of coffee and so this is what is stored into that particular **concept** of Akaka so it's basically income the **semantic memory** if you activate that note if you energize the coffee node several other nodes will energize so when I think about the coffee cup when I see about the coffee cup a lot of knowledge come to mine and these lots of colleges are connected to different different semantic networks and so one is if one is any energy the other is also in a chest and so depending on the context depending on the priority the or nearness or semantic nearness it it will decide which I energy is more and which I energize less basically something or from the **semantic memory** thing so basically let's now stick to only the idea of what a **concept** in **categorization** is so in terms of an example the idea of a dog the **concept** of a dog what does it comprised of it comprises of that reason animal so first thing is in terms of a dog a dog in an animal it has four legs so basically another piece of knowledge which is stored with you that's just four legs it has a tail it has man's best friend these are all different kinds of knowledge and all different kind of facts which are there or it barks or it eats certain kind of foods or there are different varieties of dog all this information is basically stored into a **concept** so **concept** is basically a mental bin around which knowledge is restored a number of knowledge is stored or comprised into or fitted into and so that's what a **concept** is now if that is what a **concept** is what is **categorization** basically a category is defined as a class of similar things object or entities that share a couple of factors and has an essential core so basically what do I mean by this its type of things or classes of things which which have similar objective with similar object as a better entities for example let's look at a kennel of dogs or a field of the dogs full of dogs now when you look into it then different kinds of dogs or if you have seen dog shows nine dog shows you see basically a dog show is a category and so what happens is there are different kind of dogs or dog itself is a category so with the dog category you have different kinds of dogs you would have a chameleon you would have an Aussie Shane you would have a scene but now you will have a German Shepherd and some other kind of dog different kinds of dogs which

is the Rottweiler and so different kinds of dogs are there and all of them differ in some aspect or the other so there are variations there are differences within it but they share one or two things which are essentially core for example the idea that they have bark now that is a core property so most dogs that you look into will bark although there will be an exception and so they will discussed it in the last class in in the **semantic memory** where it might be possible that one or two incidents may not be similar so that's one thing to be looked at so **categorization** includes similar things together but then what happens is they share one or two features or they share at least one essential core with it so for example the ideal acquire all signs of course is considered as Sciences so the basic question that we did or some similar **similarity** in perceptual biological or functional property basically things which are clubbed together which are categorized together they share some kind of features whether they share an essential core or they share some kind of **similarity** between them so we can always have categories where things do not share properties for example let's say the category of it house now when I say the category of an house within the house will have several things for example you will have a mother a father which are human beings then you would have chairs desks or you will have tables you will have eating things electrical appliances all of them together are there but they can be clubbed together as a category so these features then share one feature which is they are parts of a cattle category the house or the household right and so this is how it is so at times the most items in a category sash some kind of a feature on at times what happens is that is not share core features but they say characteristic features or they share some properties which are either perceptual biological functional or some other relation but something is shared upon them so items which share some kind of features together and clubbed together is what is a category or what is the process of **categorization** then what is the nature of a **concept** so as I explained to you the nature of a **concept** has to do with the idea of **categorization** and what can a **concept** actually do now concepts help us in establishing order on the knowledge base so what does it really mean it really means that through **categorization** through forms a formation of concepts we can form some kind of an order for example what is the higher **concept** what is the lower **concept** who's bigger who's smaller kind of a thing so within a number of concepts within a number of items which we have we can then assign some kind of order now remember in the **hierarchical semantic Network** we looked at something called the **subordinate** in the **superordinate** node right and so concepts shared or concepts were always the **superordinate** node and other things items when the super the **subordinate** nodes similarly when we have **concept** this **concept** allow us to basically arrange things into order for example think about musical instruments so musical instrument is a higher level **concept** within that we'll have string instruments we'll have an air instruments will have wind instruments will have other kind of instruments which can be beaten to be played for example drums and all so different kind of instruments will be there so there is an order to it so and this order of how it starts with a musical instrument - coming back to a string instrument and between the string instrument you have the guitar or the sitar and then you will have several others so there is a hierarchy or there is an order and so this order classification is important that is what **categorization** really or concepts really do also concepts help us to categorize that is what we are trying to tell you that with concepts we can put items into mental buckets so once we know that this is what a **concept** is this is what **concept** a is this **concept** B exists these concepts will have certain features right certain aspects and these aspects will help us put similar things together right and so that is what concepts really do also **categorization** helps us making prediction for example if something is coming from somewhere a four-legged animal is coming from somewhere wiggling his tail and he comes near us and his tongue is out of his mouth and his sweating kind of dripping onto food oh and he looks more or less like category or dock then you can predict that it will also bark right and so this idea that if it looks like a dog and can classify the dog you can make predictions so **categorization** helps you in making predictions of what a particular item and you might them in the category is supposed to do because it should share some features of the category so with that with all these features you can then say that if this is a dog it is going to bark and it is going to be

man's best friend and so on and so forth so this kind of conceptualizations can be developed right and so that is water **concept** really does so what is the nature of a **concept** so facts explaining the nature of a **concept** several models were developed they were different models and these models actually go ahead and develop the nature of a **concept** for example one of the model or the oldest possible model is called the **classical view** model now what are the **classical view** model says it's actually a model which dates back to the time of Aristotle and so it's dominant it was nominated in psychology - until the 1970s and then later on newer models for example Keynesian based view of **categorization** and conceptualization came in and so that took us or that took the center stage but the **classical view** is the most reliable view up till now so what does the view say it says that all examples or instances of a **concept** share fundamental characteristic features so basically what it says is that no matter what happens all instances are all different type of items which are clubbed together in a **concept** which are called which are known to be part of a category and which resemble a **concept** they share certain fundamental characteristic features okay for example if I say that a dog then all dogs should have a tail they should way they should bark and so on and so forth now in particular the **classical view** of **concept** holds the holds that the features that the concepts share together there are items which represent the **concept** share together I represent the features are represented are individually necessary and collectively sufficient so mostly the what the classical music is that most items which are grouped or categorized under a **concept** or categorize of known to be part of a **concept** or known to form a **concept** resemblance to a **concept** now they should have something called features which are individually necessary and sufficient and collectively sufficient what is the meaning of individually necessary and collectively sufficient now individually necessary means that each example must have the feature if it is to be regarded as a member of the category so basically each and every example should have the feature the kind of feature for it to be called a member of up that particular category or a member of that particular answer example looking again back into the dog **concept** so each member should have one feature that is parking so each dog should bark or it should have a tail they should wait right it should not have a drooping tail kind of a thing or it should have a certain kind of face drooping year's men's best friend or whatever you want to so it should have that thing so okay so individually necessary means that every item should have that particular feature into it and what does collectively sufficient really mean so collectively sufficient means that anything with each feature in the set automatically becomes an instant of the **concept** so if something is presented to you a new item is presented to you which has each feature of the **concept** or each property of the **concept** it automatically becomes a part of the content **concept** it becomes a category or the **concept** so in terms of looking at the triangle if we look at the individual necessary thing individual necessary thing say that this each triangle should have three sides if it has less than three sides it cannot be a triangle if it has more than three sides it cannot be a triangle and what does nefesh unnecessary sufficient mean necessary sufficient conceptualizing is so the fact that it should have three sides it should be a closed geometrical form and then it is a triangle so any figure which has three sides and which have some kind of a closed geometrical form necessarily is necessarily called a triangle whether you call it a equilateral triangle or you call it a right angle triangle or if select rule is so all kind of triangles would be there but the idea is that it should have these three sides and a closed geometric form and that's what the **classical view** say so in very simply it says that there's most **concept** they share instances and fundamental characteristics and these characteristics are individually necessary and collectively sufficient you have to look at this let's look at the **concept** of a bachelor so the features are it should be male adult and married and human so then there is a problem to this right now if I look at somebody who's 10 years old you know 10 year old person is a male or a boy who's 10 year old if is a male adult he's unmarried in his human but he cannot be a bachelor and so these are the deviations which are there and so individually necessary collectively sufficient that is the problem with it but then that's a thing for the cat the criticism of the model but before that let's look into this so then Bachelor should have these four aspects now any person any male who's an addled and who's

unmarried and who's human is actually a bachelor similarly even numbers what are the cons of the **concept** of an even number what are the features of those numbers which are even the features are it should be first an integer so can be explained in the form of P by Q first and it should be divisible by two so if we can do that then it is an even number and triangle for example it should be a planar figure so it should be a figure which should be enough two-dimensional or it should be non some plane and then a close geometric figure so figure remember from the first class where we are doing perceptual illusion and we we showed you this kind of a triangle right and so this cannot be called a triangle the reason being that this feeling that I'm doing here is done by the brain and so cannot be triangle until unless it has all the size like this I cannot be triangle and so it is not a triangle and so what a triangle to be a triangle it should be close to my trick figure and then it should have three sides two sides cannot be a triangle and so that is what the features are so that what are the implications of this **classical view** the implications are concepts are mentally represented list of features so what then this classical music is that concepts what they do is that they mentally represent a list of features which should be then so any item which comes in new should be compared at a keynote this list so if I see a dog or a triangle a dog for example if a **concept** then I have a list of features example it has to have four legs it has to have a tail wagging tail man's best friend should have should bark should have tongue out most of the time and so on and so forth and these are the number of features or lists of each other so classical music that concepts mentally represent a list of features and so any new instance is compared across that and the more fit it is the more it is also it assumed that membership in a category is clear-cut but this particular **concept** or this particular model says is that any member of any category has a clear-cut boundary the idea is that you are either a part of a category or you are not part of that category and so that is how **categorization** is done this boundary is very clear-cut so you are either a dog or not a dog right and so if you do not have the sufficiently necessary features or individually necessary features then you don't become a dog so you should have feature a let's say I have abcdefgh features which are rated for a particular **concept** of a dog then any new instance or any new element which basically is compared to this instance then has to have these features and so this it says that the mantri are very clear-cut there is no disguise in the boundaries or there is no overlap in the boundaries and the third is that it implies that all members within a category are created equal which basically means that there is nothing called ideal **concept** and unknown ideal **concept** what they say is each member of the category shall equal kind of or equal in all relations so most members within a category are equal in some 40 that's what the proposal is all about now these are the problem and so looking at dogs most dogs are equal that's what they will say well so although talking in terms of size for not talking in terms of size most dogs behave in similar manner that's what they will talk about and when I say it is clear-cut which basically means that chihuahua or Siamese cat are different because there's a clear-cut idea that they don't bark and so by barking is a individually necessary feature so Jesus since the sign is cat wouldn't bark it is not dog and since a shoe of a box it is a dog and so that kind of a clear-cut bond is and all that kind of a clear-cut boundary is out there so these are the three implications of this but then this new has several criticism so that's it's an old **concept** and so it has several problems with it first Joseph wanted people judge different members of a category differently right and so what does it really mean it means that different different members are the categorized or arranged differently that's the reason and the reason why is that people judge category membership in terms of how good a fit they are to the category right and so when I look at an audition that is a classiest dog or the dog jazz dog that I can talk about but when I look at a chihuahua it doesn't seem like a dog and so this variation of goodness of which is which element is a good fit to a category all characteristics of a category and some members not having this good fit is a problem because the implications is that it is very clear-cut each member share share more or less the same features but this goodness of it and so for some people a latte could be or a cappuccino would be a coffee but a black coffee would not be a coffee and were some other people black coffee is the ideal coffee and other coffees are variations of its own order coffee and remember those or

think about those people who are hardcore junkies in terms of coffees but they will call coffee as a coffee black tea as a tea but all other forms of TS variations of tea and not a tea right so it is variation of tea and so this differences say that then the different kind of coffee is a different kind of peas which are available are not in terms of sharing same features or not equally putted together which is the implication of the **classical view** the idea idea is that people store and refer to a list of necessary features when judging a category membership is doubtful now what does this basically say that mainly remembering a number of features and then comparing is something which people generally don't do and that is the reason why you have abstract categories you tend to have categories which are abstract which doesn't fall into some kind of a classification system and that is why when I look at categories like household or categories like marketplace now these categories then not then don't go ahead and compare this kind of relationship or this kind of mental features and that is why the problem is and that is what the problem is all about so people do not refer to these kind of lists and they make violations in terms of comparing things to a list right and the third thing is that most people cannot generate lists of features that are individually necessary and collectively sufficient to specify memberships into a category so basically what it says is that when it comes to telling people and so there are experiments done in which people were asked to generate lists which says that what should be the list or what should be the types of features which anybody or any new element in that category should have and which should which should be the most necessary features which are there people could not come up with that and that was the problem with it right so people were not able to generate this individually picture they could not pinpoint what is individually necessary and what is collecting this equation and so that is one of the problems of this particular view or this particular idea of the **categorization** model the **classical view** and so to explain further or to circumvent the problems of the **classical view** another model was proposed and that's called the **prototype** the or **prototype** model of **categorization** and so what is this model it says that like perceptual researchers conceptualist researchers also believe the idea of mental **prototype** now moving from lecture on the **perception** revit thought deal dealt with the **prototype** view or the idea of **prototype** right and this was them what I'm one of the bottom of theories and they were looking at how mentor types are formed and so in terms of **categorization** also conceptual researchers also think about mental prototypes now what are these prototypes generally these **prototype** in numbers of visual researches the **prototype** was an abstraction it had all the all the properties of the particular all the visual properties of the elements which are clubbed together similarly in the reception later researchers mental prototypes are idealized representation of some class of objects and even so these are idealized representation now one problem with the **classical view** was they were never talking about these idealized representation but this is circumvented by the **prototype** view where they say that prototypes are idealized representation so idealized representations then may not have all the features necessary or may not have but it can also be possible that prototypes may not represent or may not be a part of the cat category itself it could be something else thinking of something else right so what would happen is prototypes are basically standards or these are kind of the class or the idealized class a **prototype** of concepts are feature or aspects that are characteristics that are typical of member of the category rather than necessary anticipation so basically what this theory banks on is the typicality factor what it says is that the **prototype** view says that prototypes generally have features or aspects which characteristically or which define a particular category right and so if so for so if you want to make a **prototype** you have to look at several instances of that particular category and from the several instances of category then we'll a be able to basically bring out those things which are common so protect you is or the form of a **prototype** is similar to forming calm among among a lot of data and so there are lots of statistical technique which does that right and so here also their **prototype** we're forming a **prototype** we find the combination dough's properties which are shared by most members of the category right and so most members of the category share a feature that it shouldn't be the picture of a category and so it should it could be one feature it could be multiple feature which most members share now it

doesn't specify this thing that it is individually but it says is what it says is that if I am looking at a category and if I am looking at a number of elements in a category and if there are commonalities between them if there are common features between them the commonalities are what sum together to form the **prototype** now what do I mean by this so the meaning of this is that no individual feature or aspects need be present in the instant so to be count as a member of the category so we are not looking at any individual feature right we are not looking at one particular feature which should be shared by all people we are saying that most members have these features so not that one feature should not be present in everyone it should be that it could be that most people have a particular feature but some of them may not have it and so that could also be a reason for a particular category for example our **prototype** example let's look at cars now most cars would have an engine and would have four seats right and that's how our definition of car is but how what about the definition of a flying car and when I talk about flying car it doesn't go or it doesn't run on the road and so the sufficient feature that most cars run on a road is exactly not true similarly for the fact that toy cars may not have an engine but then still they are cars right and so we can be classified under cars and so this is this these are the variation so one of the things is that no individual feature need we present is one of the features that the **prototype** view says to be conscious and under the category but the number of more number of characteristic feature or aspects on instance has the more likely the be regarded as a member of the category so basically here we don't talk about one individual feature one defining feature but you look at more number of characters in each other so it's kind of more the merrier right so the more number of features you have which resemble you or which make you closer to a category no more typical you are to that category and the more I realize your **prototype** later right and so one way of doing it is to look at a number of messages so when we are having looking at prototyping or when we are looking at how to make a **prototype** the first thing that that needs to be done is to look at a number of variations of that category a number of variations of that **concept** and with that variations find out the commonality right which has the highest commonality once you have that commonality that commonality should be what should be idealized the **prototype** or that community should be a feature of the **prototype** because most members have it and that that is how they are defined and also the **prototype** move of category and conscious refer to the family resemblance structure of **concept** or they basically go ahead and say that this is some kind of a how do you assemble structure so what is the can reasonable structure a structure in which each member has a number of features sharing different features with different members right so that's the idea so a structure where each member has a number of features sharing different features with different members so what would happen is that there would be some commonality but then each one says other features with someone and so a good example to understand this is look at the central figure knife if you look at the central figure 9 the person here he has these mustaches and so this this is the idealized version and it is the idealized version because it has properties from all of these right so if you look into all the several variations of eight different forms of image this is called the family of view of prototyping and so here what happens is if you look into this this person doesn't has a moustache and so has a beard and has these eyeglasses this person has a beard and has a glasses but the color of the hair is different doesn't have an eyeglass beard and mustache both here it has only beard different color hair no spectacles with spectacles and so on and so forth and so this is the most idolized version the ninth version in the messiah's version because it shares a number of features with all of them hey Lou it's all about commonalities the more number of features that any element share with a particular category the better example it is of that and the better chance it stands to become a **prototype** the **prototype** is that particular element which has shares most number of features are which shares commonalities between different different items in that particular category so **prototype** is basically a some form of abstraction that includes all the characteristic for category and may or may not be actual instance of the category so it's at times what could happen is a **prototype** is developed for the **prototype** is not the actual product right the **prototype** is not the actual manifestation so I I may have a **prototype** of something I may have an

idealized **concept** over something but it never actually becomes a categorical item of the category so prototypes are that kind of a thing right because forming uploaded a with that type is not possible right and so the idealized spawns example right idea I realized one is that you should do so many things but most phones to share this for example one of the things that the phone should do it should also be able to trial teleporter but that is the **prototype** right but phones do not tell you put you in real senses so what phones do share other features of a tidally voting machine which is your idealized construction of what a **prototype** is but then the road type cannot have an existence in itself so at times what happens is the **prototype** can not be demonstrated but then the **prototype** is an idealized over again off I should have it's an idealized version of a category also prototypes are often thought of as mental summaries or averages of an instances so what that's what I was trying to explain to you what really happens is I look at all the instances and I look at the commonalities and based on that I create an average and the average of all the number of elements which is there that is what the **prototype** is it is the average it is the exact average and so averages all parts of it averages related the average point or the mean is related to every element on to onto my data and so basically that's what my **prototype** is it's a mental average also concepts they exist at many levels of hierarchy what really happens is the concepts a the different hierarchies example the top level **concept** the bottom level **concept** remember the example that I gave you in terms of musical instruments so musical instrument is a higher level **concept** and bass guitar is a lower lower **concept** because it forms or it is coming under the **concept** of musical instrument right and within the higher dimension of musical instrument is the instrument right and so these are the different **categorization** so basically **concept** existed different level hierarchies but one day of abstraction appears psychologically fundamental in terms of prototypes at the psychological level or at the fundamental level or at the **basic level** most **prototype** should have the same feature and that's what it talks about so this is the **basic level** and different from both higher levels of were **subordinate** and the lower level that is the **subordinate concept** so the **basic level** is that level in which prototypes are formed and the **basic level** is that level which most items that belong to the category share so basically then this is that level where that feature is there or that feature is shared between all items which are the core feature which is the core feature and remember the dog example so barking is one feature which most dogs should have so no matter what kind of dog and looking at whether I'm looking at the Chihuahua which is a very very small dog on a per million which is again a very small talk to the Great Dane which is a huge dog out there or st. Bernard which is again a huge dog in big doe Rottweiler which is again a kind of a temperamental dog and other kind of dogs do we look at oscillation which is a huge dog so all of them then share so there are different levels of those prototypes so the highest dog the slowest dog but then the one feature the the basic feature is working and at that they should be able to coincide or that that is the so concepts then have some kind of a **basic level** where both the the difference are at that level they should match now what is the cricket the critique of this **prototype** view now one of the critique of this perturb you is that the limit of the conceptual boundaries now where is the limit of this conceptual boundaries what should be the limit when something should be not referred as a part of the **concept** or not sure example example of a **concept** or an item of a **concept** and those that is not defined or basically this **prototype** view then as protect you says that it is a mental average and so any element can enter into it and so there are no limits or there are the conceptual boundaries or there are no clear-cut boundaries which have been defined by the **prototype** view because any new item can also share some features and so where is it that this averaging really works or what percentage of match should the new item have that is not defining so there is no limit to this conceptual boundaries or where the limit is to be drawn another thing is the typicality rating which basically says that what is the extent of typicality for example this typicality so when I see the dog the let's say up or top what birds know if you go to the West Robin would be a typical bird right but then where for india since then people have not seen Robin how does Robin become a bird or basically people will not think about Robin when you say a bird people think of what a crow or a sparrow or some other kind of a bird that you or a son

bird for example when you think about it so basically then typicality the idea of typicality rating is another feature which which should be looked at and so what happens here is that as context changes these typicality changes and so that is not explained by the **prototype** because when I say this is the **prototype** then it should be universal kind of a thing and true most items will be there so that is not explained as concepts with the with the changing of context changes the **prototype** changes also it could not be explained and the third interesting view is called the **exemplar** view and so what is the example you see it says that concepts include representations of at least some actual instances so it denies the idea that there are mental summaries are there sufficiently individual reaches which are there and this view goes ahead and says that there is some actual instant instance or there is some actual representation to which things are compared to which new items are compared and that is how they form into categories and that is our **concept** is developed right so what this theory says is that there is an example always in mind so when I think about something when I think about a car you think of a particular model of a car and that's so categorizes new instance by comparing two representations of previous resources for example example us example when I when you show me a new car what I will do is my head or my brain has an idea so here the **prototype** is actually in terms of something which is existing now the differences in the **prototype** view my **prototype** could be an instance which doesn't exist which cannot exist but in the **exemplar** view it has to exist so when you show me a new car new kind of car what I do is I compare to the idea of a car which I have in the my mind and this idea is actually a actual instance of it so I think of the Maruti Baleno which I have and I compare it this which any other car and then see whether this item is a car or a truck or a scooter or whatever it is so that kind of a thing is there so my mental **concept** is there or that mental example is then I will do the comparison the third thing is difficult **categorization** unclear are typical instances because such instances are similar to examples or different categories and so when we have our typical instance how does the **exemplar** really work so suppose let's say if I have word with clip rings how do I categorize them and so it is a problem so difficult categorizing unclear are typical instances according to this view it is very difficult of how do we categorize a typical incidence into it so what are the critiques of it the critic cyst it is the **prototype** view like the protect beauties to unconstrained it's hugely unconstrained in the sense that there are no boundaries and fails to specify which instance which eventually be stored as example as so how do I develop the **exemplar** that's the problem now with the like the **prototype** view I do not have boundaries into it and what makes an example of example the car I possess should be an **exemplar** or should the example miss the car I desire to possess or should it be something else so that's not very clear and that's a problem with this particular view also how different examples are called to mind at a time **categorization** is also a problem and so another thing is that I might have seen so many cars so basically the kind of car that I am doing the comparison to for example the car air causes how does it come to mind how does it compete with all the cars that I have known as a comparison standard so how do I bring for that standard that's another question which is out there and which this particular view doesn't explain and so this is these are the two two takes into this so in today's lecture what we did was we looked at what is **categorization** and conceptualization how does this classic **categorization** and conceptualization really work and we also looked at the different models of it the **classical view** what are the problems of the classical music fits of the **classical view** also looked at the different **prototype** view which talks about the mental averages and summaries of making the **prototype** and how it it scores over the eggs the **classical view** and then we talked about the example view which is basically an instance of the **prototype** and how that explains or that goes ahead and makes it our task easy in terms of **categorization** and conceptualizations of making concepts and categories thank you

Lecture 18: Introducing Concepts and Categories-02

■Kind: captions friends welcome back to this section on concepts and categories we will continue this section from the earlier one by revising some of the things that we did in the earlier section on concepts and categories so the things that we studied in the section on concepts and categories was we looked at what is a **concept** and what is a category and how does this concepts and categories really work and so the basic definition that we look there is concepts are kind of metal buckets where you fill in things where you fill in items instances or events and categories are basically a process of organizing knowledge into these means which are called conscious we also looked at concepts as nodes which has explained in terms of the semantic idea of **semantic memory** idea or the hierarchical model of **semantic memory** so basically then forming a **concept** allows us to ease out and form an organization scheme for world knowledge and **categorization** is the process where we taken a new instance and go ahead and fill it in two buckets mental buckets which have at the top level or which are described through the concepts basically the bucket is described through the **concept** and **categorization** the process of filling up this bucket now we also looked at some of the models of **categorization** in the last class and so one of the model was the classical approach to **concept** formation where we looked at something called the necessary and sufficient condition and what that **concept** said or what that model said is that there is something called the necessary and sufficient condition which every instance of a **concept** or a particular mental bucket should have to be categorized as a **concept** and so there were critics to it one of the famous critic being that this necessary and sufficient condition is not always validated and the idea of **cognitive economy** which says that the looking at every instance or looking at every idea of necessary and sufficient feature in particular in all instance will not be enough we also looked at the **prototype** view and the **exemplar** view where in the **prototype** view we looked at how a **prototype** is formed and how this **prototype** helps us in **categorization** and we saw that the **prototype** view scores on to the classical approach of **categorization** on several aspects and what is a **prototype** then so it's basically an idealized version of the **concept** and so the idea is that the **prototype** may or may not be in actually instance but it has mental averages or mental summaries of all instances which belong into that particular category or under that **concept** similarly we looked at also called the **exemplar** approach where we looked at how examples of a **prototype** are basically used for **categorization** and we looked at that there is something called the **basic level** of **categorization** or **basic level** of organization above which there is a **superordinate** level and below with there is the **subordinate** level and looking at the taxonomy or looking at an example here the **basic level** here is the one out of which the **concept** is form for example in the first case and the musical instrument is the **superordinate** node or the higher node the **basic level** is the guitar and within the guitar the **subordinate** node is the classical guitar or some other form of folk guitar or classical guitar so basically then what the **exemplar** view said or basically suggest is that there is a **basic level** of **categorization** and from this **basic level** of **categorization** you have a higher level **categorization** or a higher level node and a lower level node so this is basically a summary of what we did into the last class now let's continuing with what we looked at the theories of concepts and **categorization** let's look at the next theory of **categorization** and so the next view which explains what **categorization** is done how **categorization** is done and how concepts are formed is the **schema** table view and what does this view says so this view says that it shares features from both the **prototype** view and in in that the **schema** tank **prototype** store information that is abstract across instances and the **exemplar** view that the Scimitar examples of store information about the actual instance so what does I mean by this what do I mean by this so basically what it says is that this view the Scimitar view is actually a mixed view and so it borrows the good parts it borrows features from both the **prototype** view as well as the **exemplar** and so how is this schematic view the idea of schematic view similar to the **prototype** view basically it is similar because in both the schemata and the **prototype** the information are abstracted across instances so

basically what happens is Kamata says that the schematic view is similar the **prototype** view because here forming a **concept** or forming or the way of **categorization** is done through looking at an abstraction and this abstraction is generated across instances so looking at mental somebody or mental averages and how does it become close to the **exemplar** view so basically what it does is that whenever we are doing this exam **categorization** or whenever we are looking at this mental average this mental average has at the end of it or has at the top node of it the an example of the category so basically all in all the Scimitar view is a mixture of both the views or the **prototype** in the **exemplar** view and so it says that we in the schematic view which it is suggested that the **categorization** happens through abstractions or prototypes which are made from or which are generated from a number of instances looking at a number of instances and these are mental summaries and also this average or abstraction that we are looking at is an actual instance which basically means that as a **prototype** view says that the **prototype** may not be an actual instance but the schematic view says that the **prototype** that we are forming the abstraction that we are forming or the one definition the one form across which we are categorizing actually exists which means that it is an actual example of it and this is how it is close to the exact value so we skim the idea of what is a **schema** kind what is a **script** in a chapter and **memory** and so then we saw the schematic is basically a way of looking at or organizing knowledge organizing word knowledge and so schemas are basically kind of organization scheme under which world knowledge is fitted whereas **script** is basically a routine for a particular **schema** or of particular schemata so this idea of **categorization** then uses the idea that people make mental schemas or people make this mental bin or basically mental barebone structures on bases on which this **categorization** is done and how is this verbal structure really made so it is made in terms of abstractions it is made from abstractions of various incidences of the **concept** and these **prototype** that is made or these bare bone structure which is made has an actual existence onto it and so this is how the schematic view differs from both the views that have been we have studied up till now now the thing is that **schema** tates are frameworks of knowledge that has roles slots and variables onto it so basically it has this key matter that the bare bone that we are talking it not only has spaces for **categorization** but it also tells you how the **categorization** is done or basically how what are the slots for **categorization** what are the variables which decide how the **categorization** is done and the various rows that each factor play in **categorization** so that's what a schemata is all about and in the schematic view the **categorization** is done through something called a hierarchy structure so as in a **prototype** you see that there's some some factors or some features have a higher role or a higher **similarity** and the other feature features don't have a **similarity** so similarly based on how similar items are a hierarchy of closeness to the actual **concept** of closeness to the actual **prototype** is done and this is how the schematic view really works now there are several problems with the schematic view which is there in the first is that it does not clearly specify the boundaries among various kamata's so as we saw the scheme matters are actually barebone or they are actually structures on which items are categorized into so and these structures are developed or the bare bones are developed through an idealized representation or through looking at various instances of what of a particular object or a particular type of event which exists now if that is true then how does two scimitar differ or what is the difference between what is the clear boundary between two schemata is not actually very clear and so this clear-cut boundaries is doesn't exist or is not specified by the **schema** tableau we always have fuzzy boundaries for example what would happen is that if an item is classified in one that a particular schematic it could also be classified under some other schemata and so what will happen is the dog if it is classified under the **schema** of an animal is also it also gets categorized into the **pet** animal or a man's best friend and so there are no clear-cut boundaries of how this **categorization** or what is the distinction or what is the boundary line between different **schema** tasks now the other problem is the schematic view is that in the present view it's not sufficiently delineated to be empirical testing so if we don't have clear-cut boundaries if we don't have we're asking matter view begins and very tends if we don't know that then we cannot go ahead and do empirical testing to this idea because since we don't know

where the **schema** starts and where it ends and what is the boundaries between two schemas empirical testing observable testing cannot be done and so that's another problem with it since we don't know the boundary so we cannot define the boundary conditions if we cannot define the boundary conditions then data cannot be collected and so they cannot be tested now also questions like what informations leads to schemata and how they are modified plus the process of using appropriate **schema** toss are not known so how was this **schema** form first of all what leads to the formation of a **schema** or what kind of information are gathered before the formation of a **schema** that is not very clear so what kind of information get amalgamated and makes the **schema** is not clear and also if a **schema** is made and an instance comes in or even comes in which doesn't fit the **schema** how is the present **schema** modified for example let's lick the example of a dog now the idea is that most dogs the lovable creature and so supposedly we have a dog and so we have a **schema** of a dog the **schema** for dog a schematic for doggies it has four legs it barks it's a **pet** animal and so on and so forth now there comes a dog which actually goes ahead and bite someone and so this dog is not no more lovable although it has the characteristics of the particular **schema** or a particular **categorization** so how do we fit this dog into it or a temperamental dog for that example let's take a chihuahua which is a very very temperamental dog now how do I fit a Chihuahua into the idea of the dog ears dog that is therefore Marian is the most lovable dog or you have golden retrievers which have very lovable dogs so chihuahua is a very very temperamental dog so how do I fit it and so the schemata which is of a dog how does it get modified what is the way in which modifies so obviously that's not present the plus the process of using appropriate schemata is also not known so when we are classifying the dog in a wildlife and when we are classifying the dog as a dog in a home what is the process that we are using for classificatory stew classifications that also is not known and so the process of using appropriate schemata of when a dog becomes a wildlife animal and when they become **pet** and when they become man's best friend that is also not clearly defined in a schematic view so how does the **schema** changes from one context to the other or one moment to the other that is also not available and so that's a big problem with the **schema** Tavia so next on is the **knowledge-based view** and so what does this view says the idea of the **knowledge-based view** is that a person classifying objects and events just compare so what does this view say this says that when people are comparing people are making classifications or categorizations they are not doing physical comparisons but they are also thinking about the way in which a particular category is organized so it's a two-part process they are not just going ahead and looking at features and ca-- when people are categorizing they're not just looking at features of an item physical features of an item and then going ahead and comparing it what they do is they are also mentally thinking about the organization process about how the organization process work and what is the organization scheme and that's what it says is that people just don't compare features of physical as personal object to the features of the aspect of the stored representation and said the people also use his or her knowledge of how the **concept** is organized to justify the classification right and so what is the meaning of this the meaning of this is that people are not stupid so when they are doing some kind of **categorization** they just don't do the **categorization** as a machine would do by just looking at certain aspects and matching certain aspects so they are not looking at it's close to how what a **template matching** model is and **perception** so people just don't look at certain templates or people just don't look at certain features and go ahead and make the comparisons and based on those comparisons classify different new instances or of any element of any **concept** into a particular category what they do is they also think about why the organization has to be done why is he categorizing it and then why is he or how is he categorizing the whole or how is the whole process of categorizing going on and why does he do that to explain why certain instances happen to go together in the same category so this is basically one of the thing so what happens here is that people just don't do physical comparisons they think about the physical comparisons of objects and they do the **categorization** processes and that is why things which don't belong together are categorized for example let's say we have a five item is given to you and a

categorization scheme is named now these items are a child a dog you're very valuables a laptop and some money now how does these five things classify together or they are categorized together no absolutely looking into it just from the just from a third-person perspective you wouldn't think that they have anything in common what is a child has common with a laptop and where some valuables that you have and some money that you have so none of them are actually in any category or sort to propose any form of category but then if I give a name to it for example things to be saved when in a fire then it makes sense to you and so what people do is they don't just look at these four things and do the **categorization** they also think how they are related to each other and this thinking of the organization scheme then leads them to make the classifications and so this is what event actually talking about so people then think about the organizational scheme and justify the classification so in this case the four items can be classified together how they are classified together by looking at the fact that these are the things that needs to be saved when a fire is there so more previous views of conscious fail to answer so most previous views of the concepts they fail to answer satisfactorily how things in the same category go together the knowledge base you propose that people's theories or mental experience about the world are intertwined with their concepts and provide the basis of **categorization** so up till now the theories that we have looked at they don't look at instances or they don't the beauty satisfaction and answer of how people from different categories are categorized together right or people which are which have very low **similarity** how they are classified together but this theory the knowledge based theory they say the people have their own theories and men and mental explanations about the world and these theories and mental explanations help us in forming concepts and helping us categorize and some of the most weirdest of elements together then can be given a theory for **categorization** and can be categorized together so here this knowledge based view gives us the scope to categorize the most non natural or most asymmetrical objects into a kind of a symmetry that's that's the fine kind of a thing that this particular idea of **categorization** helped us now the five approach to conceptual structure the five theory that we have looked at they can be divided now categorized into suit two subtypes and this suppressor provide provided by comes a two in 1992 so what are the two categories so the five categories that we have looked up up till now they can then be or the five different models that we have looked at till now they can be clubbed into two basic categories what is it the first category is the **similarity** based category and the other is the knowledge based category so in the **similarity** based category we can classify the classical approach the **prototype** approach of the **exemplar** report and parts of the **schema** view can be these models can be put into the **similarity** based category so what does it say it includes approaches in which **categorization** is assumed to be based on **similarity** of an instance to an abstract abstraction revivification of the category and so why these four models or why these three models and part of a model is similar as I said the schematic view or it it possesses parts of the **prototype** view and parts of the **exemplar** view right and so basically how they are together or how they are clubbed together the idea how they are clubbed together is that all these use our **similarity** based system so they talk about that concepts and **categorization** or concepts are former **categorization** is done based on how a new instance or how close a new instance is or how similar and unions instance is to the available **concept** the more similar it is the more easily it is categorized into that bin otherwise it is taken to some other mean or it is pushed out of that particular bin so the more **similarity** you have to a particular **concept** the higher the chances of being categorized into that particular **concept** the more close our animal looks to a dog the higher the chances that it should be categorized in a dog category the more clothes animal looks like a cat the higher the chances that it will be classified as a cat right and so **similarity** based view so that's why and so the problem with this view then would be that are typical incidences will not be part of its for example if I have a three-legged dog or if I have a dog such as a Doberman now the Doberman dog has a very very long tail and so for its own survival sometimes the tail is chopped off and so if it doesn't have a tail right how do I go ahead and classify so this view says that I will not be able to classify the Doberman into a dog category right why because it doesn't have the absolute

similarity that we're looking at and so this view is based on this idea of abstracts there's specifications of a category the idea that things which are classified into a category should have things or should have properties which are similar now the key critic of this view or the **similarity** is meaningfulness only in certain aspects so basically the **similarity** waist view is meaningful in certain respects only in certain other respects there it is it doesn't generate meaning so in for example in in arithmetic or in in those Sciences this mu will be very nicely looked at because problems can be categorized into certain formulas a certain mathematical problem could be akin to certain other mathematical problem it means lend a solution to it and so we can look into it for example integral most integral indefinite and definite integral could be classified into a certain way of solution to it but then the for certain other kind of solutions where a new approach is needed some kind of mental knowledge is needed or some kind of work on the mental knowledge is needed or some kind of work from the person who's categorizing is needed some effort from the person who's **categorization** is needed there this this view will fall now the other category is the explanation based category and so what is this category it comprises of the **schema** table view and the **knowledge-based view** so it as I said the Scimitar view a part of it is the **prototype** view part of it is the **exemplar** view right and so the half part which is the **exemplar** view then falls under this category and so this mu has comprised of part of the Scimitar view and the knowledge based view and so what does the meaning of this or what is the proposal of this explanation based category the idea is that people using this view they base classification on meaningful relationship among instances and categories so people who use this kind of explanation based view they do not look for the explanation so they do not look for similarities rather **categorization** is done in terms of meaning of how things are related together in meaning and that is why the previous example that I gave you in which a dog a child some variable that you have and certain amount of money that you have are not clubbed together because how they are clubbed together because they are the things that you are going to save and that reason the man our dog and that could also be classified together because they form a household right and so this is how they are classified together but then looking at it in terms of similarities a dog is not equal to a man neither the man is equivalent to a bed neither is equivalent to some other household items or chair for example and so the explanation will suggest that with meaning or true meaning people can do **categorization** and so **categorization** or the property of **categorization** or **concept** formations can also be done on the basis of meaning now up till now we've looked at the five models and a combination model of how these categories are generated or how this category concepts are generated we will now look at the fact of how a new instance our new **concept** and classification of new instances are done what is the way in which people form new concepts and new **categorization** and so study was done by Bruner which we'll use as a reference here so **concept** formation requires some **basic level** of generalization so when we are using a when we're making a **concept** some kind of features have to be generalized or with all these instances all instances of a particular category have to be looked at and some commonality or some generalization has to be looked at for group grouping certain things but not others together now the process requires filling out what features are relevant irrelevant with a little feedback so basically in terms of **concept** formation we should be focusing on generalizations we should be focusing on commonalities and we should also be looking at those features which are relevant to us and those which are which are not relevant to us and only once we can do this relevant relevant kind of an identification can we go ahead and then make concepts or newer concepts so how do I make a **concept** or how do i attain a **concept** and for that purpose the process of acquiring a **concept** in was acquiring the information necessary to isolate and learn a **concept** relating to the formation of later use and transformation of information to make it usable when testing ideas and new possible instances now to show how new concepts are formed a study was done by Bruner and others and so Bruner good now in osteen in 1956 they did a study and so what was the study like so it was a very simple study in which they showed people Bruner Austin and good now they showed people a display like this right and so this play had certain cards onto it so certain cards were

there and the display had a card this kind of so I'm just drawing a part of the display so this kind of things our cards were there and these card cards had certain geometrical figures on unto it so either it had a triangle it has a circle it has a square or it had a rectangle so basically then each card had one of these features so either it had a triangle it had a circle or it had a square so each card has either a triangle circle or a square on to it in addition each card also had black white and stripped color onto it so each card was other black white or strip kind of it so basically the card was either black in color or the figure was great in color the figure inside was white or the figure inside was striped that kind of a thing was there so first feature it had a number of three of these either a triangle circle or square it had black white or strip figures then it had either 1 2 or 3 shapes into it so it could have a single triangle it could have two triangles or it could have three triangles similarly it could have one square one triangle it could have one square one rectangle I'm sorry not a rectangle one square one triangle one circle and that kind of a thing in addition it also had boundaries around the shape so each card then had either 1 2 or 3 boundaries so 1 2 or 3 figures and 1 2 or 3 boundaries so the variation was that these were the four variations these were the four aspects which varied between the cards right so most people were shown this kind of a system in which each card has either 1 2 or 3 boundaries it had either 1 2 or 3 figures it had the four figures classified as black white or striped and it had either a triangle a triangle a circle or a square onto it and so this kind of figures were shown to people later the experimenter gave a definition **concept** the experimenter asked the people a defining **concept** or it gave a definition of a **concept** that things like black circle in two borders strip figure so he said that I am thinking of a black circles a figure with a card with black circles two borders and a strip figure right so black circles two borders and strip figure and the experiment has said that in terms of classification so subjects were asked to classify of how many of these cards actually match the **concept** that this experimenter gave me so first people were shown this kind of cards and later on they were given a explanation like this of the **concept** of is the **concept** the person the experimenter is thinking of a **concept** which has black circles in it has at least doesn't say at least it says two borders and then it says it is striped in nature and so this kind of a thing was given so these cards are either black white or striped in nature and so then after this the third step first so this is step number one this is step number two in step number three people were shown up positive instance so two borders two circles which are black one another stripe so this is a positive instance right and this was shown to people and then people were asked to look at this positive instance think about this **concept** and later on go ahead and verify how many of these or what are the what all cards actually match this kind of inclusion criteria inclusion criteria for **categorization** now people each time the subjects went ahead and tested these cards each of these cards for inclusion criteria a feedback was given to them whether it was true or not and so this is what the experiment was so very simple experiment a number of cards were shown to people these cards varied on four variables one it had a triangle each card has either a triangle circle or square on to it then it that the cards were either black white or striped third it had one two or three figures onto it and it had one two or three borders on to it later on a **concept** was given by the experimenter saying that I am thinking of a card with black circles two borders and striped figure so that's what I'm thinking about and then a positive instance card was shown two subjects and later on subjects were asked to go ahead and match this instance or basically tell the experimenter which of these cards follow or which of this card matched this particular **concept** that has been given and so when the experiment was done Bruner and others found that there are three ways in which people went ahead and attained the **concept** or did this particular kind of a matching task people either use cement image scanning successive scanning or convert on conservative focusing now in simultaneous scanning what people did was all hypothesises were tested the same time so people went ahead and tested the black circles of people compared number of black circles to number of borders and number of stripe figures at the same time right so they verified all the hypotheses all the points which have been given in one single instant instance and that is called simultaneous scanning so simultaneously they verified all the hypotheses or they verified if the target card matched or

the target card had all these features or not in successive scanning what people did was they first matched whether the target card whether the card which we were matching to the **concept** had a bad circle a black circle or not if it didn't have a black circle it was left behind and the next card was moved on to so that's how people looked at so one hypothesis the time so first they verified whether people whether the cards had black circles then they went ahead and verified whether the cards that they got from the first **verification** with black circles whether they had two boundaries or not and once that was done from that group they looked at whether black circle cards with two boundaries were striped or not and that is how they did and this is called successive scanning in Kundel conservative focusing what people did was they selected the positive instance they looked at the positive instance and then they started comparing this positive instance to each card so what they did was they compared so they were not testing hypotheses they were not eliminating hypotheses they were looking at the positive instance and comparing the positive instance with each instance of the card and so what Bruner Goodenow and Osteen came up with is that this is how people our team called concepts and these are the strategies of attaining concepts they said that there are three ways of attaining a **concept** one is called the simultaneous scanning in which you go ahead and compare all hypotheses the same time then there is the successive scanning in which one after another the multiple hypotheses are compared one by one and then there is conservative focusing in which what you do is the standard instance is compared and this comparison is what basically leads to the idea or leads to the matching and in this comparison what we do is one feature at a time is compared to the target the target cards not with this so this is what I was talking about and so as you see there were stripe figures there was darker figures I mean black black figures they were white figures number of boundaries number of signs into it and so these were the cards which were use and so any instance was used and so these instants were how it was tested so Brewin others 1956 found that the effectiveness of each of these strategies dependent somewhat on the extent of the task conditions so the question is which of the strategy is better which strategy should be used in which the strategy should be used when and they found out when the strategies we're not complex the simple strategies for example arithmetic problems were very simple or very straightforward strategies people should be using the simultaneous scanning but as strategies become more and more complex as more and more complex items have to be clubbed together or categorized together people should be using conservative focusing so it's basically the task that has been given to you the idealized conditions that has been given to you how difficult it is the more difficult the more complex it is the more conservative focusing we use the more simpler it is the more successive scanning that we use or simultaneous scanning that we use so another question was how do we acquire the **prototype** in our **concept** formation so basically the the card that was positive instances that was given is actually a **prototype** so how do I get a **prototype** so people use and form **prototype** you haven't given distorted instances during learning so going back to the chapter on **perception** where we looked at a study by Cillian Posner where they gave people various random patterns of the letter M and so what happened is there were there were low distortions and high distortions pattern of or low distortions and high distortions versions of the letter M so remember if you if you remember from that chapter so this is how a M would look like and so this was the **prototype** which was never shown to people what people were shown where distortions of it so for example this is one distortion which people were shown and the other distortion could be this so several distortions were shown to people and these distortions were high distort so these were load instructions because if you join together it will form a M or in this case it from the M like this and the high distortions versions were also shown to people so people if given both high distortions and low distortions people and it was found out that people work better or people make prototypes better in in terms of moderate distortion see if you people are shown no distortions and they are asked to make a **prototype** or people are shown high distortions in both these cases the **prototype** that they form are not very convenient but if noticed medico distortions and moderate distortion the items are shown to them or moderate distortion instances of the actual **prototype** is shown to them they can

accommodate a large variation so even if a large variation if the test stimuli has a large variation into it people who have been shown distortion moderate distortion cards or modern distortion versions of the original **prototype** they are able to still classify the test cut into their **prototype** now learning about category variability may be at least as important as learning about prototypes especially if categorizations are to be made later and in instances that vary a great deal from the **prototype** so category variability is another thing how much there is a variation in the category now more variation in the category the more distorted or the more bigger the **prototype** would be or the time mentions the **prototype** would have the bigger the **prototype** abstractions would be and the higher the lower the category variability the more stronger the **prototype** is or them the more contended the **prototype** is so with category variabilities this **prototype** also varies now books 1978 they define something called non analytical **concept** formation in contrast to logical scientific focus caution and they call this as implicit learning and they require people to pay **attention** to individual exam plus storing information about representations of them into **memory** so what Brooks and others they found out that people just as we saw in **prototype** acquiring of how people acquire **prototype** what people do is people use implicit knowledge or people use non analytical knowledge to form concepts now even from the high distortions of dots of an M or hardest low distortion versions of M people were still able to make the **prototype** and M it basically means that people are able to use implicit knowledge for forming a **prototype** and how does it there because people pay **attention** to individual examples so people take all these distortions and look at these distortions and they pay **attention** to it and from that they store information about the representation of that particular thing of that particular abstraction in **memory** now later classifications are done by comparing new instances to the representations drawing analogies between the old and the new so what if a number of variations of the dot pattern is presented to people we're from somewhere high distortions or some or low distortions people will look at all of them and then store the mental **prototype** and then later on if still variations of native variations when they come in the the people compare these variations with whatever has been stored into **memory** or comparing to the **prototype** which has been stored into **memory** so in Brooks example what Brook did was we Brooks presented people with examples like this and so these were words that represented to people and people were asked to remember these words no there was one rule which was there and that what what the rule was that some of these words followed a certain kind of a **grammar** right and so these words which which are here these I'm sorry these are not words these are letter string so these letter strings are presented to people and people are asked to remember these letter strings or to predict how these letter strings are actually formed now unknown to people were the fact the some of these letter strings followed are certain rule a certain way of doing it our certain **language grammar** for example some of these words which are presented here follow a **grammar** and so as you can see this is the **grammar** this is the structure of the **grammar** so a certain rule has been followed in some of these letter strings and in some of these letter strings that tool has there not been followed for example one rule is to start here and then you can have a T to start with or you can have a V to start with so we have a T or or a V to start with and then you can either have P a number of peas into it and so this is one in which we are having a number of piece or you can have a number of X's onto it and so this is the one which has a number of X's to start with a V and number of X's and then you can proceed to either a T so this is a T or you can proceed to either a V so this is this is either a V and from there this T then follows an S and so this is one the one which follows this logic right or it it it can from T it goes to an S and s then comes back to X or it proceeds to the end so basically what the experiment was all about is that in this case these letters some of these letter strings followed a **grammar** and some of these letter strings were a random pattern which were there and so people were not told some of these people were not told that they followed a **grammar** strict kind of a **grammar** and some of the people were told that the letter strings that you are using followed a kind of a **grammar** onto it right and so later on they were asked to so some people some of the people were told that certain rule is there from which this letter strings are generated and some

people were told that there are no rules to it and so later on participants when who learned better strings that followed **grammar** rule made few errors then control participants now the thing is that participants who were told that they follow a **grammar** rule they perform or made more errors in terms of **retrieval** and then participants who didn't follow this **grammar** rule so if told about the rule people were told that there was you know a rule which was followed people who would retrieve back or learn these strings they made more number of errors and people who were asked to just learn the rule so three instances in one instance people were not told anything and they were asked to learn these words in instance two people were told that see there is a **grammar** rule and this **grammar** rule isn't followed for making these words and so each new instance then follows this rule right and the third group was there where they were shown this rule right and they were not asked to learn anything they were not stared at anything about that whether it follows a **grammar** rule or not they were shown this **prototype** they were shown this thing and they were asked to classify what happened is what was the result the people we were just own an **exemplar** people who were just shown this particular instance and they were asked to classify based on this instance or basically remember this instance based on **prototype** they perform the best but when people were said that a **grammar** ruin is there and you have to deduct the rule and fall based on that do the **categorization** they performed poorly and of course the people who were not said anything about the rule or if any rule exists implicit rule exists for learning these words things they perform the worst so that'sthat's how the result of this example turns out to be also in another example in another study people were shown this kind of a hereo graphics and a particular word related to this hereo graphics and so what was the thing here people were first shown that this is the stress stimuli and this is the response so this means wrong this means gone this means tiger this means bus and later on they were asked questions by showing this kind of a stimuli whether this flies whether this is bigger or not whether this is live or not and whether this is attacks or not this kind of questions were given now unknown to the people where the fact that each of these hey leo graphics that you see actually mean something actually mean a particular thing in English a particular word in English right and so when P when people were not shown people were not told that this kind of meaning exists they performed better they they made this idea better and they could they could just go ahead and relate this I this implicitly this particular Geographics to a particular **concept** but when people were actually told that a certain holographics mean actually something the number of errors rainbow so people who have just shown this and later on tested on to this area graphics they perform better why because they implicitly made some kind of connection they implicitly made some kind of a connection of what this thing means what this thing means and so on and so forth but if they were just told that there is some kind of a rule which is there and they perform much over sir Brooks then described five factors that encourage people to store information about individual exemplars and what are these five things so Luke says that first of all people are better off in making concepts and **categorization** in terms of exemplars and he's always stated there there are five factors which encourage people to make more examples and what are these the first factor involves learn information that distinguishes among in in among individual instances so if the task requirement is such that it makes people to learn more information that distinguishes among in individual instances if the variability is more and people learn more about individual instances they are better off making learning through our **concept** formation through through the process of **exemplar** or using exam process method for **categorization** also a second factor factor is the original learning situation so more closer the original learning situation is the more varied the original learning situation is the higher the chances that people will use exemplars for making **categorization** also some stimuli lend themselves to hypothesis testing better than others and so what happens is there are different different kind of similar stimuli for example some dogs would be called a dog that would be much better example of a dog then some other kind of dogs which are not better examples of a dog and so what what would happen here is that since some stimuli led to better hypothesizing and some not a good example a good way of **categorization** would be using the **exemplar** similarly a food

factor that in real life is that in real life **concept** learning instances may belong to a number of categories so what will happen is a particular instance in the instance can belong to a number of categories for example the dog could belong to the the animal category to man's best friend category to the household category and so on and so forth and so if that happens if the dog is sharing so many categories there's so much overlap an example is the best way or example should be the best way of **categorization** and the fifth thing is that in natural setting we learn about instances without knowing how we will be called to use for information later and so in most natural instances the immediately we don't come to test a particular instant or make a **concept** or categorize something using a **concept** so we learn a particular we see a particular instance of something and then we make a **concept** out of it but immediately we are not given the task of categorizing something according to that **concept** it may happen that or some period of time may exist or some period I may laugh before we get at instance of that category and so in those cases example as a best example or examples are the best way to go ahead and categorize so this is the end of this section and in this particular section we saw how concepts and **categorization** really work and what are the various theories which help in **categorization** and concepts and and then also look compared across the different categories we also saw how **concept** attainment is done and what is implicit **concept** the idea is that people use implicit knowledge for cancer **concept** formation and so that's another interesting thing that we looked at and at the end we also looked at the examples are the best way of forming a **concept** so in the next lecture when we meet we'll talk more about a different kind of **memory** or since they are still continuing with a **memory** example in the next section we'll talk about a different kind of **memory** thank you

Lecture 19: Basics of Visual Memory

■Kind: captions hello friends so we start this new section on visual **memory** and this has to do with how picture is perceived or visual images are performed now at the beginning of this session I should tell you that **visual imagery** was something which was not very popularly taken in cognitive science and already psychology the reason because that **visual imagery** cannot be tested it's very difficult and so it's like what behavior is just to say there is no way of testing onto it and so how do we understand visual imagining now let me ask you a question you all live in house or kind of an hostel and you have been living there for a long period of time for certain periods of time the thing is if I ask you this question all of you wherever you leave you have a window and perfectly sure that they you have and so if you have a window it would have a window sill also and so how many windows cells do you have in your window let me give me a will give you a minute to think about it so you come up with an answer let's say four five six seven some people would say I don't have a window say live by the window frame onto it some people would have glass windows and so there is no question of self let me ask another question you all live in a house or some kind of an hostel and so there are walls there I'm pretty sure most of your house have was now what is there on the Left wall from your bed and I'll give you another minute to basically answer the question so all of you have a wall in your bedroom and so what is there on the left wall of your house of your bedroom from the bed so left from the bed the wall which is there what do you have on it and so most people will have different different kind of answers you will have a poster on to it you would have maybe some shape on to it some kind of torsions onto it or something or the other now let's go back a little now when I ask you this task or when I asked you to relate this task of telling me what you have on the wall or of the question of how many windows cells that you have on the window how did you come up with an answer now most people what they did was they went back mentally and reimagined the situation now at this point of time when you are looking at this lecture when you are hearing this lecture you're obviously sitting maybe at your house maybe at the place where you have these walls or not or you are somewhere else less in a garden look sitting in the understanding ourselves sitting in looking at this lecture so when these questions were posed to you what you did was you mentally traveled back to that place and thought about the wall or made a structure of the wall so you went back from here went back to the window went back to your first to your room then looked at your room mentally and this remember all of these is happening mentally and then you've turned yourself oriented yourself towards the window counted the number of sill and then related back there to this information or you went to your room a bedroom looked at the bed and looked at identified very as the left wall and once you have identified where the left wall is then you looked at the image which is on the left wall and identified what is there and so this traveling back of you into these realms or in your house mentally traveling back through your house and counting these sails or telling me what is on the wall is basically what is the process of **visual imagery** and the process of visual **memory** so basically visual **memory** is that kind of **memory** which stores visual representation of things and so as we saw that in **short-term memory** we use acoustic representations in **long-term memory** we use the semantic associations the meaning related representations visual representations are another representations which actually help us in remembering a lot of things now the idea of visual representations of visual imagining together of visually imagining something was something which was challenged right from the very beginning in **cognitive psychology** although Allen Bayview did a lot of work in terms of visual images no stop here think about a coffee cup when you do that what really goes on in your brain now scientists have looked at the process of when you imagine something so when I'm imagining when I'm closing my eyes and imagining a coffee cup all the same processes which happen in **perception** for example taking in the stimuli through the eye transfer of this image to the visual area from the visual area which is exhibited in the occipital lobe from there into the medial temporal lobe where it will be identified where it could be categorized or pattern matching will happen and from there a feedback

given to the frontal area which will then make meaning from the input which is arriving from the middle temporal lobe so all this process happens in **perception** is similar that happens in **visual imagery** except the one process and the process which doesn't happen in mental imagery is the input of stimulus from the eye because eyes are closed so when you're imagining the exact same processes go on to your brain as when you are perceiving something and so kosslyn did a number of scanning experiments a number of work onto it a huge number of work onto it and basically related to these metal imaginaries now as I said the way to start with the debate was there but whether a visual imaginary is something which should be studied or visual codes is something which should be studied or not although it is very efficient this cause a very efficient and bizarre you in a number of situations but should we study it or not and so earliest psychologists had a problem with it exam for one instance the behaviorist had a number of problem with it the question is that visual images are personal to people right people have when I say imagine a coffee cup you're imagining on a coffee cup and this imagination is very personal to you and know where I can interfere and see a coffee cup and so there is no way for empirically testing it there is no way to testing the fact that whether you are imagining the coffee cup or not and so I cannot beat you or I cannot go ahead and anyway verify whether you are imagining a cup or not and so you can fake and so once it doesn't run this process doesn't lead to testing while one says this process doesn't lead to empirical testing the process that the process itself should not be qualified to be called a science and so initially this was not studied a lot but with the coming of the book called mental representations the dual process approach by Allen pebeo this idea of visual codes a real **memory** came to the forefront so what is visual **memory** is now exactly or what is **visual imagery** exactly it is the same thing as **perception** - the perceptual input to the eye the **visual imagery** uses the same kind of input and as we will see in this section what is visual imaginary what are the basic features of it why it is used and what are the benefits of using **visual imagery** and so visual imagine a as I said is another kind of **mental representation** for example there are four or five kind of mental representations that are there mental representations are basically demonstration or mental representations are a copy of the physical representation the physical world as it is copied into the psychological Ilham into the mind LM is mental representations and so two codes that we have looked before is the semantic code symmetric representation and the acoustic representation and with that we have the propositional representations and we also have the visual representation so we look into the propositional represent the visual representations although the propositional representation is something which we looked at before in terms of Addison's theory in the **semantic memory** where he talked about propositional thought and how propositional representations are represented into **semantic memory** so let's begin this section on visual **memory** so what is **visual imagery visual imagery** is basically information which passes through the brain as though something is being perceived when nothing actually is happening so when I gave you this task of looking at the window sill or counting the window cells are looking at the wall in your house and telling me what is on the wall what you did was you imagine these you started perceiving this right you started mentally going back to this and then thinking about it and so the process of mentally going back was similar to what **perception** is - the input from the eye all other processes are same and as I related before in this introduction that kosslyn scanning techniques or costing work on scanning where visual scanning represents or showed through **fMRI** that same areas of the brain get activated when you are visual imagining as when you are perceiving - the input so more or less the same areas get activated right and so these are **fMRI** studies which have been done on people when they are imagining something and when they are actually perceiving something so basically someone may experience sight smell sound and touch as a result of **visual imagery** when none of these stimuli xur present true basically visually majorie also similar to **visual imagery** you can have different imaginaries also for example you could also have imagery of smell so when you imagine good food when you imagine eating a pizza that is called the smell imagery or when you imagine some dog barking with auditory imaginations or touch imaginations where you have imagine that somebody touches

you or something touched you that is haptic imaginations and so visual imagine is not confined itself to the visual domain the imaginary can also happen in different domains so matter is a process which is exactly manifestation or which is basically a complete replica of the perceptual process but - the input system and visual imagining involves the use of mnemonics now visual imaginary it furthers the use of mnemonics and these mnemonics actually help us in organizing knowledge and remembering knowledge better to improve recall so if we use mnemonics these these are techniques of organizing knowledge or basically saving knowledge and so once we can do that we can also use this **mnemonic** techniques to remember information better look at this particular figure and then I ask you what do you see so I'm pretty sure by now you are um answer to what you are seeing so it looks like a piece of the cake it also looks like some kind of spongebob kind of animal which has a carrot coming out or a horn coming out of the nose and it has it's a kind of a birthday thing so this is this is like a complex picture or this is a manifestation which is there but most people would have imagined something or other out of it this is the idea of imagination when I show you this picture you think about something or you imagine something to it so it could imagine it to be anything it could be a party situation it could be SpongeBob SquarePants or it could be person having a birthday blast or it could be two legs here two hands here the eye the nose here two eyes or it could be this is the but they cap that it has and so on and so forth so and number of imaginations and number of things and when what you are thinking about it the reply that you are giving to me is based on your imagination so what is mental imaginary good for one of the best things with this mental course a **mental representation** is it enhances recall it helps you in organizing knowledge and enhancing record and one way to do it is through using mnemonics and making mental chords out of it hello water pneumatics these are basically a process of making **mental representation** individual format and then help us in organizing information in recalling them better so mnemonics involve the construction of mental picture of images which help us in increasing our chances of remembering information so as I said before it tells us in mentally creating a mental picture or something right so when I say imagine a coffee you created a mental picture of it and so that is what mnemonics also does it helps you sue and helps you to create a mental picture and what does this mental picture then do it helps in better remembering information and there are several techniques or mnemonics which are which can be used and so what we'll do is we'll use or we'll see so some of the techniques in this particular section in this particular lecture number are the first and most famous technique of mnemonics that have been used for creating mental code and for remembering things together for organizing information is called the **method of loci** now what is the **method of loci** the **method of loci** it requires the learner to imagine a series of places locations that have some sort of order to them so in this case what happens is if a large bit of information has to be imagined what I do is I create a mental image of a house or a number of locations and then tag each information into these locations for example if I have a lot of information to remember let's say I have a list of words to remember so what do I do I then remember a place I then remember or I then imagina my house and several things in this house and then go ahead and tag each of this list of words onto these places so next time when we when I think about this house that tag the word with the house come back to poor fur so imagine a case where you have been given ten items to basically bring from the shop so suppose you want to remember a list of ten items to shop and so we often when we get in this situation we forget some of these items out of the ten item at least two items we are born to forget but if you use mnemonics we'll never forget because these are visual codes of remembering so let's say there are 10 items that we need to remember and so these are the 10 items that we need to remember in a shopping list for example bananas eggs milk bread cereal cake sugar wine flowers and chocolates so 10 different items which have been given to us and we need to go ahead and remember all of this and then do shopping for this while returning from our work so how do I remember all these items no since there are 10 items and the limit to **short-term memory** is 7 plus or minus 2 so the chances are that I'll forget some of these so how do I make sure that I don't forget any of these items one way of doing this is using a **method**

of loci mental creating a **method of loci** for it and for that I have created imagination space a visual imagination so try imagining your front door but with a huge banana instead of the visual handle then when you open the door and walk into the entrance the flow is covered in eggs and you have to walk over the eggs to get into the living room so I'm just creating an image first and then we'll transform that into image then imagine the ax crackling under your feet and the mess anyway it gets much messier because when you open the living room door you're almost knocked off your feet by a river of the milk that comes gushing out your struggle over the window to pull out the curtains which have turned into two giant slices of breads then you try to turn on the TV but fail because there has been replaced by a very large packet of cereal time to have a sit down but when you collapse on the sofa you sink down into the sofa size ginger cake so the kitchen you go to a kitchen for a drink walking across kitchen floor is a bit difficult as the knee-deep in sugar and when you have reached the cattle you find it is turned into a bottle of wine now I prefer white but you can visualize whatever color wine you want to now give up and go for your mug of water unfortunately when you reach down a mile further mug from the cupboard it is filled with a bouquet of flowers and when you turn to the tap on his chocolate not water but chocolate comes out of it so I created a scenario like this I created a picture like this and see these are the 10 items which I want to remember so then this is the mental picture so there is where the banana is here where the eggs is which is crackling these are the windows curtains which are now like a bread the TV turns out to be something where a giant cereal box and so I'm sitting in the sofa which looks more or less like a ginger bird red I'm sorry and then this water turns out to be wine and chocolate comes out of the my tap and here are flowers so what I have done is I have taken all these things and put him into this figure now next time when I am although this is a very very hilarious figure so next time when I am into the shop I just have to remember this figure and this is called the **method of loci** what I have done is each location the door is a location and now associated the banana with it similarly curtains is a location on which I have tagged on the bread and similarly TV is a location on in your house which I have tagged up to be the now the serial similarly the sofa is a location on to which I have made the gingerbread and the other locations as well so this is the **method of loci** where what I do is a very famous place a place which I'm very familiar to is taken and then items which have been learned on to or which you want to learn on that is tagged on to each place on to these location and this is the **method of loci** so where it's a funny fair funny picture and it's very easy to remember a next technique that can be used for easing out or using visual code is called the technique of **interacting images** so here it's very simple what we need to do is states that the recall of concrete nouns on a list is improved when participants were told to form images of the words in comparison to when they were not given such an instruction and so the technique of interacting you may just says that when we have to learn words in a list it is better to form mental imagery of the nouns concrete nouns and then interact and make an interaction between different words in unknown for example in a bad word recall test or the pair dog and pipe dog in pipe where the item of this fad recalled test so if that is to remember what the easiest way to do is to remember the image of a dog and to imagine the image of a pipe and then to interact this picture together imagine the picture of a dog which is smoking a pipe and that is what it is the image of a dog smoking a pipe will make better recall and images of the dog and pipe kept together the reason is that it is funny and it is noble and so humor is one of the easiest way of remembering things so instead of then thinking about the dog separately in the pipe separately I can think about a dog smoking a pipe and this basically is the easiest way so dog is basically a concrete noun and pipe is basically a concrete noun so what I have done is I have created an image with dog and pipe interacting together or dog smoking pipe dog smoking pipe and this is much easier or much better way to remember so technique of **interacting images** says that if you are number upon concrete nouns or imagine these nouns and then make images of these nouns interacting figures and that is an easy way to remember things a third method which is not so visual which isn't used uses **visual imagery** for remembering can also be used and this method is called the **PEG word** method so what we do here is that it involves picturing the item with another set of

ordered cues packing them onto the queue so basically if in this case what we do is there is an ordered list that we know and we then peg the new item onto the ordered list now in this case the queues are not locations but rather nouns that come from a memorize a list so in case of the **method of loci** there are certain nouns certain locations which have been used so use the door as a location and the banana onto the load in this case we don't use locations rather we use a number of concrete nouns which come from memorize lists so those lists that we learned when we are very small in class nursery and all so those lists are there and so we can go ahead and untag or basically pick a new word onto those lists and we can remember and so a good example is here but all of you know this list so we know 1 2 3 4 5 6 7 8 kind of a thing and so one is now related to other word 1 2 is related to shoe 3 is related to 3 4 is related to door similarly 20 is related to 20 30 is related to dirty 40 related to warty 11 to lever 12 12 13 - thirsty 14 to forking 16 to reaching 70 to heaven dan80 too weighty so basically what we have done is since we know there were also some rhyme called one two buckle my shoe three four shut the door five six pick up the sticks kind of a thing so basically these are memorized lists so we know 1 2 3 4 5 6 is something which is taught to us very early on or a for Apple B for ball C for cat deferred also ABCD is a structure which we know beforehand or 1 2 3 4 is the structure which we know beforehand and so when we want to learn a layer of concrete known we pick them together we tag them together now one part the Q part of the list is coming from **memory** and it is a memorize list that target part is what we peg on to and so since that Q is easy as the Q is memorized tagging the target is easier for us to remember this paired record and so this is the third method and so this method doesn't use imagine e because what we are doing is using a memorize list whereas the first two methods use imagery the third method is not using imagery because it is using some kind of memorize list for remembering now in order to study mnemonics generally use **visual imagery** and how **visual imagery** functions there are two ways of explaining how these mnemonics really work now what is the way in which these mnemonics actually go ahead and work and so how do they lead to better recall now one of these approaches of how pneumatics actually work is provided by Allen purview in his book in his 1983 book which is mental representations and **dual coding** approach like his book talks about how mental representations are what mental representations are and so what Arendt fails in very simple words what his proposal is originated from the **dual coding hypothesis** of **memory** so he get a **dual coding hypothesis** of **memory** where which he says is the main reason for mnemonics to work so why didn't he Mannix actually work why do making mental imagery of things actually work in terms of better remembrance is because there is something called the dual going approach which is proposed by L L in patio and what is this idea that he propagated he says that the l TM contains two distinct coding systems so the **long-term memory** has not just the semantic coding system there is another coding system which is out there for representing information to the store so Allen pareo believes that the l TM uses two different codes for storing information one of it is called is in the wearable domain so one code which is used by the **long-term memory** for storing information is verbal which contains information about items abstract linguistic meaning so those meanings which are abstract those meanings which are semantic in nature those meanings which are linguistic in nature are generally stored as the visual code and the other inverse imagery the mental picture of some sort that represents the item looks like so remember in cat the concepts in **categorization** we looked at the best way of concepts and **categorization** was the **exemplar** method and so looking at **exemplar** how do you remember an example so **exemplar** a certain features certain abstract features that are verbal in nature and then it has a visual feature also for example how do we remember a car we can remember a car by its model and also certain abstract features for example it has an engine it has for those four windows it has seat for sitting it is a method of transposition and so these all these methods are or all these features are verbal in nature but the model of the car that comes to mind when you are thinking about it is visual in nature and so what he says is there are two basically codes in LTM the verbal is basically abstract linguistic meaning and the imaginary there is the mental picture of some sort of representation when the item looks like how does it

look like what is the example of it no previous idea is that picture and concrete words give rise to both horrible labels and visual images so when you imagine it something there are two codes that actually run there are two different ways or two different formats in any kind of remembrance in any kind of storing that happens in **memory** one is the visual and the other a verbal label so when I am thinking of an angry man when I am asked to remember and angry men what my LTM does it not only flashes the word angry man but it also flashed the model this angry man model or this person who looks angry also when we are looking at the word survivor I'm also my know brain is also looking at this kind of a thing when I am thinking about Elvis Presley my mind is also having an image of it and so this is the verbal label and this is the imagery label and so they combined together to form the final coordinate Ian so my LTM then stores two codes for everything a visual code and imagery code the verb code is the one in which the linguistic abstract prepositions stored on to and the imaginary code is exactly example of it so when I am thinking of a sick child this is the image that I am thinking or toddler this is what the image comes to shoes this is what the image comes to and so what L previous is these two codes interact together to form the final representation or final imagery that happens in **long-term memory** now in addition to what Aaron favio says in his **dual coding hypothesis** that there are two codes the visual code and the verbal code which actually go ahead and form the representation in **memory** there is another explanation to it which is called the relational organizational hypothesis and so what is the relational organizational hypothesis this hypothesis was presented by Bower in 1970 B I'm sorry the B is the study which I'm talking about the 1970 published two papers and one paper was not on the relational I organization hypothesis B is a reference to this paper so in 1970 he proposed the relational organizational hypothesis and what is the hypothesis say it's the theory states that imaginary improved **memory** not because images are necessarily richer so the reason why imagined improves **memory** is not because images are rich then verbal labels it is not that images are richer in the sense that it can be thought of it can be imagined or it can creates a model a visual model of it not that reason then verbal level but because imagined produces more associations between the items to be recalled now when I think about imagine about something it lends itself to a number of folks when I think about something very imagine something when I imagine about a particular picture or a particular fact and a number of hooks are made or number of connections are made number of associations are made so when I think about a sick child the word sick child will not be that much better then the idea that when I think about a sick child and think about the image of a sick child and I do that a number of associations in reform of about his condition about so many other variables which are there and so that will lead to better remembering it so when the word sick child comes in front of me it is not a better code but when a face of a sick child comes into me it will associate with it the emotion sure will and that is why it is said that pictures speak a thousand words it will give so many different associations not only if a fact that who's this child is why is he saying what is this a crike and so on and so forth it relates so many information out of it and that is the reason why it is better so forming an image typically requires a person to create a number of links or hooks between the information to be remembered and the other information and that is what I said when you imagine something when I imagine a sick child it is not just the sick child word that I'm thinking about or the idea that the child is sick when I am seeing an image of a sick child there's so many things I am imagining for example I'm imagining a positive non positive about him I am feeling a sad mood when I'm seeing a sick child also I'm thinking about what kind of disease does he have how does it look like where does he come from what are his background what has his parent like what is his age whether it is a concussion or whether he is a nation or what kind of other things which is there so thinking about a sick child or imagining a sick child will create so many associations and so many different different cues for it to be remembered then just the word we sick child because the word sick sure has a meaning related form but the picture of a sick child will create a number of cues on to it so Bower in 1970 he expelled there an experiment to distinguish **dual coding hypothesis** from was the experiment light so what he did was he gave people two letter words to start it also he gave people two letter words to start

with and so these two letter words we're either to be remembered just as words to be remembered just as images or to be remembered as **interacting images** and so in a pad recall test this is the experiment that he did and he Bauer gave people or subjects to remember word pairs right and so first group so this is the word pair that people had to remember so train and wall is the word fair that people have to remember now there are three groups which were asked to do that in the first group there is an overt who wrote repetitions of people will be shown train and a wall and they will be asked to recall this word parent so this is just one entry of the list there are several entries which other the first group will only see the train in a wall and this is a verbal code so people have been shown this formal chordal have been asked to to do wrote repetitions to remember this or to learn this by her clue to saw two images one they saw a train the other saw a wall and so these people were shown images of both the two target so mostly in my pad associate learning I have a cue word and a target word the Q word is the one which comes before the target is the one which is connected to the Q word so in my train wall my train is the Q word and the wall is the target word so when I say train you will have to say wall or when I present train you would have to present come up with an answer of what similarly in this case in the first group people were made to learn the verbal code for train involved in the second group people were shown a train and a wall which are not interacting with each other and they were asked to remember the whole list in this way a third group was there where an interacting image was made so in this case a train and a wall fused together so now I see your train coming out of the wall so this looks like a Great Wall of China and a train forming with this and this is the interacting image and so later on a recall of this WordPad list was done and what was the answer to this or what was the result of this people who do did remote did not do these associations or learn the list through horrible code they produced 30% they were efficient 30% of the time and their **accuracy** was 30% whereas people who just learned separate images their recall was not better or they recall vice-versa then the recall of people on it wrote repetitions or which learn the list through a verbal code but people who formed **interacting images** they produced the highest number of **accuracy** rate of 53 percent which basically means that when people use images and interact images them are more number of hooks are made here the images have a number of hooks but they're not connected but here what has happened is one image hook is now related to the other image hook and so a number of cues are now present and so train wall could be very easily remembered for it and so this is Bauer's experiment in which what we found out is that the more number of hooks that I have the better the reputation or better the recall of a test like this and so his idea is that why imagery is better is because it lends itself to forming a number of hooks number of remembering paradigms now what is the existence of imagery so Lee books in 68 they yes some of the best evidence that images are distinct from rubber materials or images are better than verbal materials or at least use different processes than the verbal material so liberal did a number of experiment some experiments to show that verbal materials are better than verbal materials and they use the verbal and the visual materials use different codes for tagging so in one of these experiments a letter like this was presented to people and then a strict sign was there and so what people were asked to do is to mentally move right so this is an arrow and mentally move anti-clockwise so move clockwise mentally from star to mark each corner as the topmost and bottom most that is there so you have to give an answer and there were two ways to give an answer first what you have to do is look at this F start from the Asterix sign move clockwise mentally to mark each corner so these are all corners as the topmost other bottom most corner so that is what you have to do this corner this corner this corner this corner this corner this corner this corner you have to tell me whether these corners are the topmost of bottom most and so the answers of this has to be provided in a response sheet like this either people have to say verbally yes or no or people have to mark their answers into yyy and n n so each time when you go from corner to corner whether this is the topmost bottom most you have to put a circle like this on to why on n one way of responding the other way of responding was verbally say yes and no right so this is one experiment in the other experiment people were asked to indicate for each word whether it is concrete no not not concrete not so

we were asked to relate that whether each word in this a bird in the hand is not in the bush and so relate back whether each of this particular word is a concrete not or is not a concrete not and that is what you have to do that is what was supposed to be done and same example here the same kind of response here what do you think happened what was the result of the experiment like this now the result of this experiment was people who use imaginary people who was mentally moving things together they were better off in terms of circling this yn and rather than verbally presenting an answer and people who were asked to relate back the concrete noun those people were better off saying verbally yes and no rather than say circling the YNN which basically proves that there are two different codes when you are doing something mentally when you are doing a job mentally when you are rotating mentally in the clockwise direction and finding out whether any corner is the topmost the bottom most then in those cases visually because you're using visual imagination visually marking is better produces better **accuracy** then in the case in which you are verbally verifying a sentence and finding out whether and down is concrete or not and answering also verbally and so that was the answer to date so people in the spatial location people in the F **verification** spatial **verification** usually imaginary exist it basically visual imagining is present and they help us visual imaginary helps us in remembering things better in he'll present better remembering also there are certain other tasks which were used to form to basically show that visual imaginary exists and so one of this is comparison tasks and so people may ask these questions whether a truck is bigger than a car or truck is with bigger than a bicycle or a truck is bigger than a van now if these questions are asked people were faster in telling our truck is bigger than a bicycle then telling those basically relating to those giving answers to those questions we say that truck is Rea whether it is related to a van or not so whether truck is rigueur than a van the answers were slow in comparison to those questions in which it was asked whether truck is related to bicycle the answer here is that people create mental images of a truck and a bicycle Enka do the comparison so when we are creating mental images of a truck and a van they almost look similar and so comparisons are slow a number of comparisons has to be done and so the answers are slow but when you are looking at a bicycle end and a truck there are too many different in it and so the comparison is faster and much better and much smoother then in the first case and this is basically called the symbolic distance effect it basically means that if two items are further apart or have greater distances in comparison standards they are faster verified in terms of mental imagery then when they are close together in a particular comparison standard so basically in this section then what we did was we looked at what is mental imagery we saw what mental images are and we also saw what is this kind of code as I said before we have looked at different codes and here we looked at another interesting code which the LTM has which is called **visual imagery** we also looked at some very interesting ways of helping you remember a number of things are helping you perform better in an exam by using mnemonics and I showed you three different techniques which are there one of the technique is the **method of loci** what we tend to do is imagine a place that you know very well and then are tagged on the list of words that you want to remember onto certain locations on that list our second method is called backward method in which what you tend to do is remember number of in the penguin method in wrote memorize lists and so you take whatever has to remember into particular rule is that you already remember pick the new word onto that and third model are basically a third of your for a third **mnemonic** method is the interacting imagery in which what you do is make two images and make them interact together and once you make an interacting image of two images that will be better remember that will help you to better remember is we also saw evidences of the fact that imaginary does exist and if imaginary exists that we saw that if a task is of a particular type then a particular kind of magic or a particular kind of responses better we also saw the certain reasons of what how national mnemonics really worked so there are two ways of looking into it either looking at Alan previos idea of **interacting images** of two different codes how they interact together or the idea of Bauer where he proposes the relational hypothesis where he says that **interacting images** are much better than the simple mere images simple where solve the solitude images so we will conclude

this section here will again continue with the resurrection in the next lecture for now goodbye

Lecture 20: Object Transformation in Visual Memory

■Kind: captions hello friends welcome back to this follow-up session on mental imagery and **visual imagery** so in the last class what we saw was what our visual images and what is imaginary and we looked at what are the pitfalls of using imagery and why imaginary studies we're negated in the beginning of psychology so several reasons for example imaginary cannot be verified it is very personal also imaginary mental imagery is equivalent to a picture and so that's one concern we cannot see Manatee and we cannot actually verify people's imagery also the idea that there is something called **visual imagery** code into **long-term memory** is also not proven through experiments and so imaginary studies were not taken seriously in the beginning of psychology but later on with a classical book by Alan pebeo on mentally representations and using the dual core processing model talked a lot about what Wiesel imagining and what other different things that can be done with visual imagined so one of the things that visual imaginary does for us is it makes things easier for us it makes **retrieval** of information easier for us so if information is stored in the **long-term memory** as visual code or as visual images it is much easier for us to retrieve it back the reason being that visual images are more informative and are more easily accessible and so we in the last class also told you a lot about different types of weasel imaginary techniques which help you into remembering or **retrieval** of information now some of these techniques were **method of loci** and the **peg word** method and also the method of **interacting images** now in each of these what we actually did was in the **method of loci** we made a mental picture of space a physical space that we are very familiar with and what we did was the things that we need to learn was taken in and then packed down to or tagged on to certain locations of this well-known space now in this way when we remembered the space and certain locations in the space those things that he wanted to retrieve came back to us now in the **interacting images** technique what we did was we remembered two bits of information as images and then we combined these two bits of information into an interactive image for example remembering a dog or a pipe how do we remember that - imagine your dog imagine a pipe and then go ahead and combine them together by imagining a dog smoking a pipe now it's funny it's novel and things which are no well I remembered a much better than things which are routine that's a novelty is something which the brain treats as interesting or as more as more retrievable so novelty is the reason why it is retrievable now in the third method the **peg word** method we specifically didn't use imagery what we did was we used a technique in which kind of routine ordered list which we have rote memorization of was used and things were then tagged on to this list and so basically what we could do is how these list lists are this rote memorized list it could be lists that we learned in nursery as nursery rhymes for example 1 2 3 4 5 6 7 8 or ABCD kind of a thing and then we tagged on these rote memorized list on to certain information that we want to stick in to it and also that's what we did the one two buckle my shoe three four shut the door five six pick up the sticks seven eight lay them straight nine ten a big fat hen kind of a thing and so there is a target and there is a Q and the Q is 1 2 3 4 that is the memorized list the target is the world that you have to remember now beside that we also saw several conceptualizations or several reasons which are given of how these **mnemonic** techniques so basically the technique sizer I just described up all pneumonic techniques how do they work and so one of the explanation which was given was that in terms of the fact that LTM has dual code a verbal code and a visual code and so these two codes actually lead to more number of **retrieval** cues and so they are better remembered and that is the the world coding hypothesis proposed by Allen Bayview we also looked at something called relational organizational hypothesis which was presented by power where he talks about the fact that if two images are if words that have to be learned updated as image and created as combined images then they have better remembering because we try to create a relation between these two images and have an organizational scheme for it and so that is the reason why we have better recall the last thing that we did in the last section or the last session was looking at the evidences of **visual imagery** and we saw two tasks a visual task and a verbal TAS and we

saw that preferences in terms of method of the tribal I was for visual it was visual was better whereas for verbal the method of **retrieval** over Veloz better now in today's lecture what we do is we will continue from what we learned in the last lecture and then look at some more evidence for the existence of visual imagined so one existence is that the well evidence for the existence of **visual imagery** is the fact that certain kind of visual image when it is encoded the method **retrieval** preferred for it is also visual the second evidence is **mental rotation** so what is **mental rotation** basically what people do is when they are given some mental image they're given some kind of a picture they are able to mentally rotate so one important finding for imagine is that people can do some simple rotations of images for example and they can mentally transform them for example look at this now this kind of rotation that you're seeing of a basic curve of a circle can be done mentally and so this people take in images and they can mentally rotate them and so if we can test the idea or if we can provide evidence for the idea that people go ahead and transform these images or if we can come up with solutions in terms of the fact that give them give people certain certain instances in which ask them to rotate a mental image and give them certain instances to verify whether these images will fit into the model of their transformation or whether the sequence give them a sequence out of the rotation and if they match then we can say that people form mental images and they also rotate them now what I mean by that is if people are certain mental image to think about and ask them to rotate into certain sequence for example either anti-clockwise or clockwise now once they are doing that gave them a certain sequence of rotation a certain sequence of rotation from the original angle of rotation and asked them to verify whether this is a part of their **mental rotation** now if they could verify and the time that they took to **verification** will lead us to certain kind of fact that people create materials so to start with the idea and this is exactly what one of the Li and Cooper actually went ahead and did this experiment to prove that people do for mental images and rotate but even before going that what we could see here is this kind of rotation that we are seeing here people do this kind of rotations or if you look into the other one so this is called a tesseract a tesseract is basically an four dimensional image and a four dimensional image rotation so people can also go ahead and rotate images in multiple dimensions and so this is an existence or this is an evidence for the existence of mental imagery or you would have seen images like this so these are called in very basic games these kind of images are used and so these images it's called Tetris as there is a game out there which is called Tetris and so these kind of rotations can be done or people are able to rotate so these are the standards and people are able to rotate these standards into these kind of rotational sequences they can mentally rotate that so these are questions which I you would have found in some of your exams of some of your competitive exams where you get the sequence one standard sequence and then a couple of sequences are given to you where it is asked whether this sequence what is the next sequence from this sequence and so if you look into it what is the next sequence that you should get for this sequence obviously the answer is a here because this is the one sequence which can happen so if you rotate it by say almost 70 degree towards clockwise this way on this axis if you rotate it by 70 degree you will get the sequence and so this basically the existence of the fact that you can come up with an answer a by rotating this was to true and axis by 70 degree is the fact that people are able to do **mental rotation** similarly for this kind of image you can do rotations like this now this kind of tests were done by Shepherd and Messner and what they found out in their experiment they showed that participants they so can rotate line drawings of three-dimensional objects so in this in their experiment they took participants and they give them simple line drawings like this so these kind of line drawings are given to people and they were asked in these subjects were asked in to basically verify whether which one of these line drawings can actually be the next occurrence or the next sequence from this one so this is what your Shepherd in Melser actually experiments all about and if we are able to prove that people can exactly go ahead and find an answer which basically means that people are auditing mentally rotating images and that lead to the fact that people are able to create mental imagery or the evidence of mental imagery is established so they gave people this kind of line drawings and they asked

people to rotate these in three-dimensional accesses now in each trial the subjects would see two typos trying so two type of drawings were shown to people in the first the same object with one rotated to some degree so you either have so the first one as you see here is basically the same object which is rotated to some degree of rotation so as I said if you rotate it 70 degree you will find this is a 70 degree and so same image is there until been rotated and here in this case what has happened is the original image is being rotated as mirror image so the mirror image of this is being so here the original is being rotated and here the mirror image of this diagram is being rotated to certainly certainly this was what Shepard Melser did was they took these kind of simple three-dimensional line drawings and gave people to verify certain sequences by mentally rotating this image and so one of these was the same object with so one of these experiments had rotations of the same object which had been rotated to some degrees and in the other case the mirror image in the other case miron image or reversal with or without rotation was you so the same image the image was taking a mirror image of this image was the original image was taken and that was rotate to certain degrees so either a mirror image was there or the mirror image rotated to certain degree was there in people we need asked to verify whether which of these sequences fit to the original sequence through rotation now the result of the experiment what it showed is that the amount of time that participants actually took in deciding whether the same object has been rotated or a mirror image reversal was there was directly proportional to the angle of rotation between the drawings so more the angle of rotation or higher the angle of rotation the higher the time people required because the more the angle of rotation the more kind of rotational dynamics you have to use in your in your mentally and then more time we will required if simple angles of rotations of 90 180 or 360 are used then it's easier for you to rotate it but if a angle of let's say 60 degree or the 75 degree is used now that type of rotation takes a long time because these 45 90 180 270 or 360 are huge angles that we see around us and so it's kind of purple the perpendicular angles or popular angles and so is easier for us and so what people what Messner found from the expansion measure from from the experiment is that people were able to verify but then the time period for **verification** depended on how much the angle of rotation is also there is a close relation between the angle of rotation of the drawings and the participants **reaction time** strongly suggesting that people from **mental rotation** so that is what we were doing or that is what I was actually trying to tell you that people were actually mentally rotating and so this evidence or this experiment basically provides evidence to the fact that people use mental in that people go ahead and form mental image and not only form mental images they also go ahead and rotate these mental images and so that's what factories are the evidence for mental imagery is and so this kind of images were also used so these are the ones which are used by Shepard in münster and this is **mental rotation** tasks based on canonical rotation so basically images and be really much kind of a thing so these are three dimensional line drawings and these are two dimensional line so another question which puzzle researchers was so basically what we found out up till now is that separate mendler they establish the fact that people were able to rotate an image mentally for my image was to fall and then rotate them mentally to certain degrees and that was evident from the fact that how much **reaction time** people were taking and that was actually proportional to angle of rotation but the first fact being that people were able to verify certain rotations which are available or certain rotational instances which are available and the fact that angles of rotations which which were at at an on to normal not-so-popular angle they were taking more three action times which basically means that people know we're not only forming a mental image but they were also rotate another question that was varying people or that was wearing psychologists where the fact that whether people mentally rotated the whole part of the image or part of the image so once doing the **mental rotation** task or once taking an image and mentally rotating it what was the kind of rotation that people were doing were they taking in part of the imagine rotating it or whether actually rotating the whole image and so Leland Cooper and 75 they used or design an experiment to test this hypothesis so what they did was they took in some kind of an irregular polygon task and found that **reaction time** increased linearly with the angle of rotation and the of rotation

being same for all the polygons regarding of the complexity so basically the idea that they found out the results that they found out first let's see the task that they have so this is the kind of task that they have these are the irregular polygon so you have a six point eight point twelve point sixteen point twenty four point polygon this is the standard form of the polygon and then you have the reflected from the polygon these are the mirror images so reflected from this basically mirror image so different million images were there and so these mental images or these polygons were taken in so this was the standard first six point eight point polygons and certain instances were given to people to test whether this fits into the scheme whether target rotational image that was presented to them could be possible in the rotation in the kind of rotation that they're doing to the mental images so three were used a standard form was use the reflected form was use and rotated the stimuli were given to them so for example this is rotated by sixty degree and people were asked to verify this whether this is possible with this kind of a polygon with a rotation the question that stand that was being asked or that was be investigated was the fact that do people rotate the whole image or partial image and so the results that Lynne in Cooper got from these experiments was they found the **reaction time** they increased linearly with the angle of rotation and the rate of rotation so basically the kind the the speed at which the rotation people are performing and the angle that kind of angle that they were rotating good it to so 45 90 degree way easier but if an angle of seventy five eighty or ninety five degrees are used then people were taking more **reaction time** also with this the speed of rotation dependent on this thing not also the reaction I was and in this case the rate of rotation for all the polygons regardless of the complexity so the fact is no matter whether it is an eight point polygon or a twelve point polygon or a six point polygon people we're facing difficulty all along in that with the angles of rotations we're at abnormal angles then people were taking more time now in another study so basically this proves the fact that people rotated the whole polygon instead of rotating parts of it people were actually rotating the whole polygon so this study demonstrates the fact that even if it was a six point eight point ten point or twelve point following on people were actually going ahead and rotating it and the only answer that they're getting is that the angle of rotation is the the only question or the factor which decides the **reaction time** now all the polygons were apparently the same speed it was either clockwise or anti-clockwise and so the fact here was that as the angle of rotations of the original polygon onto the target one so this is the one which is the rotated so this is my target polygon and this is my cue polygon and so once the angle if the angles of rotations are 360 240 it is easier for his angles the rotation are 60 degree or 120 degree then the difficulty arises because these are abnormal angers people don't see these angles too much and so that was the result from these studies now another study Cooper did in 76 showed that mental rotations like physical rotations are continuous in nature and so what they did in this study so they wanted to study Cooper wanted to study whether these mental rotations were continuous in nature in the sense that do people rotate in a continuous form so if I ask people if I ask someone to rotate take off mental image of something and rotate in clockwise direction do people rotate in in continuous nature and to test this they design an experiment in which they presented people with with instances from the rotation so let's say if I have an image like this and I asked people to rotate this clockwise starting from this is zero degrees and when people are rotating it I'll stop them somewhere here at some point of time let's say two minutes into the rotation I'll stop them and then I will show them a figure like this which is let's a 60-degree rotated to the origin so this is the original angle so 60-degree rotated to the original angle so this is my original figure then this is at a zero Degree then I rotate it by sixty degree let's say this is sixty degree to the original angle so this angle is 60 degree and here is my rotated image so my rotated image will be something like this and so people were given this kind of images in between to verify whether this is there or not and this is what the experiments and so they found out that people were actually able to verify that which basically means that people were continuously rotating through this polygon they were not rotating it at certain angles they were continuously rotating it and the method that they used to test this was they stopped people at certain points of rotation and presented them with a instance of rotated image and

asked them to verify whether this can exist or not or this is a part of their rotation or not and so this is a very simple task and so people were able to verify it which basically means that they were oval people were rotating it continuously any image they were rotating it continuously also cognitive psychologists they started searching how people recognize objects presented in unusual angles so basically something is presented into an unusual angle let's say at a 95 degree so how do you recognize that an object so one possibility that was available to psychologists was that mentally rotate the image till it reaches the orientation of our depiction or that distinctive jion's of the object remain visible we can recognize them with rotation so basically if something is presented at an abnormal angle how do people recognize it so one proposal was that people actually either rotated it to continuously to the refer to the point at which the rotated image is exact match with the original image of original image which is represented to you as target or what people were doing is they took the object to copy the jion's and rotated the jion's to look into it for example one demonstration could be if I put a coin onto the stable and let's say any coin onto this table and then I don't look at it from a top view I come parallel to this table so this is my table and so I put a coin here and so don't look at the coin this way I looked at I look at the coin from this way now when I look at the current from this way or from down up kind of a thing from this this line of sight then what would happen so in this cases this will be an original angle this will be an abnormal angle so for let's say if and door is like this let's demonstrate it in like this this and then when this door comes very nearby I don't Emma straight it this way so basically when it is coming near to me it becomes larger in this way and so these are abnormal angles which are there so how do I verify that so what people did in terms of whether this is the same door so this is position 1 and this is position 2 so in my position when my door is here and it is further away from me this is the hinge and in position 2 it is very near to me so this is the hinge and so it is this is how the door looks like so how do I verify it this door is at irregular angle what I really do is I either there are two ways to it I either I rotate this door into a clockwise direction till it matches this particular figure this is one view or that distinct in jion's are the objects remain visible and we can recognize them with rotation so the idea here the Gion here is our TechEd angle so a rectangle is the G on which forms this door so what I do is this rectangle is still visible to us whether it is lopsided or not whether this rectangle is bigger or smaller is not the question the question here is whether this rectangle is visible to us and so if this rectangle is visible to us we can then rotate this G on and actually think that the dories are being shut or it is being moved towards me or away from me and that is how this Steven Pinker and tar in 1989 they gave an explanation to the fact that how do people actually recognize object at unusual angles and so this is the image that was used by Lynn and Cooper and so this is the original image this is the mirror image and this is the image which is tested across various angles so 60 60 240 and 120 are angles which are abnormal angles 180 and 360 degree R so 180 would be basically a mirror image and 360 degree will be a complete rotational of image of it and so this is our people actually using and so looking at this demonstration so basically what we're looking at is this figure being rotated in a different plane and at different angles so this is basically how people go ahead and rotate it and so this is how **mental rotation** of images exist and so this **mental rotation** of images the fact that we can verify certain rotations provide enough evidence for the fact that people do for mental images and mental imagery is actually a code which can be used so let's then look at some other aspects of this mental imagining the question is what is the nature of the mental images that we are talking about what is the form of it so one of the things is that visual images they share properties with pictures the visual images that we have they have more or less exactly like a picture but what are images what kind of properties do images have and how these like and and like the properties or real picture have that has to be discussed so what is a one-to-one relation between our picture and image that is what needs to be seen in what properties they share with this picture and what properties they don't how are they like a picture how they are not like and what is the distinction is what we are going to see in terms of the nature of imagery so ronald think in 89 1989 he proposed some fundamental principles of **visual imagery** he said that **visual imagery** has

certain fundamental properties certain fundamental features which we can discuss in which we can see and these features basically correspond to the picture thing so what we did was he proposed five different fundamental principles and they took **visual imagery** he took **visual imagery** in pictures and compared them across so what are these five features of a mental image first thing is something called implicit coding so what it says is that mental imagery basically encodes some information implicitly which pictures cannot example remember the first class that I asked you in this section in this session the the earlier session on visual **memory** I asked you to think about what was on the wall on your bedroom and so there were implicit information there were things which were in your mind or there were things in the image which you never thought about so you might have never thought how many cells of the window are present but then when I ask you this question you can mentally count it which means that the information about the number of cells which your windows have is implicitly in coltan that is what this basically this principle says so it says that mental imagery is instrument in retrieving information about physical properties or objects or about physical relationships among objects that is not explicitly encoded at a previous time and that is what I was talking about though certain informations which pictures can't have visual images can have so you can have information certain information about certain facts embedded into a **visual imagery** or visual image which is not possible at a picture level the second thing is that perceptual equivalence how a visual image is equivalent to a picture it has perceptual equivalence for example it has been found that imagine is functionally equivalent to **perception** of the extended similar mechanism the visual systems are activated when objects are events are imagined as when the same object or events are actually perceived so there will be certain there were studies which were done certain new studies which were done within **fMRI** and it was found out that the same areas of the brain get activated except the visual eye system which takes in new information when you imagine something so if people are asked to imagine a coffee cup and Whipple are shown a coffee cup the way they perceive these two things together the number of areas the kind of connections that these areas have are exactly the same the same brain get activated the same kind of connections are activated and so what happens is that imagine E is more or less same as looking at a picture or looking at something which is out there so it is imagined is basically same as looking the only difference being that there is no input the input system is not that the same visual cortex is the same areas of the brain which verify an image which does object recognition which lets the frontal cortex know what it is and then come out come up with our decision of what possible objects could the visual imaginary be all those areas are the same and they say get activated the same way but know only the input system which is the eye is not activated and so in this way the visual images are similar to and picture now the third principle which thing gave in terms of a third property of a visual image is called spatial equivalence now what does it mean it means that the spatial arrangement of the elements of a mental image corresponds to the way objects or their parts are arranged in actual physical surface or an actual physical space so basically in there are some experiments done by kosslyn and so what they did was kosslyn did was he created a fictitious map and in this map what he did was he made certain positions right so he created a map like this and in this map he had created certain positions and these positions had certain kind of placeholders so you have certain things on this maps and ask people to move from point A from point A to point B and place some hurdles here and so he looked at the time that people took to move from point A to point B so what he did was he showed them map these positions and asked them to move later on they asked people to imagine this map to think about this map think about a map of America I think about a map of India start from Kanyakumari and then move towards Bangalore or start from Kanyakumari and move towards Bombay now do that and what you have to do is you're to then basically in this moment describe all the places that you venture into or going to and then I show you actually the map of India and ask you to move from point A to B kind of a thing a very simple explanation now what they found out is that the time that people took right from moving from point A to B the kind of arrangement the kind of movement that people did in terms of the actual map is the same time

and the same kind of arrangement or same kind of woman that people didn't mentally imagine it basically means that people when they move in terms of images in terms of real pictures in terms of real movement and in terms of mental imagery it's more or less same so the spatial locations in terms of moving in spaces whether in it whether through a mental imagery or through a picture for real motion or real movement in terms of space are more or less same so people took the same time from moving from point A to B in terms of mental imagery when they imagined this map and thought of the movement then similar to what people did in terms of making movements when they were actually moving from point A to B or the another experiment was done where they where costly and showed people a flower rather a tree and asked them to move from the top of the tree and start naming right so start claiming what is at the top of the tree towards the bottom to the root start naming all the things that you find and so the time that people took on the number of the elements that they kept on saying when they move from the top of the tree to the bottom of the tree or top of flower to the bottom of flower then the the number of elements then they said that the flower possessed when moving from the top of a flower towards its stem or top of a tree towards this root was similar in terms of both in in terms of actual spatial movement in actual movement of a flower as compared to the movement in terms of an imaginary when they were asked to imagine this flower and made to move which basically means that in a spatial location in terms of spatial locations both the image and flowers at sorry the image and the pictures are same the fourth property is in terms of transformational equivalence and what does it say that imagined transformations and physical transformations exhibit correspondingly dynamic characteristics in a governed by the same laws and is the same experiment which Cooper did in 1976 so what happened in his experiment where people were actually given a cube to look at and they were asked to rotate this cube through certain angles then they were asked to basically take this cube mentally imagine this cube and make the rotations and then people were tested with certain instances what was found out that the same amount of time or the same amount of **verification** was done by both or the same **accuracy** of **verification** was outputted when people were transforming mental images as where they were physically transforming an object it basically means that people while imagining we're also able to transform are also able to mentally rotate images and that is what we saw in the earlier examples also so visual images are not just static people also transform them and these transformations these rotations either on the axis or across accesses or a different excesses are similar to the physical transformation which is there and so the result from Cooper remember the result the experiment of Cooper 76 where they made people transform certain images and onto certain angles and then tested certain instances with the **accuracy** of it the same experiment can be used and evidence here and the fifth property of think of visual images is called structural equivalence now what does it means the structure of mental images correspond to the actual perceive images and so in this case Gosling and Farah and figgle they gave people certain structures certain images which had very bleak structure and certain image which has complex structure for example they showed people let's say the drawing of a fish I am not very good at drawing good drawing of fish like this and so it has very less detail or they showed people the drawing of a fish like this which has skills like this right and this kind of fins and other things here so this is a complex image because it has more details into it and these are the simple image which has lesser details into it so when given people these two kinds of things and ask and then ask them to retry more time was taken by people to verify to actually make the structural **verification** or to actually perceive this one the complex picture as compared to the simple picture now in terms of imagination also when people were asked to imagine a fish like this with these many features they took longer time to imagine a fish like this or they took more time for complex image imaginations then imaginations of simple images which basically means that in terms of structure also in terms of the fact that when people imagined they imagined with complexity where they imagined complexity with more time the more complex a picture is takes more time because more number of elaborations have to be done and so this idea of mode number of collaborations are also in present in **visual imagery** which means that the same kind of verifications in

terms of complex and simple images happen both for the the picture for the physical transformations as well as men transformations so find different kind of things to look at again reviewing quickly back into what it's all about and so the fight thing first visual images have implicit coding so it has implicit informations into it which pictures do not then perceptual equivalence perceptually speaking the same areas of the brain are active when a picture is imagined or a picture a picture is seen the third is structure especially equivalence so in terms of movement in space they are border there is the same the fourth is transformational equivalence so mental images can be transformed similar to what physically physical pictures could be and then the fifth is structural equivalence in terms of structure in terms of complexity the same kind of complexity the same kind of processes exist with both the mental image as well as the actual image now there are certain critiques to this mental imagery certain criticisms which have been there to this mental imagining what are these criticisms so there are three main criticisms first is tacit knowledge and demand characteristics so basically mental imagery has been criticized on certain factors first is the tacit knowledge people had some kind of knowledge of people gather some kind of knowledge from experimenters in terms of transformation so that's one thing and demand characteristics is certain past mental imagery tasks demand certain characteristics or demand certain kind of response from people so that is one thing and so the that's how it's looked at the second is the picture metaphor where a comparison between picture and mental image is done and the third is the propositional theory so let's look at these critics one by one the first is tacit knowledge and demand characteristics and so what salinity one they argue that the results from many majorie studies they reflect participants underlying and implicit tacit knowledge and beliefs about the task rather than their construction and manipulation of visual images so what they found out or what they told is that people they actually what they were doing is they were look they were acting upon to something called experimenter effect which basically means that it's who were asking us them to do these construct these mental images they were actually providing some kind of a clue so either mentally or either physically mentally or some form and so this was the reason why people were able to create mental images or do all kind of manipulation into mental images so think with this example of moving the coffee cup provided evidence to show silence cup so basically what was this coffee cup experiments is a very simple experiment and so what it was that if a kosslyn actually initiated this experiment are an fing calibrated onto it so what the experiment was that a coffee cup was put onto a table and it was asked we move from point A to point B so I have a coffee cup like this on - this is a table and this is position a and this is position P and I want to move my coffee cup from point A to point B now physically two ways are possible I will pick up this coffee cup from here and then move it over air to point B or I can slide the cup from point A to point B now when I do that a certain amount of time requires now when I have people were asked to imagine to move this coffee cup from A to B what they found out that they could do it very quickly they actually could do it very frequently but then people were pausing people were taking thinking about taking the coffee cup and taking the same amount of time or rather a little bit more time when moving the coffee cup from point A to B and why was this there they were the reason that thing Gail is that the experimenter were giving certain demand characteristics are certain kind of clues which asked or which made people mentally pause the coffee cup position in their head and then take that amount of time what they were thinking is or what their conclusion is that people actually thought that although this can be done quickly but this is the amount of time it should take and so I should pause the cup mentally and so these type of talks the tasks are called cognitively in imporant table which basically means that these tasks are open to influences from belief systems from the ad from your previous knowledge and so on and so forth so what people did was people also took experimental e same time in terms of movement from point A to B on a table and in terms of imagination studies also but it was not that people were had this this is because of imagination what really happened is that certain knowledge from previous situations or certain clues from the experimenters made people actually mentally pause the cup somewhere and then take exactly the same time that the that a physical movement would do and that is

what their explanation is now file in eighty-one this states the stage that tasks are that are affected by people's belief and expectations are termed as cognitive penetrable and so that is what it is so basically what happens is since these kind of tasks give the person doing it a chance to use the knowledge that this is the time we should took from it should take from moving the coffee cup from A to B and so what they do is they actually go ahead and mentally pause it so it's basically not the image construction or the image movement which is taking up the time it is basically a mental pausing the idea that people get this clue and using this clue as many positive now such task makes it obvious to participants that they out to perform and are said to have demand characteristics so trust us basically in the coffee cup moving experiment people know that it should take at least one second to second five second and so on and so forth and so what they do is although mentally they can move very quickly they pause somewhere or they take more time we cost to match the idea that this this captured this movement should be matched to the actual time also sometimes experimenter unconsciously give subtle cues to the participants now in intense and in turns and Peterson call such effective experimental expectancy effect so what happens is sometimes in these experiments what the experiment tend to do is the tends to give certain clues to to the subjects who are doing **mental rotation** for example a probable nod if they're verifying a particular angle or cue with which has been rotated a certain angle so maybe a subtle cue or a certain aha or a certain gesture which gives the subject or which provides the subject some kind of a clue of whether it is right or not and so they say that these clues are the reason why they are doing so correctly and so mental imagery is suffering from these kind of problems the other critique is called the picture metaphor so what is here so visual images are casually spoken as mental pictures so visual images are thought of as mental picture so how they are how far the statement is true for file in 73 he pointed out that pictures and images differ in several ways there are meant although mental imagery is thought of as a mental picture but there is was demarcation between what a picture is and what a mental images and so what is the demarcation first pictures can physically be looked at without knowing what it is in a picture that we are looking but mental images cannot be done so if you're looking at a picture of someone you don't know right and so in this case is so you have a friend a friend has a friend and so you looking your friends and other the friends friends picture you might not know the friend's friend and so you can very well look into it without knowing anything about it or a picture of some place that you have visited if you looking into it you might not know where this place is and you is perfectly okay to look into it but visually imaginary cannot go through those parameters you have to first know so if I am thinking about the United States I am actually know that this is the United States I am thinking about Berlin we have lived for some time so the idea is that I am thinking about Berlin which means that I have named it and so on thinking about that place so I have to know Berlin before thinking about it and forming a mental image about it whereas in picture is not necessary so first difference second difference pictures are and images are disrupted and destructible in different majors raises so pictures can be destructed or it can be disrupted and distracted in different ways for example if certain you can cut out certain things from a picture certainly so if you don't like your best friend who has now becoming enemy you can cut him out of the picture but if you try cutting out something from a mental image the mentally will image disappears so basically you cannot cut certain pictures from a mental image third images are more easily distorted by viewers interpretation so if I give you an interpretation if I say that in in a if I show you a picture of a friend two friends together which are both a common friend and then I say that this friend is a little bit shrewd now there will be a change in this picture but if I ask you to imagine a friend or imagine your friend's friend and your friend and then say a shrewd the kind of image changes the mental image will change particularly a more cunning kind of an image of the second friend would appear but in terms of picture this cannot happen so basically these distortions can happen in terms of images but in terms of pictures it cannot and so the comparison between a picture a mental image is not true because of these factors which have been outlined now also the fact that to prove that people's view or people's interpretation defines a picture a kind of this kind of setup was given now people were given

this kind of a bizarre drawing to look at and then in two different sessions two different settings or two different **retrieval** sessions in one session people were shown this and they were then given a clue that this is letter C and when a **retrieval** was done after two weeks so this is **retrieval** one after two weeks but then the same group is there so this is after two weeks then later on they come back after one month they shown this again and then given this kind of a verbal label and after this verbal level after two weeks and interpretation or re travel was done so when they were shown this and said that is letter C this is the **retrieval** but it was when they were are said that is a crescent moon this is the retriever and so basically this is what it is so images mental images are dependent on people interpretation so when a verbally this was shown in a bubble level hat was given this is the **retrieval** image but when it was shown and a beehive was what was told what was asked them or what was told to them this is what it is this is the rate reverse similarly this a dumbbell would result in this and eyeglass would this and if a verbal label of 4 was given to it or 4 was there and a 7 was given to it a 7 interpreted onto it and similarly so on and so forth so basically this image this kind of image was given to people this kind of verbal label was given to people so these verbal labels were given to people and then later on **retrieval** was done so depending on the kind of verbal label the interpretations or the **retrieval** imagery was different and so this basically proves that the imagery is open to interpretations of people the third kind of problem the critique which has been said in terms of of mental imagery is called the propositional theory so what is propositional theory so proposed originally it was believed that there are two codes that of mental images one was the vocal code or the verbal code which is from the **short-term memory** and then there is a semantic code which is basically the **long-term memory** interpretations or **long-term memory** storing so original mental imagery imagery idea is that mental images are special type of **encoding** and so later on when mental imaginary studies it was believed that mental imagery is also code for **long-term memory** now proportional theorists they say that it is not true so what proportional theory says is there I have no code so there are no ways on which a code is trained and that is what they say they say that there is only one kind of **encoding** into **long-term memory** and that is neither visual nor verbal in nature and they suggested they suggested that the experience of having a mental image is just really r.fp phenomena and so what is stored into the **long-term memory** is basically a a proposition and what is the proposition proposition are basically relationships between items for example if I write on book and table this is how remember from the first class that we did a special equivalence so this is a propositional code which says that the book is on the table so on what is on the first object the book is on where it is on it is on the table so this is how codes are because this is a preposition because it makes relationship this on is a preposition which is making relationship between the book and the table and so this is how it actually goes ahead NS is saved so basically then this is the code that is saved into the LTL rather than a visual code in a verbal code also fill in suggested that the experience of having a mental image is really just an epiphenomena something that happens with a process but does not cause the process instead is just a by-product now without the epiphenomena the process would go just like normal not necessary for process to occur so what he says is mental imagery that we talked about is just in epiphenomena it's nothing special into it and he has given several examples of seven goos for that for example he says that when a computer is calculating something it often has a flashing light which shows that it is calculating most people believe that this flashing light represents what it is calculating but actually the flashing light has nothing to do with it's an active phenomena it has nothing to do with what is going on similarly mentally imagine is epiphenomena like this because if the light blows away tomorrow or if the light is dysfunctional well it doesn't mean the computer stops working so this light is just a by-product off from whatever is happening and so light has nothing no relevance to the computation similarly mental imaginary has no relation to the computation because codes that are used in mental image is basically propositional nature so instead of **encoding** inside the **encoding** is propositional nature when concepts are stored as symbols and what is stored is not the physical relationship but a **concept** field one like the network model of **memory** so that show to you

this is the proportion and so in the proposition I have a book network a **concept** and a table **concept** and then the relation to this is the on **concept** I can basically this is the proportion which is safe this is neither verbal nor visual in nature so what is toad in **memory** in **long-term memory** our concepts and symbols and this consciousness symbol from some kind of a network and this conceptual network and this is how the storing happens in **long-term memory** so it would make sense they're trying to scan a part from a flat to the back of the board to a cabin would take less time than scanning from the flat to the emblem because when we are scanning from the back of the board to the front of the board to from the back of the board to the cabin is longer so look at this image I'm from scanning from the back of the board this is where the emblem is and this is where the cabin is so if I am if I am scanning from the rear deck to the front deck it has three levels to go from and so it should there are three prepositions to be verified and so takes more time whereas if I have to verify the emblem which is there from here it has two levels only so it is easier and so that is what it says it should it would take less time than scanning from the flag to them since you would have to move from two level of networking or two levels of the hierarchy to four levels of the hierarchy so it is possible to explain scanning times without having to use mental images and that is why the higher scanning time and lower scanning tank that people are using in terms of complex and non complex images remember Fink's idea is because of the fact that not people are making images it is because the number of propositions that they are testing the number of hierarchies that they are moving in a network is more as compared to less and so if I have to where if I come to the emblem from here from the portal it will take more time because more number of networks or more number of preposition seem to be tested then if I have moved from here to here where a number of propositions is less and so the last section has to deal with something called spatial **cognition** so basically how do we move in a special environment and so the three ways of moving in a special environment when we move in spaces when we move in enclosed spaces or any kind of space out there there are three things to be looked at first there is something called the space of the body now this is there are the parts of the body located at any particular time so while moving in a spatial location while moving in your environment we use three different kinds of information first is the call the space of the body so we move into something called the ego centric approach or the space of the body approach in this case the movement is done in terms of where my hands and legs are so left from my left my right towards the front of me towards the back of me is kind of a thing so where my body parts are and based on that I make a movement in spaces in faces which I finished fixed with locations and the second is the space around the body the area immediately around you and so it could be in terms of not in terms of the body not in terms of certain regions of the body but in terms of the area around the body so in front of me at the back of me kind of a thing so in both of these are basically they go centering right so towards my left hand towards my light hand but in front of me in back of me is kind of space around the body so towards towards the left of my left hand towards the right of my right hand is basically space around the body so two types of information both are egocentric in nature and that is what we use we also use spaces of navigation larger spaces that we walk through travel or explore so in this case what we say is where this is called the LOC entry el eccentric movement so Al LOC en PR IC el eccentric and this is egocentric so you got eccentric is making yourself as the center el eccentric is making external objects so left from the hotel right from the cabin back from something front from something so using something into the environments using certain land mass movement using landmarks in these cases I am using the body as object of movement so spaces in front and back of me or spaces relays in relation to certain body parts or I could use spatial navigation in terms of certain features or certain objects which are there into the environment which are called landmarks now our mental representations of these spaces may be distorted made neater and more regular so basically based on these we create something called the spatial map or cognitive map and these currently map actually help us in moving and this is the mental image that we make so make a mental image of the space around us we use three reference points so one is with the self it's with our self then with our landmarks and combine

them together to form mental images and these mental images are then used to move around any space out there into the environment any space into the environment so basically what we did in this today's lecture is we tacked on to what we learned in the last lecture so what we did was we verified the fact that mental imagery is there by looking at rotational Studies now with in addition to that we looked at that's how rotational Studies complement the idea that mental imagery exists and we looked at certain kind of answers in terms of angle of rotation being the only factor which which guides the rotational mechanism or the rotational time that we take in addition we also looked at certain principles of mental imagery so five different principles in terms of things theory and then we looked at certain critiques or certain criticisms of that idea of mental imagery for example demand characteristics the idea that the code that the brain uses are only propositional codes are the fact that pictures are not mental images and lastly we looked at how special **cognition** is stored so spaces around us are stored as mental images and I said there are three reference points in terms of your body in terms of the external landmarks and in terms of spaces which is in front or back of your body so thank you and we'll meet again in the next lecture

Lecture 21: Issues in Language

■Kind: captions friends welcome to the section on **language** so at the very brief outset of this section let me ask the question what is **language** or rather another question what is it that we are doing right now now when this lecture is progressing as I'm talking to you what we are doing is we are exchanging ideas and knowledge between them so on and imparting some knowledge and imparting some ideas which you are grabbing and so the medium through which these ideas that are imparting it emanates from me and it is received by you is through **language** so **language** basically is a medium of communication it's it's that medium or it is that thing which helps two people talk to each other or two people to communicate with each other now look at the title of this slide as you can see the first part of it is in English and the second part of it is in German and so what German say German part says is and surely come secondly English which basically means excuse me can you speak English a much famous thing or a much famous word that I used to use when I was research associate in Berlin and charitable in what this means is that I used to often go around talking people talking to people and asking them whether they could speak English and so why I would do that I would do that because there was some problems in communication I was not able to put forward my idea and so this is a forum or this is basically a way in German to ask if the next person that you meet knows English or not without ever disregarding them in some way so basically what **language** is its medium or it's a channel of passing information from the sender to a receiver and that is very important in **cognitive psychology** because anything in currently psychology any cognitive processes any cognitive function has to have a **language** for from through which things can be communicated and for to start the lecture I have a slide here as you can see that the person obviously in the restaurant he sees a fly in his soup but then the way he is communicating is he has mixed up the **language** and so he says there is a soup in the fly which makes no meaning and so communication or languages although there are a medium of communication but they have to follow certain rules and if - if you do not follow these rules what really happens is that the whole meaning of the sentence is lost so let us then start with the meaning of **language** so what is **language** and how is it different from communication so basically is **language** the only medium of communication is the **language** the only medium of transferring information between two objects one ascender the other is a receiver or maybe others are receiver and if they are then how is **language** different from communication now there are several communication systems out there in the world for example microwave communication it doesn't have a fixed **language** or the smoke guns that you see or the honeybee dance that honeybees do when they find honey they across certain flowers or certain sign languages out there and these mediums are a medium of communication but they are not **language** why they are not **language** because the other **language** has certain fixed rules or certain fixed characteristics which most communication mediums do not for example one rule is that in communication only brief number of ideas can be transferred and fixed number of ideas can be transferred and so new ideas cannot be generated for example if the smoke signal says that intruder is coming that is what it could do but **language** on the other hand can be used can be used to form new signs or new information that can be transferred and so the slide is in the the soup is in the fly or fly is in the soup may mean two different things the soup is in the fly which basically means the fly has drunk in the soup whereas the fly is in the soup means that the soup has a fly and so two different things the fly can have the soup and so it will mean somewhere in the stomach assuming that the fly has a stomach when the other case the soup being the fly being in the soup has a different connotation altogether which basically means that this is non eatable something which is contaminated and cannot be eaten so basically then **language** is often used as a communication system but and as I explained to you there are other communications or other systems also there which are not true languages and we'll see why **language** is different from any communication system and so some of the communication systems out there are like honey bee dances most from Martin bird songs and so many other things now what is the

difference between a communication system and a **language** a natural **language** has to have four properties first of all it a natural **language** is regular which basically means that it is governed by certain rules which are called **grammar** so certain rules tells the **language** what should be interpreted what should not be rated into what are the anomalies and how they normally is occur and how it can be correct also a natural **language** is productive which means that it natural **language** can produce in finite combinations a number of communication a number of ideas so change something into a **language** same some part of a **language** a new **language** the new idea is generated also it is to be a natural **language** is arbitrary what does it mean the lack of a necessary semblance between the word in a sentence which means basically means what a word tells and what a sentence tells or what is the meaning of a word and what is the meaning of a sentence are completely different so any word in a sentence could mean differently as against what a word would mean so a bank would mean a bank but I was standing on the riverbank means something else so here the meaning of the bank changes whereas bank could mean anything in terms of financial institution or a river bank but when I put the same bank into sentence like can I bank on you in this case the idea of a financial institution is not what is talked about and so our patroness basically means that words the meaning of words and the meaning of sentences are totally arbitrary and there is no one-to-one relationship between them also a natural **language** should be discreet the which basically means that I can take a per sentence and I can divide it into certain meaningful parts so that I can study the sentence so discreteness is a system or it's a basically property where the sentences can be divided or the meaningful sentences can be divided into recognizable parts so that I can understand what is the role of a part in a whole sentence and that is what a **language** or any characteristic of a **language** should be and this is how it differs from our communication now what is the structure of a **language** any **language** what is the structure of it now languages have a number of system which works together and evaluating a **language** requires the study of conversations it's basically exchange of ideas between two people so when two people talk which we study that we will be able to understand water **language** is so basically conversations are Brown are out or they are evil they revolve around listening and perceiving the sounds of the speaker so it's not just listening the sound of a speaker or somebody is saying and recessions also employ an idea of you should be able to understand what the speaker is saying so this is the difference between listening and understanding now different languages have different sounds called the phone names so phone name is basically a speech sound so each **language** has a different kind of a sound and this basic sound that each **language** has for example bar de anza are called phonemes now the study in which the four names these basic sounds of a **language** for example in Hindi it could be a Hui kind of a thing which are the vowels in Hindi thus whereas we call them or Kirk as wenyan we call them are in English it is AEIOU which is the verbal or ABCDEF which is a cdef kind of a consonant in english so they produce a sound for example the a sound has two phonemes into it or two phones into it the a and the e when we say a it has a and e to it and so these are the following basic speed sounds so the study in which the four named these P stands are combined in any given **language** is called the **phonology** so basically conversation is listening to and perceiving what other people say and that can be done by identifying the basic speech sounds the basic phones that each **language** has and these phones are the primary phones or the primary sounds which the **language** has and as I explained to you in the English **language** for example the word a has the a plus e sound right and so these are the two sounds that are sound the e sound so a as you say now the various phonemes are these basic sounds of a and E are combined together to you to yell a meaningful unit of a **language** which is called a morphology so the a and E then combined together to form a a which then combines with some other sounds for example eighty E which is eight or Appl e which is Apple and this is called morphology so fond of the combined together two usual meaningful words which are basically called the morphemes so these markings are generally not word but part of a word right so for example morphemes are things like word endings example making plurals so if I Appl E and if I add a F s this S which is add to Apple to make it a plural is

called a **morpheme** similarly prefixes pencils markers for example tense markers like for present continuous tense I used markers like ing or for past continuous I use tense markers like heed he walked he is walking he he will be walking kind of a thing or he walked so when I say the eet be to walk it gets convert into a past tense and so basically these are the markets now some of the more famous the smallest meaningful units for **language** are words that need to be identified in order to study the role behind each words in a sentence so basically then think of it in this way each conversation a **language** starts with identifying the basic speech sound which is called the phonemes and these phonemes are basic phones as I said a has two phonemes a and E these phones combined together to form a meaningful unit for example a **morpheme** a Martian is basically combination of phones which has some meaning for example EB in terms of the word ending or ing in terms of coordinating or things like prefixes for example any any prefix so fulfill unfulfilled kind of a thing so unhear or in here kind of prefixes which turns the change the meaning of a sentence these are called morphemes and these morphemes are then governed by certain rule or governed by certain structure which is called a **syntax** so basically the structure of a **language** is in terms of the basic speech sounds which is called the phonemes at another level with the phonemes combined together is another level which is called a **morpheme** which is collection of phonemes but which has some meanings or which has some meaningful unit of a sentence now these morphemes could be anything in terms of or endings in terms of sentence markers in terms of critical parts of a sentence and so on so forth and these morphemes are then guided by certain structure which is called the **syntax** so **syntax** is basically the structure of a sentence now a syntactically correct sentence does not itself is always good conversation so basically saying that all the objects or all the markings are all correct phonemes or if it has a sentence if a sentence has the right structure or the right **syntax** it may make meaning it may not make meaning for example look at the first sentence that we saw the soup is in the flight know probably or not probably basically this sentence has the right **syntax** is everything correct in terms of the object the subject the world everything is in line but it makes no meaning because the soup cannot be in the flight in terms of the context although soup isn't the fly basically means that the flag has drank in the soup but looking at a restaurant scenario where you see a person taking a soup and then there is a bowel in the bowel there is a there is a fly which is on to it something's basically proves wrong so looking at the context the sentence is wrong and that is what I'm trying to tell here there's every syntactically correct correct sentence may not be right now the sentence must mean something to the listener and in this in the first case the sentence was not meaning anything and so this meaning which is generated by a sentence of syntactically or offer grammatically I'm sorry not dramatically in terms of structurally correct sentence is called the **semantics** so **semantics** is the study of meaning of a sentence when you correct so basically then in a sentence you have the phones which are the basic speech sounds take these phones combined together you will found on the morphine's which are small critical units of a sentence for example it could avoid ending it could be tense markers it anything and these markings then combined together in a certain specific form to form the level of the word and these words are then governed by certain rules which is called the **syntax** now **syntax** is how should you write the sentence but then the syntactically correct sentence in terms of rules if a sentence is correct it may have a meaning or it may not have a meaning and so this idea whether the sentence has a meaning or not meaning is called the study of **semantics** so look at this here there is a sentence that I have used and I will show you the basic parts of it so the sentence here is the stranger talked to the place and so let's break this Indians into its parts so the first level from the sentence down is the phrase level in the phase level you can see there are two phrases into it we have the noun phrase and the word phrase the word phase is the phase what is a phase a phrase is a collection of words in a sentence which makes complete meaning right for example if I say The Strangers it has some meaning into it and so it doesn't need anything beside it also I say talk to the place so although there is a question who talked to the place but just by itself talk to the players has certain meaning and some phrases or parts of a sentence or how many have a sentence which has

complete meaning and so there's the level of the strength there the phrase I have an entree zina were placed at the level of the word then again breaking this sentence I have two words the and strangers and then at the level of the word on this side on the noun phrase I have talked to the place for words to look at similarly at the level of the **morpheme** I have the strange and ER s because strange is the word root word and ER s makes a number of words so strangers and similarly here talk is the word which is the core word and Edie is the marker sentence marker top so it is past continent bastions and similarly in to the play ER NS players which means the plural of it and then coming to the level of the **phoneme** so this is the level of the **phoneme** how the is stranger is spelled here an S is spelled and so on and so forth so basic base and structure of any **language** is starts with **phoneme** goes to **morpheme** which is combination of phonemes then it goes to the level of the word where morphemes combined together to form words which are meaningful units and these words are then converted into are mixed up using some **syntax** using **language** rules to make phrases and these phrases then form the sentence and the sentence then is looked for or is thought of in terms of meaning which is called the **semantics** now for conversations to work there must be some flow so any conversation which which happens between two people for any conversation to happen there should be a flow between the movement of the sentence so listeners must not only its listeners the first of a **language** or listeners of a particular conversation should not only pay **attention** and make certain assumptions the speaker must also craft their contribution in what the ways in which the listeners which makes the job of the listener feasible for example think of this sentence so somebody says I want to read watch in the other person says the ACS the NASA is sending a rocket now in this way what is happening is the two people are talking together and so one person has the meaning it's not making any meaning although two people are making sentences conversations but these conversations are not being interpreted and so for conversation to work listeners must pay **attention** not only pay **attention** to what is being said but the speaker of a sentence should make sure that whatever he is saying should be crafted or should be made in such a way that the listener understands it if you say an incomprehensible sentence the stranger may not understand the listener may not understand it and so conversation is a two-part game where there is a speaker and where there is a listener so the speaker should pay **attention** this the listeners to pay **attention** the speaker should make sentences in a way which is comprehensible and that's how conversation flows now the aspect of a **language** this aspect of **language** which basically decides what a speaker should say and how a listener should look into it is called the **pragmatics** so although meaningful sentences although **semantics** is the meaning of a sentence but just having meaning a sentences does not itself make a conversation so for a conversation to flow you should have something called **pragmatics** into it which basically is that how conversations or how things are shared between people what is it how is a listener listening to something and what is being transmitted by the center so present the **pragmatics** is the flow of conversation the kind of flow of conversation between a speaker and a listener the idea of how conversation should happen how a good conversation should happen is what is called the **pragmatics** beside this there is something called **grammar** which are sets of rules for a **language** now **grammar** has nothing to do with the rules of a good grab good **language** for example if I say the word no I ain't coming now if I say I mean coming in is not a very good work to look at and the word should be actually I am NOT coming but then **grammar** only points out legal sentences it only looks for Digger sentences and it's it's a rule of the game rule of a conversation but it never looks at or it never tells you whether a sentence is good English or good Hindi or a good **language** right so **grammar** will tell you whether a sentence is following a certain rule or not or does it is it correct in terms of rules in terms of sentence construction whether the transferrin sentence construction rules are following in a comprehension or in a conversation but will not tell you tell you whether this is the correct way to speak so **grammar** doesn't do that and that's what grammars are all about no psychologists and psycho linguist the distinguish between something called explicit and implicit knowledge of linguistic rule the idea is how do people know rules of **grammar** how do people what is the way in which people

know the rules of **grammar** or how does it grab these rules is it explicit are we do we know when we look at a sentence that we are following certain rules the glamour of it so grammatical rules how do they work are the explicit or implicit in nature so although most of us cannot state with **accuracy** the rules of English **syntax** but we can basically find out ambiguities and sentences so if there is a violation or gamma we can find into it for example look at the sentence I ran the dog Street down cat after you load the welder what does it really mean it doesn't mean anything and so if you give them give somebody who has studied the English **language** he'll find out that this is a problem there is a problem with something in the sentence and then if you ask them to correct this is probably what they're going to write back so people already know no the explicit rules if you ask somebody who knows English are studied a little bit of English they will immediately say that this sentence is wrong is somehow wrong it is not grammatically correct ask them the **grammar** rule they are not able to do it but ask them to transform this and this is the transformation that most people are going to do they say that the dog ran down the street after the very yellow cat and this is the exact interpretation right so grammatically this is incorrect but most people are able to correct it all that they don't know how this drama really comes in and so people's knowledge about **grammar** is mostly implicit so our knowledge of rules of **grammar** is therefore not explicit we often use something called prescriptive rules example what people say should follow so when we're using or when we talk or when we use **language** we generally use which tells us how we should talk or write there are certain prescriptive rules there are certain rules which goes around the world which people share between us and saying how you should talk and how you shouldn't for example if you are saying I ain't I am what you are doing this kind of sentences are not permitted or not entertained and so this is the prescriptive rule now in contrast the articulation of descriptive rule characteristics of a sentence are very hard so what people tend to do is they know prescriptive rules of using when they when they are conversing they know what is wrong and what is right but they don't know what is it in the grammatical way what is wrong or what is that and so as I said as I showed to you that if these two sentences are shown to people people will be immediately able to tell you that the first sentence is something somewhere wrong and the second sentence is right this is the right format of it but they won't be able to tell you where the problem is or what the grammatical problem is and so looking at in terms of grammatical is called the descriptive rule and prescriptive how they can do it or how they are able to tell there is a problem is because they use something called prescriptive rules of how a **language** should be studied or our **language** should follow so basically then this is how the structure of a **language** is it starts with something called phonemes goes up to something called morphemes then there is the level of the word then there is from the word there is something called the **syntax** which is the meaning of a **language** which is the structure of a **language** from the **syntax** there is a meaning generated which is studied through something called **semantics** and **semantics** is basically the meaning of a **language** but even if a **language** or even if a sentence has meaning it may not be the best way to communicate and so ways of communication is called **pragmatics** and from this **pragmatics** the certain **grammar** follows us how the **language** is looked or made undo now let's look at one by one these structures so the first structure that we discuss is called the phones or the **phonology** now if you look at German and if you look at French and if you look at English you will see that most of them are different languages why German is a very hard **language** for example somebody speaking German will show I'll sure I'm shooting them or something like very in a very miss rein of readers in which are very strong Germans are known to be very strong **language** if you look at French which is more of a subtle **language** more of a likeable **language** people will be speaking in terms some soft languages for example wait we are some other some other way of in which I don't know French French but it's a very soft **language** look at English it's more easier so hello hi kind of a thing so these the languages hawaii's German a hard **language** or a very strong **language** and French a very soft **language** or herbivore for example is a word in French which basically means that goodbye so or laughs ma laughing me for people for men and women so it's soft in nature and English is just gentle so how does this

differences arise the differences arises because the speech sound the ideal syncretic speech something these languages are different the basic sounds are very different and so fanatics is basically the study of speech sounds and how they are produced and **phonology** is the state of stretch the systematic ways of integrating these speech sounds and how altering this speech sounds for converting the **language** so basically **phonology** is the study of speech sounds of different languages and these speech sounds of how they're combined together how they are altered how they're spoken basically defines how a **language** is treated whether is a soft **language** hard **language** and so on and so forth now for the English **language** there are 40 phonetic signals there are four different take signals which are there and these phonetic segments are all the phonemes so what are phonemes these phonemes are at the speed song as we discussed in and the beginning of this chapter the idea of phonetic languages is in terms of **phoneme** systems of the phones which is the basic speech zone so phones are the basic smallest unit of a sound let make a meaningful difference for example ah II the visa sounds the Czech ones now in a given **language** now in one phone a **phoneme** of a word is exchanged for another and so I'll show you that how important they are because if I replace one **phoneme** with the other **phoneme** in a sentence then it could it would mean differently and so here is an example if I replace the D for name D funning with the T in a **language** then that leads to the duck becoming that back right and so you see the difference so if I replace one **phoneme** one speech sound or one basic sound in English **language** the D is replaced by a T the duck becomes a tuck and so duck has a meaning duck doesn't has all the top has a meaning in some way but then it's not the same thing and so these differences can arise and B and P are more or less nearby and they create something called **semantics** or they create something called a caustic confusion we have seen a caustic confusion has one reason of **encoding** in **semantic memory** and so this is what happens so this is a list of IP a list of different phonemes and so you can go to these so these are the number of vowels which are there and these are number of consonants which are there and these are the phonemes for it so if you look into the waffle these are the phonons for it this is the number of consonants and these other ponies for it and this is how the example is for example there are a as in cat the a as in arm the a in in a way and so on and so forth now the psychologists or psycho linguist they distinguish between a consonant and a vowel what is the difference between a consonant in the world now the English **language** in terms of the phonemes they have something called a consonant and they have something called a bubble now what is the difference between a consonant and a vowel no vowels generally they work without obstructing the airflow so when you produce a vowel the airflow is not obstructed in any way and simply depending on the shape the position of the tongue of Ovilus is is produced so there is no obstruction of the airflow for example when you say a e i o u it is the change on the mouth the change of the s' on the lips which is basically producing the sound and there is no obstruction of the airflow whereas in a consonants the consonants are phonemes which are made by closing at least one part of the mouth they different so basically consonants require obstruction of airflow and vowels does not require obstruction or flow for example when we say we see k-t all of them require that their which is pulling from the diagram up then which is coming from the diagram up it is blocked in some way in the mouth and this is what is the consonant is now the way a consonant is looked at the kind of obstruction of the airflow which is generated for producing a consonant can happen at four places now or it depends on four factors first the a place of articulation where is the obstruction so where is the obstruction of the airflow occurring for example B and P sound appears by closing the lips try saying me with open letter is not possible so we that's happens so when we save me or P it happens where obstructing the airflow by closing the lips similarly the s and the Z sound made by placing the tongue against the hard palate so say s without pushing the your tongue towards the upper part of the hard palate towards this part of the mouth of this part of the Mott say s3s without putting your tongue here not happening the not happening and so this is what it is and so putting the tongue or putting the your tongue at the roof of the of the mouth of the hard palate is how the snz sounds are produced the other way or the other factor is the manner of articulation

what is the mechanism for obstructing the airflow for example the word M sounds by closing the mouth while opening the nasal cavity so say m and don't close the Masri M not happening or try not using the result cavity now not happening so that is the problem here similarly the sound obstruction of the air for producing hissing sound so for example F F is produced F is produced by this hissing sound or this this mechanism of our flow airflow or the mechanism of the hissing the after sound which is there it produces the F the third factor which differentiates a consonant or which happen which makes it possible for a consonant to occur is called the voicing so what is voicing voicing is vibration of the vocal how does the vibration of the vocal chord happens now example the word s doesn't require the vocal cord to or touch a vocal cord and look into it if I say s there is no vibration but if I say Z there is a vibration out there and so these vibration of the vocal cord are also the ways of producing a particular vowel and so these are the ways in which a vowel consonant is produced and I said the basic differences in vowels there is no obstruction of airflow whereas in a consonant there is obstruction of our flow whether it is the level of the mechanism of our flow or the type of smart movement which makes their flow or the kind of vibration of the vocal cord and as you see this is basically the mod positions for producing the vowel and consonant anything else it's a funny thing yeah so to to show how ovals and consonants are different now features of phonemes are involved in certain phonological rules are governed the ways in which phonemes are combined and so these ways in which these phonemes are combined they give rise to certain foreign illogical rules now there are - there are some rules which are followed in English **language** for example if two true consonants are in the beginning of an English **language** if an English **language** or if a word in an english **language** starts with two true consonants in the first must be an S and this is one rule which is out there and this rule makes the top or Mukesh not possible right and so if we have two consonants B and K in in terms of the start of sentence or starter forward in the English **language** the first the rule says that true true consonants first has to be an S and that is how it is and that is why you don't make the top but you make stop or makes-- or skis as being the legal words now these combinations are phonological rules of how phones are there it gives to certain rules and one of the rule is the use of the to concentrate rule and that that is this is how to pronounce or pronounce approval happens in English also so basically these phones these phonetic rules also tell you how pronunciation of rules happens in English **language** for example if a word ends with the Z the S the Z the C or s or J or Z then the puerile are always in the pen in the terms of a Z for example place the pluralist places porch its porches right but if the word ending is in terms of P TK and F the plural is always in terms of and s and so these are certain rules which have been given there are certain rules which are out there which says how these spawns that we talked about or how this world wobbles and consonant that we talked about how they are allowed to mix together and follower or a make up a **morpheme** or make up a word right so if we have this PT k the ending is always Z in terms of s for example lip will not have libs it will have lips as an ending because it is ending in a P or if it in terms are telegraph it is ending in a T the plural is always in terms of a Telegraph's but if anything else beside these number which I have shown here the plural will always be in terms of said so basically these are the special ones which are there so that will give to an S ending other than that you would have the Z ending for example club it will be clubs heard heard kind of a thing so why are different **language** sounds are different why are different languages they sound different to us first they contain different sounds or different phonemes as a German has different for name example in German you have ghee or ABCDE said a gay kind of a thing or you have things like we and fear the Fatah C&W is is what is looked into and so different sounds out there similarly you have French Spanish and they each one of them has a different sound and so they appear differently they have different rules of combining these sounds for example fatter and whattcha whattcha and fatter in in German so we is pronounces V in in this case and so this is pronounced s F so when we say Father they put fatter into how it is or Schwester and in this sense for sister so basically different sounds are there and so this different sounds and this combination of different sounds are the two reasons why a german sounds different from the english **language** right so

different folds and different ways of combining or any **language** for example it could be french it could be Spanish it could be Chinese Mandarin any **language** you look into there are certain rules of combining these there are set first of all there are certain speech sounds and the certain ways or rules of command in the speech zones which make them which makes two languages different the next interesting thing to look into or the next part of a sentence or next part of a **language** is **syntax** so what is **syntax** it's an arrangement of words within a sentence how words are arranged into a particular sentence or the structure of a sentence now what the **syntax** do it tells you the ways in which different words or larger phrase can be combined to from leaguers and tenses in a **language** for example what **syntax** does it tells you how word should be arranged in in a sentence and what is the arrangement of words and then it tells you what is legal and what is not what arrangement produces legal sentences arrangement doesn't produce legal arrangement and for example the SVO subject/object were kind of a thing subject word object and sorry SVO format or the subject-verb-object time format which basically tells you what should come before that so our subject should be always followed by your work in English and then an object right and so in English Islam is going to the villages okay but in Hindi rata hey or theorem is also correct and so I can make these transpositions I can make this movement but then this is this allowance of movements in terms of German also that this allowance of movements is there for certain parts of the sentence can move but in terms of village goes to ROM is not correct why because this movement is not allowed and though this movement or this kind of changes is what's in taxes in texts are which says what in which way the words could be combined to form meaningful sentences now to explain what are the structure of a sentence mean consider the following for example look at this intense the top change the squirrel in the path now if I look into it it is in the correct sentence interest meanings now if I break this sentence this is the level of the sentence and the level of sentence if I break it it has two different parts North could refer to the first figure which I showed you it has something called a noun phrase and a verb phrase and within the non phrase then you have certain other structures which are there there is something called a determinant and there is something called a first noun and though is called the determinant which is determining the noun which is the dog so the dog which dog that particular dog if I just say dog then it is not referring to it the determinant is there and so dog could be anything but then I say third out chase the squirrel which means that I am referring to a particular dog similarly in the word phrase I have then a different arrangement altogether I have the word which connects so it is SVO format as you look into it this is the subject this is the verb and this is the object which is then so object a certain data - into it we'll come back to that in a minute then this is my subject this is my work so what does the dog do the word dog chases it is chasing down that is the action that this particular verb is doing and then I have another thing which is called the object here so the squirrel nos Y determinant here the determinant is because it is squaring it is chasing a particular square now doc choose the squirrel which means the dog just can chase any square but if I am looking referring to a particular dog referring to a particular squirrel then I have to go in with the Data Miner for it and so the dog chases the squirrel this could be a sentence but senses in this sentence has more parts of it it has something called this part which is the noun phrase here the part where does it change the square it changes the square in the park and so this is a advanced combination here or it's another part of the sentence and so this is this particular thing of breaking a sentence into its parts of the non present work phrase and within the noun phrase for how the verb and the noun acts in into or how they are divided into is what is called the the **semantics** I'm sorry the **syntax** now the diagram shows a label diagram tree and depicts what is called the categorical constituents of a sentence so **syntax** is basically what words are combined in how they are combined and so this basically what we saw is how our **syntax** actually looks into now these changes explain why certain changes are though this kind of categorical arrangement explains why certain and certain movements of sentences are allowable and certain movements are not allowable and this taking certain part of a sentence for example and this part of the cell for example I can very well take this and put it here and make this change from here to here so this

the squirrel chases the dog in the park is perfectly correct syntactical sentence although it has the wrong **semantics** in terms of meaning it is wrong but perfectly okay so I can I can take a non phrase and move it along along this structure along this constraint structure and that is called proposing so that is what with which so this category is a caracal arrangement this is a tactical arrangement of Constituent arrangement basically helps you in proposing taking certain parts of a sentence and moving it in front of the others and that it also tells why I make the square I'll move from here or the part move from here but I cannot take the chaise and move it into the front of a sentence so what can be moved and what cannot be moved or how can changes for example I can very well see that the park chases the squirrel in the dog right and as perfectly legal in terms of the **syntax** but in **semantics** it has no meaning at all and so this is what is allowed and this is not what is a lot because **syntax** threw in the exact meaning and so this constraint structure this tree structure then tells me what can be moved and what vendor cannot be and this movement of part of a sentence toward the front or back of a sentence is called proposing for example look at the two sentences so my naughty dog I am mad at is not legal whereas naughty dog I am mad at my is not legal so my naughty dog I am mad at is perfectly legal but if I try to propose something out of here what I want to take this and put it here then it is not legal at all similarly that inflated price I will not pay is perfectly legal but if I try to prop propose something for example you try to propose price price I will not pay that inflated is not legal the inflated price I'll not pay is perfectly legal and so this kind of constraint structure or tree structure tells you or tells someone using a **language** of what is possible and what is not possible now how do we concisely summarize what can and what can't be legal proposed structure or what can be the way of making a legal sentence and so one of the ways is that one of the rule is that only whole phrases can be moved you cannot move part of the sentences or part of a particular sentence from one position to another also there is something called a phase structure rule which says that or rewrite describes the way in which certain symbols can be rewritten so what happens here is that it says that certain side symbols can be related in a certain way and so if it moves something with it or certain I can rewrite a particular symbol into some other symbol and that way I can make the preposition I can make the movement similarly there is something called a transformation rule which says that structure such as those depicted in tree diagrams on to other structure example proposing and so this rule says that some structures can be moved and some structures cannot be and what can be transform what cannot be transform so there are certain rules out there for example one rule is that the only whole phrases can be moved and then the other is a face structure rule we says that certain structures in a sentence can again be rewritten or broken down into other structures and that decides what can good or not and there is a transformation rule which says that what can be moved in what cannot be moved and that says how sentence can make meaning or how can a legal sentence be formed and so as you look into it mr. Park II had a problem with his sentence and that can create problems for example it says work for will work as food not as food will basically mean that he'll be eaten up right and so there is a problem he says will work for food could be the right thing now the next part of what we look into is called the **semantics** so we looked at **syntax** in the rule of it and **semantics** is making meaning so **semantics** is basically the study of meaning and plays an important role in **language** any any **language** a meaning has to be there if the meaning is not there then there is a problem so although there is no complete theory of meaning as as of now and there are there is attempt to make a universal theory of meaning which color which captures everything but then births which in 1970 he says that any theory of meaning should have or should be able to explain several things at a minimum so any theory of meaning any semantic rule should be able to explain certain kind of deviations or certain kind of factors which is there first of all any theory of meaning should be able to point out anomaly any theory of meaning should be able to tell why and what kind of anomaly is prescribed in a certain **language** for example why can't rice a coffee I streams can take dictation why is this an anomaly the reason is any theory of meaning any semantic rule should be able to point out why is this not a legal sentence coffee ice cream can conduct a dictation right so basically

because it doesn't make meaning because coffee ice creams are not human says okay not dictation in the very idea of dictation is that it is something which which is moving which can hear a conversation and these cannot take any motion so this is an anomaly similarly self-contradiction any theory of meaning any semantic rule should also be able to point out self-contradiction in sentences for example when I say my dog is not an animal why it is self-contradictory because it is an animal right when I say it is my my dog is so and so people try to give them name this is basically self contradictory because most dogs are animal and so you cannot create them or say that this is not an animal also things like ambiguity why isn't it clear when where one intends to go when I say I need to go to the bank so same theories of meaning should also be able to point out this kind of problem this kind of ambiguity when I say I'm going to the bank it's not very clear when I am going on to the unless the context is in before it or after it for example I am trying to tell you what a situation or about my visit to some to to some water body or some ocean or some nice place and then I'm trying to say that I am going to the bank then it has some meaning so basically then any theory of meaning should be able to explain these kind of ambiguities also cinah limit is and any theory of meaning should also be able to explain why is anonymity or house anonymity exist for example when I say the rabbit is not an old enough how does it mean the same as rabbit is too young when I say not old enough how does it mean or what does it mean or what is how is it translated to the fact that it is too young so any theory of meaning should be able to also comprehend that or should also be able to explain how does this happen and the last is called the end tailmon which is when I say that that is my uncle how does it automatically generate this fact that pat is a male because uncle is a male and so pat is a male that kind of rule so any theory of meaning should be able to do that now when listeners they figure out meaning of a sentence they need to pay **attention** to more than just the meaning of individual words if you don't do that what will happen is the meaning would change so although the sentence would have the same **semantics** you will have the correct format but meanings would change and if you look into these two sentences the profits are failed the student then the student failed the professor they have entirely too different meaning into it and so any theory of meaning should be able to explain why this happened because when I say the professor failed the student it is basically failing in an exam but when I say the student failed the professor is basically not failing the professor in an exam but basically because that's not possible and that's the theory of meaning it says that this is not possible so the students don't take an examine so this is the how the world runs and so when the student failed the exam it is basically failing him in terms of certain values in terms of certain facts certain trust that the professor had so he failed the student failed the professor in terms of trust whereas when the professor failed the student he failed in terms of and academic value or an academic exam now the study of **semantics** also involves the study of truth condition so for any sentence to be truth when I'm doing a sentence when I am looking at a sentence the the theory of or the study of **semantics** or the means study of **semantics** rules should also be able to point out whether a sentence is true or not because truth conditions often decide whether a sentence exists or of what is the meaning of a sentence and the relationship between different sentences that it has so truth conditions are simply circumstances that make something true for example the dog chasing the cat is a two-sentence but the cat chase the dog is not in terms of truth conditions is not valid because cats don't chase dog in normal meaning and so any idea of meaning or any sentence or **semantics** or any sentence which has sentences like the cat chase the dog in entirely wrong although this kind of meaning can generate in an animation movie but normally seeing the dog will only chase the cat and so this is the truth condition hence our understanding of meaning of our sentences requires some steps the and RR phase or how we understand meaningful sentences they are bound by certain variables certain factors and understanding of the meaning of each word or sentence so how you make meaning from a sentence or how you make meaning from a word a comprehension or text passage is basically by understanding the meaning of each word that it has remember the professor Felder stood in this room when the professor so looking at the word looking at meaning from each word you can then say that profits are failing the student is an exam in an exam but

student feeling the professor is in terms of trust so there it is academic there it is trust so our understanding of meaning it relies on understanding the **syntax** of the sentence so whether the **syntax** is correct or not whether the grammatical rule is correct or not and understanding the truth conditions so how do we generate meaning we and generate meaning by understanding each word from or understanding the meaning of each word a sentence has next thing that we do is we look at the **syntax** and we understand the **syntax** if the **syntax** is wrong then the meaning will not be generated out of it because the sentence will not be correct and the third thing is we also look at whether in terms of the world view whether the sentence is true or not so the dog chase the cat chase the dog each word has a meaning the dog has a meaning the cat has a meaning the chase has a meaning also the **syntax** is correct because there is a subject verb and object but in terms of truth conditions the fact that normally seeing the cats cannot chase dogs or do not chase dogs that basically established the fact that it is in terms of truth conditions it is false no we looked at **semantics** which is the meaning of a sentence the next thing that we are going to look at is **pragmatics** which is basically the social rules of how **language** should be spoken or what are the ethic Cates and kind of conversations which are out there so basically in **pragmatics** we we look at the sentences what is the way what is the social way of conversing what is the require adequate switch chef so basically for example good **pragmatics** is saying conversational greetings like saying hi how are you somebody says you have to reply to him in a certain way so if somebody say hi how are you and you say something which is very rude that's not a good **pragmatics** so **pragmatics** are certain rules of **language** or certain rules of conversation so although your sentence would make meaning but how they should be converts all people should be conversing with the sentences that you're making is what is the study of **pragmatics** so suddenly 1979 he points out then in listening to another person we must understand the kinds of that they demand in different responses to us so ha Saturday says that there are certain sentences are the sentences which have been uttered which are uttered by people and all these sentences have different kind of responses and we should as a listener not only be able to understand the kind of sentence that a person is speaking we should also be able to give a reply befitting the kind of sentence which has been asked for example one of the ways or one of the particular ways of communication is called the assertive sentence in a certain sentence the speaker his or her he asserts his belief for example was like it's hot in here I am a Gemini it doesn't require any kind of response so if somebody is just asserting something or saying something now there are directives which people go ahead and speak and these directories are instruction and so you should be replying in similar to it for example the speaker saying to a listener saying close the door or don't trust him when I say close the door it requires an action from the listener and so that is the kind of thing you choose and so you do not need to speak anything when somebody says close the door you close the door and as I don't trust him you don't say anything to it you basically have to listen to it and act upon it then there is something called calm uses these are utterances that commit the speaker to some later action for example I promise to be good now it basically says that the speaker is promising or it's basically he's telling the the listener that in future he is going to do something for example being good or somebody says I will be a wing man which means that this person is going to help the other person and so this requires certain other kind of responses then the directives in an expressive described psychological stage for example in an expressive you don't really know what to do so you have to read to it example I say that I thank you for the favour you did to me this is an expression sentence this is a expression of thanks now I don't know what you could be doing with it but this is an expressive or example if I say thank you for nothing this is an a psychological state of the speaker and so you should be responding in a similar way as against a directive or a commerce sentence and the lastest declarations are speech acts in which the utterances itself are the actions now in this cases the sentences which are speak spoken are itself certain actions for example I now pronounce you husband and wife this basically says that there are certain action into it the act itself the fact itself that my pronouncing makes you husband and wife is true or sentences like you are so dead are basically and are

trends it's a declaration and the sentence itself declare a particular fact on to it and so people should be hanging around to it now according to sadly speech act theory part of a job of listener is to figure out which of the five particular utterances are there and to respond heartily so basically it's not only understanding the meaning the structure of a sentence and so on and so forth it's basically a we go a level ahead of it and then we also look at how people talk to each other and how do they generate sentences for example best example of **pragmatics** like look at the two figures so in the first case Sherlock saw the man using binoculars which means that he used the binoculars for seeing the man and in the other case Sherlock showed the man using the minor colours in the second case he saw the man who was using a by Nicholas and so this is the difference between **pragmatics** or this is the difference between house and consensus so both are the same sentence same legal constraints same structure same meaning but the **pragmatics** of to it or that's not what I meant when I asked you to carve it so what he has done is he has carved something out of it and so the subject meaning so basically the idea of how a sentence should be really constructed so in this class in this particular lecture what I tried to do is I try to show to you what is **language** what is the meaning of **language** what are the different constraints of a **language** and basically then we went ahead and looked at each constituent one by one and how do they play their role into the **language** itself or the working of the **language** itself now in the upcoming section we look into how speech is produced when a speech is how people utter languages and what are the confusions and problems in our turns of these languages so in the first part we looked at parts of and how they're combined together to form the **language** itself the second part so it is in terms of rules and structures and so on and so forth in the last in the next part that we'll talk we'll talk about errors in production of speech or how speech is per use in itself and several factors related to that so let's meet the next stuff in discuss this thank you

Lecture 22: Comprehension and Understanding of Language

■Kind: captions hello friends welcome back to this second section on **language** in the first section of **language** we saw what is **language** we saw how it differentiates from communication and we saw several parts of **language** how it is made up of what is a **syntax** what is for name what is a **morpheme** and then what is cement takes what is **pragmatics** so basically that comprises of order languages and then at the end of it **grammar** so basically what **language** is what it is composed of how these things the rules of **grammar** really work into **language** and so on and so forth now in this section will focus on to something called understanding comprehension of **language** and the production of **language** so first section was describing things about the nature of **language** and the second section is about producing **language** and understanding comprehending **language** so basically let's start this session and try and understand how **language** is comprehended how **language** is understood or how **language** is produced now **language** like any other information which is being transferred from one person to another is basically first taken in as a raw input and its transformed so the basic components of **language** are first perceived through a **language** processing unit in the brain and then it is the raw materials or the raw parts of the **language** are then processed by this **language** processing unit to form meaningful representations where these representations then mean something or has some semantic value now the steps which are followed in taking the raw **language** input and then processing it so that a meaningful representation as meaningful representation is created what are the steps into it let's look into those so the question is how does **speech perception** happen how do you perceive speech now as opposed to **perception** of words which we do one word at a time one by one so how does word conception happen or **perception** of letters and numbers happen so basically what we do is we do some kind of fixation and we do some kind of jumps and these fixation and jumps happen from the first word to the first letter of the word to the last letter of the word then there is a gap and within that gap the meaning of the word is is evaluated and then we go to the next word and similarly in terms of **speech perception** since **perception** of words happen this way in individual materials happens in the way that I just described so it's basically believed that **speech perception** should also follow something like this and the simplest way to look at house pieces **perception** would happen is to be like text **perception** that is one sound at a time should be perceived and the pauses between the spoken word to spoken words should be used to identify the meaning of the words this is how our general knowledge should be because text **perception** basically takes place I send you it takes place in terms of a word my word **perception** so the first letter of the word and last letter of the word is basically looked into this is called jumping or fixations which the eye - and based on that there is a gap and if there is no gap then **perception** of letters become difficult but this gap between letters actually help the brain to interpret the meaning of the word and so this goes on and this is how the meaning is generated so in **speech perception** also it should be we assume that you should be similar that a word by word is perceived each word that somebody is speaking now when I am speaking to you how the **perception** should work in this way that you should be perceiving each word and the gap in my speaking the time gap between two words when I speak that should be utilized for making meaning out of it that's how we believe it is but that's not how actually it really works so milna 1990 he described two problems with such a theory which says that word by word **perception** happens and meaning is generated from those kind of words and what is these two problems first Miller he found out that speech was continuous in nature the nature of speech was continuous and so how does it happen for example refer to the spectrogram so I what I'll show you I'll show you spectrogram so what is the spectrogram actually this is an output of speech so once once you produce speech the there is a air pressure so just in front of the mouth so one once the words come out of your mouth a certain air pressure is created and this air pressure is measured through a instrument and it is displayed in through a spectrograph through a CRT tube or a spectrograph and that's how what is a basically a spectrogram is so the amount of pressure which is

generated when you speak certain words against when you speak nothing is presented on an XY coordinate in terms of voltages in in in in terms of sounds n in terms of sound in Hertz and on the time axis so that's how a spectrogram really works so effective Graham is basically these pressures that I omit when speaking a word and the graph of it which is in terms of sounds in the produce in terms of Hertz and the width through a particular time so I'm just going to show you a spectrogram of the word that I am written down here so they are buying some bread let's look at this and if we find all this stops or obvious gaps in the spectrogram it would mean that speech is not continuous and so each word after each word there is a gap so let's look into it so spectrograms this is a spectrogram as you can look at and so when you look at the spectrogram it is one contained the omission of an continuous graph which is out there so on this axis we have the time and on this axis we have the measurement of sound in terms of Hertz it's so air pressure basically in terms of Hertz and so what is the result of this the result of this is that spectrograms indicate it rapidly there are pauses around each sound so once the sound is produced there are no pauses between them and rather different sounds from the same words blend into each other so what happens is when I say they are brining the bread what really happens is the end of the and the beginning of our are blended together and so there is no graph and that is what Milner pointed out he said that speech is continuous so assuming that speech is like **perception** of text is not correct one reason being that you discuss I just showed you that this sentence they are buying some bread in terms of a spectrogram how it will look like and we see that the word endings they sort of merge into each other and so there are very few chances or there are no chances for gaps between them which which negates the idea that people get timed to between spoken **language** to make meaning order now the second problem which Miller listed is the fact that a single **phoneme** when if it is changed into a spoken text the context changes for example if I remove if a toe a doe and auto can be made these words can be made by replacing the TD no.4 name so a single **phoneme** have taken out or if it is replaced or if it is misheard in certain way what would happen is the word would itself change and so one single phone a one single speech sound and **phoneme** is a very basic idea so one thing simple problem with hearing this one on would actually lead to the interpretation or would actually lead to different interpretation altogether and so what I have here is that this same for you so you have this is for toe this is for Doh and this is for auto and as you can see here that this is the time axis in terms of time as you see it's 200 400 millisecond and this is the pressure air pressure so in terms of sound pressure in terms of Hertz or cycles per second and so what you see is that the same kind of spectrograph you can see for both no.2 and auto but only the beginning changes so replacing a **phoneme** for example T replaced by a D or o the beginning of this changes but more or less it is the same right and so that's what it is one of fun for **phoneme** changes the whole idea of the word or whole meaning of the word changes no these are the two facts why it the Y sound **perception** cannot be similar to **speech perception** now another problem which is to be noted is that men and women they speak in different frequencies different accents and across different contexts they people speak in different ways so it's not about gender it's not about how women and men speak together it is also dependent on the kind of frequency so some people speak very softly some people speak a little bit loud the kind of accents that people use so it could be a raw accent it could be a toned up accent or any reason at different situations for examples in a lecture you could be whispering in some other fora when you're lecturing it could be a monotonous tone and so on and so forth so across context across different different situations people both men and women use different kind of you know speaking or speak differently and so this could be another problem which could lead to the fabulous **speech perception** is not like text **perception** so then if it is not like text **perception** how is speech perceived then what is the way in which pitch **perception** happen and the answer to this is that **speech perception** is generally categorical in nature and that's what it is the answer is that we come specially a cube with pursed systems which perceives speech in an efficient way in a very efficient way and that is called **categorical processing** so what happens here is that certain categories are automatically made and so when we speak the speech is automatically without awareness

pushed into these categories and at how we actually go ahead and process speech we pay **attention** to certain acoustic properties of the speech and ignore certain others so based on the caustic properties we can tell whether a person is male or female remember experiments from the **attention** section where we saw that from the unshadowed ear people were able to tell things like the tone of wise and the gender of the person who's speaking and so these things are automatically **attention** is paid onto it automatically and so people were able to tell from the Nanshan down here also so basically in **categorical processing** what really happens is there are certain categories and these within these categories when you when a person is speaking automatically we perceive or we force the spoken speech into these fixed categories and this happens without awareness we are not really aware of the fact that we are doing it so we are not perceiving it in a continuous manner we are not perceiving it in the breakdown manner but what we do is certain categories arise so what is a noun what is the work kind of categories are different different categories or what could be the constraint category and so we look into speech or with the hear speech and we take parts of speech and they we then go ahead and then push it these parts into different different categories out there and that happens without of any kind of awareness we are the kind of explicity into it so **perception** of speech is also affected by something called visual cues and these are known as context effect so certain kind of visual cues in the environment they also affect kind of **perception** of speech that we do know the an experiment was done to show that these different things that context affects **perception** of speech and so Warren what it did was he presented a statement like this to people now the statement was the state governess meant with their respective it should actually be legislators right and so that is what it is about where the word ledge is the word s would the fundamental come he produced a coughing sound so the statement transcends the state government met with their respective legislatures concerning in the covering convening in the capital city so this is what the basic sentence was and so what he did was in the world legislature where this **phoneme** would be a cough sound was produced a light cough salsa produced and it was amazingly people were not able to identify the cuff sound only one out of twenty people actually they reported hearing the cuff sound nineteen people didn't didn't actually report the cough sound and they could hear the word legislature very nicely nicely and the honam s here legislature so funny mess and this basically is what is called the **phoneme restoration** effect so in which one hundred twenty so that is what it is so this sentence of presented and 120 milli second portion was replaced the this was 120 milli second portion where a **phoneme** was there and a cuffing sound was induced so one after twenty percent listening to this actually found out that there was a missing **phoneme** and most of the 19 people restored this one him and that is called the **phoneme** distortion effect and that demonstrates that the the percept the **perception** that we actually do is a categorical in nature so we don't look for look through word to words we actually force implicitly or without awareness we force part of sentences into preformed categories and that is our **perception** of speech really happens so the answer to this or the result of this experiment were that people were capable of using a great deal of information to predict what the correct sound of the missing segment was and as I said that people were able to restore this phone him they didn't notice this break which was there and because they were doing **categorical processing** they were not even aware of it that this been replaced by a cough sound now Warren Warren in 70 they demonstrated this by presenting people with one of the four sentences so they're people represented people with these four sentences now each one was the same recording with the exception of the final word has been spliced off and each contained a missing segment as indicated in the Asterix on so basically what Warren did Lauren wanted first to find out whether people are able to restore this one him or not or perform this **phoneme** restore restoration which basically means that are people capable of implicitly filling up gaps into spoken speech to demonstrate that what Warren Wan did was he presented four different sentences to people with one phoning being spliced off and what he wanted to see was whether people restored these phonemes or not so there were four sentences present to people with a missing segment and each of this in each of this the missing segment is the key to making meaning of the

sentence so what were the sentences the first sentences was it was found that eel was on the exel the second was it was found that star eel was on the shoe and so this star which you see here is basically the missing phone him here and so I'm pretty sure by now you would have come to know the answer for all of these and so this is what was replacing in in worst case this eel was an Lda to an excel in the second case this eel was related to a shoe in the third case it was related to an orange and the fourth case it was related to a table now take a second and tell me what the eel should actually be and I'm pretty sure that you would have come up with an answer so in the first case when the neck seal was presented the context help people to replenish back or to fit in the right **phoneme** and so people perceived really there but in in the second case where it was the shoe people didn't say wheel but people perceive the EE L the the the part where the **phoneme** was placed off as heel not wheel the wheel was for axle because the axle is the context and so wheel was replaced in if shoe is the context then what was replaced was heel but in the third case when orange was the context when orange was what was being referred to the missing portion became peel whereas in the fourth case when table was what was the context or what was the whole sentence was referring to it was meal correct and that's what it is and so depending on the sentence participants reported hearing wheel heel peel and meal so basically they restored this morning's of WH the H H in heel the P in peel and the M in meal this kind of thing is that which basically again proves the hypothesis that people perceive speech sound automatically and implicitly and how they do it they do it through something called **categorical processing** right as a **speech perception** is basically not like text **perception** people don't wait over speech segments and the gap between CC segments are not utilized for making meaning rather the **perception** happen in the categorical automatic manner now with the **perception** of speech or speech itself there are certain errors which are there when we produce speech also so certain areas can also arise when production of speech happens so so beside perceiving speech from others we also produce speech and for others to comprehend the and process the speech fear so it's not only always listening we all sometimes also talk so we produce speech and so this is called speech production now speech production can need to switch production itself can lead to certain kind of errors which are called **speech errors** interactions in which the speaker intends to say quite clearly where the speaker makes some solutions substitutions or reorders the stimulus in a certain way so **speech errors** then are certain substitutions are certain reordering of the statement which makes the sentence not legible and so these are happening this happens because we do some kind of a reordering in sentence productions or we do some kind of a substitution in sentence production and those are called **speech errors** for example Mary keeps the food in her West now when we say whisk actually this is wrong and so what are happening so you would see these kind of things happening in smaller children's and so what have happened here obviously you can see is the desk the DS replaced by the phone in V and so what has happened is Mary keeps wood in her desk so instead of the desk the production itself it becomes reversed and it happens to most of us we do these kind of problems another is another sentence which shows this kind of speech production error is this one where we sometimes make this kind of errors also where what we have done is it's exchange of words sometimes we replace phonemes are sometimes the exchanged words also so look at the sentence we sit around the song and sing fire and so it happens to most of us you'll realize that people who can speak English they can have these kind of errors in to it and so here what has happened is the song is replaced by fire so the actual sentence should be we sit around the fire and sing songs and so out of some substitution in speech production what has happened his fire goes to the end of the sentence song comes to the middle of the sentence and our sentence becomes we sit around the song and sing fires which has which is syntactically correct it is semantically correct but makes no meaning at any indian aid so in terms of **pragmatics** make no meanings so in terms of **grammar** rules is perfectly correct but it has no meaning and so this is an error of speech production and this error arises because we substitute certain words for certain other words similarly Gerrit 1990 88 while studying such these errors found that the word substitution has two broad categories of errors so this

substitution of words this exchange of words can then also be divided into two different classes two huge classes out there and what are these classes one of what substitution can error have never solution error can happen in terms of meaning relations so in terms of meaning relations in the sense that I can substitute a word by another word which is equivalent to the meaning of the first word so here replacing finger or walk with a run is what meaning relations are there so finger and toe are more or less like similar in meanings of what finger does is what a toe would does and walk and run are more or less same in meaning similar in meaning exactly not the same but then similar meaning and so people could actually do this kind of word substitution error and these word solution substitution errors are meaning related errors or people can go ahead and form form relation errors so form relations errors are generally very fewer it is very few that you see these kind of errors and so what you do here is instead of a guest you would say good so the guest has come the goat has come right or mushroom is replaced by mustache so let's go to the feeling and pick up fresh mustache instead of the mushroom so form relation errors can also exist and so Garrett says that these are two kinds of **speech errors** or speech related errors that can exist and so these speech related errors can also create or are also categorized into form relations and meaning relations okay so this is up till now we have seen that in the production of speech also so not only that writing **language** or written **language** has some kind of errors while producing speech also we create a number of errors and so what we saw up till now is how speech is produced and this production of categorical speech the production of speech itself which has which is categorical in nature this can also lead to certain kind of errors which the producer of the speech is making and which confuses the listener and so he would be hearing something else right and so the question now is how do we comprehend the sentence right so when somebody writes a sentence how do we go ahead and read this sentence so how do people understand and recover their meaning from sentences so given the fact that a certain sentence is written in fact in front of you how do you make meaning from such a sentence the answer to this lies to the fact that people pay **attention** to syntactic constraints however they do do not process sentence clause by clause so what people generally do is that they look at a sentence and they look at parts of the sentence or they look at the syntactic constraint of a sentence right but then they do not process the sentence Clause by Clause so what really did they do they actually look at a sentence look at the Sun send the constituent of a sentence make a gist out of it and that is how they process a sentence they actually don't read the sentence cross by clause and that is the way in which people go ahead and process sentences so what after reading a sentence what people do is they discard most of the words in the sentence and just keep the meaning the central representation or the central meaning in terms of a gist representation so comprehending a sentence involves dissolving it's possible ambiguities so in sentence comprehension also certain kind of errors can occur and so it has been found that people are very good in resolving ambiguities if a sentence and we saw that in the first section also that even if the sentence is a certain problem people they are able to resolve the ambiguity for example the first sentence that we looked at the very yellow dog chase something something we saw that people are able to people all they don't know the descriptive rules of **grammar** of how to correct the sentence but they know the prescriptive rules and they know how to correct this so in terms of sentence comprehension also people are very good in resolving these ambiguities now in the interesting thing here is that we normally don't notice ambiguities and only rarely we come to notice them when do we know ambiguities only when certain kind of sentences arise or a certain kind of certain parts of the sentence have some confusions in to us now look at this sentence the sentence says the horserace past the bond fell now if you look into the sentence what happens is initially the sentence is alright the horserace past the bank till the point that you come to fell and fell is ambiguous and these kind of sentences are called **garden path sentence** so you it leads you to a certain direction of meaning but then the last word creates a problem and so what is the problem so how do we go ahead and solve this what is the this is obviously an ambiguity so how do you solve this one way to solve it is the horse weight raced past and put a comma here the bond fell two parts of the sentence two

sentences to look at na makes meaning or the horse raced past the bond fell can be broken not into this two constants and so till the point of time that you're reading the horse is past the bond it is okay but a fell the word fell itself is a problem and so here is the problem so you can solve it by this way and this type of sentences actually lead to ambiguities or look at this now this sentence is the cotton shirts are made from comes from Arizona not from from the beginning of the sentence if you read there is a but the pertinent meaning that you see to it but at a point a certain point certain water arises which creates confusion for example the cotton shirts are made from comes from Arizona now there is a problem here the cotton shirts are made from expects certain input as some form of it alright what fine form of cotton I am destroying that kind of cotton qualifying the cotton example so simplifying this sentence so basically if you want to make meaning of the sentence and so the here is where so we will decide the cotton shirts are made from made from comes from Arizona is where the problem really comes in and so you want to solve this sentence the ambiguity ambiguity is this part of the sentence I just want to solve the ambiguity what we really do is cotton shirts the if we put a comma here the cotton shirts are made from comes from Arizona then the sentence losses is an ambiguity and becomes a more meaningful sentence but reading from from on words in to say in this way creates an ambiguity and so this kind of sentences creates ambiguity but with normal ambiguity because what happens is the people have already started reading the sentence and so it has made people started thinking in a particular direction but then from the from on words or from the ban on words the next word which appears creates so this kind of sentences make people that assume confusions or make people think about confusions and they need to then solve the ambiguity so these sentences are then called garden path sentences the reason being that it starts making you think in a certain way and then at some point of the sentence it pushes you into a different movie right so results from several studies they suggest that when we process ambiguous sentences all the meaning of the sentences are temporarily available so when we are looking at these kind of ambiguous sentences and M because sentences which has multiple meaning and the results from several experiments suggest that all meanings of the sentences are available for example if there is a sentence saying that this room means full of bugs now both the meaning of the bugs the bugs as in terms of cockroaches or pets or bugs in terms of listening devices are available to us depending on the context we then decide after three **phoneme** lengths which word meaning will be accurate and so results from several studies basically suggest that so basically what it suggests is that when we are reading an ambiguous sentence all the meaning of the ambiguous word the word which is creating ambiguity are temporarily available through an automatic **bottom-up process** so while reading a sentence every meaning remember the sentence where we use Bank in two different ways in in terms of financial institutions and in terms of riverfront now both the meaning are available if a sentence is this Bank is creating confusion in a sentence both the meanings would be available at the time of processing the sentence now quanta fix effects do not immediately restrict the listener to most appropriate reading of the word so basically when we are reading a sentence the various sentence is created the context in which a sentence is created it does not stop the reader from interpreting all the meanings of the ambiguous word right instead for a short period all meanings are accessible to them now three syllables after the presentation of the ambiguous word only one meaning remains active so what happens is the context doesn't decide what is the meaning of ambiguous word of the ambiguous word which is created in the word which is creating ambiguity in ambiguous sentences what really happens is three syllables after the processing of the ambiguous word all meanings are available only after these three syllables has elapsed the exact meaning or the meaning which befits the context then remains and all other meanings are discarded and so this such as has been done on several kind of sentences and so since this we are not dealing in detail in **language** so I'm not bringing in those examples which will show you how these experiments are done and how it was basically found out but this is the result of it so three syllables after the presentation of the ambiguous word only one meaning remains active which is the meaning which is more up most appropriate to those contexts suggesting that people

resolve sentence ambiguity fairly quickly so no matter how ambiguous the sentences people are able to resolve these ambiguities and how do they do this resolving of ambiguity the fact is we read the sentence to the point they come to the fact that the word which is creating ambiguity and when they are reading the word this is creating ambiguity context is the word the context of the sentence doesn't help them till three four names or three syllables after the presentation of the word all meanings are accessible but only after three syllable presentations which is a very very brief time only after that the appropriate meaning or the necessary meaning or the most optimal meaning which befits that sentence it remains and all of the meaning goes away and certain experiments are done to create that and so what does it actually tell you it tells you that people are able to read up and because sentence is very fast and resolve ambiguity very fast right so although all meanings are available but then people are very good in resolving ambiguities although I mean they may not know the **grammar** they may not know what is wrong with a sentence but they may that they can correct the sentence so they have this feeling they have this implicit feeling that is that is what we saw in the first lecture they said they have this idea of what is wrong but they may not know the **grammar** of where it is wrong okay so up till now we have looked at **speech perception** and sentence comprehension very good let's now look at what is the comprehension of a text message or how text messages are comprehended now for understanding how people process takes messages first we need to know how people actually need so what is the way in which people read and just in carpenter 1987 use something called an eye tracker to monitor the eye fixations for written text so what they did was they use something called an eye tracker SOI tracker actually tracks the movement of the eyeball as it move across written words and so from that they had some idea of how people actually me now results indicate that reading consists of a series of fixations and jump across text with average fixation and jumping 250 and and 10 to 20 20 to power minus 3 second the basically 250 to 10 to 20 20 millisecond kind of a timeframe so basically then what they found out from the study is that people just don't read text the way is with what happens is the eyeball tracking experiment suggests that people jump from word to word and they then have fixations so when they're looking at a word they look at the first word they fixated they looked at the last letter of the word they stopped there for some time make meaning out of it jump to the next first next letter or next word the first letter the next word last letter of the next word look at the context find meaning next makes the jumps and make another jump into it and so this kind of jumping and fixations how text is Percy now just in carpenters model assume that as soon as readers found encounter a new word they try to interpret it and assign it a rule and this is called the immediate immediacy assumption so in takes passages what just in carpenter says is that when we are reading something in these fixation happens between the first letter and the last letter of the word and then we wait there for certain time the is called a fixation time and then we jump to the next word which is there now during this period when we make this jump this fixation and jump during that period we immediately encounter when we encounter that what immediately gather the meaning of the word and that is called the immediacy assumption so as soon as we get a new word we try to interpret what the word is the author also formulated something called the imine hypothesis and through water is it which holds that interpretation of each word occurs during the time it is fixated so basically what happens is when reading a text there are certain jumps in the eye actually does and so if you are reading at words like w RDS and like these are the two things that we are looking now I will first fixate here stay here then look at the last word d s then there is a context to it since it is the **language** context in some words are related to context so this we are teaching this I'm teaching this in in a lecture on **language** and so the context is there and so this w word starting with w ending with s or ending with D would be a word that is how the immediacy of immediacy assumption is and so immediately that is what it is and the imine hypothesis is that as the fixation happens and I have before the jump this word is interpreted this word is matched with a segment or matched with a representation into the brain and a meaning is extracted out of it and so these are what it is immediacy says that an interpretation is as is done as soon as the fixation happens and a rule is assigned to it and the imine hypothesis says that

meaning is also extracted at the same point of fact so it is not only the role assigning to new word what is the role of that new word whether it is a word whether it is a noun whether it is a pronoun what kind what form it is doing a sentence or what is the role that is playing in a sentence that is called the immediacy assumption the i-mine hypothesis is that the meaning is also extracted as that very time it's a fixation time so fixation tells you what each word means what is the role it has in the sentence and what in hand and how to interpret this there's this role in the imine I possess it says that also the meaning of the word is extracted at point RAM and that is how we did so these through these fixations and jumps across words to words is how we actually read and that is how comprehension of text messages happen now justin carpenter they argued that a number of variables influenced fix a influences fixation duration so the time number the amount of time that the eye fixates on to a particular word is influenced by a number of variables and so what are these variables for example word length the bigger the word length the more the fixation has to be right also what frequency how frequently you see even this word so words like apricot words like hello where are was which people don't encounter in everyday life they require longer fixations then word like cat dog hand food that kind of words which are more common the flood frequency the usage of these words are more common and syntactically and semantically ambiguous word so if words are syntactically and semantically ambiguous for example bank two meanings of bank or bugs - meaning of the word bugs or multiple meaning of the word bug or the position of the word is such that it can also be an injective and it also be a noun that can happen so the word could qualify it could cruel mean too many things into it in those cases the fixations are long or the fixations are a larger duration and the jumps are then quicker not quicker it's takes more time to jump and the more the longer fixation because that determines how much time you have to spend into it why because then you have to first if we have words which are ambiguous we have to first assign a role to it because that tool will tell you what this word means and what this word how this what is driving the sentence right and so basically that so longer the word the more complex the word lesser the frequency more the fixation period siltation keenan 73 they found out the semantic factors influence the reading task there are certain semantic factors which also go ahead and influence text comprehension or reading tasks they showed that two sentences of equal length might be diff differentially difficult to process and so they said that the cementing the meaning how much meaning a sentence carries how much ideas that the sentence carries that will also decide how easily a sentence can be read so those sentences which has a number of ideas on to them they will be difficult to understand they will take more time to comprehend or comprehension of sentences which have multiple ideas embedded into them will take more time than sentences which have simple ideas and so they presented something like this so the word is Romulus the word is Romulus the legendary founder of Rome took the woman off of the Sabine by force and so there are two or three ideas into it and so you look into Romulus woman and by force are three ideas and then you have Sabine is a woman there these are two other things and so the number of ideas are limited here are three ideas and so it takes some time to process in comparison to that if you look into it it is the same length the second sentence is also the same length but here the number of ideas are more and so it takes more time to process and what is the sentence Cleopatra's downfall lay in a foolish trust in the fickle political figure or the Roman world and so if you look into it there are multiple ideas into it or multiple segments into it now because of what because of trust of Cleopatra Cleopatra fell down and within that within the trust what kind of trust there is a foolish trust there is a fickle trust and there is the part of the world so that is related to the Cleopatra and figure in in one idea and Cleopatra falling down is the other idea so trusting of paper ties one idea and falling down of Cleopatra is another idea and within that there are certain systems which are there and if we look into it it has multiple ideas right and so if it has multiple ideas it will take more time to process and so the more meaning more ideas that have the more meaning is to be interpreted the more meaning of sentence has the more time it will take to be interpreted and the more fixations are required into it and the longer duration of comprehension for such words now another factor which influences the processing of text has

to do with relationship among sentences what sentences or how sentences are related to each other so the more the sentences are related to each other the larger the relationship or if those two sentences are not related to each other then it will take more time to be process now halav and and clark 1974 describes something called a given new strategy and what is this given new strategy they say that whereby listeners are listeners and readers divide sentences into two parts the given part and the new part so when they are listening into a sentence when people who listen to a sentence they are listening in to us and hence they immediately divide the sentence into two parts one is the given part the other is the new part now the given part of our sentence contains information that is familiar from the context of the preceding information or the background information so when you are listening to a sentence what you tend to do is break this intends into its two part first part is the given part the given part has all the informations on given partners all the inputs from preceding context are from where the borrower over effect from previous sentences and people are familiar with it the new part contains unfamiliar information new part of the sentence they can contain unfamiliar information so listeners first search **memory** for information current corresponding to the given information and update **memory** by incorporating the new information offer as an elaboration the given part so basically then let's look at a sentence and then see how does this thing really work so basically what I'm doing is I'm creating two sentences and with a given part of a new part so if I write this sentence we got some beer out of the car this is the first part and the second part is the beer was Wow so sentence number one sentence number two right this is the given part and this is the new part similarly we checked the picnic supplies and then again the same this is the given part and the beer was warm now in which case do you think people will take more time to comprehend yes you are right in the first case people will take lesser time to comprehend this sentence the reason being that in the given sentence in the given part of the sentence the mention of beer is there and so it makes a relation with this sentence and so the comprehension is easy so make basically they got from where are the currents so they were wrong it's easy to interpret but in this case the mention of beer is not there in the first sentence and so you don't know how the new part connects to the old part and so then first you have to make the assumption that beer is part of the picnic supplies first you have to make this supposition assumption only then you can comprehend this sentence in this case it is easy because previa contexts provide you that idea and so if this is a given new sentence people will take lesser time to verify this one then this one because the first part of the sentence contains the referral to the new part right so this is basically what they say so in a given new strategy the more different the given and new parts are the more difficult comprehension turns out to be now when they brew and and Gustafson in 1990 they offered three conclusions from research on reading text first the **mental representation** is a construction by the reader which differs from and goes beyond the information of the text itself when you read something and that is why I always advise my students to read books what happens is when you read something you make a new world out of it a new construction you go beyond the book and imagine things which is not written even in the text and so people own this mental imagery and so in the in the last chapter and mental magic we saw that people have this mental imagery make these mental representations it may so once you read up texts people make certain imaginaries or certain texts or certain kind of formats which are personal to them and so they go beyond the what is written in the world and that becomes personal to us and so these personal things this these personal influences representations that we make or text help us in understanding something so when people read something they create a **mental representation** which is personal to them and it is beyond what is written on to the text the second is the representation is coherent when we read text the representations have to be coherent in the sense that it is linked in ways right so one part of the sentence text should always be related to the second pattern that is coherency is there so even if we read elect if you are reading a novel we read the back part first in the front part in the middle of the novel we still make this coherency the fill up gaps is there in this coherency is always maintained and the third is this readers attentional resources are limited we cannot be very ambiguous in writing sentences if we write very very

ambiguous sentences the actions our **attention** span of the resources is limited text comprehension will be flawed so when writing a text for comprehending we should be assured of the fact that first whatever we are writing people are there people will misinterpret it or people will create their own construction sort of it so it should be very limited it should be should be written in a way that you should give the chance for lesser representation varied representation also the representation that we form the writing that we do should be coherent it should not be all over the place having all kind of meanings into it and third that it should be written very strictly in ways that respect people's **attention** 'el resources which basically means that the idea should be clearly spread out if it is not then people's **attention** resources are very limited and so you keep on jumping from one's meaning to others to write them so this is how next text comprehension happens now let's look at an interesting fact which is stories how do people comprehend stories we have looked at how people comprehend text now let's look at how people comprehend stories so what is the way in which people actually read stories and so forth reading story it has been found that there is something called story grammars Sasori gammas are described as the way in which people comprehend large integrated pieces of text using **script** schemas and **grammar** of the **language** so while reading a story while reading a novel while eating some kind of artwork or fiction or nonfiction when reading any kind of large text which they need to comprehend they use certain kind of dramas which is different from the kind of **grammar** that we saw in the first part of this lecture what is it then so what our story grammars these are similar to scripts in that both have variables or slots data filled in differentially by different stories so basically story **grammar** is kind of a **script** right it's kind of a block and so what this block does is it this block says that each story should have certain fixed factors into it right so what story Lama suggests is that people expect certain kind of fixed factors so for example when you are looking at a story he story should have a protagonist the person who's the one who's acting the main component of the story or the main person of the story they should have a setting where the story is setting from where it is coming from where it is based on a plot or what really happens in the story what is the baseline of the story backgrounder story conflicts and resolutions then if you are writing a story there has to be some kind of a conflict and then the storyteller resolve this conflict and all these are expectations that people have and these are called story grammars so every story should have things like this these type of grandmas on to it now story grammars are stimulated to syntactic grammars in that they help us identify the units of each role plays in the story so basically like what drama does for us what **syntax** does for us because **syntax** tell you in a sentence what is the subject what is the object in what is the verb and what role each object is playing in a sentence and that helps you into making a legal sentence similarly story grammars actually tell you the main constituents of the story what is the protagonist who is the protagonist what is the plot what is the setting what is the tone what are the conflicts which are there what are the resolutions so that you come to in in one brief look into it what the story is about or what the story is going to tell you right and so these these are fixed factors it is screwy Gramma's are like syntactic rules for drama now story grammars they provide a framework with which to expect certain elements and sequences and to fill with default values things are not explicitly stated so basically a story **grammar** then tells you that any story has to have these things now if it is not if something is not explicitly mentioned here we fill it implicitly to make a story let's look into what story grammars is so this is basically a story of the three little pigs now obviously when I say a story of the three little pigs there are certain expectations that people have what will this story have right and so first of all there should be character the bird organist and the protagonist is there has to be first of all if it is a story about three little pigs so three little pigs have to be there that is the main thing and then if the three little pigs are there just three little pigs going around is not going to solve our problem so conflict has to be made and a for a conflict and a willingness to be an antihero has to be put in and that is what the world face so story dramas then give you this idea that only the hero is not going to make up a story so some kind of a villain has to be brought in and P hero is to brought in because int you know in hero when they collide will produce the kind of problems or the kind of difficulties contradictions which has

to be resolved later on because no stories work around this way and so I have now created well now obviously if there is a villain and there is a hero then there is some kind of problem has to arise because they don't cannot heroes and villains cannot live together and so what is the problem here the wolf wants to eat the Three Little Pigs now we have a story in condition so we have a problem we have certain characters into it we have all the characters into it and we have a title of it so let's then write the story certain events ain't waiting so we can relate to what what I had done is we have put in some so wind blows the house will blows down the straw house then will blow down the steak house will can't blow down the brick house will climb the chimney and so these are the three kind of houses these pigs have they leave in and so once the wolf goes ahead and blows it the the pigs create a new kind of a house why they created a new kind of us because they are afraid of this particular wolf and they don't want to be eaten and so this is the one this is the is a set of events you can create any kind of event just as the look into it and then the resolution is will falls into a pot of boiling water and sort of or the open door and is never seen again so the resolution isn't the goal who is going to eat the things so the resolution is these pigs were actually boiling water and the world comes down a chimney falls onto the water burns himself and says that no one not going to do this again lunch is lesson and goes away and never returns back again so this is our story is created this is how situations is create a certain contradictions of greatness on and so forth on this side you will see the Saudi structure and recall and so I'll leave it to you to actually read so basically there are different kind of things in a story for example what it is what is the event structure of the event episode beginning development simple reaction and so on and so forth so basically then most stories have these kind of filling gaps and so this is how a story drama really works and so what is the need for a story **grammar** because it gives us certain kind of **insight** of how our story's going to work and then if something is missing we can always go ahead and fill that particular thing now looking at something called gracious maximum conversation up till now we have looked at something called speech how it is perceived what is Rama what isn't misperception all those things but there are certain makes in the conversation there are certain rules of conversation which needs to be followed and so Gration gives these mixing so grace 1975 believed that for people for con to converse each must do more than just produce utterances that that is phonological sound they must syntactically and semantically produced on which are appropriate for example if you look into two sentences I just heard that yo got promoted today isn't that great if a is saying and me says the Salt Lake City is locate located in Utah and the third is Charles Darwin is the revolution they are not conversing at all there is no conversation although all the sentences here a semantically correct syntactical e correct have meaning but they are not following conversation so there are certain rules of conversation so what is it what is the and again the a comes up with answer then what is the square root of 34 BC is that chocolate ice cream is sweet so basically this is not how a conversation is is actually progressing and so Bryce described the speaker in a conversation should actually produce some kind of a maxim so follow some kind of a rules into it the four Maxima are there first is the quantity whenever we speak we should speak in terms of quantity we should not speak more we should not speak less but we should speak enough for example somebody asked you that time and you say 4 o'clock this is not quantity because what o'clock would be morning 4 o'clock would be evening so you should have given enough information not too much information but enough information then quality quality in the sense that whatever you are saying should have some kind of a context some kind of a quality into it if there is no quality that you have into your speech if you have no context into your speech or know the meaning into your speech then it will not be processed in the right way then you have maximum of a maximum of relation if you are speaking a sentence and two parts of the sentence make no relation so I am going to the I am going to the the market and Sun the sun is shining on my head and the two sentences have no relation with each other so when you are saying two sentences or when you're saying as intense both the parts of the sentence should have some relation and so this maximum relation should be always followed it should not be awarded you should not be monster and also people should have or speak sentences

which have relationships or which express relationships into each other and for is the maximum manner we should always have good manners if you are speaking in the wrong manner so if you are speaking to somebody who's elder than you and you say come here sit down that's not a manners to purchase to be done similarly speaking to people who are smaller than you so certain kind make sense or to be followed if it is some somebody you senior to use hold it please use words like please or some kind of respect has to be given and these are the Maxim's which is a similarly if you are referring to people who are below of your age of similar ages okay but of below age then you will have to use some kind of a mannerism into it if you do not then you are not moon to be a nice social person and so a violation of rule a and B is there then people are uncooperative and obnoxious so just telling you something like giving parts and bits of information or no quality information no meaning information at all then people are known to be uncooperative and obnoxious and people violating rule C are known to be bizarre because if we don't say relations if we just keep on saying things then they are bizarre in nature okay then so what influences this **language** have an other cognitive process up till now we have seen what is the **language** and about and how it is related to different cognitive processes so what is the influence that these **language** or the **language** has on different cognitive processes so there are two ways to look into it first **language** and other cognitive processes operate independently one view see is that and the second is **language** or other cognitive processes are dependent on each other and both of this view one view says that they are separate **language** has nothing to do with other cognitive process and the other will say that they are integrated together and both of these are further explained by two popular hypothesis of **language** in in **language cognition** one of these hypothesis is called the modularity hypothesis and so what does hypothesis actually say so it was proposed by John federal ossified Jerry Fodor in 1982 85 I mean argued that some cognitive process in particular **perception** and **language** they are modular so they process or they do processing in a modular formula modality of a process so what is modularity then modernity for process says that it is domain-specific operates specifically with certain kind of inputs so **language** then is modular because it operates only through certain kind of inputs if you give the wrong input **language** is not going to work similarly with **perception** if you give the wrong input **perception** is not going to taking place and some **perception** to take place a certain kind of input has to be given and the second is informationally encapsulated which basically means that operates independent of the believer and other people's information available to the processor so no matter what people believe **language** is independent the processing of **language** is independent what we say is obviously independent dependent on the context or what we believe but the processing of **language** if somebody says something to you how it is processed that is in informational informational its encapsulated which means that **language** processors don't depend on people's belief system or their expectations to process things and so this is how languages independent or **language** is not dependent on the cognitive process so basically then modularity hypothesis or fordoes view is that **language** is modular in nature and that is why it is not related to any other because it has specific in words defined and formally specific inputs is going to work and also the fact that it processing is not dependent on people's belief systems they are expectation their views and so on and so in direct opposition to this there is the woof Aryan hypothesis referred by a by example or woof Benjamin Whorf was actually a chemical engineer and so he gave the idea that **language** has to be related with other different kind of cognitive process which is out there and so what did he say he said that **language** and other cognitive process are strongly related he believes that **language** development has a lot to do with the kind of atmosphere or the kind of environment for somebody is born with and so the environment in which you want shapes your **language** right somebody coming from a very rich environment will have a different kind of **language** in compare in comparison to people who are coming from lower socioeconomic backgrounds people who are coming from those backgrounds where the **language** is not English and so they're thinking their **language** their whole idea would change because the environment forces it and so he believes that **language** as one grows up learning and speaking

organizes and directs the way when perceives the world organize information about the wall around that happens because the world around you gives you a different nature of the world interaction with you or your interaction with the world gives you a different kind of feeling a different kind of thought process and this thought process will then change the way **language** is there and so when you see somebody from the street the way he speaks the way he is thought processes and somebody from very a light background when you speak is look at you at his speech and the way he speaks both are different and so this is the interpretation of difference the differences which is there so basically then in this section on **language** what with this does we continued from the earlier section where we looked at what is **language** and what are the parts of the **language** and we proceeded by explaining how **language** comprehension is done or basically how long how speech is produced so we looked at some water speech production and how it is produced and what are the errors can that can arise in production of speech we also looked at comprehension of text messages and comprehension or sentences and grammars we then looked at some principles of better communication of how people should be speaking and then finally we looked at a very popular debate the modularity versus the non-modal arity or the debate between the fact that **language** is like other cognitive process and so it is interacting with them against the debate that **language** is different from other cognitive process it should not be interacting so this is the basic overview that we did in **language** we studied what is **language** how it is in produced what are the rules for it and so on and so forth later we saw it is the production of **language** the errors that happens with the production of **language** the comprehension how comprehension of spoken and written **language** happens and then label the errors that can arise there and then we looked at how stories are perceived and to end with we looked at a famous model versus war Farion debater into a **language** comprehension **language** combination thank you

Lecture 23: Introduction to Problrm Solving

■Kind: captions good morning friends so in today's course what we are going to do is we are going to take a new topic on **cognitive psychology** which is called **problem solving** and so **problem solving** is like the culmination point of any cognitive process now when I say this **problem solving** is the culminating point of any cognitive process what do I mean by that now it's that process which comes right after something is encode something is perceived something is being attended to and then a problem is generated look at any problem in this world look at the problem **perception** or looking at a tree or looking at a visual world around you now the process starts by looking at the **distal stimulus** then forming the **proximal stimulus** then forming a percept out of it which is the process of **perception** this percept is then basically given **attention** which is the process of **attention** a sustained **attention** and through its certain characteristics of the perceptor processed after that next process comes in is basically called storing this material that you are storing this visual information which is in terms of **mental representation** of the visual nature making or storing this representation into short first in the **short-term memory** and then storing it to **long-term memory** now once that gets stored into **long-term memory** it is then analyzed and the analysis of this percept for generating meaning is basically problem-solving basically what you have seen or as we as I describe this process the in the Mentalist presentation of what you saw as a **distal stimulus** the meaning part of it the analysis part of it is called **problem solving** and so looking at something in the environment is the start of the cognitive process and the point at which you need to decide or before deciding the point at which you you ask this question of what is it that I am looking at is called **problem solving** and so basically what is the problem here the problem here is what are you looking at now problems are everywhere if you look around anything that you do any cognitive process raises a problem and so it's the next to last step in problem-solving that is by the three sections of **problem solving decision making** and **reasoning** are called the higher cognitive functions why are they called the higher cognitive functions because these processes actually make you do the interpretation of the stimuli up till now whatever we have looked at in terms of **perception memory attention** and **categorization** and **language** all these processes were required to take in information from the environment and then encode it or form mental representations of it the process of problem-solving a process of **decision-making** in the process of **reasoning** are the three processes which actually help us in making interpretations of things making interpretations of what the visual stimulus is or what whatever problem is being put on to the cognitive system so then let's look at the first slide which I have over here and so these presents some basic problems in life and so what you see here are natural problems and I'll also explain to you a natural problem of how what a problem is in a few minutes so in the first slide as you see the fish has a problem of having a problem with the fish food which is outside its box and so the product the **problem solving** technique that it is using is basically to receive that it needs food represents this **mental representation** in the food is realizes the food is outside the place in which the fish is the aquarium in which the fish is and so decides to use this time anger to get the food similarly there are other funny problems like example the one you see in the right extreme right top is the idea of solving a problem and so we generally use this idea of thinking outside this box and so this person has now come out of the box as the literal the metaphorical box that I'm talking about and so he has come out of the room from where the problem is you know he doesn't know how to get in and so that is the problem for him I so he thought that he should think outside the box and so he came out of it and the last one and are the bottom slides you see this cat having this problem of food which is in in in this particular box and so what the cat decides is to use this kind of a problem-solving technique although the easiest technique would be to hit this bottle and get it out but the cat decides to go on into this so basically through these funny pictures I am we are demonstrating water problem and so in your life also in your experience of everyday day-to-day experiences you have a lot of problems and so one of these problems could be taking an exam or it could

be to do your Academy co-op and so when one of the things that happens in the Caddy mix is is writing a paper or writing a dissertation and so that could be also a problem so this kind of a thing the writing a dissertation requires certain kind of input or certain kind of processes which you should go through to solve this problem and this idea of how do we solve of writing a problem how do we solve this problem of writing a dissertation or basically a paper for some course that is a that could be considered as a problem and so solution to this lies in identifying something called the **initial state** which is the state at which you identify that what is the problem and defying something called the **goal state** which is the state or which is the point at which you have written the paper and certain rules and certain hindrances to it so certain rules of what you should do in what you should not do and basically our hindrances in terms of what can be done or what are the problems in your face so taking the idea of a problem of writing a paper for your carrying course this problem has an **initial state** where you are worrying about writing this paper we are thinking about writing this paper and this is the **initial state** the **goal state** is that state where you have finished your paper and you're happy or you have got a B or C or whatever into that paper maybe an a I don't know depending on your intellect you would get any numb any letter grades out of it so that's the **goal state** where you are finally successfully it in the paper and then there are certain rules the rules are that you have to follow certain sections certain certain kind of knowledge and or certain kind of strict adherence to for example you cannot copy things from the internet you have to think about the genuine problem you certain ways if it's arithmetic problems you use certain kind of an arithmetic rule or if it is a problem from psychology use certain kind of knowledge from psychology which suggests what can happen in what cannot happen and then basically certain kind of inferences interesting is the term of the fact that it may be you are hungry at this point of time and so that is an inference or the fact that you have to go with your friends out today evening and say we do not get time to write the paper so these are the hindrances or innocence could be a number of things and so basically then this is how the problem really work so this is how the demonstration of her problem is so right so basically any problem will have these four parts and so let us discuss this in detail so what is a problem then what do we call a problem and as I said as I explained to you with an example of what a problem is generally a problem consists of several basic elements one the first part of it is called the **initial state** this is situation and the beginning of a problem so this is the state or this is the point at which a problem starts right this is the state value this is the point do you realize that you have a problem and this is the point when you start creating mental representations of the problem or **encoding** the problem as such or realizing that there is a problem or you have come to a point or you have in in your life or in your cognitive process where something is creating hindrance of something is creating disturbances and so now you two need to solve it so the first state is realization that you have a disturbance and this disturbance should be cleared out and that is what the **initial state** is matching the **initial state** is a state which is called the **goal state** now this is that state which you arrive after you have solved a problem and so this is that point which will come to you once you have solved the problem once you have acquired the solution to a problem and then you become happy you become elated or whatever you do and whatever feeling you have once you solve the problem that is called the **goal state** and then there are two set certain sub States for exact of rules and they said constraints that must be followed so certain rules for example this what can be done and what cannot be done what is allowed and what is not allowed to solve the problem and a set of obstacles for example hindrances that has to be overcome so yeah as explained in the in the paper writing problem the hunger that you are feeling or the fact that you have to go with your friends today evening all these things are called hindrances or set of obstacles which are thought they so basically then a problem definition of a problem has DNE problem has these four points on to it now obviously the question now comes in that what are the type of problems how many types of problem can exist and so there are two basic categorizations which are out there we will just look at these two basic categorizations the first **categorization** is the difference between a well-defined and **ill-defined problem** what is a **well-defined problem** a well-defined problems are those

problems which are cleared and structured and they have very clear and structured **initial state** goal states and constraints and all are understood for it for example a problem of writing a paper now when you are thinking of writing a paper it's a very clear problem you know that there is **initial state** the worry that you are having the final state where you want to reach and you know all the hindrances which is out there and so these are called very defined problems so those problems which present themselves in a very concise way where you know where you start from and where you want to go and you also are able to see the set of rules which you need to follow and the set of that you need to that that you will face in the way is called a **well-defined problem** for example think of a problem getting a license now when you want to get a license there are said this the **initial state** is you don't have a license the final state is getting a driving license and then there are certain rules for example certain rules have to be followed you have to give a particular example our learning exam and then after the learning test you have to give the full test and in that year to scroll the driving so you have to learn and so on and so forth and certainly Indian says the influences may be to study for the exam to want to pass the first learning test and then to pass the second learning test to make enough number of practices the fact that you don't own a car and then you want to learn car so basically then the fact that you have to go to someone who has a car and then get his permission so these are the kind of hindrances which is out there and so these are well-defined route in addition to the well-defined problems are ill-defined problems now what are these problems these are fuzzy and abstract problem that is the initial **goal state** and constraints have gaps in understanding leading to difficulty in assessing the solution now in this case what happens is there is a **initial state** there is a final state but you don't know how to go from the **initial state** and the final state or then you don't know how to what are the hindrances in it for example let's say that tomorrow I want to stand in an election and become a minister now if this is the **initial state** I don't know what the final state would be of course becoming a minister the final state but I don't know how the process is what are the endurance is what are the steps to be taken and how do you become a minister or what is a minister what kind of minister should I become and all those questions are there and then the kind of rules that I have to follow so what are the rules of nominations when they are there on what basis is my nomination can be rejected how to stand from someone how to impress people to so that they vote for you since it's a democracy so people vote for you and so many other things are there so I don't know on these in Integra C's I don't know all these hindrances which are going to be there and all the rules which out there and so this is an **ill-defined problem** for example about a person like me who's an academician to become a minister there is a set up itself becoming a minister itself is a really defined problem or going to the moon is an in defined problem I don't know how to go to the boat right where should I apply and what should I do to go to the moon so this is an **ill-defined problem** now in addition to well define an **ill-defined problem** there is another **categorization** out there which is in terms of routine and non-routine problem and that is routine problems are those which you are familiar with the type of process that you have to follow and non routine problems are problems where you don't know what steps to follow so basically in terms of well-defined and ill-defined you can also have a match between routine and non-routine problems so routine problem is one that can be solved by applying well-practiced procedures right for example writing exam making tea so in making P you know what to do so start with boiling the water pour sugar into it to put milk into it boil it for some time with tea leaves and then you get a tea so this is a **well-defined problem** where you know number of steps that has to be there the kind of hindrances that is their **initial state** where you have water and other materials other raw materials were from which tea is made and a final state where you finally get that liquid which you call tea and enjoy and so this is a routine problem opposed to this is basically a non routine problem something that you don't face every day for example let's say you have an accident and in this accident or you have now got a twisted ankle now this is not a everyday thing for you and so this is a non routine problem so how do you go ahead and treat this or how do you go ahead and take precautions and so this is a non routine problem because you don't take this every day or the idea of writing a research

paper because that doesn't happen every day you don't do it or so on and so off and so often and so that you are familiarity with this so twisting an ankle treating a twisted ankle or any other kind of problem which is not well-defined which doesn't happen too often for you and so you're not familiar with the process is called a protein problem so basically then to categorizations one is in terms of well-defined and **ill-defined problem** and the other in terms of what is a routine problem and non routine problem so basically when this problems rezoned there are certain ways to solve this problem there are certain answers to solving this problem but there are certain challenges out there which happens or which arise when you solve problems there are certain challenges which is out there for example **problem solving** is the last hurdle to most cognitive process that I have been saying all along and it takes a long time to accomplish up till now you have seen those processes cognitive processes which help you in to encode information make mental representations realize that some information is there store it and then basically that's the end of it right understand in which format is appearing and it and store it in some format and so on and so forth this is the step **problem solving** is the step where you realize what the problem is or realize what is to be done with whatever has been stored and what is to be generated out of it right so that is the problem-solving position and the solving problem does take more time it takes more time because you have to realize what what where you are and where you want to go right if you got if you have a problem you know you should have know where you want to go if you don't know where you want to go then it is not a problem-solving technique or this is not **problem solving** so you always know where you want to go or what is the **goal state** in a problem so assessing **problem solving** in terms of **accuracy** rate provides a rather draws estimate of **problem solving** proficiency so why am I saying that that **accuracy** is not a good idea **accuracy** does not demonstrate how do you solve a problem now a person could be very familiar with the problem it could be a very routine problem and so his **accuracy** would be very high but then that doesn't guarantee the fact that he the way he is solving a problem how he is solving a problem or what is the way in which is solving a problem just with the fact that since he has seen problems like this and so basically he solves it much faster or more much accurate doesn't give any sense of what the problem is or how the solving technique is so being accurate is just a chance factor right and so trans sector **accuracy** has nothing or very less informative about the problem-solving techniques the other way to look into it is measuring solution times which provides some useful information but doesn't shed much light on the result of processing now how quickly you solve a problem again how quickly you solve a problem says how can you move from the **initial state** to the **goal State** but it doesn't tell you whether you realize the hindrances whether you got through the real illnesses maybe you are too lucky and the fact that a number of hindrances doesn't arise so you haven't considered all the hindrances or you are you are lucky to get a set of rules which actually take you through the **problem space** we'll come to the idea what a problem spaces in in the later section but maybe that is the reason how you solve the problem and so basically solution times are no guarantee to how the processing or what is the kind of processing which is happening so neither **accuracy** nor **reaction time** which are the two major things in terms of behavioral research remember lecture one on methods of doing **cognitive psychology** and one of the methods were the behavioral methods in the behavioral methods we were looking at something called speed **accuracy** trade-off or Sweden **accuracy** as being two major ways of solving problems in **cognitive psychology** are dealing with product **cognitive psychology** related problems and so both **accuracy** in **reaction time** doesn't give you too much information about the solution process of how does it happen so basically then there are other ways of also how problems under should happen so these are the two ways of solving the problem but it doesn't tell you how the problem solution really happens and so solution of problems of how people actually went and solve a problem it's really a difficult thing to assess and so one of the ways to do - basically whenever methods which I use to basically come up with the solution was called the idea of a **verbal protocol** but even before we go there I have a problem for you and let's see how do you solve it so this problem says that look at this problem and I'll give you a minute to actually read that and then basically

come up with an answer so leave this problem and let's see if you can answer it I will give you half a second to read this problem maybe one second to read this one minute to read this problem and come up with an answer so starting now here is your 30 second period for reading the problem and coming up with an answer so the 30 second period is up and I'm pretty sure that most of you would have been able to solve this problem now this is a famous problem by Greeks and maker gari and was published in Journal of renewal psychology at some point okay do you know the answer to this problem the answer is no and so those of you congratulation who who could have solved this problem and those of you could not take this away there is no way that this is possible this kind of arrangement is possible and so they also found leak and McGary also found that only fifty percent of the people could solve the problem and fifty percent could not in if it is rushed through time but given enough time most people were able to solve it the answer is no it cannot be done and so this is basically the idea of our problem so basically in terms of **reaction time** that **reaction time** doesn't say how problem is represented because I really don't know how did you represent this problem and so maybe you have seen a problem like this before ensue come up with an answer and the **accuracy** of it also doesn't know how you solve this problem because I don't know how did you make the representations of it so basically then what is the way in which I can understand how a referral program is represented or how what is the process of representing a problem or how did people solve a particular problem and one important way in which we can do this is something called the **verbal protocol** so what is the **verbal protocol** now in order to understand the process of **problem solving** researchers made extensive use of something called **verbal protocol** if you remember **structuralism** there was something called **insight** and **insight** is much much a close brother to Verburg protocol and inside studies what was done is people were asked to do certain kind of psychological tasks and then they were are when they were doing this task they were asked to basically relate how the task was being done and this this **insight** generation was something like this purple protocol is a similar procedure in which people are to report on how they generated the solution to a problem so which are reports so basically whatever protocols these are reports generated by problem solvers and they think aloud during the solo process so as you're following the process are solving a problem people are made to go ahead and report loudly how they're solving it for example the problem of how do you act two plus two right now how do you go about it what is the way of addition and so what you could do is first I'll think about to then I think about something called addition addition is basically this then I take the other two I use the operator addition I go ahead and add this now I come up with an answer now I verify this so this is basically how our protocol really works it so all the steps that you take in doing addition of two plus two or any other problem solution what you have to do is you have to tell in a stepwise manner what is going on or what is the process that you are using and so this is what is work per protocol you see a problem here I see and so what is the problem here there are several problems with this kind of a protocol and the first problem is that everyone doesn't have the verbal ability required to affect accurately what they are thinking it's very difficult to verbalize what people are thinking and so the I the problem here is that the ability of people to basically narrate what they're thinking is a major concern here people are not able to narrate what they're thinking and so verbal protocols fail or the idea of loudly speaking out thinking aloud what they're doing fails because people don't have the verbal ability they cannot generate or they don't have the right verbal device or right horrible requirements and enough words to speak out the problem the second problem is there is no way to assess the **accuracy** of what the verbal reported somebody would may say something else and they must be doing something else and so there is no way to tell if people are lying or cheating or deliberately run deliberately in in deliberately if they are coming up with problems or if they are inputting the right technique of how they're solving the problem and so there's no way of **accuracy** testing and the third is the very act of thinking aloud it may interfere with or change the way and not solutionist on and so since I'm doing a dual task right I am also going ahead and loudly **shadowing** the process of problem solution which is basically loudly also speaking it back so it's a dual task and so this dual task may in the

nature of the task itself the idea that I have to use also verbal **memory** or verbal **information processing** verbal representations are basically repeat this thing back may create problems because what will happen is this idea of verbally repeating it things back may interfere with a very idea of the solution with a very idea of how the thought process of solution really works and so these are the problem with the verbal protocols of course so these are the problems with the solution of problem now another problem or another way why problem solution cannot cannot be packed on towards difficult to look at how problems are solved is because problems are of varied nature and so the complexity of the **problem solving** presents another challenge for researchers the fact that there are different kind of problems and there are device set of circumstances and from right from solving a math problem to solve in the verbal problem to solving a traffic problem to solving n different problems and so with these such a circumstances there are a number of problems that may arise and so all these problems may have different different solutions and so that could be one of the factor or determining factor why problem solution or looking at problem salvation could be a problem in itself now man 1992 he distinguished five different types of problem that can arise and so we look into these problems one by one and then we look into the act of the problem itself the first kind of problem that mark in 1992 defined is called the transformational problem and so what is the transformational problem it's a problem in which the solver is presented with the **goal state** and and he knows the **initial state** and then what he has to do is to find out appropriate strategies to use those strategies and reach the **goal state** so you'll know the **initial state** you know the goal States and you have to do something or use the appropriate problem-solving technique problem-solving state and eventually reach the **goal state** so these are transformational problem you have to transform from in each state to the **goal State** using certain rules or certain by navigating certain hindrances and I went that the second fire type of problem are called arrangement problems what are these problems they these problems present the user all with all the necessary elements which are required for solving the problem what is the role of the receiver then the solver of the problem he has to then arrange all these elements together in a way that gives the final solution so you know all the you are given all the problems solution elements or you are giving all the parts of how to solve a particular problem the only job of a person here is to basically arranged them rearrange everything so that the solution comes out of it the third type of problem which arise here or which which people in always face or which might define in his in his **problem solving** or types of problem definition is called an induction problem and what is the induction problem involves giving the solver a series of examples or instances sopra is basically forming an analog in solving problem by analog so induction problem a number of instances or exemplars are given to you and using this you have to figure out the pattern or rule that relates the instances together so you have a number of example number of instances are given to you and you know you have to find the pattern through which these instances are connected and once you know these rules one you once you form or I understand this rule you will be able to find the final solution so induction problem certain things are given to you certain kind of examples or instances are given to you you have to find the rule of how they are connected and this rule will actually give you the final solution of how the how to reach from the **initial state** to the final state in deduction problem what happens is certain premises or conditions are presented to people and they need to determine whether a conclusion fits the premises so certain kind of provides a certain kind of truth statements are given to people to start with and then what people need to do is to they have the conclusion in front of them and they have to look at the fact that given the fact that the premises are true the conclusion czar also true so this kind of reduction problem we'll see in the next section which is called thinking and **reasoning** so in terms of **reasoning** and the **judgment** and **decision-making** the next section that we're thinking will see these kind of problems basically these kind of deduction problems or this kind of premise rigid problems are **syllogistic reasoning** and the **inductive reasoning** and so we'll look at that in the next section but for now what deduction problem does is some certain number of truth value certain number of crude treatments are given to you and what people need to do is to find out that

given the fact that certain two truth statements are out there whether the conclusion fits these two treatments or whether the conclusion fits the premises in certain way and the last kind of problem which are there are called the divergent problem so what are these problems they operate as you know in a way then a solver here has to give a number of solution whatever comes to your mind a number of solution and so that is how the problem solution really works or this kind of problem really works so divergent problems and those problems there are a number of solution exists and the solver can then give you a number of solutions so let's look at these problems one by one the first kind of problem so five different problem the first kind of problem that you want to look at is arrangement problem look at these words here can you rearrange the letters to form an answer the answer here is VI ki ng very good for those of you who could answer it so this is an arrangement problem everything is given to you here what you need to do is rearrange the letters so that you get the answer not diverting problem as I said it lets the solver give a number of solution and so as you think the problem here could be think as many used objects so bricks could be used for meeting someone it could be used for making a house bricks can be used for as a weight as a substitute for weight it could be paperweight it could be used in selling something as in terms of standard weights are so many other things right and so these are divergent problems this is a deduction problem so certain premises are there and these premises so all professors are carrying people this is the premise and this is true the second premise are all carrying people are good this is again a premise which is true and so now you have to validate this conclusion which is drawn from these two premises the fact that would be accept the statement that all professors are good whether all professors are good follow from these two statements and these kind of problems are called the deduction problems right so certain premises are given to you and a conclusion has to be validated against the premise the fourth kind of problem is called the transformational problem and so what is here I mean transformation problem you have the **initial state** and the **goal state** known to you and you have to use a particular strategy to arrive from the **initial state** to the **goal State** and so as you see this is called the Tower of Hanoi problem and so you have to the fact here the requirement here is to move these three workers from position a to position C so same arrangement here the only thing is you can make you can move or these circles one at a time so one movement is allowed you can move from here to here or here to here kind of a thing right also the fact that a smaller circle will never go beneath down on or a smaller circle cannot be beneath a larger circle so that is the kind of thing so how do you arrange from here to here and so you are know that this is the **initial state** this is the final state and so by moving one at a time and this constraint using this constraint that a smaller circle cannot be banette or cannot be below a larger circle you can solve this problem this is called a transformational problem and the last kind of problem is called the induction problem and this is look and look at the number in your sequence and then come up with and solution so this I will explain to you this is rather interesting so take your time solve this problem I will give you the solution to this problem let's wait for it and because I'll use this problem as some other at some other places also so can you find out a rule through which these numbers are connected there is a rule remember so if you go about 5 minus 4 is 3 or 4 5 8 minus 5 is 3 5 minus 4 is 1 4 minus 1 is 3 you will never come to the solution because there is a certain rule to it I'll tell you the rule but with that will happen as we progress into this section so how do I solve a problem what are the various approaches to solving a problem the rest several approaches to solving a problem for example the first approach is **behaviorism** solving problem and associative learning so two approaches I will describe here one is the guest art across the other is the behavioral approach comes from the first section itself introduction to already psychology where we saw what is **behaviorism** **behaviorism** is basically understanding simulus response relations so basically what we haver is said is there is nothing called the cognitive system there is nothing called the human mind there is a behavior there is a response and making connection between these behavioral response or behavior in response is what **behaviorism** is all about so look at each behavior look at each response find the most optimal response the most optimal response is the correct response to the stimulus and this is associative running

associating a response to a stimulus so **behaviorism** presents a solution to **problem solving** or gives a way to solve problem and so the people who started explaining **problem solving** s as be various as **behaviorism** where early behavior is like ile Thorndyke and so turned like 1800 he conducted his first estimating study of **problem solving** using cats so what he did was he was looking at feline animals and so he took this salon animal put it into some kind of a box a capture box and the cats doesn't like to be captivated into box so what he did was he took this cat put it into a box and enclosed the box within the box was a lever the cat had to realize to press this lever now look at the the easiness of it what cats do when they and when you enclose it into some kind of a structure or in some kind of a cage it tries to scratch it and so this lever was placed in such a way that it needs to be scratched down or bend down from the power and so the scratching ability would give the cat the idea to scratch this or bend this paw and so later sooner or later the cat will realize that if I scratch here so where the liver is so this is my box and so the cat is put here right and so here is the lever the cat initially goes ahead and scratches everywhere right but once it scratches here there is a door which opens and the cat can not escape out of it so the cat has to realize that this is what and this is the solution to the problem so Trond I was interested in knowing whether the ability to solve the confinement problem would appear suddenly as an **insight** of gradually through the process of trial and learning so what he was want what he wanted to see is that whether cat did something for learning the solution that I if I scratch the right button window door will open and I will get out of it or whether the cat sits and through inside through some kind of magic or through some kind of a internal problem solution will the cat to be able to know that I scratch this and so what they found out then the cat learned by learn the idea that if I press this button the door will open I can move out of it through something called trial and error and what is trial and error what the cat did was it's crashed everywhere and so if it scratched a number of times slowly slowly it realized and if I scratch at this button what is going to happen is the door is going to open and I'm going to come back of it and so that is what he realized now turn like describe the learning process with which Islam there's something called the law of effect and so what is law of effect it states that if a response leads to a satisfying outcome the connection between the response and the situation will be strengthened but if it leads to not satisfying outcome this is deleted so scratching while satisfying or scratching on levers catching on things which was coming out of the cage was a successful response was the cat scratched on everything which was protruding out of the cage but then it started not scratching on polished surfaces right so this this this idea of sketching on to those objects which are protruding out or protruding in or which was shiny in the intron dyes case lead to the freedom of the cat that is this **Association** increase in assertion with this fact that I should scratch on those objects which are shiny on those objects which are illuminated in some way or this look which looks like a leaver will actually make me escape is what is called the law effect and so what a lot of effects is that the cats learn that when I do a certain act and if I am positively rewarded I'll do this if I'm not personally rewarded those acts I will not do I said this is what the tonic box is all about and what you can see here is the number of trials is the cat took and the number of time required to escape and as you see initially it is very high so this is trial and error and so as the cat kept on doing it later on the time required was in terms of 100 seconds right in the by the 7th plus 17 trial the cat was as fast as you put into it million hundred second it will escape but initially it was requiring 600 seconds to find this idea and so if you can see this is where the cat is this is what the arrangement is all about and here you see the latches and so it needs to scratch we scratched on the right surface with the door we open and that's what it was it and so what we every said is that the solution to problem is always in terms of trial and error in a position to the guest artist which believed that by forming perfect associations between responses and similar says can a problem be solved so problem solution is basically making up new associations and rewarding those associations which give us a reward or learning those institutions which give us a reward guest artists oppose this view they said that **problem solving** is basically **insight** related process which basically means that it is not the actor doing something which solves a problem **problem solving** happens because people restructure

representations people restructure the problem in certain way meaningful representations and these restructuring of meaningful representation is how the solution generated so against all psychologists they believe that mind has an inherent tendency to organize incoming information so what the mind could go and we saw in **Gestalt** Theory what get start is proposed is certain **Gestalt** is wrong and what does this get yes tartre really say it's say that the mind organizes stuff we structures information restructures mental representations and this is this restructuring which gives solutions which make easy solutions right so what the mind does is it always organizes and restructures information right now defining features of condition thus rather the defining problem solution as mindless playing out of associations that gradually build up over time they believe that it involves restructuring or reorganizing the problem so they believe that its solution to problem doesn't come from time again and again making associations mindless associations between the fact that some time I will get rewarded or at some point of time I'll get rewarded or some of my behavior will get rewarded they said it is not the solution the solution is the solution from power to problem comes from the fact that when I look at a problem and not only look at a problem i restructure it in certain way I rearrange the problem elements in certain way such so that the answer comes to it that is water solution to our problem is all about now a pioneer to the study a pioneer to **gestalt** get started school was well google collar and so what he did was he found a popular way of basically or he found a way to test this idea that **insight** leads to problem solution so basically for guests our psychologists then problem solve a solution involves a process of restructuring where my problem elements are suddenly recognized and seen in a new way so what happens is they believe that the elements of a problem the problem itself is restructured into a mind and it is it is kept on restructuring till the point of time that a solution suddenly appears to you and this is called **insight** or in some books it will be called as the experience so this our experience leads the so what happens is that problem solution has a stage where people don't do anything they just sit passively and reorganize the problem or rotate the problem in subtle way mentally restructure the problem in certain way till a solution happens so against the fact that what behavior says that problem solution is all about making or doing acts and learning a solution what they believe is it is mental restructuring which leads to reorganizing of information which leads to problem solution so the sudden successful restructuring of a problem element is termed the inside and this is the major focus of **gestalt approach** and so how they tested it they kept certain monkeys and with crates and then an appealing item so this is what an appealing item is here is the appealing item here some banana were actually taken in in they were held on to the roof and certain crates were given to these monkeys and they were made to say what corner found out is that initially the monkeys try to jump to the roof and take the banana now that was not happening because no matter how hard the monkey jumped they were not able to reach the roof so what did they do they kept trying everything right initially they kept trying everything and then after a period of time this sat there with the idea now the solution is right there in front of it there are number of crates which can be put over one and another and a kind of a ladder can be generated so what these monkeys did was they sat there for a certain period of time and then later on after a certain period time has elapsed suddenly the idea came to them or they developed this idea by restructuring this problem and so this thought the boxes in this way and now it is like a ladder and so he climbed on the ladder and takes the food so this is this wait time is what is called the **insight** or this wait time is what leads to the **insight** and the solution to the particular problem out there now **cognitive psychology** could also suggest orders autosuggest solution to problems so for any psychology also is a way of solving problem in what is the suggestion here now just as computer solvers computer solves a problem by executing programs that use certain information stored on to it humans also solve problem by applying certain **mental process** to representations so what does computer to computer program takes in information in certain way arranges this in in a certain way the the certain kind of processing on manipulations and come up with a solution similarly what earlier cognitive psychologists like Neville he suggested that humans also saw problem in this way the problem solid solution happens by applying certain processes applying certain rules are

applying certain methodologies to already story play representations so already stored representations which are coming from **perception attention** and so on and so forth the basic processes these representations are basically taken in and certain processes are applied on to it and these processes lead to least two four if I'm finding the solution to a problem so basically they gave these cognitive early cognitive psychologists gave the idea of something called the general problem solvers so what is the net problems or the **general problem solver** is a conceptualization of the computer program we saw the problem and so what are the **general problem solver** does it takes in a problem breaks us into subproblems and then by solving this sub problem one by one attains the final **goal state** and so this is what the general problems are where does and so this is about naval also said son Alan Simon in 1972 they originated the conceptualization of **problem solving** as a step-by-step process from an initial to the **goal State** so what I do here is if I have a bigger problem if I have a problem of writing a research paper what I do is I break it into smaller problems smaller sub goals right and so each sub goal I take the initial problem divided into in certain parts and then each part now has a goal in a sub goal and then what I do is start from the first part such initially from the first sub part reach to the sub goal then start to the second part sample so writing a paper could be first going to the library and sitting there achieving that once I do that then reading something in the library then understanding something then finding a new problem then reading a little bit of what that problem then coming up with the answer of what the problem is then discussing that with certain professors or looking at certain materials that and finally writing the paper so the writing the paper is now broken down into several steps and similarly well this is what Neverland Simon did they said that larger problems can be broken down into smaller smaller problems and can be solved by step-by-step process from the **initial state** to the final say they did so within the framework of a computer program which is called the **general problem solver** or the **GPS** now what is the **GPS** the **GPS** is basically a model of human computer **problem solving** sorry human **problem solving** now one that can be applied to any problem so basically it is equivalent to computer problem solution so basically the **GPS** is a model of human **problem solving** which is basically made or which is basically developed out of the computer **problem solving** things now this approach approach minimize the distance between the **initial state** and the **goal state** by breaking the problem down into a series of sub goals so what I do is if a bigger problem is there I know the **initial state** I know the **goal state** and I don't know how to solve it I take the initial problem I break it into its part as I said making the writing paper problem into his part so first I'll go to the library first and achieve that thing so initial goal is starting from here final goes in which in the library once I reach the librarian see so many people are not motivated so first problem solved in terms of motivation of writing it then I will read some books the more I read books the more ideas will develop and when that happens when I have more a number of ideas with me I will then consult some professor or some kind of a TV program or some kind of internet websites where more information is given from there I could write a problem so basically this this whole writing a paper problem is not broken down into sub parts this is what it is said here in the **GPS** also I minimize the distance with from the goal - back to the initial problem by creating sub problems or by creating sub goals and it is easier for a person to reach from reach the goal through sub goals then to reach the goal on its own now the sub goal analysis is accomplished through the application of something called operators so how do you reach the sub goal though there are certain operators now what is operator is basically a fancy word which is out there for **problem solving** techniques so one subway technique is going to let's say the library and that there is one thing into it and then there is reading a book so how do you read a book there is a certain technique of motivating yourself and then there is another technique and so on and so forth so basically these problem-solving techniques taking a bigger problem and solving it require several techniques a bigger problem could be how to find out how do people process first **language** right so if people have dual **language** proficiency let's say I am a proficiency of English in Hindi now something is spoken to me in English how do I respond to this device take this thing take this word in English and search for solution to it in English or do I take the problem the

word which is being spoken to me in English translate back into Hindi search for the solution in Hindi then come up with the solution in English and then respond it back now this is the problem how do I break it in a several problem first is a behavioral part second is the cognitive part the third is the mirror physics part so first I do some behavioral analysis to see how the solution is and then another sub part is there so part of it looking at behavior response so kind of design problems in terms of several problems could be developed in which a certain kind of a sentence is given to you an answer is looked into into so several kind of things is then we look at the speed and **accuracy** of how you are doing it then we'll design a cognitive experiment or maybe a brain related experiment **EEG** ERP experiment where we look at the time process or the brain potential what happens in the brain when you are solving this problem and by combining these two things together these two subproblems I can finally say how do people think or what is the way because the behavioral will only give you the result what happens in the brain you don't know so you use brain imaging techniques to tell you what happens in the brain when this kind of problem dual **language** or single **language** people solve a problem so basically that is what it is and so these using those two techniques is called using the operators so these techniques are applied at the micro level to reduce the differences between the current state and the **goal state** so what could happen is since I have created a bigger goal I mean from starting from the initial point to the final point is as the bigger the problem says the bigger problem so is I'd not divide the **problem space** into subproblems space and so the I can use operator s both at the level of micro level which means navigating from navigating within the subproblem in the in the earlier case that i defined so i can use certain operators or certain so use certain techniques for studying only the behavioral part of it and the electrophysiological part of it this is called the micro level analysis and then to reduce the distance between the initial in the **goal State** or I could use a bigger bigger technique which is techniques of summation of results right reduction technique or some other technique by you through which I can take the two results that I have the behavioral result and the electrophysiological result match them together and game of the final solution so I can use operators both at the micro level where each part of the problem how each part of the problem is solved or sub problem is solved and then use techniques at the micro level of integrating these results together to give us the final solution of how do people who have two **language** proficiencies really think so **GPS** is a notion of **problem space** which basically refers to the problem solvers **mental representation** of the **initial state** the **goal state** and all possible intermediate sub goal states and the operator that can be applied to these so basically **GPS** talks about **initial state** a **goal state** and subproblems and certain operators into it so look at this definition or look at this picture this is how the solution will really look at this is an idea of a human **GPS** general problem system so as you look into it this is the problem solver this is the noticing or review mechanism external environment **long-term memory** immediate **problem space** transfer goals and constraints this whole thing is called the **GPS** now how does it really work this is my **GPS** system so this is my **initial state** and then I have so many inputs these are operators on to it this is my **goal state** what I do is I break my problem into several parts so this is the behavioral part this is electrophysiological part this is the cellular biology part of all surrender biological mechanism part where I look at certain cells of how they respond to certain neurotransmitters which will how they will work and this may be another imaging this so this is electrophysiology and this is a brain imaging and from all these I will combine the results together to come up with the solution of how people who are two **language** proficient think or how do they come up with solutions so basically then this is what it is this is how the idea offer **general problem solver** is really so in this today's lecture what we did was we talked about what is a problem first of all definition of a problem and we thought different different or definition features out there then we looked at a number of problems a number of types of problems which exist out there and we discussed these types one by one and after that we looked at certain approaches to solutions of problems we look at the **gestalt approach** the cognitive approach as well as the behavioral approach and we saw how these what these approaches has to offer as solution the problem in the upcoming lecture will look into how a problem is actually solved and then we

look into something called **creativity** in inside of how do they go with **problem solving** thank you

Lecture 24: Factors Influenceing Problem Solving

■Kind: captions hello friends welcome back to this lecture on **problem solving** now this lecture is a continuation from the previous lecture which we in which we saw what is **problem solving** what is the nature of **problem solving** and what is the problem comprise off and after that looking at what the problem comprises off we also looked at whether different kinds of problem to exist and we looked at Mayer's definition of number of type of problems which could actually exist so we looked at five different kind of problem and we saw how these problems differ from each other what is the nature of these problems and how do they differ from each other in what respects then we looked at something about how to solve a problem so we looked at two basic approaches to the solution of problems the first approach that we looked at was the approach that behaviorist took in terms of Skinner's idea of context idea of law of redness and how does a problem solve in terms of associations which are made between correct responses and the consequences of it so that was one approach to solving problem in terms of the behaviorist in terms of the behavioral psychology and the next approach that we saw in last class which helped us understand how to solve problem was in terms of the **gestalt approach** which talked about problem solution as reorganization of the items or materials available in a problem or so what this approach said is that problem solution is about reorganizing the problem elements in such a way that a novel solution comes up and so if they talk about something called the eye experience or the **insight** and so this is what in brief we saw in the last class and in addition to this we also saw the cognitive approach to solution a problem and there we discuss the idea of renewal and saya Simon's based on novels in Simon's idea of what **information processing** theory is the idea of a **GPS** the **general problem solver** and so this **GPS** basically then talks about how to break the problem into its subproblems and then arriving at the largest solution of the problem in terms of integrating smaller problems or inter finding solutions to smaller problems of the larger problem now what do I mean by this take a large problem break up into smaller problems then arrive at solutions for smaller problems integrate them together and then arrive at a final solution of the larger problem and that is what the idea of the **general problem solver** was and the **concept** of a **general problem solver** now what are we going to do in today's class now in today's lecture we will see what is a solution to a problem how does solution to a problem arises and not only that we'll also see what are the problems in solving a problem what are the hindrances which are out there in solving a problem so when problems or a solution occurs or when we try to solve a problem what are the kind of obstacles that we face and how to go about these obstacles how to take care of these obstacles in addition we'll also see the several factors which go ahead and help us are basically better or increase or decrease problem solution so let's starts to raise lecture by something called representation of a problem so solving a problem by using any of the ways either it is a guest artists approach or using a behavioral way requires the first step of it requires to encode a problem now as I said problem formulation itself or the idea of a problem itself is to understand some kind of hindrance some kind of a uneasy State and representing this uneasy state as a mental presentation is what problem is all about so basically the first step in problem solver solution is **encoding** the problem or basically making a **mental representation** of the problem that one is facing and so several problems or several hindrances or obstacles can arise at this very first phase where somebody might not be able to represent a problem or form a good representation a **mental representation** of a problem now remember in **long-term memory** one of the problems that we faced or one of the things that we looked into is something called forget and one of the rules of **forgetting** or one of the theories of **forgetting** was in terms of the fact that the problem or the information which was thought to be encoded in the **long-term memory** well they were encoded and that was basically due to poor **encoding** or due to non **encoding** of the problem **forgetting** happens a similar situation arises here one of the problems or one of the obstacles in **problem solving** could be at the very first stage of problem solution which is **encoding** or forming of **mental representation** and that is called problem representation

so let's go ahead and look at what is problem representation now **problem solving** it involves the process of converting a presented information into some type of internal **mental representation** so as I said the first step in the problem solution is to take up a problem or to take up the elements of a problem and then to do an actual **encoding** of it in terms of **mental representation** that is the first step so no matter what problem you face whether it is writing a paper or going somewhere or eating something no matter what the problem is first you need to understand this problem first you need to understand the related items in this problem or what is the problem the question that the problem is posing to you and then to go ahead and do exactly presentation or do some kind of **mental representation** of this into the mind now within the framework of the **GPS** problem solver a problem representation involves correctly specifying the **problem space** that we correctly identifying the **initial state** as well as the **goal state** and the operators which must be applied and within the constraints of the problem so basically then problem representation in terms of the **GPS** solver in terms of the **GPS** sequencing is about finding out the **problem space** defining the **problem space** or laying out the boundary conditions of the **problem space** so what we get to do here is that if this is the idea of my **problem space** problem representation basically involves defining the boundaries of this problem so this is my **GPS** solver this space and not only defining the boundaries of this space but it also leads us to understanding or identifying what is the **initial state** where is the problem starting and where is my sub goal so these are the several sub goals which are there this is my **initial State** or I guess I would write and this is my **goal state** GSI right so basically identifying this is what are the parameters of these is and what are the parameters or attributes of this **goal state** so what will the problem look like an **initial state** is what will the problem look like at the **goal state** also in addition to it a problem representation also involves finding out the constraints what are the constraints so given the fact that this is the boundary condition and these are the attributes of the initial and the final state the said there will be several constraints in solving this problem so what are the several constraints now one constraint could be how we can divide the problem into smaller parts so what are the constraints or sort of the rules into it and further to that it may also define about the different operators through a different solution for example this **goal state** or this sub goal may be solved by using behavioral approaches but this **goal state** may be solved by using the **gestalt approach** right or this cold state could be solved by using another kind of approach which is that and so each state or each sub goal here may be solved by using different approaches but then what I'm getting at final is this is the **initial state** of the sub goal and this is the final state of the sub goal so here I get the solution and then I take from this solution and this solution now becomes my secondary **initial state** and then I move on so this is what it basically involved so in in general what the problem representation really means is to find out the **initial state** of the problem face find out the **goal state** of what the problem will look like identifying the constraints the hindrances which is out there which are they're ready to block you for solving the problem and also in addition to that the type of operators driver solution mechanisms type of solution on analysis that you will use to solve the problem so basically this is what is problem representation for example look at this problem now I will give you one minute to look at this problem and come up with a solution let me see if you can come up with a solution to this so starting now I'll wait for one minute so coming back to the problem I hope that you are able to solve this problem most of you would have been able to solve this problem so basically it's it's the problem talks about two moms going up and down or one mob monk going up a particular Hill starting at sunrise and reaching there at sunset and then starting at sunset and reaching towards the downhill position from where he starts at the sunset so basically what the problem ask is to find out to verify whether there is a point which the monk is going to cross both while ascending the mountain and descending them now you realize that what is the solution to this do you understand what is the solution to this and so most for most people who start writing mathematical equations to solve this problem you will not be able to solve this problem by this and this is what I mean by problem representation which is basically understanding what a problem is and I am presenting you the solution the easiest possible way to solve this problem is using a visualization

and so what I'm going to do is show you a visualization of the menthe problem and so if you do the visualization like this again coming back to the solution so if this is the problem the solution is doing a visualization and as you see if I do a visualization these two people go up and down the mountain and so there is a spot where they are going to meet and so the easiest way to look at this problem is doing the visual visualization or visual interpretations of it and so if you use any other interpretation or nyrians or operator to solve this problem you may not arrive very easily at the solution of this problem right so this is basically what is problem representation to identify what the problem is and to understand the sequence to understand how it should be solved now I give you another problem try to solve this problem now give you 30 seconds it is a very easy problem I will give you 30 seconds to solve the problem and let's see if you can solve this problem right so here goes the time and so on more than 30 second over so basically then did you solve the problem and most of you would have been able to now what we realize here is that in this particular problem what we need is we need a mathematical representation so in comparison to the visual representation that we used in the last problem the kind of representation the kind of solution the kind of operator that we are going to use in this particular solution or this particular problem is the mathematical operator and so in this way the problem gets solved easily so if you do a visual representation of this you are not able to solve a problem and so this is what problem representation talks about it talks about that at the very initial phase we have to understand how to solve the problem or represent the problem in the correct format now rigidity in representation now problem solutions to problem what hindrances the problems or a solution may arise because people get rigid about solving a representation or people become rigid in representing a problem so the initial representation of problem is basically it's critical and failure to represent these kind of representations will lead to number of factors now one of the first factor is that sufficient **attention** has not been given to the problem so when a problem is presented to you and if you do not pay sufficient **attention** to parts of the problem or to the problem as a whole what would happen is that you will not be able to identify the key elements of the problem and this will lead to you identifying the problem itself and representing it in the correct form and once a problem is represented in the incorrect form a solution will not be most probably the solution will not be arrived at now the second problem or the second hindrance is that problem elements have not been understood what would have happened in this case is that even if you put some **attention** to it or more **attention** to a problem you did not understand the various elements of the problem or what is it asking you so basically it is like looking at a question and not understanding what the question is asked you and so some people make questions like that which which has a lot of so basically in most open-ended questions or open textbook questions you get the question in such a way that the question revolves around certain things but there are certain key elements that you need to realize and once you realize these key elements you'll be able to solve a problem and so another problem element or another factor which helps in **problem solving** or which hinders **problem solving** is not understanding or not realizing the various elements of the problem and what relation they have to each other the third problem that could arise or third obstacle that could arise is previous experiences similar problem may not have led to **encoding** the problem elements in a rigid manner so what would have happened in since some examples of the present problem and this is basically called analogous **problem solving** so when solving a problem due to past experiences what we tend to do is not look at a problem in a particular way but think of the problem in some fixed manner and that is basically a reason for fingers in **problem solving** so instead of thinking independently thinking about a problem independently realizing a problem or independent focusing your **attention** on to certain elements of the problem we tend to use a **heuristic** approach to it matches the problem to some already stored format of how it was all previously and that could lead to problem not being solved or it into incorrect representation of the problem now some of the major entrances to **problem solving** arises from something called mental said so what is it it is a tendency to rely on habits and procedures use in the past and this is what is mental said so basically if I know or basically if I think of solving a problem in a particular

way I tend to basically use that format and take a new problem and fit it into that format and if I do that what happens is the solution becomes difficult so this this basically idea this basic idea of using a **mental set** or using a previous habit of solving a problem in particular may leads you to leads you to incorrect representation of the problem and thereby leads to solution other incorrect solutions of the problem now look at a question here and so this is the problem which I asked so there are six eggs in a basket six people take one of these ejects now how is it that one egg is still left in the basket now most of you would worry about the fact that how does this happen but then hidden within this is the representation problem and mental said says that six people took six different takes so how is it that one the one egg is still in the basket the solution to this region you have to realize that was the last person who took the egg also took the basket with him and then it is possible that the last egg is still in the basket and what when when you to think of this in terms of six different people the mathematical variable then you think six people going away taking six different eggs and don't realize that the sixth person can also take a basket with it and so there is whether **mental set** is all about so the mental said they tends to affect the representation phase of a **problem solving** as past experiences leads to inappropriate representation of a problem so what mental sets intends to do is that as I said there are certain past mental representations and these mental representations are these inappropriate ways of representing a problem actually makes you think of a problem a solution in a particular fixed way and that leads to incorrect incorrect solution to that particular problem another factor which is of importance or which is of any importance to solutions of problems or in terms of representation a solution to the problem is something called **functional fixedness** what is it please what does it mean it refers to people's tendency to view objects in a narrow fixed sense right in terms of the typical functions of an object so in **functional fixedness** what we tend to do is that we just look at something in a particular way in a particular manner so a spoon for example should be looked at or is looked at only as a spoon something which is for eating we do not think of a spoon as a lever and so what we can do is you can use you can think of a spoon as a lever and then use it to raise the the upper part of a team or if you Tim can and there is a lid to the tin can we can wedge the spoon between the tin can and the lid and apply pressure to it basic physics basic simple physics and that can help us in opening the lead of a tip but then we don't realize it and **functional fixedness** says that a spool is thought of as a spool and so it should not be doing anything else or using the down part of a toothpaste the bottom of a toothpaste as a tongue cleaner nor do - I'm sorry tooth brush and so the tooth brush is thought of as the tooth brush and so we do not think of it as a tongue cleaner and so but there are certain toothbrushes are available and so **functional fixedness** what it says is that certain objects are seen as that object are having a fixed function and it cannot be seen as having another function or more functions out of it so one development perspective says that functional for a fixed Ness they pick up or they start or they hold on to people as they grow older in age so younger children are more flexible in terms of **functional fixedness** they can they are not harmed by or they are not troubled by **functional fixedness** but people as they grow in age they become this narrow sense they approach at this narrow sense of how a functional or water functional fitness fixed nurses now look at this particular problem this is called the two string problem now I will give you 30 seconds to look at this problem and tell me how to solve this and I can bet you that due to **functional fixedness** you will not be able to answer this question and though once you are able to see this problem pretty sure that you are not able to solve this problem the reason here is that **functional fixedness** the certain items which have been mentioned the chair the plier and all those have been represented as represented in your **memory** in a fixed way and so you are not able to solve this problem but a solution is there of this problem and this is the simple solution what cart has to do is to climb up into the chair hold one end of the string onto the other end of string he has to tie the pillar make a swing out of it go and stand on this chair and then basically hold on to this string now if if you put a pillar here it will come back to you and then you are holding on to this and so you can go ahead and then tie it together that is what is so he has to recognize that he can use the pillar as a novel function as a weight for the

pendulum as I said this is the swing that it will do so try the pillar here it will work function as a pendulum and you can always go ahead and solve the problems and so what what people don't understand is the pillar is used the plier here is used for mechanical things for basically use or tightening of screws and all and so you don't realize that the player can also be used as an object for swinging or for making a pendulum and that way you can tie up that and so this solution problem to solution arises from the fact that you are not able you are functionally fixated by certain functions of these objects now another interesting thing the basic problem which can arise with solutions of problem is something called stereotyping so Serio typing threat occurs when a member of a negatively stereotyped group fuse that the stereotype might be used as to judge their behavior thus resulting in a negative **judgment** so here in this kind of a problem solution or present problem representation what would happen is that since a certain groups have been stereotyped in certain way they are not able to register the problem or represent the problem in a correct way and so problem the problem if not represented in the correct way could not be generated and so Queenan Spencer 2001 they use men and women to solve math problem by using either word word representation of it or a numerical representation of it and this is what they form so this is the kind of problem that is out there and so this problem represents a statement or basically it's a word problem and so this kind of problem was given to people know so this is a word problem it can this problem can also be represented in this particular format so either you represent the form of the problem in this format which is now a word problem or I could use as oblique problem or an equation problem or equation format of it so two representations are that and so it was believed that or there was a need or this study basically proposed that women would solve more of numerical problem then in in word so what are the same problem in terms of what when it is related in terms of word for then when it is related in terms of an equation and this is the results from Queen and Spenser study it was found that the males and females so word problem more males there and less males were able to solve the word problem and more measures were able to solve the numeric format so this is the numeric format of the problem and this is the word format and in terms of the fact that now it was proposed that more women would be solving the word problem more women would be about ending the word problem and solving it much easily but the solution represents that only eight women out of the total number of people which has been given this solution only eight preferred the word format for it most women for men or they use the numerical format or they applied the numerical format and through the numerical format they were much easier for them to arrive at a solution and so this is basically Jindal industry typing are basically stereotyping so basically due to CBO typing initially it was believed that women would be more comfortable in solving the words representation of the same problem but then this is not what the solution is and so this is another reason for representation problem representations or obstacles in problem representation not solutions to problems now once the problems has been successfully transformed from an externally presented information into an internally presented information the next phase of **problem solving** involves searching for and testing evaluating of solutions so once a problem has been encoded in the right format a right representation of it has to be done now a **problem solving** process has to be evaluated or it has to be searched off how to solve this problem now within the context of Numan enzyme is 1972 IP approach **information processing** approach from the solution amounts to traveling through the **problem space** so basically in terms of problem representation it is identifying the **initial state** what is the **initial state** of a problem so in a **GPS** solver in a **GPS** prob **problem solving** space the problem representation is basically at this point where we need to identify what is the **initial state** of the problem or how it the problem should be represent in the **initial State** whereas the solution to a problem requires us to move to this state from the initial to the **goal state** and realizing how to move from the **initial state** to the **goal State** now there are several ways of how to solve a problem and the first interesting way our first important way of how problems can be solved is using the algorithmic approach or using algorithms so what our algorithms then algorithms are a set of rules that can be applied systematically to solve certain type of problems so these are a set of rules these

are a certain list of rules which when applied in the correct format which when are applied in the correct sequence or which applied which one applied sequentially will finally always give you the solutions so that is the best part of an **algorithm** an **algorithm** is a set of steps set of steps bounded by rules now once you use these set of steps once you pass through all the set of these steps which have been there then you really finally arrive at a solution and that is guaranteed so almost all gorillas guarantee a solution to it right now mathematical formula is a good example of an **algorithm** so you have to go through the number of steps which is there a formal solution you cannot jump and if you go through a number of steps you are always going to solve the problem and remember from a real classes of how an **algorithm** is written I'm terribly sure that most of you had mathematics till plus 12 and so if if that is what it is then you do know what in **algorithm** is so basically then how to define an **algorithm** for making T the idea is that you have to describe a step two step process from right from taking water into a pan going deleted or adding something into it and in the step-by-step manner you will always end up by making that T so **algorithm** is a set of sequences or a sequence of steps which when followed in the correct order it always guaranteed a solution and so one way of finding solution to problem is using the **algorithm** and so the tool for example look at this problem two short sides of a triangle at three centimeter and four centimeter so what is the hypotenuse it has to be 5 so a square plus B square is C square that is how we look into it and so then this way it has to be 5 centimeter and so this is an **algorithm** that is a correct way of doing it and so we know that this is how it has to be solved so algorithms are very powerful problem-solving techniques applied correctly an **algorithm** is always giving you the correct solution if there is an existing look at this initial problem that we encounter now if we need an **algorithm** to solve this problem there will be several obstacles or there will be several states or there will be several steps into it for example what would happen is it will lead to since it is 2 3 4 5 6 it is 6 so to lead to basically 6 factorial number of representations which is there now what do I mean by this what will happen is that initially I will just change the last two letters so this the next letter is n I then I will change the third letter and so one letter at a time I can upon changing and then arrive at a problem this is basically using the **algorithm** approach so in **algorithm** approach all the representations are all changes within this letter all arraignments different arrangements will will be there and from that final solution will be outputted now if you do that you would have to do these many number of representations of these many number of variations into it to finally arrive arrive at the answer that this will lead to whitening as the answer now there are several shortcomings to using the **algorithm** as I said now if we requires these many steps if it requires these many variations which are out there if it requires us to do these many variations for arriving at a solution then it is a it's a huge task now when it is a huge task so basically then there are some shortcomings to it and so what are the shortcomings first the exhaustive nature make them overly Tydeus in quite impractical what would happen is since 6 sec toriel times doing it these many times doing all representations all for all number of variations on this so each letter being varied in not only in relation to itself but in relation to all other letters will be huge and it is to exhaustive and so it will require basically a computer for you to do this and so if but but if you do this the correct answer is one of these sequences and so one of the problems or one of the hindrances or one other shortcomings of using the algorithmic approaches that it it is very tedious in nature it requires a number of variations and so it is quite impractical nobody has that much time to go up and look at these many variations no simple code breaking password breaking how do you do it one of the ways of doing a password hacking is basically using or to basically do ethical hacking is to use some people the brute force now in the brute force what happens is all combinations that exist in to the dictionary is used by the computer to arrive and the password now if we do that and if it is assuming that it is a seven-letter password then we will have to do almost innumerable number of representations right from a in the first place to a 26 different representations of the first place 26 different representations a seventh place second place till the seventh place and then in relation to each other also using each representation so huge numbers and so this is one problem or this is one shortcoming of the algorithmic

approach what is the second shortcoming there simply aren't any for most of the problems in our daily life so what would happen is is algorithms are the best way to come up with a solution and they always guarantee a final solution what it turns out to be is that for most problems on our daily life we don't have an **algorithm** out there and so we don't have an **algorithm** out there then solutions are difficult and then it becomes difficult for us to come up with a solution because there is no **algorithm** there is no true method of doing it why because each new problem daily life problem is a new one it is not an exact replica of some problem that we have seen and so that is one thing another interesting way of coming up with solutions to problem is using a theorist now what is a yearís these are general strategies or rules of thumb that can be applied to various problems so basically then one way is using the **algorithm** another way of solving a problem is using a shortcut and this shortcut or mental shortcut or rule of thumb is called a purist so this serves as a shortcuts through **problem space** what Urist are it and these are fixed ways of solving a problem or these are fixed ways of how do we go ahead and bypass the all great **algorithm** some steps of the **algorithm** so these are water Kuras so given the strength and limitation of human problem solver along with the fact that most problems are ill-defined they are relatively large problem spaces **heuristic problem solving** is more effective the reason is since the **problem space** is huge since any problem requires a number of big **problem space** and a number of subproblems into it what people tend to do while solving problem is using a theorist what does the here is do it gives you some kind of a shortcut so what is the shortcut it is in terms of bypassing certain types of the **algorithm** now since we know that if we come to know that two three four or five of an **algorithm** always gives a particular answer for example mixing yellow with green gives a particular kind of color and if we know that we need not repeat this each time or mixing the Tilly's into boiling water and boiling water always gives a certain kind of color or a certain kind of taste and flavor we need not repeat this again and again and so we can jump by stating that if you mix tea into it this is the kind of color that we are going to get and so this is basically what is called the **heuristic** approach of problem-solving for example how do we solve the same problem so in terms of algorithms we saw that we'll do n number of manipulations we'll do these six the different letters are there and we can manipulate we can do all number of arrangements out out of it and then come up with a problem a good purist of how to solve this problem is that thinking of gkv k and i KN are not possible so English words do not start with this kind of beginning and so we can eliminate these kind of or these different starting with it and so these are eliminated and we can sell since use more huge and then come up with the final answer that VI ki ng is the final solution because this is the URIs it says that if in your understanding or in your in your life you haven't seen in your readings you haven't seen any words starting with gk v k VG or i KN and so these can be eliminated this can be ruled out also look at this particular things now it's a daily shopping experience and if you do that basically or if you would think that you are shopping what most people tend to do is something called roundup and so one easy way to solve this problem is rounding this to 20 and rounding this to $30 and rounding this to $30 so 30 30 60 70 80 is not answer so $80 is what you are spending but already the approach says is that we have to add up 9.95 0.98 0.97 generally speaking most people will say that I spent up eighty dollars so if you're purchasing three the item it is $80 which is then what I have done is I have rounded up and this rounding up is called the nearest or this is the problem or this is the mental shortcut so the fastest way is to basically go ahead and round up and this is what is called the **heuristic** way of solving a problem a third way of solving a problem is by using a **means-ends analysis** now what is a means and analysis now basically the **general problem solver** which was done by Newman in Simon it basically utilizes a URIs known known as something called **means-ends analysis** now what is the MA it involves basically breaking a problem so if this is my problem state **initial state** and this is my **goal state** and what then happens is that means ends analysis says that I take this problem and draw a break into four different parts and each of it which is solvable and then what I do is I then this becomes this part then becomes my **initial state** one and this is my **goal state** one so I first think of solving this part of it and then when this **goal state** now becomes my

initial state to problem two I solve this part of it then I solve this part of it and then I solve this part of for example think about launching a rocket now if I want to launch a rocket this is a huge problem now I can then divide this rocket launching into several steps for example thinking about the physics of it thinking about what material should be used what kind of fuel should be used so they would do people who will calculating the fuel calculating how the rocket moves up calculating things like what what is the material that should be used and so several people are this launching or rocket can be broken down into several types rise from the physics to the chemistry of it the physics of it and several other systems in which should be tearing through the electrical appliances and so on and so forth now I solve this one part one part by one part and then I come with the final solution or a simple thing that if you have a car it is not starting now if the car is not starting brake this starting of the car into several sub part for example check the key checked some the ignition check the battery and so on and so forth so several solutions are there and I arrived from one one solution say to the nurse solution says the first step is to raising up the bonnet to identifying where the battery is to testing the battery to basically then testing some other facts the electrical circuit stressed in the fuse and then finally arrive at solution of this problem how to start a car if which is not starting so then the MEA it involves breaking up the problem into smaller several sub goals in which accomplishing each sub goal moves the solver close to the final goal of the problem solution as I said if you are tailed the if you solve this huge problem if you make the shield problem into smaller problems and solve it one by one or solve parts of it one by one you will finally arrive at the solution as the term implies in MEA the solver systematically attempts to devise means to get to the each sub goals end ma can be effective way of transana the effective way for solving a transformational problem now look at this problem and if I want to solve this remember this is the solution to it thank so from state one where it was here to state to where it was here there is a solution to it and you can see the solution so on the right hand side I have the solution so this is how the solution really looks like from moving from this state to this state this is the number of solutions and so it realizes or you have to realize that the problem has to broken down into sub parts there are certain constraints and so if you break the problem into sub parts you will be able to solve this problem as it is now when you actually another way of solving a problem is using analogous problem right and so what we do in analogous problem is that we think of a problem when reaching at the solution of a problem we think of a problem which is similar to the problem which is at hand and then we tend to borrow certain features or we tend to look at the similarities with the problem which is at hand and then think of the problem which which we are thinking about which is represented in our mind and then solve it for example I'm giving you a problem now the problem and I will give you a solution to it so the problem is that there is a person and he has to be treated with lasers now the fact is that if this person is given a high-voltage laser a treatment what would happen is or a high-energy laser treatment what would happen is the single focus I mean single ray of laser light is given to you with high energy what will happen is the organ which is diseased in in in this case that will disintegrate or it will burn up and so what is the solution or what is the solution to this problem how should we treat but then this organ can take in small or can only take in small laser energy values right so it cannot take in a high-energy laser it can take in small energy laser if you if you give in a high energy laser from one direction what will happen is the instead of curing the problem instead of curing the cell which is disease the whole organ will burn up so what is the solution the solution here is using several lasers which are low powered from several different directions now what will happen is the since the whole organ is not surrounded so it is basically in this case what has happened is if I give one laser with a high value high value laser this organ will burn up so what I do is I use several different lasers which are smaller in energy from several sites so then since it is each laser has a very low value it is not going to burn up the organ it is only going to hit upon the particular cells which are diseased and also the fact that the total energy of all these lasers will be the energy which is required to one of their cells so I can then distribute it across no analogous problem is there is a military officer who was going to go ahead and capture a particular fort now the only problem is the fort is bounded

by certain bridges now he require a certain amount of army and certain amount of tanks to be taken before fighting the war but the fact is that each bridge is going to or each breach has a certain limited capacity of carrying the track now he has to take ten tanks and but each bridge can only take one tank and a few soldiers onto it so how do how does he go about solving this problem and all of us - what we have though if he can relate the his problem the major or the commander of army can relate his problem to the problem of this this particular laser thing what he could do is that he can use several bridges and each bridge will now hold one tank and then each of it will arrive at the enemy fort and can actually win the war and so this is how analogous problem or analogous **problem solving** really works now in analogous **problem solving** there are two things to look at once in terms of comparison one is the company called surface versus structural features so what are the primary steps in solving problems by using analog involving the use of mapping the original problem - the new problem basically what happens here is that in terms of what we saw that you have to understand the commander of the army has to understand that how his problem winning the enemy fort is similar to the medical problem that I defined before so he has to go ahead and map these two problems together or basically make a one-to-one relation between his problem and the problem which Virtua which was the medical problem now yeah can her geek and holic they termed this mapping process a **schema** induction he says that this mapping or this identification of one-to-one identification of the elements of the medical problem and the elements of this war problem how they are related together is basically called **schema** induction now I ability to notice map and develop schemata depends on particular type of **similarity** so we look at similarities and the more **similarity** a particular section of the problem particular part of the problem has to the new problem which is out there the more similar the more easier it is for us to solve now these two can be of two types so we look at two types of similarities one is called the surface feature similarities now in the surface feature similarities what happens is specific elements or the problems are Mac if two problems sure they share specific surface features we believe that it is same and then we go ahead and basically map the problem together for example paper jams know if there is a printer and all of you are very familiar with what a printer is so if you have a printer at home and it it gives up paper jams then if I take you to a new printer and if I introduce you a new printer you will be more or less similar you will understand that the back of the printer is there the roller is and so that is where the fact where paper jam is and so it will take you very less time because most printers work in the similar way it uses the same kind of roller mechanism and so you will always look forward for where is the back button or owning a printer and so you know that the the central switch will be big and it will give you the affordance of pushing down and so no matter what printer I am giving you with the idea that the most printers require this kind of this kind of build up you understand that pushing this button is what is going to help and so this is called surface **similarity** mapping in structural features although the surface features although two problems may not look like each other but then there are certain unique features are the certain meaningful features within the problem itself which would look identical and that will make solution of the problem easy and so in the case that we described below the commander has to realize that although the other problem was a medical problem but the same kind of solution do exist for example in this case in understanding the structural features two problems are structurally similar they may look quite different on the surface and underlying similarities would be there for example looking at the two sentences one is active wise the other is passive wise if we look into it the dog chased by the dog chased the cat or the cat was chased by the dog both of them although structurally they are different but meaning wise in terms of the meaning in terms of the **semantics** of it they mean the same thing and so both the problems are structurally similar now another interesting thing to look at is in terms of how do expert solve problems so experts have something which are called expertise which are exceptional knowledge performances is a certain specific problem domain area so now it's Ron's 1950s experts are now viewed as the product expertise is basically seen as a product of innate capabilities but nowadays in today's view expertise is seen as more of a skill memorizes so according to Erickson and

Paul since 1988 skill the **memory** theory there are several differences between how an expert or how unknown novi solves a problem first experts have highly rich semantic networks it basically means that the semantic network the thinking network the network through which they solve problems are highly developed or highly connected with each other experts have quicker and more direct access to long term **memory** so most experts can easily access long term **memory** because they have done this **problem solving** this kind of a problem over and over again and so they have quick access to long term **memory** then in terms of what our nowise could do and information is more easily encoded onto **long-term memory** by experts and the speed of the **encoding** improves it this now if for an expert he can understand quickly the problem the the parts of the problem quickly represent the problem in the correct format and quickly find out the **problem space** also find out the correct operators and can solve it for novices the problem is the problem arises the hindrance either right from the very beginning where the representation of the problem as reacts as a hindrance and even if the representation is there they might not find the correct operators and so this is the difference between an expert enough nowise **problem solving** now what are the advantages when expect problem solver the core of the **problem solving** is **memory** the **long-term memory** that allows for storage know of domain knowledge and general information of specific episodes that **working memory** there allows for quick and efficient online processing of the information so basically studies by Chase and Simon 1973 and DeGroot 1988 at a point that experts can instantly recognize problem configurations based on extensive knowledge and experiences so what experts tend to do is they quickly repair understand a problem when they look at a problem they are very easy in understanding or they very quick in identifying the problem configuration which means that problem representations is very easy an expert they will very quickly identify a problem now Ericsson and Creech in 1995 they proposed the idea of long-term **working memory** to explain expert advantage in in online processing of problems now the theory states that experts can bypass limits of the **working memory** by using the information or the **working memory** to direct access to LT M so what experts can do is they can bypass these **working memory** storage capacity problem and they can using the **working memory** directly access the **long-term memory** and so this is the way how experts are better also expert problem solvers use general strategies that differ from those are the no wise release to searching the **problem space** in a forward fashion then reaching from the good so what experts tend to do is they when given a problem they don't think about the **goal state** and and travel back from the **goal state** to the problem state what experts tend to do if they use novel strategies they use effective strategies and they always move in a forward direction from the **initial state** to the **goal State** and that is why they are faster in solving a problem then coming from that goes straight to the problem state experts are much better picking up on structural features of the problem and recognizing analogous problem so experts are not fooled by the surface feature of a problem they always look at the meaning they always look at the at the conceptual level how a problem is different from some other problem and so they are easy they are easy and understanding the structure these structural problems and identifying analog of solutions to it and Lamar and Ziegler in 1995 proposed that adaptive strategy model which distinguishes between experts and Noailles and what does he say the strategy existence experts have more number of strategies in disposal since they have solved problems in so many ways or they're solve so many problems they have something called the newer strategies or more number of strategies at the disposal then then no wise also strategies based based rate experts in a given domain of knowledge which strategies tend to work and select those strategies so in a given domain of knowledge for example looking at math expert they tend to know which formulas are going to work will formulas are not going to work and so there tend to be very choice e in using those strategies or using those formulas which are going to work and not those formulas which are not going to work so they don't do foolish acts what they tend to do is only those strategies are used by experts which tend to work and they use those strategies over and over and the third thing is strategy choice refers to express advantage in discerning which strategy should be chosen for a specific

circumstances so they're very quick in identifying which strategy should be chosen over which other strategies and how these strategies choice works with in circumstances work with environment so what is the relation between environment and statically and they're very quick in identifying which strategy is going to work and which strategy is not going to work or which strategies to choose and which strategy to not choose and for this strategy execution they first do that expert advantage in actually carrying out the strategy in terms of speed and **accuracy** so basically even if they know our strategy no wise would not be able to basically implement this strategy with speed and **accuracy** but experts what they tend to do is since they have used strategy a number of then no war strategy is going to work and they have a number of strategies at they use they also can execute the strategy fast with higher speed and IR **accuracy** and that is how they differ from the otherwise but then this is not the only thing there are certain disadvantages also for example nowise are better at understanding randomly arranged problems and experts know I can focus on two problems or elements of the problem which experts will not focus on since experts are always directed at solving a problem they always look at those factors in a problem which require a solution or which give them a hint towards the solution and so they tend to overlook other features or other answers in a particular problem also nowise remember more information about specific problems and experts and this is termed as immediate amid intermediate effect now what happens here is that no wise since they look at a problem study the problem wholly they understand all parts of the problem right what experts tend to do is pick up only those configurations which need solution or which have solution and so one once you ask both the expert as well as the Noailles about basic questions relating to what the problem was no wise will tell you more about what the problem was and not the solution faster but experts will not be able to tell you what the problem the overall problem was they'll only look at features of the problem and how to go ahead and solve it and experts may actually we saw actually solve problem using **mental set** as compared to no wise users no experts make this **mental set** of a solution of how a particular solution should arise our particular problems will be represented in solution should we approach that and they use mental said whereas no wise don't have these **mental set** and so they can have varied solutions they come up with multiple solutions although if not correct it may not be correct but they can come up with these things and so they are different from experts and so in what we did in this today's lecture is we looked at the different type of problems solutions which exist what are the problem hindrances to the problem solutions which are there need a comparison of various **problem solving** techniques and at the end of it we also looked at what is the difference between or how experts and novices before with each other in terms of solving a problem thank you

Lecture 25: Insight and Creativity

■Kind: captions welcome back to this lecture on **problem solving** and this is the third part lecture and **problem solving** it's a continuation from the first two parts that we did know in the first part of **problem solving** lecture that we did we looked at what is a problem and we defined definition of what **problem solving** looks like so basically **problem solving** is higher order cognitive function a higher of cognitive process which starts after the basic ugly processes of **perception attention** learning **memory** has taken place and so what **problem solving** involves is to solve problems that have been or to solve those mysteries which have been generated through the other cognitive processes so basically making meaning of those mental representations which have been made through the other processes now on the first lecture we also looked at what are the different types of problems which exist so what are the nature of **problem solving** and what are the approaches that we use in **problem solving** and so we looked at two basic approaches to **problem solving** in the first lecture the first was the behavioral approach where we looked at things like something called the law of effect where we looked at how behavior is conceived **problem solving** and so what behavior is tend to say is that **problem solving** is basically process where number of acts are done a number of trial and errors are done and these trial and errors then leads to solution and so **problem solving** is basically understanding these assists are making associations between the correct act leading to the correct solution so basically it sits in terms of the law of effect which basically says that how associations are made between the correct act and the correct solution or the correct output so that's that's what problem-solving is all about and we define Skinner's idea of how he used cats and and understood problem-solving as a trial and error process and what he said is problem-solving is all about generating solutions by doing an act over and over again and then finding the right solution to it now the second view of problem-solving was the guest artist view which did not support what behavior is said in terms of what problem-solving is what they supported or what they went ahead and looked at is problem-solving is more of reorganization of the mental representations which are created by the lower quality process now when I say lower cognitive process and higher cognitive process we look into all those cognitive process or we classify all those cognitive processes it starts with **perception attention** learning problem and **language** and then **memory** all those processes are called lower cognitive process because these pregnant cognitive processes may eevee presentations so they look at the problem they look at any any kind of information which is impinging on a neural system and they make mental representations higher art of cognitive processes are problem-solving **reasoning judgment** and **decision-making** so these generate solutions are these acquire solution by manipulating the mental representations so that's one distinction which is there so basically their guest artists believe that problem solution arises by restructuring the mental organizations were manipulating the mental organizations and they believe with eternity that there is something called an inside the idea of an **insight** which leads to mentally present which manipulations are mentally presentation in the right order with generates the solution and so this is the some of the things that we say on the first lecture in the second lecture we moved ahead of it and we saw the various solutions that exist to mental locks in **problem solving** or hindrances in **problem solving** and so there is have some of these solutions to define an analogue he using **problem solving** hindrance sorry using heuristics as ways of **problem solving** or using the algorithmic approach to solve problems and so these are the some some of those solutions issue a described in the second lecture on **problem solving** of what are the different ways of solving a problem so we looked into those and we did a comparison of these and we looked at which is the better one and which is the non better when we solve some of the problems which is out there we also looked at problem representations as challenge to **problem solving** so what are the factors which affect **problem solving** so we saw some some basic challenges in terms of how problems re presented and some of the representation problems basically what representation of a problem means is at the very outset of when a

problem arises how do we encode the problem and so we looked at those processes also and some of the challenges in prominent representation is **mental set** or **functional fixedness** and our stereotypes is something else that we saw in the earlier classes on **problem solving** beside that looking at the challenges to **problem solving** in terms of problem representation in terms of the problem process we also looked at problem solutions or **problem solving** as a **GPS** device a **general problem solver** so basically how humans solve problem and how it is equivalent to computers or machine solving problem so we looked at that and we looked at the **concept** of **GPS** the **general problem solver** and we looked at how this solution really works or how does the **GPS** really works in addition we also looked at something called expert versus nowise **problem solving** we looked at how experts solve problem and how no Isis solve problem and what is the relation to an expert problem solver and no ice problem solver I saw that experts use a number of strategies different strategies the right combination of strategies they also know when to use what strategy and also know which strategies work in comparison to what NOAA and that is why they're called experts they also tend to no problem configurations but then being an expert in **problem solving** also has its downside and so the wrong side was some of the obvious problems that expose don't tend to know is they don't grapple or they don't go ahead and register some of the basic obvious factional problems since experts are focused on to solving a problem they are not aware of several other problem facts which is out there and so they tend to miss on some of the facts in in terms of in comparison to no vices so that is where we ended the last section now given the fact that experts are not able to conceive some or some of the basic information or some of the extra information additional information which is given in problems it would suggest that experts would be poor in **creativity** they would not be able to come up with creative solutions since experts are at most of the time engaged in coming up with solutions which exists from coming up with true solutions of a problem it could hinder **creativity** is what the general notion would be out there but as you look into it you will find out that this is this is the opposite of it it has been found out that experts with with the very fact that since they can come up with solutions or come up with true solutions they are very good in **creativity** so let's discuss that here so in this section what we will do is we look at and look into what is **creativity** and we look also into something called process of **creativity** and look at how **insight** helps into **creativity** and that is what we'll be doing in this section so how do we define **creativity** what is **creativity** and how does creative products occur what do you call **creativity** so is it and it is is it that somebody has long hair looks like an artist is a creative person or somebody giving a solution which is very different than what most people would think about is a creative solution what is a creative solution and how do people attain **creativity** we often define people in terms of creative inputs in terms of this is a very creative person and this is a very non creative person kind of a thing and so people who are called nerds are known to be non creative in comparison to people who are out to go out of the way to come up with solutions or think out of the box to come up with solutions and we call these people as non-trade so in this section we'll try to see whether there are certain personality dimensions to **creativity** and what is **creativity** basically what kind of solution to a problem is called creative solution so basically well as 1926 he proposed that the process leading up to a creative breakthrough can be described in terms of four stages so what he says is that creative solutions or noble solutions of any problem it it requires four steps it goes through a process of four steps by the way **creativity** is generally coming up with solutions which are not only more informative but it should be also workable so creative solutions are those solutions which are not only distinct it shouldn't be that these that the solution that you are coming up a solution that you come up to a problem is called creative when it is distinct when it is different from the general solutions which are already there at hand it that's not the only definition now that's not the only the inclusion of what a creative solution is it should also be informative it should also be giving some kind of results it should be result oriented so not only solutions which are different are called noble solutions are called creative solutions the solutions in terms of novelty should also vary in terms of in four informativeness which give means that it should give you more information it should give you a wide

variety of information and it should not only solve the problem it should solve the problem in we and that is not pretty solution sir so then what Wallis's says is that there are four processes in **creativity** or there are four steps in **creativity** and the how are these steps really related so the first step in **creativity** is something called the **preparation** stage and in this particular stage the solver gathers information and makes initial attempts to solve a problem so given the fact that a particular solution is given to you a particular problem is given to you for which you have to define a solution at the first stage you will look at all the information available and this is the stage where problem representation happens you look at the problem at hand you understand what the problem is try to understand what the problem is gather basic informations or basic bits of information from the problem make mental representations of it and then mentally represent or mentally make a map of the problem at hand this is the **preparation** stage and you come try to come up with initial solutions to this say so in the **preparation** stage this is basically representation problem representation stage where you look at a problem gather basic information from problems at the **incubation** stage the which is the next stage what happens is a period of productive inactivity happens so what well in valid systems the first stage is **preparation** where you gather all the information from a problem and try to get initial solutions then a second stage comes in which is called the **incubation** stage where a productive inactivity happens so here people don't respond or don't give up any solution it is basically they are sitting and thinking over it that's that's that's the thing so in terms of colas monkeys an experiment we saw on the first lecture on **problem solving** in terms of colas monkey this four stages can be described so the first stage when colas monkey monkeys we're trying or attempting to get the banana which was on the ceiling and they were doing all kinds of fact was the **preparation** stage where they were trying to find out what should be done or they were trying all their maneuvers all their knowledge into getting the bananas the **incubation** stage was the stage where most all the monkeys or the monkey today trying for the banana this sat down and they started thinking this is the **incubation** stage we're productive in inactivity of productivity happens a period of relative inactivity happens now what happens here is the the what the monkeys were doing is they were mentally restructuring the problem they were looking at what was available they looked at whatever is available to us in in this particular cage and how to reach the banana the third stage is called the elimination stage a stage where the solver arrives at the solution or a critical **insight** this is the stage where you first get the solution or the first appearance of the solution happens so after this is this **incubation** stage an elimination stage happens in the elimination stage whatever happens in **incubation** all those mental processes which happen during elimination they release an **insight** or they give rise to an inside they give rise to a probable solutions and the four stage is a **verification** stage in which the observer or in which the person who solving a problem takes up the solution or the accept the **insight** which is developed from the **illumination** stage or which arises from the elimination stage and verifies this solution or **insight** with the problem and tries to come up with the solution so basically then two important stages in this step that is the **incubation** and elimination so I don't gain corresponding back to cold as monkey the **incubation** stage was the stage in which the monkeys are not doing anything and just thinking or restructuring the banana problem and the elimination stage was the stage where the monkey who was trying the heart he came up with the solution that piling the crates one over the other another could be like a stair or it could give us some kind of an height and that was the elimination which he required and the **verification** stage was the stage in which he actually went ahead piled all the crates which were available into the cage stood into it stood over over the last and got to the banana so these are the four stages which a well as defined since these are the four strangers in problems audition as he says appears now Wallace theory has not been accepted in in entirety however it has been pointed out that to major this theory of malice has not been taken out very nicely by cognitive psychologists they they have not accepted it and as a hands down maneuver what they have done is but what they believe is this these stages do arise but from what well as the study comes out is two important questions on **problem solving** first what is the nature of inside right so the solution that I am talking about

the solution that wallahs talked about in the third stage of elimination that **insight** what cognitive psychologists wanted to do is to find out what is this **insight** and what is the nature of it what are the barriers to it what are the various factors to it or it doesn't improve or does it decrease under certain circumstances and what does it look like so that kind of thing they were interested in and second is a sudden breakthrough in **problem solving** is a reality can these be encouraged by a period of **incubation** so the second thing they wanted to know is that buying increase in the **incubation** time can we come up with more rigorous solutions or more noble solutions right so we wanted to establish this relationship between the time spent in **incubation** and the kind of **insight** through nature of inside the kind of **insight** that I generate the quality of the **insight** that I generate and the time I spend in **incubation** they wanted to establish a relation between that they wanted to understand if increasing or decreasing the time of **incubation** leads to and generation leads to the more certainty of generation of the **insight** which further solves of the problem so these are the two questions that most kinetic psychologists generated from Wallace's study so basically then what is an **insight** let's look at what an **insight** is so **insight** involves basically making this realization that a problem solution occurs and what was the key idea needed for that solution and the problem with inside is that theories believe that **problem solving** to be an incremental process and not a sudden realization so the idea that **insight** leads to **problem solving** is the fact that this **insight** which solves a problem comes out all of a sudden right and so what most theories believe is that this kind of realization of all of a sudden of a solution does not exist rather what most cognitive psychologists talk about the theorists believe is that solution to problem comes by trying over and over again or doing something called and trial and error so **problem solving** is more or less like come at doing acts towards attaining the final goal or incremental acts or incremental behaviors which successfully leads to the final goal another problem lies with the clear definition and non experimental proof or the existence of inside the **gestalt** psychologists so guest stars were the first people who came up with the idea of what an **insight** it and they said that this **insight** is basically comes out by restructuring of the mental representations which are made of the problem and so the non-existence of experimental proof of this **insight** is another problem which exists outside so basically then the idea is that this **insight** has to be tested and so there are several attempts which have been done to test what is the nature of this inside and whether the inside generates from this kind of restructuring whether this this and so on also the fact that and whether this **insight** is suddenly nature is it comes as it comes suddenly or does this **insight** that we talked about which is the problem solution of a problem does it come by mentally not by material cheering the problem or spending a time of **incubation** out there but it's how does it occur and they believe that it occurs through this incremental process of **problem solving** or in in mental **problem solving** in a step-by-step problem-solving process so Gil hi and Murphy in 2005 they made an observation that even if **insight** was a reality it could not be applied to all types of problems so they believe that there are even if the fact that in the period of **incubation** exist and even if the fact that this period of **incubation** leads to a period of elimination which generates something called inside which is probable solution to a problem they believe that there are problems in this world which can never be solved by an **insight** and this type of **insight** doesn't work for all type of problems and they pointed out that there are two different kinds of problems which exists in this world one is called the insightful problem the other is called the anon insightful problem so this is the definition or this is the classification of Gillihan Murphy now what Gillihan Murphy says that the non insightful problems are those they are likely solved by and they require analytical step-by-step processing so what Gillihan Murphy says is the non insightful problems are those problems which are which are analog e-base which are **reasoning** related problems which are mathematical proof problems and these problems will not be solved or cannot be solved through this sudden realization or this sudden **insight** and what they require for solution is trial and error they realize that or they predict that non insightful problems the solution to those kind of problems arise from incremental working towards or incremental efforts towards arriving at the final solution it is like the **GPS** problem solver and so what we

tend to do is to solve a larger problem we keep on solving the problem and each time improving on to it and then are changing the final solution in comparison there is something called insightful problems are those with solutions appear suddenly and so what gilma and Murphy says is that the fact that non insightful problems are not the only thing which is out there there are certain things called insightful problems also and so certain problems are exist which basically give solutions or which in which solutions are occur or solutions come up but certainly all of a sudden and you don't really have to do incremental processing or if you don't have to do incremental step-by-step processing of the problem so how does **insight** help in problem-solving what is the way in which **insight** actually leads to **problem solving** so there are two key assumptions of how in **insight** actually leads to solving of a problem and the first is that it involves a mistaken assumption that once removed will clear the way to successful problem-solving and so what it is believed is that **insight** as a problem solution works in this way it makes two assumption first there is a problem there is an inherent problem in insightful problem solutions arise with the fact that there is some kind of a mistaken assumption to start with so people have started with some kind of a mistaken representation of mistaken assumption and when this mistaken assumption is or taken away the solution appears suddenly and so what inside tends to do is that it remove this mistaken assumption right and so what they believe is that sometimes our problems will not get solved because there is a problem with the assumption itself people assume in certain ways people assume certain kind of facts or people use some kind of facts for problem representation which has a mistaken identity and with this mistaken identity is removed in some way or is is made aware people are made aware of this mistaken assumption in some way and when they remove this the solution occurs so this is the first way in which **insight** helps you to solve problem and the second is that the solver is with the solution the solver is hit with the solution suddenly and that was termed as I and the second fact the second key assumption in insightful problem-solving is the fact that the solver who solving the problem he suddenly gets this realization that the solution appears and this sudden realization is called a experience so this is that part in which it's believed that the sudden sudden realization of solution happens so basically two assumptions in the insightful problem-solving the first assumption is the fact that the solver there understands of mistaken identity or understandably a mistaken assumption which he used for creating the problem and as soon as he understands that this mistaken identity is removed or replaced a solution appears and the second that this solution which appears to the solver in the elimination stage of **problem solving** it happens suddenly it doesn't happen over a period of time over an incremental time but it appears suddenly to the solver is suddenly the restructuring of whatever the process whether it is mental restructuring or mental manipulation and mental representations due to that a sudden realization happens so it has to be sudden it has to be out of the blue and only then this is called the eye experience and these leads to the solutions of problems so let's look at two in non insightful problems and two non insightful problems and let's see what is the whether the the fact that we have Delta till now in terms of mistaken identity and in term terms are non mistaken identity and whether there is a difference between insightful and non insightful problems so I will give you one minute to basically go ahead and look at these problems and arrive at the solution and then I will further give you the solutions of this so beginning now I give you one minute time to look at both these problems and arrive at a solution right so I guess that you have enough time you had enough time to look at both these problems and are you aware of the solution so in the first case the solution is 59 days right because the lilies are doubling every 24 hours and so on the 59th day we half the pound and the sixteenth day since it is doubling so basically on the sixteenth day there will be full of the pawn will follow be full of lilies and some mistaken assumption is what we tend to do is very tend to create a mathematical expression of it remember met the representation problem that we saw in in the first lecture itself how the problem representation creates problem solution interest problem solution and that is what it is so this realization itself that it is doubling itself and so on the 59 day when it doubles in the 60th day the lady in the lake becomes completely covered with the least so on the 59 day it will become half so what we are doing

is we are having the process and so that's how it actually goes I'm pretty sure that you have also the answer to the second one and the answer to the second one is that he has to unwound the row so in sit is half the distance so what he has to do is unwound the rope and once you have in one Wanda row once you find out that the rope is composed of two strings which are bound together right so most of your ropes have these kind of stringing together and so once you unwound it it will become two pieces they're back to back attach them and then he can use it and so this is the solution and so as I am pretty sure that these are the solutions which has appeared to you suddenly and so these are called insightful problems now we have two non insightful problems and easy as you would require that in these problems there is nothing called inside like what you need to do so thirty Seconds to read this problem please read this problem and come up with a solution and so as you would find out that both these problems require to create mathematical equations and once you created mathematical equations so the second problem is a permutation problem and the second first problem is also kind of a permutation problem so both of you are familiar with the concepts of computation and combination you will find out that you have to create a mathematical equation for solving these problems and so the it cannot be done through inside right and sudden realization of the solution is not there you have to create some kind of mathematical expression for it solve for that expressions all for both the X and Y's and you come up with the solutions and so this is the difference between insightful and non insightful problem in terms of insightful problem a mathematical equation can be generated or it you require to generate the correct expression and then solve for the expression and the solution comes in or correct the understand the correct representation in terms of combinations or permutations and based on that arrive at a solution where there is in your insightful problem you don't have to do that you just think of the problem and then is there is one light bulb which which appears on your head and that is the solution so the mistaken identity is the fact that 60 is full so if it is doubling that 59 will be half that kind of a thing or the fact that these ropes are intertwined together so if you break it apart we can attach it end to end and then can escape the prisoner can escape the cell so these are the basic solutions to both insightful and this is the nature of insightful and insightful non insightful problem so the idea of removal of mistaken assumption which is there in inside the problem now the difficulty of insightful problem is that they involve one key but a errant proposal which leads for the kind of problem that you are facing in solving a problem right now one of the most studied problem which presents this kind of mistaken identity or which leads you to assume this mistaken identity and that further leads you to non solution of this problem is called the nine dot problem right so I'm going to present to you something called the nine dot problem and your job is to basically solve it so basically what is the nine dot problem you have to connect all the dots with one continuous line so these are continuous line and what you have to do is to basically join all of them with one continuous line and the problem requires or the problem is made in such a way that there is one or some key and error relating **perception** or assumptions and these key error perceptions or relations basically hinders you to solve the problem so basically how do you connect all the dots with one continuous line and as you can see no matter how you try it's very difficult now people who would have solved it may have seen this problem before and that is making them solve it but other people they'll realize that solution to this problem is not easy no matter how what you do no matter which configuration you look at basically joining these dots with one line is not possible this is the solution right and so as you realize that this solution exists and this solution is the only solution to this particular problem so you start from from here and then this is how what you do this is what you do this is I mean this is what you do and then you come with the final solution and this is what the solution is all about right so what prevents you from solving this problem what is the mistaken identity the mistaken identity that according to Karen or Carlson three sources or difficulty are evident in the nine dot problem the three three different kinds of problems can arise now generally speaking the problem is the problem which makes you not see the solution is the fact that you want to create this or you want to stay within this boundary once people want to stay within this boundary or four or this idea of staying within the boundary they will never

be able to solve it so in this as you see we violated the idea of this boundary and so we are able to solve it and so this is the mistaken idea so Kershaw and Olson they they basically suggest that there are three sources of problem or there are three basic sources which leads to non solution on the nine dot problem and what is it first the perceptual factors relating to vestal organization principle so guest law organization principle says that this **mental representation** most **mental representation** have an organization principle they are organized together and these organization principles help you into coming up with solutions so the idea of **similarity** the idea of perceptual **similarity** the idea of figure-ground all those things leave you to stay within the boundary stay within the boundary of the line dot so these are the dots that I'm looking at and so most guests will again start perceptual factors are staying within boundaries of per se or perceptual organization lets you stay within the boundary and once you stay within the boundary you will never be able to solve this problem you have to think it in a different manner you have to break this assumption of the boundary conditions that it is a square the idea that this is a square and these dots which are out there actually form a square you have to break away from that and only then you can solve this problem the second is called the process factors they - **information processing** demands of the problem now sense the problem is presented in such a way it looks more or less like a square and the demands of the problem is in in terms of a particular way of solving it a particular demand is out there of thinking in terms of this boundary condition and solving it and the idea that it should be a continuous line this leads to that this basic facts leads you to solving problem in a particular means or thinking about the problem in a particular way and so not able to solve the problem and the third is the knowledge factor our prior knowledge dictates the fact that these kind of solutions don't exist or these kind of solutions cannot be done no matter how you try to connect it in whatever way it's not possible or if you even if you try connecting it this way it is not possible and so these prior knowledge or this prior **mental set** which is out there with problem similar to this that creates the problem in solving it so basically both perceptual factors press process factors and knowledge factors add up together previous knowledge which you have about solutions like this or problems like this that hinders you solving the problem so any or all of these factors might come into play and prevent from **problem solving** or the nine dot problem the second thing that we saw is the our experience the experience of sudden **insight** or sudden fact that a solution exists now another defining feature or the **insight** experience in a certain tangible feeling of discovery usually described as the eye experience now this the issue of what someone is thinking as they relate to metacognition so this idea of AI experience is related somewhere to something called the metacognition what is metacognition it refers to a person's knowledge of their own thought process so the idea of this our experience of whether you come up with this sudden AHA experience or the sudden realization the solution is there to test this we should be very sure that people are able to monitor their thoughts and that is called metacognition so can people actually monitor their thoughts while solving the problem and to solve this particular fact of whether people are able to monitor their thoughts when they are solving the problem matter of metcalfe and why way they conducted something called the warmth rating problem what is the warmth rating problem now what they did was they took two groups of people for one group of people they gave an insightful problem and the other group of people they gave a non insightful problem now both of them were given something called the hot cold plate and what is the hot hot cold plate they were asked mostly as most of these people were asked to monitor their thoughts or to monitor the process of how they were solving the problem and they were asked to basically report to the experimenter whenever they were approaching to the problem so the more clothes they thought they were so coming to the solution of the problem the more warmer they would say that they were feeling so if you if doing something if and an attempt to solve a problem leads you to believe that you are nearer to the solution you say and what I am one degree moment two degree warm and three degree warm and so on and so forth but if you do not feel that you are closer to the problem you say I am cold that is what the people have to say so basically the more closer you think you are to the problem the more warmer you will say you are

and the more distinct you think you are away from a problem the more colder you would say you are so what could be the output of this kind of a result you would believe that in generally in terms of non insightful problem people would increase their mom's rating because with number of attempts with number of the solutions or a number of attempt at making the solution they become closer and closer to the problem and so the warmth rating will be increasing that is what the answer to this the experiment was for insightful problem the hindrance or the the typicality happened for the insightful problems in insightful problems it is believed that the solution comes suddenly and so people would mostly say that they are cold and suddenly they will say that they are mom as the intuition comes in or at the elimination stage at the stage of elimination where the **insight** is generated from the second **incubation** stage people will say higher degrees of what this is not what happened and so as expected that people will stay cold or people will say they are cold for for a longer period of time and suddenly they will say that they are feeling very warm and this is the generation of the solution this is what did not happen and so the from this warmth rating itself it was very difficult to to say that the sudden realization happens and so this **verification** of the sudden realization of I experienced that was sort of not proved or disproved by this metruff and why we experienced or met laughs and why we could not find out any support for this evidence now in **insight** is also been taken as an intuition right it's believed to be an intuition so the issue of whether inside problem-solving involves special unconscious process suddenly restructuring or the problem elements or removal of sudden mistaken assumption serious controversy in the problem so basically is it intuition that leads to this reason reconstruction of problem and and there's this institution leads to a solution and whether this **insight** is the institution there is still controversy in the problem that this restructuring happens so this kind of controversy still exists in the the field whether **insight** and intuition are the same thing and whether this institution is intuition that people talk about is exactly restructuring of the problem another prop fact that we saw in the beginning of this lecture was whether **incubation** leads to **insight** whether more time given to intuition leads to **insight** so one of the more controversial notion was within the study of the problem of **incubation** the idea that taking a break from **problem solving** leads to quicker solution then continuing effort so whether if I give you more time to solve a problem or if you spend more time in the **incubation** stage whether it will guarantee a solution or not whether that was being tested and whether this is equivalent to solving using continuous process of solid solving a problem the ideas to break is that brakes allow for the elements to or the problem to beliefs reorganize for unconscious process to continue to work on the problem and this unconscious is limited by conscious work on the problem so the question was whether allowing you more time how does it leads to **insight** or how does it leads to this insightful solution and the solution which has been proposed is that taking break from problems these two more unconscious process and these unconscious process leads to mental restructuring or the problem and providing the solution now to test this it people were given this kind of a these kind of problem let me see if you can solve this problem so there are four problems in given to yes to you in in in front of you these are called rebuses by the way rebus problems and can you find a solution to this so 10 seconds to basically find solutions as you would see that the solution to this problem is in the first case the solution so no matter how much I time I give you the the **problem solving** of these cases if you are not seen these rebuses before it is very difficult for you to solve this problem and so giving enough time giving more time is not going to solve only the fact that if you have dealt with these problems before can you actually solve it and so as you see the first box basically expresses just between you and me that is what so just is between you and me and that is the solution I am pretty sure that you are able to call the second one and the second one says reading between the lines so reading is the word reading is between the lines and that is what it is so the answer is e reading between the lines that is the solution to it third one I am pretty sure you have guessed by now it says degrees below zero so these are the degrees which are there and is a below zero that is what it is are you sure you're able to solve the last problem the soft-rock no matter how much time I give you it is not going to be solved because there is a mistaken assumption out there

and it basically says fly-by-night fly is by night or the side of night and so these are the sorts and so several terms have been done to find out whether increasing the time leads to solution dog **incubation** time increasing of **incubation** time means to correct solution it isn't founded that's not what it is true so then what kind of factors actually leads to solution correct solutions and Smith suggests that **incubation** effects do occur but only under certain circumstances so increasing time is going to work as generating solutions for insightful problems only under certain conditions namely when a problem is doable and when the solver is blocked by some way from which the solution so only under two conditions if a problem can be solved only in though stages you can be able to or even be able to find our solution to problems and the second is when the solver is really blocked by some kind of an idea so he's trying again and again and he's blocked by some some kind of a problem mistaken identity only then these problems can be given more time into **incubation** stage is going to solve it but if the problem has no idea or is represented the problem in a wrong way or there is some other problem in Durance to the **problem solving** it is not going to work so Smith offers a contextual view of **incubation** which is basically the **encoding** principle or **encoding specificity** principle in Reverse what does it really mean if you basically go to **encoding specificity** principle it in in terms of long term **memory** it basically means that the more closer our solution context is to the problem context the easier the solution is in terms of insightful problem-solving in terms of using **insight** as a problem-solving method a reverse is true which means that the more diverse the context the more time you give to people and the more diverse the context in solving problems the better the **insight** is so **insight** as a solution to a problem occurs only when you change the context in the same context is that people will be thinking in the same manner so if you change the context the thinking change is the restructuring of the problem of how occurs or a new line of thinking arrives and because of this new line of thinking the solutions can be appeared too and this is what Smith 1995 view of contextual problem is all about so the contextual view of **incubation** stage that when **problem solving** is time the solution will come more easily if there is a contextual change and that's what I was trying to tell you now what is **creativity** now creative individuals are able to think outside the box what does it mean to come up with new ideas views and old problems in a fresh perspective and connect seemingly disparate problem solutions so these are those four people who can come up with new ideas view old problems from a new perspective and then even with old problems they can connect the old problem or think of the problem in a new way so what is **creativity** the experts they believe that **creativity** generally agree that creative solutions have two components any creative solution any **creativity** has two components to it one is called the novelty and the other is called the appropriateness now as I described at the beginning of this chapter any creative solution has to be new it should not be a copy of the solution which exists and it should be appropriate in the sense that it should give you enough information it should give you appropriate information for a solution only novelty is not going to solve the creative insights so creative solutions are novel different from previous solutions and usually unexpected the solution of creative problem must also satisfy the constraints of problem at hand that it that is it must fulfill a need to be sensible and useful so just coming up with some novel solution to a problem is not **creativity creativity** is also appropriateness which means that it should typically solve the problem in the easiest possible manner an optimal solution to be exact now one of the most cited framework for describing and investigating **creativity** was proposed by Rhodes who suggested that **creativity** can be informed by a focus of several dimensions so basically what's wrote says is that **creativity** can be mapped on the basis of four dimensions or four things which are out there and what are these first things person for **creativity** four things are important the first thing is a person **creativity** to some extent is related to the aspect of a person or personality is believed that **creativity** occurs for a certain kind of personality and so what is this kind of personality creative people are taught to exhibit a number of personality trait like they have broad interest they have a number of they appreciate complexity and they can tolerate ambiguity so creative people not only are those people who have just number of interests out there but they have tolerance for ambiguity which means that they can

tolerate ambiguous solutions or they can tolerate ambiguous situations and they love complex things out there right and they have higher self-confidence and they are risk-taking people so those people who are risk aversive are never creative people who are is taking a more creative in nature so one of the factors will define three TVT or creative solutions or how you come up with greatest solution is the kind of person that you're using the personality factors nor simply believes that **creativity** is not always the product of a particular comfortable environment in fact **creativity** depends on diverse and sets of life experience so the more life experience you have the more tolerable you are to be ambiguity the more broad interest you have and the more self-confidence you have and the more risk-taking you are then you are creative the second process or the second factor in **creativity** is called the process so **creativity** also refers to specific set of processes which are there to contradictory ideas about cognitive processing in **creativity** has been preferred one is called one view is that it's a special process and abilities and like the ability to quickly restructure problem information and connect seemingly remote possibilities so basically one view of **creativity** believes that **creativity** is a special process and it is the ability to restructure problem and how this restructuring happens by connecting seemingly remote probabilities so those probabilities are those possibilities which are not thought of before they are brought in those answers which were not thought of as an answer to a particular problem they are thought and they are included into the new solution that is how **creativity** works and the another view is that creative thinking is a product of garden variety of cognitive processes such as **attention** in **memory** and this first so first view is basically using those solutions which are not there and restructuring problem and taking those solutions which have been given before but we somehow didn't explain the solution using all of them together and restructuring the problem and forming a solution to it the second view believed that creative solutions are basically through higher cognitive processes or different to increase in different cognitive processes like **memory** and **attention** so the more **memory** and **attention** you have or the varied kind of every intention processes that you employ the more creative you will be the creative cognitive approach Smith 2003 they argued that the answer is probably that creative thinking can be the result of either the two so basically they believe that it is not just restructuring which leads to **creativity** it is also a different kind of common are multiple cognitive processes and all of both of them combined together to actually generate **creativity** and so stand back and Davidson they cite three processes as important in reaching creative insights and what are these processes one is called selective **encoding** the other is called selective combination and third is called selective comparison so these three processes are and selectively **encoding** information selectively combining information in selectively comparison of information leads to a person to doing different cognitive acts and then restructuring problems in a different way and then also using those solutions in a not been used before or have been neglected before and coming up with creative solutions the third factor in **creativity** is called press it refers to the notion that creative behavior does not occur in vacuum rather there is a subject to various external pressure and contextual factor process so basically the context and the external environment has a lot of role to play into **creativity** now creative acts are also products of interpersonal disciplinary and so-called social cultural environments so it's believed that it is not only that internal factors cognitive process relates to it is also diverse setting diverse kind of factors which are out there so different kind of social environment different kind of interpersonal accel exchanges all these contextual factors these two **creativity** and the fourth factor or the fourth part of this creative in how **creativity** develop is the product which refers to the outcome yielded by the creative process be it a painting poem design or a new technology so what kind of autumn outcome that I'm generating the analysis of **creativity** from this perspective is a challenge as its evaluation requires sometimes of objective standard and so basically it says that the kind of product that is generated out of **creativity** it also requires certain kind of to evaluate it right and so basically what it is that the kind of product that is generate out of **creativity** that requires certain evaluation standards a different kind of evaluation standard for it to be generated and so there are different kind of creative out output it could be in terms of the painting or it

could be in terms of mold process it couldn't be in terms of an old product and each of them will require a different standard for evaluation or a different kind of evaluation which is out there now there is a taxonomy which has been proposed for creative process and products and this has been done by titration 2004 they proposed a useful scheme that imposes some order in the desperate research into **creativity** so nitrogen at creative insights as our **creativity** works on two different domains one is called the processing mode and the other is called the gnomish the knowledge domain so on two accesses that which try to look at creative processes what he says that any creative product or any creative process can be mapped on to two-dimensional space and this two-dimensional space has a dimension or something called processing mode in which mode the processing is being done of creative solutions or in which mode then the the creative solutions have been generated and the second node is second domain is called the knowledge domain or the second axis on which creative products is to be analyzed is called the knowledge domain so in terms of processing domain refers to whether **creativity insight** is there is a result of deliberate or if a full search or whether it emerges spontaneously as an X and expected flash of **insight** so in in in terms of the processing mode in terms of the processing mode access or the processing mode dimension there are two ways to look at **creativity** right **creativity** can arise or creative solutions can arise from a thought about from a well thought about process by using most of your **memory** into it or using most of our cognitive process and generating a solution or creative solutions can occur suddenly after the blue and you do not do enough cognitive processing into it and so on this dimension of the processing dimension the dimension varies from a deliberate effort full cognitive processing to a sudden realization of the solution or a sudden up coming of the solution on the knowledge domain or basically on the axis of knowledge domain on the on the dimension or knowledge domain it refers to the nature of the creative **insight** whether it is characterized by more cognitive or more emotional breakthrough so basically what type of output it is creating whether the output that the creative solution is producing is more cognitive in nature or more emotional in nature and so basically there is this kind of a table has been created and it's very easy to understand this table there is a two axis on to it on this table one is the processing mode and one is the knowledge domain so within the processing mode you have the spontaneous processing mode where is believed that creative solution happens all of a sudden or just a deliberate solution mode in which you believe that creative solutions occur from deliberately thinking about it or doing efforts and coming up with it in terms of the knowledge domain in terms of the fact that what kind of solution do what kind of creative solution are we coming up with there are two things it could be more cognitive in nature which basically means that it could be more thoughtful in nature it could be more information knowledgeable in nature or it could be an emotional product or it could be an emotional output that that is there and so as you look into it a more cognitive and more spontaneous idea or **creativity** in in terms of more cognitive and more spontaneous is cokecake you esteem of what benzene ring so what happens is although the solution came to kql in his night sleep but he was thinking about the solution for a longer period of time and so it was more cognitive and more spontaneous in terms of thinking the deliberate mode was mapping the human genome so people so people who created a human genome they were working for this over and over again and they were doing deliberate efforts to solving this and out of this deliberate efforts out of this continuous effort to solving this problem and using a lot of knowledge a knowledgeable way of mapping the human genome or creating the DNA sequencing came up in terms of emotional product the van Gogh self-portrait is actually a spontaneous output to it so all of a sudden when God draw this idea of self portrait or the idea of Mona Lisa which is more emotional in nature and then it creates it was created all of a sudden but those insights which are generated out of psychotherapy those emotional insights or reason for emotional problems which are created or which are generated or which are evaluated in terms of psychotherapy sessions are a deliberate mode because the psychotherapist has to work with the person and he has to work backwards right it is not suddenly spontaneously this person will say that this is the emotional problem that I have or this is the problem which is the reason for all the emotional hindrances

that I have been facing in my life what they will say is that building up using a free recall procedure that the psychotherapists go step by step to to zeroing in into that emotional fight or that emotional discrepancy which leads to problems now I have so I'll leave you with certain problems which are out there and some of these problems are are insightful some of these firms are not insightful and you can actually look into it so basically as you look into it this is this is possible because a child could have actually fallen out on the second floor right and this this is possible because the mother could have been the surgeon so I'm just giving the dancer this is most people know it and if you are familiar with the same sense if there is an episode which talks about this candy base and how the Simpson father Homer Simpson he goes ahead and solved this problem so I'll refer I'll give you a hint to look into it I think in season 26 or 24 they gave a solution to this and this is possible because he was not having the truck with him and so he can walk so these are the same different number of insightful problems a non insightful problems out here which you have a solution so what we did in this lecture is we continued with what is inside we try to explain you what is inside and I also tried to include in introduce to you what is **creativity** and then looked at different kinds of how **insight** really works into finding solution to problems and what is the role that inside has into solving problems then I went up and explained what is creative creative process and the four different parts of the creative process and at the end of it I explained to you that hierarchy that taxonomy which basically goes ahead and look at in in any creative product and maps it into two domains or Maps it into two 2-dimensional X's one dimensional being in terms of the knowledge domain so where it is going to improve the creative product where it is going to affect whether it is creating knowledge or whether it is Khaimah creating emotional support or emotional output and in terms of the processing mode whether on an effort was done to create the solution or whether the solution arrived on its own so this is what we did in this / - day's lecture and this is the last section on **problem solving** so until we meet again thank you

Lecture 26: Reasoning - 01

■Kind: captions hello friends welcome back to the section on **reasoning judgment** when **decision-making** no this particular section on **reasoning** and gentleman and the one which is going to follow after this **decision making** will be one of the last sections that we'll be building in this course on **cognitive psychology** now in the last lecture on **problem solving** I explained to you how **reasoning** this is in making **judgment** and problem-solving are called higher order cognitive functions what is the reason why they're called higher order cognitive functions one reason is because these processes help us in making choices in making decisions or in actually evaluating those mental representations or those information which has been stored by the primary cognitive processes so basic primary quality processes like **attention perception** learning thinking **memory language** all these cognitive processes actually taking information raw information from the environment and does something onto it and store it into **memory** or store it into some kind of a store as a **mental representation** now processes like problem-solving **reasoning** this isn't making and thinking actually use this information which has been stoned to make meaningful out of outputs out of the information which has been stored and so these meaningful outputs are then turned into a behavior because of these meaningful outputs a behavior is generated so basically the idea that we saw in the first car itself the idea how of cognitive thinking or **cognitive psychology** really works is basically to take an information from the environment to use some kind of cognitive operators onto it and arrive at a particular conclusion arrive at a particular decision and from this decision deciding the way now si psychologists believe that there's something called stimulus there is something called response and there is one-to-one matching on stimulus and response and nothing exists between the stimulus and the response the cognitivists believe that there is something called the organism or the O which they call in their paradigm this organism has a mind which basically is involved into producing the output or making decisions based on incoming information which then decides the response so basically stimulus response a response to a particular stimulus is not **Association** which is formed through trial-and-error it is a well thought process and so this well thinking or the idea of how well thought process really works is the culmination of that thing is **reasoning judgment** and **decision-making** so what is basically **reasoning judgment** and this is in making so we'll take this in making into the next section in this section we'll just discuss about **reasoning** and **judgment** so what is basically **reasoning** and what is **judgment** now if I say that barking dogs seldom bite and then I say that Tony is a dog and if I conclude from this that Tony is going to buy is not going to buy it because barking dogs seldom bite is what the premises this is what **reasoning** is all about so we think is about making inferences from some information which is given to you or making conclusions out of or concluding from some inferences **judgment** is a process which is happens after **reasoning** and in which we either we make some cave use some kind of a top-down model to judge whether the conclusion that we arrived at is worthy of choosing or not and then the third process of **decision-making** comes along which basically goes out and says that among the possible outcomes which are available to us through **judgment judgment** basically what it is what it does is **judgment** actually gives us a number of outcomes from a number of experiments or a number of information that has been gathered so any information which has been processed to the basic cognitive processes it leads to some kind of conclusions and so this conclusion accumulating this conclusion or deciding which conclusions are of use or not is **judgment** this isn't making is the is a process where we look at these number of conclusions and make the choice final choice of a **judgment** to go with or waive **judgment** not to go so as I said we look into **reasoning** and decision design meant in this chapter and or in this lecture and in the forthcoming lecture or upcoming lecture we'll venture into **decision making** now most books will talk about **reasoning judgment** this is making a single chapter but I have done I what I've done is I've split it into two parts so let's then venture into the idea of **reasoning** and **judgment** and basically understand what is **reasoning** and what is **judgment** so the process of complex

thinking as I said the process the higher order cognitive processes it involves three processes of **reasoning judgment** and this is an making so here goes the definition **reasoning** involves evaluation of a conclusion based on solely given information given the fact that certain kind of information is given to you given the given the fact that the primary basic cognitive processes provide you with some kind of mental representations which which represent certain kind of information making an evaluation of conclusion or drawing some conclusion from it and evaluating that conclusion whether it is valid or not whether it is true or not is what is **reasoning** so **reasoning** is a process where we go ahead and do the evaluation of the conclusion in terms of its **validity** and in terms of its true **validity** for statement is whether the statement follows from whatever premises have been given no premises and statements which are given before a conclusion so generally in how **reasoning** is looked at is there are a couple of premises given to to start with and then these premises are known to be are are tested to be valid or not and the job of a reasoner is to look at these premises and then basically go ahead and conclude whether the conclusion which follows after the premise is either valid or not and I also look at the truth conditions whether it's true or not so that that's the job of a reason and so that's what **reasoning** is all about it's about evaluation of a conclusion and how this evaluation is done is based on solely on the fact that certain given information is given a is provided to and these given informations are what I call the premises so we never actually go ahead and judge the truth of the premises so given the fact that given information is given to you certain information is provided to you we never go ahead and judge the truth valid of these statements or these information and these information the given information is called the premises we only go ahead and test whether the conclusions is logical in sense whether whether the conclusion which follows after a premise is logical in a particular sense or not and this logical being logical is what is called **validity** so **validity** is basically if the conclusion logically follows from the two premises or the two premises leads to a logical conclusion which we are seeing as a controller so I will elaborate that as we move into the lecture the second part of this chapter will be look into something called **judgment** so what is **judgment** its **reasoning** is applied to a given information to arrive at a conclusion and so here what we do is whatever **reasoning** we use from the **reasoning** whatever evaluation we do those reasonings are then applied to come up at certain conclusions and that is what is **judgment** so given the fact that certain information is there and from that certain information we arrive at a conclusion or number of conclusions this process is called **judgment** so using the **reasoning** to arrive at conclusions is basically called **judgment** and then what is **decision-making** is the evaluation of a given information is done to arrive at **judgment** and based on the **judgment** our choice among several possibilities is done this is what is this isn't making so **reasoning** leads you to validate or basically arrive at a conclusion **judgment** goes ahead and look at this and use this **reasoning** to arrive at to give information about conclusions or to arrive at different conclusions and these conclusions are then provided to the **decision-making** operator in the brain or **decision-making** current a process in the brain which then looks at the number of conclusions which are there and based on the need of the are based on the kind of requirement that that is available at that point of time chooses the most optimal decision so there are several choices there are several conclusions given to us they say making is a process of making a choice and this choice is done in based on terms of cost-benefit analysis there's ever loved ways of looking into it when we run into the process of **decision making** he'll explain it to you there so basically the same aching is making the choice right so these are the three process which is there let's then go ahead and describe these processes one by one so the first thing that we should be looking forward is how does this whole process of **reasoning judgment** and **decision-making** really go about and Daniel Kahneman and Amos Tversky someone of some of the famous researchers in this area they basically say that people should be focusing on errors and because while they should be focusing on errors because the focus of an error will provide you in into the fact of how your **reasoning** how is your **reasoning** going on and how is your **judgment** and decision process going on so basically if you want to study **reasoning** and **judgment** they should be not looking at judgments which are correct we should be

looking at errors in **judgment** and **decision-making** because these errors in **judgment** judgments and **decision making** will be giving you more information as to what should be done or what is wrong with a particular **judgment** process or what is the most optimal way of **reasoning** and **judgment** so the emphasis in research and **reasoning** and **judgment decision-making** has been on mistakes that people made and then can men Nobel prize-winner makes in 2004 2002 or four I are not very sure about that but he says he believes that errors provide us with input in furtiveness informativeness now what does it really mean when we look at when how a person does job correctly it doesn't give you much of an answer because he does the job correctly and so that's it what should be what should one be focusing on is how an error is done or what should we one be focusing on is what kind of error exists in completing a job because this focus on error will tell you what not to do and so in any job or in any kind of decision system it is better to find out what not to do then to find out what to do because what to do is what is required but then for our for arriving at what to do we have to look at the hindrances and then look at the ways of getting around with hindrances and so that's what Daniel Kahneman says he says that focusing on errors give us more information now the conditions under which are thinking fails us really important aspects of cognitive processing and inform us how the process of **reasoning judgment** in **decision-making** process really works so basically that's what he conclude he says that by looking at errors in either **judgment reasoning reasoning** and **decision-making** we actually come to know a lot about what is the way to particular **reasoning** or what is the process of a particular **reasoning** or a particular decision and so that provides us with what not to do or how to process because these errors will also tell us how the system looks like now think about a think about a field in which a lot of people are playing football now there is one way to one of them uses a correct method of a goal and goes ahead and scores a goal now if you only concentrate on that of how he scored the goals or how is code a particular goal will only be knowing that how a world is code but if we look at all the errors that he did while scoring the goal will come to know what not to do in a match and how to win matches because this kind of problems are this kind of errors into a wire **judgment decision-making** doesn't succeed we will tell you more about the system of what are the hindrances which are out there because if you if you never face a hindrance you will only know one part to a goal but facing hindrances facing difficulty and it happens with lies also if you face difficulties in life these difficulties actually tell you what life is all about so when I tend to take a tell you that I have a lot of life experience what it really means that I have seen a lot of negatives in life and I have not always been successful and being unsuccessful is a good thing because the more you being successful the more you learn because doing the correct thing doesn't give you opportunity to learn learning doesn't happen because then then you always reach the goal and so there is no learning in it and so that's what our Nobel Prize winner Daniel Kahneman talks about so how exactly do we know that a given chain of **reasoning judgment** in **decision making** is in error what is the process in which which tells you well whether a particular process of **reasoning judgment** and **decision making** is erroneous now there are several approaches to it but what can men and Tversky actually proposes is there had two ways of particular thinking one approach is called the normative approach and which describes how we out to think in a given situation so these are the rules right so normative approach are the norms which are set in by society it says that this is what we tend to do so normative approaches are those approaches which are what people should be actually doing while the second approach is called the descriptive approach describes how we actually think and descriptive approaches is what people actually do so normative approach and different and descriptive approach the difference between them a is normative approach our rules are ideal conditions of what should be done in a particular situation and descriptive approach is actually the thing or actually the process which is applied to reach at a conclusion because nobody is ideal or no system is ideal and so of descriptive approaches are the optimal solutions which lead you to the final result whereas descriptive approach is the the normative approach is the ideal process or the ideal condition of doing a particular task now let's look at a problem so given the fact that there is a family and so this family has had the chance of child rearing and

so for the past four or five pregnancies a woman has let's say six boys right then comes a boy along which is the same then comes a seventh child now what do you think if I ask you the question what do you think so the condition is that I have a family or I have a woman who has produced offsprings six boys in number and then she is also ready if she is now ready for a seventh offspring so what is the chances or what is it that the probability that or rather or what do you think will be the seventh child and so what happens is the normative approach say is that whether it is a girl or a boy is independent of the the the probability of what happened before and so in in terms of normative approach each case or each boy or girl being born it has a probability of 50% half the chance of being born and so the seventh in fact the seventh child in the case of the seventh child also the probability of a girl or a boy is fifty-fifty whereas the descriptive approach will say that since seven six boys have already been born the chances of is there something novelty would happen and so it there are higher chances of a girl now the thing is this higher chances of a girl it doesn't it's not supported by the theory of probability because each each event of childbearing is independent of each other and there is no relation to it and so the seven should also be independent to each other but people miss does this kind of a fact and they say that or they conclude that where the the seventh child will be more of a girl simply because of the reason that six children which have been one before where boys and so this is the difference between what a normative approach and descriptive approaches normative approach is what norms are and so norms of probability say that the chance girl or boy is still 50% in the seventh child but descriptive approaches how people actually think and so people miss culture this so they say that or they believe that the chance of a girl increases because six boys have already been born in a row and so this is the difference of what a normative and descriptive approaches further to it there is a **concept** of **bounded rationality** and so what does it say so Barron 19 1999 believes that rationality is not necessarily the same thing as **accuracy** and that irrationality is not the same thing as error when I see a judgement or when I say a particular **reasoning** is rational what do I mean by that what is rationality how do we describe rationality and so what Baron says is being rational is not same as being accurate so if you do a job nicely if a do do a job correctly it may not always be rational you may not always be rational in doing it similarly if you create errors or if you get into errors by doing a particular job if the way of thinking that you employed may not be irrational in nature and so this rationality and irrationality is basically independent of the **accuracy** of performing a job now rationality involves choosing the method that help us attain our goals so basically then rationality says that use those methods whether it on the on the effectiveness of it so those methods produce results and using those methods to attain goals is what rationality is rationality it doesn't talk about whether **accuracy** is affected by it or not and so that's that is what is the difference between it so we can reason well but still have a decision which will work which may not work out and so given the fact that we use best of a **reasoning** and still the conclusion may not work in our favor or the the job output or the situation output may not work in our favor and sometimes the reason badly and still the conclusion is in our favor and so there is no way to actually to suggest what is rational and this is related to the **accuracy** or doing a particular job now the simple notion that there are limits to our power of reason is what is called banded rationality and so bonded rationality basically says that human beings since they are bounded by this structure they are not calculators or they are not people or there are not physical calculators or calculators in terms of physical sciences they are not computing machines and so they have limits in terms of thinking and **reasoning** and so they make errors and that is what the whole idea of **bounded rationality** it says that rationality the idea of rationality of what is rational and what is rational depends a lot about the thinking process and what who is making the decision in terms of computer a decision a particular decision may be rational or not but in terms of humans that same decision when the computer has given may not be rational right and so this kind of rationality is of how we define rationality is what is called **bounded rationality concept** that people are bound by limitations on through the power of **reasoning** so basically then how do people reason and for answering that question it was proposed that human beings actually go ahead and use two modes of **reasoning** the first mode is

called the **heuristic mode** which is called the system one mode and in this the process used for thinking operate quickly and without much deliberation so the first way is quick **reasoning** process where a huge T is used as mental shortcut is used so you just remember from problem-solving chapter Urist is basically a thumb rule a rule of thumb a shortcut of an **algorithm** so that is what a erases and so in **heuristic mode** people use some kind of a shortcut don't think too much and reason automatically in a position to that there is an analog **analytical mode** of thinking which is called system to thinking and where the process are relatively slow deliberate in control so there is one kind of thinking there is one kind of **judgment** and **reasoning** process which is automatic in nature which is called system one and there is another kind of thinking for **reasoning decision-making** process which is more deliberate and so depending on what how much value the disease and in reasons have to you you either use the deliberate mode or the analytical board the **heuristic mode** or the analytical goal the ethical mode is more cognitively demanding than the **heuristic** one and that is what since you have to do a number of alterations a number of calculations so **analytical mode** are more demanding more engaging and and so they require a lot of **working memory** too so that was what about thinking was that is a brief description of what thinking is let's not jump into what is **reasoning** so what does it **reasoning** really mean there are two types of **reasoning** as I said **reasoning** is basically looking at conclusions or drawing conclusions from given pieces of information that is what **reasoning** is so basically then two type of **reasoning** exist one is called the **deductive reasoning** the other is called the **inductive reasoning** and we'll start with our section or will will be sticking to **deductive reasoning** in this particular lecture so what is **deductive reasoning deductive reasoning** is very simple it involves determining if a specific conclusion is valid based on general principle or assertions or premises so basically there are certain premises which have been given now when I say valid a valid statement may not be true right and so the thing that I was explaining at the beginning of the section in **reasoning** in **deductive reasoning** it might happen that the premises that the values that have been given to you that the previous knowledge that has been given to you they may be valid bit but may not be true and so in **deductive reasoning** we do not think about the truth values of a statement truth values is how are whether a statement is true or not so what we tend to do is we look at a number of conditions and we look at a number of premises which have been given to you previous knowledge is given to you and from these previous knowledge we try to arrive at a conclusion and that is what **deductive reasoning** is all about so **deductive reasoning** problems involved a large degree of constraint and the conclusion is easily assessed using algorithmic approach so in **deductive reasoning** what we do is we are bound a lot of rules a lot of constraints are there and then we use the algorithmic approach to arrive better at a result and since you use the algorithmic approach always a result is out there so basically what **deductive reasoning** tend to do is look at previous knowledge as focus on previous knowledge is based on that arrive at a conclusion and how do I arrive at a conclusion by using an algorithmic approach and using certain kind of Endurance's or certain rules which are out there now there are two primary types of **deductive reasoning** which are there and events in 2002 use two classification systems or **deductive reasoning** the first is called the **syllogistic reasoning** so what is **syllogistic reasoning** look at the statements which I mean given to you so it said there's all students are bright and then all bright people complete assignments on work therefore these are the premises so I have three mice one and then I have premised two and then I have premised the conclusion see here so P wise one I'll say this is valid right and then I'll say this is valid and then what I have to do is reason whether the conclusion that I draw from these two valid conclusions for those two valid premises premised two and premise one whether this conclusion which I draw from it is valid or not now let me explain to you all students are bright now if I look at the truth condition if I know the truth value of this it may not be true so we call our stress or not right and we all know that right so the fact is that in **deductive reasoning** we never look at the truth conditions we always look at the **validity** and so all students are bright is a valid statement right similarly all right people complete assignment on work may not be true but then it is valid and since both the conditions are valid we try to

engage in this kind of a conclusion which says that all students complete assignment on work how does it conclude and what we want to check is whether this conclusion comes from these two statements and so in this case it does right and so that is what is did a **syllogistic reasoning** so then let's break this up and understand what a **syllogistic reasoning** so **syllogism** generally consists of two premises and a conclusion and I said mostly logistic reason the examples will have two premises to start with premise 1 and 2 and these premises are then followed by a certain conclusion the promisor conclusions may begin with the universal quantifier all or particular quantifier son as you saw in these statements it says all students all students all students or it could also say some students have right all students all right so basically most of these **reasoning** examples they start up with these kind of quantifiers which are either summer or also the terms within a **syllogism** may be stated positively which is all ARB or negatively all AR not be so the terms or the **syllogism** the statement itself could be both negative and positive if it is positive it will be all A's or B's if it is negative it will be always or not B's it could also be some a as a czar B's some ways are not B's and so on and so forth so depending on the quantifier and depending on the Silla how the conditional **syllogism** is or the environmental **syllogism** is a particularly large statement will be stated in a positive or a negative reference frame so syllogisms are either valid or not that is it the conclusions either does or does not hold given the premise and that is what we said that I that I said so more syllogisms are tested on their **validity** whether the conclusions followed from the statements we do not test the truth conditions from reading repeating again we do not test the truth condition but this is an error most people try to test this in a truth condition and so there are errors in making this kind of **reasoning** this kind of **syllogistic reasoning deductive reasoning** which are all **syllogistic reasoning** and that is one reason for error which may appear on which may appear to us and so that is that error has happened because we were testing the truth condition so varied arguments then imply that the conclusion does not followed from thus follow from the premises and so any valid conclusion basically says that these conclusions are following from the however it says that nothing about whether the premises are true or not and that's what we have been discussing with that truth it may not be a part of it that the truth value of an argument depends on what the valid it is the argument form and the truth of the premises right so a condition a particular **syllogistic reasoning** may be valid but may not be true for example look at this all professors are comedians all comedians are funny so I have premised one here and then I have to be my Stu here so pretty much one says that all professors are comedians and similarly all comedians are funny is premised number two and therefore I make the conclusion we generate sort of fate and so what is the conclusion that all prophecies are funny now looking at it this is a valid **reasoning** but in terms of truth conditions most professors are not comedians neither the fact that comedians are all comedians are funny and so in terms of truth conditions it will fail but in terms of **validity** this statement will pass similarly all our is our B's all C's are B's therefore all agencies similarly that kind of a problem may exist for this kind of a **reasoning** for this kind of **reasoning** no arranges our apple no lemons or oranges therefore no apples and lemons is so what what do you think is this whether it is valid or not whether this conclusion is drawn from it or from the premise so I have prima 1 and I have premised 2 so what you need to tell me is whether this conclusion is can be drawn from the particular premises or not so are these conclusions value valid based on the respective premises the if not then if you are not able to answer this correctly then it there are certain errors which are making you draw internet control jewels and so what are the reasons or what are those factors which are making you draw incorrect conclusions one of the first first factors is called atmospheric effects so what are the atmospheric effects atmospheric effects says that the quantifier used in the premise combine to form atmosphere which in which the valid is rather the conclusion is racist now as we looked into it a premise has a quantifier it starts with a quantifier the quantifier could be all or some and so given the fact and we also looked at the environment right so whether a quantifier is positively stated or negatively stated and so what it says here is that certain quantifiers may have certain kind of frosty Winston's or negative instance and this positive instances or this positive effect of the quantifiers leads to

people over judging or misjudging the conclusion to be falling in line with the environment of the premise let's say for demise is positively stated premise two is positively stated so people are more likely to validate those conclusions which are positively stated then to validate those statements which are negatively stated and so this is the one one of the reasons for the kind of error that can happen in **syllogistic reasoning** a positive universal atmosphere is produced by a positive instance and this produces in an erroneous tendency to claim that universals Universal and positive conclusions are valid that is what we looked at so basically if two premises are stated in a positive way and that leads us to follow that the chances are very high that people will go ahead and validate a conclusion which is framed in a positive instance in a positive atmosphere than in a negative atmosphere and so this negative atmospheres or this kind of errors can exist the second kind of error which can exist in terms of **reasoning** in **syllogistic reasoning** is that believe biased so what is the believed bias our believes about truth interfere with the ability to assess the argument for **validity** and that's not what I have been telling you so our belief system tells us that professors are not comedians and so they're itself you make an error what we need to do in **syllogistic reasoning** is look at **validity** and not the truth conditions but most people get stuck with the truth condition of it and then decide that professors are not funny and so this statement cannot be valid and so this cannot be true and if it is not true the conclusions can never be true and so that is the problem which is out there and so this kind of error is called the believed bias for example look at these statements so all intelligence beings are Simpson fans all dolphins are intelligent beings therefore so this is premise one positively stated the is premised to positively stated and this is my conclusion which is again positively stated and but the moment you give this conclusion to people they'll say that this is not valid why for statement one they look at the **validity** for statement two they look at the **validity** but as soon as they look at this dolphins are intelligent beings it is valid and it is true too but when they conclude dolphins are Simpson fans they use their knowledge of truth condition they believe that dolphins don't get a chance to look at citizens and so there is no way our dog feeling is actually going to be a Simpson fan and so that leads them to say that this is invalid whereas this is a perfectly valid statement because this statement here somehow interfere with people's belief systems now the tendency to allow believes to interfere with the evaluation of conclusion in **syllogistic** the arguments has been termed as **belief bias** so this kind of problem that arise is called the statements all smart people are reasonable now premise one positively stated all Democrats are smart people positively stated this is the conclusion and so we're looking at statement one and two we look at the **validity** of the statement but when looking at all Democrats is nibble and so given the fact that you are not a Democrat you will never believe it and so you the interfere your belief interferes with the idea or shakes or basically goes head and monitors this statement this conclusion in terms of what you believe in what you don't and based on that it it evaluates this statement not in terms of **validity** but it gains the terms are truth and so it says it is not true which is wrong this is the first kind of **reasoning** out there the second kind of **deductive reasoning** which is available to us or which has been shown by events or in 2002 is something called **conditional reasoning** so what is **conditional reasoning** in the second form of **deductive reasoning** called **conditional reasoning** it's called the if and then **reasoning** this kind of **reasoning** is called if and then **reasoning** and involves evaluating whether a particular conclusion is Aled given a certain condition or premises hold basically it you have a statement to start with which has an antecedent and which has a consequent and following that as another premise so you have a scenario where you have even then statement and again it is about **validity** and truth conditions and from there you are given a premise and you have to look at the pre mines look at the statement which has been given to you which is in the if-then format if antecedent and consequent and then you have to conclude whether the premise which follows after the incident statement whether it leads to the right conclusion or a valid conclusion for example let's look at this if someone likes we need the puh they are sensitive person so this part of the statement is called the antecedent and this part is called the consequent antecedent because it starts before consequent is this so this is someone lies we need the

Pugh is the antecedent it is the first statement right and then because of that they are sensitive person is the consequence so someone likes Winnie the Pooh makes them a sensitive person this is what the statement is so if this then this right and then there is a second statement there is a second premise which says that Mary lies Winnie the Pooh now given the fact that this is premised one this is premised to whether this conclusion is valid or not so therefore Mary is a sensitive person what do you think and so what you would realize is that this is true this kind of conclusion is true because someone lining within the Pooh they are sensitive person and Mary likes Winnie the Pooh so she has to be sensitive because most people who like Winnie the Pooh a sensitive person and so this kind of **reasoning** is called **conditional reasoning** now another version of the **reasoning** problem is if someone likes winning the Pooh they are sensitive person mary is a sensitive person now in the first case I have affirmed the antecedent which means that I have validated or I have my premise has validated the antecedent in the second statement I am validating the consequent I am validating the reason then part of the statement the consequent part of the statement and then it follows therefore Mary likes Winnie the Pooh what do you think whether this conclusion is drawn from or is a correct conclusion and so what you will come to know is that this is not the correct conclusion so this is my premise 1 this is my premise 2 this is not valid so not valid right **conditional reasoning** conclusions can be evaluated quite easily even applies a set of logical rules so how do we go ahead and then know which statements is correct and which statements and not how to go ahead and and look at particular conditional statement or find out which conditional statements are valid or not so most conditional statements so if let's take this conditional statement if someone likes Winnie the Pooh which is the antecedent then they are a sensitive person which is the consequent for scenarios can develop out of it right the from this four scenarios from any other this is the first statement which is there and then any premised that I draw from it a second premise can go any of the way what do I mean by this I can have the premise and stated in two different my second premise what does Mary do or if somebody likes Winnie the Pooh and so he should be a sensitive person is the statement and then Mary is either sensitive person or like Winnie the Pooh both of these conclusions most of these can can be put into four scenarios the first is affirming the antecedent affirming the consequent and I can go denying the end citizen and denying the consequent so for truth basically this is in terms of technology in terms of a truth table so since this is not a logic class or not bringing in that logic but this is these are the four scenarios or four conclusions which can be drawn from it so given the fact that if I have a statement like this this statement leads to four different scenarios out of that Mary likes Winnie the Pooh therefore Mary is sensitive person right and so this is correct this is valid why it is true and in terms of logic though this is called the modus ponens oh and what is this basically affirming the incident antecedent is always giving valid statement so if I if I have a if-then **reasoning** and in that if-then my promis-- validates the antecedent or affirms the antecedent it will always be true let's look at that statement in which my pre mice are forms the consequent Mary is a sensitive person therefore Mary likes Winnie the Pooh and so this control is not valid because all sensitive persons do not like even in the Pooh it is the other way around if somebody likes Winnie the Pooh only then they are sensitive persons but then there will be hundreds of people who are sensitive and may not like when in the Pooh and that is why it is not valid and so a firming the consequent is never valid and so these are mental shortcuts that we need to use or that we need to basically apply to these kinds of **reasoning** to come up with certain conclusions the third is denying the antecedent Mary does not like Winnie the Pooh therefore Mary is not a sensitive person again this is not valid because denying the antecedent doesn't really mean so if somebody doesn't like we need the Pooh it necessary doesn't mean that they are not sensitive because all sensitive people don't like Winnie the Pooh it is the other way around most people who like Winnie the Pooh are sensitive but it doesn't say anything about whether how sensitive people believe or the fact that whether all sensitive people like Winnie the Pooh so that kind of statement conclusions cannot be drawn and so denying the consequent is always a not valid thing but then denying the consequent always leads to a valid conclusion and so if Mary is not a

sensitive person and of course she is or he this person Mary is not going to like Winnie the Pooh because people who like the way the Pooh are always sensitive and so somebody doesn't like Winnie the Pooh so they are not too sensitive and so this is called the modal Stern else and therefore Mary doesn't like me the Pooh so two conditions affirming the antecedent or denying the consequent in our if-then **reasoning** in a **conditional reasoning** will always lead to those conclusions two valid conclusions anything other than that will lead to invalid conclusions in **conditional reasoning** so people run into a lot of problem of difficulty when judging the **validity** of conclusions derived from if-then statements now one tendency that people run into is by conditional which basically means that if P then Q is wrongly interpreted by people as if Q then P so if we need likes somebody likes Winnie the Pooh there are sensitive person if P then Q people also judge a backward conditioning or a backward linkage to it which basically means that all sensitive persons like me the Pooh where has this that my statement does not at all say that people who are sensitive like Winnie the Pooh it's a forward direction of statement it says that if P then Q it never says that if Q then P and so when people would tend to do make errors is because they go ahead and judge it in a backward sense saying that if Q then P not to test how people to about or get caught into this kind of structure from Wasson design a selection task and so what I did was the classic question of the task requires the reasoner to decide which of the four cards need to be turned over in order to determine the following evening statement and so what is the statement if a card has a vowel on one side so if then statement this is the antecedent and this is the consequent if a card has a vowel on one side then it must have a even number on the opposite side so this is the statement I need to test if a card has a vowel on one side then it must have an even number on the opposite side and so these are the four cards which are there one two three four now my question to you is which of these statements which of these cards two cars do I need to flip over and check for finding out whether this treatment holds up not and the statement is if a car has a word on one side then it must have a even number on the opposite side if I want to check this rule what should I do what should people do or what what will you do to test this so the error here appears is that most people tend to flip the a card and the two card and this is where the problem is remember what I said you have to either affirm the antecedent or denying the consequent and so if you want to come with this if you want to come up with this solution you have to turn card number one which is affirming the antecedent or card number four which is denying the consequent the consequent is an even number so denied that and denying that will remove the card or turn the card which is an odd number and check whether it has a woven on to it or not and so this is the right solution but as you will find out and as you would have done you would have turned the even number card and the woven card and so that is the problem if P then Q kind of a thing so the selection tendency or wst worsen solution test is called revealed something called a confirmatory bias and people generally do this error because this is called the confirmatory bias which refers to a tendency to seek out or notice evidences that that is constraint with a particular hypothesis and so we're testing this hypothesis we believe that if we let let's check the bobble thing and let's check the card with the wobble on to it and the even number on to it and this is what is called the confirmatory bias and I have an evidence that would be inconsistent so we always look at two positives into it to consistence in into it or to consistent statements into it and so the consistent way a statement is since if-then says a wore well any human number we'll go ahead and check the wobble and the event number but what we should be doing is checking the wobble and checking an odd number denied and accepting the a fervent affirming the antecedent and denying the consequent so there are several rules or models of **deductive reasoning** explainations of how we reason deductively generally fall into two different caps one say is that there are strict or rule-based accounts now what what does it say it says that people possess certain representations equivalent of logical rules so people in their mind they have these certain logical rules or representations of logical rules in their head and that makes them think or - make them go ahead and perform on the **reasoning** task or come up with solutions on to the **reasoning** task so if a conclusion is valid we use these logical rules there is another format or there is

another way to look into it there is another Rule two it or another model which is out there which says how people go ahead and reason and that is called the mental model view by Johnson in layer and what does it say it says that we first form mental model based on information that premises and our own previous experiences have so looking at the information which is given in the premise and looking at our own personal experiences we form mental models next we search for a mental model in which the premises would be true but the stated conclusion would be false so what we do is once we make a model out of it from the information from the premises which is given in our past experiences we then look at our **memory** for all those models which are available to us in which the premises is true but the conclusion is false now you're successful if you find a model where the premises are true and the conclusion is fault which is given in front of you then we select that model otherwise we don't need to so what we tend to do is we use a mental model approach we look at the information which is given to use our past experiences create mental model and then go ahead and validate or non validate a particular statement so what we did in this class is we looked at what is thinking what is the process of thinking how does thinking really work and all the details about the thinking process the nature of thinking process and why should we focusing on a rat for unto it and after that we looked at a particular kind of **reasoning** which is called **deductive reasoning** so we looked at what is **deductive reasoning** what what causes errors in **deductive reasoning** and specifically we looked at two types of **deductive reasoning** which is out there one is the **syllogistic reasoning** so we look at what is a **syllogistic reasoning** how does it function what are the chances of error into it and then we looked at something called **conditional reasoning** in which we looked at how does the **conditional reasoning** work and what are the solutions to **conditional reasoning** Hart is the most valid way to come up with conclusions in **conditional reasoning** and errors to it and then to end it up we saw how or what models are used for **reasoning** or what are the different models which are available out there which people tend to use for **reasoning** to progress or for people to reason so in the upcoming lecture we will discuss another format of **reasoning** which is out there which is called in the the **reasoning** and in addition to that we'll look at certain **judgment** procedures how does **judgment** really work so given the fact that we have reasoned how does **judgment** progress and bring up valid conclusions and from then on we look into the process of **decision-making** which actually helps you choose among the various judgments which are out there thank you

Lecture 27: Reasoning - 02

■Kind: captions hello friends welcome back to the second section on **reasoning** and **judgment** now in the last class we looked at what is **reasoning** and how does **reasoning** differ from something called **judgment** and this is in making and we define **reasoning judgment** and **decision-making** as higher order cognitive functions which really mean that **reasoning judgment** and **decision-making** into an including problem-solving these are the culmination point of any cognitive process or any cognitions now what do I mean by this as you discussed over and over again the basic processes of **perception attention** learning **memory** they take in information from the environment and create mental representations and store it into the **memory** what problem-solving and **reasoning judgment decision-making** do is make meaning out of these representations by manipulating the representations in specific patterns or in specific ways so that is what are these things do now what is **reasoning** the **reasoning** is basically using certain well-known principles or using certain rules of law to validate certain premise so that is what **reasoning** is all about so these thing is making information or waking from based on some previous evidence making meaning of some mental representations or making meaning from somewhere from information which is stored into the **memory judgment** is basically a step ahead of what **reasoning** is and so what **judgment** is basically is to look at evidences provided by **reasoning** and to make a DC to make number of choices available or basically to judge or to gather evidences against all the alternate interpretations which are available for a particular **mental representation** or a particular information which is in **memory** and this is in making is a step ahead of **judgment** in which what we tend to do is we tend to choose among all the alternatives which are available through judgments so what **judgment** tends to do is it sorts of filters all the available interpretations which are **reasoning** and **decision-making** then goes ahead and makes a choice of what particular kind of interpretation from the data that you want to keep last class with these informations we also look at something called focus on errors and so why focus of areas that need it and we looked at the fact that focus of error is the study of focus of errors deals because errors tell you what is the amount of or what is the number of hindrances which are available in making correct interpretations and so looking at errors basically provides you with a lot of information of what to do and what not to do and so knowing about what not to do actually trains help you train in a particular way of thinking so that is why we need to focus on errors now beside that we also looked at two different kinds of objective **reasoning** which was there and these were syllogisms are **syllogistic reasoning** and something something called **conditional reasoning syllogistic reasoning** and **conditional reasoning** so these two types of **reasoning** that we looked in the last class now in terms of syllogisms what we tend to do is we tend to validate the existence of a conclusion from a premise so promise' statements in insula justic **reasoning** we are given certain statements which are called premises now these premises are valid or invalid so we have to decide whether they are valid or not and so based on this **validity** we judge whether a particular conclusion following two premises or two statements which are there logically follow or not that is what we do in **syllogistic reasoning** so in in terms of **syllogistic reasoning** in in another word what we tend to do is couple of statements are given each of the statements are valid and so what we then need to do is based on what the statements are we need to define whether the conclusion that is coming out of it is valid or not and that is what is **syllogistic reasoning** then we looked at some factors which produce error and **syllogistic reasoning** for example atmospheric effects bias beliefs and so on and so forth besides that we also looked another kind **deductive reasoning** which is basically called the **conditional reasoning** in **conditional reasoning** what we have is we have a statement which has an antecedent and a precedent and these antecedent and precedent part of a statement is in the form of if and then and based on the statement are looking at the statement what we tend to then do is go ahead and validate the existence of a conclusion so basically a statement is given and the statement is there now if then form and we then look into it after looking into it there is a conclusion which is given and we need to then look at the

conclusion and a second premise is also given so based on a statement and a premise we have to decide whether the conclusion logically follows or whether it is validated and we looked at four optional things which can happen so once we have a statement in the event form which has an antecedent dinner and a consequent following that we have as a premise now the premise can with the with the statement with the engine statement the premise can make four different kind of possibilities one is the premise can affirm the antecedent it can deny the antecedent the premise can affirm the consequent and the premise can't deny the consequent and based on that a conclusion is given and we have to basically go ahead and tell what whether the conclusion is valid or not now the conclusion validly the conclusion there are some shortcuts are hearest in in this kind of a **reasoning** and that is tolerance and what the spoons and what does it really mean so every time we have found the antecedent or we denied the consequent the conclusions are valid so these are the out of the four possible responses these two will always be true so if the premise is validating the antecedent or denying the consequent in those terms the conclusions which follows from the premise and the statement will always be true so that's what our conditional statement or occur are a conditional logic or a **conditional reasoning** really works beside that we also saw several factors which go ahead and produce errors in conditional billing now one thing is in **deductive reasoning** we always look at **validity** of a statement so validator first statement says that the fact that whether the the conclusion is a logical conclusion or a logical following from the premises or the precedence which has been given we never look at the truth conditions so there's a difference between a truth conditions and and a valid condition a valid condition may or may not be true but a truth condition is always valid and so that that is the one difference and so in these in **deductive reasoning** we never look at that another interesting thing that we that we have to understand is that **deductive reasoning** is basically coming from general to specific so you have a general conclusion some general conclusions given to you and then you have to go ahead and testify or verify specific statements and that's the format of water or **deductive reasoning** is in today's class what we're going to do is they're going to look at another form of **reasoning** which is called **inductive reasoning** or inferential **reasoning** and so **inductive reasoning** what we tend to do is we tend from specific we need to generalize statements so specific instances are given and from those specific instances we have to come up with a general conclusion and so one primary difference between inductive and **deductive reasoning** is **inductive reasoning** it is basically coming from general to specific so it's more of less like a **top-down process** and so what happens here is that there are certain general conclusions which are given and a specific thing has to be our specific statement has to evaluate against it in comparison **inductive reasoning** goes the opposite way certain specific statements are given to you and so from those specific statements you have to then go ahead and follow a general conclusion and so that is what **inductive reasoning** is all about so this is one difference which is that now in terms of **deductive reasoning** we already look at the **validity** of the conclusion and so that is the core of deduct **reasoning** whereas an **inductive reasoning** we look at conclusions in terms of the strength so conclusions are never 100% valid 100% true and so we always in **deductive reasoning** we look in the strength of a conclusion so how strength II or what is the probability of the conclusion to be holding true and so that is what we do in **inductive reasoning** so **inductive reasoning** is making inferences and all around the world all around you you will always look at in mind active raising so another kind of **reasoning** is **inductive reasoning** let's then go ahead and understand what **inductive reasoning** is all about so **inductive reasoning** we reason from specific pieces of data or information towards a general conclusion for example statements like Barney is a dog who box and from that and then Fallon is a dog who also box robin is another dog which also box and from that if we conclude that all dogs go ahead and bark is basically what is **inductive reasoning** so there are three specific statements which are there I have shown you three different or I have narrated three different instances of it so all three dogs that I have mentioned actually bark and from that if we go ahead and conclude that all dogs or most dogs actually about is what is **inductive reasoning** so this is from coming from specific data to general data specific to general and so

that is the kind of **reasoning** that we need to do here so unlike **deductive reasoning** where conclusions are labeled as valid or invalid on in terms of absolute certainty we do not look this in **inductive reasoning** what we tend to do **inductive reasoning** is it leads to uncertainty inclusions that vary in strength so from the very outset itself **inductive reasoning** implies the fact that the conclusions will be uncertain they will not be certain at all in terms of **deductive reasoning** the statements are always certain and so the what we need to do is find out the validating of the statement but in **inductive reasoning** we know that the conclusions are never fully certain and so what we need to do **inductive reasoning** is to find out the strength of certainty which is there so we never get a hundred percent certain statements for example look at the three statements which have been given here now the first statement says that Professor X get upset when asked if she will issue a paper extension and then there is another statement which says professor why won't accept late papers and a third statement which says that professor said takes 20% off each day if a paper is late now there are three statements which have been given and so an **inductive reasoning** we now need to conclude and so the more of this conclusion from these three statements out there could be that professors most professors actually do not like late papers as you see professor X gets upset why won't accept late papers the thread would deduct some amount of marks from your late papers and so it is believed from here we can conclude that most professors actually do not like a late paper and so that is what is **inductive reasoning** right so if we zan and cortman in 1984 describes some characteristics that seem to defy **inductive reasoning** so that's how we don't elective **reasoning** so these people they gave some characteristics that seem to typify what **inductive reasoning** is all about now the product of in the **inductive reasoning** it's not necessarily correct inductive arguments are evaluated in terms of strength rather than in terms of their **validity** so basically it's saying that inductive statements what what it says is the inductive statements are not always correct in nature inductive arguments are not always correct in nature and so assuming that most people or most professors will not allow late papers will lead you to not submitting a late paper and so even if there is a chance for a late paper submission you might miss it so if you make this kind of statements or if you make this kind of conclusions the strength of a conclusion being very weak that most professors do not go ahead and accept late papers then even if there is a chance for you to give a late paper you will not attempt it because this is what the conclusion is and you you kept the strictness you paid the conclusion in a strictest possible sense and so you are not availing that thing because there are those three pieces of data and from these three pieces of data you are making a genuine conclusion and so even if there is a chance for a late paper you might not follow it and might not get a chance to submit a late paper which might have happen from any reason and so that is what it says so one of the things is that arguments their inductive arguments inductive arguments are evaluated in terms of this strength rather than the in terms of the **validity** so it's not in terms of **validity** conclusions from negative arguments are not evaluated in terms of whether they're valid or not whether they are logically following or not it is evaluated in terms of whether it is certain or not or whether it is necessary correct or not and the second point to be noted here or the second factor a characteristic of **inductive reasoning** then we noted here is that pointed out that **inductive reasoning** there is a need for constraints on the kind of conclusions that you would draw not since **inductive reasoning** is coming from specific to general all kind of conclusions can be drawn from it and so we have to be very constrained of what conclusion we are drawing an **inductive reasoning** and so this is one thing that we are to note we cannot draw very weird statements or very weird conclusion sort of it example1 weird conclusion that can be drawn from this particular thing or this particular statement which has been given at the top here example look at statement 1 2 & 3 is that professors with the name XY and Z will not accept late papers and this is not true because if we draw this kind of a conclusion that these kind of professors which have named XY and Z or has an x y&z on the name they will not accept late papers is not a valid conclusion and so there is no truth into it there is no certainty to into it and so this kind of constraints has to be looked at or these kinds of constraints have to be we have to be very aware of these things so in

inductive reasoning two things you have to when making conclusion graph to be very very or we have to be worried about two things first that we do not actually look ahead at the strength at the **validity** of our conclusion we look at the strength of the conclusion right and the second thing in **inductive reasoning** we are to be very sure about is that we should not be making two unconstraint conclusions so if we make to one constraint and strain conclusion then it's not a good to be worried about this now there assert what type of rules or instances support **inductive reasoning** what kind of processor support mental structure support inductive really so researchers disagree on whether induction is based on formal rule based processing or more complex bond experience we not **heuristic** processing so there are a division of people or they're a division of psychologists who who differ between what kind of mental processes work into it so there are groups which believe that there is a strict rule driven or a strict formal logic driven way of looking at how **inductive reasoning** work and there is another group of psychologists who believe that it is more context one that's more about a curious which is there now the rule-based view term as the strict or stylistic view say that **inductive reasoning** involves special processes and representations that operate in the abstract outside any real-life context and so what this view believes that the rule weighs view believes that **inductive reasoning** basically follow a rule based system and so there is a logic to it there is a rule based system to it and this deal based system works in an abstract manner outside any real context to give the conclusions whereas income in the directrix opposition to this the context based view or and lose view as it is called they contend that **inductive reasoning** involves updating the strength of one's belief based on recall of specific instances and so the news view or the contextual view says that **inductive reasoning** is based on how quickly can you validate your belief from past experience and that will give the strength to the conclusion that you are drawing and so this is what the difference is one group looks at the strictness of logic or sect or they believe that rules and logics are followed and that leads to this kind of a **reasoning** or **inductive reasoning** the other group believes that it is our beliefs or it is our past experiences which strengthens our belief on the logic on the conclusion that we are deriving from **inductive reasoning** and that leads to a good **inductive reasoning** or those processes leads to **inductive reasoning inductive reasoning** is present everywhere and that's why I'm written the omnipresence of **inductive reasoning** omnipresence is a phenomena where something is present everywhere and so that's what we tend to do here so only presence of **inductive reasoning** really means the reductive **reasoning** is present everywhere and we look into it yes **inductive reasoning** actually is present everywhere in most problem-solving or in terms of most problems are in strategies **inductive reasoning** is the idea about it two examples of specific city of the omnipresence of **inductive reasoning** is it is used in **categorization** so remember from the section on **categorization** in **concept** we look at something or **categorization** so what do we do in **categorization** mostly in **categorization** what we tend to do is we look at specific elements and from the specific elements we find the commonality and from terms of the commonality then we go ahead and then form a category so those factors which are common in many specific instances from the rule for the **categorization** and that is basically coming from specific to general so induct and that is what is **inductive reasoning inductive reasoning** is coming from specifics to general so **inductive reasoning** provides another view or thought with phenomena known as something called typicality can be viewed and so in not only in terms of **categorization** the answer to **typicality effect** that we saw in **categorization** and in somatic **memory** can be explained in terms of **inductive reasoning** now look at the statement which has been given here there are two statements that I have one is Robin is susceptible to this is a therefore all words are susceptible to disease a this is one statement and the other statement is Turkey's are susceptible to disease B and all birds are susceptible to disease and sorry this should be a is your character and should be correct here the thing is in which of the statement do you think people are going to becoming more definite conclusions so which of the statement has more conclusive or has more strength in terms of certainty and so you will see that most people then turn out to the fact that statement one is more true the reason for why the statement one is more true is

because Robin is a more typical bird and so it is more typical word and so the conclusions are more valid and so it's more of a it represents more of a bird and that kind of **inductive reasoning** or that kind of a specificity from there we draw the conclusion and so drawing conclusions in terms of this is better then in terms of this where turkeys are more typical bird and so in this case the conclusions drawn from statement one is has more strength in terms of believability now subjects rate argument one is more likely to be true because Robins are seen as more typical birds and turkeys that's what I've been telling you that statement one is verified by more number of people simply because of the reason their problems are more typical birds now another interesting phenomena observed in **inductive reasoning** about **categorization** means this may be something called the diversity effect so just like as we have seen in the **typicality effect** there is something called diversity effect so what is diversity effect not look at the two statement which have been given here and tell me which of these statements are more likely to be true or more likely to happen so we have statement one which says that robins are susceptible to disease why sparrows are susceptible to disease why therefore all birds are susceptible to disease why other hand we have these statement which says cardinals are susceptible to bcz turkeys are susceptible to disease ed and therefore all birds are susceptible to disease it so which of the statements do you think that should be true and if you are like most general people you will believe that statement two is more true then statement1 the chances of the believability or statement true is more than statement one now the reason the reason here is that because cardinals and turkeys are two end of the extreme so we're starting for Cardinal to turkey a whole new range of birds coming to it but in number robins and sparrows they are very close together kind of close together birds which are there and so in terms of believability effect we believe that if you want to look at all birds then cardinals and turkeys this statement to end to hold more weight or tend to generate more believability in subjects then statement1 so in this case people rated argument to a stronger because cardinals and turkeys depends more diverse set of birds and that's what i've been saying since cardinals and turkeys are on the two extremes of it they're more diverse kind of words different kind of words so more different kind of words are tend to have a particular disease we believe that most birds have it but since the sparrows and robins are the same kind of same bird so people don't believe it and so this is another fact with something called **inductive reasoning** the second fact so **inductive reasoning** is not only in terms of zatia nor the defendant only solved the **categorization** problem **inductive reasoning** is also seen in **problem solving** for example one kind of problem-solving which is called analog in using **problem solving** using an analogy basically this **inductive reasoning** is used so another set of cognitive process that depends critically on **inductive reasoning** is problem-solving most specifically solving problems by analog II now remember the radiation problem and the war problem or the war attack problem major problem war major problem that we have known the thing is in in these cases the problem was solved by using analog machines in the since in the radiation problem it was solved by using many lasers of smaller strength or smaller value which summated the the amount of heat that is required the amount of the strength of the laser that is required for curing a disease by using many lasers the similar thing was used for the by the major by making an analogy to this radiation problem and so he then took a number of Tanks by different different bridges and so attacted on the wall so this kind of a thing so basically what does this analogy that we that we saw that they major did in terms of the medical problem in the radiation problem what he tend to do is he tend to look at two specific instances and from there he deduce this thing that one way of using a major strength or one way of increasing strength of a particular thing is using small bits of and sources or many sources with small energy so many sources of smaller energy is equivalent to one source of bigger energy why because many sources will sum it up to bicker to equalize the energy which is out there and so this is the generality which is there so one of the thing that happens in terms of **inductive reasoning** is **inductive reasoning** is also available or is also used in problem-solving so that's a end to the **inductive reasoning** thing now the next step in the section is about **judgment** so what is **judgment inductive reasoning** involves arriving at general conclusions based on specific pieces of what

might be called data so **inductive reasoning** is basically arriving at control general conclusions based on specific data segments which are out there so from specificity to generalization is what induction is all about so but what is **judgment** then so **judgment** is an extension of **inductive reasoning** have seen no a 2001 they define that **judgment** is the human ability to infer estimate and predict the character of an unknown event so in **judgment** what we tend to do is **inductive reasoning** will only provide you with the conclusion but how what we infer from the conclusion what we estimate or what we predict from the conclusion is what is called **judgment** the **judgment** is that particular branch or that particular higher cognitive process which does what is which looks at the data which looks at the conclusion which has been provided by **reasoning** and from there it makes inferences or it makes predictions and so **judgment** is a process of making predictions and inferences from data or from conclusions which have been drawn through **reasoning** from data this data is available out there through the basic cognitive processes and so this **reasoning** are of two types they're deductive or the **inductive reasoning** and so **judgment** is basically inferring or making conclusions out of it and once the conclusions are available choosing one of these conclusions based on cost-benefit analysis based on how much you want to gain or how much you want to lose is what is called **decision-making** which we'll see in the next section so let's go ahead and look at what is **judgment** so then **judgment** is a process of making educated guess based on limited information along with our previous knowledge expectations and beliefs so basically stereotyping so what **judgment** tale says or what **judgment** follows or what **judgment** predicts is that they are judgments are basically educated guesses which are based on some kind of limited information if all the information is available we will not make limited guesses so as you know mostly in the world we don't have all the information available with you and so there is always uncertainty and so **inductive reasoning** is also done in terms of uncertainty when we know don't know the premises or we don't know the valid is the premises new do **inductive reasoning** from specifics to general si that kind of a thing and so when we want one we make educated guesses on the limited computer on the information which is available or due to limited computational power of the brain that we have we make judgments right along with and how do we do it we do it in the basis of the knowledge that we have the previous knowledge bit that we have some kind of expectation that we believe and some kind of believes that we hold on to and this is also equivalent to stereotyping stereotyping is making some kind of a prediction or making some kind of a category for certain kind of people or labeling people by a certain category or a certain characteristic is what is video typing for example we tend to do all kinds of stereotype schools RT breeches are also T this kind of people is that that can educate people are more nerdy so this kind of people is that and so all these kind of stereotyping that we tend to do women are core in mathematics men are very good in mathematics and so on and so forth so this is basically stereotyping Serio typing is classifying people based on certain labels or certain characteristics and that is what **judgment** is all about so given the fact that whatever knowledge we have whatever information we have and whatever experience and believe we have we merge all them together and then make a give a statement and this statement is basically what is called **judgment** so basing **judgment** on **memory** so there are three kinds of purists or there are three kind of mechanisms which are used for making judgments and these judgments are dependent on **memory** and so the first kind of **judgment** that we tend to do or the first kind of furious that we use in making judgments are called the **availability heuristic** it all about the **availability heuristic** eight that we base our estimates of likelihood or probability on the ease with which we can think of example so as example if I give you this question that in your whole life a number of people that you have known what is the probability or what is the one letter of the English **language** from which their name starts and so from all the people that you know now immediately your answer will be S or R or T or whatever it is s and r-mo to base a sample which is now the thing is it is basically called you have fallen to something called availability or s the reason that you know that s is the most commonly used name out there and to most names are existing or s and they even they in the English dictionary most words are there from s and so you believe that s and are more

common names and so this is this has to be it and that is what color of a list is all about availability here this speaks that the more clothes that we have the more number of thing examples that we can think about a particular problem that that is what is call of ability units and so you making predictions based on that is what is called availability theorists right and so that that is the definition of it so the ability here is is dependent on two main sub processes the **availability heuristic** in defining what availability orestes if we tend to make judgments based on the fact that how much example is available to you or the ease with which we can think about an example that is called availability or s the easiest way that that is there which we can think of an example if that is how we make a **judgment** that is availability or s right so it's basically why do we need curious for judging the we need jurists for making **judgment** because the three things the data may be huge first of all the data available to you is huge and so we cannot go through and organic approach finding at the **judgment** second thing we may lack the computational power since the data is used we do not have enough computational power to go ahead and compute the outputs of it and so we use a here is and the third is that computational power is there the data is huge and it is ambiguous in nature and so we tend to use units for viewing it's not cut and so that kind of thing is that now the availability here is it depends on two means of processes which is there the first is called biased and coding now why does the **availability heuristic** be there the **availability heuristic** means there because it could be an **encoding** problem or it could be a problem lead to do travel there as an **encoding** problem it **availability heuristic** an happen because an overestimation of certain facts in **memory** are done because you tend to overestimate certain things right now as I asked you the question of how many people that you remember from us English **language** the idea is that since you most people that you meet or this knowledge this is given to you is that s is what you think to meet or re is what you tend to know more people this kind of our representation in **memory** basically makes you use the materials now this in turn makes the bias retrieved from the **memory** as the information stored in **memory** is bias example media over estimation right so one of the things for probability interest is bias **encoding** and this bias **encoding** could happen from media also in a day to day world we look at a lot of media we look at a lot of news a lot of information which is out there and so these informations they are most of the time they are biased and the fact that since media feeds us with so much information and there is an effect which is called **recency effect** right so the medias and speeds up with feeds are with so much information we tend to believe it or for example one of the thing that can happen in availability grace is the media has been trying all along these days that for example looking at here what is happening is the weather is going to be foggy and so even if it is sunny since the media keeps on saying this over and over again and everywhere we talk to people we tend to believe that it will be a foggy weather since it is winter and so on and so forth even if the weather is very sunny the conclusion that it is winter or the fact that it is winter and that media is overestimating something that leads to the fact or using of or that's lead us to make use of the ability erase and predict that or basically think that the weather is going to be all foggy and so one of the reasons is this thing that arguably viewless is bias **encoding** now VAS **encoding** also can happen with its as I said there is something called a **recency effect** or these days the media if you go into the media it is all about political we all believe that most people are at this point of time are turning towards Hinduism are turning towards the construction of the Mandara that kind of now the media is saved in the information to us and so we believe that all over India this is what is going to happen that most people are running towards Hinduism or favor the RAM band-aid or whatever this kind of information is there and so this is what **availability heuristic** since the media is feeding us this information this is the claim that we have made this is the idea **judgment** that is we have made and so we also turn ourselves to that kind of a thing or align ourselves to it because we tend to believe that this is what it is now although if you look into if we go all on our own and start looking at people's belief system or what people believe all over and they are talking to different different people we may not find the kind of representation that media is giving us right and since we are trusting the media the kind of representation that they are giving based on that we

believe that this is what the nature of or this is what the mood of the country is based on what they say another kind of effect another kind of reason for why **availability heuristic** says success is something called bias retrial and what does it mean it says the availability can lead us to astray from sampling process bias in the sampling process what does it mean so at times the **retrieval** of information is also bias from **memory** and that leads to the availability and or that's leads to the kind of bias which is there for example try the following and state whether there is more of number two statement or more of number one statement tell me six-letter words that have a letter n in the fifth letter or a six letter word where the fifth letter is an end how many do you think will exist so whether you think this is more or the second is tell me words that fit the pattern I n G now which do you think is going to be more and so obviously the answer that most of you are going to come up with is this one is more the second is more than the first now if you realize that this is ing ending right and so if you believe that this is more you are wrong the reason being that this is a subset of the second a subset of first so all ing words will be six letter words with a fifth letter ending and ending but there are also other words which have a fifth letter than n and may not have an ing ending and so one two is actually a subset of the letter two or the statement two is subset of one and so they make it can never be more than this and so this is another problem which is there a biased **retrieval** itself since the way we retrieve information itself is biased that can be too the use of availability theorists and the third thing is called illusory correlations and that can also lead to the **availability heuristic** in and what does it really mean in terms of illusory correlation it says that primarily coincidences two events will seem to be linked when they are really not and that is what is called illusory correlation so if two events tend to happen together and just because of just because of a coincidence we tend to rate them together for example we often hear state saying statement that cricketer was not playing very good in the first match he didn't place where you go to the second match he didn't play very good third match and fourth madman and he hit a century and so this kind of a correlation saying that three times he didn't play good and the fourth time he essentially and so that was making the correlation between three three after each three time you perform better that type of idea that if you perform three times bad the fourth time is always good is there a reason any correlation making correlations that really so because each time the hit the play is the probability of what he tends to do the probability of how the output turns out to be is only 0.5 either it does good or does bad and so each event is independent of itself so basing the our idea that after three bad runs the fourth run is going to be a good run here is already correlation or something another example is Sports Illustrated jinx the same thing that I was talking about that if some some something bad is going on or if some particular model or a sports person appear on some particular journal sports magazine and since this magazine is known to ruin the life of many people if a person who is doing good in sports appear on that particular magazine front cover then his life is going to get bad is what is availability here rest and so this is irizarry correlation and so this kind of correlations are also the reason for **availability heuristic** now another kind of thing that we need to know another kind of information that we hear is that we tend to use is called the recognition hearest and so what is recognition Yunus is often used when we have faced with two alternatives one that we recognized and the other that we don't cognise and so in cases that the information which is available to you through **reasoning** one information we recognize very nicely the other recombination we don't have an idea about we tend to pick up that recognition we tend to become that data point that conclusion which is available to us or which we are familiar to and that is called the recognition here is so in this case what happens is if two conclusions are drawn from a body of data and through proper **reasoning** we tend to make judgments favorable in terms of the one statement which we are close to or which we recognize better then the one we don't recognize better and this is called the recognition URIs and that recognition here is the leading cause for the availability availability forest or the availability heuristics that we tend to use in making **judgment** now another kind of heuristics that we tend to use in making **judgment** is called the representative representative Urist and what is this it is basing **judgment** on **similarity** at times we tend to make judgments we tend to make conclusions draw

conclusions from evidences based on the fact of how similar a particular set of data looks to us so when we are trying to place a person in a particular category or **judgment** we rely on representative here is rely on representation here is the degree to which people represent some basic idea of that object for example if we see somebody who's six feet too tall and he's walking immediately then we will tend to think that this person would be a basketball player because somehow it is related it is similar that people who are taller basketball players similarly if we see and we have several kind of examples of somebody use Y it's somebody who's fair somebody is not fair and all of these are related to certain such kind of similarities which is out there and so this is what is represented a representation here is so what does a definition of in this says it's rely on your positive your is the degree to which an object represents a basic idea the more object or more event represents in our **memory** a particular kind of category the more representative we believe it is to me the more **similarity** we believe it to it and that is what is called **similarity** restore her presentative your wrists now why do we get a fall to the idea of representing here is or why do you do we actually make errors in representation or repel why do we use representativeness one basic problem which has been or one basic reason why people tend to use **similarity** here is or representative Eurus is because they ignore something called base rate fallacy now what is based rate fallacy the main straight fallacies any event has a certain rate of occurrence a certain base rate of occurrence and so if we over represent that we note that base rate we tend to fall into this base rate fallacy so people mostly commit the base rate fallacy and what is it ignoring the rate of occurrence of a particular category in a particular sample right and so let's say if we know a family of people in which two people are tall let's say the son in the father is tall we tend to believe that everybody plays a basketball in that family now that is called base rate fallacy the event of father and son or the instance of other and son are independent of each other and that doesn't say that everyone is tall in the family and so we cannot say it is a family of basketball players even in any basketball family it not everybody will playing it and the idea that everybody is a sports person in that particular family is wrong and that is called base rate fallacy that is how often a certain event tends to occur so when we tend to ignore the fact that how often a particular event is going to occur a certain event is going to occur that is called the base rate fallacy thereby getting a biased of **similarity** for example consider the classic demonstration by Kahneman and Tversky so Kahneman and Tversky presented a problem and I'll show you that problem so he gave this statement to people and give the following instruction now panel of psychologists have interviewed in an instant first energy test on 30 engineers and 70 lawyers and all successful in their field now on the basis of this information thumbnail descriptions for each of these individuals have been written for each description please indicate the probability in the person described is an engineer from one to hundred so a certain kind of statement is given and in that statement there are certain base rates which are given and that is what people tend to over rate now base rates are there are 30 engineers seventy there are lawyers out there and the total they are hundred and what we have to tell is that a description is given to you and this based on this description you have to tell me whether the person I am describing is an engineer or not and then rate it in one of an hundred and so what is the descriptions subject were then given the following description this is the description which is given jack is a 45 year old man he is married and has four children he is generally conservative careful and ambitious he shows no interest in politics and social issues and spends most of his time on many hobbies which include home carpentry sailing and mathematical puzzles this kind of a person profile is given to you right and so then you are asked to do this so subjects were a turn on the rate of probability that Jack was an engineer what do you think happened what is the probability that Jack was an engineer and immediately you will say that it is above 50 percent or 50 to 60 percent what you are ignoring here or you are making an error this is incorrect the thing is this words that are used here in terms of his hobbies which says that is thus carpentry sailing mathematical puzzle makes you believe that he is an engineer but you know that in this particular thing seventy percent people are lawyer and so there is never a chance that you will have 70 person or they will have 50% chances that he is an engineer there

are only 30 engineers out there and so they cannot be extending 50 percent right and so that is the error that you're doing because the base rate is only 30 engineers and 70 lawyers and so no matter what this statement says in terms of his thing of carpentry selling and mathematics because we tend to do what we tend to do is we intend to ignore the base rate and make **similarity** statements that people who are who does carpentry or people who does sailing or does mathematical jobs tend to be engineers right why can't I rather do with it is because of stereotypes we believe that this is the stereotype of water engineer isn't this is the video type of lawyers and so what this is what the result they also got so when they gave when not given the profile the probability that a randomly drawn name was an engineer was 30 up 100 or 30% higher when the profile was given most people in to judge that a random name was a engineer bus 50 person and this is their air because they ignored the Baystate then if there are 30 engineers the property cannot be 50% no use of representative URIs and the confirmation tendency to ignore base rate may relate to something called racial profiling so when we tend to say that certain kind of races are our people who are blacks tend to do more tend to be more teased or tend to be do more violent acts this is basically called racial profiling or rate people who are blacks has to be more criminal in nature this is basic because we ignore the base rate we believe that if some people some black people do it we tend to over extrapolate it and believe that any black that we see we tend to believe that he is the one who's doing this kind of thing now another interesting thing or another reason why this kind of representatives theorist is used or the representative jurist tend to fail us in making **judgment** is this conjugation fantasy and what does the conjugation fallacy really say the conjugation fallacy is another cause for bias which are caused by stereotyping what does it say so Travis Qian can men demonstrated this fallacy at work so then the conjugation fallacy is if there are two events a and B and they occur together they co-occur together what is the probability of both happening together so let's first look at the statement and then I'll explain you how this bias really happens so this is the statement that Kahneman and Tversky gave people so Linda is thirty one-year-old she is single outspoken and very bright she is majored in philosophy as a student she was deeply consumed with issues of discrimination and social justice and also participated in anti-nuclear demonstrations so this this is the basic profile of this person now the subjects were asked to decide whether it was more likely that she was a bank teller or a bank teller who was active in feminist movement and so without looking at anything in return the slide after this tell me what you think most people will agree that statement two is correct that Linda is a bank teller who's active in feminist movement the reason being that the last line of this says that these fields consumed with discrimination and social anti-nuke mission and so on and so forth and so you believe that second is having the more probability but you are wrong the reason being that why can't a bank teller have all this thing and so the chances that she is a bank teller is higher than the chances that he's a bank teller who was active in all these things because any bank teller can have this right now this is what is called the conjugation policy or the conjugation fallacy which is out there now the probability of statement 1 was 0.5 and the probability of statement 2 was 0.5 and when we Club them together when we conjugate them together the probability always has to be 0.25 and that is called the conjugation fallacy when the conjugate to events together the probability will always be lower so bank teller who is activate from an ism will have a cost probability of 0.25 whereas a bank teller independently you're a feminist has a probability of 0.5 and so this conjugation people don't understand this conjugation and they tend to realize that this will be more whereas this has to be less because that's what probability says right so that's what it is now the next thing that we look into is another reason for this bias and that is called misinterpretation of cluster events now when given our event that has two ways of working out such as a coin flip people tend to misconstrue with a random sequence should look like so people what they tend to do is they tend to misconstrue it they can't tend to misrepresent what a random sequence is all about or how randomization really works and in this case if I give them if I give you to judge whether this is more probable or this is more probable you tend to tell me that this is more probable because it looks like more random whereas if you look into it both have an equal probability of occurring

and none of them are random events so this has a probability of 1/12 upon or 12 probability and here also is the same a single event happening that is what it is and so both events are non random in sequence but since this looks more random to you you misconstrue that this is more random and you believe that they as a higher probability of occurring and this is the Miss calibration that you tend to do then this is because this looks more subtle this looks more arranged and so cannot be a random sequence now examples of misinterpretation of cluster events what is the interpretation of this misinterpretation of cluster events one example is called the hot hand and what is it the tendency to miss perceive even clusters as indicating non-randomness may underlie when sports fan Thomas Horton waters what is hot hand hot hand is basically a feature in which we believe that if some player is going very well he is burning hot white like the Sun and so in the next match also is going to do well so if we are looking at if you're looking at coldly batting and we believe that in one to three match he has done good we believe that the fourth match also is going to do good because he is in that streak that winning streak and so that is the problem each match is independent of itself and each player is independent of itself and his own probability and the chances that because he did good in three matches the fourth match is going to do good is something called the hot time similarly is called something called gamblers fallacy and so what happens in gamblers fallacy we tend to believe that if we are failing if a gambler is failing or if we are not getting good results then the next result that is going to be good is going is or coming is going to be good because it is due in US and that's called called gamblers fallacy so if you are playing let's say a card game and three times you have failed we believe that the fourth time you are going to win the reason being that it is this winning is due on your three failures will lead to the fourth inning where as we all know that each event is independent of itself and each has an independent probability of 1 upon 6 in terms of adye's 1 upon 53 in terms of a card game or so on and so forth and so that is what the difference is all about and the last fear is that we need to see here is something called the **anchoring** and adjustment your s and what is **anchoring** and adjustment theories in many cases **judgment** people start with an idea or standard in mind and the initial estimate of the first impression tends to make ourself as overly biased so in **anchoring** and adjustment here is what we tend to do is before making judgments we anchor ourselves we we look inside ourselves for our experiences and leave that this is what the value should be and then we tend to then make judgments based on the anchor that we have kept so we tend to look into ourselves and believe that this is how it something is going to be so if let's take an example so if you ask your parent that you give a statement to a parent saying that this is what I spend each day they will anchor that when they were your age what is the amount of money that they used to spend and based on then they'll make adjustments and say that you are overspending because when they were young and they were at your age they used to spend let's say 10 rupees per day and based on that they made some calculations of what over the years that have been given and so they'll say that your daily spending should be 50 rupees but you know that 50 rupees is not what is good and so this **anchoring** adjustment is another problem with a **heuristic** what they tend to do is what they tend to do when they were when they were little like you they would make that statement or they will make that an anchor and then make adjustments and based on that decide your pocket money or decide decide your daily spending and that is what is called **anchoring** and adjustment units so the URIs involved in these judgments is terms **anchoring** and adjustment a good example of this is something called the spotlight effect what is spotlight effect in spotlight effect what really happens is somebody is wearing so if I wear a taunt loss and when I enter the room I believe that everybody is looking at me and I become overly conscious the fact is that nobody looks at me so when you wear a pink shirt or very very yellow shirt to class you're made to wear a yellow shirt to class you believe that everybody is looking at you - and this is called the spotlight effect and this is a good example of **anchoring** and adjustment the reason is that you anchor yourself saying that I am since I'm wearing this disgusting clothes and so everybody is looking at you and then you make the adjustment that everybody's looking at you you make yourself the anchor and then you make this adjustment everyone's looking at you and so you overestimate the number

of that people look at you and a number of experiments are done where it was people were made to do this kind of a disgusting wearing cloth going into into a room and they were later asked how many people noticed you and the people were actually asked how many people noticed it and it was founded there was a huge gap of how many people actually noticed them and how many people did the person wearing that is going or actually thought we're looking at and this is called the **anchoring** and adjustment purist now our judgments can also be biased or our judgments can also be faulty in terms of certain biases which are out there so biases and judgments can also arise from the fact that at times we are not good at estimating how much we know or when we knew it now a couple of biases of this type is called one of the biases called the hindsight bias and what is hindsight bias people often seem to know for sure of what something has occurred that they knew things would work out just the way they are and this is the tendency of what is called I knew it all along hindsight bias and some so at times a hindsight bias of how what people think that they knew is what is the bias or what is the kind of error that can happen in terms of judgments and it is more evident in civil suit so in civil suit a judge has to determine the probability defendant would have foreseen the what would obtain people would have done so let's say that the plaintiff did some kind of a or defendant did some kind of an gross negligence and this class knowledgeable negligence leads to some kind of a problem to a plaintiff now the plaintiff then goes ahead and files a defamation suit against the defendant for certain kind of money or certain kind of compensation now the plaintiff will go to code as well and not the court person who's the judge has to decide whether the defendant would have seen that this kind of whatever act he did from that this kind of problem would have arisen to the plaintiff or not and so most judges then fall to this error of hindsight bias and believe that the defendant could have very easily a certain that if he did something this is the kind of problem don't have arisin and would have made the plaintiff suffer and they rule in favor of it and so this basically what is hand-side bias it is basically the fact that people say that I knew it all longer the bias that people think that they knew all along and they didn't look into it though that kind of a thing is the hindsight bias another kind of bias is something called miss calibration of confidence so at times we tend to miss calibrate of confidences about something and we take make errors for example the fact that we overestimate the extent to which we knew something was going to happen demonstrates our insensitivity to what we knew and what what we knew and when we do it and sometimes we tend to make our confidences we tend to overconfident of what is going to happen and when it is going to happen for example any kind of thing that we can take so if an experiment were done in which people were given a statement to verify and they were asked to first be positive that whatever statement was meant to be verified they have to be first-rate their confidences and then based on the confidences the truth values generated so a certain statement was given to people and they were asked to rate this statement as correct or not and their confidence ratings were done so it it was found out then people who said that they were 100% confident that they knew the statement was true it was founded the **accuracy** was 75 percent only and people who read who we're lower in terms of confidences we had more accurate accuracies and so this kind of errors do arise and that is called Miska liberation of confidences all right so in this particular section then on **reasoning** and **judgment** what we tended to do is they tended to carry forward from what we did in the last section in the last section we looked at what is **reasoning** and we looked at something called **deductive reasoning** in terms of conditional raising and **syllogistic reasoning** in today's lecture what we did was we looked at what is called **inductive reasoning** and what what is the format of an **inductive reasoning** and what are the possible errors which can be there from there on we move into something called **judgment** which is making basically making meaning or based on the evidences providing some kind of estimate that is what **judgment** is all about and so we looked at what is **judgment** and we looked at three basic types of heuristics which can affect **judgment** the **availability heuristic** intuitive viewers and the **anchoring** and adjustment theorists and in addition to that we also looked at what is biases the kind of biases that can be and these biases how they can affect our **judgment** - for example one is called the hindsight bias I knew it all wrong kind of a thing so this this

you tend to have happen to know all the time when you fail you tend to say that I knew it all along and so that kind of a bias is always there because you tend to say that you knew it all wrong and then there was second bias which was misinterpreting confidences of how confidence you are on to something so this is the end of the section on **reasoning** and **judgment** when we meet next we will be discussing a section on **decision making** and we'll be carrying on some of the things that we have learned there learned here on to that and we look at how people make choices of judgments that has been done so when a **reasoning** is done and out of that **reasoning** certain judgments or certain kind of judgments are evaluators that kind of facts are brought forward and certain kind of rules are laid out how do we choose among the alternatives which are available after **judgment** that is what a **decision-making** that is what we going to see in the next section or this is their making how choices are made among the alternatives which are available after **judgment** and **reasoning** thank you

Lecture 28: Classical Theory of Decision Making

■Kind: captions hello friends welcome back to this last section of **decision-making** which could be the last **concept** with that we'll discuss in this course on introduction to **cognitive psychology** in the previous lecture we discussed complex thought process and under that we discussed two basic phenomena in complex thought process namely **reasoning** and **judgment reasoning** and **judgment** when they finish up they lead to the process of **decision making** and that is the topic that we'll be discussing in today's lecture so what this lecture will be comprised of is **decision making** what the term implies what processes are involved in **decision-making** how it is done what are the various models of **decision making** and at the end of it the various other cognitive processes which interact with this isn't making to effect idea or to give rise to final choice so before we go into the process of defining what is **decision-making** let's first revise a little bit on to the topic of **reasoning** and **judgment** so the last class on topics of **reasoning** and **judgment** we thought and I showed to you how this is the last step in any cognitive process so within this whole lecture series we have been studying various cognitive processes starting from **perception pattern recognition** in **perception** to **attention** then leading to the process of **memory** and following by processes of **language** thinking and problem-solving and then coming to the higher order cognitive process of **reasoning judgment** season making now what does **decision-making** actually imply and the classes are in the lectures of **reasoning** and **decision-making** we saw that **reasoning** is basically a phenomena which leads to **verification** of facts based on current knowledge so basically when the lower order cognitive processes which are basically **perception attention memory** and and and these basic cognitive processes when they produce an input or when they produce a kind of a knowledge representation a **mental representation** these mental representations needs to be further analyzed and process further analysis and process of the mental representations is generally done by the process of thinking **reasoning judgment** and **decision-making** so in in **reasoning** what we tend to do is the **mental representation** the solutions which have been generated through or the kind of output that has been generated to the basic currently processes they are evaluated or they are considered in terms of the present information in terms of the given information at that point of time so basically **reasoning** is a process of finding evidence for the conclusions that are been generated or there have been there are generated for the kind of conclusions which have been drawn from the basic of the cognitive processes now from **reasoning** we move into there are the second process which is called **judgment** so basically in **reasoning** we define there are two kinds of **reasoning** we have something called **syllogistic** and **conditional reasoning** which is which are part of reductive **reasoning** and then we have **inductive reasoning** which has the ephors kind of a statement two kinds of reasonings are out there now the **reasoning** process basically goes ahead and finds out evidences or provides evidences of to support the conclusions which have been generated as an output in **judgment** what we tend to do is looking at the conclusions we go ahead and judge whether whatever the conclusions have been brought up in which of those conclusions are valid in which of these conclusions are not valid so let's say that the basic order cognitive processes gave us a number of conclusions and through the process of **reasoning** we then go ahead and provide evidences for the existence of each of these conclusions which have been derived so in a cognitive process or in a let's say in a cognitive pipeline in the process of **perception attention** and **memory** we have generated an output now this output or this kind of **mental representation** which has been generated from the **basic level** property processes have to be supported by evidences and **reasoning** is a process through which we go ahead and provide the evidence of why these conclusions are valid and not **judgment** is a process through which we look at those conclusions which are valid in those conclusions which are not valid and so these are something which we have done in the previous classes what is this isn't making this isn't making is a process which is a step ahead of **judgment** in **judgment** we look at a number of conclusions we look at a number of outcomes from the **basic level**

processes in the process of **decision making** we made a choice so out of the number of given available inclusions to us through **judgment** and **reasoning** we then make a choice according to our needs according to our requirements of which other choices which are the options or which are the conclusions which have been drawn from the **basic level** our current processes should we go ahead and used and this choice process is actually a very very risky process the reason being that once we make a choice once a number of options are available to us here and we make a choice there are always a chance that the choice will backfire meaning which that if there are four different options or four different conclusions which can be drawn from a particular **mental representation** or four different interpretations of a particular **mental representation** choosing one representation or over the other basically puts up puts us into a situation where if the choice that we are making is wrong we could have a wrong judge wrong decision and that could that could harm us in some way so basically **decision-making** is then a very complex process and this isn't making what we need to do is we need to make choices also another interesting thing that we should remember here is that when we are making choices in **decision-making** when we are making the same choices most the time that these choices have to be made in uncertainty or with certain amount of risk thus the fact that human beings are not calculating machines and human beings do not have all the information available for making a choice selection order the number of options which have been given to us so we tend to make these selections which come out of **judgment** through a process or through in a state of uncertainty in a state of risk and so what we tend to do as humans is we tend to minimize this risk or uncertainty we try to make decisions which do not backfire on us so let's then take a look at what is **decision making** and how do we go ahead and make decisions because these decisions are the ultimate or the final point of any cognitive process because these decisions finally decide how the goal that we have said how do we attain a goal so basically then this is in making is a process of making a choice one of the alternatives which is been a from the process of **judgment** and this choice has to be made mostly in dams or or in in situations where there is considerable amount of risk present and considerable amount of uncertainty passes and for that we need to understand how this decision process really works and the various models of **decision-making** which are available so let then start with a very basic model of **decision-making** how do people actually go ahead and decide let's take a very simple situation for example a situation could be in terms of identifying major course that you would like to take when you enter a college when people or when when students freshman Center college the very basic question which is in front of them is to chose a major is to choose a stream of study and so how do people go ahead and make this choice of stream of study so basically what they do is they look at the number of courses which are available and they look at the goals that they have what they want to want they want to become in their life and based on the goals that they have based on their personal values based on the number of courses which are available and based on so many other variables they make a final choice and this choice once made cannot be reversed that has to be understood that once you have selected a particular kind of a year so what are the what is the flow chart or how does our decision actually made or how is the decision actually made so this is a simple **decision making** flow chart which explains how this isn't really what so first the first thing is to in any **decision making** process the first step is to set a goal first of all in making a decision in making decisions among to exist you people have to set some kind of a goal so let's take this example of choosing a major or choosing a particular branch of study and then evaluate this particular flow chart so let's say that somebody X enters a college as a freshman and then he has to make a choice of which a particular subject to take or which particular subject choose for that process first yes to set a goal this goal is to - the first step in **decision-making** is setting a goal so the person who's making a choice has to first set a goal of what he wants to become and this is the first step so let's say somebody wants to become a doctor he or she will not opt into course which has mathematics and physics onto it and will opt into courses reach her closer to Medical Sciences and closer to biology and those kind of courses so first in **decision making** process if you want to become a doctor or if somebody wants to has a goal of

becoming a doctor into the medical field or doing work with primates or animals and in that case he has to choose that kind of a set that goal first now once that goal is set once once somebody says that goal he has to then make plans and gather information this is the second step so before say picking up a major or before taking up a stream of study first step is setting a goal and somebody events they have set up a goal that they want to be in a medical field that they want to do work on people they now have to make plans the making plan step basically evaluates of what has to be done what are the requirements of this kind of a study in a medical field so you have to then make plans of what is to be done and what is not to be done in terms of making plan is which universities to select which universities are offering this kind of courses which are what is the rating of these universities which is the best kind of rating within the medical field which branch that you want to select and so on and so for that kind of planning has to be done for this planning to be efficient or for this planning to proceed you have to go through a step which is called gathering information and so if within this step what people tend to do is to look at a number of universities which are available which offer these courses in medical so and in number of universities are there what kind of courses that they offer what are the specializations are there who is a good teacher what is the ranking of the University and all those kinds of information has to be catered and this information says then has to be started along several dimensions or several requirements which people might based on their values based on the likeness for example one of the things that people could decide is I want to be a doctor but I don't want to supposedly say practice dentistry and so I want to work with basic anatomy or I want to work with dermatology in these cases they have to select the University of that requirement or somebody would say that I would I would like to work too hard so I'll take a medical university and so they wouldn't be going for top class institutions and going for secondary institutions and so on and so forth other people would like to specialize in certain areas and so they would have to select hospitals or Institute's which specialize for example if somebody wants to work on to brain in India they would like to go to NBR C and so this kind of thing has to be there so then you have to collect a number of institutions which work on to brain and you come to know that in India there are three or four institutions which offer you medical degrees or medical specializations in brain and so you were to apply to that and that is the process which is there so first in any **decision making** process the first step has to be setting up a goal once the goal has been set we gather information or people gather information and then make plans according to this information and this process of setting the goal gathering information and making plan is cyclic in nature in the sense that as more and more information comes to you your plants keep on changing and so does the goal keep on shifting and so this has to be done until and unless people are team a certain level of certain it is that this is what I want to do and they get satisfied with the incoming information or the information that the FHA once they have decided to go to a particular Institute and be in a particular kind of select a particular kind of major for their graduation degree they have to then structure their decision the structuring or decision basically requires you to follow what path do you have to follow to carry out this decision for example in structuring decisions in terms of choosing a major is basically understanding how do you go and role what are the kind of activities that you are going to do there and what are the kind of tests that you are going to do afterwards and then which is the kind of practice that you want to do and so this is structuring this isn't how are you going to take up this decision how you're going to basically carry on with this decision and then based on this structure you know decisions making looking at all the finer points looking at all the pluses and minuses from all the Institute's of your choice or all the Institute's of your liking taking all the benefits of one Institute and the non benefits of that Institute and comparing it with benefits and on benefits of all available Institute's of your liking you then go ahead and make a final selection so basically this is how the **decision-making** process really works so this is how the flowchart of this is in making processes it starts up with setting up a goal then fetching more information about then making plans of how effective this information is or making up plans of how to go about in our team this code once that is done we need to structure our decision which basically means that we need to recalculate the

plusses and minuses of all the options which have been available so all the things check coming out of **reasoning** and **judgment** all the conclusions which are coming out of **reasoning** and **judgment** we need to go ahead and balance them on some some way and then make a final choice and make a final selection and so this is the final step which is generally and this is making so although all of these steps require **decision making** but this is where the final step of **decision making** is all about so what is then how do we proceed with **decision making** so as we studied in the section on **reasoning** and **decision-making** that **judgment** the process of **judgment** which comes before **decision-making** it suffers from a lot of errors for example the three most errors which affect the process of **judgment** is the error or of using hereis such as the availability here is the representative furies and **anchoring** in **judgment** theorists now what are these these are some kind of mental shortcuts these are some kind of shortcuts which people use for making gentlemen now since all information is not available to people at all point of time people use this kind of here is in terms of the availability here is people trust on their **memory** so based on the **memory** whatever comes first to their **memory** people make conclusions or people make **judgment** based on that for example the fact that if I ask people or if I ask anyone what is the most common name in India people will come up with which word for example is is the most common name but represents the most common name in India people will come with the idea that it is s or R this comes from the fact that the dictionary has more number of words under SNR and so this is called the availability Harris because this comes to us or this comes to the person if people look into the **memory** these are the two letters which comes to the name so what people will tend to do is that they will tend to know or they'll tend to recollect the name of all the people that they know and since most people are from Essen iron in somebody's let's say in somebody's nearby area and so that is what they tend to do that it is basically SNR and so this is this is turning on or this is basically the use of our laboratories in comparison to this or in contrast to this basically is the representative Furious which affects **judgment** and this here is is basically the one in which people look at data and compare it with some representative sample examples if you see somebody who's 6 foot 2 inches tall 6 foot 3 inches tall the first thing that comes to mind is he has to be a basketball player because he is tall and so most basketball player at all and so this has to be basketball cat and this is representative Yunus because what we tend to do is we tend to use this basic shortcut of taking this tall person and matching him him with everybody who's tall and this idea somewhere is inside our head that everybody who's tall has to be there and so these are some of the kind of errors which are there for example there are other errors or miss calibration of confidence and the errors of spotlight effect and so these are the errors inject now despite these errors which are coming from **judgment** that occur when we make a judgement these **judgment** from an important part of the database for the process of **decision making** judgments are very important although **judgment** can be routed or judgments has afflictions or **judgment** can be affected by a number of errors but judgments are important to us why because they form the very basic data or they provide the very basic data for **decision-making** now this isn't making in truth generally a choice between alternatives as I said now the process of **judgment** gives us a number of alternatives gives us a number of conclusions which are there is a number of options which are there what **decision-making** does is to make a final choice or to make a final real liking for a particular choice now with the increase of number of alternatives now there is another problem as the number of alternatives increase as the number of options increase the probability or chance of alternatives being wrong also increases leading to the increased risk or unsecured choice what does this really mean it means that if a number of Fortnum's are available we have to do a number of comparisons and so if let's say if a process of **judgment** leads to many number of alternatives then the chances of the alternatives number of activities being wrong also becomes more and more or the probability of those alternatives become more and more and this chance or this probability that a particular that some of these alternatives are wrong will lead to increasing the risk and uncertainty let's say that a process results in only two options now if there are two options the chances of some one of these option being wrong is 1/2 and so the chances of the

other option being right is always half now if we increase this to four the chances of a particular option being right becomes 1 by 4 and the it being wrong becomes 3 by 4 and similarly if we keep on increasing the number of alternatives the chances of an option being wrong increases and this increases the risk and uncertainty as number of options increases the uncertainty in risk also of increases try to understand this in the present consumer society with only a few choices available a few brands available for us to pick up we are very sure of what to buy and what not to buy let's look at final brands the number of brands of fennel which are available now since there are only couple of brands of fennel which are available and so we don't have to be very certain we don't have to be very choosy and so these kind of products are called low involvement products and people don't have to do go ahead and get themselves involved and the choice is very easy but look at beauty products now in beauty products there are a number of changes which are available and every day there is a new choice which is out there and so as a new as new and newer choices Gibbs gets increased into or newer and your choice gets included into what really happens is that the chances of failure also increases because people are not really certain the risk increases as people have not seen a new product or not tested a new product the chances that it will fail also increases and so this increases that risk or this increases the uncertainty now there is something called the **threshold approach** of choice thing to be remembered is that as the number of choices increases as the number of alternative increases the risk of our risk and uncertainty of failure also increases or the risk or uncertainty in making decisions also increases now there is threshold to your choice which were proposed by someone called Clements and what it says is if a decision depends on likelihood of an event happening the attractiveness of the option should increase as the probability of the other event increases so what what it really means is that if a particular event or a particular decision depends on an event another event happening and if the second event becomes more and more sure then people will choose that option let's take a shopping experience let's say that somebody wants to buy something and this buying of this product depends on let's say whether a kind of rebate is given onto it and so as you become more and more sure that so you buying a product depends upon whether something is free with it or not so as you are promised more and more that things are becoming free or things are being given free and for you to buy that particular product as the people advertise as every teaser makes more sure that you get free things out of it your choice or your likingness or your probability of buying the particular product a also goes on increasing so what it basically says is that if an event is there if a particular buying of a product is dependent on some other event which could be whether something is offered with it or not and as the conformity or as it becomes more certain that when you buy a product a product with which you think comes free with it that certainty of it becomes more and more you will buy product a and not buy product P because product a is offering something which is free and so the more certain you are that it is being offered free the more chances that you will buy product a instead of P let's look at this in this way so let's say that there are two products a and B and these products have some associate product C right it could be that you are buying a washing machine and this washing machine has an additional accessory which is C which is required for and so product a is product both product a and B require this accessory C no product a announces that C comes free with a this accessory C comes free with this washing machine this C could be a dish loader or any and any other thing which is there now the more certain or the more confirm that the manufacturer of a becomes with the fact that they are providing C free product a will be chosen more or product B because pure product B is not a thing C so is an event of if you buying a washing machine depends on whether C comes free with it it or not and as C becomes more and more certain you start buying product it more and more and that is what is called the threshold Choiza or **threshold approach** to choice so basically it's stated as is a likelihood that another event happening that the attractiveness of an probability of other event also increases now once that probability reaches a minimum level of certainty the alternative will be chosen and this is called the threshold of culture choice so as producers of product a confirm that they are going to give C for free you choose product a so till the

point of time that a and B both say that they are going to offer C but you don't have an idea you don't have any confirmed news you have both these options for buying a and B washing machines but the moment that manufacturer of a say is that it is certain it is a minimum level of certainty is provided or given the fact that they say that some charge some minimum charges will be incurred from you and then C will be given to you for a very very minimum value but B doesn't say anything about the accessory see the product manufacturer of product B doesn't say anything about the product the buying of products C you will buy a because a is not saying that some amount of money will be taken from you I mean your Martin C will be given and so this is certainty and so you will choose this particular thing and this is called the threshold choices or this isn't making no decisions which involve **overconfidence** and **judgment** are attained a minimum level of certainty to easily read into choice of longer and dividing choices now what does it really mean now when once we become overconfident with a particular **judgment** once we display some kind of **overconfidence** we increase or we get in the minimum level of certainty is reached at a very early stage and so at this point of time the wrong or unrewarding choices can come now going back to this incident in which a and B are two washing machines or companies which are offering washing machine and C is an accessory let's say that I trust a so much and the fact that since a is a company which has always been offering something for minimum rates or with giving things for free and so with this trust I have an over confidence that a is going to launch a machine with manufacturer or a product a is going to launch this machine and I am confident am more confident with the fact that it is going to give free because previous seasons or previous a buying of this product shows that this particular manufacturer gifting three I am then leading to something called **overconfidence** and this will lead to wrong choices because it may so happen that although you get C for free but a may not be very good and so **overconfidence** actually leads to attaining off or selecting up punk choices so if people are overconfident in making a choice if people ignore certain facts and becomes show over confidences in in accepting a number of choices out of **judgment** they turn on or they basically invite something called faulty choices no basic question which comes with it is that is there a minimum level of certainty which varies with culture so is it true that certain cultures are overconfident than certain other cultures and so there is there was a study done by Yeats Lee and shamoto suka in 1996 where the Asian culture and the you and the and the Western culture was compared and so in this study what really happened is the **overconfidence** effect was tested in people from us and Taiwan and so a situation was given to people from both these situations or both both these countries in which they were told or they were given this profile that a particular two medicines have been invented for a particular kind of two different kinds of diseases which are there and so this since these two diseases are very similar in nature very similar in in in symptoms these are the two different kinds of medicines that they work on to it and then they were asked to give this medicine basically provide these medicines for these two kinds of diseases so basically - this is a different kind of diseases or two kind of diseases were thrown in which had similar symptoms and two kind of medicines were prescribed for or two kinds of medicines we're given or were suggested for these two kinds of diseases and then it was people from both these cultures were asked to go ahead and diagnose what is the looking at a profile of a patient and so several profiles were given to the evaluators from both the countries and they were then asked to basically then diagnose the kind of disease that they have then give the medicine this is the kind of setup which was that and so people from both Taiwan and us actually went ahead and did that what really happened what was the result of an experiment like this it was found out that in general people were more confident and in the sense that people showed **overconfidence** in terms of assigning a particular disease to a particular symptom so particular disease are assigning a particular medicine to a particular disease so they were overconfident that but in general the results showed that people from Taiwan were more confident and made more number of errors in terms of assigning the wrong medicine to the wrong kind of disease remember initially I said that what happened here was that the two diseases had similar symptoms and so two different medicines were given to two different kind of diseases but then people from

Taiwan may were more confident and made more errors than people from the Western culture of the United States why could this be what is the reason for this kind of **overconfidence** and this kind of a more error-prone s and so what are the reasons for this was right of the fact that people from Asian culture from Taiwanese culture they depended more on their **memory** and so they searched more of their **memory** and use more of their **memory** in terms of finding or the symptoms or assigning the symptoms whereas people from the Western cultures they referred to written manuals and so they were more correct whereas people from the Asian cultures they they depended more on their **memory** of what they have learnt and assigning a medicine and so one of the reason is this has to be one of the reasons just provided was this kind of a dependence on **memory** versus dependent not written literature or written directives and following these directives which is the result for minimal level of certain it is with various cultures now this isn't making in **decision-making** there are several models which have been used and so one other there are two different kinds of model which have been used in **decision-making** one is called the **normative model** and the other is called the **descriptive model** so in the **normative model** as we saw in in the chapter on or in the section on **reasoning** we saw there are two different models of **reasoning** also one is called the normative models and the other is called the descriptive models the normative models are the models which are ideal which are the ones which should be followed and these are rational models so normative models are those models which actually should be followed in a situation or those models which have been prescribed or those model which are prescribed for any kind of an investigation or any kind of use and descriptive models are those models which actually are not ideal models but which make a goal attainment easy and so these are the models which kept people actually follow so let's say if there are if there is a problem which needs to be solved there it can be solved in two vases one is using analog e or one is using algorithmic approach ends in algorithmic approach using the **algorithm** is basically a **normative model** because here everything will go in always a fraction and a result is expected out of it whereas in the **descriptive model** people use **heuristic** approaches people use heuristics to apply or to come up with a solution and so they they do not follow these ideal solutions or they use some kind of a rules of thumb to arrive at a solution and so these are the two different kinds of models and so one of these normative models way of **decision-making** is called the **expected utility** model or the EU model to be brief now what is this model this is a model which have been borrowed from economics so acquire mix are interested in the factors involved in choice and what's the type of model which describe a rational choice and behavior so thus basically in an **expected utility** model the idea is that how do we can amidst want to know what are the factors with which makes people choose between alternatives so given the fact that you have a number of water keys which comes out of **judgment** how do you go ahead and make a choice and once make a choice what is the model which describes a rational choice now what is the rational choice a rational choice of choice which provides you the better good which basically means the rational choice is that choice which provides which leads you to goal attainment with minimum or losses and so a rational choices that choice which has which maximizes gains and minimizes loss to maximizes gains in the sense that you get more gain or out of it but even if you don't gain then you do not lose too much so what a rational choice is choice which maximizes which tends to maximize gains and minimize loss a situation like this now what are the wireless stablished theories of **decision making** is something called the **expected utility** theory which comes from economics and what is this theory so the theory states that when faced with some type of uncertainties when because in **decision-making** it's always uncertainty because we don't have enough of information available to us so we make decisions under uncertainty and risk so the fact that when people make decisions and uncertainty we make our decision based on two factors generally when we make decisions and when this decision is under uncertainty when we once we don't know all the information which is available we don't know all the facts which are available to us we tend to make decisions with two factors with two different under two different factors one is the **expected utility** of the outcome so any decision which is rational according to the **expected utility** choice or **expected utility**

theory is to be made in terms of the how much the **expected utility** of the earth of each outcome is now we'll define utility now in a while so the one one factor which defines a rational choice a choice which gives maximizes ghee and minimizes loss is the one which provides the expected U which provides the maximum utility and the second is what is the respective probability so the probability of the outcome versus in x the **expected utility** the outcome will define a rational choice so basically **expected utility** theory says the **expected utility** is equivalent to the utility of the outcome into the probability of all outcomes and so this has been done so we have done for the iith outcome it can be four and number four and so if there are three outcomes let's say IJ and K we find out the **expected utility** for I at item for JH item and for K at item and based on this we then go ahead and tend to make the choice in which we have the maximum utility and the highest probability of the outcome being offered now utility refers to whatever end what is utility how do we define utility so utility refers to whatever end a person would like to achieve it is basically what do you want to get out of a particular decision it is that particular thing that you want to get at the end of the season me it could be happiness it could be money it could be something else satisfaction or whatever it is no broom 1991 they suggest that utility refers to the amount of wood that comes out of a decision so basically once we make a decision what is the benefit that we are getting out of it what is the good that we are getting out of it so utility is basically defined in terms of the greater good which comes out of making a decision now thus while making a decision we weigh the good that might come out with each alternative against the cost of that alternative so when we make a particular kind of a utility assessment of a particular outcome we look at what is the good that comes out of it and what is the payment that we are making what is the sacrifice that we are making to get a particular good of that water is the point of particular kind of a sacrifice that we are making in acquiring or in selecting a particular alternative and what is this or one selected what is the good that this alternative is going to provide to us we also access the probability of each of the alternative occurring we also look at where if there are four alternatives what is the probability of an alternative happening let's say that if there are four alternatives one alternative or the probability of happening which is 0.8 80% of times and one alternative has the or the probability of happening which is only 1% time now the fact that if there is the alternative which has one of the probability of occurring upon there's only one person that should not be chosen because it will never occur or it will never be available and so we should not select that thing and so that particular alternative so not only the utility say if an outcome presence or if your outcome gives you greater utility if it is very good for you but then the chances of this thing particular happening is very less we should not take that outcome we should take only that outcome which provides utility in **Association** or in addition to its probability also a probability of occurrence also which means that the availability of it so if a particular choice is available is vastly available and provides even if less good then some some other alternative which are very less formative occurring we should choose that and that is called the rational **decision making** so whether or take is provide the best combination of good and the likelihood will be choosen for example let's consider this example let's look at look at these two options which are there now if you flip a coin it turns up head and you get $40 now if you roll a dice it comes up four you get a $50 no which is the choice that you are going to make if you look at the **expected utility** if you look at through the lens of **expected utility** here you're getting a more higher dollar amount then in this case but then people generally go ahead and take this see the utility is 40 here the utility is 50 here but the probability of if you can if you calculate the **expected utility** the value is 40 here right and the probability of getting a head is 1/2 whereas the probability of getting a 4 is 1 by 6 and so in this case if you look into if you calculate the **expected utility** or the second is less or the expected good that is going to get the appropriateness of this is lesser than this because the probability of 4 is 1 by 6 a probability getting a 4 in a throw of dice because there are 6 faces of a dice and getting a 4 is 1 by 6 and so in this case this is more lucrative than this although this is a lesser dollar remark now which options would you choose and definitely the option that you are going to choose is the one which is the first one and so this the **expected utility** tree now

generally speaking this is a mechanical theory it's an economic theory or mechanical theory of choice behavior or **decision-making** and so generally speaking how people make decision is they violate this **expected utility** and the violation comes in terms of in variances of choices let's look at what is in variances of choices now what are the normative predictions that need to be expected from the utility theory is that our choice should be invariant now one of the predictions which comes from this **expected utility** theory is the fact that the choice that we make for a particular option should be invariant which means that you want if we choose a under situation one then we should stick with our choice and say under situation - but that's not what happens let's say that there are two in the previous case only given the fact that you choose the first one or the second one or the choices you take the choice of rolling a dice and flipping a coin and you choose the $50 option given the fact that if situation changes how does the situation change the fact that it's the fact this becomes an even number if some situation changes in the fact that you know that the dice is biased in such a way that it will always fall towards even numbers now if this situation changes even then you should consider or you should go with this dice or the fact that you come to know that the coin that is being flipped is somehow biased is in the it will always in it will give more number of heads than tails then you should not be varying your choices and then why is this kind of a thing there or why is this kind of an invariance not being followed the people don't follow this invariance or they change change their outcome or this change there this is in process and this is called the in variances or people show in variances of choices and this is what one of the violations of **expected utility** theory now what is it that a decision makers choice should not depend on the way a choice is presented so if a choice is presented in a negative frame or positive frame which means that if it's positively worded or negatively worded people should not go ahead and make flip between their choices if you make a particular choice if you have concede all the information if you make a particular kind of a choice you should not be going ahead and flipping between choices that should be one of the things so example if I prefer choice a or choice B in situation 1 then I should prefer choice a or choice B in situation number 14 as long as a and B are identical in two situations so if situation changes if some other things in the external environment changes people should not be changing their choices but this is what doesn't happen and this is called **preference reversal** or in variances of choices now people often switch their preferences from one outcome to another as situation changes people go on changing their choices and so this is one of the problems with the following off of the **expected utility** theory I mean the expected entry theorem the fact is and given the fact that a particular option is chose particular kind of utility to you it is highly utilized good providing you higher good and it the probability of it offering is also the same if situation changes you should not be changing your options but people do change so people often change their preferences of one outcome over another based on how these ought comes are presented demonstrating irrational irrationality and this is basically what is called irrationality this is what people show in terms of irrationality now consider the **preference reversal** which is shown so live latest in and Slovak 1971 they showed something called a **preference reversal** and they showed this violation of **expected utility** through a phenomena which is called **preference reversal** now the general procedure involved having subjects look at two different Gamble's and decide one which gamble they would like to play and number two how much the gamble was worth so given the fact that people were given six different choices and so in each choices I show you the choice which has been given to that there are six different choices to people and each each one of has a winning percentage so there is there is a probability of this gamble happening and then there is a utility of it happening and but the situation changes so understand that each of these choices have a particular utility and a particular probability now the fact is two different options or two different things are there in case you are the one who's playing this gamble or who's playing this kind of a game and the other case you are the one who's owning the gamble and you want to sell this gamble or you want to other people to gamble on this in which in the first case which gamble are you going to choose in the second case which gamble are you going to choose let me give you one minute time to basically go ahead and

make your choice if you are the one who's playing which gamble or which of these choices are that that you are going to basically accept or which are the these choices are the one which are going to play but if you are the one who is the owner of this gamble and want other people to invest in this which is the choice that you are going to choose now as you would understand that if you are the one who's going to play most people choose this gamble because there's a 99% choice as a probability of winning so you win only for dollars but the chance of winning is higher so you the chances of winning are higher people actually when they play a gamble this is what they bet into but if you are the one who holds this gamble if you want other people to to basically go ahead and put an e or play your gamble we will look into something like this in which case 33% that you win 16 dollars so you will now look at the higher amount of money right so it is the money value in which so you put more money on to a higher gamble and so the very fact as it appears to be the fact that people should be choosing this in both the cases that is not how it is one use once you own the gamble you would want other people to get attracted by the money value but the fact that is the higher money value has a lower probability of winning is the one gamble that you will choose but when you are playing it you will play that gamble which has higher certainty but lower gambling money so this is how **preference reversal** happens so one was I am playing I will always look at a gamble which has higher certainty of being played or higher certainty of giving me even if it is giving me lower values but when I am selling a gamble I will always think of a gamble in which higher value money is there but the chances of winning it are very less and that is what we need to do and so in this in in this case I'll this will be the particular gamble where I own so while owning I will own this but while playing I'll play this and so this is called preference labor self no let's just in in slow way expected the choice of which gamble to play would be influenced by the probability of winning whereas the choice of selling price for the gamble would depend on the partial dollar amount to be won so if you want looking at the amount of dollar that is do we want that and is the decision factor in selling a gamble but when you are playing the gamble it is the probability of winning which decides you how or why you are going to play a particular gamble now why is this **preference reversal** irrational what is the reason for the irrational irrationality the **preference reversal** phenomena demonstrates the inadequacy of **expected utility** as a **descriptive model** of **decision making** the utility model fails to provide a good description of how we make choices the utility model doesn't reflect how humans actually go ahead and make choices the utility model is very good in terms of economic factors in terms of prices in terms of other non living entity but when can when it comes to humans it is very difficult to understand how people make choices now in many circumstances because assumes as much human rarely have them because it assumes as much know the **expected utility** theorem it understands or it assumes that humans have all the information necessary for making a choice but it is to be noted that humans do not have all the information which is available or with humans do not have all the information available for making a choice at any point of time so you must rarely have all the available information necessary to make a decision even if they did they do not have they lack the ability to combine and weigh all the information accurately and so this is one of the reason or this is one of the factor in which or this is one of the reason why there is this **preference reversal** so in terms of machines in terms of economics in terms of prices the **expected utility** theory or theorem is the one which fits but with humans it doesn't fit in the sense that humans do not first of all have all the information for making the choices and so most of the times they are irrational in making the choices or they show irrationality and the other reason is even if they have all the information present or possible they are not able to combine all the information to make the right kind of a choice because this computation this calculation requires a lot of other factors to be there and so this is the reason why people do this **preference reversal** or switch choices now in addition to the **expected utility** theory there is something called the multi attribute utility theory and what is the multi attribute totally a theory it's an extension of the **expected utility** theory now what if choices differ on many dimensions now given the fact that in **expected utility** Theory there was just one dimension on which the choice was wearing now if the choice is varying on number of dimensions

or number of variables we use something called a multi utility theory or a multi attribute really a theory example choosing a major in college now if you're choosing a major in college it is not the goal of becoming a doctor that is making you take a medical field there could be several other things for example how close is the campus from your home how pleasing is the campus in this so many other variables which lets you decide a particular College a particular medical college to go to and so once we are doing that there are multi attributes for multiple dimensions on which the final decision is dependent so major differences in many ways as you are interested in them your job market after graduation the faculty etc and so number of variables will actually go ahead and decide not just one goal of becoming a doctor that how should one make the choice first of all in multi attribute utility Theory as against the expected into theory where there is one choice and one utility there if there are multiple utilities and multiple probabilities based in people need to first make decision down it's important dimension we have to understand those dimensions which are important to us and those dimensions which are not important has to be thrown out or given lower voltages determine the relative weights importance of each dimension each dimension which there for example one other dimension that people are for example for me other dimensions are choosing colleges how near it is to my hometown and that was one of the highest and I gave it the highest weight and so other things got lower weight weight is basically how much preferable that dimension or that particular attribute is to me and so we give the kind of weights then we have to list all the alternatives of which are available with their weights and run the ranked the alternatives around each dimension and then multiply each ranking by the appropriate weight which leads us to choosing the alternative with the highest value for example look into this suppose this is a this is a particular scenario which I have created so then the the criteria on which you want to select a school is interesting topic job prospects faculty in Department and requirements this is the important ways that you give to it and these are the majors that you would like to choose and so once you look into this the each one of it has multiplied by the weight will give you a final summary score this is the multi attribute theory this is the equal weighted criteria and this is the top criterion based on that you will then finally be able to decide which college you would like to go and which college you would like to avoid and so this is the multi attribute theory which is again a **normative model** of **decision making** thank you

Lecture 29: Prospect Theory of Decision Making

■Kind: captions so welcome back friends we are now into the second lecture on **decision making** and as you know **decision making** is a complex process compressed quality process so it follows from **reasoning** and and **judgment** and so in the last lecture on **decision making** what we saw is what is **decision-making** and so the definition that I provided you there is that this isn't making is make basically a process of making a choice among a number of alternatives which have been given by the **judgment** process now when we make this choice in in **decision-making** this generally happens in terms of or in in the presence of some kind of a risk or some kind of a uncertainty and the reason that people make or the reason for these uncertainties that people make decision into is the fact that people are humans that humans do not have all the information which is necessary for calculating a choice for calculating the effectiveness of a choice and so they tend to run into irrationality or irrationality of choices beside that we also discussed the idea of something called **expected utility** theory and this expected neutral theory which has been borrowed from economics basically suggests that people's choice which are which should be rational or ideal is basically dependent on the utility of a particular option and the probability of happening of this option now what do I mean by this no generally this is in making models that we tend to study and at this level are two in nature the first kind of **decision-making** models are called the normative models are the prescriptive models and the prescriptive models are those models of **decision-making** which are supposed to be followed or which are supposed to be utilized in making a decision so given the fact that **judgment** has given us four or five different choices which are in front of us the **normative model** provides us certain guidelines which needs to be followed for making choices and if we follow and it is a guarantee that if we follow these guidelines the choice all will be rational know what is a rational choice a rational choice is the one which give bridge guarantees that ain't meant of a final goal and it gives us the maximum gain or it maximizes our gain and also minimizes our losses so it is a choice in which the gain the chances of gains are there but then the transfer losses are very minimum and so rational choices are those choices which provides maximum utility in the presence of maximum probability of occurrence of an even know an **expected utility** theory the idea of expectancy says that joy those choices which provide us the maximum utility then for a particular option and with the availability the highest probability of that option happening is the most rational choice what do I mean by this given the fact that there are four or five different choices which which come out of a particular **cognition** now what we tend to do is rate the utility of each choices each choices present a certain goodness or sources are certain benefits and certain downside of it so what we tend to do is look at the benefit of each choice and the downside of it and equate them this is called utility or the the use of a particular choice with that each of this option will also have a probability of occurence for example some of these choices tend to occur more frequently than the other number of choices or the other kind of choices so an irrational choice are according to the **expected utility** theory the rational choice will be a choice which has maximum occurrences the highest probability of occurrences and also the highest **expected utility** now given the fact that some choices have very high utility and very low probability we should not make this choices we should only make those choices which which has a decent amount of probability of occurring and also give us a decent amount of utility so if a choice has highest probability of occurring but it gives us the very low utility we should not chose this choice on the other hand if a choice has highest utility and lowest occurrence we should not make that choice a rational decision-maker will make a choice which gives them a certain level of utility and also a very good level of probability of occurence and so this is what the **expected utility** theory really works now the problem is this **expected utility** theory are always violated by humans the reason being that humans don't have enough information are not calculators not statisticians to calculate all the options on which to be the choices to be made and so they go under or they violate these principles of **expected utility** and they create violations of them and one of the violations that we discussed in the last class is called

preference reversal which basically means that if if people tend to prefer choice a in certain situation they reverse their choice in certain situation B or C know the reference reversal theory says that if somebody likes choice a or somebody likes option a better than B then no matter how the situation changes it should stick with it but then as humans we do not stick to that which keep on changing our preference from situation to situation and so this is a violation or the **expected utility** theory now why does this happen the reason I've given before this happens because humans do not have all making the computations and neither even if they have the information they are lazy or they have some kind of a deficiency in organizing all this information and coming up with the best choices and so humans then do not follow the normative or the prescriptive model of generally do not follow the prescriptive model of **decision-making** so then what do they follow they follow a model of **descriptive model** of **decision-making** what is the **descriptive model** the **descriptive model** of **decision-making** is a model which generally humans follow while making decisions and it has it at times tends to give originality or choices of irrationality so people then tend to make choices which are low yield and tend to then suffer and so this this kind of people's choice is making or people's **decision making** are not always rational they are irrational and under this irrationality is not too much of a harm to people and so what are the theories which has been proposed by the Nobel prize-winner with Travis key and Kahneman in 1979 what's called a **prospect theory** this is a **descriptive model** of **decision-making** and it takes into account of the fact that humans are different from computers and they don't generally have these expected utilities and probabilities combining and making decisions onto it so then let's look at what is prospectively now one of the propeller alternate is to the **expected utility** theory is provided by a Nobel Prize winner Daniel Kahneman with his colleague and Amos Tversky in 1979 and which is called the **prospect theory** what is the **prospect theory** say the **prospect theory** it's a **descriptive model** of **decision-making** that attempts to describe how we make decisions and why our decisions violate the **expected utility** model and so what is the statement of this theory the statement of this theory says that decisions are not valued based on an absolute value of the end result the way we make decisions generally are not valued if stability is not valued as in terms of what the end result would be or what the final value attained after the decision choices should be but this theory passed proposed by expectedly theory instead we value decisions based on the amount of gain or loss from what we have right now so people don't look at decisions or people don't look at the utility of a decision or the desirability of a decision in terms of the end value in terms of the final value that you're going to get but people make decisions in terms of the value that they're going to get from where they are from the reference point of where they are so people make these value decisions in terms of the amount of gain or loss that they are going to have from the point of time or from the point where people are right now so let's take that take the fact that if playing a gamble gives you 10 rupees and if you lose that or if you don't if you win that gamble yet if you lose the gamble you get you lose ten rupees then in terms of expected in to be theory the Tetra please is the final value and so that is what the this isn't criteria should be followed on to that but then we as humans we decide this this ten rupees the value ten rupees is not the same value on which you make decisions what we tend to do is even the fact that if you have five rupees in your pocket what you tend to do is that the expected value or the expected value of this outcome then becomes five rupees because this is because if you have if you lose still you will have five rupees which is with you and so it doesn't matter too much so from where you are standing from there you look at the final price so winning ten rupees when you have five rupees is not that lucrative in terms or the fact that if you lose the ten rupees right and so that is the difference which is there so in terms of **prospect theory** what people tend to do is make these evaluations of final value of an outcome not based on the final value from the starting point but from the reference point from the point of how much they have at the point of making decisions so when you're making a decision at that point of time how much good that you have or how much a backup that you have from there we make the assessment of gains and losses on to any gamble it also adds that gains and losses are on different scales of value so one thing that the problem that the **prospect theory** says is

that people do not make evaluations of gains and losses right from the beginning to the end value of a gamble of n value of a decision but what they tend to do is these decisions are based or based on the fact that how much people have previously are people how much good people have before they make a decision so somebody who has money and whose gambling he will gamble more or less depending on how much money he has but somebody who has no money in his choices will be different so people do not make this kind of a calculations in absolute terms but rather from a reference point the second prediction of the **prospect theory** is that the guy and losses are on different skills or value for example the scale of value the in terms of gains people perceive gains as lower value but losses has higher value so losing ten rupees note gives you more losses or lets you feel that you have lost more then in terms of gaining ten rupees given the fact that if today morning you get a ten rupees note from somewhere you the amount of happiness that you get will not be equivalent to the fact that if you lose ten rupees and at some point of time so the amount of feeling the amount of happiness the amount of happiness the gain will not be equivalent to the sadness that you get out of losing that value and so this is the kind of things or this is the kind of predictions in the **prospect theory** actually goes ahead and predicts now the value of your attached to gain increases more slowly as a function of the size of the gain then does the negative value we place on the loss as a function of the size of the loss and so what it basically says is that the gain as the amount of gain increases the function of gain increases very slowly so larger gains will slow I will show you smaller increase in happiness whereas even smaller losses we show larger increase in sadness so if even so if you gain fifty rupees the amount of happiness that you get and if you gain hundred rupees the change in happiness from gaining fifty rupees and hundred rupees will be sloppily in contrast to the fact that if you lose ten rupees and if you lose five rupees in terms of the drop in terms of the sadness that you get in losing ten rupees then in five rupees the drop is very sharp that then in terms of gains so people will be more or less same happy when they gain 100 rupees in fifty rupees but if they lose five rupees in ten rupees the amount of different sadness that they feel will be more and that is what one of the predictions of this particular theory is now basically what this theory says is that we feel losses more act acutely than we feel gain and so as humans in terms of **expected utility** theory if we gain ten or if we lose 10 rupees the value of that money is still 10 rupees but then and so we should feel the same amount of happiness and sadness but as humans you know that we losing money gives us more pain and so losses are more dramatic or gives us more sadness then if you gain 10 rupees and that is one of the more prediction so we feel more losses acutely so the losses are felt more accurately then we feel guess now the psychological pain associated by losing a dollar 50 is greater than psychological pleasure of gaining a fifty dollar and that is the basic **concept** or that is the basic standpoint of the **prospect theory** as against the **expected utility** theory which says that gains and losses and in terms of value say 50 rupees is lost a 50 rupee is gained in both of them the **expected utility** that we get or the rational choice rational decision maker will not be hindered by his preferences but a **prospect theory** says that if 50 rupees is lost we feel more sad and we don't want to spend more money out of it but if we gain 50 rupees we are not that much happy and so this is the invariance which is there or this is the kind of difference which is there now **prospect theory** predicts that people will be especially aversive to losses and will show differences in preference depending on how our enemies are presented or frame so basically another interesting feature of this **prospect theory** is the fact that people do not like losses they are averse into loss so people do not like losses at all and so given the fact that if two options if a particular option is framed differently is what it differently people will change their preferences so if an option has a gain frame which means that it talks about everything with certainty people will have a different kind of a response or people will choose a different kind of an option whereas if something is framed in a negative sense or something talks about losing some statement talks or some options talk about losing people will have an entirely option to choose from and this basic tract is called framing so look at this graph this graph basically says that this is the reference point and in terms of these are the losses and these are the gains and this is the value so in terms of value

if you see that for higher values as the value goes high the gains are very less but then even for very smaller values the loss drop is very high so people in general they do not like losses and losses are perceived as bad losses of perceivers ever see people want to avoid losses and people but on the other hand gains even if a very high amount of money is given to you you gain a high amount of money you will not become happy so the amount of happiness that comes with gains is not equivalent to the amount of sadness which comes with losses so a couple of facts to be remembered once this the other the fact to remember is that people hate avoiding going into losses and so there are different scales of it and the third thing is that if frame differently if the same option is framed differently people are presented differently people Reavers their choices sue them let's look at something called framing what is framing framing is the term which is used to describe the effects on our decisions of how our scenario is presented let's say an option is given to you and the presentation is the option is reworded one so something and let's say the option of winning of 50 rupees or winner or or let's say that there is a particular decision that has to be made and out of this this is a certain kind of choice is that certain kind of alternatives come up now these alternatives can be framed in a game frame which means that everything is positive or it can be or everything that the alternate is talk about is certain or it could be in a loss frame in the sense that everything that the alternate talk about is uncertain in nature and in those cases people tend to reverse their preference so **prospect theory** it predicts that our preferences will change whenever our reference point Changez if we are in a grave frame if they are doing evaluations if we are choosing alternatives and options in a grain frame then in those cases the kind of decisions that we make and if we are doing evaluations we are choosing options in a lost frame those two decisions will be different the reason being that gains are taken in in a different in different way than losses gains are not that pleasant as losses are - more sadness so decisions can be influenced by how information is presented and this is what is framing framing is basically using a say statement or basically reframing rewording a particular option now if information is presented in terms of a positive or a gain frame we will be more likely to avoid risk we will be risk aversive and pick up a sure bet so if something is given to our some option is given to us in a gain frame in a positive frame or in with certainty that something is going to happen we are we become risk aversive so in terms of the fact that if certain options are given to us which show certain amount of certainty which have a gain frame which has some positive outcome out of it we tend to avoid losses in those cases as against those so however if the same information is presented in a negative frame in a loss frame a certain amount of information is given to you or certain information which is available in a particular option is in a loss frame which means that certain ideas certain level of certainty is not provided to you we will be taking more risk so people take more risk when a loss frame is there when people know that something is negative when people know that they are losing they take more risk but when people know that they are gaining maybe people know that there is no loss to it people always select sure bets so this is what framing is all about so it is all about how do we go ahead and make this kind of a choices or this kind of a thing so in a gain frame people avoid risk right and they become risk aversive in a loss frame in a frame where people know that certain kind of losses are there people take more risk so people become a more risk flown people take more risk in terms of negative in terms of negative situations in terms of positive situations people always prefer a sure bet consider so I'm presenting a study by a very famous study by Travis Kane camera in 1981 and this is available everywhere and this study what it does it is a sample study to show that framing has this effect or framing can lead to this reversal of preferences among people and so what is this subject in this particular study subjects they're shown two different kind of options which was there so two different kind of options are presented two or two different scenarios represented to people let's look at the option so the optional scenario says that imagine that the United States is preparing an odd break for an unusual Asian disease so basically this is what it is now which is expected to kill 600 people so the situation says that United States is basically preparing a situation or preparing themselves for an Asian outbreak of disease where they have already calculated the 600 people

are going to die now two alternative programs for combating the disease have proposed so there are two options for combating these disease are there so 600 people are going to die no matter what this is the certainty with it now assume that the exact scientific estimate of the consequence of the program is as follows so there are two scientific options there are two or scientific estimates that there are two options to look at and then you have to choose which are the option you would like to go with now if you adopt program a 200 people will be saved if program B is adopted there is one-third probability that 600 people will be saved and 2/3 probability that no one will be saved this is one way one thing to be looked at so 600 people were going to die out of it and so two programs induced to save more number of people in option A in using the option a what happens is that there is your sort fact that 200 people are going to be saved if program a is rapid but if program B is adapted then there is one third of a probability that 600 people will be saved and two-third of probability that no one will be saved this is one option so one group of subjects were present this kind of an option another group of subjects represented a similar option and they were asked to make the choice whether they will go with program a a program B now in the other group of subjects were presented a similar kind of situation in which a prostate ate that program C is adopted 400 people are going to die but a program D is adopted one-third probability that nobody will die and 2/3 probably there 600 people will die now look at these two options which I think so basically the situation is that there is a scenario which has been presented the u.s. is suffering from some kind of an attack 600 people expected to be die dead and so they create a hypothetical set programs two hypothetical situations for combating with these diseases and so these are the options so as you look into it the first option the first two options program a and B we went to first kind of subjects as in a in a game frame and in the C and D are in the lost frame what do you expect to happen it'll happen is that when things are presented in the gain frame most people select option a which is 200 people will be saved rather than selecting program B which says that one-third probability that 600 people will be saved in 2/3 probability that no one will be saved but look at this look at the C and D option it is presented in a loss frame and here it says that if C doctor is R opted 400 people are going to die and where D is adopted one-third probability that nobody will die and 2/3 probably that 600 people will die if you look into it there is no change from A to B right it is the same but if provided in a gain frame people select option A whereas if provided in the lost frame people adopt B as the answer and this happens this shift or reversal of choosing D over C or choosing a over B is from the fact that it is framed in a different way it is presented in different a positive or negative frames so in a positive frame people select those option which presents alternately so 200 people save certain thing I'll go with it but the fact that if it's a lost frame if wanted people are going to die on own why not go with the rescue option which says that one-third property that nobody will die so one-third probably that nobody will die is going to be the same thing it's not going to be different in any way 200 people is what is the one-third probability that is there and so people select the option whereas in the first case people select a option and this basic demonstration shows that people show this kind of a **preference reversal** or reversal of choices for yourself which options to go with depending on which frame it is being presented to remember the invariance theorem which counsel expected interim it says that if people select a in first case they should select C in the second case although they are the same but that's not what happens people reverse their choices and this happens because these options have been presented in different frames or in different wordings in a gloss frame and in a gain frame so in a gain frame in the first case what happens is people become more risk-averse see they do not want to taste this so they go with a certainty option they go with option where two hundred people are going to be dead but in case where is the last frame people go with a more risky option they take more risk so although they will even if we calculate is going to be the same so take a more risky option and in this case they take the B option which is more risky but it is going to they think that it is going to say more people but it is almost the same and that isn't very very interesting finding of this **prospect theory** which says that gains and losses are valued differently or they are seen of person differently by people now another interesting phenomena or another interesting output of the **prospect**

theory is something called **psychological accounting** it is another interesting output of the **prospect theory** or one of those features or one of those ways in which people make decisions and what does it says so this principle says that people will make different choices depending on how the outcome is felt or perceive basically how people feel so it is about feeling it is about emotion it is about how people like a particular option so the way they feel about an option that decides how people are going to make the decisions and so people feel feelings about a particular option makes the final choice of the decision and not the utility so people do not go ahead and make statistical calculations of probability in utility in making decisions rather what they do is they use a **heuristic** approach of how they feel about a particular choice and based on that they make the decision and actually make the choices of what to do with and they consider that to be rational now look at there are two situations which I have presented to you to explain **psychological accounting** first first of all let us look at the first case now imagine that you decided to see a play for which dollar 10 a ticket and you enter the theatre you discovered that you have lost dollar 10 bill would you pay $10 for the ticket to the plane the so the first option is that you have lost a $10 bill which means that you have lost the money and so you enter into a theatre you want to go see a movie you enter a theatre and then in after entering a theatre you lost a $10 thing which is also the cost of the ticket so are you going to pay for the ticket are you going to buy the ticket and there is a second situation in which decided do you decide to see a play for which admission is $10 as you enter the theatre you discover that you have lost the ticket so you bought the ticket and now the ticket is lost the seat was not marked and the ticket cannot be recovered there is no since you bought the ticket the seats and and unfortunately the seat are not marked and there is no way to recover the ticket would you pay $10 for a ticket for the play so in which of these chances and which of these cases are you more happier or are you more likeable to buy a second ticket what is the answer what do you think is going to happen now given the fact that if these are the choices which have been presented it has been found that choice 2 in choice 2 people are more reluctant people do not want to buy a new ticket the reason being that they believe that $10 the value $10 although the ticket has retained their value at only $10 has been already assigned to the and so is the ticket is lost there is no point in buying a ticket because the $10.00 from the psychological Money account that they have that has which was assigned to the ticket has been lost but in the first case the idea that $10 bill the $10.00 rupees or the 10 rupees for which the ticket is worth that got lost and so that the probability of losing a 10 rupees is far greater than the probability of losing a ticket to a particular play which is of $10 and so the fact that in in case one in the first option you are more likely to go ahead and buy the ticket then in the second case because here the value of 10 has been assigned to the ticket and the ticket has been lost and so the value 10 has been assigned to the ticket and in your **psychological accounting** says that I do not want to pay more money to this particular ticket and so this is what Daniel Kahneman and Tversky also found out in Drammen and Kowalski in 1981 original study the subjects were less willing to purchase a ticket in scenario two now why could this be this could be because of the fact that the $10 that the spend in option 2 will be from the ticket option from the psychological account which is assigned to the ticket well in the first case the probability of losing a $10 is very high so even if you lose a $10 bill it it could have been lost anyway and so you will buy a ticket but in the second case the ticket which was further for the $10 the ticket was lost and so the value was assigned to the ticket and so since you have lost the ticket you are not going to put any more money onto it and so this is called the **psychological accounting** another interesting fact or another interesting phenomena which is a follow-up of **psychological accounting** or **prospect theory** is something called **sunk cost** it is another feature or rather phenomena which shows how people show irrationality in making decisions they are not rational decision-maker and so what is **sunk cost** now in the **sunk cost** effect is another interesting variation of the notion of **psychological accounting** what does he say this effect was demonstrated by our keys and lunar in 1985 in one of the X and so let's look at the experiment so in this experiment what happened is subjects were to imagine that they purchase tickets for two different ski trips so basically two different ski

trips were then subject they actually brought ticket for both these trips now one ticket a trip for Wisconsin costed $50 while the other ticket was a trip to Michigan which costed $100 now the scenario made it clear that the trip to Wisconsin was preferable because it would be more enjoyable so basically then there are two trips that you buy a ticket to a ski trip one is to Wisconsin and the other is is to Michigan the Wisconsin cost of the restaurant is katikut is $50 the cost of the Michigan ticket is $100 both are ski trips that are there but then this information is provided to you that the Wisconsin trip is more desirable it is more fun and so it is both preferable so now you have decided to go to both the places now comes a complication where is the complication a complication arises where the two trips are fall on the same weekend now unfortunately both the trips tend to come on the same weekend or the same dates and the tickets are nonrefundable you cannot go ahead and they find the particular ticket my question is which other tickets are you preferring to go to or which of the trips are you going to go to and which is the trip that you are going to avoid no rationality suggest that this constant trip is more exciting although it is of lower value so people should go for this one and leave the trip to Michigan which is expensive hundred dollars but is that what happens no this is exactly the opposite of it people actually prefer the non enjoyable trip because it is expensive and so people actually go to the Michigan trip and so this is what the result of Emerson Travis keys or Arkansas main study is that they found out that people take the Michigan the Michigan trip although it is not enjoyable but it is expensive and so this is a phenomena which is also seen in everyday well maybe what we see is that people throw in money after bad money so if people lose money they tend to throw in more money so if there is something that you have which is not working and so when you see that this thing is not working and you put more money into it for it being repaired and it doesn't get repaired you tend to put more money on to it people do not get into it so some you buy a particular product it gets defective you give it to someone for repair now after the parrot is still not working you tend to put more money into it to getting repaired rather than buying a new thing because it has already costing you more and so you tend to make some more money on to it and this is the exact same phenomena here happens people tend to put more ornate on to the expensive option then the expensive option rather than choosing an option which is a more enjoyable but lesser value so these are what is this is to do with what is **prospect theory** and how does **prospect theory** really was another interesting theory or another interesting fact in **decision making** is the role of effect in BC's and making or the role of emotion in **decision making** so positive and negative outcome a positive and negative effect which comes with an outcome is a feeling of goodness and badness which comes with an outcome they give us different different feelings with us and with predictable implications on disease and we make so our decisions are not just in terms of factual information it is also related to the kind of feeling that we get after taking a particular decision after taking a decision or after choosing a particular alternative how do we feel about that alternative also plays a large role or plays a major role in how people make decisions or how this is in making it done so affect us is an important determinant of **decision making** and can have a sizable impact on **psychological accounting** process so it's basically the feeling that you have the kind of pleasure or pain that you get out of taking a decision that depends or that decides a lot about how you pick diseases now has a and rot the stage 2004 they made this point by highlighting an important dimension of choice that interacts with effect which they term as scope they basically came up with this dimension of choice which interacted effect and we call this the scope which refers to the sweep of a decision or action how much impact will it have so basically Hattie and not rotten stage in 2004 they invented or they came up with the whole this a whole new dimension which is called scope which refers to the sweep of a decision how do sometimes diseases are made in terms sometimes of positivity or negativity or sweep and this is what they discovered so consider these particular two things suppose you give $10 to save one endangered tiger it feels good now how much would you give to save forces in endangered tigers given the fact that you often see people coming to you saying that see these kind of animals are endangered they have to be saved in some way and so we need some kind of money from you to save these so if they

then they show you some basic thing give us $10 and we we are going to save some endangered animal out of it let us say panda tiger or whatever it is and so you give some kind of money into it the fact that for one it is okay but then he says that you are going to save for Tigers are you going to pay more that's the question now how much are you going to shell out for and saving for endangered tigers or what is the way in which your decision process really works the answer to this question so if you are giving $10 for saving one tiger whether you're going to give $40 for saving four Tigers is the question and so the answer to this depend on whether the subjective value derived from saving tigers is somewhat multiplicative so basically the kind of value that you derive that or the kind of utility that you derive by saving tigers how much that is so is it multiplicative or not so saving one tiger if you are and so this basic the decision depends on two different modes of **decision-making** so basically if one tiger requires $10 for saving and if somebody asked you to save for Tigers are you giving $40 for saving the tiger and that depends upon how much utility or how much subjective value derive from saving these four Tigers now the author proposes a dual process view of the relative impact of scope and subjective value or **decision-making** they say that people make these kind of decisions under two different modes of **decision-making** and what is this mode there is a particular mode which is called a deliberative mode in which map onto the conscious **reasoning** so when you are into a deliberative mode or when you are into a conscious mode of **reasoning** then if somebody says that saving one Tiger is $10 saving for Tigers $40 then you would provide $40 because you reason that one tiger will require $10 of saving then for Tigers will be 10 into 4 which is 40 but if you are functioning on to an effective mode if you are functioning into one of those in a way that you feel good about saving Tigers then saving one tiger is enough for you and so if you are functioning into an effective mode we should map on to an unconscious **reasoning** in this case you say that I don't care I have said Tiger that is the answer and so one time beat one tiger or four Tigers I have saved an endangered species and so $10 is more than enough and in those cases you don't shell out more money so in a deliberative mode you are more prone to giving more money because you do more calculations and it maps or from the conscious part of the brain and so you tend to shell out more money because you do a calculation but if it is the feeling which comes with the decision of saving Tiger if you feel good by just saving Tiger if that is the underlying meaning out of it then whether it is 1 or 4 it doesn't matter and so in those cases you are take a different kind of decision altogether so when we are in a deliberate **decision making** more we value things by calculating 4 is greater than 2 and so saving for is much better than 2 while when we are in an effective **decision-making** mode we value things by feeling help Tigers so saving one tiger is helping Tiger and saving for tiger is helping tiger so whether I say 1 or 4 is basically the feeling which is out of it it is not that some when you say 4 Tigers the feeling will be greater it is not multiplicative but when we are in a deliberative more we understand they're saving one tiger and saving for Tigers are saving two tigers and four tigers four tires is more then two tigers so let's give more money and so their decision will be based on that kind of an imperative or that kind of a system but when in an effective mode once you are in an effective mode it is the feeling which is of interest to you it is the fact it is the pleasure that get when you say what iger's is the kind of appreciation that you get when you say what I got and that is all it matters so it doesn't really matter whether you're saying one Tiger or you're saving for Tigers and so it is saving Tiger which is of importance to us so in deliberately **decision-making** mode as cop increases subjective value increases corresponding when in effective EC and making mode Scoob doesn't matter nearly as much as affected by the presence or absence of first stimulus so scope is basically the sweep of the decision so in the deliberative mode the scope has very less value than in the effective mode it is all about the scope which is our there and so people's **decision-making** then vary in terms of whether they are in the deliberative mode or in the effective mode and their decisions are affected by the kind of feeling that they generate in the affective mode it is the feeling which basically decides what kind of decision that they are going to make or what kind of imperative that you are going to use or what kind of disease and options that you are going to do but in terms of the deliberative mode it is quite different and so this is the last section

onto a course on **cognitive psychology** and in terms of **decision-making** so basically what we did in this particular class is that we saw a extension of one of the popular theories of **decision-making** and this is the descriptive theory so in the earlier class what we did in the earlier lecture what we did we looked at a class of theories which is called the normative of prescriptive theories which work in terms of **reasoning** and rational decision makers or ideal decision-makers and it it is called **expected utility** theory or it is called the multi attribute utility theory both these theories work on the fact that this isn't making is basically done in terms of what is the utility of a particular option choosing a particular option and what is the probability of that option coming in and multiplying this gives us the final basis on which an option should be choosing these are the prescriptive theories or these are the normative theories which are there we saw that human beings since they do not have a lot of information available to them and even if the information is available to them they cannot be working as calculators or mental computers so they make it now choice additionality and this irrationality basically happens because human beings have certain limitations and so we as humans follow a descriptive theory are more worldly theory or **decision-making** and one of these theories are the scene making which is descriptive in nature is something called the **prospect theory** so what is the **prospect theory** the **prospect theory** is a theory which basically goes ahead and says that gains and losses that people have have different values so gains are higher gains I have lesser pleasure lower losses have more pain into it so gains and losses come in different frames they evaluated in different frames and gains and losses are thought of differently also **prospect theory** proposes the fact that losing and gaining are basically dependent on the frame in which a particular option is presented so if a particular option is presented in a gain frame then people's choices will vary whereas if a particular option is presented in a loss frame the people's choices will also vary according to it so reversals of preferences or reversals choices of option depending on which frame the information is presented in gain frame people are more risk-averse if people do not want to take risk people always select certain options whereas when something is in a loss frame people are risk-taking they like risk they take risk and so once something is in a negative frame when in the last frame people want to take risk and put more risk into it and so that is the kind of **decision-making** that they tend to take and then we discussed two other outputs out of it or two other phenomena related to the **prospect theory** one is called **psychological accounting** it says that how you feel about a particular situation how you feel about a particular option makes you decide whether you like that option or not in contrast to whether it is multiplicative or it is whether things are rational or not so it is the basic feeling that people get of our part of it the second the option or the second output to it or the second theory to it was the **sunk cost** theory which says that people always what they tend to do is take the non desirable option they always put more money after the bad option so if even if something cost more people tend to take the more unlikable or the more enjoyable journey because people put more money on to sunk money that is the way that they like so they don't do these calculations given the fact that if they calculate they will always go for the lesser enjoyable trip but they tend to go to the more higher trip which has higher money do it because they think that says larger money has been there let's go with this thing that is one of the things now in the last section of this particular lecture we saw how effect of or basically how getting pleasing or a non pleasing feeling out of an decision basically helps you into **decision making** or basically as you do making decisions and so we found out that when people have a deliberative mode people are in a conscious mode then they tend to do this calculating and they are more helping it is not about feeling that they think about it is not about the effect which decides which help them decide a particular decision which they which doesn't rule the decision but when people in an effective mode the kind of decisions they tend to take is entirely different but and and they are affected by the sweep or the scope of the decision so basically effect then itself has a major role to play now in terms of brain physiology it has also been found that this risk options of framing has been proved or the the effect of framing on **decision-making** has also been proved in terms of certain brain studies so in certain brain studies it has been it was found that people who were

presented options in a positive frame so if some option was given to them in a positive frame and people selected and went in the positive frame people selected those options which was certain the brain activity was very less so in if things were in offer an option if an alternative was presented in a positive frame and people selected certain options people selected those options which were not presenting risk people showed sir brain activity then if they selected those options which presented some kind of a risk on non certain options in contrast to this so basically less activity was there when they selected certain options there then if they selected those opt options which were non certain in a positive frame but in a loss frame it was quite the opposite in a loss frame when people selected those options which were risk producing less brain activity was there then people when people selected those options which were certain so in a lost frame risk-taking led to lesser activity of the brain and more common activity of the brain which means that decisions a more thought true this is answer more like through then in cases in in the other case in the positive frame where certain option selections are selecting those options which was certain which we get trying to which were very definite which are very certain or which were sort of very fixed those in those cases the activity of the brain was less than the wind was much calmer then selecting those options in which it was not in this the options were not certain so in this way human beings make decisions and in this way human beings produce decisions or make decisions and and they differ from calculated machines or economic **decision making** so basically using these processes of **decision making** either using the **prospect theory** or using the other things human beings come up with diseases and then they make this this is an under uncertainty and under risk and based on that when a certain decision fails they reevaluate it make another decision make another choice and then keep on learning from it this is how the higher order cognitive function **decision-making** really works and this is the last lecture in the series on **decision-making** or in this whole course which is of **cognitive psychology** so until we meet again thank

Lecture 30: Course Summary

■Kind: captions hello friends welcome to this last lecture on the course on introduction to **cognitive psychology** now over the past twelve weeks we have been dealing with several areas of **cognitive psychology** starting with introduction to **cognitive psychology** the chapters on **perception attention** then the chapter on **memory** various forms of **memory** then into **language** and thought thinking process **problem solving reasoning** and **decision-making** ending finally with this is in making so this last lecture I have dedicated to basically summing up what we did in this 12 weeks so what I will do in this particular lecture is briefly get back to each section that we did along this 12 weeks and review what we did in all of these lectures and basically also sort of refresh you into what each topic that we wanted to do or that covered so basically it's a review kind of a lecture and it will be very very brief every free touching into our topics that were content of each lecture for each week so the lecture was designed in basically four modules the total scores of **cognitive psychology** comprised of four modules and within these modules we had different different lectures each module had two to three subtopics and they had their individual lectures and all of them comprises two packet of 30 lectures which then was divided into a 12-week period referring back to what is **cognition** so what is **cognition** basically **cognition** is a mental activity and so committee psychology with it studies how these mental events or mental representations which are basically the representation of any event or any object in the external world how they are manipulated and studied and so in this course on **cognitive psychology** what we tend to do is study how these manipulations are maintained is done by the mind or the brain and so that is what we will do in this whole lecture there's an outward or an outline of this particular review lecture so basically this course already psychology comprises a set of four modules the first module that we did had two lectures onto it and the module was called introduction to the study of **cognitive psychology** and it had lectures on the history of the **cognitive psychology** and also on the methodology with it in module two we covered some of the basic cognitive processes which are applied on to mental events on to mental representations and the cognitive processes that we studied in our in this particular course was basically **perception** and **attention** so we will be then evaluating these two or studying or reviewing these two cognitive processes and also I think we redid **memory** onto it the third module was basically organization of knowledge so the first two modules basically tell you the first model basically focuses on what is **cognition** and what is cognitive knowledge and what is the background of **cognitive psychology** the history of **cognitive psychology** and how does it develop what are the research methodologies in it and that kind of a thing the second module covered on two basic cognitive processes how an input is taken and this input which is taken from the various senses and organs how they are converted into meaningful organizations or into knowledge or mental representations so basically it is the pre-processing it is those processes those cognitive processes or those cognitive work benches where draw information which is captured through our primary senses they are pre processed before meaning is generated out of it in module 3 we looked at organization of knowledge so given the fact that the basic cognitive processes are able to extract information from the raw input that is given up by the senses these input has to be safe somewhere at to be kept somewhere in in a particular format which is called **mental representation** and so this third module were basically focused on to how this **mental representation** is stored and where it is stored into what is the format and how it is accessed what are the limits of it and so on and so forth so is basically the organization of knowledge into the human brain and the fourth module was when the knowledge is stored once the knowledge has been attained by the basic cognitive processes and they have been stored or organized into the **memory** how are they accessed and what how are meaning generated out of it or how is this knowledge successfully used is what the fourth module looked into and so what we'll do from now on from the next slide onwards we will take up each of these module one by one and then explain to you what we covered there and how it relates to the next module or the next chapter so under module 1 in the week 1 what we did was we

looked at the history and the study of human **cognition** so the first week which first and second week which are part of the first module there we looked at briefly at the history of **cognitive psychology** how does karate psychology come into being what does it study what are its proponents what are the basic principles governing **cognitive psychology** and after that we also looked at the various research methodologies which are available for doing **cognitive psychology** for someone who's interested in **cognitive psychology** what are the research methodologies that you're going to do so we started a journey into this section of history and study of human **cognition** by studying or by basically reading about the basic debates and philosophy for example the debate about nativism and empiricism or the debate about mind body problem not these debates about empiricism an atavism or mind body problem or nature-nurture issue they describe or they form the basic core of their **cognitive psychology** comes in because it was the philosophers before the coming of the real science of psychology that they wanted to study what is the difference between the mind and the soul and what I wanted to study what is the mind and they were interest these philosophers are the ones who were interested or on how human beings do things that they do so they had no idea what the mind is or the brain composed of and at that point of time they wanted to study why individuals why humans are different and so they proposed these theories now for example the empiricism and nativism theory they emphasizes one of it emphasizes the role of heredity in producing knowledge or in gaining knowledge and the other emphasizes the fact that human beings are born with a blank state of mind but later in life through experience they develop knowledge so knowledge is not something which is passed on from heredity it comes on through the interaction with people interaction with the environment so these debates basically move forward the idea of what is the basis of **cognitive psychology** further on we went ahead and studied the basic schools of psychology we looked at the basic schools of psychology schools like **structuralism functionalism Gestalt** school the school of **behaviorism** and the cognitive school and all these schools what they did was they provided us the basis or the reason how **cognitive psychology** developed as a field so these schools the study of these schools gave us an idea of what is **cognitive psychology** what is the content of **cognitive psychology** and how these schools mattered in the coming after the field of **cognitive psychology** especially the **cognitive revolution** the idea of qualitative regulation or the creative school they for the first time introduced the fact that **behaviorism** believed that stimulus response is how a particular behavior is related to a particular kind of a stimulus and so **behaviorism** believed that there is nothing called the mind there is nothing called the brain there is nothing called mental activities or mental events and so each response that a person does each behavior the person does is directly related to the upcoming stimulus or to any external or internal stimulus and so for the first time when the **cognitive revolution** comes in people started focusing on the Oh in SR and so the over introduced which basically meant that stimulus s do not directly respond to directly response attack reproduce behavior but rather there was an organism or there were a human mind which controlled responses so stimulus responses were not kind for mechanistic rule but in an organism or a human mind controlled the response and this is for the first time the karate Revolution produced the idea or the cognitive school of thought produced the idea that the human mind is responsible for producing all kind of inputs further on we looked at the definition of what mental events are we looked at what are mental representations for example what mental representations are mental representations are basically representations or these are **encoding** of any event or any person place or thing into the human mind so anything that happens in the external environment has to be steady in the brain in a particular format in the mind in a particular format and that about mental representations are and so these representations are have a format and a particular kind of **encoding** so the format describes what the **mental representation** in which way it is encoded and the content of mental represented describe what it wants to say and we explained it in that chapter in terms of both the special as well as the pictorial definitions or the special and the propositional kind of representations of how particular information can be represented in the human brain so that is what we did in this section further than that

we also looked at several other schools of thought or several other paradigms which went ahead and described how **cognitive psychology** progress for example we looked at the evolutionary school we looked at the connectivist school the **information processing** paradigm and so on and so forth in addition to that we also looked at why study the brain if you want to study soccer **cognitive psychology** and the **reasoning** that was provided there was because study of the brain for studying for very psychology providers not only identify ability effects but also it gives us the adequacy effect it basically means that studying the brain will tell us what functions are responsible for or what kind of mental activities are a small related to what kind of brain activations and that way our relation can be mashed up in the second week we went ahead and studied theories and researches in human **cognition** in this particular lecture we looked at the various research methodologies that has been used for creating or studying **cognition** and the three primary research methodologies that we studied was just the methodology of behavioral approach the correlational approach the casual approach and the modelling approach now in the behavioral approach we primarily use three different type of measurements the first measurement being the measurement of **accuracy** the second measurement being the measurement of **reaction time** the third measurement being the measurement of **judgment** and these three measurement or these three criterias can be used to the to measure or to design experiments in **cognitive psychology** besides the behavioral methodology for studying **cognitive psychology** there is also something called a correlational method of studying **cognitive psychology** or cognitive phenomena or mental events and this correlational methodology uses the input from devices like such as the electrons of the graphs or the mental the MDG the magnet or insula graphs for the the magnitude in circular graph of the electroencephalograph they provide us brief idea about what happens at the brain when a particular mental event is being processed or particular mental event is being taken care of in addition to the eg & M eg there are devices which are called the positive emotions tomography damar eyes NF MRIs all of which provides us what happens or tells us what happens at the brain when a certain mental activity is happening and so they gave us kind of a correlational structure so if some change in a mental event happens correlational change or us simultaneous change also happens in the brain and these equipments provide us the idea of what happens into the brain when a particular cognitive activity happens beside that explain to you the causal network where we looked at three methodologies the methodology of using neuro physiological studies in which patients were studied for finding out what are the reasons for or how is brain related to the particular cognitive activity also looked at an equipment called the **TMS** which can specifically inhibit certain areas of the brain and study how a particular area is related to a particular mental activity and a third way of using drugs or certain kind of inhibitors neurotransmitter inhibitors for studying or inhibiting certain kind of brain region and studying the function of that brain region in addition to that the last method that we use in this particular section was the method of modeling in which what we studied is there are different computer models can be generated and these models are two in nature the box model and the neural network model and both of these models can be used to basically represent the mental activity of how a mental activity gets processed or how a mental event gets processed so this is what we did in the first two week and this is what we studied in the first two chapters in victory we looked at object **perception** and recognition and so in this particular section in object **perception** and recognition what we looked on - or what we studied was we looked at both the classical and the **gestalt approach** to **perception** and in the classical approach it was said that there is something called distal and **proximal stimulus** and that is how **perception** really happens so basically the idea of this chapter was to study how perceptions are developed or how information which is passed on from the sensation that is taken into the brain and that is coded into the brain to the process which is called **perception** and so two theories were studied the classical theory said that whenever something and we basically studied the visual **perception** we specifically studied the visual **perception** although at a later point we also described a little bit of auditory **perception** but we focused ourselves into visual **perception** because that is the easiest sense to study and so in terms of visual

perception we studied the classical theory which says that there is something called a **proximal stimulus** and there is something called a **distal stimulus** and these two stimulus is interact together to form images onto the retina which are further carried on to the specialized object recognition area or the brain in the occipital lobe and from meaning is generated we also looked at certain theories of other object **perception** for example the bottom up in the top-down theories now within the bottom of theories we looked at something called the **template matching** theory we also looked at something called the **feature analysis** theory and the **prototype** matching theory the difference being that in **template matching** theory it proposes that there are huge number of temples in the brain and that helps someone in recognizing a particular object in the external field of vision whereas the **feature analysis** theory goes ahead and says that object recognition is not based on these kind of human templates the reason being that so many templates cannot be stored in the brain so it is instead of the number of templates stored in the brain it is the pattern it is the feature of a particular object which is of use for doing object recognition whereas the **prototype** theory says that nine of the **feature analysis** truly not nor the **template matching** theories enough because so many templates cannot be stored in the brain and neither the **feature analysis** Theory tells us what a feature could be so the **prototype** model says that human brain creates a **prototype** and this **prototype** is what actually goes ahead and helps us in doing bottom-up processing so what is what am i processing in bottom-up processing we start with the very basic knowledge we fought with the very basic inputs which are coming from the visual senses and from there we try to do the object **perception** in a position to the so coming from general to specific or looking at a number of information which is out there and combining the all this information to understand what an object is is called **bottom-up process** so it is from bottom of the pyramid going towards top of the pyramid in a position to the bottom of process there is something called the **top-down process** and in the **top-down process** we actually studied how the experience past experiences of human beings help us in object on identification in within the adoption process we studied something called **perceptual learning** studied **change blindness** and word security effect all of these displays that the human experience is enough in creating **perception** and lastly in this section on **perception** recognition of the **perception cognition** we study something called **direct perception** which is the theory which is directly opposing to the theory of classical **perception** and **gestalt perception** so what does the theory say it says that human beings don't need to do anything onto the incoming stimulus rather the incoming stimulus is has enough information the incoming light from an object in the external field has enough information when it falls into the retina and that is enough to create any kind of any **perception** which is needed in this fourth week we ventured on to the chapter on **attention** and attentional processes and **cognition** and so in this second section we specifically select studied **selective attention** although so what is **attention** basically **attention** is focusing yourself into a particular information so focusing your senses into a particular information and we looked into the **selective attention** within the **selective attention** we discussed several theories which are our **selective attention** for example one of the theory that we discussed was the **filter theory** we also looked at an attachment theory of attenuation we looked at the **late selection theory** we looked at kanbans theory and **schema theory** now all these theories differ from each other in certain ways for example plate selection theory says that the bottleneck that is there in terms of **attention** is not at the start of attentional process but at a later part of an attentional process whereas the **filter theory** suggests that whatever calls whatever information falls under the perceptual system it is filtered right there at the beginning of the attentional process the blade selection theory says that this filtering happens at the end of it and there are several **reasoning** or there are several line of experimentation which has been provided for the support of these theories similarly the **attenuation theory** says that nothing is filtered out in an attentional filter what happens is only the volume of information which is available is toned down similar to that there is cannulas theory which talks about alertness and certain propositional requirements and certain other kind of a motivational needs which all come together to decide how **attention** is focused on to something and the

schema theory is direct victory which basically goes I didn't say is that whatever is not perceived is never looked on - and that is the reason why **attention** is an all-or-none kind of a phenomena beside that we also looked at how **attention** becomes on what conditions attentions become automatic and how this autumn a city of **attention** actually helps people it was the end of the chapter we discuss something called the **psychological refractory period** it basically suggests or which basically is the time gap or the time required for somebody to process to incoming informations so if two incoming informations are given to someone who's processing up already so two incoming informations occur simultaneously to a person this person will be delayed in replying to the second information and this delay is what is called the psychological refractive period in the fifth week we looked at **encoding** and retrieving of **memory** traces and so in this particular week we looked at what is human **memory** so we discussed the popular **modal model** of human **memory** which talks about **sensory memory** then the **short-term memory** and **long-term memory** beside that we also looked at different kind of sensory memories which are available for example the **iconic memory** and the equation brain and also the haptic **memory** we looked at what is STM and we looked at the various features of STM before that we also discussed in detail what are the features of an icon and what are the features of an echo and how they are distinguished and we discuss several experiments of what in equal what an icon is of a **sensory memory** in terms of the **short-term memory** we looked at what is **short-term memory** what is this capacity what it can do what it cannot do part of the kind of **forgetting** that it it happens from **short-term memory** the coding in **short-term memory** which is basically in terms of the the **semantics** in acoustic coding we also looked at how **forgetting** happens we discussed two experiments on Ron and Peterson in Vega Norman of how **forgetting** actually happens in short-term maybe towards the end of it this section we looked at what something called **working memory** so **working memory** which is an improvement on short-term memories and we looked into that so what is basically **working memory** it is an ruminal **short-term memory** whereas the preposition goes that **short-term memory** is a store which is non dynamical in nature **working memory** proposes a store which is more dynamically nature in the sense that it has three different parts it has something called a **central executive** there's something called a **phonological loop** it has something called a visual spatial sketchpad and so as against the fact the **short-term memory** is not directly connected to long term **memory** it cannot directly interact with long term **memory working memory** at all points of time are connected to the long term **memory** so something called the episodic wafer and I provided you reasons of why this improvement was necessary and what this improvement actually meant so this was the first part of selection in the second part of lecture we looked at what is **long-term memory** so in the first part of the lecture we looked at what is STM what is **sensory memory** the coatings are **forgetting** the variables which are affecting it and so on and so forth in the second part of this lecture on lecture 5 between lecture 5 and lecture 6 there was another section on **long-term memory** where we look specifically on what is **long-term memory** and how does Kody happen in **long-term memory** so basically the coding that happens in **long-term memory** is semantic in nature we also looked at the various **forgetting** to use of long term **memory** and some popular theories that we discussed was a **decay theory** the **interference Theory** the poor **encoding** theory and so on and so forth we also discussed what is the capacity of **long-term memory** and so what it can store how it can store what it can process and what it cannot we discuss two basic types of recoil method two basic types of the tribal method from **long-term memory** so if an information is stored in **long-term memory** how is it pushed back or how is it Detra back and we discussed recall and recognition being the two method of retrieving information from **long-term memory** recall being a method in which a person has to remember from his **memory** and write something back in recognition some distractors are provided with the correct answer and the person has to identify the correct answer and discriminate it from the distractor so these are the two methods of each other we also looked at something called context effect and state effects in **long-term memory** and we showed how varying the context and keeping the context same while

encoding in **long-term memory** and while they travel in **long-term memory** the benefit of **memory** is seen we also looked at the physiological state the person days or the mood the person is also helps in better remembering of **long-term memory** or worse remembering of **long-term memory** we also here included the distinctions of **long-term memory** in terms of the declarative and procedural types so here itself we saw how **long-term memory** is defined in two parts the relative in the procedural type in the declarative type the old memories are crap which are conscious in nature whereas in the process of procedural type the implicit memories are kept and we also discussed how this **declarative memory** is further broken down into its semantic and episodic form and how the procedural memories are broken down into its classical conditioning habit and priming in several other forms of **procedural memory** so this is what we did there at the end of this section we looked at reconstruction of **memory** or how high **memory** reconstruction is there how false **memory** is formed and what is called eyewitness testimony so basically what we understand is that **memory** is never true whatever we store has some form of reconstruction onto it and so at the end of section we looked at the experimenter by lock test and several of the experiments for example the DRAM paradigm the DS door dragon egg timer paradigm which basically shows us that the **memory** the information which is kept in **memory** is never accurate it is never a representation of the events that have happened and it is always a reconstruction which is there and we try to focus here or we try to tell here I try to evaluate and show to you that **memory** is never true it is all it is always a reconstruction of the the information or the manipulation of information which was originally stored in the next section of **memory** in general knowledge what we did was we started the section with an introduction to the distinction between the episodic in the semantic form so in the last chapter we saw how **long-term memory** is broken down into it's declarative and procedural format and within the procedural format we have the cementing in the episodic store so this section started by distinguishing between what is **episodic memory** and what is **semantic memory** and this section was particularly dedicated to **semantic memory** so **semantic memory** is that **memory** which holds in knowledge information of all the fact and things like that daily routine events all those information are stored into something called a semantic store or semantic a part of the **long-term memory** whereas in the episodic form or in the episodic part of the **long-term memory** we have events life events which are stored and there are several different types of **episodic memory** which are there and so our Associated memories for example there's something for the autobiographical **memory** there is something called the flashbulb **memory** and we discussed that at the end of the last section in this particular section we looked at **memory** for general knowledge which is **semantic memory** so we looked at what is **semantic memory** and what are the different models of **semantic memory** the particular models that we discussed in this particular chapter was the hierarchical semantic model there are the HSM and the HSM what we found out is that the various knowledge is which are stored in **semantic memory** they are connected to hierarchies or they're connected by something called **nodes and pointers** and so a new information has a **superordinate** node and **superordinate** the way information is coded is in terms of categories and concepts and so these concepts and categories are how information is stored on to one another and this the **hierarchical semantic Network** model basically explained how information this general information is stored as hierarchies within the **semantic memory** beside the hierarchical semantic model we also looked at something called the **feature comparison** model which was an improvement on the hierarchical semantic model and said that since the there was some problem the criticism with a director semantic model in terms of **verification** of sentences **verification** of information which is present in higher the semantic model the **feature analysis** model says that information is not stored in terms of hierarchies rather it is stored in terms of **feature analysis** so based on the core features and based on the characteristic defining and the characteristic features information is actually stored into this hierarchical networks or information is stored into the **semantic memory** we also looked at something called the **spreading activation** model which says that whenever one **concept** or whenever one particular information bit is excited this

information this excitation or this kind of a **retrieval** is spread on two concepts which are related to each other and how multiple concepts which are somehow related to each other are activated when particular node is activated so how does the spreading happens or how does energy spread from one node to the other node so these are some of the models that we study into and beside that we also studied the AC T model the Andersons is it the model which talks about three different kind of **memory** system in the **working memory** system the **procedural memory** in the **declarative memory** and what this model said is **declarative memory** is entirely different from priscila **memory** in the sense that in **procedural memory** we have some kind of propositional thoughts and propositional structures whereas the **declarative memory** the common form of **memory** is looked into and the **working memory** was a connection between these externality procedural type of **memory** so this was an end to this section on **memory** of general information in the next section was based on defining what are concepts and categories and so what why we wanted to study concepts and categories is because in the earlier section on **semantic memory** we saw that the node consists of concepts and categories so what does **concept** concepts are basically representations of worldly knowledge so however something is represented what is the way in which particular information is represented in the human brain or human mind is what concepts are and categories are similar concepts or things which have similarities together how they are clubbed together is what is called categories so we looked at how categories and concepts are formed we also looked at the various nature or various models of **concept** for example we looked at the **classical view** which says that how the way in which concepts are form or how we categorize things together is in terms of the fact that the more closely two things appear together identical they are clubbed together in a position to this the **prototype** view **concept** in **categorization** suppose the suggest that there is an abstraction which is made out of it because it happen that two things may not be similar but they are clubbed together for example a bird which doesn't has feather will will be more of a bird than any other kind of an animal and so this kind of violations led to the proposition or the **prototype** model which says that people make abstractions people when they look into a certain kind of a **concept** or certain kind of an object free people when forming categories they make abstractions of what the category element is so from that **prototype** they actually go ahead and then include other members into that particular category then we came to something called the the schematic view an **exemplar** view in the **exemplar** view what we looked at is that these prototypes that we talked about so basically what people do is when making **categorization** they look at all the elements which are present in the category and from that they they create an abstraction they look at the commonality and based on the common oolitic they create an abstraction now in the **prototype** view the idea was that this abstraction may or may not exist in real life but the **exemplar** view suggests that the abstraction that we have made in the **prototype** view does exist in real life so when we think about fruits I am thinking about a particular fruit maybe an apple and so this is the **exemplar** view it says that when I say the category fruit the idea that fruit is something that is eerie that you eat it has seeds it as a particular taste and so on and so forth but you also think about a particular fruit for example in orange or apple and so on and so forth and so this is what this particular **concept** of **exemplar** view suggests in addition to this this chapter also explained how concepts are really formed what are the way in which concepts are formed and so in this we discussed successful scanning conservative focusing and some other methods of forming a **concept** beside that we also looked at how people acquire prototypes and once they acquire prioritize how do they use to form this kind of a knowledge system that we have discussed in the semantic that is useful in forming and organizing **semantic memory** in the eighth week we discussed on to something called the visual and spatial nature of **memory** and so in this we looked at what is visual **memory** so visual **memory** is mental imagery it's basically similar to the fact that when I ask you to think about something by closing your eyes and when I ask you to imagine something what do you imagine so the same kind of thought the same kind of imagination that you do is what is called visual memories we also looked at some mental mnemonics which are useful which are mental shortcuts or

which are mental aids which help us in remembering things and some of the mechanics that we discussed here where the **method of loci** the method of **interacting images** and the method of backwards now what this pneumatics tends to do is they tend to organize information in such a way in such a visual manner that people tend to remember them better and this remembering them better makes their **memory** better or makes better **encoding** of their **memory** in addition to this we looked at two propeller theories of the visual **memory** of how why visual **memory** is better than any other kind of **memory** and the one method or the one theory was the **dual coding hypothesis** and the other theory was the relational organization hypothesis where is the **dual coding hypothesis** suggests that whenever something is committed to visual **memory** it has two codes and auditory code in a visual code and since it has two cores it is better remembered the relational organization hypothesis say is that if the better learning is not because of the fact that any **memory** which is committed to visual **memory** has two codes it is because of the fact that there is a relation established between the number of **interacting images** and so number of images if an image is taken and if two two things have to be remember two bits are for three bits of information has to be remembered and if you make imagine of these three bits of information and make them interacting together the interacting image is better remembered then separate images and so this is what the theory goes on and says and then we looked at theories of **mental rotation** which basically goes ahead and says that it is not only that we store static **visual imagery** we also are able to do mental manipulations rotations from mental images and there are several theories out there there are several experiments to prove that people are able to mentally rotate an image mentally rotate and particular information or **mental representation** in addition to that we looked at what is the nature of **visual imagery** and we discussed some of the some of these nature for example the nature of implicit **encoding** perceptual equivalence special equivalence and so on and so forth and this is where we ended the section on visual and spatial **memory** the next section was about studying **language** or understanding what **language** was and in this section we started by first identifying what is **language** and how it is different from communication so **language** has a particular format it has a particular **grammar** it is arbitrary Nature has a particular structure and then it can be used to create new sentences whereas communication is bounded by certain limits so **language** and communication differs in the fact that communication is restrictive in what information it can pass but **language** is not restrictive because it has several features which helps us in to communicating a number of ideas across a number of people and it is the universal method of combining two people or multiple people together we also looked at the structure of **language** here now in the structure of **language** we looked at how Western **language** is nearly built so we looked at what is the **basic level** the phonemes level the morphemes level the level of the **syntax** the level of the **semantics** the level of the **pragmatics** and so on and so forth so at the funding level at the word level how information is we looked at the **phonology** and **phoneme** level from there we motored morphology where the basic speech sound are combined together to give the first words and then or the first sentences and from there we came up to the level of the **syntax** which is how sentences are written in in particular **language** for example we discussed the English **language** here so we looked at the **grammar** or the **syntax** so our sentence can be completely syntactic but it may have no meaning and so generating meaning out of a sentence is what is called **semantics** so **semantics** is the study of meaning of a **language** a **language** could have a perfect way or a perfect **syntax** which means that it could be grammatically correct but it could produce no meaning and so this production of meaning is what is **semantics** from **semantics** to moved on to something called **pragmatics** which is the social rule of how our **language** should be used and so this is what we did in this section other than that we looked at how **language** comprehension happens and with **language** comprehension we also looked at how text comprehension and sentence comprehension happens now in **language** comprehension it is believed that **language** unlike text is continuous in nature and so changing one **phoneme** can change some can change something else so **language** comprehension is basically categorical in nature in the sense that people form categories and

they understand the **language** which has been broken since **language** is not broken down whereas in text **perception** what really happens is that there are something called circuits and fixations of the human eye that leads to **perception** of text in terms of processing sentences sentence **perception** happens in formal just way so people read a particular sentence or a particular paragraph and generate a list out of it are generated gist out of it and gist is what is remaining with people so meaning is or just is what remains and all other words and meanings are taken away from it so this is what the **perception** Apple when we looked at several errors in speech production different different kinds of errors which can be there and how they are compensating for **language** or how they show what languages is or capable of and what is or not capable of in the end of it we looked at something called story grammars which are basically how stories are perceived so any story **grammar** is basically a way in which a story is made and how it is read and how the meanings are generated so story grammars are basically a structure of any story and so this ended the section on human **language** skills next came the section on thinking and problem-solving and so in this particular section we started by first defining what a problem is we looked at two different kind of problems the problems of the ill-defined and **well-defined problem** and the routine and non-routine problems we looked at both this kind of problems which other out there and what is the difference between that we also looked at different kinds of problems which exist for example we have the analogue assess the arrangement problem the divergent problem the transformational problem the arrangement problem and the induction problem and so on and so forth so we looked at different kind of problems which could be out there we also looked at the approaches to the solution of problem for example the behavioral approach where we looked at trial and error is the reason of how a problem is solved and we also looked at the **gestalt approach** in which it is the phase of induction and the phase of waiting which leads to the final answer so a stage where people just don't do anything and they wait and that leads to the I experienced in that gender is the solution we also look at various variables which affect **problem solving** for example problem representations it could be mental said **functional fixedness** or zero types we can help in the formation of ill-defined problems or how what can the errors be there in problem formation we looked at various strategies and problem solutions for example we looked at the algorithmic strategy we looked at the strategy of heuristics and we looked at the means-end strategies so there are different strategies how a problem can be solved and at the end of this we looked at what is **creativity** what are different kinds of **creativity** which is out there and we looked at two dimensional view of mapping **creativity** next section that we discussed was a section on **reasoning** and in this section we looked at how the **reasoning** and **decision-making** the last two sections are basically sections in which we looked at the higher order cognitive processes up till now up till before **language** or till **language** it was the basic coronary process which was taking information from **perception** sensation **attention memory** and they were creating raw materials from the sections on thinking when problem-solving **reasoning** and **decision-making** interpretations are made of these mental representations and so in **reasoning** we looked at what is **reasoning** so basically **reasoning** is an evaluation of conclusion which is given which is based on a given meaning so we looked at what is **reasoning** and how the **reasoning** is used or how evaluations are generated beside that we looked at different kinds of **reasoning** so the two kinds of **reasoning** which are there which is called **deductive reasoning** and **inductive reasoning** within the **deductive reasoning** we have the **syllogistic reasoning** and we have the **conditional reasoning** we also looked at **inductive reasoning** which is another form of **reasoning** how we come from general to specific so this is the kind of differences which we looked into we also addition to that we also looked at several criticisms of these **reasoning** processes in addition to that we also looked at what is **judgment** how does **judgment** really work and what is the meaning of **judgment** so basically **judgment** is **reasoning** the applied to a given information for generating a particular conclusion we looked at several errors in **judgment** into this section for example some of the errors that we discussed was the error on availability or Esteve labelled an error on the representativeness and the **anchoring** in what these errors

really mean is that these errors lead to faulty judgments or lead to inconclusive judgments so **availability heuristic** a Houston which leads to improper **judgment** which basically means that we trust people trust their **memory** more and based on that they make the judgments similar to that the representative curious gives the fact that since a particular information a particular line of omission which is available from **memory** since it represents or it it is very representative to a certain kind of category people make these kind of **judgment** errors and similar to that is the **anchoring** in effect or **anchoring** error when Indian dition to that there are several other biases which can appear in **judgment** for example the hindsight bias the bias of illusory correlations or the by of conjugation fallacy and several other kind of errors it's there in the last section we discussed something called **decision making** so this is the last process which happens after **reasoning** in a **judgment** once **reasoning** and **judgment** is there a number of options are available to choose from and so this isn't making a process where we go ahead and look at the number of options which are available to us and make a final choice now whenever we they make this choice this change is always under certain kind of risk and certain kind of uncertainty and so as humans we always wish to make the best choices which are there and which is rational so what is the rational choice the rational choice is the one which gives you which maximizes your profit but minimizes your loss and that is what we tend to do so this is what our **decision making** was all about now we looked at two different ways of making decisions first was the normative view where we looked at something called the **expected utility** theorem and the violations of it now in the **expected utility** theorem we looked at what is **expected utility** and we calculated that expected you or the **descriptive model** of **decision-making** suggests that **decision-making** has to be made in terms of what is the utility of particular option and what is the probability of that option happening and we looked at this kind of a calculation and the option which gives us the maximum utility and has the maximum probability of occurrence is the one we select and we looked at that this is how a rational decision maker should make the decisions but with humans enough information is not available or information is not available and so there tend to make irrational choices or irrational interpretations so what they tend to do is they trying to do violations of this **expected utility** by doing something called **preference reversal** or aversion shifts which basically means that if they choose an option one in a particular situation they keep on reversing or they keep on shifting the choices in a situation to situation 3 and so on for whereas the **expected utility** Theory believes that if option a is chosen and the condition 1 the say option should be chosen under n number of different conditions which stress to occur and so we looked at some descriptive theories of **decision making** the theory that we looked at is called a **prospect theory** which basically says that the gains and losses which comes out of a particular decision are mapped on to different dimensions larger gains has show smaller increase or smaller increase in feeling whereas smaller losses show higher increase in pain and so losses and gains are interpreted differently by human beings also the fact that not only these gains and losses of how it is the way a particular game and our particular option is presented to you whether it is a nickname frame or a loss frame well decide what option are you going to choose from a number of options which are given to you so given the fact that a particular option is given to you in a gain frame people choose your bets but if something is given to you in a loss frame people are more risky so people are more risk aversive in a gain frame whereas people are more risk prone in a loss frame beside that we also looked at something called two phenomenon of **psychological accounting** and **sunk cost** effect which are basically demonstration or extensions of what **prospect theory** suggests it suggests that people to throw good money after bad money which basically means that people value those options which have taken a large part of their money and but still are unenjoyable whereas they do not go for better options options which have taken in lesser when you are but are more enjoyable similarly when people have to spend on things which have been assigned a certain psychological account people will not like to invest onto those options whereas if some particular loss or a particular gain has not been psychologically accounted for people would like to spend more money on to it the last section we looked at how effect it interacts with **decision making** and

what is the role of effecting **decision making** and when they looked at there are two modes one is the effect mode and the other is the descriptive mode in the descriptive mode people more conservative more people are tend to think in non effective way and tend to donate more whereas in the effective mode people make decisions in terms of how they feel about it and so they are more conservative and they are influenced more by the sweep of or the scope of the particular effect so this in total is the number of things that has to be studied that any course on **cognitive psychology** has to happen and as you can see we started our journey from the history of community psychology moved on **perception attention** then **memory** several forms of **memory** then **language** thinking and **problem solving** this is in making visual memories and so on and so forth so this course was an attempt for from my side to basically introduce you to a very basic knowledge of **cognitive psychology** although cocaine is like oh this is just an introduction and this is just the starting point of a **cognitive psychology** any course of **cognitive psychology** each chapter of this section can be expanded further and can be studied further into more basic fields and once you do a course on **cognitive psychology** the other fields that opens up are cognitive neuroscience or fields of neuroscience cognitive technology and so on so on and so forth so the whole field of cognitive science then opens up with you because security psychology is an inherent part of any cognitive program or brain and behavior program in any Western University or any University is out there so I hope that you enjoyed this whole 12 weeks ride with me enjoyed each part of it there were points at which you may not have understood you're free to some concepts you are free to write back to me and consult with it I'm probably sure that you enjoyed this whole write for 12 weeks I'd like to then say a final goodbye to you and hope you do good in your exams goodbye and thank you